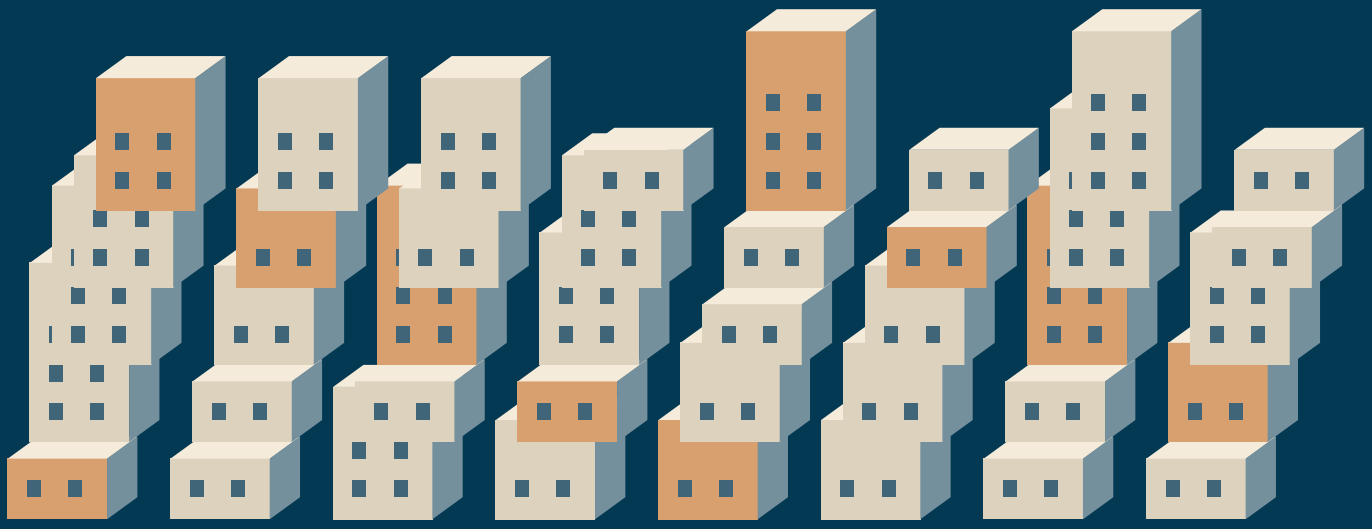
HOUSING THE NEXT BILLION

Land, finance, institutions, and technology for affordable housing



ABHAS JHA

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Revised discussion edition · 28 October 2026

About this discussion edition

Housing the Next Billion examines affordable housing as a coordination and information problem embedded in political economy. It connects land, planning, finance, infrastructure, construction, and long-term management, with AI and digital technology assessed against specific decisions and their effects on residents.

This discussion edition, revised on 28 October 2026, contains an introduction and twenty-five chapters. Comments, corrections, criticism, and suggestions are welcome. Practical procedures and transfer tests are my proposed applications of the evidence unless identified as findings of a cited study. Stakeholder tables set out interests and risks to investigate locally, not established motives of named people or institutions.

Affordability is assessed through housing payments, transport, services, maintenance, and access to livelihoods and opportunity. The proposed measure of safe, affordable, occupied home-years means years in which stated safety, service, affordability, and occupancy conditions are met. It is an analytical proposal, not an established universal statistical standard.

The contents link to chapters. PDF bookmarks also identify sections and source notes. Bracketed citations refer to the source notes within the same chapter; numbering may therefore repeat across chapters. Source links lead to the cited publications. The Word edition preserves editable text, headings, and tables.

Authorship and publication notes

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Use of a large language model. I write with a large language model in the loop as an editor, researcher, and thought partner. All judgments, errors, and omissions are mine.

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Introduction: how housing systems work

1 Define success by the lives households can sustain

Affordable housing policy should be judged by whether people can obtain a suitable home, use the opportunities around it, and afford to remain there. Delivering this outcome requires land, infrastructure, construction, finance, and long-term services to work together. Governments also need reliable information about household needs, rights, costs, and results. My argument is that coordination and information problems are embedded in political economy: the people who control decisions do not always bear the costs of delay or receive the benefits of wider access. I came to this argument through programmes in which delivering houses and improving lives were not necessarily the same thing.

My early experience taught me how easily a programme's construction target can displace the preferences of its occupants. I began working on affordable housing more than three decades ago, implementing Indira Awaas Yojana in India in the early and mid-1990s. I remember tribal families who wanted to build mud houses that they regarded as beautiful and comfortable, with much more space than the scheme's standard provision. Their proposal did not get through the local approval process. That may have been a local error; I do not treat it as evidence of a national prohibition. The broader problem was measurement: completed houses told us little about satisfaction or changes in health, education, and livelihoods. My later visits to Chennai raised a related question about what housing programmes enable families to do over time.

The neighbourhoods I visited in Chennai made me question why serviced plots had lost support as a housing approach. In Ambedkar Nagar and Arumbakkam, people had built homes and businesses through sites and services, creating vibrant neighbourhoods. Yet a later visit cannot reveal how long residents endured poor access, who left, or which institutions would be needed to repeat the result. Chapter 3 examines those qualifications as well as the achievement. This combination of observed progress and unresolved implementation questions shapes how I assess the cases throughout the book.

I write as a practitioner concerned with policies that ordinary institutions can deliver repeatedly. My work has taken me from public administration in India to urban development, affordable housing, and digital technologies, with a focus on how technology and AI can improve services for poor households. I am interested in how policies are funded, approved, procured, and maintained through changes in staff and leadership. I want governments to deliver at scale while families retain meaningful choices about where and how they live. That requires a definition of affordability broad enough to capture what families actually pay and receive.

Affordability includes recurring costs, usable services, and access to livelihoods as well as the price of the dwelling. Add housing payments, utilities, transport, maintenance, and fees in a household cash account. Record travel and care time, service interruptions, and uninsured risks separately; do not add hours to currency or count the same repair twice. Test irregular incomes and seasonal expenses, and show residual income after essentials by household type. If

time or risk is monetised, disclose the assumptions and also show the result without monetisation. The next question is whether the nearby opportunities are ones the household can actually use.

Physical proximity is useful only when residents can access the jobs, services, and social support they need. Check skills, travel costs, route safety, school admission, disability access, opening hours, childcare, and informal work. A locally chosen 45–60-minute travel screen is not a universal entitlement threshold or proof of opportunity. Assess whether households in the bottom 40 percent can enter and remain, including deposits, documentation, discrimination, displacement, and access to shared facilities. Track people who leave the project boundary as well as those who stay. These household-level tests help interpret national and global estimates without confusing different forms of housing need.

Headline housing totals describe different problems and cannot be added into a single measure of demand. UN-Habitat's World Cities Report 2026 estimates that up to 3.4 billion people lack secure, safe, adequate housing, including more than one billion in informal settlements and slums. It reports a deficit of 288 million units in 2023, up from 251 million in 2010, and estimates that 44 percent of tenant households spent more than 30 percent of income on rent in 2023. The last denominator is tenant households, not all households. [1,201] Turning these measures into a programme requires local estimates of household formation, replacement needs, and usable supply.

Local demand must be estimated by household type, income, tenure, and location. India's urban population is expected to approach 951 million by 2050, while a 2025 World Bank assessment estimated a need for more than 144 million new homes by 2070. UN-Habitat's May 2025 summary estimated Africa's deficit at roughly 70 million units and reported that 53.6 percent of urban residents in Sub-Saharan Africa lived in slums. [3,4] Such estimates do not identify what students, seasonal workers, extended households, older residents, or people unable to form independent households can afford. Separate new demand from replacement needs and verify whether vacant homes are usable. The response must address adequacy as well as quantity.

A low price cannot compensate for unsafe, inaccessible, or unsuitable housing. Assess secure tenure, basic services, habitability, accessibility, location, and cultural suitability, including crowding, ventilation, damp, sanitation, safe energy use, and room for care or home-based work. These dimensions follow the international adequate-housing framework; local legal duties and technical standards still require review. [209] This book focuses on urban and peri-urban systems, especially in Africa and India. Rural housing, emergency shelter design, structural engineering, and jurisdiction-specific legal practice require specialist guidance. Within that scope, the central task is to identify why the inputs needed for adequate housing fail to come together.

2 Diagnose the institutions and incentives behind the constraint

Housing interventions work only when their complementary inputs are available in a feasible sequence. Land without roads, water, and drainage needs further investment before occupation. Infrastructure without development rights may remain underused. Mortgage liquidity cannot help a household that fails income or collateral tests, while a construction subsidy may

raise land prices when supply is fixed. Examine land, planning, trunk and local services, construction, producers, household and developer finance, allocation, tenure, and maintenance together with transport and jobs. The binding constraint will differ by city and household group, and removing it may expose another.

Information gaps raise costs when institutions cannot verify rights, capacity, payment ability, or performance. Developers face uncertain titles, approvals, and demand. Lenders spend money checking informal earnings and collateral. Households need to assess products, developer promises, and future service charges. Governments may lack reliable records of occupancy, buildings, tenancies, and utility capacity. These gaps can reduce formal transactions and leave fewer observations for estimating prices and risks. Better information can help, but only if the actors controlling the relevant decisions have authority and incentives to respond.

Political economy determines whose costs and claims enter housing decisions. Existing residents can vote, attend hearings, and organise locally, while prospective residents often have weaker channels. Owners may seek to protect amenities and property values; tenants may fear displacement; municipalities may resist growth that adds service costs without revenue. Identify these interests through contracts, budgets, decisions, and separate interviews rather than assume motives. Record who gains from development rights, infrastructure, subsidies, and maintenance spending, and who can challenge the allocation. These conflicts also explain why coordination cannot substitute for a decision about public resources.

Some housing constraints require subsidy, redistribution, infrastructure investment, or a different site. Coordination cannot make a very poor household pay the full cost of adequate housing, and information cannot remove a physical water shortage or make an unsafe site suitable. A housing assessment must identify who can make the required choice and who will pay for it. The same discipline applies to affordable plots, rental rooms, redevelopment, and homeownership. Choosing among them requires attention to the institutions and markets in which they will operate.

Africa and India offer comparisons of mechanisms rather than a single transferable housing model. India combines national programmes, domestic capital markets, a large construction sector, and digital public infrastructure, but local land administration, staffing, services, and rental markets vary widely. Many African cities combine rapid growth with limited mortgages, small municipal revenue bases, incomplete land records, and extensive production by households and small landlords. A policy requiring salaried work, undisputed title, and deep mortgage markets will exclude people who lack them. Incremental building can widen access, but fire, flood, sanitation, insecurity, and exclusion still need public action. Technology should be assessed against these specific constraints.

3 Use technology and evidence to improve decisions

Digital technology and AI are useful when they reduce the cost or improve the reliability of a housing decision. Imagery, registries, and computer vision can help locate parcels and structures; workflow records can reveal approval delays; alternative data may assist underwriting; building information systems may reduce errors and waste. Rental platforms and

digital payments can reduce search costs or create transaction histories. Each application needs five questions: which constraint changes, by how much, who captures the gain, who gains power, and what new risk appears? Large mapping programmes show why outputs must be distinguished from their subsequent use.

Mapping at scale does not establish that rights have been resolved or that households can use them for finance. India's SVAMITVA programme reported drone surveys covering about 329,000 villages and 31 million property cards prepared for 187,000 villages by March 2026. On 4 August 2026, the ministry reported 32.4 million cards prepared and about 27.2 million distributed. Preparation, distribution, recognition of rights, and borrowing are separate outcomes, and rural coverage does not demonstrate urban applicability. [7,8] Inheritance, tenancy, customary claims, women's rights, and disputes on public land may remain unresolved after mapping. That makes rules governing data use part of the housing intervention itself.

Housing data needs correction, appeal, and use restrictions before it is shared widely. A new address can enter a tax register, a mapped structure can become an enforcement target, and a payment history can affect lending eligibility. Linking records can improve service planning while increasing the power of an agency or investor over occupants with weak documentation. Specify permitted users and purposes, prohibited secondary uses, and accessible remedies. Maintain common definitions and check uncertain matches, including when imagery and administrative records disagree. Evaluation must then test whether the resulting decisions improve household outcomes.

A recommendation should connect evidence of results to an explicit explanation of what may transfer. Examine a documented improvement, a comparable failure or adverse consequence where available, the mechanisms that may explain the difference, and the conditions required for adaptation. Paired cases are rarely controlled experiments: cities, populations, periods, and policies differ. Keep modelled calculations separate from observed results and record missing comparisons as evidence gaps. In this edition, the transfer tests are developed within each chapter's argument rather than relying on a separate box. Their conclusions must also survive the move from a pilot to a larger programme.

Scaling requires testing whether finance, institutions, markets, and political agreements can sustain greater volume. State the mechanism and outcome supported by the evidence, the required legal powers and demand, and the adaptations needed for local tenure, informal work, exclusion, and capacity. At ten or one hundred times the pilot volume, test prices, networks, staff, supply chains, and changes in participation. Identify who pays for operations and renewal after pilot funding, who bears contingent liabilities, and who could block expansion. Name the responsible agency and its remedy for failure. Those questions should determine the data collected before implementation.

4 How to judge a housing programme

Before committing public money, a government should establish whose housing conditions will improve, why the proposed intervention will work, and what it will cost. A programme may complete its buildings while leaving these questions unresolved. The

assessment must therefore examine household circumstances, the alternatives available, and the obligations that continue after construction.

I would use eight tests to decide what to fund, change or establish before proceeding. They apply to new construction, settlement upgrading, serviced land, rental assistance and housing finance. The evidence required will vary with the intervention and the scale of the commitment.

1. The programme reaches the households it is intended to serve

Define the intended households before choosing the housing product. Record their incomes, income variability, household size, tenure, savings and livelihood needs. Distinguish households that can afford a serviced plot from those that need a rental room, an improvement to their existing dwelling, or continuing assistance.

Eligibility rules need to be tested against these circumstances. A programme can admit informal workers in principle while excluding them through requirements for regular payslips, deposits or loan approval. Examine eligible people who never apply and applicants who withdraw or are rejected. The income profile of successful applicants alone cannot establish that the programme reaches those with the greatest need.

A sound design explains how each intended group can obtain and retain suitable housing. If a substantial part of the target population cannot enter the programme, the product, subsidy or delivery mechanism needs to change.

2. Households can meet the full cost over time

Affordability includes the deposit, repayments or rent, utilities, service charges, maintenance and travel. Assess what remains for food, education, health and other essentials. An expenditure ratio should be accompanied by a household budget. The same ratio can leave a better-off family with adequate resources and a poorer family without enough to eat.

Test difficult months as well as average earnings. Specify what happens when income falls, a repair becomes necessary, or a household needs to move. A longer loan term or lower interest rate cannot resolve every mismatch between the price of a dwelling and the purchaser's resources.

Highly concessional credit still requires a repayment capacity that many poor households lack. A 2008 evaluation of Chile's Progressive Housing Program reports that its publicly provided, interest-free mortgage component was eliminated in 1996 following high arrears associated with beneficiaries' low incomes. Marcano and Ruprah, 2008.

3. The chosen intervention performs better than feasible alternatives

Compare the proposed programme with other ways of addressing the same housing problem. These may include improving water and sanitation in existing neighbourhoods, making land available with services, supporting incremental construction, expanding rental

supply, or providing targeted assistance. Compare costs over a common period and specify differences in safety, space, service quality and access.

Thailand's Baan Mankong programme illustrates a design that accommodates several options. Its official framework channels subsidies, loans and technical support through community organisations and allows on-site upgrading, changes to settlement layouts, reconstruction and relocation. This establishes that a programme can support different responses to different settlements. It does not establish that every community-led intervention is cheaper or more effective. Community Organizations Development Institute, programme description.

The appraisal should explain why the proposed mix is preferable locally. Where a claim depends on uncertain costs or demand, test how far those assumptions can change before another option becomes preferable.

4. The location supports livelihoods and access to services

Assess housing locations within the area where people work, trade, study and obtain care. Include travel time, fares, reliability and safety. For a household enterprise, relocation may also change access to customers, suppliers, storage and working space.

Cheap land can produce an expensive place to live. The OECD's 2013 review of Chile described how the search for low-cost land pushed subsidised housing towards peripheral locations and reinforced segregation. It reported that 60 percent of Santiago's social housing built between 1990 and 1998 was concentrated in three of its poorest municipalities. These are historical findings, but the appraisal question remains relevant: who bears the costs created by the location decision? OECD, 2013, p. 109.

A proposed location should have usable services and access to livelihoods, or a funded and credible plan to provide them before occupation. Employment targets attached to a master plan are insufficient evidence of future jobs.

5. The public subsidy is explicit and its distribution justified

Count the full public contribution. Include the opportunity cost of land, infrastructure, tax concessions, below-market finance, guarantees and future operating obligations. A low recorded construction subsidy can conceal substantial support elsewhere in the public accounts.

Identify who receives the benefit. Some may reach occupants through lower housing costs. Some may accrue to landowners, developers, lenders or better-off purchasers. State which transfers are intended and how the programme will limit unintended gains. Where mixed-income development is expected to finance housing for poorer households, show the cash flows and test them against lower sales and higher costs.

Then compare the proposed reach with the scale of need. Show how many underserved households the available budget can assist and what provision is made for those outside the programme. Intensive support for a small group may be justified, but its opportunity cost should be visible.

6. Delivery and maintenance fit the institutions and resources available

A credible programme identifies who will resolve land claims, procure works, connect utilities, allocate housing and maintain shared assets. These responsibilities need staff, authority and funding. An agreement between agencies should specify what each must deliver and what happens when it does not.

Resolve access rights and service obligations before construction. Construction should not depend on unresolved access rights, unfunded drainage or a utility connection that nobody has accepted responsibility for providing. Formal title is one form of evidence about land rights. It does not replace investigation of occupation, tenancy, customary claims or disputes.

Scaling requires a separate assessment of delivery capacity. Establish whether contractors, materials, infrastructure networks and administrative teams can support the proposed volume. A successful pilot provides limited assurance when its staffing, subsidy or access to land cannot be sustained across the wider programme.

7. Improvements protect residents' rights and safety

Establish who occupies and uses the land before deciding how it will change. Include tenants, informal occupants, businesses and people whose livelihoods depend on the location. Specify how the programme will address displacement, compensation, temporary accommodation and access to suitable alternatives.

Hazard assessment must influence the design and location of housing. Where occupation is unsafe, identify feasible protection or relocation measures and their costs. Where risks can be managed, restrictions should reflect that evidence. A map alone cannot resolve either engineering uncertainty or people's rights.

Track what happens to existing residents after improvements. Better services or greater development potential may increase rents and land values. The assessment should establish whether intended beneficiaries remain, move voluntarily, or lose access. An increase in property values is insufficient evidence of improved conditions for poorer residents.

8. Results can be verified and the programme can be corrected

Measure occupation by intended households, housing costs, service reliability, safety and access to livelihoods. Keep these outcomes separate from contracts awarded, money spent and units completed. Count temporary construction employment separately from sustained changes in employment and earnings.

Evidence may support some benefits without supporting others. A March 2026 study of 4,000 households in eight selected Ethiopian cities found better service access in matched comparisons near new arterial roads, but no statistically significant improvement in formal employment. Non-random exposure and differences between households limit causal interpretation. These findings support examining service gains without assuming that they establish an employment effect. Haile, Lamson-Hall and Molla, 2026.

Set review points before committing the full programme budget. Identify which findings would trigger changes to locations, prices, eligibility, financing or the intervention itself. Assign an authority that can make those changes.

Reaching a decision

For each test, record whether the proposed design is demonstrated, unproven or fails. Missing evidence leaves a claim unproven. Evidence that intended households cannot afford the product, or that essential services cannot be provided, requires redesign. Record the supporting evidence and the uncertainty alongside the judgment.

These findings should not be averaged into a single score. Strong performance on procurement cannot compensate for unsafe siting or unaffordable payments. Some uncertainties can be resolved through a limited pilot. Others, particularly those affecting safety, rights or irreversible expenditure, need to be resolved before proceeding.

The assessment should end with a short decision record: the option selected, the households it will serve, the full public commitment, the conditions still to be met, the responsible agencies and the point at which results will be reviewed. This provides a basis for approving a programme and for changing it when the evidence requires.

Organising the assessment

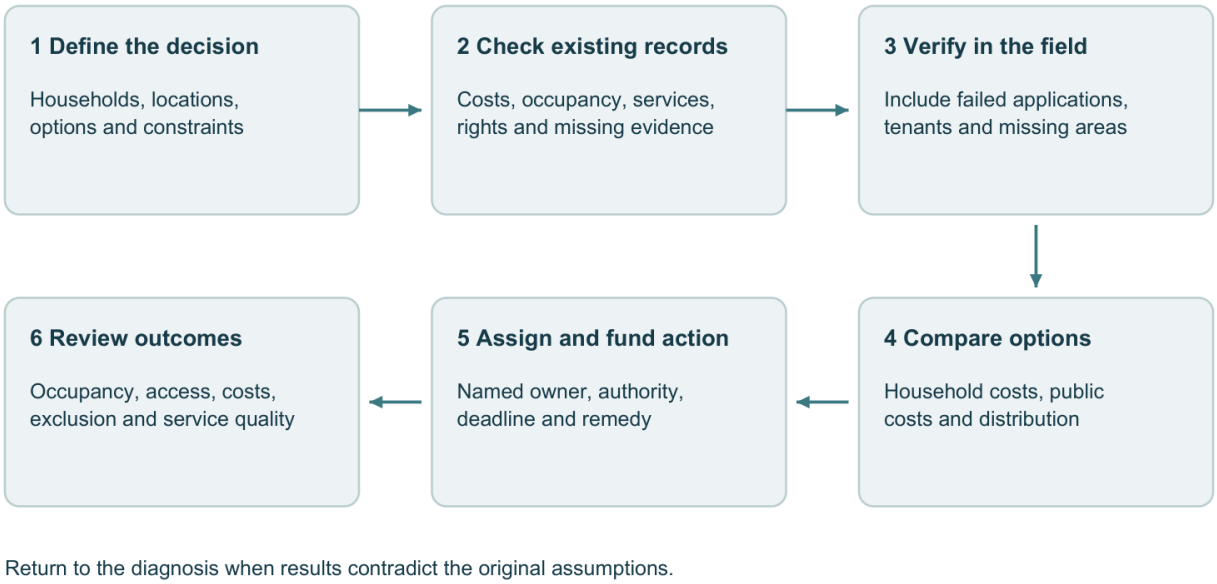
Assign responsibility for the assessment and for acting on its findings. Name a housing or planning lead, a municipal finance counterpart, and an owner for evidence quality. Combine census and survey records with permits, completions, utility connections, transactions, rents, spending and loan performance. Keep household, dwelling, parcel and project identifiers distinct, and record dates, coverage, missing observations and data owners.

Fieldwork should follow households and decisions through the housing system. Include rental rooms, informal settlements, owner-occupied homes and subsidised estates in central and peripheral locations. Hear from migrants, informal workers, women-led households, people with disabilities and eligible non-applicants. Record payments, deposits, irregular earnings, journeys, interruptions, moves and failed applications. Choose sample size for the comparison and precision required: a convenience sample can reveal problems but cannot estimate citywide prevalence. Interview tenants separately from owners and frontline staff separately from managers, using safe follow-up and aggregate reporting.

An initial assessment can use a small multidisciplinary team and existing records before commissioning a major survey or platform. As a proposed diagnostic sequence, a team covering housing or planning, municipal finance, data and social assessment could spend two weeks tracing completed, stalled and rejected cases. This is not a universal standard or an evaluated programme. Prepare a feasible-location inventory, reconciled household and public costs, and an action register recording evidence, owner, funding, deadline and review trigger. Use documented files and a spreadsheet if they suffice. Technology procurement should follow a demonstrated operating need and include acquisition, data, maintenance and staffing costs.

The book follows housing decisions from usable supply to sustained neighbourhood access. Chapters 1–4 examine land, rules, services, and producers; Chapters 5–7 examine tenure, finance, and subsidy; Chapters 8–12 examine institutions, growth, risk, and incremental change. Chapters 13–24 develop the information, finance, contracts, and operating responsibilities needed for execution. Chapter 25 brings these decisions together around inclusive neighbourhoods. Throughout, track approvals, construction, occupation, household costs, rights, service quality, and maintenance separately, including temporary versus sustained employment effects. Use the relevant chapter to prepare a decision rather than commission every suggested survey. The first decision is whether land can become serviced housing in a location households can use.

Figure I.1. From assessment to a funded decision



Proposed workflow. The fieldwork and comparison are designed around the decision, rather than around a predetermined technology.

Sources and evidence notes

The chapter distinguishes sourced facts from the book's analytical judgments. Quantitative claims are drawn from the public sources below; access dates: September 2026.

1. UN-Habitat, World Cities Report 2026: The Global Housing Crisis: Pathways to Action [source]

5. Ministry of Housing and Urban Affairs, Government of India, PMAY-U dashboard [source]

6. PMAY-U Newsletter, 15 April 2026 [source]

7. Ministry of Panchayati Raj, Government of India, SVAMITVA Scheme [source]

8. Press Information Bureau, SVAMITVA progress data, 2026 [source]

9. OECD, Urban Housing [source]

10. Centre for Affordable Housing Finance in Africa, Housing Finance in Africa Yearbook [source] [additional source] [additional source] [additional source]

[201] UN-Habitat (2026). World Cities Report 2026. Executive Director’s introduction and affordability findings. Global estimates use different units and definitions. [source]

[209] OHCHR and UN-Habitat. The Right to Adequate Housing, Fact Sheet 21/Rev.1. Adequacy dimensions; apply local law and standards. [source]

1. Access to land

Land policy should increase the supply of serviced homes near jobs while protecting existing rights and sharing the gains from development. Releasing hectares is only one step. Housing also needs permission to build, infrastructure that can support occupation, and an agreement that households, landowners, developers, and public agencies can fulfil. I would assess a proposed land reform against three requirements: whether it makes the site usable, whether it allocates benefits and risks fairly, and whether its delivery commitments are funded and enforceable. These requirements begin with the difference between land that is available on paper and land on which people can live.

1 Make land usable for housing

A parcel contributes to affordable housing only when its rights, location, development permission, and services work together. A cheap site can become expensive once households pay to reach work and the city extends water, drainage, and transport networks. Conversely, a costly site may support more affordable homes if its location reduces household costs and its infrastructure permits more floor space. The appraisal must therefore separate the components of land cost before comparing sites.

Land cost reflects the raw site, its legal development potential, its services, and its access to opportunity. The first component is the price of undeveloped land. The second is the value added by legal conversion and permission to build. The third reflects roads, drainage, water, sewerage, and power. The fourth is the location premium associated with access to jobs, schools, markets, and social networks. These components interact: a road may improve access, but it does not settle a disputed title or secure permission to build. Their combined effect is best assessed through the usable floor space and household access that a site can provide.

Measure buildable floor space and household access alongside land area. With adequate infrastructure, a hectare that allows mid-rise apartments or smaller plots can accommodate more households than one restricted to detached single-storey homes. Height limits, parking requirements, and minimum plot sizes affect that capacity, as do the space needed for roads, drainage, public facilities, and safety. The useful comparison is how many households each option could accommodate within a locally defined travel-time and cost threshold, at what fiscal and household cost. Evidence from African cities shows why those costs must be considered together.

Evidence from African cities links high urban costs partly to fragmented land markets and weak connections. The World Bank's Africa's Cities study found that urban households in 39 Sub-Saharan African countries paid 20–31 percent more overall than households in countries at similar income levels; a separate city comparison put the premium at about 31 percent. The report associated part of the difference with fragmented land markets, poor connections, and infrastructure arriving after settlement. UN-Habitat's 2025 summary estimated a housing deficit of roughly 70 million units across Africa and reported that 53.6 percent of Sub-Saharan Africa's urban population lived in slums. These figures measure different problems and should not be

combined into a single estimate of need. [1–2] For a land team, the operational question is how the sequence of settlement and servicing contributes to those problems.

Servicing land after settlement can require costly changes to routes and existing uses. In 2024, the World Bank estimated that Kinshasa, then home to more than 15 million people, was adding roughly 2,000 residents and five hectares of urban area each day, much of it without drainage and other basic services. Where buildings already occupy a proposed utility or road route, the city may need a different alignment, a negotiated agreement, or compensation. Teams should compare those options before considering displacement and include replacement housing and livelihood costs in the appraisal. [3] Preventing this problem also requires understanding why households cannot obtain serviced land through formal channels.

Restricted formal supply can redirect demand into informal subdivisions without reducing the need for housing. World Bank work on Ethiopia found that formal urban land supply failed to meet demand and that discretionary allocation helped push households towards peri-urban informal markets. It estimated a need for nearly 400,000 new urban housing units annually to meet new demand. Older evidence from Addis Ababa found that about 30 percent of the housing stock was considered illegal in 2004, alongside a large informal secondary market and slow regularisation. [4–5] These findings support investigating a possible reinforcing process: limited formal supply raises entry costs, settlement spreads informally, and later servicing becomes harder. Choosing a response requires identifying which rights and users a proposed reform would affect.

2 Agree how development changes rights and distributes gains

Land reform needs an explicit agreement about who contributes, who benefits, and whose existing rights must be protected. Government decisions on conversion, infrastructure, and permitted floor area can increase land values. Owners, developers, public agencies, tenants, and informal users have different claims on those gains and different exposure to the costs. An ownership register alone will not show all those interests. The first task is to establish who can approve, refuse, fund, or appeal the proposed change.

A stakeholder assessment should document actual powers and losses rather than assume motives. Existing owners may benefit from restricted competing supply, while peripheral owners may gain from new roads and development rights. An agency may reserve land for a future public use; a tenant may face removal when an owner's claim is recognised. Future residents and renters may have less access to planning decisions than current owners. These possibilities must be checked through records and consultation, including separate opportunities for less influential residents to be heard. Once the affected interests are recorded, the team can test whether a development agreement offers each party a credible reason to participate.

Land readjustment

Land readjustment can make development feasible when owners accept smaller serviced plots in exchange for land needed for public uses. Gujarat's Town Planning

Schemes reorganise private parcels as an area urbanises. Owners receive reconfigured plots; the scheme reserves land for roads, amenities, and other purposes, and may levy betterment charges to recover part of the investment. A World Bank guidance note describes a legal ability to take up to 50 percent of scheme land, with typical deductions around 40 percent. Its statutory menu includes 15 percent for roads, 5 percent for parks, 5 percent for social amenities, 10 percent for affordable housing, and 15 percent for a land bank. [6] Those percentages describe an instrument; its performance depends on whether the promised services and resulting plot values materialise.

Ahmedabad's experience shows that the timing and financing of services can determine whether readjustment delivers at scale. By the mid-1990s, schemes could take years, infrastructure lagged, and the value-capture formula failed to keep pace with costs. Reforms after 2001 allowed infrastructure to start after draft approval, increased the land available to finance it, and permitted private contractors to prepare schemes. World Bank research reports that more than 200 square kilometres were developed through Town Planning Schemes between 2000 and 2009, compared with about 100 square kilometres in the preceding 25 years. [7–8] Faster development, however, does not establish whether residents without ownership titles shared in the benefits.

Ahmedabad's gains in land development coexisted with gaps in inclusion and service funding. WRI's case study reports that readjustment enabled roughly 33,000 low-income homes, but tenants and informal occupants often lacked a place in negotiations. Some amenities remained unfunded, and owners resisted where appreciation was unlikely. These findings do not establish a causal effect on affordability. [101] They do show why a transfer test must examine both the value offered to owners and the protection offered to other residents.

The transferable principle is a credible exchange of land and future value, supported by funded services and protection for non-owners. A city considering readjustment should independently value the contributed land and returned plots, test demand, secure infrastructure finance, and provide a process for objections. It should also specify what tenants, informal occupants, and livelihood users receive, since they may have no ownership plot to exchange. Copying a deduction percentage without these conditions leaves the main implementation problems unresolved. Collective ownership adds a further question: who has authority to make that exchange?

Customary and collectively owned land

Development on collective land requires verified authority and a clear distinction between ownership, use, and development rights. Mexico's ejidos and comunidades, Māori land, African customary holdings, and newly created community land trusts have different legal histories and rules. A leader's agreement cannot be presumed to bind every rights-holder. Housing for existing members also differs from housing for incoming low-income households, whose eligibility and rights require a separate agreement. The legal routes available in Mexico illustrate why tenure must be examined before a financing or development model is selected.

Mexico's agrarian law provides several routes to development, each with different consequences for community control. The Ley Agraria distinguishes settlement land,

common-use land, and individually allocated parcels. Article 45 permits specified association or use contracts, including renewable third-party use terms of up to thirty years. Article 75 provides a conditional route for contributing common-use land to companies in which the ejido or its members participate. Articles 81–82 allow full ownership of eligible allocated parcels after the required assembly and registration procedures. These routes do not themselves provide urban planning permission, infrastructure, or an affordability obligation; a transaction needs specialist review of the land category, approvals, and applicable law. [201] The experience of regularisation shows what remains unresolved when these other requirements are left outside the tenure process.

Registration alone has not resolved the planning and servicing problems of urbanising collective land in Mexico. Qualitative research in Guadalajara finds that PROCEDE, FANAR, and RRAJA improved mapping and registration while unplanned subdivision and informal urbanisation continued. The authors identify a mismatch between federal agrarian administration and metropolitan planning, together with procedures designed around rural land. This is an explanation from case research, not a causal estimate of each programme. [202] It supports examining whether housing development can combine retained collective ownership with practical support for finance, services, and approvals.

Māori housing experience suggests that organised development support can matter as much as access to a loan product. Biddy Livesey's research describes Mangatawa retaining Māori land while developing housing around social and cultural objectives; the first ten homes in a planned thirty-home phase were completed in March 2012. The study draws on project-manager interviews and does not establish long-term affordability or a causal effect. By contrast, the Auditor-General's 2011 review recorded only one Kāinga Whenua loan since its February 2010 launch, with two more at final approval. Low incomes, restrictive eligibility, consent requirements, and limited advice constrained uptake. By the 2014 follow-up, eleven loans and nineteen infrastructure grants had been agreed; these are historical approvals, not completed-home totals or current performance. [203–205] The comparison points to preparation finance, representative authority, and technical assistance as conditions to test when adapting a collective development model.

Thailand's Baan Mankong programme offers a related approach to organising residents, although its collective tenure is not equivalent to customary ownership. CODI describes communities negotiating collective purchases, land-sharing arrangements, or long-term leases with support for housing finance and infrastructure. Its programme description does not independently establish impact. [206] The relevant question is whether an accountable resident body can negotiate, allocate benefits, and manage debt with technical support. That organisational test should guide the choice among leases, community development, pooling, and joint development.

The development route should match the rights the community wishes to retain and the obligations it can manage. A lease can preserve collective ownership while defining occupancy and development rights, but needs clear rent-review, renewal, succession, and default terms. Community-led development can preserve allocation control, provided preparation,

accounting, and technical supervision are funded. Pooling can exchange land for services and reconfigured plots; joint development can exchange rights for homes, rent, or equity. Each requires independent valuation and remedies for non-delivery, and each remains subject to local law. Before inviting a partner or lender, the community and public authority need a shared record of the rights that the agreement must respect.

A community rights register should record overlapping uses and claims before bids are invited. It should cover collective authority, individual allocations, tenants, seasonal users, access routes, livelihoods, inheritance claims, and disputed boundaries. Women and less influential members need independent opportunities to challenge it. The parties should agree the lawful consent process, independent legal advice, valuation, distribution of proceeds, and treatment of rights that cannot be met through a cash payment. Financial terms can be disclosed without exposing personal records. With these matters established, lenders can assess the rights and cash flows actually available to support the project.

Finance must be secured against enforceable project rights without assuming that the communal estate can be mortgaged. Ask lenders to price the proposed lease or cash flow, fund preparation and trunk services explicitly, and test delays, low occupancy, cost overruns, and partner failure. The agreement should specify how affordability continues and who can replace an operator without extinguishing protected rights. If the parties disagree about what a lender may take after default, the transaction is not ready. Similar discipline is needed when the land contribution comes from a public authority rather than a community.

Public land and land-value capture

Public land should be valued as a subsidy even when its transfer involves no cash payment. A parcel may have alternative uses: sale, lease, a school, infrastructure, or continued community use. Giving it to a housing project therefore has an opportunity cost that belongs in the subsidy account. For a well-served site in a prosperous area, compare continued public use, affordable rental, and mixed-income development before treating market sale as the default. The next step is to compare the benefits of those options with both the public contribution and the household costs of a cheaper peripheral site.

A public-land appraisal should show the homes, access, and fiscal obligations that each option would deliver. Begin with ownership, occupancy, disputed claims, valuation, and a stated public purpose. Compare sale, long lease, joint development, readjustment, community use, and affordable-housing reservation. An expensive site may justify its cost through better access, but that benefit must be demonstrated. If the preferred use differs from the most profitable market use, selection and contract terms must secure the intended public benefit.

A competitively awarded public lease can retain ownership while attaching enforceable housing obligations to development. The authority can specify affordability, occupancy, delivery, and default terms, including a lawful replacement or termination mechanism if delivery fails. This may preserve some future appreciation in public hands, but it requires monitoring and enforceable contracts. Land-sale receipts, by contrast, are one-off capital revenue

and should not be treated as a continuing source for services. The wider financing question is how much of the value created by public action can realistically be recovered.

Land-value capture can help finance infrastructure, but its results depend on local markets and institutions. A World Bank comparison reported positive economic outcomes but mixed equity effects in São Paulo, and few observable economic or equity gains in Addis Ababa and Hyderabad at the time studied. The authors identified differences in tenure, planning, financial instruments, property markets, and implementation; the comparison does not isolate the effect of each condition. [9] It supports testing the conditions for collection before incorporating expected receipts into a development budget.

A value-capture charge needs a defensible valuation, a legal basis, and a level that leaves development feasible. Estimate the land value without the public intervention, state the legal claim on the increment, and establish capacity to assess and collect. Publish the charging formula and intended use of receipts. Test the charge against project economics and household costs: excessive charges can suppress development, while project-by-project negotiation creates room for discretion. Funding new infrastructure also requires a separate plan for maintenance. These calculations depend on land information that is accurate enough to support decisions and open to correction.

3 Use information to secure delivery

Digital land systems should be judged by the decisions they improve and the rights they protect. Searchable records can make transactions easier to trace and can connect land administration with planning and infrastructure data. They can also reproduce errors or omit women, tenants, customary users, and informal occupants. An authoritative digital record can make those omissions harder to challenge unless correction and appeal procedures are accessible. Administrative rollout figures therefore need to be read separately from evidence of better housing outcomes.

India and Kenya demonstrate the scale of digital administration, but their reported activity does not by itself establish additional housing supply. India's DILRMP dashboard, cited for August 2026, reports computerised records of rights in 626,237 villages, or 95.56 percent of the total, with 5,229 computerised sub-registrar offices and 4,837 integrated with land records. The Department of Land Resources reports more than 408 million Bhu-Aadhaar parcel identifiers and guidelines for 2026–31. Kenya's Lands Ministry reported on 16 February 2026 that its national stamp-duty module had collected KES 1.946 billion over two months. Its projection of more than KES 20 billion by 30 June 2026 was a forecast, not a verified outturn. The approximate US$15.1 million and US$155 million equivalents use the Central Bank of Kenya's 2025 annual average of KES 129.3 per dollar. [10–13,207] To assess housing benefits, evaluation must follow changes in costs, delays, disputes, and usable land.

Measure whether digital records reduce transaction costs and make serviced sites legally usable. Track search and approval times, disputes, correction requests, and the distribution of benefits, alongside the number of records digitised. Test whether affected people can obtain records and appeal without digital access. A shorter transaction is useful only if the

resulting rights are valid and the land can support the intended housing. Imagery and AI can help investigate physical conditions, but they answer a different set of questions from the registry.

Geospatial analysis can screen sites and structures, but it cannot establish ownership or consent. Imagery and models can estimate built floor area, compare it with permitted capacity, locate potentially vacant parcels, identify roof materials and settlement growth, and screen access to jobs or exposure to floods and heat. Accuracy and update costs vary by task and data source. Field-check a sample, label inferred attributes, and keep them separate from legal records. The next test is whether this information changes a specific release, servicing, or investment decision.

Technology should be evaluated against a defined administrative decision and a measurable result. For example, compare the time taken to release an eligible serviced site before and after a new information process, while recording other changes that could explain the result. Automated valuation may support property-tax collection, but requires documented methods, corrections, and accessible appeals. A dashboard or a larger assessed tax base is not evidence of additional affordable homes. A practical appraisal should therefore bring legal, physical, financial, and household information together in one site decision.

Prepare a site decision that agencies can implement

The land agency should assemble a parcel-specific options sheet with the people and agencies needed to deliver it. Bring together rights-holders, tenant representatives, planning officials, utilities, and the municipal finance team. Record who can approve, refuse, fund, or appeal each step; for collective land, verify representative authority and hear minority claims separately. Include existing users and prospective low-income residents in the assessment of benefits and losses. The options sheet should then make the costs and deliverability of each alternative comparable.

Each shortlisted site needs evidence on rights, buildable capacity, service costs, and household access. Obtain the title or tenure instrument, overlapping claims, occupants and uses, permitted floor area, transaction evidence, hazard exposure, utility capacity, and actual journey costs. Separate gross area from buildable area after public facilities, roads, drainage, and protected uses. Require utilities to confirm connection costs and timing in writing. For employment-location screening, Chapter 13 explains the use and limitations of call-detail records and the field checks they require. With this evidence, the team can compare feasible development routes rather than choose one from a general policy preference.

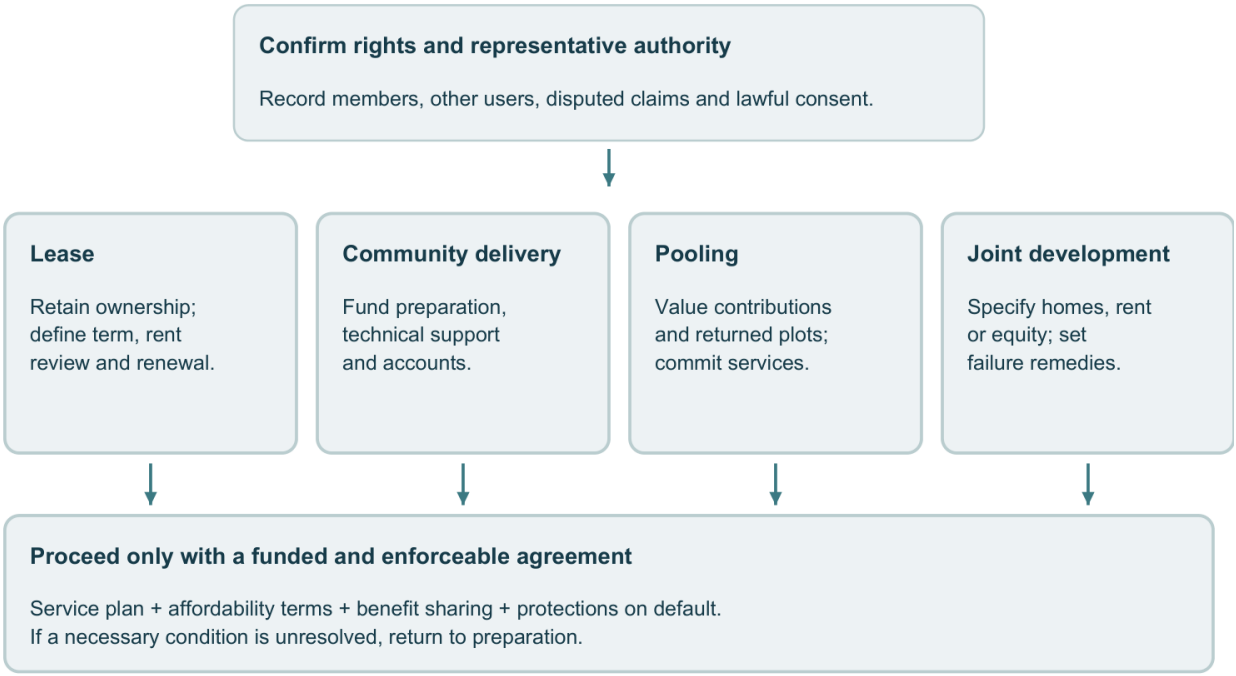
Choose the development route only after comparing its occupied homes, retained rights, fiscal cost, and delivery risks. Depending on the site, the options may include upzoning, a public lease, readjustment, community development, redevelopment, serviced plots, in-situ upgrading, or expansion. Estimate the additional low- and middle-income households each could accommodate within the local job-access threshold, including losses through demolition or displacement. In customary-tenure settings, map use rights alongside registered ownership; in Indian cities, check state approval powers and tenant eligibility separately. The preferred option must then be translated into funded commitments rather than a list of expected benefits.

The agreement should allocate delivery obligations and downside risks as explicitly as it allocates gains. State what owners receive, what infrastructure utilities will supply, what developers must build, and what residents will pay and retain. Prepare the group-level distributional assessment described in Chapter 14, with evidence, uncertainty, and funded remedies. Include lawful compensation, livelihood protection, appeals, and any provision for existing residents to share in appreciation. Secure trunk infrastructure and rights-of-way before relying on the value those investments are expected to create, and add household subsidies when supply can respond. These commitments provide the basis for deciding whether a first phase is ready to proceed.

Release a first phase only when unresolved conditions have named owners and enforceable completion requirements. The readiness decision should identify disputed claims, service commitments, valuation assumptions, financing gaps, and the officer responsible for each. Use conservative land receipts for capital works and a separate recurrent revenue plan for services; expected sales are not committed cash. Chapter 22 develops the contract provisions for public-land transactions. Before authorising expansion, check whether the first phase delivered the homes, services, and protections on which the agreement depended.

Scale should depend on demonstrated delivery and the capacity to repeat it. Track net additional occupied affordable homes near jobs, household charges, service completion dates, resident retention, owner gains, and displacement by tenure group. For collective projects, also record retained community assets and income, benefit distribution, and unresolved claims. At 10 or 100 times the initial volume, test whether valuation, appeals, representative institutions, infrastructure finance, and service delivery can keep pace. These tests establish whether land release is producing usable housing capacity; Chapter 2 examines how planning rules determine how much of that capacity can be built.

Figure 1.1. Choosing an agreement for collective land



Proposed decision structure, subject to local law. The four routes are alternatives to compare; none removes the need to resolve rights and infrastructure.

Sources and evidence notes

Evidence. Quantitative claims and program descriptions in this chapter are drawn from the sources below. Judgment. The concepts of four land prices, the affordable floor-space frontier, the coalition sequence, and the five-question technology test are analytical syntheses. They should be tested against city-level prices, tenure systems and political institutions before use.

1. World Bank, Africa's Cities: Opening Doors to the World (2017) [source]

2. UN-Habitat, Strategic Plan 2026-2029 adoption: Africa housing deficit and slum share (2025) [source]

3. World Bank, Kinshasa: Harnessing Nature-Based Solutions for Urban Resilience (2024) [source]

4. World Bank, Why should Ethiopians care about urbanization? (2019) [source]

5. World Bank, Ethiopia Investment Climate Assessment, urban land evidence (2007) [source]

6. World Bank, Access to Land in South Asia: Guidance Note (2024), Gujarat Town Planning Schemes [source]

7. World Bank, Planning, Connecting and Financing Cities - Ahmedabad evidence (2013) [source]

8. World Bank, Transforming Cities with Transit - Ahmedabad case [source]

9. World Bank, Urban land value capture in Sao Paulo, Addis Ababa and Hyderabad (2020) [source]

10. Government of India, Digital India Land Records Modernization Programme [source]

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12. Government of Kenya, State Department for Lands, digital transformation update, September 2026 [source]

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[201] Cámara de Diputados. Ley Agraria, consolidated text with reform of 14 November 2025, articles 44-46, 75, 81-82. Legal routes, not a project approval. [source]

[202] Reyes Carranza, M., Hoefsloot, F., Datta, A., and Agraz Joya, R. (2026). Urbanizing Commons Lands: Ambiguous tenure regimes and the challenges of land regularization in Guadalajara Metropolitan Region, Mexico. Land Use Policy, accepted manuscript. Qualitative case research; repository abstract and manuscript record. [source]

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[204] Controller and Auditor-General (2011). Government planning and support for housing on Māori land, Part 6, paragraphs 6.4-6.8. Historical lending and programme-design findings. [source]

[205] Controller and Auditor-General (2014). Government planning and support for housing on Māori land: Progress in responding to recommendations, paragraph 1.7. Approved loans and grants, not completed homes. [source]

[206] Community Organizations Development Institute. Baan Mankong Urban. Programme description of land negotiation, collective finance, and upgrading; not an independent impact evaluation. [source]

[207] Central Bank of Kenya. Bank Supervision Annual Report 2025, table 15. The 2025 average exchange rate is KES 129.3 per US dollar. Currency conversions in this chapter are the author’s calculations. [source]

2. Planning rules and development rights

Planning reform should make additional housing economically feasible while retaining enforceable safety and service requirements. The sequence is to identify which rules constrain usable supply, compare reforms through their effects on project viability and land value, and fund the services and administration needed for delivery. More permitted floor space is an intermediate result; the outcome is additional homes that intended households can occupy. The starting point is how individual rules alter what can be built on a site.

Planning rules affect housing costs by determining which buildings can be developed profitably on each site. Allowing apartments on a plot previously limited to one house changes both the supply of housing and the value of the land. Planning rules determine the amount and type of legal floor space on urban land. Floor-area ratios, plot sizes, setbacks, height limits, parking rules, use restrictions, and approval procedures affect how much housing may be built. They can protect light, safety, drainage, heritage, and infrastructure capacity. Restrictions that exceed those needs can also limit supply and contribute to higher rents, land values, commuting costs, or informal construction. Residual land value provides one way to test a rule's effect. Start with expected building revenue and subtract construction, finance, taxes, infrastructure, professional fees, marketing, and a required return for risk. The remainder is the amount available for land. Additional usable floor space may increase that residual, while costly parking, large plots, or delays may reduce project viability. Whether these changes affect household prices depends on competition, demand, and the supply response. The economic test must be paired with a check that the permitted building can actually be delivered.

1 Identify which rules constrain usable supply

Development permission and housing supply

Planning reform must align development rights with infrastructure and feasible construction. Uncertain network capacity, parcel rights and approval times can prevent a project from proceeding. Because permission also changes land value, those who control decisions have strong incentives to influence the rules. Parking, plot sizes, and building envelopes are practical places to begin that check.

How planning rules affect housing costs

Rules intended to protect amenity can exclude affordable building types when their costs exceed their benefits. Depending on rents and other costs, a developer may build fewer homes, change location, raise the asking price, or abandon a project. Test the parking requirement against observed car ownership, access needs, and alternatives at the proposed site. The same is true of minimum plot sizes. A household that can afford 40 square metres of dwelling on a 60-square-metre plot is excluded if the legal minimum plot is 200 square metres. A minimum unit size can ban the room a migrant worker can afford. A setback can make a narrow parcel undevelopable. A ban on mixed use can separate a street vendor's home from her livelihood. A

cap on floor space can push the marginal household farther from work. These rules interact with finance. An unauthorised extension can make collateral unacceptable to a lender, and planning noncompliance can prevent a formal sale. An informal landlord may still build while accepting enforcement risk. Where affordable building types cannot obtain approval, households may face a choice between expensive compliant housing and unauthorised alternatives. The scale of that effect requires local evidence. Where households cannot obtain a suitable compliant home, their building decisions reveal what the formal system is failing to provide.

Informal construction can reflect both weak enforcement and unmet demand. When thousands of households add rooms, divide plots, convert shops, or build extra floors, they reveal a demand for forms of floor space the formal system supplies too slowly or too expensively. Upzoning can remain unused if water, sewerage, drainage, streets, or transit cannot support additional residents. Networks can also remain underused where legal density is too low. Discretionary exceptions may make approval less predictable and increase the value of an individual decision. Assess whether the permitted floor space is serviced, financially feasible, and available through a transparent process. Indian floor-space policies illustrate how these constraints interact in high-demand locations.

Floor-space regulation in India

Indian evidence shows why additional floor space must be evaluated alongside transport, infrastructure, and project economics. World Bank analysis has noted that Mumbai historically permitted floor-space indices around 1.33 to 4, while limits in Tokyo, Hong Kong, and Shanghai could reach 10 to 15. These figures cannot be transferred directly between cities because infrastructure, street widths, fire access, and transit differ. They show how planning limits can constrain floor space in high-demand locations. A World Bank simulation for Bangalore suggested that relaxing density restrictions could reduce the city's radius from about 12 kilometres to 8 kilometres and save households about 4 percent of income through lower transport costs. These are modelled outcomes, not observed effects of a reform. The proposed mechanism is that more central housing would reduce demand for peripheral locations and some travel. Local demand, transport, and construction assumptions need testing before using these magnitudes. Delhi's transit-oriented development framework, as described in the cited World Bank sources, permits higher density on qualifying large parcels in transit influence zones, combines land uses, mandates smaller units, and reserves floor space for economically weaker sections. It also uses land pooling for public space and transit plazas. This is a policy design example; implementation and any subsequent revisions require separate assessment. The design also has to work for owners who cannot assemble large sites and residents whose claims are weakly documented.

Land assembly requirements and social exclusion can prevent households from benefiting from additional development rights. Affordable-housing obligations may make a project unviable if they exceed the available residual land value. Uneven exceptions can add uncertainty. Implementation therefore needs clear rights, predictable approval periods, funded infrastructure, disclosed obligations, and a lawful process for fragmented ownership. Planning reform also needs an assessment of social exclusion. Caste and class can affect access to rental

and ownership housing in India. Informal settlements, unauthorised colonies, and subdivided peripheral land may provide access when formal housing is unaffordable or unavailable. Regularisation can improve security but may also change expectations about future enforcement. Enforcement decisions must examine impacts on households with weak documentation and provide due process, rather than treat legal status as a complete account of their rights and needs. African experience adds the problem of rules that fit only a small part of actual housing production.

Planning rules and informal development in African cities

Planning reform must make safe incremental production feasible for the households and landlords already building much of the city. Small landlords, households and informal developers build the rest. The gap is large enough that policy cannot close it through public construction alone. Kenya's earlier housing assessments illustrate the arithmetic. The World Bank estimated a deficit above two million units, formal production below 50,000 units a year against a target of about 200,000, and a majority of urban households living in slum conditions. The exact figures move over time; the structural point remains. These figures put the annual target at more than four times reported formal production. They justify investigating which rules, costs, and finance constraints keep ordinary builders outside the legal market. Dandora in Nairobi provides a historical sites-and-services example. Households received serviced plots and later added rooms and floors as income and demand changed. Incremental development allowed construction spending to be staged and added rental supply. A fixed original building envelope could restrict those additions. Any permitted expansion still needs to be checked against structural safety, fire access, drainage, and service capacity. The costs they face extend beyond materials and labour to the infrastructure and administrative requirements attached to a legal home.

Servicing and compliance costs can limit the benefit of a cheaper construction method. CAHF's cross-country work estimated that materials and labour often accounted for roughly 45 percent of the cost of a typical 55-square-metre house, and infrastructure for around 14 percent. Administrative compliance, land, and finance made up much of the rest. A materials innovation should therefore be appraised against its share of total delivered cost, including any additional transport, finance, or maintenance expense. World Bank-supported urban programs reported more than 2,700 kilometres of roads and 2,700 hectares of serviced land delivered across 117 cities between 2018 and 2024. A right to build becomes economically meaningful when pipes, roads, drainage and public services can support the building. Where those costs push housing away from employment, planning becomes a question of access to work.

Density and access to work in South Africa

South African evidence links dispersed settlement patterns to the need for housing closer to employment. Apartheid located many Black households far from economic centers. Three decades later, the geography remains unusually dispersed. New World Bank research using administrative, census and satellite data finds about 9,300 fewer residents per square kilometre than expected within five kilometres of commercial centers relative to twenty global peers, roughly 60 percent below the global median. The same South African research finds that greater

distance from Johannesburg business hubs is associated with lower employment and household asset wealth. This association supports examining access costs but does not, by itself, identify the effect of moving a household closer. Transport, skills, job availability, and household selection also need to be considered. Upzoning can have different effects across neighbourhoods. Where jobs and infrastructure are available but apartments are restricted, more permitted housing may widen entry. On an occupied low-income rental corridor, higher development value may also increase demolition and displacement pressure. Compare added homes with losses of existing low-cost housing and track outcomes for original tenants and new entrants. Accommodating more residents in accessible locations also requires a credible municipal service budget.

Municipalities need a way to finance the network obligations created by additional housing. Metros must maintain water, sanitation, electricity, roads and waste systems. National government captures large tax bases while municipalities carry much of the service burden. A municipality asked to permit more residents needs a credible way to finance the networks those residents use. The country's new Metro Trading Services Program makes this institutional link explicit by tying national performance-based grants to improvements in metro trading services. Housing supply and municipal finance cannot be separated for long. International zoning reforms help distinguish changes in legal capacity from changes in actual supply.

2 Compare reforms and allocate development value

International evidence on zoning reform

Planning reform can increase supply where rules prevent building types that would otherwise be viable. Its effect on rents depends on the amount built, the timing of completion, demand, and changes elsewhere in the housing market. Minneapolis is useful because it changed several rules over time. The city reduced and then eliminated parking requirements, made apartment construction easier near commerce and transit, reduced minimum lot sizes, and later allowed duplexes and triplexes more widely. Between 2017 and 2022 its housing stock grew about 12 percent while rents rose about 1 percent; the rest of Minnesota added about 4 percent to housing stock while rents rose 14 percent. Pew estimates that nearly 21,000 units were permitted in Minneapolis over that period, 87 percent in buildings with twenty or more units. Most Minneapolis permits in this period were for larger apartment buildings. The reported construction associated with duplex and triplex legalisation was limited. This pattern suggests testing corridor rules and parking costs alongside smaller-building reforms; it does not isolate the contribution of each policy. Construction cost, finance, parcel dimensions, and demand may still prevent an approved project from proceeding. Other reforms show why the geographic coverage of new rights and the building types they enable also matter.

Smaller plots and broader development rights can expand supply, but the response depends on the market and the reform package. After reducing minimum lot sizes, from 5,000 square feet to as little as 1,400 in parts of the city and later citywide, nearly 80,000 townhouses and other small-lot homes were built, according to Pew. Smaller legal parcels created a product between an apartment and a detached suburban house. These comparisons are observational, and the policy packages overlap. They support testing whether added legal

capacity allows more housing in high-demand locations. They do not establish a universal supply response or show how much of an observed rent change came from a particular rule. Auckland and Chicago show why coverage and location need testing. Auckland's 2016 reform covered about three-quarters of residential land. Using an extrapolated linear counterfactual, researchers estimated 21,808 additional permitted dwellings over five years; that point estimate is sensitive to the counterfactual specification. These were permits, not net completed homes. Chicago's narrower transit-area reforms raised property values without a detectable increase in newly permitted dwellings in the initial five-year study. The markets, periods, and methods differ, so this is not a controlled test of breadth alone. Test whether enough competing sites become viable, alongside demand, finance, and infrastructure. [102,103] Those differences make the distribution of benefits and costs central to securing support for reform.

Interests and constraints in planning reform

Planning authorities should test each group's stated concerns against documented costs, rights, and service limits. Owners may seek to protect asset values and avoid construction disruption, congestion, or service overload. Developers need feasible rights and approval dates; their interests may differ over competing supply. Landowners may receive gains from added capacity. Utilities need to fund expansion. Tenants may benefit from added homes while facing displacement from redevelopment. Establish each agency's legal duties and resources. Publish the criteria and use of discretionary exceptions rather than infer officials' motives. People who might move into an area often cannot vote there or attend its planning hearings as easily as current residents. Existing owners and tenants have established local channels of participation. A capacity assessment should therefore include households excluded by current supply, alongside the documented costs borne by residents. Test stated objections and benefits against evidence. For neighbourhood character, identify the particular amenity or heritage value affected. For affordable housing, state the eligible incomes and full monthly cost. For infrastructure capacity, obtain engineering estimates and a funded expansion option. For regularisation, identify recognised claims and affected occupants. This allows a proposal to be assessed on its effects rather than its label. A workable agreement must then connect those concerns with the benefits of additional housing.

Support for reform depends on sharing its gains and addressing the costs borne by current residents. Tenant protections and targeted anti-displacement measures can bring existing low-income residents into the coalition. Infrastructure funding can reduce homeowner resistance grounded in service quality. Include affluent and employment-rich areas in the capacity calculation. Count demolition losses and entry by lower-income households. Development rights create value that can help meet those obligations if the required contributions remain feasible.

Allocating the value of development rights

An increase in permitted floor area can raise land value. The public authority should estimate that increase and state how infrastructure costs and affordable-housing obligations will be allocated. The city can grant the right freely to encourage rapid supply. It can auction additional development rights. It can charge for infrastructure. It can require affordable units. It

can use land readjustment to obtain streets, parks and public sites. It can tax some of the uplift later through property taxation. Each instrument changes the project's economics and the politics of reform. Use residual land value to assess the feasible obligation. Estimate the revenue from the proposed building. Subtract construction, finance, infrastructure, taxes, professional costs, marketing and a competitive developer return. The residual is what can support land acquisition and public obligations. If the city demands more than that residual, the project stalls or the obligation is evaded. If it demands nothing where rights create a large uplift, it transfers public value to the landowner. Publicly owned sites require the same viability test and additional scrutiny of the terms on which public value is transferred.

Public-land decisions should disclose the value contributed and the housing obligations obtained in return. An auction to the highest bidder maximizes the immediate receipt. A long lease with density, affordability and delivery conditions can purchase a stream of urban outcomes. A negotiated allocation requires transparent selection and conflict-of-interest controls. The minimum information set should be public: parcel boundaries, valuation, permitted rights, bids, beneficial ownership, milestones and subsequent changes to the contract. Publishing those obligations is easier when parcel rules and approval records can be checked consistently.

3 Implement and evaluate the reform

Digital planning rules and AI

Digital planning systems can make rules and exceptions easier to check when they retain accountable human decisions and protections for occupants. A parcel has an area. A code specifies setbacks, height, use and floor area. Infrastructure networks have capacity. Transit has travel times. Hazards have spatial footprints. A city can combine these layers into a parcel-level statement of what can safely and legally be built. A usable development-rights system should show the applicable rule for each parcel, its legal source, infrastructure constraints, the approval path, and recorded exceptions. Test these outputs against actual files before relying on automated advice. AI can lower the cost of several tasks. Computer vision can estimate existing buildings and floor space from imagery. Models can flag parcels where actual use diverges from records. Network models can estimate where additional density would overload drainage or water. Generative tools can test whether a compliant building envelope fits a parcel. Workflow data can reveal which approval stage creates delay. Publishing parcel rules and approval times can make exceptions easier to examine. The same map can expose informal occupants to enforcement before their rights have been considered. Planning reform should set data-access rules, limits on secondary uses, and remedies for incorrect records alongside the new approval process. They will improve delivery only if the underlying rules and infrastructure commitments are workable.

Implementation constraints

Implementation must address the full set of binding requirements and the timing of their costs. Publish infrastructure costs and feasible charges before land prices absorb the new entitlement. Planning, utilities and building control must share a dated delivery schedule; appeals

must not become an alternative indefinite queue. Track permits, starts, completions and net occupied stock separately; rents relative to a credible comparison; affordable entry and involuntary moves. Planning changes can leave important constraints untouched. A higher floor-area ratio may have little effect if parking, height, or setbacks remain binding. Weak demand, unfunded utilities, unviable affordability obligations, or continuing parcel-by-parcel negotiation can also limit construction. Digitising the existing process does not resolve these constraints. Test complete building designs and actual applications under the proposed rules. Construction disruption usually precedes completion, while any effect on rents may take longer and be difficult to attribute. That timing can make support for reform harder to sustain. Publish expected construction and service schedules, provide remedies for disruption, and distinguish near-term delivery measures from longer-term housing outcomes. That diagnosis should determine where a reform starts and what it measures.

Start where demand is sufficient, infrastructure exists or can be funded, and planning restrictions prevent otherwise viable housing. Allow additional floor space through predictable rules subject to safety, environmental, and resident protections. Remove requirements with high cost and weak demonstrated public benefit. Publish parcel rights, fund network upgrades, and protect tenants exposed to redevelopment. Measure permits, starts, completions, rents, land prices, travel times, and displacement to identify any remaining constraint. The assessment tools below turn those conditions into a comparison of feasible building options.

Planning policy assessment

Question	Operational measure	Failure signal
1. Where is demand strongest?	Rents/prices, vacancy, job access, informal intensification	High prices + crowding + outward growth
2. What is legally buildable?	FAR/FSI, height, lot coverage, plot/unit minima, parking	Physical capacity exceeds legal capacity
3. What can networks carry?	Water, sewer, drainage, transit, schools	Paper upzoning without service headroom
4. Is a project viable?	Residual land value under current/proposed rules	Rights rise but no starts
5. Who is excluded?	Minimum legal unit cost by income/social group	Code prices out target households
6. Who may be displaced?	Tenant/owner/informal occupancy map	Uplift falls on occupied low-income sites
7. Who captures uplift?	Land value before/after reform; public share	Large windfall with no supply/public return

Question	Operational measure	Failure signal
8. Who can veto?	Map of interests, risks, and decision authority	Reform relies on actors who lose without compensation
9. Can rules be audited?	Parcel rules engine, approval clock, exceptions log	Digital process preserves hidden discretion
10. Did capacity become homes?	Permits, starts, completions, rents, commute, displacement	Legal capacity rises; occupied supply does not

Implications for housing supply

Planning reform should be assessed through usable capacity and household outcomes rather than changes in permitted floor area alone. Planning law determines what may be built, while infrastructure and access affect whether those rights are valuable and usable. Land and planning assessments should record both. Restrictions can increase scarcity costs borne through rents, crowding, informal construction, and longer journeys. Additional rights can also raise land values and create service or displacement costs. The policy task is to permit feasible housing, fund its infrastructure, allocate the resulting gains and obligations, and protect affected residents. Assess those outcomes separately rather than treat a higher permitted floor-area ratio as success. A field assessment can establish the project and household evidence needed for that judgment.

Prepare the local decision

Test proposed rules on real parcels, building types, applications, and household outcomes. Select common building types and assemble parcel dimensions, current rules, achievable floor area, land prices, construction costs, expected rents, approval times, and connection costs. Follow a sample of applications through every office, including withdrawals and refusals. Calculate project viability under current and proposed rules, counting demolished homes. Auckland's supply evidence and the qualifications in this chapter provide a reason to test completions, not assume that additional permission will be used. Ask small builders and larger developers to explain which requirement changes their investment decision, then test their assumptions independently. Utilities must identify specific capacity limits. Record objections from neighbouring owners and tenants separately, including evidence of drainage, access, or displacement risks. The planning authority should publish which concerns require a rule change, funded works, or a project-specific safeguard. Apply the proposed rule change to well-served higher-income areas as well as low-income districts. Publish feasible additional capacity by area, after safety, service and rights constraints. Test whether minimum plots, parking, or discretionary exceptions make the apparent capacity unusable. Model any affordability obligation against project viability. Track net completions and affordable occupation separately: extra development

rights alone do not establish access for poorer households. Once the design is feasible, a limited digital pilot can test whether the approval process delivers it reliably.

A planning pilot should retain human responsibility and distinguish faster approvals from additional occupied homes. Encode a limited set of frequently used planning rules and test the results against approved and rejected applications. Keep a human planner responsible for ambiguous rules and exceptions. Pilot the revised process in clearly defined areas, tracking application cost, approval time, starts, completions, replacement losses, and rents against suitable comparison areas. Investigate a rise in approvals without starts before extending the reform. Chapter 3 examines the infrastructure commitments needed to turn that permission into a serviced neighbourhood.

Additional development rights should be broad enough to create competition and practical enough for builders to use. Auckland's map is not a transferable template. The relevant conditions are demand, viable building types, service capacity, and predictable approvals. Concentrating rights on a few sites can allow owners to capture the gain without producing much housing; granting rights where construction costs exceed achievable rents will also disappoint. When permits increase tenfold, check utility queues, contractor capacity and cumulative demolition. A citywide rule requires citywide service finance.

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3. Infrastructure and serviced land

Approve housing locations only with a credible plan for the infrastructure and services their occupants will need. Site cost, network investment, access to work, and operating finance belong in the same appraisal. Experience with serviced plots and upgrading shows why delivery capability and the order of investments matter as much as the chosen engineering standard. The first decision is whether the proposed site can support affordable occupation.

Service capacity and connection costs should be confirmed before a housing site is approved. A low land price is not a saving if the project must finance a new trunk main or households must buy expensive water while waiting. My experience of Chennai shows why that decision must also consider how a neighbourhood develops after its initial construction.

1 Assess the whole cost of a serviced location

Infrastructure and housing affordability

Chennai and the case for serviced neighbourhoods

The Chennai neighbourhoods I visited demonstrated what residents could build over time, while raising questions about their early conditions. I remember visiting Ambedkar Nagar and Arumbakkam in Chennai and seeing vibrant neighbourhoods developed through sites and services. The experience made me question why we had moved away from an approach that gave households a place to build and room to adapt. My visits are personal observations. The wider evidence comes from studies that revisited settlements and examined their development over time. Those studies identify limitations that my visits did not reveal. In 2015, Sumila Gulyani and colleagues revisited 15 sites in Chennai and Mumbai. Her published account reported thriving neighbourhoods in all but one, with fewer than 10 percent of plots vacant. Small plots, economical street layouts, a range of plot sizes, and space for businesses and amenities helped make gradual household investment possible. This was a selected group of projects observed decades later, not a representative evaluation of every sites-and-services programme. [214] The experience of the first residents is essential to judging whether that outcome can be repeated.

Early isolation and the later decline of sites-and-services lending complicate a simple account of success. The Ambedkar Nagar study in Chennai describes early difficulties associated with remoteness and weak access to employment. Among surveyed occupiers, original allottees accounted for 76 percent in 1998–99 and 44 percent in 2017–18. These are survey-composition figures, not a measured rate of forced displacement. Assessing how the settlement developed and what happened to its original beneficiaries requires separate evidence. [215] A World Bank review documents a shift towards housing finance and sector-wide policy reform: sites and services together with upgrading fell from more than 70 percent of shelter lending in 1972–86 to 15 percent in 1987–2005. The review also records deteriorating project ratings during 1982–91. These are categories of lending across countries and do not establish that Chennai itself

failed or explain the causes of the lending shift. [216] Understanding the policy retreat requires examining the institutions expected to finance, manage, and support gradual development.

My explanation for the retreat centres on institutional incentives and the difficulty of sustaining support through a long development period. My reading is that three institutional problems help explain the retreat. First, the time needed for a neighbourhood to develop can exceed the period over which a project is judged. Low early occupancy may indicate a poor location or an area where services and employment have yet to develop. Second, the approach demands coordination over land, trunk services, transport, allocation, household finance, and later maintenance. Providing serviced plots requires agreements among these agencies. Third, a completed dwelling is easier to count and attribute to a provider than years of investment by households. That creates a possible bias in programme design; it is not evidence about the motives of any particular official. A new sites-and-services programme must address the costs households incur while waiting for transport, services, and nearby employment. Upfront plot and construction payments can exclude the poorest. Collection and maintenance failures can prevent replication. The appropriate response is to test these conditions at the proposed site, not to assume that a successful Chennai neighbourhood settles them. A renewed programme would need to address those conditions before asking households to invest.

I would support a new serviced-neighbourhood programme only with a funded transport and service timetable, a credible route for households to build safely in stages, rental options, and a plan for families unable to self-build. Track original allottees, incoming buyers, and renters separately. Follow household costs and reasons for leaving alongside plot occupancy. The transferable principle is to organise affordable access to serviced land while letting households invest over time. It cannot depend on their surviving a decade of avoidable isolation. The first commitment to price is the infrastructure that makes staged building and occupation possible.

The cost of serviced land

Separate trunk and local infrastructure costs before comparing housing sites. Trunk systems, including arterial roads, bulk water, treatment plants, substations, main drains, and mass transit, serve many developments. Local infrastructure, including neighbourhood streets, pipes, sewers, lighting, and public space, connects individual parcels. A third layer is operations: pumps must run, drains must be cleared, bills must be collected, assets must be repaired. If the first affordable-housing project must pay the entire cost of a trunk main that will later serve wealthy developments, its units become artificially expensive. If government pays everything from general taxation, landowners may receive a windfall. If a utility charges full connection costs upfront, low-income households may remain unconnected even when a pipe passes their door. One useful measure is the additional infrastructure cost per square metre of legal floor space. Estimate the public and private investment needed for another 1,000 square metres of housing at a given location and density. Identify the upfront payer, repayment source, and expected receipts from land value, tariffs, or taxes. Report capacity thresholds separately because a small addition may require a large network upgrade. The number of households sharing those networks will affect both costs and the capacity required.

Density affects network costs. A kilometre of pipe serving 40 homes costs more per household than an otherwise comparable pipe serving 400, until capacity upgrades are required. Compact development can therefore lower per-household network cost. Density beyond funded capacity can instead increase outages, sewage overflows, and flooding. Compare engineering designs rather than assume costs fall continuously as density rises. Ethiopia's municipal programme illustrates an attempt to finance infrastructure while strengthening the institutions responsible for delivery.

2 Match infrastructure delivery to local institutions

Municipal infrastructure and capacity in Ethiopia

Ethiopia's reported infrastructure outputs demonstrate the scale achievable through a multi-city programme, but do not establish the full housing outcome. Between 2018 and 2024, Ethiopia's Urban Institutional and Infrastructure Development Program financed construction and rehabilitation of more than 2,700 kilometres of roads and development of 2,700 hectares of serviced land across 117 cities. World Bank programme reporting attributes improved living conditions for more than 6.6 million people and 1.15 million jobs, including 915,000 temporary and 237,000 permanent jobs, to the programme and associated investments. These are reported outputs and benefits; they are not estimates against a no-programme comparison. [124] Participating Ethiopian cities also worked on financial management, audits, planning, and own-source revenue. The mayor of Arba Minch reported a 262 percent increase in municipal revenue over five years. The cited account does not specify whether this figure is adjusted for inflation. The programme linked investment support to institutional development, but the revenue figure alone cannot establish its causal contribution. [124] The proposed mechanism is that predictable transfers reward institutional improvements, allowing cities to prepare and execute investment plans. Better revenue and asset records may then support maintenance and later investment. Assess each step through budgets, audits, delivery dates, and service performance. The reported construction totals do not establish that this full process occurred. The appraisal must distinguish those construction outputs from sustained employment, functioning services, and resident access.

Employment and service claims need measures that distinguish temporary outputs from lasting benefits. Serviced land does not automatically become affordable housing. Allocation rules, lease prices, building standards and credit determine who ultimately occupies it. A road can enable formal development or accelerate speculative land conversion. The reported outputs demonstrate delivery at scale; they do not isolate the programme's causal effect on institutional capacity or establish who retained the benefits. [124] Sustained service needs a separate test. The IDB's follow-up assessment of 88 Favela-Bairro II settlements found uneven durability: roughly one-third had problems with street paving and drainage, and two-thirds had poor street lighting and sewers. These are conditions in upgraded settlements, not an experimental estimate of the programme's long-run effect. Ethiopia's delivery figures do not establish that it has avoided such maintenance problems. Performance-linked funding should verify operating services and resident access as well as completed works. [124,110] Indian land-pooling experience shows how the timing of those benefits also affects participation.

Land readjustment and delivery times in India

Readjustment agreements depend on a credible timetable for delivering the infrastructure promised to landowners. Owners contribute part of their original holding for streets, amenities, and sale plots, and receive reconfigured parcels. Their smaller plots may gain value if services and accessibility improve. This can reduce cash acquisition needs, but the agreement must also address occupants and users who do not own land. World Bank documentation reports that three Ahmedabad schemes allocated 25–40 percent of their land to infrastructure and public facilities. Betterment charges recovered part of the increase in value. Participating owners may gain despite the reduced plot area, but that depends on the value of the returned plot and the costs and timing of delivery. A deduction percentage alone does not establish fair treatment. The same World Bank review found that Ahmedabad schemes typically took about ten years from initiation to final government approval, despite a statutory expectation of three to four years; some took fifteen. Land-value capture depends on credible infrastructure delivery, transparent obligations, and timing that owners and lenders can finance. A charge payable now against a benefit delayed for a decade imposes different risks from a charge linked to a funded three-year delivery programme. Amaravati illustrates why participation figures must also be separated from outcomes for all affected groups.

High landowner participation does not establish adequate treatment of tenants, workers, and other affected residents. The World Bank's December 2024 Amaravati factsheet reported that about 90 percent of the required land had been assembled through pooling, with roughly 29,000 farmers contributing agricultural land in exchange for serviced residential and commercial plots. The programme planned 320 kilometres of arterial roads, 1,280 kilometres of neighbourhood roads, and water, wastewater, stormwater, power, and telecom networks. The master plan reserved 22 percent of residential area for affordable housing. These were commitments and plans, rather than completed housing outcomes. The same factsheet describes land acquisition under national law where negotiated approaches are not used, and livelihood support for landless workers. Assess those groups separately: landowner participation cannot establish consent, adequate compensation, or livelihood recovery for every affected resident. [4,5] Once development proceeds, municipal institutions must be able to operate and renew the services it requires.

Infrastructure maintenance in South Africa

South Africa's metropolitan reforms link infrastructure finance to the reliability and financial performance of municipal services. The World Bank's November 2025 programme announcement reported that South Africa's eight metropolitan municipalities served more than 22 million people and accounted for about 85 percent of national economic activity. Its accompanying assessment described declining reliability in water, sanitation, electricity, and waste services, with weak revenue collection, underinvestment, and inadequate asset management. The economic-activity share is the announcement's estimate; the source does not provide its calculation. [6,7] The Metro Trading Services Program changes the fiscal incentive. A US$925 million World Bank Program-for-Results supports a broader government effort; national grants to metros are released against verified improvements in governance, revenue management,

asset management and service delivery. Failure to achieve results means the grant is not released. A housing programme can fund connections without providing the utility's continuing operating and renewal budget. Low collections, network losses, and deferred maintenance can undermine the service to new homes. Appraisal should therefore examine utility finances and measured reliability alongside the number of connections financed. South Africa's reforms link national urban objectives to municipal service responsibilities. Performance-based transfers can give metropolitan governments a financial incentive to maintain networks while accommodating growth. Housing supply policy therefore requires an explicit municipal financing arrangement. Private finance can contribute only if the allocation of repayment and operating risks is credible.

South Africa's Credit Guarantee Vehicle is intended to attract private infrastructure finance. A World Bank programme approved in March 2026 projects that it could help mobilise about US$10 billion over ten years. That is an expected financing outcome. Guarantees can change the distribution of financial risk, but tariffs, procurement, project preparation, and oversight still need to support repayment and reliable service. The same discipline applies to proposals that finance infrastructure from increases in land value.

Infrastructure investment and land values

Public investment may create land-value gains, but the amount recoverable must be tested against local demand and project costs. The size and distribution of that gain depend on demand, development rights, and adverse effects such as noise or displacement. When public spending creates an increase in value, appraisal should identify how much goes to owners and whether some can lawfully help finance services. That creates five broad choices. Government can recover value through property tax; levy a betterment charge; sell or lease public land after infrastructure raises its value; auction development rights; or require developers to finance specified infrastructure. Each has different incidence and administrative demands. Value capture depends on enough development value remaining after construction, finance, and other obligations. Charges that exceed project viability can reduce development; charges below the feasible level leave more publicly created value with owners. Estimate residual land value under alternative market conditions and disclose the distribution of costs. A fixed recovery percentage cannot replace that appraisal. The choice of how to dispose of serviced public land determines how much of that value supports affordable access.

Selling serviced public land to the highest qualifying bidder may increase immediate revenue while favouring a higher-price housing product. A discounted allocation can instead create a subsidy with weak delivery conditions. A long lease may preserve ownership and impose affordability, density, and delivery obligations. Compare these options using enforceable terms, monitoring costs, and treatment of current occupants. Making that choice requires an explicit account of the parties gaining value and those responsible for delivery.

3 Fund and sequence services before occupation

Interests and responsibilities in infrastructure provision

Actor	Possible interests	Potential risks or gains	Decision or influence	Possible agreement
Existing landowner	Maximum uplift	Infrastructure-created land gain	Holdout, litigation, political pressure	Predictable rights + transparent value-sharing
Developer	Serviced site, certainty	Avoid trunk costs / delay	Can defer project or change segment	Published connection charges + delivery clock
Utility	Cost recovery, manageable demand	Connection fees, tariff revenue	Can delay/no-connect	Capital plan + tariff/transfer settlement
Municipality	Growth without fiscal stress	Rates and fees vs service liabilities	Planning, permits, local capex	Transfers/ value capture tied to serviced floor space
National government	Housing/jobs/ growth	Tax base and political credit	Intergovernmental transfers	Performance grants + national trunk investment
Informal resident	Service without displacement	Location and tenure value	Protest, electoral pressure	In-situ service + tenure safeguards
Contractor/ broker	Contracts and predictable terms	Procurement and payment risk	Bids, claims, delivery capacity	Open procurement + benchmarked unit costs
Future resident	Affordable access	Benefits from added supply	Almost no current voice	Represent through housing targets/access metrics

Infrastructure agreements should connect each party's expected gain to a funded obligation and a delivery date. Examine land ownership and transactions before and after proposed routes become public, disclose selection criteria, and record conflicts of interest. These checks can identify preferential treatment or speculative responses. Do not infer either from a rise in land value alone. Cost recovery distributes costs among households, landowners, taxpayers, and future users. A uniform connection fee may be simple to administer yet unaffordable for low-income households. Low tariffs without operating support can leave the utility underfunded. Free basic services may protect households while requiring separate capital and renewal finance. Show those transfers explicitly. The cases describe different exchanges: Ahmedabad owners contribute land against promised serviced plots; Ethiopian cities receive infrastructure support linked to institutional requirements; and South African metros can receive grants after verified results. For each arrangement, identify when costs are incurred and when benefits arrive. Long delays or unfunded commitments can make participation or continued service difficult. In informal settlements, those obligations must be established without assuming that all tenure claims have already been resolved.

Services and tenure in informal settlements

Service provision and tenure protection need to proceed together where claims are disputed. Requiring full title before connection can leave entire neighborhoods without water or sanitation. Providing permanent infrastructure can be interpreted as recognition of occupation. Connection decisions therefore need an explicit statement of their effect on tenure rights. A practical approach separates service entitlement from final adjudication of property ownership where law allows. Basic water, sanitation, drainage and fire access protect public health and the wider city. Digital enumeration can record occupants and service accounts without converting every record into title. Tenure regularization can proceed through its own due process. In-situ infrastructure can raise rents and land values. Owner-occupiers may gain; tenants can lose. Structure owners may capture public investment through higher rents. Upgrading therefore needs a tenant strategy: rent monitoring, additional supply, targeted support, rights during redevelopment, and protection against opportunistic eviction. Protecting residents also requires knowing what reliable services will cost them after upgrading.

Operating and maintenance costs

Unreliable services create household costs that tariffs alone do not measure. Intermittent water requires storage or alternative purchases. Poor drainage can damage possessions. Unreliable power may require batteries or generators. Broken streetlights can affect perceived safety, and poorly managed waste sites can expose nearby residents to pollution. Include these costs where they can be measured, and disclose uncertainty. For every network, identify its owner, expected asset life, maintenance standard, annual operating cost, replacement provision, tariff or transfer source, and service target. Missing information on these items is a gap in the project appraisal even if construction is fully financed. Capital grants can fund assets without sufficient operating revenue. Results-based finance may help if it rewards measured reliability, sound collections, and asset management, with independent verification. Track the duration and quality of household service alongside kilometres of pipe installed and public cost.

Such outcome measures need agreed definitions and adequate monitoring. Transport adds another recurring cost that can reverse the apparent advantage of a cheap site.

Transport and household affordability

Compare peripheral and central sites using the full household budget: rent or mortgage, utilities, transport, expected maintenance, and uninsured climate losses. Include travel and care time separately unless a transparent valuation is used. Lower land or purchase costs may be offset by higher recurring expenses. This creates a coordination rule: large affordable-housing sites should pass a job-access test before public money is committed. Measure the number of jobs reachable within 45 and 60 minutes by the modes low-income residents actually use, at peak times, and include fares. A proposed transport line counts only when funding and delivery are credible. Transport investment may raise nearby land prices and increase displacement pressure. Consider affordable-housing provision before or alongside it through public land, inclusion requirements, social rental, lawful acquisition, or value-sharing agreements. Acquisition requires due process and protections for affected rights-holders and occupants. Compare these options before land-price increases make affordability commitments more costly. A reliable site comparison therefore depends on combining housing, network, transport, and land information.

Digital infrastructure records

Shared infrastructure records are useful when they resolve a specific conflict between planned housing and available capacity. Inconsistent identifiers or missing updates can delay decisions. Diagnose which record or coordination failure affects the project before commissioning a shared platform. A shared geospatial operating picture can connect parcel, development-right, network-capacity, hazard and project data. Satellite imagery and AI can detect growth before official maps catch up. Digital twins can test network loads. Smart meters can reveal losses. Mobile payments can improve collections. Remote sensing can verify whether roads and drains were built where contracts say they were. A shared information system should support a specified investment decision. For example, compare where an additional 10,000 low-income households could be accommodated, counting land, networks, transport, existing occupants, and hazards. Identify the required works, responsible agencies, funding, and capacity dates. Test whether the system improves those decisions over existing records. Those records should be used before site and subsidy commitments make changes expensive.

Sequencing infrastructure and subsidies

Assess infrastructure and transport before fixing the housing site, design, and subsidy. The budget may later need to absorb roads and utilities while households still face poor access. Check those requirements before committing to a beneficiary count or site design. Begin with access, network capacity, and existing rights. Compare places where jobs, land, and services can be coordinated. Where land assembly is needed, use lawful procedures with resident protections and costed alternatives. Test density against network design, agree funding for trunk and local systems, and provide operating and maintenance revenue. Then choose tenure, subsidy, and

finance that households can sustain. This appraisal can show when low-priced public land has high infrastructure and transport costs, or when a proposed private development depends on unbudgeted public works. It also provides a basis for negotiating land-value contributions before new services change prices. The appraisal framework below makes those commitments visible before approval.

Serviced neighbourhood assessment

Gate	Question	Metric / decision
1 Access	Can low-income residents reach opportunity?	Jobs within 45/60 min + fare/time burden
2 Trunk capacity	What bulk investment is required?	Water, sewer, drainage, power, roads/transit headroom
3 Local networks	Who funds on-site and neighborhood works?	Cost per serviced parcel / legal m²
4 Density	Is infrastructure being used efficiently?	Households or floor area per network kilometre
5 Timing	Will services arrive before occupancy?	Critical-path dates by agency
6 Incidence	Who ultimately pays?	Taxes, tariffs, developer charge, land value, household price
7 Value capture	Who receives the uplift?	Pre/post land value; public recovery share
8 O&M	Who keeps the service working?	Collection, losses, maintenance plan, replacement reserve
9 Informality	Will service delivery displace residents?	Tenure/tenant map + safeguards
10 Politics	Who can block or redirect investment?	Map of interests, risks, authority, and potential agreements
11 Digital power	What new visibility is created?	Data rights, purpose limits, appeal, enforcement protocol
12 Outcome	Did infrastructure create affordable access?	Housing+utilities+transport cost; service reliability; occupancy

Limits of the evidence

Programme outputs and proposed appraisal methods support different kinds of conclusions and should remain separate. These sources establish scale, institutional design and observed delivery. They do not prove that every reported job, land-value increase or service improvement was caused solely by the cited program. The serviced-square-metre framework, the sequencing rule and the Serviced Neighborhood Test are judgments derived from urban economics and operational experience. They should be populated with local engineering costs, land values, travel-time data, utility financials and household expenditure before project decisions. Fieldwork must supply the local engineering and financial evidence needed to use those methods.

Prepare the local decision

A serviced-land appraisal should establish network capacity, connection costs, delivery authority, and access for additional households. Obtain network maps, measured capacity, condition assessments, connection queues, outage records, tariff receipts, and the cost of reaching each candidate site. Distinguish trunk works, local distribution, household connections, and recurring operation. Check whether reported capacity exists during peak demand and the dry season. Use the Ethiopia and Gujarat cases above to examine delivery and delay, rather than infer service quality from kilometres of infrastructure built. Walk the proposed connection route with utility engineers, municipal works staff, land officials, and residents. Record rights-of-way, adoption standards, payment responsibilities, and the authority that can resolve each obstruction. Ask tenants about charges paid to informal providers and the cost of switching. The finance team should reconcile the housing programme's budget with the utility's capital and operating plans. Ask the utility to substantiate a claim that a prosperous area cannot accommodate more homes: identify the limiting component, measured capacity, upgrade cost and delivery date. Compare that cost with peripheral network extension. Budget the selected works and address existing service deficits; do not assume every central site is cheaper. Make the resulting capacity available under published connection rules rather than reserving it informally for existing owners. Technology can support the resulting commitment register, but the final check is whether services function after connection.

Use a shared site record to assign and verify the actions needed for functioning services. Combine asset records and geotagged inspections in one dated network inventory. Test remote readings against field measurements before using them to allocate capacity. Approve a service package only when connection dates, funding, operator acceptance, and household charges are documented. Verify pressure, drainage, billing, and occupied homes after connection; a completed pipe without a functioning service remains an incomplete output. Chapter 4 turns to the producers, construction methods, and costs that determine what households can build on serviced land.

An infrastructure commitment is credible only when the operating agency can sustain the promised service. Clear asset ownership, funded maintenance, and access for repair crews connect construction to reliable use. A performance-grant formula cannot supply missing

verification capacity. Handover without an operating budget, disputed boundaries, or demand beyond design capacity breaks that connection. Model the cumulative utility load and replacement bill for 50,000 homes. Replicating a 500-home layout does not replicate bulk-water capacity.

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4. Construction costs and productivity

Reduce construction costs by addressing the constraint in the local production system while protecting safety and occupant choice. A new building technology will help only if demand, approvals, logistics, skills, and finance allow it to perform at commercial scale. Procurement should reward verified quality and lifetime cost, and households should retain choices that meet those requirements. My early experience with rural housing explains why a volume target alone is an inadequate specification.

My experience of Indira Awaas Yojana exposed a gap between the volume delivered and the housing families wanted. In the early and mid-1990s, my experience of implementing Indira Awaas Yojana was dominated by the imperative to deliver houses in volume. The NITI Aayog library abstract of the Planning Commission's 1992 quick study reports 1.06 million houses constructed nationally during 1985–92. The study covered 56 villages in 14 states; that field sample is distinct from the national administrative total. [214] I remember tribal families wanting mud houses that would be beautiful, comfortable, and considerably larger than the standard provision. Their preferred designs did not get through the local approval process. I associated the refusal with the Public Works Department's technical framework, but it may have been a local error rather than a requirement of the scheme. I cannot identify the specific rule. The families knew what they wanted from their homes. We focused on how many houses we completed. Later national policy explicitly recognised appropriately specified mud and bamboo houses meeting durability requirements. The June 2013 IAY guidelines also rejected a compulsory standard design and provided for beneficiary choice. These provisions do not establish what applied locally two decades earlier, but they rule out treating my recollection as evidence of a permanent national prohibition. [215] The distinction is supported by field evidence on choice and suitability, rather than by treating my recollection as a national rule.

1 Define the cost and quality problem

Indira Awaas Yojana and the houses people wanted

Restricted household choice requires evidence on satisfaction and use as well as completed buildings. A 1996 Development Alternatives pilot survey examined nine cases involving Scheduled Caste widows in two villages near Rohtak. Conventional construction and the junior engineer's model largely shaped the houses. Its author reported greater satisfaction where beneficiaries exercised their own choices, alongside weak technical advice and workmanship. This small survey concerns a different population and does not verify my encounter with tribal families. [216] The broader problem in my implementation experience was the limited attention given to life after completion. We did not systematically follow income, health, schooling, or occupants' satisfaction to understand what the housing support had changed. This is a statement about the work I remember, not a claim that no evaluation of IAY ever measured these outcomes. A house count was necessary for managing delivery. It was insufficient for judging the programme's value to the family. An approval process that permits suitable alternatives is a practical response to that problem.

The remedy is to give engineers a reliable approval route and households a real choice. For an earth-building option, require a locally competent technical institution to establish performance requirements for the relevant hazards, moisture exposure, foundations, roof connections, and maintenance. Prepare costed details, train masons and inspectors, and specify which evidence releases each payment. Compare usable space, indoor comfort, repair burdens, and resident preference with the conventional option. No material deserves approval merely because it is traditional, and no household should be compelled to use it. Keep the volume target, but pair it with a funded follow-up sample. Ask whether families occupy the homes, find them comfortable, can afford repairs, and can still reach work and school. Record alternative designs proposed and reasons for refusal. Track household income, illness-related expenses, and school attendance where the programme has a plausible route to changing them. Do not attribute every improvement to housing: other services and wider economic changes matter. Chapter 13 sets out how to establish a baseline and a credible comparison before construction begins. The alternatives must then be compared against all the costs of delivering and occupying a home.

Components of housing cost

Construction savings improve affordability only to the extent that they reduce the household's total delivered cost. Their effect on affordability depends on their share of total cost, whether savings reach households and whether the method can be used reliably at scale. In CAHF's benchmarking of a typical 55-square-metre house, materials and labour have accounted for about 45 percent of total cost, while infrastructure has contributed around 14 percent. The shares vary sharply by country and city. Older city-level benchmarks found construction near 40 percent of total development cost in Nairobi and Kampala, while it exceeded 60 percent in some other markets. Land, infrastructure, statutory compliance, finance, holding costs, taxes and marketing fill the rest. An illustrative calculation shows the limit of a component saving. If construction represents 40 percent of the delivered price and a method reduces that cost by 20 percent, the direct saving is 8 percent before other changes. The buyer receives that full saving only if it passes through. Higher land prices, developer margins, or additional financing and implementation costs can reduce it. The size of the affected cost component provides a first test of the likely saving.

Estimate the cost share affected, the feasible reduction, and the share expected to reach households. Then test demand and production volumes under which the saving persists. The product of those factors can be a screening calculation, but expansion probability is an assumption, and the factors may interact. Compare the result with savings from approval time, finance, and other large costs. That calculation requires an account of the linked production systems behind a completed home.

Construction inputs and productivity

A finished home is the output of several linked production systems. Materials must be made or imported. Contractors need labour, equipment and working capital. Designs must satisfy codes. Developers must hold land while approvals proceed. Banks or investors must finance the interval between spending and sale. Utilities must connect the site. Buyers must trust the product

enough to purchase it. A construction change can affect other costs. Prefabrication may shorten site work and reduce defects, while introducing factory overhead that becomes costly at low production volumes. Locally produced earth blocks may reduce transport and embodied emissions if the material and design suit the site, but lenders may require evidence of durability and resale value. Standard designs may speed approval while limiting later extensions. Include these effects in the comparison. Distinguish three levels of production. Repeated designs can reduce design and training costs. Firms can spread equipment and overhead across projects. A wider market needs multiple suppliers, contractors, lenders, inspectors, and repair providers. A demonstration establishes experience with one design or firm; evidence on supply chains and repeat demand is needed before inferring wider market viability. African cost comparisons show how the constraints differ across those systems and locations.

Institutional differences in African construction costs

Differences in materials, logistics, and institutions can produce large cost differences for a similar house. CAHF's comparative work reports substantial variation in the cost of a standardized house across African cities. Its early bill-of-quantities exercise priced the same specification from under US$30,000 in Dar es Salaam to more than US$60,000 in Nairobi. Labour accounted for more than one-third of construction value in some high-cost markets and more than half of the construction component in Nigeria. Finance and holding costs were especially high in Maputo; sales taxes were unusually high in Nairobi. The cost of an identical specification can vary with labour markets, taxes, logistics, approvals, and finance. These historical comparisons support a local cost breakdown rather than a single construction-cost assumption for African markets. Check which component accounts for the difference before selecting an intervention. Rwanda illustrates the logistics channel. World Bank analysis estimated that import taxes plus transport and logistics could raise construction-material prices by as much as 43 percent. Benchmark housing costs were about 30 percent higher than in South Africa, with building and infrastructure costs important contributors. The analysis identified alternatives such as autoclaved aerated concrete, compressed stabilized earth blocks and modern clay products, while stressing that scale remained limited. Lower-cost local alternatives still need an approval and supply chain that can support repeat use.

A local material may lower transport costs and foreign-exchange exposure while requiring new testing, inspection skills, contractor training, or lending criteria. Adoption therefore depends on evidence that it can meet performance requirements and be supplied, financed, and maintained reliably at the intended volume. A consistent cost record is needed to establish whether those conditions produce a delivered saving.

Construction cost monitoring in Kenya

Huruma's pilot cost record illustrates why unit prices need their quantities, dates, and accounting scope attached. The cited Huruma upgrading account in Nairobi reports KES 2.85 million for four pilot units (about US$22,000) and KES 717,485 per two-bedroom unit (about US$5,550), alongside KES 16,491 per square metre (about US$128), as of December 2024. The US dollar figures are my conversions at KES 129.3/US$, the Central Bank of Kenya's fourth-quarter

2024 average. The first two reported costs do not reconcile: KES 2.85 million divided by four is KES 712,500 (about US$5,510). The difference in scope or rounding remains unresolved, so these figures are unsuitable as a reconciled tender benchmark. [3,201] Interior works, including plumbing, plastering, windows, doors, electrical works, and kitchen slabs, represented 29 percent of construction cost. Masonry walls on the two main floors represented 17 percent. On those reported shares, halving masonry costs would reduce total construction cost by at most 8.5 percent before changes elsewhere. Compare that estimate with options to simplify wet services, finishing, or procurement. Record quantities and costs by component so a final contract price can be checked against the work delivered. A national programme needs the same accounting discipline across its procurement pipeline.

Larger housing programmes should publish enough cost and performance information to distinguish efficiency from concentration and cost drift. Large pipelines can create economies of scale in materials and contracting, yet they also create procurement rents and concentration risk. If a small group of developers or suppliers wins repeated packages, scale economies can become market power. Procurement should therefore publish unit-cost benchmarks, bidder concentration, variation orders, delivery times and defect rates. India's technology programme provides another test of what reported construction outputs can establish.

2 Match construction methods to demand and producer capacity

Construction technology adoption in India

Public demonstrations can establish experience with a construction system without proving comparative household savings. Six demonstration projects in Chennai, Rajkot, Indore, Lucknow, Ranchi and Agartala have tested different industrialized and alternative systems. The public sector plays the role of first buyer: it creates reference projects, gives regulators experience with unfamiliar systems, and lets engineers observe construction and performance. Evaluation should continue after a demonstration project. Assess whether developers adopt the system without a special grant, local suppliers can support it, contractors can repair it, and lenders, insurers, and households accept it. Cost comparisons must include transport, royalties, factory utilisation, and financing. World Bank results reporting states that the Tamil Nadu Housing and Habitat Development Project built roughly 15,000 affordable homes between 2020 and 2025 through public-private market interventions, including a state-sponsored investment fund. It reports use of resilient-design standards and green certification. Related reforms addressed planning permissions, mixed-income development, private participation, finance for poorer households, and resident involvement in upkeep. These reported outputs do not establish comparative construction savings or lower household bills. [5,6] Adoption therefore needs a project-specific assessment beyond the demonstration.

A construction method must be assessed alongside permits, finance, maintenance, and local demand. Compare its installed cost and service life under those conditions before committing to production. For factory-based production, that assessment must also test whether sufficient orders will materialise.

Demand and industrialised construction

Industrialised construction exchanges some site uncertainty for factory, logistics, financing, and utilisation risks. In principle they reduce waste, improve dimensional accuracy, shorten site time and reduce defects. Their fixed costs create the central economic condition: utilization. Consider an illustrative factory with capacity for 5,000 units a year. With 5,000 funded orders, it can spread fixed costs across that output. With 800 orders one year and none the next, unit costs and financing needs increase. Before supporting a plant, test procurement commitments, the cost of unused capacity, and whether the operator can serve other customers. These volumes are a scenario, not observed project results. This creates a role for the state that differs from subsidy. Government can publish a multi-year pipeline, standardize performance specifications, pre-approve systems, and aggregate demand across agencies. It can also avoid locking the market to a single proprietary system. Use open interfaces, multiple qualified suppliers, and performance-based codes to support competition. Speed has a financial value. A shorter construction cycle reduces interest during construction and returns developer equity sooner. Where finance is expensive, time savings may be more valuable than material savings. The pro forma should price both. The comparison with conventional building must use all-in delivery costs under realistic demand.

Pilot unit costs and industrial production economics answer different questions. The UK House of Lords inquiry documented the collapse of Ilke Homes and House by Urban Splash and the closure of L&G's modular operation. Witnesses cited insufficient orders and wider business and economic problems. These are different technologies and markets, not a matched comparison with Huruma. They show why factory utilisation, financing and all-in delivery cost need testing before a government commits to scale. [3,114] Even a verified production saving must pass through the developer's land, finance, and sales decisions before it reaches households.

Developer returns and cost savings

Developers bear coordination and financing risks that influence both production costs and the distribution of savings. A required return compensates investors for capital and risk. Observed margins may also reflect market power or scarce rights, so appraisal should not treat every margin as a necessary cost. Compare assumptions with competing projects and bids. This explains why an apparently cheap technology may be rejected. A developer may prefer a familiar system that costs 5 percent more if contractors, banks, valuers, buyers and building officials understand it. The expected value of fewer delays and lower tail risk can exceed the quoted saving in construction cost. For affordable housing, the policy goal is therefore to reduce risk that serves no public purpose: uncertain approval time, unclear title, unpredictable connection dates, arbitrary standards, slow inspections and volatile subsidy payments. Lowering these risks can reduce the return a developer requires and widen the set of viable projects. Competition then determines incidence. In a constrained market with scarce development rights, a cost reduction can capitalize into land. In a competitive supply market, developers have stronger incentives to pass savings to buyers. The same construction innovation can therefore produce different affordability outcomes in two cities. A delivery strategy must therefore identify the producers and financing arrangements able to carry those risks.

A housing target needs a production plan that identifies who can build, who can finance work before receipts arrive, and who remains responsible after occupation. I would not give a large developer, a household-builder, and a social landlord the same support package. Their cash cycles and obligations differ. The comparison below is an appraisal framework, not a ranking of which producer is inherently more efficient. The business-model comparison below distinguishes the functions each type of producer must perform.

Producer business models and entry barriers

Table 4.2. Comparing producer business models. Proposed diagnostic; validate costs, financing terms, and constraints in actual project accounts.

Producer and cash cycle	Likely constraint to test	Support and condition to test
Household-builder: savings or remittances finance successive stages; occupation may precede completion.	Small purchases, interrupted work, technical advice and safe extensions.	Serviced tenure, technical assistance and inspected stages. Avoid a repayment schedule that assumes a salary.
Small contractor: pays labour and suppliers before receiving certified payments; may own no land.	Working capital, late payment, bid security and contract size.	Prompt payment and proportionate lots and securities. Retain safety, quality and completion checks.
Small landlord: builds against future rents, often on an existing plot.	Legal additional floor area, utility capacity, construction finance and vacancy.	Approval and connection support with staged finance. Test debt service after vacancy, repairs and tenant protections.
Larger developer: commits land and equity, then combines borrowing, sales or institutional investment.	Land assembly, uncertain approvals, infrastructure timing and demand at the intended price.	Predictable permissions and competitively allocated serviced sites. Verify additional affordable supply before extending concessions.
Cooperative or community organisation: pools contributions and public support; members share decisions and risks.	Collective authority, uneven contributions, professional support and internal accountability.	Recognised rights, technical support and transparent accounts. Protect minority members and residents who cannot contribute equally.
Social or long-term rental provider: capital funding precedes a stream of rents and operating costs.	Patient capital, allocation duties, collection and renewal funding.	Match grant and debt terms to rents and lifecycle costs. Fund any operating gap and enforce affordability through ownership changes.

A housing programme needs each production function to be performed, even when several functions sit within one firm. These functions may sit within a developer or in separate organisations. Combining them can reduce handoffs, but it can also restrict entry if one firm controls both developable land and access to services. Before promising a pipeline, establish which rights and connections another qualified producer could obtain on comparable terms. Why might formal firms move up-market? A fixed approval cost, minimum plot size, expensive finance, or long delay must be recovered from the homes eventually sold. Higher-price products may cover those costs when lower-price products cannot. Scarce serviced land can also reward holding land rather than building it. These are mechanisms to test in local project accounts, not a universal explanation of developer behaviour. Compare entrants and failed projects as well as established firms. Before awarding a large pipeline, test whether smaller contract lots, shared infrastructure, standard approvals, or staged finance allow more qualified firms to compete without weakening safety. Track bidder entry, completion time, final price, occupied units, and the firms that stop bidding. Tracing actual projects shows where a missing function or financing gap prevents delivery.

Follow money through a small number of completed, stalled, and abandoned projects. Record deposits for land, approval spending, supplier payments, certified bills, interest, sales receipts or rent, and the months of maximum cash exposure. A viable final sale price can conceal an unfinanceable construction period. Compare applicants who failed to obtain finance with those who built; accounts from successful firms alone will miss barriers to entry. The cases in this book support different interventions. Chennai's sites-and-services experience shows households building over time on a serviced layout, while the evidence in Chapter 3 also records long gestation and uncertainty about who retained access. Singapore's core-contractor arrangements in Chapter 25 illustrate a reciprocal pipeline and capability commitment; they do not prove that guaranteed work is competitive. South African social rental illustrates why a capital grant needs an operator able to sustain affordable rents. Use these comparisons to select the missing function, not to declare one production model universally superior. Cape Town's support for smaller producers illustrates one way a city can address an administrative barrier.

My conversations with Hugh Cole drew my attention to Cape Town's practical assistance for small housing producers. My first personal interaction with Hugh Cole, Director: Policy and Strategy and Chief Data Officer for the City of Cape Town, was in June 2025, when I learned more about Cape Town's efforts to support affordable housing. We continue to work with the city. Cape Town's Local Planning Support service provides small producers with assistance: guidance on affordable rental flats, development rules, design guidance, prototypical layouts, and location maps. Its published resources show how a city can make the approval route more usable. They do not establish the number of additional completed homes caused by the service. [217] An adaptation should test which applicants reach compliant occupation, what they spend, and whether the new rooms remain affordable. A public procurement pipeline requires a separate test of whether its packaging allows capable smaller firms to participate.

For a proposed public pipeline, compare at least a bundled delivery option with smaller lots or independently serviceable phases where feasible. Include the public cost of managing more contracts. Require open eligibility, proportionate qualification criteria, and

disclosure of common ownership where lawful; smaller lots alone do not establish competition. If public support merely increases the land bid or the incumbent's margin, revise the award terms. If smaller firms cannot meet essential standards even with feasible support, do not relax those standards to meet an entry target. Those firms are only part of the wider network of households, landlords, and tradespeople producing low-income housing.

Households, builders, and small landlords

Support for small producers should simplify compliance while retaining independently verified safety protections. A household may add rooms or floors over several years using savings or remittances. Measure this production locally; it can be missed by permit and developer records. Its financing, safety, and service needs differ from those of a large formal project. Building in stages can take longer but allows expenditure to follow irregular household income. It avoids a large mortgage. It can adapt the dwelling to a shop, workshop or rental room. Households buy materials at retail prices, technical advice is weak, work stops when cash runs out, and hidden structural defects can accumulate. Support for household and small-builder production can include accredited mason networks, tested structural designs, pooled purchasing, staged loans, digital bills of quantities, and inspections at critical stages. These are options to test for local cost and accessibility. Preserve independently verified standards for structural stability, fire, sanitation, drainage, electrical safety, and accessibility; simpler administration must not remove essential protection. The choice of production method also affects the mix of labour, skills, and output.

Industrialized construction can reduce some site labour; incremental construction sustains large networks of local trades. The correct comparison includes productivity, wages, skills, safety and total housing output, rather than maximizing construction employment per house. These trade-offs should inform standards that protect occupants without excluding workable building methods.

Safety standards and affordability

Building standards should be tested against the safety benefits they provide and the costs they impose. Requirements that households cannot finance may discourage compliance or formal production. That affordability problem calls for review of avoidable costs, suitable designs, and targeted support. It does not establish that health or safety standards should be reduced. Use risk-based regulation to distinguish essential protection from avoidable prescription. Structural stability, fire escape, sanitation, drainage, electrical safety, hazard exposure, and accessibility need enforceable standards. Assess finishes, parking, and other amenity requirements separately. Room dimensions can affect ventilation, crowding, and disability access, so changes require technical review. Performance-based rules can admit alternative materials when independent testing demonstrates equivalent safety. Standards can affect competition as well as safety. Require evidence for specifications that exclude alternative products or add professional steps, and disclose conflicts of interest in code review. Provide a transparent route to demonstrate equivalent performance. Assess each requirement on its costs and benefits rather than presume

the motives of producers, professional bodies, or officials. The cost comparison must continue beyond construction to energy, repairs, and climate exposure during occupation.

Carbon, resilience, and lifetime costs

Occupancy costs can offset a low construction price. Thermal design affects cooling expense and heat exposure; drainage affects asset damage; roofing affects storm resistance; and locally unavailable parts can increase repair costs. Compare upfront and recurring costs over a stated period, together with safety and resilience outcomes that should remain minimum requirements. Tamil Nadu's reported use of resilient design and green certification provides a design example to investigate. It does not by itself show lower lifetime household costs. Measure rent or debt service, energy, water, transport, maintenance, and expected uninsured losses after occupation, and examine whether resident organisations have the resources needed for upkeep. [5,6] A unit that costs 5 percent more to build may be cheaper over twenty years if it reduces energy and repair bills enough to offset the higher initial cost. A green premium that households cannot finance upfront still needs a financing instrument. Lifecycle efficiency and liquidity have to be solved together. That longer comparison helps determine whether a proposed technology addresses the main local constraint.

3 Align procurement, finance, and implementation

Interests in the construction supply chain

Actor	Possible benefits	Decision or influence	Illustrative concern	Feasible reform
Cement/steel producer	Price, market share, tariff protection	Standards, lobbying, distribution	Jobs and quality	Competition + transition support + performance standards
Large developer	Margin, speed, risk reduction	Can change segment/location	Affordable projects are not financially viable	Serviced land + predictable approvals + competitive pipeline
Small builder/mason	Work and cash flow	Can remain informal	Formal rules cost too much	Simple code + training + staged finance

Actor	Possible benefits	Decision or influence	Illustrative concern	Feasible reform
Contractor	Equipment and workforce utilization	Bids, claims, adoption	New systems are risky	Multi-year pipeline + training + shared performance risk
Professional body	Fees, liability, professional control	Codes and approvals	Safety	Risk-based rules + open equivalence path
Municipality	Safety, fees, enforcement power	Permits and inspections	Capacity constraints	Digital permit clock + standard designs + risk inspection
Household	Low monthly cost, adaptability	Purchase/ acceptance	I need a home I can extend	Lifecycle cost + incremental design
Politician	Visible units, jobs, contracts	Budget/ procurement	New technology will solve shortage	Publish total unit economics + post-occupancy evidence

A demonstration building does not establish that a construction method addresses the main local cost constraint. Before subsidising a factory, compare the expected savings with the costs attributable to land, infrastructure, finance, and low production volumes. Industrial policy adds another coalition. Tariffs on cement or steel can protect domestic jobs and firms while raising housing cost. Import liberalization can lower input prices while exposing local producers. Local-content mandates can build a supply chain or create protected rents. The housing ministry rarely controls these choices; trade and industry ministries may value different outcomes. Changes to tariffs or specifications can concentrate losses among established suppliers and workers, while spreading savings across future buyers. Publish cost evidence, test alternatives, qualify several suppliers, and assess training or transition needs before changing protection or standards. The timing should follow the evidence and supply conditions; gradual removal is one option rather than a universal rule. A low factory-gate price is insufficient if transport, land or building form confines cheaper homes to remote sites. Public procurement can test the complete cost and performance proposition before committing a large pipeline.

Public procurement and market development

Procurement should compare viable local options and disclose costs, competition, variations, and defects. Publish the cost and duration of purchase guarantees; avoid indefinite subsidy to unused capacity. The procurer must verify quantities, structural performance and defects, and retain the ability to procure alternatives. Track comparable all-in cost per usable square metre, delivery time, defects, energy and maintenance bills, and household access to jobs. Public procurement can shape the local construction market where government purchases are substantial. Lowest-price awards may encourage bids that depend on later claims. Prescriptive specifications may exclude suitable alternatives. Very large packages may exclude small firms, while small unpredictable annual orders may not support factory investment. Compare these risks with actual bidder, price, and delivery data. A better procurement design buys performance: delivered cost, time, structural and fire safety, energy performance, defect rate, maintainability and adaptability. Bidders can choose the means. Framework contracts or multi-year pipelines can create demand certainty while keeping several suppliers in competition. Open cost data make collusion and drift easier to detect. Private financing changes the contract's risk allocation but does not eliminate the underlying costs.

Public-private partnerships add financing and delivery capacity, yet they do not make costs disappear. Private capital prices land risk, approval risk, demand risk and government-payment risk. Transfer a risk the private party cannot control and the bid price rises. The contract should allocate each risk to the actor best able to manage it. Construction finance must therefore be assessed against the actual timing of expenditure and receipts.

Construction finance

Construction finance covers expenditure before sales or rent receipts begin. Its cost rises with time and uncertainty. A two-year approval delay before construction can be as economically important as an expensive material. A six-month reduction in build time reduces financing costs most where interest rates and required equity returns are high. Small builders face a different gap. They need working capital in thousands of dollars, not project finance in millions. Banks often cannot underwrite an incomplete structure on weak title. Supplier credit, micro-construction loans, savings-linked finance and milestone disbursement can fit the production cycle better. Capital markets enter further upstream. Long-term finance can fund rental portfolios, factories, warehouses and developer pipelines. Cheap capital for an uncompetitive supplier or an unviable project postpones the loss; it does not remove it. Reliable records of that expenditure also provide the basis for useful digital construction tools.

AI and digital construction tools

Construction data can improve cost control when comparisons account for specification, location, timing, and scope. Drawings, quantities, schedules, procurement records, photographs, inspection reports, invoices, defects, and change orders provide data for construction analysis. Select digital tools for a defined task, such as checking quantities or tracking changes, and test their error rates and costs against current practice. Digital tools can

help compare quantities across designs, flag unusual bids, estimate schedule risks, identify clashes, and analyse change orders. Site images may assist progress checks, but cannot establish every hidden defect or material property. A small contractor may prepare quantities from a standard design; a lender should still verify the design, costs, and completed milestones before disbursement. Begin with a usable project-cost record where one is missing. Track land, infrastructure, materials, labour, approval time, finance, taxes, margins, change orders, and defects. Investigate why comparable homes differ in cost before using those records to automate procurement decisions. Differences in specification, location, date, or accounting scope may explain an apparent 30 percent gap. The assessment below combines those records with household costs and delivery risks.

Construction technology assessment

Gate	Question	Metric / decision
1 Total delivered cost	What does the serviced, completed home cost?	Land + infrastructure + approvals + build + finance + taxes + margin
2 Cost share	Which component does the innovation change?	Component as % of total cost
3 Magnitude	How much does it change that component?	Audited counterfactual, not vendor claim
4 Time	Does it shorten approvals/build?	Months saved x cost of capital
5 Pass-through	Who receives the gain?	Buyer/renter price vs land value/developer margin
6 Scale	Can it work beyond a pilot?	Supplier depth, pipeline, utilization, competition
7 Finance	Will lenders fund it?	Valuation, collateral, construction finance terms
8 Standards	Can it obtain approval predictably?	Performance-code route and inspection capacity
9 Adaptability/O&M	Can households repair and extend it?	Local skills, spares, lifecycle cost
10 Jobs	What happens to labour demand and skill?	Output per worker, wage, safety, transition plan

Gate	Question	Metric / decision
11 Politics	Whose income, costs, or authority would change?	Map of interests, risks, authority, and potential agreements
12 Household outcome	Did shelter become cheaper?	(Housing + energy + transport + maintenance) / income

Investment criteria

Assess a construction technology using the full developer and household cost comparison. Include time, finance, infrastructure, maintenance, and later adaptation, then test plausible order volumes and cost pass-through. If construction costs fall without a comparable household saving, examine land prices, market conditions, margins, and added costs before changing the subsidy. Applying that test requires distinguishing measured cost evidence from assumptions in the appraisal.

Limits of the evidence

The cited cost and output evidence supports local testing rather than a universal ranking of construction technologies. Several construction-cost benchmarks are snapshots from different years and should not be used as current tender prices. They establish mechanisms and variation rather than a current ranking of cities. The cost-share calculation, production-volume tests, and proposed stakeholder assessment are analytical tools drawn from the chapter's evidence and project economics. They need current quantities, land and infrastructure costs, financing terms, approval times, sales or rental evidence, and post-occupancy performance. Scenario probabilities and pass-through assumptions should be reported explicitly rather than treated as measured constants. A field comparison can supply the missing production and household evidence.

Prepare the local decision

Compare completed projects and realistic order volumes before claiming that a construction method reduces household costs. Compare bills of quantities, paid invoices, labour hours, transport, wastage, financing charges, defects, and repair costs for comparable building types. Separate factory price from delivered and installed cost. Record project size and repeat orders when comparing conventional and industrialised methods. The India technology cases above should lead to tests of local production volume, skills, and supply chains, rather than a comparison of demonstration-unit prices. Interview small contractors, manufacturers, building inspectors, workers, lenders, and eventual maintenance providers. Establish who bears certification costs, warranty claims, late payments, and shortages of replacement parts. Procurement staff should identify requirements that unnecessarily exclude competent smaller firms and retain independently verifiable safety standards. Test digital quantity checks or construction scheduling on completed projects with known costs and defects. Inspect hidden

works physically; photographs cannot establish reinforcement quality or material strength. For a new construction method, commission a limited batch with independent testing and a funded repair obligation. Compare cost per compliant usable square metre, time to occupation, and defects after use. Expand only if delivered savings persist at realistic order volumes. Chapter 5 examines whether households can obtain and retain the resulting homes under different tenure arrangements.

Industrial construction needs a credible order pipeline and verified lifetime performance before it merits public support. Repeat orders, adaptable designs, logistics, quality control, and local repair capability determine whether factory production can reduce costs. Demonstration buildings do not establish commercial viability. Intermittent demand, fixed factory costs, warranty obstacles, and omitted transport or finance costs can reverse an apparent saving. Test tenfold volume and a demand shortfall together. Include factory failure, spare-part supply and the cost of switching contractors.

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5. Tenure, rental housing, and ownership

Offer tenure options that fit household incomes, mobility, and exposure to risk rather than treating ownership as the universal destination. Compare rental, incremental ownership, and supported housing through access, continuing costs, and enforceable rights. Then assign maintenance and public support to institutions capable of sustaining them. The analysis begins with the different obligations bundled into each tenure arrangement.

Tenure determines who can occupy a home, on what terms and with what rights. It affects mobility, wealth exposure, maintenance incentives, access to credit, political voice and the ability to absorb a shock. Ownership bundles shelter with an asset. The household supplies equity, often takes debt, bears repair costs and gains or loses with the property's value. Rental unbundles shelter from the asset. The tenant buys use for a period; the landlord holds the capital and residual property risk. Social rental adds a public subsidy and an institution charged with preserving affordability. Informal rental may lack some formal protections while often achieving a price and location that formal markets cannot match. Tenure support should fit household risks and preferences. A migrant with uncertain work may need a room near jobs. A household expecting to remain may prefer ownership. An older resident may prioritise stable charges and an enforceable right to stay. Compare these choices using realistic payments and access to services. The choice deserves explicit appraisal because public programmes often favour ownership without comparing the alternatives.

1 Match tenure to household needs

Tenure, mobility, and household finances

The policy preference for ownership

Ownership support should be assessed by who can qualify, occupy, maintain, and benefit from the asset. They also create demand for loans, land, and development. Those features may help attract support from recipients and providers. They do not establish the reasons a particular government chose ownership; that requires evidence from its programme design, budget, and stated objectives. Rental support should be measured through sustained occupancy, repairs, predictable charges, and access to jobs. These outcomes depend on management after construction and may receive less attention than completed units or titles. Compare actual budget allocations and household tenure needs before concluding that policy favours ownership. Ownership subsidies can interact with supply restrictions. If subsidised demand rises while little housing is added, part of the benefit may increase prices. Existing owners may support restrictions that protect amenities or property values. Test those responses before expanding buyer subsidies, and compare support for rental supply and other tenure options. Mortgage subsidies, tax preferences, discounted public land, and infrastructure support often favour owners. Renters receive fewer asset-linked benefits. Assess the distribution of support by identifying who qualifies, who can wait for allocation, and who receives the resulting land-value

gains. For many urban households, rental housing offers a more attainable route into a suitable location.

Rental housing in African cities

Rental is central to some African low-income housing markets. A World Bank review found that 92 percent of residents in the Nairobi slums studied were rent-paying tenants. It described poor services alongside high rents. This historical finding supports examining how scarce well-located space affects rents even where water and electricity are inadequate; it should not be treated as a current citywide tenure estimate. Slum upgrading that assumes residents are owner-occupiers can transfer public value to structure owners while tenants pay higher rents. Tenure regularization that recognizes only one claimant can erase subtenants, room renters and informal commercial users. Compensation tied to title can exclude poor households without title. Kenya's 2023/24 Housing Survey provides a contemporary base for this debate. It was designed to measure tenure, affordability and housing conditions across all 47 counties. Collect rental-market data alongside mortgage data: rents, deposits, vacancy, lease length, landlord type, utilities, arrears, eviction, and the cost of moving. Small landlords are important to that supply because their production model differs from a completed development for sale.

Small landlords can add rooms or floors to land they already hold and spread construction spending over time. Knowledge of local demand and nearby management may reduce some costs. Finance, inspection, and tenant protection may remain limited. Compare their actual costs and returns with other providers rather than assume that all can accept a lower return. Those advantages need to be assessed together with the risks that upgrading creates for existing tenants.

Small landlords as housing producers

Incremental rental production can lower entry costs, while improvements may expose current tenants to higher rents or displacement. Management can be on site or nearby. Savings, remittances, or a rotating savings group may finance the extra room, with rent repaying the investment. These advantages vary by ownership, service access, and building condition. Formal policy can improve this production system without converting every landlord into a corporation. Cities can publish pre-approved designs for room additions and backyard units; utilities can offer bulk or simplified connections; lenders can make small construction loans against staged inspections; building departments can focus on structure, fire, sanitation, drainage and electrical safety. Suppose a public program pays for sanitation, drainage and structural improvement in a rental settlement. The asset becomes more valuable. If supply remains fixed and tenants have little bargaining power, the landlord can capture part of the subsidy through higher rent. Tenant protection and additional supply must therefore accompany upgrading. The household cost of losing a location becomes clearer in evidence on relocation.

Relocation costs in India

The Ahmedabad relocation experiment shows why better buildings do not necessarily compensate for lost livelihoods and networks. A randomised housing lottery in Ahmedabad

offered women working in the informal sector the opportunity to move from slum housing to improved subsidised homes on the urban periphery. Fourteen years after allocation, winners reported better physical housing but no detected improvement in family income or human capital. They also reported greater isolation from family and caste networks and reduced informal insurance. Thirty-four percent never moved into the subsidised housing; a further 32 percent of winners moved in and later left. The design estimates the effect of the housing offer in this applicant population. [3] The Ahmedabad exits show that improved physical housing and subsidised rent were insufficient to retain many winners. The study links the result to geographic isolation and weaker networks; it does not separately identify every effect of commuting, customers, schools, or childcare. Appraise those location-dependent costs before offering relocation, and interview households that decline or leave. [3] Housing policy often prices land and construction while treating social networks as free. For poor households, networks can substitute for formal insurance, childcare, job search and credit. Their value can be large precisely because formal markets are weak. Where a site remains unsafe, relocation still requires an appraisal of those losses and the alternatives.

Some locations may remain unsafe after feasible risk-reduction measures are considered, including floodways, unstable slopes, and contaminated sites. Railway-reserve status alone does not establish that displacement is necessary. Where relocation is justified, compare alternatives, follow applicable due process, consult affected residents, and fund replacement housing and livelihood support. Assess job access, travel cost, school continuity, network disruption, and enforceable occupancy rights before moving households. Housing choice can also be restricted by discrimination even when a safe and affordable dwelling exists.

Discrimination in housing access

Access to housing depends on social acceptance and allocation practices as well as the legal tenure offered. Landlords screen tenants. Owners form associations. Brokers steer households. Lenders price risk. Planning rules sort by income through minimum lot sizes, parking requirements and limits on apartments. In India, caste and religion can affect access to rental and ownership markets. In South Africa, the geography of apartheid still shapes distance to economic opportunity. In both settings, income alone cannot explain residential sorting. Financial support may be insufficient when a household faces discriminatory screening or cannot find a suitable landlord. A formal title may improve security without improving job access. Social housing in accessible areas may widen choice, but its actual allocation, services, and treatment of neighbouring residents must be assessed. Existing residents have established local channels for influencing housing decisions; future residents may have few. Developers can organise around projects, though their preferred product may be beyond low-income households' budgets. Include renters, younger households, employers, transit providers, and public finance officials in assessing added capacity and its costs. Do not assume that any group has a single position. Rental policy must therefore address the terms on which households can actually obtain accommodation.

2 Compare the effects of rental and ownership policies

Rental policy in India

India's rental policy response needs to be evaluated through occupancy, household costs, and responsibility for operating gaps. The COVID-19 migration shock exposed a gap in India's ownership-heavy housing system. The Affordable Rental Housing Complexes framework was designed for urban migrants and poor households who often choose informal settlements or peripheral areas to keep housing costs near work. The program uses two channels: convert existing government-funded vacant housing into rental complexes, and allow public or private entities to build, operate and maintain rental housing on their own land. Both channels require years of management, maintenance, and reletting. PMAY-U 2.0 has carried rental forward as a distinct Affordable Rental Housing vertical. As of April 15, 2026, the broader PMAY-U 2.0 program had sanctioned more than 1.361 million homes, while the ministry continued to promote affordable rental for migrants, homeless people, industrial workers, working women, construction workers and other vulnerable groups. The ministry also emphasized locating partnership housing near mass transit and within cities. Test whether target rents can cover management, maintenance, vacancy, utilities, debt, and replacement reserves after the capital subsidy. If they cannot, the programme needs an explicit operating contribution, a larger capital contribution, lower costs, or a different financing structure. Record the gap before occupation and assign responsibility for funding it. South Africa's social-rental model provides a different institutional approach to accessible rental housing.

Rental housing and spatial inclusion in South Africa

Social rental requires location standards and capable operators whose performance is verified after occupation. Social rental projects are targeted to restructuring zones and well-serviced areas of economic activity. The state uses a capital grant to make the project viable for lower-income tenants, while accredited social housing institutions and other delivery agents manage the stock. SHRA's undated About Investment page reports an average construction cost of R426,000 per unit, including land and bulk services (about US$23,700, my illustrative conversion at R17.96/US$, a cross-rate calculated from the Central Bank of Kenya's 2025 annual averages). That page also displays an older grant amount. SHRA's capital guidance and 2025/26 performance plan instead specify a standard grant of R426,308 (about US$23,700 at the same reference rate), with adjustments for the income mix; the plan dates this level to April 2023. These are different measures and vintages, not a current cost-to-subsidy ratio. Obtain a dated project budget and applicable grant decision before appraisal. [6,7,211] Social housing institutions need the staff, systems, and revenue to collect rent, resolve arrears, allocate units, maintain buildings and common areas, manage security, and meet income-mix obligations. SHRA's accreditation and regulation link grant eligibility to institutional requirements. Verify operating performance alongside accreditation status. Its income and payment requirements also define which households need a different form of support.

Eligibility requires the ability to pay rent. The poorest households may still fall below the social-housing market. Social rental can bridge a large gap between informal renting and market

housing, while deeper poverty requires income support, public housing, shelter assistance or other instruments. For households already renting, protection from sudden shocks must be balanced with maintenance and future supply.

Rent regulation and housing supply

Tenant protection should account for residents' investment in place and landlords' options to withdraw or convert housing. They learn a commute, build local networks, place children in schools and find nearby work. A sudden rent increase can destroy those investments. Rent regulation therefore answers a real insurance problem. A San Francisco quasi-experiment found that rent control increased covered tenants' probability of remaining at their address by nearly 20 percent relative to the comparison group. Affected landlords reduced their rental supply by about 15 percent; this is not a 15 percent fall in the city's entire rental stock. The authors' December 2017 working-paper model estimated a 5.1 percent increase in citywide market rents from the supply response. That model-based figure is from an earlier version, not the 2019 published article. Local treatment estimates and model-based citywide results are different kinds of evidence. [9,109] Rent rules change returns to supplying, maintaining, or converting property. Where owners can withdraw homes from rental use, controls may reduce the regulated stock. Outcomes depend on exemptions, enforcement, conversion rules, market demand, and new construction. The San Francisco result therefore requires local testing before its magnitude or policy implications are transferred. That trade-off calls for comparing instruments that insure tenants without making continued rental provision unviable.

Compare ways to protect tenants from rent and income shocks while financing maintenance and supply. Options include longer leases, predictable indexation, eviction due process, targeted allowances, and temporary hardship support. Their costs and effects differ by market. Where stronger rent controls are adopted, monitor conversions, repairs, vacancy, new construction, and rents outside the regulated stock. The San Francisco result applies to its particular rules and landlords' options. It supports separating protection against income and rent shocks from the return needed to maintain and add rental homes. Before adapting the instrument, establish whether owners can convert or withdraw units, which tenants receive protection, and what options remain for newcomers. [109] Ownership provides another form of security, but suits a different pattern of income, risk, and mobility.

When ownership suits household needs

Ownership can be suitable for households with stable income and a long horizon. Mortgage amortization turns payments into equity. A paid-off home can reduce housing costs in old age. The household gains control over alterations and can use the property for a business or rental room. In markets with secure title and liquid resale, the asset can store wealth. A household can become overleveraged. A peripheral home can leave workers far from new employment opportunities. Climate risk can destroy an uninsured asset. A weak condominium association can leave common infrastructure unfunded. Title can be secure while resale demand is weak. Homeownership concentrates household wealth in one neighborhood and one local economy. Ownership subsidies should therefore be tested against a rental counterfactual.

Compare the present value of down payment, debt service, maintenance, taxes, transaction costs, insurance and transport with the cost of renting an equivalent location. Include the value of mobility. A mortgage payment alone is an incomplete comparison. Households unable to finance a finished home may instead build ownership gradually.

Incremental ownership

Incremental tenure can match how households earn and build if rights and essential services are secured at each stage. A household may acquire a plot, occupy a core unit, add rooms over time, rent part of the structure, formalize tenure gradually and connect to services in stages. This pattern is common because household income is volatile and long-term mortgage credit is scarce. Incremental building allows households to stage expenditure and pause after an income shock. A completed rental room may help finance a later addition. Compare those benefits with slower completion, exposure to weather during unfinished work, technical risks, and the cost of repeated small purchases. Incremental ownership is an option to assess against feasible alternatives, rather than an efficient choice in every case. Policy can improve the path through serviced plots, expandable core designs, clear building envelopes, staged title or occupancy rights, small construction loans and inspections at critical structural points. The aim is to make the informal production logic safer and more legible without imposing the full cost of a finished formal product on day one. Public investment in those areas must still distinguish owners from tenants and other occupants.

Distribution of upgrading benefits

Upgrading benefits need to be tracked by tenure group so that public support does not finance tenant displacement. An informal settlement is rarely a community of identical owner-occupiers. It contains tenants, structure owners, land claimants, shopkeepers, subtenants and brokers. Public investment changes the value of each claim. Drainage, roads, water and tenure security can raise rents and land prices. In-situ upgrading can preserve location and networks while increasing rents or land values. A title issued only to a structure owner may change the tenant's negotiating position. Owner-only compensation can exclude renters and livelihood users. Record those claims and specify funded protections before works or tenure changes begin. Every upgrading project should therefore begin with a tenure census rather than a parcel map alone: who occupies, who owns the structure, who claims the land, who pays rent, who runs a business, and who would receive compensation? The project should then state explicitly how much of the public investment may be capitalized into private rents and what will protect tenants. The feasible tools include temporary rent stabilization during works, additional rental supply, tenant relocation rights, return rights, compensation for documented tenants, and community land or lease structures where appropriate. The objective is to prevent the public subsidy from financing displacement. Publicly owned land can help retain affordable access if its use remains subject to enforceable obligations.

Public land and housing services

Public land can help retain affordable housing in accessible locations when long-term obligations are enforceable. Sale may increase immediate receipts; a discounted transfer creates a subsidy that should be recorded. A long lease can preserve ownership and require affordable housing over time. Compare the alternatives with their monitoring, renewal, and enforcement costs. The contract should state the subsidy and its conditions. Price the land at market value, record the concession, specify the affordable-rent or income mix, require delivery milestones and maintenance standards, and define what happens at expiry. This converts a one-time land transfer into a measurable purchase of housing services. Transit, rezoning, and infrastructure may increase public-land value. Selling before those investments can transfer part of the subsequent gain to buyers, depending on what prices already anticipate. Compare early sale with retention, leasing, or reservation for rental and mixed-income housing, including the public cost of each option. Choosing those obligations requires understanding who gains now and who bears costs over time.

3 Support access and continued occupation

Interests in tenure reform

Actor	Possible interests	Potential risks or gains	Decision or influence	Possible agreement
Tenant	Low rent, location, security	Rent shock, eviction, moving cost	Exit, protest, vote	Supply + due process + targeted support
Small landlord	Yield, asset value, simple rules	Compliance cost, credit constraint	Can stay informal / stop investing	Legal pathway + finance + minimum safety
Institutional landlord	Stable long-term yield	Regulatory and vacancy risk	Can shift capital	Predictable rules + targeted capital subsidy
Homeowner	Asset value, neighborhood control	Price decline / congestion fears	Planning hearings, elections	Infrastructure + citywide rules + credible design

Actor	Possible interests	Potential risks or gains	Decision or influence	Possible agreement
Developer	Viable margin and absorption	Land, approval, sales/rent risk	Can change segment/location	Predictable rights + competition
Housing agency	Visible delivery, budget	O&M and project failure	Project selection	Tenure-neutral outcome metrics
Informal broker	Fees for access or claims assistance	Opaque fees or allocation; verify locally	Local information and networks	Transparent allocation + enforcement + local legitimacy
Future resident	Access to opportunity	Excluded by today's scarcity	Little present voice	Represent through citywide housing targets

Tenure reforms distribute benefits and costs over different periods. New rental supply may take years, while a change to tenant protection can affect next month's payment. Density may add service demand before revenue arrives. A landlord may face inspection costs before finance becomes available. Show these sequences in the implementation and funding plan. Coordinate obligations with the support needed to meet them. A landlord formalisation programme may need finance and a lawful approval route alongside inspections. Tenants need accessible due process before rules change. Municipalities need service funding as density increases, and social housing institutions need finance tied to continued management. These are proposed implementation conditions, to be costed and tested locally. Policy documents should describe who will be able to enter and remain after reform. Terms such as clearance or neighbourhood protection need explicit tests for displacement, service quality, and access by households currently excluded. Test who can enter protected neighbourhoods, not only who remains. Include migrants, young households and applicants facing discrimination. The municipality's recurring service costs are part of that distribution and need their own financing arrangement.

Municipal finance and rental housing

A municipality bears the service cost of new residents: streets, drainage, waste, water interfaces, public space and local administration. The tax gain may accrue partly to national government. If local revenue does not rise with population or density, resistance to new rental housing can be fiscally rational. The metropolitan economy gains when workers live closer

to jobs, while the host jurisdiction bears local service costs. National or metropolitan transfers can compensate the locality and reduce exclusionary incentives. Property taxation can also turn added floor space into recurring local revenue. The proposed mechanism is that added rental homes increase service demand, predictable local revenue finances the required capacity, and reliable services may reduce opposition to further housing. Test each step through occupancy, service costs, receipts, and recorded objections. Revenue alone cannot establish public support or a supply response. The tenure agreement must also specify who funds and performs maintenance inside the development.

Maintenance responsibilities under different tenures

Maintenance responsibilities differ across ownership, rental, and common property and must be funded accordingly. Rental concentrates maintenance with the landlord. Social rental adds a public interest in preserving the asset. Each tenure therefore needs an operating model before construction begins. For a rental project, calculate target rent, collection rate, vacancy, management cost, routine maintenance, major-repair reserve, utilities, insurance and debt service. Stress-test arrears and inflation. If the affordable rent cannot cover sustainable operations after the capital subsidy, write the operating subsidy into the fiscal plan. Unfunded repairs should be reported as a financing gap. For condominiums and subsidized ownership, the equivalent risk is the owners' association. Elevators, roofs, pumps, facades and common pipes eventually fail. A low-income owner who can afford the mortgage may not afford a large special assessment. Reserve funds and simple, enforceable governance are part of affordability. Digital records can help administer those responsibilities if they also protect occupants whose tenure is informal.

Digital tenure records and resident protection

Digital tenure systems should improve contracts and services without making vulnerable occupants easier to exclude. Governments may not know who occupies a unit. Tenants may not know comparable rents. Lenders may not see a small landlord's cash flow. Owners may lack proof of regular payments. Digital records may help address these gaps when households can access them and the responsible agency can correct errors. Digital leases can record rent, duration, and obligations. With suitable permission, payment histories may inform underwriting. Geospatial records can support inspection and service planning, and complaint systems can track repairs. AI may help classify requests or flag patterns for review, but estimated rents and suspected discrimination require validation. Provide assisted and offline routes, and do not treat missing digital records as proof that a tenancy is invalid. Data rights therefore belong inside housing policy. Purpose limitation, correction rights, explainable decisions, audit trails and separation between service enrollment and punitive enforcement can determine whether digitization brings informal households into the city or makes them easier to remove. The policy assessment should test those protections alongside changes in supply and household cost.

Tenure policy assessment

Start tenure reform with usable contracts, accessible disputes procedures, and measures of resident outcomes. Do not require documentation that makes most tenants ineligible. Budget any public insurance explicitly and distinguish temporary income support from permanent rent subsidy. Rent adjudication, building inspection and benefit administration must work together without requiring tenants to risk retaliation. Track tenant retention, new entrants, evictions, repair quality, rental-stock changes and total household cost by income group. The comparison below helps select the arrangement suited to the intended household.

Gate	Question	Metric / decision
1 Income risk	How volatile is household income?	Monthly variance; shock buffer; informal/formal earnings
2 Duration	How long is the household likely to stay?	Expected years in city/neighborhood
3 Mobility value	What does moving preserve?	Job turnover, migration, household change
4 Entry cost	Can the household fund deposit/equity?	Cash required / liquid savings
5 Location	Does tenure preserve access to opportunity?	Jobs within 45/60 min; fare + time
6 Asset risk	Can the household bear ownership risk?	Title, hazard, resale liquidity, maintenance
7 Supply response	Who can add units?	Small landlord, developer, social landlord, household
8 Tenant power	Can renters enforce basic rights?	Lease, deposit, eviction, complaint mechanisms
9 Incidence	Who captures subsidy?	Tenant/owner/landlord/ developer/landowner shares
10 O&M	Who funds 20+ years of upkeep?	Collection, reserve, management, operating subsidy
11 Inclusion	Who is screened out?	Caste/race/class/gender/migrant outcomes
12 Fiscal fit	Does the city gain resources with residents?	Local revenue vs marginal service cost

Gate	Question	Metric / decision
13 Digital power	What new visibility is created?	Purpose, appeal, audit, enforcement boundaries
14 Outcome	Did life become more affordable?	(Shelter + utilities + transport + maintenance) / income; exit; job access

Investment criteria

Choose tenure by minimizing the household's total risk-adjusted cost of shelter and access. Compare ownership, market rental, social rental and incremental tenure at the same level of job access. Count entry cash, monthly payments, utilities, transport, maintenance, moving costs and asset risk. Then test subsidy incidence and the long-run operating model. If the program creates an affordable unit that the target household cannot afford to occupy, maintain or reach work from, change the tenure or location before increasing the subsidy. The available evidence supports this household comparison while leaving important local parameters to be measured.

Limits of the evidence

Evidence on tenure and relocation must be applied using local incomes, rents, access costs, and maintenance obligations. The quantitative evidence in this chapter comes from primary government sources, World Bank material and peer-reviewed or NBER research. The Ahmedabad lottery provides causal evidence on one relocation program; its magnitudes should not be mechanically transferred to other cities. The San Francisco rent-control study identifies a credible supply response in one regulatory setting; the mechanism should be tested against local landlord options and supply constraints before policy transfer. The Tenure Portfolio Test, the political-economy map and the risk-allocation framing are judgments that synthesize the evidence. They are intended as decision tools. A city should populate them with household income panels, rent and vacancy data, job-access measures, landlord economics, local fiscal data and actual maintenance costs. People without a usable home require an additional response beyond choosing between ordinary rental and ownership products.

Homelessness prevention and supported housing

Households without a usable home need a distinct response. Count rough sleeping, temporary accommodation, and insecure stays with other households separately, using outreach and service records as well as surveys. Protect confidential information about domestic violence and health. Ask why the last tenancy ended and whether arrears assistance, legal help, a deposit, accessible accommodation, or continuing support would address the cause. Canada's At Home/Chez Soi trial tested Housing First with support for people experiencing homelessness and mental illness in five cities. Over two years, participants receiving the intervention spent an

average 73% of their time stably housed, compared with 32% under usual services. This supports rapid access to housing combined with appropriate support for that population; it is not evidence that a dwelling alone resolves every need or that the same effect will occur elsewhere. The report also identifies participants whose additional needs were not adequately met. [210] Adapting supported housing requires securing the accommodation and continuing services together.

A local adaptation needs actual homes, willing providers, continuing rental assistance where required, and accessible support services. Begin with an outreach-to-placement pathway and name who will fund support after initial placement. Compare sustained housing, returns to homelessness, household choice, and total programme cost with existing services. Do not require a person to complete a digital application or demonstrate mortgage capacity before receiving emergency assistance. Clinical support arrangements require qualified health services. The field assessment should establish which households need that support and what the other tenure options can deliver.

Prepare the local decision

Compare tenure support through entry costs, involuntary moves, repairs, and income left after housing and travel. Record who owns the land, owns the structure, occupies the dwelling, pays rent, and has an enforceable right to remain. Collect deposits, notice periods, repair obligations, arrears, moves, and household preferences under realistic costs. Include households who refused or left a subsidised home. The Ahmedabad relocation evidence above makes livelihood access and support networks necessary parts of the interview. Speak separately with tenants, resident owners, absentee landlords, employers, women's groups, and dispute-resolution providers. Ask the responsible agency to identify the legal route for enforcing each proposed tenancy or occupancy right. Compare rental assistance, incremental ownership, and full ownership for different household cash flows and expected lengths of stay; do not assign a tenure solely from income. Offer assisted and offline registration with a receipt the household can retain. A rental register should record obligations and remedies without making every occupant's address public. Test whether households can find a suitable home, obtain repairs, and challenge an incorrect record. Review entry costs, involuntary moves, repair completion, and residual income after housing and travel; title issuance alone cannot establish security. Chapter 6 examines how financing can match those household circumstances without creating unsustainable debt.

Tenant protection and the revenue needed to maintain rental homes should be designed together. Clear protected tenancies, enforceable rights, and accessible administration can insure occupants against disruption. A copied rent ceiling can instead exclude newcomers or induce conversion out of rental use when maintenance revenue and supply incentives are ignored. The test must cover sitting tenants, future tenants, and the retained rental stock. At national scale, quantify the fiscal cost of targeted support and monitor conversions, repairs and new supply outside the protected stock.

Sources and evidence notes

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6. Housing finance systems

Choose a housing-finance intervention only after identifying whether the binding constraint is income, deposit, loan terms, lender capacity, or suitable supply. Longer funding can improve affordability, but it cannot make every household creditworthy or create serviced homes by itself. Compare instruments by the additional households reached and the risks transferred to borrowers, lenders, and government. Start with the relationship between the housing payment and the timing of household income.

Housing finance spreads a large upfront cost over time. A bank, housing finance company, cooperative, or microfinance institution advances money and collects repayments. This can allow a household to buy or improve a home before saving its full price, provided the payments and other housing costs remain affordable. Consider an illustrative US$10,000 loan. At 14 percent for 12 years, the monthly payment is about US$144. At 8 percent for 20 years, it falls to about US$84. A household able to devote US$90 a month to debt can borrow roughly US$6,260 at the first terms and US$10,760 at the second. Better tenor and cheaper capital expand borrowing capacity by about 72 percent in this example. If the cheapest adequate, well-located home costs US$20,000, a household with US$90 monthly debt capacity still has a substantial funding gap in this example. Extending the term may reduce payments while increasing the period exposed to income, interest-rate, and property risks. Finance can address payment timing and liquidity; the remaining gap needs lower costs, savings, income, or subsidy. Calculate the cheapest adequate housing product first, then ask which financing friction keeps the household from reaching it. Select the financing instrument after identifying that constraint. The first step is to distinguish the financing barriers that different instruments are designed to address.

1 Identify the financing constraint

Housing costs and the timing of income

Five constraints on housing finance

Housing-finance policy should separate low income, missing savings, short funding terms, information gaps, and credit risk. A resource gap exists when the household cannot afford the efficient cost of adequate shelter. A liquidity gap exists when lifetime affordability is adequate and the deposit or construction stage is too large. A term gap exists when lenders fund themselves short and cannot safely lend long. An information and credit-risk gap exists when lenders cannot observe informal income, collateral or repayment capacity at reasonable cost. A currency and rate-risk gap appears when long-term funds exist only in foreign currency or at rates households cannot absorb. Match instruments to the diagnosed constraint. Capital grants can address a resource gap. Savings matches, smaller deposits, or staged loans may address liquidity. Refinance institutions and suitable long-term funding may address maturity mismatch. Guarantees can share defined credit losses, while better underwriting can reduce uncertainty. Local-currency funding can reduce currency mismatch. None removes the need for a housing

product and repayment schedule the household can sustain. Using an instrument against the wrong constraint can increase costs without improving access.

A credit intervention should follow a diagnosis of the specific barrier to sustainable housing access. A government that creates a refinance company for households whose main problem is a down payment may lengthen tenor without creating a borrower. Diagnosis should precede instrument. Mortgage-market size alone does not provide that diagnosis.

Mortgage markets and household access

A large mortgage market can reflect high incomes, secure property rights, low inflation, deep capital markets and abundant housing supply. A small mortgage market can reflect low incomes, informal work, expensive homes, weak title, shallow long-term funding or all of these at once. Raising mortgage-to-GDP is therefore a weak policy target on its own. The Central Bank of Kenya reported 30,762 residential mortgages at end-2025, worth KES 307.2 billion (about US$2.38 billion), with an average loan of KES 10.0 million (about US$77,300). The US dollar figures are my conversions at KES 129.3/US$, the bank's 2025 annual average. Average mortgage rates were 13.5 percent and average maturity 11.5 years. Nine institutions held 90.6 percent of lending. Non-performing mortgages were 16.3 percent of the gross mortgage portfolio. These figures describe the surveyed formal mortgage market. [1] Those figures describe a formal mortgage market that remains small relative to the number of urban households. A larger mortgage book can help middle-income borrowers and developers. Reaching lower incomes requires smaller loans, lower transaction costs, better information about irregular earnings, and housing products whose prices fit those earnings. Kenya's refinancing institution illustrates an intervention directed at lenders' funding cost and maturity.

2 Choose instruments that fit lenders and households

Mortgage refinancing in Kenya

Mortgage refinancing can address long-term funding constraints, but administrative growth figures do not isolate its effect. The Kenya Mortgage Refinance Company (KMRC) was established to address funding cost and maturity constraints. It raises long-term funds and refinances primary lenders, including banks and savings and credit cooperative organisations (SACCOs). Stable funding can help those lenders offer longer terms or fixed-rate periods. Verify what changes in the retail loan rather than assume the funding benefit reaches borrowers. The World Bank's February 2026 appraisal reports that by December 2025 KMRC had refinanced about KES 25.3 billion (about US$196 million, my conversion at the Central Bank of Kenya's 2025 average of KES 129.3/US$) across 5,148 mortgages. It reports average repayment terms rising from 8.3 years in 2021 to 12.3 years in 2025, nearly 48 percent of refinanced mortgages going to women, and SACCOs' portfolio share rising from 7 percent in 2022 to 15.2 percent at end-2025. These administrative comparisons do not isolate the effect of refinancing. [2] The remaining borrower and housing-supply constraints must be assessed separately.

Refinancing alone may leave informal earners and lower-income households unable to obtain a suitable loan or home. The same appraisal reports that affordability remained a constraint for many low- and middle-income households and that informal earners needed products beyond conventional refinancing. It describes the Kenya Mortgage Guarantee Trust, initially capitalised at EUR 4 million (about US$4.52 million, my indicative conversion using the euro-dollar cross-rate from the Central Bank of Kenya's 2025 annual averages). The intended function is to share credit losses on lending to underserved borrowers. Its additional reach and realised losses need separate evaluation. [2,201] The publicly disclosed appraisal dated 27 February 2026 proposes US$210 million for mortgage refinancing, US$125 million for land administration reform, and US$135 million for sustainable financing instruments, alongside technical assistance. These amounts describe the appraised design. The document lists an expected approval date of 20 March 2026; it does not establish a signing date. Linking finance with land administration may address complementary constraints, but subsequent approvals, implementation, and housing outcomes require separate records. [2] Tanzania's experience allows comparison with a programme that also supported housing microfinance.

Wholesale housing finance in Tanzania

Tanzania's wholesale-finance programme demonstrates institutional development and lending reach rather than a complete measure of household welfare. The IDA-supported Housing Finance Project established the Tanzania Mortgage Refinance Company and a Housing Microfinance Fund. The first supplied medium- and long-term liquidity to mortgage lenders; the second supported smaller incremental housing loans. The results below describe market changes during the programme, rather than its isolated causal effect. World Bank reporting describes a roughly sevenfold nominal expansion in Tanzania's mortgage market, from TZS 77 billion in 2011 to TZS 508 billion in 2018. These equal about US$49.5 million and US$224 million at the respective annual-average rates of TZS 1,557.4/US$ and TZS 2,263.8/US$ reported by the Bank of Tanzania, my conversions. The number of mortgage providers rose from four in 2010 to 33 by project close in 2019. Rates fell from above 21 percent to about 15 percent, alongside favourable macroeconomic changes, and maximum terms extended from roughly 10–15 to 25–30 years. The project reported about 5,000 mortgages and 2,000 housing microfinance loans; women, including joint owners, represented about 34 percent of beneficiaries. [3,113,201] Its funding model must be assessed alongside the distinct barriers addressed by each lending product.

Wholesale funding and microfinance serve different constraints and require separate performance tests. IFC later invested in its equity and bonds. This demonstrates access to funding beyond the original credit line. Continued operation still depends on the cost of funds, loan quality, investor demand, and sufficient revenues. Tanzania used separate products for borrowers with different needs. Conventional mortgages require sufficient verifiable income and capacity for a long repayment obligation. Housing microfinance can support staged improvements with smaller loans. Evaluate the demand, cost, and repayment burden of each product; creating two funding facilities does not itself establish adequate coverage. These are programme-reported results, not a causal estimate of poverty reduction. Kenya's experience shows why the same funding channel may still leave informal earners unable to borrow. That limit does not mean refinancing failed at its liquidity objective. It means that income verification,

repayment capacity and the cost of a suitable home require separate responses. Evaluate each product against the barrier it is intended to remove. [113] (See sources 1–3.) A regional funding institution adds questions about whether local legal and lending conditions are sufficiently compatible.

Regional housing finance in West Africa

A common currency can support regional refinancing without making collateral enforcement or borrower risk identical across countries. That creates an opportunity a small national mortgage market cannot easily create alone. The regional mortgage refinancing company, CRRH-UEMOA, pools demand for long-term funding across lenders and countries. A US$155 million IDA-supported regional project enabled more than 12,000 people to become homeowners and helped CRRH-UEMOA raise an additional US$385 million through regional and international bonds. World Bank reporting says 86 percent of beneficiaries were first-time mortgage borrowers and three-quarters were low- or middle-income. Almost 100 banks and microfinance institutions received support. Regional refinancing can pool demand for funding and give smaller lenders access to longer terms. It also requires consistent loan information, servicing, and credit standards across participating institutions. Property and enforcement law may differ within the region. Test those differences at the loan level rather than assume a shared currency produces equivalent collateral protection. Informal-income lending raises a different information problem even in a deeper financial market.

Lending to informal earners in India

India's affordable-housing lenders illustrate how field assessment can replace some missing income documentation. Traditional salary-based underwriting misses households that may have stable cash flow without formal payroll documents. National Housing Bank data report outstanding individual housing loans of INR 36.07 trillion (36.07 lakh crore) in March 2025, about US$426.5 billion at INR 84.576/US$, the Reserve Bank of India's 2024/25 annual average, my conversion. Housing finance companies held roughly 19 percent. NHB reports that about 45 percent of their housing-loan portfolios consisted of loans up to INR 2.5 million (25 lakh; about US$29,600 at the same rate), compared with smaller shares for public and private banks. Loan size alone does not establish borrower poverty or informal-sector status. [5,201] Earlier World Bank support to NHB demonstrated the model. A low-income housing finance project refinanced about 15,000 mortgages by 2017; 82 percent went to borrowers working in the informal sector. The average beneficiary household earned about US$2,753 a year. The project helped a new class of affordable housing finance companies use field-based income assessment and smaller loans. That assessment must still establish capacity to repay rather than confuse absent records with adequate income.

Informal-income assessment can address missing documentation where a household has sufficient repayment capacity. Field visits and consented records may help establish turnover, expenses, residence, payments, inventory, and household cash flow. Such checks have costs and error risks. Compare them with simpler methods, and separate difficulty documenting

income from income too low or volatile for the proposed loan. An interest subsidy can then be assessed against the remaining affordability gap and the supply of suitable homes.

Credit subsidies and housing supply in India

India's interest-subsidy rules define an eligible market but do not make all households within it able to buy. Under the cited PMAY-U 2.0 Interest Subsidy Scheme rules, eligible households may earn up to INR 900,000 a year (9 lakh; about US$10,640), borrow up to INR 2.5 million (25 lakh; about US$29,600), and buy homes priced up to INR 3.5 million (35 lakh; about US$41,400). The subsidy is 4 percent on the first INR 800,000 (8 lakh; about US$9,460) for up to 12 years. The maximum actual release is INR 180,000 (1.80 lakh; about US$2,130), in five annual instalments. Dollar figures are my reference conversions at the RBI's 2024/25 average of INR 84.576/US$, not current exchange-rate quotations. Apply the operative eligibility and disbursement rules to each application. [7,201] The policy lowers debt service for a borrower already close to the ownership threshold. Eligibility still depends on access to a lender, an eligible property and enough residual income to service the rest of the loan. The poorest households remain outside the instrument when the property price is far above their payment capacity. The distribution of the subsidy also depends on whether housing supply expands in response.

Supply conditions influence how much credit support reaches borrowers and how much increases prices. Global evidence shows that cheaper credit and interest subsidies can capitalize into house prices when housing supply is constrained. Swedish evidence found near-full capitalization of subsidized financing into asset values. U.S. evidence using conforming loan limits also finds that easier credit raises prices at the margin. The effect strengthens where new supply responds slowly. When designing buyer support, test whether the relevant submarket can add suitable homes and how quickly. With little supply response, some subsidy may increase sellers' receipts through higher prices. Publish the expected incidence, track prices and completions, and compare support for rental or incremental housing where ownership remains unaffordable. South Africa's first-buyer programme illustrates the need to follow support through to completed purchases and continuing affordability.

Housing finance for middle-income households in South Africa

First-buyer assistance should be evaluated through additional sustainable access, not approvals alone. South Africa's First Home Finance targets eligible first-time buyers earning R3,501–R22,000 a month (about US$195–US$1,225, my reference conversion at R17.96/US$, calculated from CBK's 2025 annual cross-rates). The Department of Human Settlements' 15 September 2026 release reports close to R2 billion disbursed (about US$111 million at that rate) and more than 32,000 subsidy approvals since 2019. Approvals and completed purchases are different outcomes. The release prints the private-investment figure as R10.6 without a complete unit; that number cannot support a financing ratio until its unit is clarified. [11,12,201] The scheme can reduce a deposit or mortgage balance and can support eligible construction on serviced sites through approved builders. It is most useful where the contribution closes a defined affordability or liquidity gap. Test whether the remaining payment, upkeep, and transport costs fit the household budget before treating approval as access to sustainable ownership. Measure

whether the subsidy enables an additional household to buy or build, or mainly reduces debt for a household that would have proceeded without it. Track approval-to-purchase conversion, a credible estimate of access without support, arrears, resale, and the incomes of recipients. For households building in stages, a smaller incremental loan may be more appropriate than a mortgage.

Microfinance for incremental housing

Housing microfinance can match staged improvements, but financing reach does not establish welfare impact. Low-income households often build in stages: foundation, one room, roof, second room, rental unit, services. Their financing need may be US$500 or US$3,000 at a time. Housing microfinance offers smaller, shorter loans for staged building or improvement. Underwriting may use household cash flow and business information, with security other than a registered mortgage. Health, resilience, or rental-income benefits depend on the work completed and its quality. Verify the improvement and repayment burden rather than infer the benefit from disbursement. World Bank programme reporting records a Tanzanian commercial housing-microfinance market of TZS 30 billion within four years of the fund's development (about US$13.3 million, my illustrative conversion at the Bank of Tanzania's 2018 annual average of TZS 2,263.8/US$). This is a reported market outcome, not an isolated causal estimate; the reference conversion does not establish the exact observation date. [113,201] Separately, IFC reported US$44.1 million channelled to 41,505 families in the Kyrgyz Republic for housing construction and improvement over a decade, and US$15.7 million to 15,434 families for energy-efficiency upgrades. These totals describe financing reach, not measured welfare impacts. Its shorter repayment period creates a distinct affordability risk.

Short tenor creates its own affordability risk. A US$2,000 loan paid over two years can have a higher monthly burden than a much larger mortgage spread over twenty years. The product should match the stage of construction and the household cash-flow cycle. Repeat lending should follow verified improvements and repayment, rather than becoming permanent short-term debt. Guarantees address lender risk rather than this mismatch between payment timing and household capacity.

Guarantees and risk allocation

Partial guarantees should change lender behaviour while retaining incentives to assess and manage credit risk. The public sector agrees to cover a defined share of losses; the lender retains enough risk to preserve underwriting discipline. The instrument works best when the problem is risk perception, thin data or the cost of entering a new market. It performs poorly when the borrower simply lacks payment capacity. Guaranteeing a loan that a household cannot service shifts an expected loss onto the public balance sheet. Guarantees also create incentives. Lenders may select borrowers they would have financed anyway, a problem known as low additionality. They may relax underwriting if the public share is too high. Government should therefore price the guarantee, cap coverage, require lender risk retention, publish cohort performance and compare guaranteed borrowers with a credible non-guaranteed counterfactual. Kenya is moving in this direction with the Kenya Mortgage Guarantee Trust, designed to provide

partial default protection for lower-income and informal-sector borrowers. India also operates credit-risk guarantee arrangements for low-income housing. The decisive test is whether these tools expand access with acceptable realized losses. Longer-term investor funding requires a further set of protections and performance tests.

Secondary markets and investor confidence

Secondary-market instruments can diversify lender funding, but announced transactions do not establish lower borrower costs. Pension funds and insurers may demand long-duration assets, but a maturity match alone does not make housing loans suitable investments. Compare credit risk, liquidity, prepayment, inflation, and legal protections before expecting lower funding costs or longer fixed-rate loans. Investors need reliable loan files, underwriting, servicing, performance data, and enforceable rights. Recovery procedures must include borrower notice and due process. Weaknesses in these arrangements can raise funding costs or exclude loans from a pool. Standardisation should preserve material differences in risk rather than conceal them. In December 2025, IFC announced a commitment of up to INR 3 billion to residential mortgage-backed securities issued by Grihum Housing Finance, its first RMBS investment in India. This is about US$35.5 million at the RBI's 2024/25 average of INR 84.576/US$, my reference conversion. IFC reported more than 85,000 customers and an average loan of about INR 1 million (10 lakh; about US$11,800 at the same rate). The transaction seeks to diversify lender funding; the announcement does not establish a realised fall in borrower costs. [10,201] Their design must preserve responsibility for the quality of the loans being financed.

Securitisation may diversify funding and reduce its cost, but can separate loan origination from the ultimate bearing of credit risk. Require clear underwriting and servicing standards, lawful loan-level disclosure, suitable risk retention, and surveillance of arrears and prepayment. Evaluate the funding and borrower outcomes after the transaction. Public capital should therefore be allocated according to the additional access each instrument produces and the risks it retains.

Concessional finance and market development

Use concessional resources against an identified constraint and account for both expected cost and downside exposure. Compare these uses against the constraint they address, the additional housing service expected, and their full public cost. Private finance mobilised is one output rather than a complete measure of benefit. The West African project shows one form of leverage: US$155 million of IDA support helped CRRH-UEMOA mobilize US$385 million in additional bond finance. Tanzania used US$100 million of IDA financing to create institutions that later accessed private capital. Kenya is using public and development finance to support KMRC, a guarantee facility and the next generation of land and finance reform. Private finance raised can overstate a programme's additional benefit. A guarantee may support a large loan portfolio while exposing the public budget to losses; a refinance line may replace funds banks would otherwise have obtained. Measure additional sustainable housing access per unit of expected public cost, and report the downside exposure separately. Guarantees require that distinction even when no immediate cash payment appears in the budget.

For a simple proportional guarantee, expected credit loss is exposure at default multiplied by default probability, loss given default, and the guaranteed share. Estimate these over the same period, account for recoveries, and add administration and funding costs net of guarantee fees. Caps, first-loss layers, and correlated downturn losses require a portfolio model rather than that simple formula. For concessional credit, estimate the present value of the funding concession against an appropriate risk-adjusted benchmark. Report expected cost and stressed contingent exposure even when no payment is due immediately. Household credit also needs a supply of homes from producers able to finance construction through completion.

3 Allocate risks and protect borrowers

Finance for housing producers

Housing finance must cover viable production as well as the eventual purchase or occupation of homes. Affordable housing also needs finance before the household arrives: land acquisition, infrastructure, construction, rental portfolios, small-landlord improvement and working capital for material suppliers. A developer facing expensive construction finance may require a higher selling price even when mortgage rates are low. A small landlord may have tenants ready to rent an extra room and no access to a US$2,000 construction loan. A social rental provider may need twenty-year debt because the asset repays capital slowly through rent. This is why a housing finance system should map funds by stage: pre-development, construction, take-out, mortgage, incremental improvement, rental operations and major maintenance. A gap at any stage can stop the chain. Buyer finance cannot compensate for a developer who cannot reach completion. The allocation of that support changes the interests of lenders, developers, landowners, and households.

Interests in credit allocation

Actor	Possible benefits	Potential risks or gains	Decision or influence	Possible agreement
Borrower	Earlier access to a home	Debt burden, rate risk, asset risk	Take-up, default, political voice	Transparent terms + suitability + shock options
Bank / HFC	Interest income, customer relationship	Credit loss, funding mismatch	Underwriting, pricing	Refinance + risk sharing with lender retention
Mortgage refinance company	Scale and fee income	Market and liquidity risk	Eligibility standards	Standardizatio n + market funding path

Actor	Possible benefits	Potential risks or gains	Decision or influence	Possible agreement
Developer	More buyers, faster sales	Absorption and construction risk	Can shift segment/location	Supply-linked finance + transparent eligibility
Landowner	Higher bid capacity	May receive credit-driven price gains	Land supply, holdout	Value capture + abundant serviced supply
Investor / pension fund	Long-duration asset	Credit and duration risk	Pricing and allocation	Loan data + standards + local-currency instruments
Government	Visible homeownership, jobs	Fiscal subsidy, guarantees, bailout risk	Budget and regulation	Publish full fiscal cost and additionality
Renter	Possible future buyer; may face price spillover	Excluded from owner subsidy	Weak voice	Tenure-neutral policy + rental finance
Fintech / data platform	Fees, data and market power	Model risk, cyber risk	Access to data/model	Auditability + portability + fair-lending tests

Credit allocation should be judged by additional affordable neighbourhood choices rather than lending volume alone. More credit may raise land bids where sites are scarce, while new supply can change that effect. Record who receives support, who bears repayment and fiscal risk, and which groups participate in the decision process rather than assume a common industry or homeowner position. An interest subsidy provides a payment and a recorded loan, while planning or land-tax changes can involve concentrated near-term costs. These differences may affect support for the options. Compare their documented approval process, fiscal cost, and housing outcomes before attributing a programme choice to political convenience. Refinancing can align lenders' demand for stable funds, borrowers' need for manageable terms, and investors' demand for suitable assets. That alignment still requires housing prices and repayment burdens the target households can meet. Track whether institutions remain focused on the intended income groups as volumes increase, and whether subsidy costs or credit losses rise. Count the neighbourhood choices a loan makes affordable after transport, services and risk, rather than the loan volume alone. The household benefit also depends on what happens when repayment becomes difficult.

Arrears, foreclosure, and borrower protection

Default arrangements must protect contractual reliability while recognising the costs of household displacement. A system with no enforceable collateral raises lender risk and interest rates. A system that evicts rapidly after a small shock can create severe social harm and political backlash. The design problem is to separate temporary distress from persistent inability to pay. Responses to distress can include rescheduling, temporary forbearance, term extension, insurance, voluntary sale, or targeted assistance. Lenders need clear recovery rules, and borrowers need notice, due process, advice, and an opportunity to cure arrears where feasible. Establish lawful shock-response procedures in advance, assess their fiscal and credit effects, and provide for households that cannot regain sustainable payments. The World Bank's February 2026 appraisal cites seven KMRC-participating lenders reporting an average realised mortgage loss rate of about 2.6 percent after recoveries, despite higher non-performing-loan ratios. This sample illustrates that arrears and final lender losses differ. It does not establish losses for the whole market or for newly included borrowers. Strong collateral recovery can still impose severe costs on displaced households, which a lender-loss measure does not capture. [2] Preventing distress requires testing interest-rate, currency, and income risks before approval.

Interest-rate and currency risks

A low initial payment is useful only when the household can survive the path of rates and income. Variable-rate mortgages transfer interest-rate risk to borrowers. Foreign-currency mortgages transfer exchange-rate risk to households whose income is usually local. Long maturities amplify both when the reset is large. Kenya has moved modestly toward more fixed-rate lending: the Central Bank reported fixed-rate mortgages rising from 14.1 percent of loans in 2024 to 24.3 percent in 2025. Most mortgages still carried variable rates. Refinance companies can help lenders offer longer fixed periods when they obtain matching long-duration funding. Keep household debt in the currency of household income. Match lender asset duration with liabilities. Price fixed-rate options transparently. Stress-test payment-to-income ratios under rate and income shocks before approval. Digital underwriting can improve that assessment only if its data and decisions remain reliable and contestable.

AI and digital underwriting

Alternative data may improve credit assessment, but predictive performance does not establish fair or sustainable access. Digital records can supplement field assessment where they are relevant, reliable, and lawfully used. They should not make applicants with little digital activity automatically ineligible or substitute transaction volume for disposable income. IFC's account describes Vastu Housing Finance as combining field verification with a digital platform using more than 1,500 data points from credit bureaus, employment, and previous loans. It reports that around 80 percent of borrowers are self-employed and most are first-time homeowners. These are programme or company descriptions, rather than an independent comparison of model accuracy, discrimination, or long-term repayment. Test whether digital checks add reliable information beyond the field assessment before transferring the approach. [8] Fair-lending governance should include reason codes for rejection, correction rights, model

monitoring by group, limits on unrelated data, independent audit, cyber protection and a human appeal path. A model that improves prediction can still worsen access if it optimizes profit against groups with thin data. Implementation must match the product to the household and preserve remedies for error and distress.

Implementation constraints

Match finance to how households earn, occupy, and build, while testing the political incentives sustaining the programme. Price guarantees and subsidy per additional sustainably housed household, including losses and servicing costs. Lenders must assess irregular earnings and handle arrears; the regulator must monitor concentration and consumer treatment. Track additional eligible borrowers, full repayment burden, arrears, location, gender and income distribution, and public cost per sustained outcome. Implementation requires agreements on standards, funding, and risk. Lenders may need to change loan files and servicing practices. Pension funds may reject housing instruments with unsuitable prices, ratings, or liquidity. Governments must reconcile lower-cost credit objectives with risk-adjusted funding costs. Fixed borrower payments also need a credible allocation of long-term interest-rate risk. Establish each party's constraints from terms and records. Changing an established buyer subsidy affects households and providers that have planned around it. Supply reforms may deliver benefits later and to different households. Assess transition arrangements, existing commitments, and the cost of retaining or replacing the subsidy. A programme's political durability does not establish that it is the most effective response. Supply readiness and legal protection should be prepared alongside the financing intervention.

Prepare supply, legal protection, and finance together, with sequencing determined by local constraints. Improve land records and accessible dispute resolution, establish fair recovery and consumer-protection rules, and standardise origination and servicing. Use refinancing where term funding is scarce and guarantees for defined risk gaps. Test incremental loans for staged builders. Develop capital-market funding when sufficient performance data and oversight exist. Target subsidy to the remaining affordability gap and publish who benefits. The assessment framework below makes the remaining affordability gap and public exposure explicit.

Housing finance assessment

Gate	Question	Metric / decision
1 Product	What is the cheapest adequate housing option?	Price/rent at viable location and safety standard
2 Payment capacity	What can the household sustain through a shock?	(Debt payments + utilities + transport) / disposable income
3 Gap type	Which constraint binds?	Resource, liquidity, term, credit risk, rate/currency risk

Gate	Question	Metric / decision
4 Supply response	Can more homes appear if credit expands?	Permits, completions, serviced land, vacancy
5 Underwriting	Can irregular income be observed cheaply?	Cost per loan, approval time, default by segment
6 Tenor	Does lender funding match the loan?	Asset-liability maturity and repricing gap
7 Currency/rate	Who bears macro risk?	Local-currency share; fixed/variable share; stress test
8 Guarantee	Is public risk sharing additional?	New borrower share; realized public loss; lender retention
9 Capital markets	Can investors trust the asset?	Standard data, servicing, arrears, rating, liquidity
10 Subsidy incidence	Who captures cheaper finance?	Borrower payment vs house/land price and margins
11 O&M/asset risk	Can the owner maintain the property?	Maintenance reserve, insurance, climate exposure
12 Politics	Who benefits from preserving the current system?	Map of interests, risks, authority, and potential agreements
13 Digital rights	Can the borrower understand and challenge decisions?	Reason codes, correction, audit, appeal, offline access
14 Outcome	Did finance widen adequate housing access?	Additional low-income households housed per public dollar

Limits of the evidence

The reported programme results and analytical financing tests require different interpretations and local inputs. Kenya mortgage-market data come from the Central Bank of Kenya and the 2026 World Bank appraisal; Tanzania and WAEMU results come from World Bank results reporting; Indian market structure comes from National Housing Bank; PMAY-U 2.0 parameters come from MoHUA; current Indian digital-underwriting and RMBS examples come from IFC; South African First Home Finance figures come from the Department of Human Settlements. The global capitalization evidence comes from published empirical studies.

Judgment. The five-gap taxonomy, sequencing rule, political-economy map and housing finance assessment synthesize this evidence into an operating framework. Each requires local borrower cash-flow data, housing prices, lender costs, default and recovery data, supply indicators and fiscal-cost estimates before policy decisions. Fieldwork should establish whether support changes borrower terms and reach rather than merely replacing existing funding.

Prepare the local decision

Trace household cash flow, lender costs, defaults, and financing alternatives before selecting public support. Obtain anonymised loan-level applications, approvals, rejections, pricing, fees, maturity, arrears, recoveries, and refinancing terms. Add interviews with eligible non-applicants and informal earners. Separate lack of repayment capacity from missing documentation, expensive funding, and unsuitable collateral. The Kenya, Tanzania, and West African cases above require examining how wholesale funding changes the retail offer and who receives it. The finance ministry and regulator should compare lenders' stated constraints with underwriting files and funding costs. Include smaller lenders, informal earners, incremental builders, consumer advisers, and the guarantee administrator. Assign responsibility for interest-rate risk, currency risk, defaults, and complaints before committing public funds. Test cash-flow verification against independently checked borrower records, using permission appropriate to the data. Compare any model with existing underwriting on cost, processing time, errors, and borrower outcomes. Historical repayment is observed mainly for approved applicants, so it cannot establish performance for everyone previously rejected. Require lenders receiving public support to report new access, effective borrowing cost, and arrears by loan cohort. Reconsider support if it only replaces existing funding without changing terms or reach. Chapter 7 examines who ultimately receives the benefit when that support takes the form of a subsidy.

Housing finance should address the constraint that prevents additional households from obtaining suitable homes. Long-term local-currency funding helps when lenders can originate and service affordable loans and suitable homes are available. A refinance corporation cannot resolve insufficient income, unaffordable deposits, or disputed collateral. A mortgage-to-GDP target can reward larger loans without showing whether excluded households gained access. Stress-test default correlations, investor demand and liquidity during a downturn. A small subsidised portfolio does not establish a sustainable national market.

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7. Housing subsidies and their beneficiaries

Design subsidies around a specified household constraint and verify who ultimately receives the benefit. The amount paid by government is not necessarily the amount retained by the household. Supply conditions, land values, landlord participation, and continuing operating costs affect incidence. A subsidy appraisal should therefore trace the payment through the housing market before choosing its recipient and instrument.

A housing subsidy reaches households through changes in prices, rents, borrowing costs, developer returns, and supply. The government may pay a buyer, developer, lender, municipality, or landlord; the household benefit can differ substantially from that payment. A mortgage interest subsidy can lower a borrower's monthly payment and simultaneously increase the amount she can bid for a scarce house. Free public land can lower a developer's cost and increase the developer's margin if the sale price remains unchanged. A tenant subsidy can reduce rent burden while the extra purchasing power raises market rents. A road built for a housing project can raise the value of adjoining private land. An explicit grant appears in the budget. A land concession, tax preference or state guarantee can be larger and absent from the reported subsidy total. Subsidy incidence explains why the recipient named in the budget may differ from the ultimate beneficiary.

1 Identify the gap public support should close

How subsidies reach households

The benefit of a subsidy depends on how prices, costs, and supply respond to it. A transfer's legal recipient and its economic beneficiary can differ. The division depends on which side of the market can adjust. When housing supply can expand, a demand subsidy can produce more homes. When well-located land and permissions are fixed, extra purchasing power competes for a fixed stock and part of the gain is capitalized into land and property values. Empirical work finds stronger capitalization where housing supply is more constrained.[1] Estimating that benefit first requires a complete account of the public contribution.

The full cost of public support

Public support includes land, tax preferences, guarantees, infrastructure, and operating commitments as well as cash grants. Cash grants reduce the household's or developer's direct cost. Interest subsidies lower the price of credit. Guarantees shift credit risk to the state. Public land transfers reduce land cost. Infrastructure spending supplies roads, utilities and transit. Tax expenditures lower the tax burden on ownership, development or finance. Operating subsidies cover the difference between affordable rents or tariffs and the long-run cost of running the asset. Each should be converted to a present-value fiscal cost. A public parcel transferred at a nominal price has a subsidy equal to its opportunity value less the price paid. A guarantee has an expected fiscal cost based on probability-weighted losses, fees, and administration. Correlated defaults also require a separate assessment of severe fiscal losses in a

downturn. A tax exemption is foregone revenue. A maintenance liability is future expenditure. Once these costs are recorded, government can compare instruments on equal terms. The value of that support should then be compared with the duration and quality of occupancy it produces.

Compare subsidies through their full public cost and the sustained housing benefit obtained. A project on free land can look cheaper than a project that buys land, even when the public gives up a larger asset. A guarantee can look costless until a downturn. A low social rent can look sustainable until the roof, lift and pipes need replacement. Public finance should record the full lifecycle subsidy when the commitment is made. The useful denominator is household-years of adequate housing. A US$20,000 subsidy that produces twenty years of safe, well-located occupancy has a different value from the same subsidy attached to a peripheral unit that remains vacant, is sold informally, or deteriorates after five years. The choice of instrument depends on which financing or resource gap prevents that benefit.

Four types of financing gap

Subsidy design should distinguish inadequate resources from liquidity, information, and financing constraints. A resource gap exists when even the cheapest adequate housing in a viable location costs more than the household can sustainably pay. The remedy requires a real transfer, cheaper inputs, or a smaller housing product. A liquidity gap exists when the household can afford the lifetime cost but cannot fund a deposit, connection fee or construction stage. Savings matches, staged finance or deferred payment can work. A credit-risk gap exists when lenders cannot observe or price irregular income, collateral or repayment risk. Guarantees and better underwriting can help. A supply gap exists when land, planning, infrastructure or market structure prevents enough suitable housing from being produced. A government that uses an interest subsidy against a supply gap raises demand without fixing quantity. A government that gives a deep capital grant to a household facing a short-term liquidity problem overspends. A government that offers mortgage credit to a household with a resource gap can create debt stress without resolving the affordability gap. A long-term programme can then target support to the gap that remains after avoidable constraints are addressed.

For a long-term program, address avoidable supply constraints, define an adequate housing product, measure what households can pay, resolve liquidity and information barriers, and subsidize the remaining resource gap. These steps may need to proceed together. Immediate assistance for unsafe housing, homelessness, or severe rent stress should not wait for a supply reform that will take years. Track prices and new supply to test whether support reaches households. India's different programme channels illustrate how that choice changes eligibility and the route through which benefits flow.

2 Compare who benefits from each instrument

Subsidy instruments in India

India's housing subsidy channels differ in who qualifies, who delivers the home, and who may capture part of the benefit. The official dashboard, last updated on 30 September

2026 and accessed in October, reports 12.768 million houses sanctioned, 12.103 million grounded and 9.907 million completed under PMAY-U, with Rs 1.806 trillion in central assistance released and total investment of about Rs 8.96 trillion.[2] These are administrative counts of approvals, construction starts, and completions. They do not establish occupancy or household welfare. PMAY combines several delivery channels. Beneficiary-led construction supports households that already control a site. In-situ redevelopment ties subsidy to slum land and developer economics. Partnership housing works through projects. Credit-linked subsidy works through lenders. PMAY-U 2.0 adds a distinct rental vertical. The channels differ in eligibility, delivery arrangements, and the opportunity for benefits to accrue to landowners, lenders, or developers. The interest-subsidy channel makes this distinction particularly visible.

Interest assistance changes borrowing costs, while supply conditions determine whether part of the saving raises land or house prices. The current PMAY-U 2.0 Interest Subsidy Scheme illustrates the mechanics. Eligible households with income up to Rs 900,000, loans up to Rs 2.5 million and properties up to Rs 3.5 million can receive a 4 percent subsidy on the first Rs 800,000 for up to twelve years; the maximum subsidy released is Rs 180,000, paid in five annual installments.[3] For a household close to the ownership threshold, this can change the decision. For a household that cannot pass underwriting or whose affordable home costs far more than its borrowing capacity, the same instrument does little. Beneficiary-led construction has a different incidence. When a poor household already possesses a viable plot, a grant can turn an incomplete structure into usable housing with relatively little land-price leakage. The household controls the scarce factor. In a developer-led project on scarce urban land, part of a buyer subsidy can appear in land bids or sales prices unless new supply can enter. Evaluation must follow the benefit to households rather than stop at sanctioned support.

Evaluate programme channels by additional access within income groups and by the household's actual control over the asset. Ask how many beneficiary households would have built, bought or rented without the subsidy; how much public land and infrastructure they received; and whether location, occupancy and household welfare improved. India also shows how subsidy design can alter intra-household distribution. Female ownership or co-ownership rules can strengthen women's claims to the asset. The gain depends on whether the legal right changes control over sale, mortgage, inheritance and household bargaining. Administrative compliance should be separated from substantive economic power. Kenya's dedicated levy raises the related question of how contributors and beneficiaries are connected.

Housing subsidies and public finance in Kenya

Kenya's housing levy requires a transparent account of who pays, who receives support, and which service obligations follow. Kenya's Affordable Housing Act 2024 created a dedicated Affordable Housing Fund and imposed a levy of 1.5 percent of an employee's gross salary, matched by the employer; people with gross income outside payroll are also subject to a 1.5 percent levy under the Act.[4] The levy raises distributional questions: who pays, who receives support, where projects are built, and whether contributors obtain a housing benefit. Official Affordable Housing Board data report KES 73.2 billion collected in FY2024/25. Collections from July 2025 through January 2026 add about KES 44.8 billion, taking the recorded total for July

2024-January 2026 to roughly KES 118 billion.[5] The Board manages the fund and supports affordable-housing delivery.[6] Earmarking can support multi-year planning. It can create a more predictable stream for land, infrastructure and construction. It can support multi-year procurement. It signals commitment to developers and financiers. It also raises hard incidence questions. A formal employee who contributes for years may never receive a unit. A household outside formal payroll may benefit from social housing or upgrading. Employers bear a statutory matching cost. Developers and contractors gain from a predictable public pipeline. Counties face infrastructure and planning obligations. Project-level accounts are needed to test whether the resulting fiscal agreement is delivering its stated benefits.

A credible subsidy account should connect collection and procurement to allocation, occupation, and maintenance. How much comes from public land? Which income group occupies the unit? What is the transport burden? How much private capital is genuinely additional? What share of levy resources goes to households below market affordability thresholds? Kenya's digital Boma Yangu platform can make allocation more transparent. It can also turn a complex distributive choice into an opaque scoring system if citizens cannot see the rules. Contributors need a public audit trail from levy collection to land acquisition, procurement, allocation, occupancy and long-run maintenance. South Africa’s experience shows why location and future household costs belong in that account.

Subsidised housing and location costs in South Africa

Subsidised units, social rental, titles, and serviced sites deliver different benefits and require different cost comparisons. The Department of Human Settlements describes government-subsidized housing as a fully built house provided free to qualifying beneficiaries, who then remain responsible for municipal rates and service charges.[7] First Home Finance serves households above the free-house threshold and below easy access to commercial mortgages; current program guidance targets incomes between R3,501 and R22,000 per month.[8] Millions of households have acquired formal housing assets since 1994. The location of subsidized homes also creates costs. Cheap peripheral land allowed more units per rand while placing households far from jobs and reinforcing apartheid-era urban form. Public money saved on land can return later as transport subsidies, household travel time, trunk infrastructure and weak municipal finances. The 2026/27 national program illustrates the continuing fiscal weight of housing. The Department announced targets of 39,058 housing units, 2,878 social housing units, 21,918 title deeds and 25,186 serviced sites, backed by more than R15 billion in departmental allocation.[9] Complete units, social rental, title registration, and serviced land require different subsidy arrangements. Serviced sites can reduce the initial public contribution while shifting construction and occupation costs to households.

Compare serviced-site support with completed housing using the household’s full cost and ability to build. Current policy allows qualifying households below R3,500 income to receive serviced sites free, while higher bands pay differentiated amounts.[10] This can shift the state from building every finished house toward supplying land and services, while households add the structure over time. The gain depends on access to construction finance, safe incremental standards and location. The incidence test should compare the full public cost of a free peripheral

house with a well-located serviced plot, social rental unit or upgrading intervention. Include transport, service charges, and maintenance in that comparison. Chile's experience provides a further test of whether physical improvements produce broader household gains.

Housing supply and location in Chile

Chile's subsidy experience shows that expanded delivery and better services can coexist with weak wider household outcomes. OECD data show the share of households without housing or living in substandard housing fell from 23 percent in 1992 to 10 percent in 2011.[11] OECD analysis finds that pressure to produce large numbers drove social housing to the cheapest peripheral land. Between 1990 and 1998, 60 percent of Santiago's social housing was concentrated in three of its poorest municipalities. Some peripheral settlements faced very long travel times and weak services. Housing production reduced the measured deficit, but its concentration on peripheral sites also created access and service costs.[12] An Inter-American Development Bank evaluation of Chile's Progressive Housing Program found poor targeting, high under-coverage and leakage. The program improved access to water, sewerage and electricity, yet produced no discernible gains in poverty, school attendance or occupational outcomes and worsened overcrowding.[13] In this evaluation, improved physical services were not accompanied by measured gains in those household outcomes. The implication is to include location and neighbourhood opportunity in the benefit purchased by the subsidy.

Chile's experience supports assessing location and neighborhood quality alongside the dwelling. A demand-side voucher is more likely to improve access when developers can build in enough places, transport and services are coordinated, and the subsidy reflects the true cost of inclusion in high-opportunity areas. Public land is one way to secure that location benefit, provided its contribution is valued explicitly.

The subsidy in public land

Public land can create a substantial unrecorded transfer in housing. A ministry can report a project cost that excludes the market value of the site because no cash changed hands. The public balance sheet still gave up an asset. Every public-land housing transaction should therefore show five numbers: market value before the project; price or rent paid by the developer; infrastructure added by government; affordable-housing obligations; and the value of the land after rezoning or infrastructure. Value the land concession and public works separately, and avoid counting an infrastructure cost again through the land-value increase it creates. A direct sale maximizes immediate revenue when competition is strong. A discounted transfer lowers project cost and creates risk of private capture. A long lease can preserve public ownership while trading land value for rent or income restrictions. Competitive tenders can ask developers to bid the lowest subsidy required for a fixed affordability and location outcome. The timing of public investment and acquisition influences how much of that value the authority must pay for.

Acquisition and allocation decisions should account for the land-value effects of announced public investment. Where lawful and financially justified, government should secure land or development rights before transit, infrastructure, or upzoning raises values. Acquisition still requires transparent valuation, consultation, compensation, and protection of

occupants. If it sells first and invests later, taxpayers finance a gain captured by the buyer. If it acquires after announcing the investment, it buys back value created by its own decision. Tax preferences create another public transfer that can be less visible than a cash allocation.

Tax expenditures

Housing tax expenditures need the same distributional scrutiny as direct subsidy programmes. Mortgage interest deductions, capital-gains exemptions, reduced property taxes and tax-favored savings can move more money than direct housing programs. Their beneficiaries often have higher income because the household must own a valuable asset and, in many systems, face a tax liability large enough to use the preference. U.S. evidence illustrates the distributional risk. Sinai and Gyourko estimated a highly skewed geographic distribution of owner-occupied housing tax benefits; in many large metropolitan areas the top quarter of owners received 70 percent or more of the subsidy flowing to the metro.[14] Even where tax systems differ, assess indirect subsidies using the same incidence and value-for-money tests as budget expenditure. Tax subsidies also build coalitions that are hard to unwind. Beneficiaries experience the preference as part of the normal tax system. Removing the preference raises recipients' tax bills. This helps explain why regressive housing tax expenditures can outlive targeted programs for poorer households. Guarantees likewise create public exposure that may remain outside immediate expenditure figures.

Guarantees and contingent liabilities

Guarantees should be evaluated by the lending or investment they change and the expected public losses they create. A partial mortgage guarantee can expand credit to informal workers. A municipal or project guarantee can attract infrastructure finance. Cash payments may arise only when a guarantee is called, but the fiscal risk exists when the commitment is made. Appraisal should estimate expected losses net of fees, administrative costs, and the cost of bearing risk. Stress tests should separately assess correlated defaults during recessions. A guarantee with a low average default rate can generate a large public call when house prices, employment and bank balance sheets all weaken together. A guarantee that covers a loan the bank would have made anyway transfers risk to taxpayers without creating housing. The test is the marginal borrower or project: did the guarantee change approval, tenor, interest rate or capital mobilization? That test must also examine which income and asset groups receive the benefit.

Mortgage guarantees can distribute benefits unevenly, so conclusions from one institutional model require local testing. A model of U.S. government-sponsored mortgage guarantees finds that eliminating the implicit interest subsidy can raise aggregate welfare by reducing a transfer that favors higher-income, higher-asset households.[15] The result depends on the model and its institutional assumptions; distribution should be tested separately in other mortgage systems. Supply constraints help explain another route through which support may accrue to people other than the intended household.

Supply constraints and subsidy capture

Subsidy incidence becomes easier to predict when the team identifies the factor that cannot expand. The comparison below identifies how different constraints can redirect the benefit.

Scarce factor	Likely capture channel	What to measure	Policy response
Well-located land	Higher land bids and residual land value	Pre/post land prices; developer land cost	Expand legal supply; reserve public land; capture uplift
Planning permission	Scarcity rent in permitted projects	Price gap between permitted and comparable sites	By-right density; reduce discretionary approvals
Developer competition	Higher margins / weak pass-through	Bid concentration; profit margin; price-cost spread	More entrants; open-book cost benchmarks; smaller packages
Mortgage funding	Lender spread or rationing	Rates, fees, approval by risk group	Wholesale liquidity; competition; transparent guarantee pricing
Rental supply	Landlord rent increase	Voucher/upgrade rent changes; vacancy	Add supply; rent reasonableness; tenant protection
Utility capacity	Connection queue / developer payments	Connection time and charge	Fund trunk capacity; publish queue and capacity
Public land	Developer or buyer windfall	Market value versus transfer value	Explicit land subsidy and clawbacks

The extent of subsidy capture depends on local supply conditions rather than a universal rule about demand support. Assess the intervention's scale, safeguards, and initial tenure conditions. A voucher that helps an existing renter pay the same market rent can have little effect on aggregate demand. A subsidy that creates a new buyer with much greater bidding

power in a tightly constrained ownership market can have a larger price effect. Incidence is an empirical question guided by supply conditions. Evidence from Sweden finds strong capitalization of subsidized interest rates into house prices. U.S. evidence using mortgage-market segmentation also finds that easier credit increases house values at the margin.[16][17] These studies do not provide a universal coefficient for Africa or India. They establish a mechanism worth testing whenever governments inject purchasing power into a constrained market. Those price effects also influence the political interests that form around a programme.

Homeowner interests and housing supply

Subsidy programmes can create constituencies whose interests differ from those of future entrants. An ownership grant turns a renter into an asset owner. The new owner gains from neighborhood investment and from scarcity that supports property values. Over time, an owner-heavy electorate can become a constituency for restrictive planning, low property taxation and protection of incumbent asset values. Government subsidizes ownership. Asset values rise. Owners resist new supply. Scarcity raises entry prices. Government expands buyer assistance to restore affordability. If these responses occur, additional public support can raise prices without resolving restricted supply. Pro-housing coalitions, often called YIMBY ("yes in my backyard") groups, try to reverse this asymmetry by organizing renters, younger households, employers, developers, construction labor and transit interests around more supply. One representation problem is that potential future residents may lack a local address and a vote in the jurisdiction that decides whether their home can be built. The fiscal system can change the coalition. Infrastructure grants and local revenue sharing can compensate jurisdictions that accept density. Property taxation can give municipalities recurring revenue from additional floor space. Citywide or metropolitan targets can reduce the ability of one neighborhood to veto benefits that accrue across the labor market. Even an eligible household may remain excluded if landlords, sellers, or allocation practices prevent access.

Discrimination and access to subsidies

A housing subsidy has value only if the household can convert it into a dwelling. Discrimination can break the conversion. Landlords may refuse tenants by caste, religion, race, ethnicity, migration status, occupation or family composition. Brokers can steer households. Lenders can price neighborhoods or income sources differently. Housing societies can impose social screening after formal eligibility has been met. The Seattle Creating Moves to Opportunity experiment tests whether search assistance improves outcomes beyond financial support alone. A bundle of information, short-term financial help, customized search support and landlord engagement increased moves to high-opportunity neighborhoods from 15 percent in the control group to 53 percent in the treatment group. Full customized support produced far larger effects than information and financial incentives alone.[18] Indian and African subsidy programmes should measure the share of eligible households that obtain housing. Among eligible households, who actually secures a unit, lease or loan? How long does it take? Who is rejected? Which neighborhoods remain inaccessible? Where law and data permit, outcomes should be broken down by social group, gender, migration status and income source. Count successful access to

suitable housing alongside approved benefits. Evidence on search assistance shows why the content and delivery of that support need careful attention.

A tested assistance package cannot be reduced to a leaflet without evidence that the simpler intervention will work. Its value lay in helping families complete a move that the voucher already financed. Chile's peripheral-housing experience provides a separate warning: better physical conditions do not guarantee access to opportunity. Subsidy design should therefore follow households through search, landlord acceptance, occupation and continued access to jobs and schools. [108] (See sources 11–13.) The same household-level tracking is needed when public upgrading changes the value of an occupied neighbourhood.

Upgrading and displacement risk

Upgrading can preserve access and networks while creating new displacement risks for tenants and other occupants. Its welfare advantage depends on whether the site can be made safe and adequately serviced. It also raises the value of land and structures. That gain can displace the households the project intends to help. The incidence depends on tenure. An owner-occupier gains an improved asset. A structure-owning landlord can gain through higher rent. A tenant receives better services and may face a higher rent. A shopkeeper can gain foot traffic or lose the premises. A land claimant can capture value without living on the site. Every upgrading project should therefore create a pre-investment tenure map: occupants, tenants, structure owners, land claimants, businesses and vulnerable households. The project should state who has return rights, who receives temporary relocation support, how rents will be monitored, and what new rental supply will be added. Record occupancy before investment changes land values or eligibility incentives. Sites that remain hazardous require a separate decision about risk reduction or relocation.

Hazard-related relocation must account for the household's livelihood and access losses as well as replacement shelter. Upgrading where risk is manageable; relocation where life safety or critical infrastructure requires it. Compensation should include the economic value of location: travel, customers, school continuity and networks. A replacement unit alone can leave the household poorer. Whichever location is chosen, the municipality will inherit service responsibilities that need funding.

3 Fund and administer the subsidy over time

Responsibilities across levels of government

Capital subsidy and municipal service finance should be agreed together across levels of government. A national ministry can maximize unit delivery by building where land is cheap. The municipality then carries the operating and maintenance cost. The household pays through unreliable services when that liability is unfunded. The apparent national subsidy understates the local fiscal burden. A sound project should calculate the present value of municipal operating costs and identify the revenue source. Capital transfers can be linked to density, service standards, asset-management systems and local revenue effort. South Africa's

current move toward performance-based metro finance points in this direction. The broader principle is that national housing policy should compensate cities for the marginal cost of accommodating low-income residents in well-located areas while giving cities a recurring revenue stake in added floor space. Without that alignment, exclusionary local behavior can be fiscally rational. A jurisdiction that bears the service cost and receives little revenue has reason to resist housing. Intergovernmental finance can fund those service costs and reduce the fiscal incentive to exclude new residents. The development also needs an account of recurring costs within the housing itself.

Operating subsidies and maintenance

A capital subsidy does not fund every cost after occupation. Appraisal must identify how management, routine maintenance, and replacement will be paid for over the life of the home. Public and social rental require rent collection, routine maintenance, security, major repairs and a replacement reserve. Condominium ownership requires fees and collective governance. Serviced-site programs require utilities to maintain networks. Upgraded settlements need drains cleared, roads repaired and lights replaced. A program that cannot fund these functions risks delivering an initial benefit followed by declining housing quality. Chile's experience with low-quality peripheral housing and many global public-housing failures show why capital efficiency should be measured over the asset's life. Every investment should therefore publish a twenty-year operating model. State the annual cost, payer, collection assumptions, reserve, asset owner and fallback if collections fail. The net present value of any gap is part of the subsidy. Recognising those commitments makes reform harder to conceal but easier to evaluate against the interests sustaining it.

Interests in subsidy reform

Actor	What it can capture	Veto / leverage	Illustrative position (not a quotation)	Possible agreement
Low-income household	Capital transfer, lower rent or payment	Vote, protest, take-up	Housing is a right	Portable benefit tied to access and occupancy
Excluded poor household	Often little	Weak formal voice	The scheme is not for us	Rental, incremental and simpler eligibility channels
Landowner	Capitalized subsidy and infrastructure uplift	Holdout, land supply, litigation	Land reflects market value	Predictable rights + transparent value capture

Actor	What it can capture	Veto / leverage	Illustrative position (not a quotation)	Possible agreement
Developer	Margin, pipeline, discounted land	Can delay or switch product	Affordable housing needs viability	Competition + open-book cost + milestone subsidy
Lender	Interest, fees, guarantee protection	Underwriting and origination	Risk is too high	Risk sharing with additionality tests
Landlord	Rent uplift after subsidy or upgrading	Tenant selection, supply	Maintenance costs are rising	Supply + rent monitoring + tenant security
Homeowner	Asset appreciation	Planning hearings, elections	Protect my investment	Infrastructure + citywide supply rules
Municipality	Rates/fees or unfunded service liability	Permits, utilities, local plans	Growth must pay for itself	Transfers linked to serviced affordable floor space
Politician/ broker	Allocation and contract rents	Beneficiary lists, public land, enforcement	I unblock access	Rules-based queues + public exception logs
Digital platform/operator	Control over allocation data and scoring	System rules and access	Technology removes discretion	Explainable rules, appeals, audit and open standards

Subsidy reform needs evidence of additional benefit and a clear account of the groups affected by changes. A developer hires staff and buys land around a program. A bank builds origination systems. A homeowner treats the benefit as part of the asset's value. A broker learns how to navigate the allocation list. A public agency builds its identity around delivering a particular product. Reform threatens sunk investments and future rents. Even a technically sound

reform can face opposition when its immediate costs fall on organized groups and its benefits arrive later. Redirecting ownership grants toward rental can reduce benefits expected by aspiring owners and developers. Pricing public land can raise costs for firms receiving concessions. Reducing a mortgage tax preference can impose losses on owners who have capitalized it into purchase decisions. Tighter targeting can remove support from existing recipients before new beneficiaries receive it. The feasible sequence often begins with transparency. Price public land. Publish subsidy per occupied household, location, income band, vacancy, resale and operations and maintenance. Add new tenure and supply channels. Protect some existing commitments. Shift new budget growth toward better instruments. Use sunset and evaluation clauses so programs must demonstrate additionality to justify continued funding. The household's resulting neighbourhood access provides a practical test of that benefit.

The subsidy should purchase usable neighbourhood access for the target household. Ask what neighbourhood the subsidy purchases and whether the bottom 40% can enter without losing access to work or support networks. Administrative data can help test this outcome if eligibility, exceptions, and subsequent occupancy remain open to scrutiny.

Digital subsidy administration and AI

Digital administration can improve targeting and traceability when allocation rules and exceptions can be challenged. Digital systems can reduce duplicate claims, speed payments, verify milestones and identify vacant subsidized units. Subject to lawful data access and local validation, AI tools can help assess irregular cash flows, flag applications for review, reconcile property records, and select cases for audit. Geospatial analysis can test whether subsidized units are near jobs or exposed to hazards. Computer vision can help screen construction progress, with physical inspection for concealed work and safety. Test costs and error rates against manual or rules-based processes before claiming savings. The governance rule should be stronger where the benefit is life-changing. Applicants need reason codes, correction rights, a human appeal, audit of group outcomes and an offline route. Automated fraud detection should never become automatic exclusion. Digital allocation can reduce patronage only when the scoring rule, queue and exceptions are published and open to challenge. Linked price and occupancy records can also reveal who captures the benefit after allocation.

Technology can also improve incidence analysis. Where lawful and necessary, authorized analysts can link subsidy, parcel, price, rent, occupancy, and transaction data to test whether support raised supply, reduced household burden, or capitalized into land. Publish aggregate findings while protecting identifiable household records. That analysis should inform adjustments to contracts, services, and access support.

Implementing subsidy reform

Adapt the subsidy to the barriers households actually face and measure continued access after allocation. Fund administration and recurring benefits, and state which costs continue after external support ends. The housing agency must complete lease-up and resolve landlord problems; planning agencies must address supply constraints. Track lease-up, sustained residence, total costs, school and job access, landlord refusals and benefits accruing to non-

recipients. Reform timing must also recognise that supply-side changes and immediate assistance produce benefits on different schedules.

Step	Action	Reason
1	Define the household outcome	Use adequate, occupied, well-located housing as the objective
2	Diagnose the gap	Separate resource, liquidity, credit-risk and supply failures
3	Fix avoidable scarcity	Expand legal serviced supply alongside assistance; do not delay urgent relief
4	Price all public inputs	Land, tax breaks, guarantees, infrastructure and O&M; become visible
5	Choose the narrowest instrument	Use the cheapest tool that matches the diagnosed gap
6	Design for pass-through	Competition, supply response and transparent procurement limit capture
7	Protect inclusion	Measure conversion barriers, displacement and social screening
8	Fund the operating model	Prevent capital subsidy from becoming a deteriorating asset
9	Publish incidence	Show who gains by income, tenure, location and actor
10	Adapt when the bottleneck moves	Reassess the constraint before renewing support

Immediate support and longer-term supply reform should be compared through the timing and distribution of their benefits. Cash subsidies can be delivered sooner. Appraisal should compare both the timing and distribution of benefits, and test whether continued demand support is raising prices in a supply-constrained market. The assessment framework below records those choices before commitments become difficult to change.

Subsidy incidence assessment

Gate	Question	Evidence / decision rule
1 Problem	Which gap are we solving?	Resource, liquidity, credit risk, supply, service or location
2 Counterfactual	What happens without the subsidy?	Additional households, units or housing quality caused by the program
3 Full fiscal cost	What does government truly give?	Cash + land + tax + infrastructure + guarantee + O&M; present value
4 Targeting	Who qualifies and who converts?	Take-up and completion by income, tenure and inclusion group
5 Supply response	Can quantity expand?	Permits, serviced land, vacancy, starts, completions
6 Incidence	Who captures the gain?	Household vs landowner, developer, lender, landlord and local government
7 Location	What access does the subsidy buy?	Jobs within 45/60 min; transport cost and time
8 Quality	Is the product adequate and adaptable?	Defects, unit size, resilience, ability to extend or rent
9 O&M;	Will the housing service last?	Operating budget, collections, reserve and asset owner
10 Neighborhood	Does it deepen segregation or displacement?	Income/social mix, school/job access, tenant exit
11 Politics	Who can block or capture reform?	Actor-rent-veto-coalition map
12 Digital rights	Can applicants understand and appeal?	Reason codes, correction, audit, offline route
13 Value for money	What outcome is bought per public dollar?	PV subsidy / additional household-years of adequate housing

Limits of the evidence

Official programme figures describe delivery and finance, while the proposed tests assess the distribution and durability of benefits. Chile's housing outcomes come from OECD and IDB evaluations. The global evidence on capitalization, tax subsidies and mobility comes from peer-reviewed or NBER research. These studies identify mechanisms in specific institutional settings; they do not supply a universal pass-through coefficient for Africa or India. The four-gap taxonomy, present-value public-cost framework, subsidy sequence and Subsidy Incidence Test are analytical judgments. They should be populated with local land values, household payment capacity, project pro formas, municipal operations and maintenance costs and market supply responses. Their purpose is to identify the recipients and beneficiaries of each transfer before the subsidy becomes politically permanent. A field appraisal must connect those records to what households receive and pay.

Prepare the local decision

Compare subsidy options through the housing, access, and remaining household costs they deliver. Assemble eligibility rules, applications, refusals, take-up, payments, land contributions, tax relief, guarantees, and operating support. Estimate the full subsidy per occupied home and per household served. Survey eligible households who never apply or fail to secure housing. The Chile and South Africa cases above make location costs part of the benefit calculation; a lower purchase price is insufficient evidence of a lower household burden. Treasury should reconcile costs with housing agencies and municipalities. Ask lenders and developers which requirement prevents a qualifying household from proceeding, and check their explanation against rejected files. Hear tenants and applicants outside formal employment separately from successful recipients. Assign a responsible agency and budget to each assistance or appeal step. Compare the subsidy required for usable access in a well-served area with the full cost of peripheral provision, including transport and recurring services. State how many lower-income households could enter under each option and how long affordability lasts. If an inclusionary obligation is replaced by an off-site payment, disclose what access is lost and what equivalent, enforceable benefit the payment buys. A lower subsidy per unit is not sufficient justification. Monitoring prices and occupation can then test whether the intended benefit survives implementation.

Use administrative data to follow support through to prices, occupation, and household access. Use record matching to flag possible duplicate payments for review, retaining the reason and evidence for every exclusion. Audit both wrongful payments and eligible households missed, including those with incomplete digital records. Trial the proposed instrument in a defined market and track prices, rents, take-up, location, and sustained occupancy. If sellers absorb the benefit through higher prices, revisit the supply constraint and subsidy design before increasing the allocation. Chapter 8 examines the institutional coordination needed to deliver those outcomes across the housing system.

Subsidy design should cover the combination of payment and access barriers preventing a household from leasing a suitable home. A usable benefit, available homes, willing landlords, and tailored search assistance operated together in the mobility evidence. Copying a voucher amount or application portal omits those conditions. Remote locations, few participating landlords, and constrained supply can leave approved applicants without a workable tenancy. At 100 times pilot size, measure landlord recruitment costs, counsellor caseloads, rent responses and competition among recipients.

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10. South Africa Department of Human Settlements, serviced-site eligibility. [source]

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13. Inter-American Development Bank, An Impact Evaluation of Chile's Progressive Housing Program, 2008. [source]

18. Bergman, Chetty, DeLuca, Hendren, Katz and Palmer. Creating Moves to Opportunity, NBER 26164. [source] [additional source] [additional source] [additional source] [additional source] [additional source] [additional source]

Operational use: In a government or MDB review, ask the team to calculate the present value of every public input, identify the scarce factor likely to absorb the subsidy, and state the number of additional household-years of adequate housing expected. If the answer cannot be produced, the subsidy is not ready for scale.

[108] Bergman, P. and coauthors (2024; revised working paper 2023). Creating Moves to Opportunity: Experimental Evidence on Barriers to Neighborhood Choice. NBER Working Paper 26164. [source] [source]

8. Coordinating housing delivery

Housing coordination requires enforceable decisions, funded handoffs, and accountability for the completed service. Creating a committee or consolidating agencies will not resolve missing authority or finance. Compare institutional arrangements through the decisions they enable, then specify who can act when another organisation misses a commitment. The delivery sequence reveals where those responsibilities need to be connected.

Delivery depends on the reliability of the whole sequence of required decisions, not the performance of one agency. Improving each agency separately will not resolve failures between them. A land agency records a parcel. The planning authority grants floor space. The utility has no funded trunk connection. The developer holds land while interest accrues. The housing agency announces beneficiaries before the transport plan is settled. A lender rejects informal income that another public agency already observes through payments and taxes. A building is handed over while the municipality has no maintenance budget. Every institution can point to its own completed task. The household still does not receive a usable home. Delivery can fail when any required decision remains unresolved. Consider an illustrative chain of eight critical decisions. If the decisions are independent and each succeeds 90 percent of the time, the probability that all eight succeed is about 43 percent. At 95 percent per step the joint probability rises to 66 percent; at 98 percent it reaches 85 percent. These are illustrative calculations. Actual decisions are often dependent, so their joint success rate must be measured. The example shows how several required steps can reduce end-to-end delivery. Investment must therefore address the step that prevents the rest of the sequence from producing a usable home.

1 Assign authority across the delivery process

Coordination across agencies

Strengthening an already effective step will not resolve a different binding constraint. Cheaper mortgage money cannot cure disputed land. Faster permits cannot cure missing sewer capacity. Better construction cannot cure a relocation site that separates households from work. The state has to manage the chain and measure whether its components produce a usable home. Government coordination has four practical functions in managing that dependency.

Four functions of government coordination

Coordination should establish common information, resolve authority, and turn repeated negotiation into clear rules. First, it defines a single household outcome. The target is an occupied, safe, serviced dwelling with access to work and a funded operating model. Unit starts and budget disbursement become intermediate indicators. Second, it establishes a common, dated evidence base. Agencies need the same parcel identity, ownership and occupancy record, planning rights, infrastructure capacity, hazard information, approval status, subsidy status and construction progress. When each agency works from a different map or beneficiary list, meetings cannot repair the underlying information conflict. Third, it sets decision rights and

clocks. Every handoff needs an owner, a rule, a deadline and an escalation route. Coordination fails when all agencies may comment and none must decide. Co-locating offices will not remove a delay caused by incompatible approval rules. A genuine integration reform converts recurring negotiation into rules and auditable exceptions. Those arrangements need financing incentives tied to the outcome the agencies must deliver together.

Public funding should reward the agreed delivery result while fitting the responsibilities of each institution. Budget incentives influence agency decisions. A housing agency rewarded for unit counts may prioritize cheap land. A utility rewarded for financial health may delay low-revenue connections. A municipality that bears service costs and receives little new revenue may resist density. Intergovernmental transfers, capital grants, tariffs and subsidy rules need to reward the same outcome. These four jobs are separable. Singapore performs them through unusually integrated public institutions and state control over land. Ethiopia uses performance grants to align municipal incentives. Odisha's Jaga Mission creates a mission structure and community partner. Thailand's Baan Mankong assigns implementation responsibilities to community cooperatives. South Africa is using performance-linked fiscal transfers to repair service institutions. The institutional form differs because the underlying state is different. A responsibility map can reveal where the agreement still depends on an unresolved handoff.

Mapping responsibilities in delivery

Link	Required decision	Common failure	Potential beneficiary of the failure
Land	A legally and socially usable site is identified	Conflicting records, acquisition delay, hidden occupants	Brokers, insiders, speculative owners
Planning	Enough floor space is legal	Discretion, sequential reviews, obsolete standards	Scarce permit holders, incumbent owners
Infrastructure	Trunk and local capacity arrive on time	Separate capital plans and budgets	Owners of already serviced land
Procurement	A competitive supplier is selected and paid	Bid concentration, claims, payment delay	Connected contractors and intermediaries
Finance	Capital matches project and household cash flow	Title, tenor, risk and income records do not align	Bankable households, incumbent lenders

Link	Required decision	Common failure	Potential beneficiary of the failure
Allocation	The right household gets the right tenure	Opaque lists, patronage, weak demand evidence	Brokers, politically connected applicants
Occupancy	The household can actually use the unit	Remote location, high transport, service gaps	Agency still counts delivery at handover
O&M;	Assets keep working for decades	No owner, reserve, tariff or staffing plan	Capital agency avoids future liability

The responsibility map is a diagnostic of decisions, budgets, and authority rather than a substitute for resolving them. Fragmentation can create income from privileged access. Unclear handoffs can create demand for someone who knows how to move the file, secure an exception, identify a public parcel, accelerate a utility letter, or influence a beneficiary list. Such intermediation can add costs without resolving the underlying administrative problem. Integration reform can change the value of information, access, and discretionary powers. Assess which decisions and revenues move between institutions, who may object, and what rules and review procedures will govern the new arrangement. A software platform alone cannot resolve conflicts over statutory authority or budgets. Vietnam and Indonesia show how those responsibilities can be divided between national programmes and neighbourhood decisions.

2 Match coordination arrangements to implementation conditions

Scaling upgrading in Vietnam and Indonesia

My work in Vietnam and Indonesia centred on delivering upgrading at scale while retaining room for local choices. As the World Bank's East Asia and Pacific urban manager responsible for Indonesia and Vietnam, I worked with programmes that faced a recurring problem: reaching large numbers without requiring every neighbourhood to accept the same solution. I regarded Vietnam's upgrading work as an outstanding programme. That is my assessment as a practitioner; it needs to be tested against evidence of who benefited, who moved, and which institutions could sustain the improvements. The two countries offer different ways to combine local decisions with public finance and city networks. The Vietnam Urban Upgrading Project operated in four cities between 2004 and 2014. Community Upgrading Plans linked resident priorities to the work of design consultants and utilities. Local investments were accompanied by primary and secondary infrastructure, and the package included housing microcredit and relocation where necessary. The World Bank's case review reports approximately 3,000 households relocated. The project emphasized in-situ upgrading, but relocation remained necessary for some households. [214] The programme results need to be read alongside the municipal obligations that community implementation cannot replace.

Reported upgrading benefits do not remove the need for continuing municipal spending and coordinated investment. The Bank's project brief reports about 7.5 million people benefiting directly or indirectly. That figure includes the wider reach of infrastructure investments; it should not be read as 7.5 million households receiving new homes. Neighbourhood improvements depend on the networks serving them. A local drainage plan cannot work if the downstream system is absent or overloaded. [215] Indonesia's National Slum Upgrading Project, supporting KOTAKU, used a national platform with local government and community institutions. Its January 2024 completion report records 8.9 million direct beneficiaries, above a revised target of 7.2 million but below the original 9.7 million target. Its definition includes people benefiting from services and people engaged as project workers. These administrative counts demonstrate reach under that definition; they are not a controlled estimate of income gains or proof that all residents retained access. [216] The Independent Evaluation Group identifies an unresolved weakness in Indonesia: local budget limits constrained operations and maintenance after project closure. It also reports difficulties aligning the implementation and budget cycles of upgrading and housing support. A successful community process does not remove the need for municipal spending or coordinated investment. [217] The division of responsibilities should follow which decisions residents can make and which require wider authority.

The shared principle is to distinguish decisions that residents can make well from obligations that require wider authority. Residents can identify where access is blocked, what service fails, and which layout would disrupt daily life. The city must decide network capacity, engineering interfaces, and continuing service obligations. National government can fund a repeatable process, common safeguards, and technical support. Each level needs a defined task and a budget. For an African or Indian adaptation, I would start with the connection between the neighbourhood plan and the municipal capital programme. Identify the utility officer who accepts the connection design, the budget that pays for the trunk works, and the operator who inherits the asset. Put these decisions before procurement. Let residents choose among technically feasible options and give renters and people without formal title a way to contest exclusions. Expansion then requires enough facilitation and technical review to preserve the quality of those local decisions.

At ten times the scale, facilitation and engineering review can become the constraint. Count the caseload per facilitator, the time to resolve a disputed alignment, and the share of completed assets with funded maintenance. Expand the support system as well as the works budget. Community participation cannot substitute for those resources. This is the difference between repeating small projects and sustaining a national programme. Performance-linked municipal grants offer another way to align resources with verified delivery capacity.

Municipal performance incentives in Ethiopia

Performance-linked grants can connect municipal funding to institutional improvements when compliance is independently verified. The World Bank's 2026 synthesis of urban performance grants reports that participating Ethiopian cities increased local revenue from about US$68 million in FY2020 to US$156 million in FY2023. The same model has

reached approximately 150 million people across low- and middle-income countries. In Uganda, participating municipalities recorded an 83 percent cumulative increase in own-source revenue; Ghana's participating secondary cities recorded 148 percent.[1] The 2018-2024 Ethiopia Urban Institutional and Infrastructure Development Program paired institutional conditions with substantial capital funding. It operated in 117 cities, financed more than 2,700 km of roads and 2,700 hectares of serviced land, and the World Bank reports improved living conditions for more than 6.6 million people. Arba Minch reported a 262 percent rise in municipal revenue over five years.[2] The design seeks to align incentives. Central government wants cities to budget, audit, procure and maintain assets better. City leaders want roads, markets, public space and serviced land. The performance grant makes institutional reform the price of capital. Independent verification checks whether the conditions for funding have been met. The strength and fairness of that incentive depend on its size and on municipalities' ability to respond.

Grant thresholds from a synthesis should be tested against local fiscal conditions rather than copied mechanically. The World Bank's 2026 synthesis argues that performance-linked allocations generally need to reach at least 20 percent of the discretionary capital or development grant available to a local government before the incentive becomes strong. This is a program-design rule of thumb, rather than an estimated threshold that applies to every local government.[1] These figures are administrative program results rather than randomized estimates of causal impact. They show large changes in revenue, systems and infrastructure during program periods. The mechanism - tying meaningful resources to verified institutional performance - should be assessed against local fiscal capacity, verification costs, and the risk of excluding weaker municipalities. Odisha illustrates a different coordination problem: bringing residents' knowledge into land and upgrading decisions.

Local knowledge and implementation in Odisha

Odisha's Jaga Mission illustrates why formal parcel administration needs residents' knowledge of occupancy and local conditions. Odisha created a state-wide mission around land rights, upgrading and community governance. The official program covers 2,919 slums. World Habitat reports a target population of roughly 1.7 million people; by the 2023 award cycle, 707 slums in 30 cities had been upgraded and eight cities declared slum-free.[3] The mission joins three actions that usually sit in separate silos. Eligible households receive land rights. Settlements receive infrastructure. Slum Dwellers Associations participate in planning and are framed by the state as a fourth tier of governance. The state supplies law, finance and engineering. Residents supply information about occupation, local networks, drainage failures, vulnerable households and the social cost of relocation.[4] Community institutions can supply information that official records omit. A cadastral map may show structures. Residents know who rents, who sublets, who runs a shop, who arrived recently, which path floods, and which claimant would be excluded by a title issued to the wrong person. Resident participation still requires safeguards against exclusion and unaccountable local representation.

Mission structures need representation, records, grievances, and external review to remain accountable. Railway land, forest land, major hazard areas and private ownership can fall outside the mission's effective authority. A housing mission cannot resolve another agency's

legal mandate through meetings alone. Cross-government disputes need a formal escalation path with a decision-maker above the participating agencies. Community power also needs checks. A Slum Dwellers Association can represent residents and can reproduce local hierarchy. Migrants, tenants, women or minority groups may have weaker voice. Representation rules, public records, grievance systems and external appeals provide ways to review decisions and challenge exclusion. Thailand's community-led approach provides a further test of how authority and assistance can be combined.

Community-led delivery in Thailand

Community-led delivery depends on organised residents, finance, technical support, and safeguards within local organisations. The Community Organizations Development Institute channels infrastructure subsidies and housing loans through organized community cooperatives. Communities survey housing problems, negotiate land, manage budgets, select technical support and oversee construction. The state finances and sets the frame; organized residents perform a large share of implementation. Current program summaries report more than 130,000 households reached in more than 1,300 communities across about 200 cities by early 2026. The historic subsidy package averages around THB 80,000 per household, with infrastructure support, a house construction grant, technical assistance and community management support combined with repayable finance.[5] Community savings groups and cooperatives create a local balance sheet, a negotiating counterparty and a source of information. Collective tenure can protect households who would remain too weak as individual claimants. In canal and railway settlements, the same organization can negotiate with public landowners while redesigning the settlement to preserve infrastructure access. The state gives a legal route and finance to an organization that has information and enforcement capacity at the block level. The risk is capture within the cooperative. Transparent budgets, elected committees, grievance channels and external technical review are essential because local organizations can also exclude residents. Ahmedabad's resettlement experience shows why those capacities cannot simply be assumed after construction.

Participation needs to influence decisions before residents inherit obligations they cannot manage. Associations were asked to manage services after relocation without comparable authority, preparation or clear public-service responsibilities. Baan Mankong places organised residents in land negotiations and spending decisions, supported by public finance and technical help. This institutional comparison is not a matched impact evaluation. It supports building residents' capacity and authority before delegating duties, while retaining safeguards against local exclusion. [111,112] Performance systems must therefore measure service and inclusion outcomes as well as institutional activity.

Performance-based support in South Africa

Performance-based support should use measures that remain valid descriptions of the service being delivered. Chapter 3 describes the service constraints that housing inherits from municipal networks. South Africa's Metro Trading Services Program addresses the institutional side through results-based intergovernmental finance. Here the implementation question is which

agency must change its operating and financial practice to meet the conditions. [6] The World Bank operation provides US$925 million inside a larger government program, with US$2.31 billion of national contribution. Metro grants are released when verified institutional and service results are achieved. The targets include revenue management, asset management, governance, procurement and service quality. Disbursement depends on the applicable verified results and program rules.[6] Housing appraisal must account for the utility's operating capacity. A new connection has little value if the utility cannot maintain pressure, bill users, repair pipes or replace assets. Service reliability should therefore be assessed alongside the dwelling. Results-based finance creates its own risk: gaming the metric. Linking finance to an indicator can reward improvements in the measured value without equivalent improvements in the service. Good design requires measures that reflect the service, independent verification, periodic revision, and enough flexibility to stop measuring a proxy after it ceases to describe the problem. Post-construction evidence from Rio shows why that verification must continue after handover.

Construction and maintenance institutions in Rio de Janeiro

Rio's upgrading experience demonstrates the need to verify operators, recurrent budgets, and asset condition after construction. The second phase upgraded 88 favelas between 2000 and 2008. A ten-year follow-up by the Inter-American Development Bank found that the treated favelas had, on average, returned to maintenance conditions similar to the control group. Sewerage, garbage collection and electricity scored significantly worse in the treated group. Residents reported sewage overflows, accumulated garbage, electrical failures and broken streetlights.[7] The study identifies several possible contributors to deterioration. Rio's favela population grew nearly 27 percent between 2000 and 2010 compared with 3.4 percent in the rest of the city. Dense topography raised maintenance needs. Violence and deliberate destruction affected access and infrastructure. The assessment supports stronger maintenance arrangements, but does not isolate their effect from population growth, topography, or violence.[7] Capital approval and recurrent service budgets operate on different schedules. Roads, drains, and public spaces require staff and repair funding after construction finance ends. The Rio findings support identifying the responsible operator, expected service demand, and recurrent budget during project design, then checking whether these arrangements continue after handover.[7] Singapore offers a different institutional arrangement for coordinating those obligations over time.

Integrated institutions in Singapore

My conversations with Cheong Koon Hean highlighted the coordination of buildings, services, and public space around daily life. In my conversations with Cheong Koon Hean, then chief executive of Singapore's Housing & Development Board, I learned more about how HDB coordinates buildings, services, and public spaces around residents' daily needs. I understood its approach to be evolving towards closer attention to neighbourhood use and inclusion. This is my recollection of those conversations, not a quotation or a claim that Singapore had previously ignored neighbourhood planning. The practical implication is to review the connections between investments. A comfortable home loses much of its value if a person cannot reach the clinic, use the footpath, or afford the service charge. Technology can help an operator detect and correct these failures, provided someone has the budget and responsibility to

act. Chapter 25 follows this approach through HDB's environmental and estate-management programmes. Singapore illustrates a highly integrated public housing system. The Housing & Development Board has built more than 1.25 million flats and houses close to 80 percent of the resident population.[8] Public housing is planned together with transport, amenities, jobs, land use and long-term estate management. Singapore's continuing spatial planning shows how that coordination extends beyond the housing block.

Integrated spatial planning requires housing, transport, amenities, and management to share a population forecast and timetable. The current Master Plan 2025 continues this logic. URA states that new homes, including public housing, will be developed close to job nodes, transport and amenities; by the early 2030s, eight in ten households are planned to live within a ten-minute walk of a train station. Bidadari, for example, is planned for about 10,000 households with extensive walking and cycling links to two MRT stations.[9] Current HDB policy also treats asset stewardship as part of housing. In FY2024 HDB spent S$532 million on upgrading programs and S$570 million on ancillary residential functions including facilities management and planning and building administration. In 2025 it completed about 19,600 flats, while its 2026 program planned another 19,600 BTO flats.[10][11] Adaptation in African and Indian cities requires attention to the surrounding institutions. Singapore combines control over land, planning, public housing finance, construction pipelines, estate management and transport coordination in ways few governments can reproduce. A similar agency would need compatible fiscal, land, and legal powers, along with safeguards for affected residents. The transferable idea is to require a coordinated spatial delivery agreement: every large housing program commits land, transport, utilities, schools, public space and long-run management against the same population and timeline. A one-stop approval office is useful only if it resolves the underlying decisions needed to deliver that agreement.

One-stop approval systems

A one-stop shop should reduce discretionary handoffs rather than add another office above them. The device can work when it changes workflow. Historical World Bank evidence on permit reform shows systems that consolidated reviews, imposed time limits, recorded the officer responsible and escalated delays to supervisors. Evaluate the workflow through decision times, rework, and the availability of review.[12] A one-stop shop fails when it adds a new coordinating office while each underlying agency keeps the same veto. The applicant submits one form and waits for the same sequential approvals behind the screen. Concentration can also increase the value of the gatekeeper. If exceptions remain discretionary, the new office becomes a more valuable point of influence. Assess the approval process itself. Record how many agencies can stop the project, which decisions run in parallel, how long each clock can run, which reasons justify a refusal, and who hears an appeal. Publish the exception log. Integration should reduce the number of discretionary junctions. Its records must also represent occupants and uses that a title-centred system can miss.

Recording informal housing and occupancy

Housing administration needs occupancy and service information alongside registered ownership. Informal settlements contain people whose claims do not align neatly with those fields: tenants, subtenants, structure owners, customary claimants, traders, recent migrants and households sharing a structure. A parcel database can be accurate and still miss the people most affected by a project. The integrating state therefore needs two linked records. The parcel record describes land, legal rights, hazards and planning. The household and use record describes occupancy, tenancy, livelihoods, vulnerability and service needs. Neither should automatically overwrite the other. Infrastructure raises asset values. If only owners appear in the record, tenants disappear from compensation and return rights. A title can improve security for one claimant while increasing rent for another. Enumeration before investment preserves evidence of existing occupancy and claims before land values change. Service provision can often be separated from final title adjudication. Water, sanitation, drainage and fire access can be justified by public health while land claims continue through due process. Making title a precondition for basic service can turn an administrative dispute into a health hazard. Changing which records govern a decision also changes institutional power and requires explicit accountability.

3 Make shared responsibilities enforceable

Authority and interests in coordination

Actor	Institutional interest / rent	Veto	Illustrative concern (not a quotation)	Reform agreement
Housing agency	Unit count, budget, political credit	Site and subsidy rules	We need speed	Reward occupancy, access and lifecycle cost
Land agency	Control over valuable parcels and records	Transfer, valuation, title	The land is strategic	Public inventory, valuation, timed escalation
Municipality	Local control and fiscal balance	Permits, roads, waste, local plans	Growth will overload services	Recurring revenue + performance transfers
Utility	Financial health and network control	Capacity and connection	There is no capacity	Joint capital plan + funded O&M;

Actor	Institutional interest / rent	Veto	Illustrative concern (not a quotation)	Reform agreement
Developer	Margin and predictable pipeline	Can defer, litigate, switch segment	Risk makes the project unviable	Clear rules + competition + fast clocks
Contractor	Claims, utilization, payment certainty	Bids, variation orders	Costs changed	Benchmarks + prompt payment + open change orders
Treasury	Fiscal control and leverage	Transfers, guarantees, budget	Housing is one spending claim	Tie money to end-to-end verified results
Community group	Security, services, local voice	Legitimacy and local access	We know the settlement	Formal role + representatio n + appeals
Broker / patron	Information and exception rent	Access to lists, records and officials	I can unblock the system	Rules, public queues, auditable exceptions
Digital operator	Control over data and workflow	Access, scoring, architecture	One platform will solve it	Open standards, audit, correction and appeal

Coordination reform redistributes authority and should make representation, decision rights, and review procedures explicit. Changes to authority can therefore create institutional disagreement. Moving land approval into a housing mission can transfer authority from the land agency. Creating a metropolitan utility can transfer duties from municipal departments. Publishing queues can reduce reliance on intermediaries. A shared parcel database can expose allocations that depended on ambiguity. City leaders need specified benefits, such as capital funding. Utilities need a credible route to recover costs. Communities need secure standing. Developers need predictable rights. Treasury needs independent verification. Future residents need representation through citywide targets because they cannot appear at today's local hearing. For an institutional reform, record each new decision right, the rule limiting it, and the person or body able to review it. Check whether residents can use that review route in practice. Tenant and migrant representation must be designed before land and allocation

decisions are settled. The same review must test apparently neutral rules for unequal effects on access.

Institutional bias and housing access

A delivery process can exclude households even when each agency follows a formally neutral rule. Income documentation can exclude informal workers. Credit models can penalize neighborhoods that correlate with caste or race. Homeowner hearings can block rental housing through parking, density or design objections. Public-land allocation can favor groups with political access. Informal settlement committees can exclude migrants or tenants. The integrating institution should track conversion at each gate. Who receives land rights? Who obtains planning approval? Who passes lender underwriting? Who is allocated a unit? Who remains after relocation? Who bears a rent increase after upgrading? A final beneficiary list hides where exclusion entered the chain. This suggests an inclusion ledger alongside the financial ledger. For every gate, record eligible households, approvals, rejections, reason codes and appeals. Where lawful and necessary, use protected data to test outcomes by gender, caste, race, class, migration status, and tenure. Limit collection, access, and retention, and publish aggregates that do not identify households. The aim is diagnostic: find the gate where apparently neutral rules generate unequal access. Some barriers arise because the service obligation and the public funding sit at different levels of government.

Intergovernmental fiscal arrangements

Intergovernmental agreements should match each housing-related service obligation with authority and funding. National government captures broad tax gains from urban growth. Municipalities bear much of the marginal cost of local roads, drainage, waste, public space and service interfaces. A mayor who sees added low-income housing as a service liability can resist density for rational budget reasons. Performance grants can change the agreement. Ethiopia uses capital to reward stronger city systems. South Africa now links metro transfers to service performance. Kenya's earlier urban support program used performance-based grants across participating municipalities and reached about 3.6 million urban residents, according to the World Bank's 2026 cross-country synthesis.[1] Housing-specific transfers could reward legal serviced floor space in high-opportunity locations, fast rule-based approvals, affordable rental supply, occupancy, property-tax effort and asset maintenance. The indicator should sit within the recipient's control and connect closely to household welfare. Transfer design should also account for spillovers. A municipality that accepts dense rental housing can bear local costs while the metropolitan labor market receives the gain. A metropolitan or national grant can compensate the host jurisdiction. The agreement should specify the service obligation and the transfer that funds it. The operator responsible after occupation needs resources and workable management arrangements of its own.

Institutions for operations and maintenance

Maintenance requires an accountable organisation as well as a budget allocation. Someone must own the drain, inspect the roof, collect the rent, procure the repair, enter the

settlement, answer the complaint and carry the political responsibility when the service fails. Every affordable-housing project should identify six items before capital approval: the asset owner; the service standard; the staffing model; annual operating cost; major replacement reserve; and revenue source. For social rental add tenant management and arrears. For condominium ownership add association governance and reserve contributions. For upgrading add municipal access and security. This requirement changes project economics. A design with lower capital cost and high maintenance can be more expensive over twenty years. A peripheral site with cheap land can require a costly new utility district. A low service tariff can be unaffordable to the utility. Lifecycle affordability belongs in the appraisal. Digital workflows can support that organisation when records correspond to decisions it can actually make.

Digital workflows and accountable decisions

Stable identifiers can connect housing records and make decisions and handoffs easier to trace. One parcel or project identity can link ownership and occupancy, zoning, infrastructure capacity, environmental risk, permit status, procurement, construction progress, beneficiary status, financing, occupancy, complaints and maintenance. Reduce repeated requests for the same information. Once a title or utility capacity statement has been verified, other authorized agencies should consume the same record. Date each version and record changes in rights and capacity. The second gain is workflow. A useful system records who owes the next decision, when the clock started, what rule applies, which evidence is missing, what deadline governs, and who authorized any exception. Managers can use the record to locate delays and check their causes. AI tools may reduce some information costs if validated locally. Potential uses include helping staff to reconcile parcel records, classify applications, forecast approval delays, detect duplicate beneficiaries, compare bids with cost benchmarks, identify construction progress from imagery, flag payment anomalies, predict maintenance failures and cluster citizen complaints. Automating those decisions increases the importance of correction and appeal.

An automated rejection can repeat a mistaken or discriminatory decision across many applications. A geospatial system that identifies informal occupancy can support service planning and can make eviction easier. A fraud model can detect duplicate claims and can penalize atypical households. Purpose limits, reason codes, correction rights, human appeal and audit belong in the housing design. The resulting records should help government identify when the main delivery constraint has changed.

Identifying constraints during implementation

Implementation needs repeated diagnosis because resolving one bottleneck can expose another. Once title improves, finance may become the bottleneck. Once planning is liberalized, trunk infrastructure may bind. Once infrastructure expands, developer capacity or construction finance can become scarce. Review spending as the main constraint changes. The integrating institution should therefore run a quarterly constraint review. For each major city or program, identify the next binding gate using observed data: time to developable land, approval queues, network capacity, construction cost, unsold inventory, rent burden, mortgage rejection, occupancy, commute, service reliability and maintenance. Budget and policy should follow the

constraint. When supply is blocked by planning, spending more on buyer subsidies risks price capitalization. When permits are abundant and developers cannot obtain construction finance, liquidity facilities can matter. When completed units are vacant, investigate their location, price, condition, tenure, and service access before financing more of the same product. Vacancy in one segment does not establish adequate supply for every household. This is where a shared information system becomes economically useful. It enables government to stop treating every housing shortage as a shortage of money. Changing the response may disturb existing authority and benefits, which must be addressed openly.

Implementation risks

Institutional reform needs lawful authority, published decisions, and review because it changes established interests. Public land inventories threaten opaque allocations. Parallel approvals reduce agency vetoes. Shared data expose inconsistencies. Performance grants may withhold resources from municipalities that fail the conditions. User charges can anger residents. Property-tax reform can raise taxpayers' bills. Maintenance budgets compete with politically attractive new construction. Three design moves improve feasibility. First, pair the loss of authority or resources with a specified benefit: capital grants for municipalities, faster approvals for developers, tenure and services for communities. Second, sequence reform so affected actors can adapt: phase new standards, grandfather defined commitments, and move new budget growth toward the better system. Third, publish performance and exceptions. Public reporting allows residents and oversight bodies to challenge departures from those rules. Conflicts over land, utility finance, and municipal powers may require decisions beyond the authority of technical staff. The lead institution needs lawful authority to resolve disputes, and its decisions need published rules and independent review. Delegation should follow demonstrated capacity and funded public obligations.

Implementing institutional reform

Build participation and technical capacity before delegating funds, while retaining public responsibility for network services. Budget facilitation, audits and recurrent support as programme costs, rather than assuming unpaid resident labour. The lead agency must settle inter-agency disputes; community organisations must disclose accounts; utilities must deliver their commitments. Track time from agreement to occupancy, unresolved handoffs, representative participation, service performance and exclusion from membership. The choice of institution should follow these tasks rather than precede them.

Step	Action	Why this order matters
1	Define the household outcome	Use occupied, serviced, accessible housing as the end result
2	Map the delivery chain	List every decision, data source, owner, deadline and veto

Step	Action	Why this order matters
3	Identify the binding gate	Target the constraint identified by the evidence
4	Name one accountable integrator	Give convening and escalation authority
5	Set clocks and decision rights	Turn recurring negotiation into rules and timed exceptions
6	Align finance	Make transfers and budgets reward the same result
7	Create the shared record	Use one parcel/project identity with versioned evidence
8	Build local and community capacity	Add information and legitimacy where formal systems are weak
9	Fund O&M; before new capital	Prevent deterioration from becoming the next housing crisis
10	Audit distribution	Track who gains from land, subsidy, approvals and digital decisions
11	Review the next constraint	Move policy when the bottleneck moves

Identify the decisions, powers, and resources required before creating an authority, committee, or platform. Choose the institutional arrangement after identifying the decisions, powers, and resources needed. The assessment below tests whether those arrangements form a complete delivery sequence.

Housing delivery assessment

Gate	Question	Evidence required
1 Outcome	What household result defines success?	Occupancy, housing+transport cost, service reliability, job access
2 Integrator	Who owns the end-to-end result?	Named office, decision rights, escalation power, public blocker list

Gate	Question	Evidence required
3 Land	Is the site legally and socially usable?	Title, occupants, claims, hazard, public-land value
4 Planning	Are development rights clear and rule-based?	Parcel rights, approval clock, reason codes, exception log
5 Infrastructure	Will capacity arrive before occupancy?	Utility capacity, capital plan, connection date, O&M; owner
6 Finance	Is the full capital stack closed?	Public subsidy, private capital, guarantee, municipal liability
7 Procurement	Can the market deliver competitively?	Bidder depth, cost benchmark, payment time, change orders
8 Tenure/allocation	Will the target household use the product?	Demand, eligibility, tenure fit, waitlist, exit risk
9 Inclusion	Who can be screened out or displaced?	Tenant map, social-group incidence, grievance and appeal data
10 Neighborhood	Are jobs, schools and transport part of the product?	45/60-minute job access, fares, school/service access
11 O&M;	Who keeps assets working for 20+ years?	Asset owner, tariff/rent, staffing, reserve, service standard
12 Political economy	Who gains, loses and can veto?	Actor-rent-veto-coalition map and sequencing plan
13 Digital power	What is automated or newly visible?	Purpose, access, explanation, correction, appeal, audit
14 Verification	Who checks claimed results?	Independent protocol and public reporting
15 Learning	How does policy change when the constraint moves?	Quarterly bottleneck review and adaptation rule

Readiness ratings should expose critical failures rather than average them away. Any unresolved issue that can block occupancy, safety, or long-run service must have an assigned and funded remedy before approval. A decision-maker cannot waive statutory or life-safety

requirements through a risk rating. Do not average the scores. A strong building cannot compensate for disputed land; a strong subsidy cannot compensate for a two-hour commute. A scale decision requires evidence that the whole sequence can operate under accountable leadership.

Investment criteria

An affordable-housing program is ready for scale when land, planning, infrastructure, finance, procurement, tenure, neighborhood access and operations and maintenance form a credible sequence under one accountable integrator. Tie part of public finance to verified institutional performance. Put community institutions inside the system where official information is weak. Record discretionary exceptions and automated decisions, identify the decision-maker, and provide audit and appeal routes. That judgment must distinguish documented programme results from the proposed diagnostic framework.

Limits of the evidence

The cited programmes provide evidence to test the coordination framework, not proof that one institutional form will work everywhere. The quantitative evidence in this chapter comes from current World Bank, Government of Odisha, World Habitat, CODI, IDB, HDB and URA sources. The evidence differs in strength. Performance-grant results are mainly administrative and program data. Favela-Bairro provides a long-term comparative maintenance assessment. Jaga Mission and Baan Mankong provide strong institutional case evidence with less causal identification. Singapore provides a high-capacity benchmark rather than an estimate of a portable treatment effect. The illustrative probability model, four coordination functions, responsibility map, constraint review, and housing delivery assessment are proposed analytical tools. Government and MDB teams should use them to identify unresolved decisions before committing additional funds. Fieldwork should trace actual projects and unresolved handoffs before further funding is committed.

Prepare the local decision

Follow decisions from agreement to occupation and measure resolved handoffs rather than meetings or registrations. Trace recent completed and stalled projects from land approval to occupancy. For each handoff, record receipt date, missing documents, decision date, legal authority, and expenditure released. Compare formal processing rules with what staff actually do. Use the Odisha and Thailand cases above to examine whether residents' information reaches officials with authority to change the project. Convene the officials who sign decisions and the staff who prepare them, alongside utilities, municipal finance, developers, and tenant representatives. Identify which delays arise from missing evidence, incompatible rules, unfunded duties, or disagreement over rights. Give each unresolved handoff a named decision-maker and an escalation route with a lawful mandate. Start with a shared dated case register. Add automated reminders only after deadlines and responsibilities are agreed. Audit a sample of records against correspondence and site visits; a closed task may still leave the next agency unable to proceed. Test the revised handoffs on a small project cohort and measure waiting time, rework, disputed

decisions, and time to occupied serviced homes. Do not count meetings or portal registrations as delivery improvements. Chapter 9 follows those responsibilities into the maintenance and continued affordability of the completed stock.

Decision authority should sit with institutions that possess usable information and can be held responsible for delivery. Prior organisation, secure land arrangements, real authority, and paid technical support explain why community-led approaches can work. Forming a committee at handover does not reproduce them. Tenant exclusion, unfunded duties, and no remedy against failed leadership can make delegated responsibility ineffective, whatever the organisational chart says. Going from one settlement to 200 cities requires intermediary organisations, technical staff and independent oversight; local trust cannot be manufactured by decree.

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Operational use. Put the 15-gate Housing Delivery Chain Test on the wall at concept review. Require one named owner, one dated piece of evidence and one escalation path for every gate. The exercise usually finds a housing problem hiding inside an institutional handoff.

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9. Maintaining affordable housing

Preserve existing affordable housing by funding the repairs and management needed to keep it usable. New construction adds little net benefit if a comparable stock becomes unsafe, vacant, or unaffordable. Assess losses, match management arrangements to tenure and incentives, and protect residents during renewal. Conditions after occupation provide the first evidence of whether those arrangements are working.

Formal completion does not establish that a home remains safe, serviced, and affordable after occupation. Cooperative housing societies are responsible for collecting fees and maintaining common services. The study discussed below examines how residents and committees meet those obligations when pumps, lifts, or other shared assets fail and some households cannot afford the charge. A 2024 study by Robert Chaskin, Mouleshri Vyas and Manish Jha examined two such colonies, Majas and Lallubhai Compound. Lallubhai comprises 70 buildings and roughly 100,000 residents in a poorly connected part of the city. Majas, a smaller 16-building settlement, sits closer to transport and amenities. Researchers found that formal housing brought new rights and new obligations. Cooperative societies were expected to collect fees, handle disputes and repair buildings. Some residents distrusted their leaders; some leaders used utility disconnection to compel payment. An initial maintenance endowment of INR 20,000 per resident was meant to support upkeep, yet interviewees described funds that were inadequate, delayed or never received.[1] Affordable housing can be lost when existing dwellings become unsafe, vacant, disconnected, too expensive, or unavailable to the intended income group. Housing accounts need to measure these losses alongside new completions. Housing accounts must therefore track losses from the usable stock as well as additions to it.

1 Measure housing losses and continuing costs

Costs after occupation

Additions to and losses from the housing stock

New construction can be offset by deterioration when repair and replacement obligations are left unfunded. A city adds homes through new construction, extensions, conversions and upgrading. It loses effective affordable housing through physical failure, vacancy, tenure insecurity, rising rents, displacement, hazard, and the withdrawal of subsidies. Some losses are reversible. A vacant unit can be repaired; a seized building can be recovered. Others, such as a household pushed away from its job market, can impose lasting damage. For a defined household group and period, usable affordable housing capacity at the end equals capacity at the start plus additions, less losses through deterioration, exclusion, vacancy, or change of use. Track this stock in occupied dwellings meeting stated standards. Over time, measure household-years of adequate, well-located shelter as the service produced. A sewer failure, rent increase, or flood can reduce usable affordable capacity even when the building remains. Capital grants buy assets at a point in time. Operating institutions determine how many years of shelter those assets produce. When a government rewards a housing agency for delivery at handover, the incentive can favor lower

initial costs without accounting for later lift replacement or other repairs. The eventual bill falls on a cooperative, landlord, utility or household with far less bargaining power. Rising land values can conceal that deterioration while changing who can afford to remain.

Land values can rise while buildings depreciate. Better transport, safer streets, improved drainage and growing employment can make a neighborhood more valuable. It also invites higher rents, redevelopment pressure and exclusion. The neighborhood can become more successful as an asset and less accessible to the households the project was meant to serve. A replacement-reserve plan must therefore be tested against both operating cash flow and household affordability.

Financing replacement reserves

Regulatory benchmarks can support a maintenance stress test, provided hypothetical calculations remain distinct from project forecasts. Its 2025/26 Annual Performance Plan reports an earlier sector assessment in which rent collection fell from 90 percent in the second quarter to 85 percent in the third; vacancy improved from 8.6 to 7.2 percent. The regulator's benchmark was at least 95 percent rent collection and no more than 2 percent vacancy. These are historical quarterly figures reported in a plan published in 2025, rather than current October 2026 operating results.[2] Consider 100 hypothetical rental flats with a monthly face rent of US$100. With 7.2 percent vacancy, 92.8 homes are billed on average; collecting 85 percent of billed rent yields US$7,888 per month. At 2 percent vacancy and 95 percent collection, the same buildings yield US$9,310. The gap is US$1,422 a month, or US$17,064 a year. These are hypothetical rents with actual reported sector ratios used as stress-test assumptions; they are not a South African project forecast. A funding gap must be resolved without treating tenant security and provider viability as separate problems.

Operating shortfalls threaten repairs and financial survival while poorly designed responses can threaten tenants. Reducing maintenance can conceal the shortfall for a few years. It raises future repair costs and weakens tenant willingness to pay, and further reducing revenue. Aggressive collection can improve accounts and also punish households suffering income shocks. The regulator therefore needs a dual test: financial survival of the provider and security of the tenant. Subsidy design can help align affordability with the discipline needed to maintain the asset.

Figure 9.1. Rental revenue under different vacancy and collection assumptions

Assumption or result	Reported Q3 conditions	Regulatory benchmark
Units / face rent per month	100 / US$100	100 / US$100
Vacancy	7.2%	2.0%
Collection of billed rent	85.0%	95.0%
Monthly cash received	US$7,888	US$9,310

Assumption or result	Reported Q3 conditions	Regulatory benchmark
Annual cash received	US$94,656	US$111,720
Gap from benchmark	US$17,064 a year	—

Illustrative calculation. SHRA historical ratios are used only as scenario inputs; no exchange-rate conversion or claim about a representative rent is made.

A maintenance subsidy should fund an auditable service obligation and a credible whole-life cost plan. A project may need a larger capital write-down so that sustainable rents can cover routine operating costs and a lifecycle reserve. Some households may also require income support. A maintenance reserve should be held in a ring-fenced account with approved uses, audited balances and a replacement schedule. Assigning maintenance to residents requires an estimate of affordable contributions, external support, and repair costs. The developer has incentives at design stage that differ from the operator's. Cheaper materials can lower the construction bid and increase future repair bills. A developer that also operates the buildings for twenty years internalizes more of the trade-off. Performance specifications and long-term warranties can partly replicate that incentive when construction and management are separate. At appraisal, finance ministries and lenders should demand an audited whole-life cost model, rather than accept the lowest initial price. Owner-occupied buildings require a collective institution able to perform those tasks.

2 Match management arrangements to tenure and incentives

Collective management in owner-occupied buildings

Collective ownership needs practical support for accounting, decisions, collections, and repairs after handover. The exterior wall, roof, staircase, lift, water pump and drain are shared assets whose maintenance requires collective decisions. A resident can withhold fees and still benefit when neighbors pay. A committee can divert money. An owner who plans to sell soon may oppose spending that preserves the building ten years later. A tenant may have no vote although she bears daily service failures. The Mumbai study interviewed 89 residents across Majas and Lallubhai. It found uneven trust and participation, contested elections, reports of favoritism, and heavy demands on unpaid leaders. The study describes relocated households being asked to assume the legal burdens of common-property management while facing income insecurity and incomplete state support.[1] The policy fix begins before occupancy. Residents need an accurate asset register, an engineer's schedule of likely replacements, simple accounts in the local language, independent signatories, grievance procedures and a funded handover period. The authority should verify that the original endowment has reached the society and that the elected committee can procure and supervise repairs. Technical support should taper only when those tasks work in practice. Membership and voting rules determine whose priorities that institution represents.

Rules on membership and voting determine whose interests a housing association represents. Voting restricted to titleholders can exclude tenants, women not named on deeds, and migrants. Existing discrimination can also affect cooperative decisions. Residents need a way to appeal beyond the managing committee. Building management also depends on neighbourhood networks whose failure can make individual homes unsafe.

Neighbourhood drainage in Nairobi

Drainage and sanitation require operating agreements across households, utilities, roads, waste services, and wider catchments. In April and May 2024, heavy rain in Nairobi demonstrated how neighborhood drainage and sanitation affect household safety. Water ran through settlements built along rivers and low ground, carrying waste from blocked drains and damaged sanitation systems. On 9 May, Nairobi City County estimated that 20,968 households had been affected and opened camps across seven sub-counties. A 2026 World Resources Institute study links the damage to rainfall interacting with settlement location, paved surfaces, encroached wetlands, waste blockages, weak drainage and split responsibility for risk management.[3][4] A household can repair its roof and still be flooded because an upstream drain is blocked. The landlord can improve a compound toilet and still face contamination because the downstream sewer fails. The city can install a drain that works on opening day and fails after two seasons because no agency keeps the channel clear. Environmental resilience is therefore an operating relationship among households, solid-waste collectors, water utilities, road departments and basin managers. Upgrading projects should record those duties and funding sources against each asset.

Infrastructure and tenure interventions need an asset record that names the operator, service duty, and recurrent funding. The program reports about 1.36 million beneficiaries, 23,397 titles and 98 km of roads under its earlier phase. KISIP2 combines tenure regularization with infrastructure upgrading and institutional capacity under national and county arrangements.[5] These output figures do not, by themselves, establish the lifetime condition of drains or the fate of tenants after an area improves. For each upgraded settlement, counties should agree an asset boundary with the water company and other providers: who owns the drain, who pays the sweeper, who clears illegal dumping, who repairs after a flood, and which budget releases emergency funds? A geotagged asset ledger should record an enforceable service duty, responsible operator, and recurrent funding source for each asset. Where risk still requires relocation, the agreement must also protect rights and livelihoods.

Relocation for climate safety also raises rights and livelihood questions. A home in a river corridor may impose unacceptable risk; leaving it in place can be indefensible. Demolition without safe, accessible alternatives can move households into another floodplain or force them into higher rents. Risk maps should trigger graduated decisions: protect in place where feasible, retrofit where risk can be reduced, and relocate where life safety demands it. Every route needs tenant enumeration, compensation, a feasible destination, lawful claims procedures, and transparent control of vacated land. In safer locations, public improvements can create a different risk by increasing the rent households must pay.

Repairs, rents, and landlord incentives

Neighbourhood improvements can raise rents and require a local assessment of who captures the benefit. Who receives the gain depends on tenure and competition. An owner-occupier enjoys the service and the rise in asset value. A landlord can raise rent or enjoy lower vacancy. A tenant may gain a cleaner street and lose the home when the lease ends. A structure owner in an informal settlement may capture public value even without holding the land title. Marie Huchzermeyer's study of Nairobi slum upgrading showed how commercialization of shelter and services complicates the distribution of upgrading benefits. Andrea Rigon's ethnographic work on a Nairobi upgrading program found that participatory enumeration and community structures could entrench existing inequalities between landlords and tenants. These are qualitative institutional findings, not measured citywide estimates of rent pass-through.[6][7] Nor does every upgrade immediately raise market values. A study of Rio's Favela-Bairro found no statistically significant average increase in dwelling values, while some initially disadvantaged properties appreciated. Legal tenure, security, access and the wider market all affect capitalization. A government should test the local incidence of improvement rather than assume a fixed coefficient.[8] Tenant enumeration and an explicit protection strategy should precede the investment.

Before paving the street, enumerate tenants separately from structure owners. Record current rents, lease terms, deposits and households' contact details. Offer tenants notice, temporary relocation support where works require it, and a route to dispute arbitrary exclusion from return lists. Where rents rise quickly, expand legal low-cost rental nearby and consider a time-bound protective rule paired with supply expansion. Monitor who remains two, five and ten years later. The reform coalition must include small landlords as producers as well as tenants as consumers. Offer owners simple permits, staged loans and bulk service connections for added rooms in return for minimum safety, transparent rent terms and inspected work. A policy that threatens the entire informal rental business will struggle to survive; one that ignores tenant rights can spend public money to improve the bargaining position of landlords. Vacant housing presents another case in which existing buildings do not translate automatically into usable homes.

Housing vacancy and location in Mexico

Mexico's recovery programme illustrates why targets, intermediate actions, and homes returned to use must be counted separately. The OECD's 2024 economic survey reports that roughly one-seventh of the national housing stock, between five and six million units, was vacant, based on national data. The estimate includes housing of many kinds; it must not be equated with abandoned subsidized developments. Still, weak service access, distance from work and violence were important factors behind abandonment. Some social developments were in locations where households could not sustain the costs of living and reaching work.[9] Repairing the stock has proved hard. In Tlajomulco, the municipal Renta tu Casa initiative had rented only 90 homes over three years against a reported stock of 17,000 vacant units. Agreements signed from 2020 to July 2023 aimed to recover about 131,000 abandoned dwellings nationally, yet the OECD noted slow progress caused by property disputes, municipal capacity

limits and the politics of three-year local mandates. These are targets and partial implementation results, not recovered-house totals.[9] Recovery also depends on restoring the neighbourhood conditions that make continued occupation feasible.

Vacant-stock recovery needs dedicated responsibilities, budgets, and measures of sustained occupation. Deteriorating services and security may cause further departures. A recovery plan must rebuild the service environment and address security, mobility and legal claims alongside repairing walls. Public money should enter only after a credible estimate of how many households will occupy the recovered homes, at what ongoing cost, and with what prospect of staying. Recovering abandoned housing requires property claims to be resolved, buildings and infrastructure repaired, and prospective residents' demand assessed. Those tasks differ from procuring new construction. The Mexico evidence supports giving recovery programs their own budgets, responsible agencies, and measures of sustained occupancy.[9] Vienna provides a contrasting example of institutions designed to retain and maintain housing over a long period.

Long-term housing institutions in Vienna

Austria's cost-rental institutions provide descriptive evidence of lower rents under a distinct long-term housing arrangement. Limited-profit housing associations hold or manage more than two-thirds of the country's social and affordable housing stock. They set rents on a cost-recovery basis that includes finance, management, maintenance and refurbishment. Profits are capped and earnings retained for housing purposes under a system of audits and regulation. The associations have an institutional reason to preserve the asset rather than extract the highest possible current rent.[10] The OECD's 2026 review reports that, among dwellings constructed since 2011, rents in Austria's for-profit sector were 74 percent higher than in limited-profit dwellings; in Vienna the gap reached 91 percent. This is an observed comparison across segments, not a causal estimate of what one particular tenant would save if the same apartment changed ownership.[10] Their transfer depends on the financing and reinvestment institutions that sustain that arrangement.

Cost rents require institutions able to collect income, reinvest, enforce obligations, and fund upkeep. Housing Europe's Housing2030 case study describes building-specific renovation funds alongside retained surpluses used to invest in additional housing. Revenue earned from a property can return to the property when major systems wear out. A provider with a stock of buildings, patient capital and regulated earnings can plan repairs and investment over long periods.[11] The Austrian model also requires decisions about eligibility and allocation. Long-standing tenants may gain valuable security; outsiders may wait. Broad eligibility sustains a middle-class coalition and helps avoid deep concentrations of poverty, yet lower-income households still need access in high-demand areas. Public support, tax treatment and land allocation need scrutiny. Its transferable principle is the duty to preserve a long-lived affordable stock, backed by audited reserves and a stable provider balance sheet. Copying cost rents without institutions able to collect, reinvest and enforce would leave the proposed cost-recovery system without a workable operating basis. Comparing Vienna and Mexico helps identify the different obligations needed to preserve use and affordability.

The Vienna and Mexico cases support examining long-term operating obligations without treating their different measures as equivalent impacts. Vienna's providers collect and reinvest income over the life of their stock; recovering vacant Mexican homes requires resolving location, services and legal claims as well as repairing buildings. Their outcomes cannot be attributed to ownership form alone. A local adaptation needs a viable operating budget, usable location and enforceable responsibility for renewal. The rent comparisons above are descriptive, and Mexico's vacancy total includes housing beyond subsidised developments. [9–11] Those obligations redistribute responsibilities among residents, providers, and government after occupation.

3 Fund renewal and protect residents

Interests in neighbourhood management

Neighbourhood management must account for the people controlling services and the residents affected by their decisions. Landowners, developers, lenders, and permit authorities shape development; managers, associations, utilities, and maintenance contractors determine much of its subsequent performance. Check whether tenants have enforceable service rights and a route to challenge decisions affecting them. Four recurring conflicts determine whether affordability survives. Maintenance: the person who must pay today may leave before the repair pays off. Appreciation: owners and landlords can capture a public improvement that tenants helped finance through taxes. Renewal: a city wants to rebuild an aging or risky block, while residents fear permanent displacement. Entry: current owners can use building rules, zoning, security practices and political pressure to exclude lower-income households or new rental supply. Association rules, landlord screening, and enforcement can discriminate by race, caste, or class. A residents' association can demand parking and low density in the name of order; the same rule raises the entry cost for poorer households. A cooperative board can refuse subletting to certain groups. A landlord can discriminate through screening. Enforcement should be checked for unequal treatment of informal settlements and other development exposed to the same hazard. Credible inclusion policy must trace decisions to people, not stop at formally neutral rules. An operating agreement should bring those actors together around affordable, reliable use.

Table 9.1. Interests and responsibilities after handover

Actor	Rent, risk or objective	Veto point	A possible agreement
Incumbent owner	Capital gain; low fees; neighborhood control	Votes, association rules, redevelopment consent	Reserve fund + transparent renewal gain + protected return
Tenant / migrant	Affordable rent; stability; job access	Weak formal veto; can exit or protest	Notice, appeal, rent data, alternative units and return rights

Actor	Rent, risk or objective	Veto point	A possible agreement
Small landlord	Yield from improved location; low compliance cost	Can increase rent or withhold rooms	Finance for safer additions, enforceable leases, fair inspection
Utility / municipality	Cost recovery, service duty and local fiscal balance	Tariff, connections, repairs, local approvals	Ring-fenced O&M; and transfers tied to occupied serviced units
Developer / renewal contractor	Construction margin, land uplift, density rights	Project viability, finance, claims, delay	Competitive tender, open-book value, milestones, resident protections
Resident committee / broker	Fee control, allocation information, local influence	Records, membership, building access	Audited accounts, rotation, independent appeal, tenant voice
Future resident	Affordable entry into an opportunity area	Usually absent from local proceedings	Metropolitan housing targets and by-right supply

A workable management agreement should include residents, landlords, utility workers, municipal finance teams, employers, and transport providers where their decisions affect services. It must assign maintenance duties and budgets after handover. For renewal or clearance proposals, publish who benefits, who must move, who can return, and who receives any increase in land value. Physical integration is lost if low-income buildings become unsafe or residents cannot pay common charges. When ordinary maintenance is no longer sufficient, the agreement must extend to renewal and resident protection.

Renewal and resident protection

Renewal needs enforceable terms for temporary accommodation, return, costs, and disputes before redevelopment proceeds. Buildings require seismic strengthening, flood protection, new pipes or replacement. Neighborhoods need more floor space as household size, migration and jobs change. Governments then face a choice among rehabilitation, selective redevelopment, densification and relocation. Each route creates winners and losers. Rehabilitation works when the structure remains safe and the service system can be renewed. Selective replacement works when some buildings are beyond repair and temporary accommodation can be secured. Densification can finance better infrastructure through added development rights. It can also create a residual claim on land that prompts developers to remove lower-income

residents. Peripheral relocation may lower immediate land costs while destroying access to work. A renewal contract should therefore contain a return agreement before a developer is selected: precisely which occupants qualify, how tenants and owners are treated, how long temporary housing is funded, what happens if the developer stops paying, and who bears the cost of delays. Land value and added floor space should be independently assessed. Release development rights or subsidy in stages against verified rehousing and service outcomes. Residents need a right to see the contract and a forum outside the developer's control for disputes. Negotiation must distinguish non-negotiable safety requirements from preferences and distributive claims.

Existing owner coalitions can also block beneficial density. A pensioner fears a levy for a new lift; an owner hopes for a high redevelopment payout; renters want security; neighbors oppose taller buildings; municipal officials may welcome land-value gains. Transparent valuations and a sharing rule for uplift make bargaining possible. The state should distinguish safety rules that cannot be traded from design preferences and distributional claims that can be negotiated. Climate change adds another reason to reassess whether continued repair remains sufficient.

Climate risk and maintenance

Future exposure and household operating costs should inform maintenance and renewal choices. Heavy rain, heat and changing flood paths shift the service standard. A drain sized for an earlier rainfall pattern may fail more often. A roof that was cheap to install may trap heat and raise the household's electricity burden. A wall that survives dry seasons may crack after repeated wetting. Climate risk changes the expected lifetime of assets and the amount that must be set aside for replacement. The World Bank's April 2026 review of resilient, green and inclusive buildings reports that Tamil Nadu's 2020–25 housing and habitat operation built approximately 15,000 affordable units while using resilient and green design and reforms to housing institutions.[12] That experience points to a useful appraisal rule: price energy, cooling, flooding, insurance, repair and transport over time. Compare initial costs with households' recurring costs and expected losses. The preferred option must also account for its different effects on owners and tenants.

Assess how each risk-reduction option affects owners and tenants. Stronger drainage raises land value. Flood zoning can make some plots unbuildable. Household retrofits may be affordable to owners and unreachable to renters. Public investment can inadvertently speed displacement. Risk reduction needs a tenure plan, a safe location option and a source of money for continued operation. Reliable post-handover information is needed to detect those risks and assign action.

AI, maintenance information, and resident rights

Maintenance information should connect reported defects to accountable repairs while protecting residents from harmful secondary uses. A planning system asks what can be built. A neighborhood management system asks what is failing, who is affected, what repair is due, how rents are changing and whether intended residents still have access. These questions are repeated every month. A simple digital asset register can assign each pump, drain, lift and streetlight an owner, condition, inspection date, repair budget and public complaint route.

Household surveys can record service quality and affordability. Satellite and street imagery can flag physical change; water and power data can reveal abnormal service loss; maintenance histories can help predict failures. AI can sort complaints, compare repair bids, detect unusual work orders and forecast reserve needs. The savings must be established through a pilot with independent verification. When a housing committee uses electricity or water disconnection to enforce payment, a digital arrears system can make service disconnection easier to implement. Data rights must specify permitted uses, notice, error correction and independent appeal. In informal areas, separate the service register from enforcement action so that seeking a water connection does not automatically create a demolition case. Its value depends on both service performance and residents' ability to obtain a remedy.

The decision tool should be audited at both ends: have repairs become faster, and have residents become more secure? A low average repair time says little when the most vulnerable households cannot report a problem or appeal a decision. The management agreement should specify how those results will be reviewed over the asset's life.

Neighbourhood management agreement

Management arrangements should match the building type, resident means, and actual cost of the promised service. Combine affordable contributions with transparent targeted operating support and funded replacement obligations. Operators must inspect and repair; municipalities must provide networks; regulators must enforce service and affordability commitments. Track occupied safe home-years, failures and repairs, reserves against forecast need, total fees, arrears and displacement after renovation. Governments should append a twenty-year management and affordability agreement to every major housing or settlement-upgrading investment. It is a plain institutional agreement among the asset owner, operator, local government, utility, financiers and residents. It links physical performance to social access. The questions below form a test for concept review, completion review and each five-year renewal review. A baseline allows later reviews to distinguish progress from deterioration and exclusion.

Table 9.2. Neighbourhood management assessment

Gate	Question to put to the government and operator	Required evidence
1. Unit of success	How many adequate, affordable household-years will be sustained?	Occupied, adequate, serviced units by target group, tracked over time
2. Asset boundary	Who owns every common asset and off-site connection?	GIS-linked register, handover certificate, named operator
3. Cost	What is the 20-year cash requirement?	Repairs, staffing, energy, insurance, climate allowances, renewal schedule

Gate	Question to put to the government and operator	Required evidence
4. Reserve	Where is the replacement money held?	Ring-fenced account, actuarial schedule, independent audit
5. Affordability	Can the target household pay total recurring cost?	Rent/fees + energy + water + transport; shock scenarios
6. Collection and rights	What happens when a household cannot pay?	Arrears protocol, hardship support, limits on service cutoff, appeal
7. Tenant incidence	Who receives the value of public improvements?	Baseline and follow-up rents, leases, owner/tenant map, displacement
8. Inclusion	Who can enter, remain, and return after works?	Rules tested by income and relevant social group; tenant return rights
9. Hazard	Does the site stay safe as risks change?	Climate and structural risk map, trigger levels, financed response
10. Renewal	Who decides when to retrofit or rebuild?	Independent condition survey, votes, valuation, temporary housing plan
11. Local finance	Can city and utility fund recurrent services?	Tariffs, local taxes, performance grants, service contracts
12. Data power	Can digital records be challenged?	Purpose limits, public metrics, correction, offline access, independent appeal

Rating rule: a red gate that can interrupt water, safety, affordability or lawful occupancy blocks approval until a remedy is funded and assigned. A green construction schedule cannot offset an unfunded asset reserve.

The contract should begin with a baseline: household roster by tenure, rent and fee levels, service conditions, maintenance backlog, flood exposure, asset ownership and job access. By completion, identify every operator and fund the first years of upkeep. At year two, compare actual collections, vacancy, rents, repair time and displacement with the baseline. At year five, adjust the tariff, subsidy, design or provider. At year ten, commission an independent physical survey and plan renewal. Financing should follow performance over time. One option is to retain a modest share of the capital subsidy until a verified handover and operating reserve exist. Another is to tie municipal transfers to sustained occupancy, service reliability and inclusion, with protection for cities serving poorer households. Development banks can finance

asset-management systems and rehabilitation alongside new units. Appraisal should value sustained use of existing homes alongside new delivery. The strength of the available evidence determines what those comparisons can establish.

Limits of the evidence

Different source types support different maintenance conclusions, so evaluation should retain their limits and measure continued affordability. Mumbai's housing-society findings and Nairobi's accounts of participation are careful qualitative studies that reveal mechanisms; they do not estimate representative effects for all India or Kenya. South Africa's rent and vacancy numbers are reported sector monitoring data from a specific historical period. Mexico's vacant-stock estimates describe the wider market and cannot be treated as counts of abandoned subsidy units. Austria's rent gap is a descriptive comparison across different market segments. The Nairobi flood record documents a damaging event and the institutions that mediate risk. The worked financial example and the Neighborhood Management Agreement are analytic judgments offered for project use. The propositions can be tested. Randomly phase technical help to new housing societies and compare repairs, collections and grievances. Compare upgraded settlements with matched non-upgraded areas while tracking tenants as well as owners. Link asset condition to rent, vacancy, health and commute outcomes. Audit digital complaint systems for unequal resolution. And calculate the public subsidy per year of housing that actually remains affordable to the intended group. The shared test is whether both building condition and household access are being preserved.

Preserving affordability requires attention to both building condition and access as land values change. Track whether lower-income residents can remain, return after works, and enter the area. The field assessment should establish these conditions for the existing stock before choosing an intervention.

Prepare the local decision

Inspect assets, service records, fees, complaints, and tenant outcomes together before setting a preservation budget. Inspect a sample of occupied and vacant buildings by age, tenure, and management type. Record component condition, estimated remaining life, repair history, service interruptions, charges, collection rates, reserves, and reasons for vacancy. Include tenants who moved out. The Mumbai evidence and the Mexico vacancy discussion above require distinguishing a repair problem from unaffordable management charges or unsuitable location. Building managers, resident owners, tenants, utilities, and the municipality should identify who can authorise and pay for each repair. Verify reserve balances against bank records rather than approved budgets. Ask residents with disabilities whether lifts, paths, and water interruptions prevent them from using their homes. Give assets and work orders stable identifiers. Link each complaint to inspection, payment, repair, and a resident check that service resumed. Test predictive maintenance against a simple age-and-condition schedule before buying it. Adopt a funded repair sequence that starts with safety-critical failures, then review downtime, repeat defects, collection performance, and household departures. Falling complaint numbers require

checking whether reporting became harder. Chapter 10 examines how metropolitan growth decisions affect the locations and services available for additional homes.

Housing policy should finance safe, affordable occupancy over the asset's life. Stable operator revenue, renewal reserves, a useful location, and enforceable resident rights sustain this outcome. Vienna's ownership form cannot compensate for remote sites, ambiguous maintenance duties, or fees below necessary costs without support. Portfolio planning must account for the possibility that many buildings require major repairs together. Aggregate the replacement cycle and shortage of competent managers across the whole portfolio; many estates can need renewal in the same year.

Sources and evidence notes

Sources cited in the manuscript. Endnote numbers identify the evidence supporting the adjacent passage. Statistics are presented with their source period; old program figures are not treated as current stock estimates.

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10. Metropolitan growth and coordination

Metropolitan housing decisions should compare locations through household access and the full cost of services across municipal boundaries. The administrative boundary of the housing project rarely contains its labour market, transport network, or infrastructure obligations. A workable strategy combines location appraisal, development rights, and an agreement on shared costs. The household's daily journey shows why the scale of coordination matters.

A metropolitan housing strategy should compare homes through the access and recurring costs households experience across boundaries. Imagine a worker living on the edge of Nairobi. Her employer, landlord and bus operator all serve the same labor market. The planning office that controls the next apartment block may sit in another county. The road agency can widen a highway; the water utility must still connect the site; the county must maintain its drains. Without coordination among these agencies, the worker can face higher rents and fares, longer journeys, and unreliable services. A Nairobi study measures the access problem. A World Bank study using job-location data, the Digital Matatus network and household surveys estimated that the average resident could reach fewer than 10 percent of Nairobi's jobs on foot within one hour, and only around one-quarter by minibus. Homes with better job access commanded higher rents. Poorer residents and residents of informal settlements faced the weakest access [4]. Limited access can reduce the jobs available to a household and raise the cost of reaching them. A household may find cheaper shelter farther away and lose the saving on travel, foregone work, or both. Urban growth increases the number of decisions that must be coordinated to protect that access.

1 Compare locations across the urban region

Growth across municipal boundaries

Growth forecasts should inform shared housing and service decisions while representing renters and prospective residents. An OECD-led assessment projects Africa's urban population rising from roughly 717 million in 2020 to 1.4 billion by 2050, with urban land area more than doubling under its spatial scenarios [1]. For India, a 2025 World Bank assessment projects urban population approaching 951 million in 2050 and estimates that more than 144 million new homes will be needed by 2070 [2]. These projections use different definitions and horizons; adding them produces an indication of scale, not a harmonized dwelling forecast. Both point to the same practical task: millions of sites, streets, pipes and legal floor areas need advance planning and timely provision. The metropolitan agreement is a deal among the places that gain from growth and the places that must host it. A central district gains workers and customers. An outer county gains land development and perhaps property taxes. A transit agency gains riders. Incumbent owners can lose quiet streets, parking and exclusivity; landowners close to new infrastructure can gain a windfall. Low-income renters need access to the city yet have the least influence over the decisions that govern their future address. The physical options are infill, serviced expansion, and new development centres, each with different costs.

Three approaches to urban expansion

Metropolitan strategies should compare infill, planned expansion, and new employment centres using common demand, access, and infrastructure assumptions. Infill adds homes within the built-up city through subdivision, small rental buildings, redevelopment, and construction near jobs and stations. Its constraints include costly parcels, multiple owners, restrictive rules, and limited network capacity. The built-up core can also be hazardous. Where added density requires expensive pipe replacement, the public cost must appear in the comparison. Expansion prepares land beyond the existing footprint. New employment centres can reduce dependence on the old core when firms and workers have reasons to locate there. Each option needs a comparison of household costs, safe buildable land, and funded services. The choice also depends on how firms use the city.

Housing and business-location decisions should be assessed together. Firms can benefit from shared suppliers and infrastructure, better matches with workers, and knowledge exchange. These advantages depend on connectivity and compete with congestion, commuting costs and land prices. Grover Goswami and Lall's review explains why cities can develop several employment centres rather than a single dominant core. The World Bank's Africa's Cities report likewise argues for coordinating housing, transport and industrial investment. This evidence supports examining where firms can operate productively and workers can afford to live. It does not establish that greater residential density or a designated business district will generate growth. [226, pp. 2–5; 227, pp. 20–23]

Location requirements differ across firms and activities. The literature reviewed by Grover Goswami and Lall shows that some services benefit from close contact in central locations, while land costs and goods-transport requirements influence manufacturing location. The patterns vary by industry and country; evidence from developed economies cannot be applied mechanically elsewhere. For a proposed growth area, I would therefore test demand for business premises, reliable utilities, supplier and freight access, and housing affordable to the expected workforce. Existing supply chains and credible firm commitments should carry more weight than employment targets attached to a master plan. Compare a housing-first or industry-first sequence with coordinated investment before committing public capital. [226, pp. 14–16; 227, pp. 20–23]

Expansion planning should connect a citywide road network with workable neighbourhood layouts. Angel's published planning framework combines protection of environmentally sensitive land with an arterial network and subdivision of the larger areas between arterial roads into smaller, walkable blocks. It recommends using existing roads and property lines where possible. I would organise this as a division of work: public authorities secure the connecting routes and protected areas, while landholders, communities and developers work with planners on local streets, plots and public spaces. The layout must respond to drainage, terrain and existing occupation. This is a planning approach to assess locally, not evidence that a geometric grid by itself produces inclusive growth. [228, pp. 131–132]

Reserving a route and constructing its infrastructure require separate decisions and budgets. The Ethiopia Urban Expansion Initiative divided plans into phases intended to accommodate five years of growth. Its implementation sequence included engineering surveys,

compensation, demarcation and construction of simple roads. Some lightly settled land was reserved for later use; in occupied areas, residents were assisted in reorganising plot boundaries. The authors reported funding and institutional constraints as well as progress. These examples support adapting the sequence to local occupation and resources. They do not establish that all future road land can be acquired immediately or that reservation guarantees later servicing. [229, sections 'Plan development' and 'The creation of an arterial road grid']

Land arrangements must protect people whose claims are not recorded in a formal title. Before agreeing on road contributions or plot changes, identify owners, tenants, customary rights-holders and livelihood users, and explain how benefits, costs and remedies will be allocated. The World Bank's ESS5 guidance distinguishes people with formal rights, legally recognisable claims and no recognisable legal claim; assistance differs between these categories. It also calls for consultation and a grievance mechanism where ESS5 applies. I would require a documented agreement, a responsible authority and funds for the resulting obligations before treating a reservation or subdivision scheme as ready. Chapter 11 applies the corresponding risk and rights tests to protected land. [230, pp. 7–12]

These choices produce distinct forms of building. Four-storey walk-ups, row houses, courtyard rentals, high-rise flats and serviced incremental plots have different land and construction economics. Lifts, fire systems, foundations and common-area maintenance can make a taller building costly; large parking mandates and setbacks can make a low-rise one consume too much land. Materials policy, industrialized construction and developer finance matter after the city establishes where a dwelling can be built and who is likely to occupy it. Land scarcity and transport costs help explain why a cheaper building form can still produce a costly household location.

Land scarcity and transport costs

High housing and transport costs can reflect scarce usable development opportunities rather than a lack of raw land. The World Bank's Africa's Cities study finds that, after accounting for national income levels, housing costs in African cities were about 55 percent higher than in comparable cities elsewhere; transport prices were about 42 percent higher [3]. These are historical cross-country price comparisons, not present-day market quotes. Fragmented urban land markets make serviced locations scarce. Weak transport makes a distant location a poor substitute. A housing ministry can lower a unit's construction cost and leave this combined scarcity untouched. The price of a central apartment reflects several things: floor space, building quality, neighborhood safety, services and access to jobs. Some of that premium pays for genuine scarcity, including fixed land and expensive infrastructure. Part reflects rules. A height limit, minimum plot area, wide setback or parking mandate can prevent the city from adding homes at a location where households would pay to live. A developer then bids against other developers for the few legal projects, and the right to build becomes valuable in its own right. The price of a peripheral plot may also exclude infrastructure costs borne elsewhere.

Peripheral land should be appraised with the public network cost and the household's actual pattern of work and care. The central public budget may finance a cheap house on cheap land. A distant municipality must add schools and maintain drains. A national agency

builds the trunk road. The household pays higher fares and loses time. The land bill looks small because part of the cost has been assigned to other balance sheets. When cities appraise projects this way, peripheral development can appear cheaper in procurement while imposing higher total costs. The practical measure is generalized housing cost: the present value of shelter payments, utilities, transport cash outlays, time spent travelling, service unreliability, climate exposure and relocation risk. Governments should calculate it for distinct income and tenure groups. A dual-income family has a different job-access map from a single worker; a domestic worker with several clients values a different network from an office commuter. A worked comparison makes the effect of those recurring costs explicit.

Comparing development locations

A consistent present-value comparison can reveal trade-offs hidden by initial land and construction prices. Consider three hypothetical ways to house 10,000 low-income households. Each option supplies an adequate dwelling. All figures below are illustrative US dollars per household. Housing cost includes the land and top structure; public infrastructure cost is counted separately; monthly mobility combines fares and travel time valued using an explicit shadow wage. Assume a 20-year horizon and a 5 percent annual real discount rate applied at 5/12 percent per month. The present-value factor for a one-dollar monthly cost is 151.5. The table below applies the same assumptions to three hypothetical locations.

Option	Housing cost	Public infrastructure	Mobility / month	Total 20-year social cost
A. Infill near jobs	$26,000	$4,000	$80	$42,122
B. Serviced extension	$20,000	$8,000	$180	$55,275
C. Remote cheap land	$16,000	$12,000	$280	$70,427

Illustrative model only; it describes a method for appraisal, not costs observed in an African or Indian city. Totals add housing, public infrastructure and present-value mobility costs. Time valuation is an explicit assumption and should be sensitivity-tested.

Initial land savings can be outweighed by recurring household travel costs. The distant option saves $10,000 in land and construction relative to infill. It requires $8,000 more public infrastructure and imposes roughly $42,427 in present-value mobility costs, compared with $12,122 for infill, a difference of $30,305. The full cost per household is about $28,300 higher. At 10,000 households, the difference approaches $283 million. Change the travel-time valuation and the result changes; so report a sensitivity analysis. A city that counts only the housing agency's capital budget can favor sites with lower capital costs but higher total costs. The household cannot pay a large up-front premium for good location even when that location would save public money and commuting costs over time. The state may need to assemble land, finance enabling infrastructure, use a land lease, or support rental housing. Where the expected benefits

justify the cost, identify who can finance the initial expenditure. Evidence on actual household moves provides a further check on what the calculation omits.

Observed relocation outcomes reinforce the need to assess livelihoods and access alongside physical housing improvements. The US Moving to Opportunity experiment found later earnings gains for children who moved young; the Ahmedabad housing lottery improved dwellings without measured income or human-capital gains over the long follow-up. The populations and interventions differ. Together they show why the appraisal must test usable opportunity and the cost of losing networks, rather than assume either relocation or proximity benefits every household. Chapters 18 and 25 examine the household evidence in more detail. [106,107] Nairobi shows why those connections often require agreements across several jurisdictions.

2 Learn from metropolitan growth and reform

Metropolitan coordination in Nairobi

Nairobi's functional housing market extends into Kiambu, Machakos, Kajiado and Murang'a. Kenya created the Nairobi Metropolitan Area Transport Authority in 2017 to coordinate integrated public transport across these five counties [5]. Its metropolitan council includes national transport and treasury officials as well as county governors. The arrangement addresses journeys that cross county boundaries: daily commuters and traffic cross boundaries that each county cannot manage alone. Transport coordination is necessary and partial. Housing sites, zoning, development approvals, land rates and local roads remain tied to other authorities. A county can approve a large estate on its side of the boundary. The national highway may carry commuters toward Nairobi. The water utility faces a new connection bill. No single investor has a contract to deliver the entire housing-and-job-access outcome. A corridor agreement can align those fragmented decisions around common population, service, and employment assumptions.

A metropolitan agreement should connect growth forecasts and infrastructure commitments with transparent land terms and resident representation. It would link station investments to legal rental and ownership floor space within a defined catchment. County housing targets should follow corridor capacity and job access, with transfers compensating counties that serve a wider labor market. The political obstacle lies in who gains. A peri-urban landowner can capture the uplift when a new interchange or pipeline is announced. A developer values rapid approval. An incumbent neighborhood can oppose small apartments while still employing the workers who need them. Minibus operators may fear the loss of routes or income under transport reform; they can also become part of a regulated feeder and operating system. A feasible agreement needs specified benefits for these actors, transparent land transactions and a direct voice for current tenants and future residents. An accessibility atlas can turn that agreement into a set of feasible sites and preparation tasks.

A land and accessibility atlas should identify feasible sites, required works, funding commitments, and the households expected to benefit. Use it to compare options and

assign preparation tasks; update it when service or cost assumptions change. India illustrates how divided urban powers and weak municipal finance complicate such agreements.

Urban growth and municipal finance in India

Municipal finance determines whether additional development is treated as a source of opportunity or an unfunded service burden. For metropolitan coordination, the immediate task is to reconcile investment needs with each municipality's ability to prepare, fund, and execute works. A regional growth plan should identify who pays and who can deliver each cross-boundary connection. [2,6] The fiscal institutions constrain the housing response. The Reserve Bank of India's 2024 review of 232 municipal corporations found continued dependence on transfers and grants, with own revenue inadequate to meet recurrent spending for most corporations [7]. A World Bank assessment put urban property tax receipts at about 0.15 percent of GDP during 2011-18 [6]. A metro that cannot recover a reliable share of the value created by additional floor space can view growth mainly as a service obligation. The division of legal powers also affects which agency can assemble and service the required land.

Urban land instruments must be assessed within the actual division of planning, acquisition, and infrastructure powers. Planning, land administration, major transport, utility investment and municipal service functions often reside in separate state or local bodies. A new rail line can raise land values well outside the local government that pays to maintain the surrounding streets. A municipality may seek development charges at permit approval because the state controls the property-tax framework. A developer may then encounter multiple charges, delays and uncertain rights; buyers face the cost at the end. Ahmedabad's town-planning scheme provides an institutional tool for planned expansion. The mechanism readjusts parcels, reserves land for streets, parks and public uses, returns smaller but better-serviced plots to owners, and can recover part of the increment through betterment charges. A World Bank review of schemes in Ahmedabad found land contributions of 25-40 percent in the schemes studied, together with a typical period of about ten years from initiation to final approval against a statutory expectation of three to four [8]. Later World Bank guidance notes that Gujarat law can permit deductions up to 50 percent and that reservations for affordable housing form part of the tool [9]. These legal ceilings and specific case figures should be kept distinct. A workable owner agreement must still account for people whose claims are absent from the cadastre.

Readjustment can align owners' interests with servicing while leaving tenants and livelihood users needing separate protection. It gives the city a means to lay down a street grid before prices surge. Its weak point is time, title complexity and the treatment of people who hold no recognized plot. A tenant, informal structure owner or small shopkeeper may lose a livelihood during readjustment while registered owners receive the uplift. The right to return, rental supply and a verified occupancy census must therefore enter the land process at the start. The cadastral map alone shows a partial distribution of claims. South African spatial evidence shows why the locations made available by these agreements matter for inclusion.

Access to central locations in South Africa

Persistent spatial separation makes accessible social rental and public land important metropolitan policy tools. A recent World Bank study using census, administrative and satellite data found that areas within five kilometres of commercial centers in South African cities had about 9,300 fewer residents per square kilometre than predicted against twenty global peers, roughly 60 percent below the peer median [10]. The housing market supplies much of its lower-income population at a distance from the main job centers. That mismatch reflects decades of forced segregation, capital investment patterns, land markets and present-day regulations. A policy of peripheral construction can leave this structure intact. Central land may be held for low-density uses, industrial parcels, parking or expensive development; older low-income districts may be exposed to eviction, poorly funded services and weak safety. Increasing allowed density near employment centers can widen supply. It also raises the value of sites, and redevelopment can displace long-standing tenants. South Africa's social rental and public-land tools have a clear metropolitan role where they secure below-market homes in areas that already contain jobs and high-capacity transport. Additional central housing alone does not determine which households can occupy it.

A coalition for spatial inclusion must address allocation and affordability as well as the quantity of housing. Employers benefit when workers spend less time commuting. Transit agencies benefit from all-day ridership and connected station catchments. Municipalities can receive more rates from improved land. Incumbent owners may oppose apartments or low-income tenants; developer margins may favor expensive units. Central-government capital grants, explicit inclusion covenants on public land, credible utility upgrades and published tenant protections can convert some opposition into a negotiated agreement. Land-use reform that leaves allocation to the highest bidder may produce more dwellings with limited change in who can live centrally. Auckland's reform provides evidence on the supply response to broad metropolitan development rights.

Metropolitan zoning reform in Auckland

Auckland offers evidence on broad upzoning, with estimates that remain conditional on the study's counterfactual and local context. In 2010, seven local territorial councils and the regional council were merged into one Auckland Council with regional powers and local boards [11]. Its 2016 Unitary Plan allowed more medium- and higher-density housing across approximately three-quarters of residential land. The 2026 peer-reviewed study by Greenaway-McGrevy and So used a synthetic comparison of other New Zealand urban areas and estimated that Auckland rents on new tenancies were 23 percent below their 2024 counterfactual without the reform. The authors also estimated that permit rates more than doubled relative to a synthetic comparison at the height of the building boom [12]. The study reports about 127,000 dwelling permits during 2017-24 and an estimated 18 percent increase in Auckland's dwelling stock across the eight years. These are observed totals, while the 23 percent rent result is a model-based causal estimate. Synthetic-control studies depend on the similarity of the comparison cities; this paper examines sensitivity to donor selection and pandemic-era changes. The estimate supports a rent effect under the study's counterfactual assumptions; its size remains sensitive to how Auckland

would otherwise have developed. It is no guarantee of the same coefficient in Lagos, Mumbai or Nairobi. The institutional coverage of the reform is part of the mechanism to examine before adaptation.

Broad, common rules can reduce fragmented decisions, but their effects still require credible measurement of housing outcomes. A regional council could set common rules across the commuting area. Central government required a single plan. Developers gained many possible sites instead of a small negotiated list. The reform reduced the scarcity of legal development opportunities; the study's results are consistent with greater supply reducing rent pressure. Public submissions and local resistance were substantial, and local boards preserved a role for place-specific needs [11]. Wide coverage helped prevent every neighborhood from treating extra homes as someone else's obligation. African and Indian cities face different constraints: title disputes, incomplete utility networks, small informal landlords, weak construction finance, uneven enforcement and lower household purchasing power. Their supply response may be smaller if upzoning changes permission without changing serviced land. The transferable mechanism is the breadth of reform, the common metropolitan rule, and the measurement of completed homes and rents against a credible counterfactual. São Paulo adds the question of how transport-related development rights translate into inclusion.

Development rights and inclusion in São Paulo

Transit-oriented rights and obligations should be calibrated before land transactions absorb the expected gains. São Paulo pursued transit-oriented development under its 2014 Master Plan, allowing denser construction along transport axes. Research published in 2023 found more development and smaller units in the corridors, yet insufficient inclusion of low-income households because property prices increased [13]. Transit investment and development rights can produce marketable central housing; scarce access and high land values can still screen out poor households. The design choice arises before a developer buys the site. Governments can reserve public land near planned stations, require a share of regulated rental or ownership units, charge for additional development rights, and use the proceeds to finance infrastructure or housing support. Each choice changes the residual price a developer can pay for land. A requirement calibrated after land changes hands can stop a project; an obligation announced in advance is more likely to be reflected in land bids. Longer-term occupation and tenure protection determine whether the intended households retain access.

Inclusion also needs tenure protection. Renters and informal vendors near a new station can face higher rents and relocation long before any affordable unit is delivered. A pre-project occupancy survey, phased construction, temporary rent support and return rights can protect the households who helped create the market for transit in the first place. An inclusion target should measure who occupies the completed homes after five years, as well as the share of units reserved at approval. Metropolitan finance must support those commitments while sharing costs across jurisdictions that benefit from growth.

3 Share costs, authority, and implementation responsibilities

Financing metropolitan growth

Publicly created land value and intergovernmental transfers can help align the benefits and service costs of metropolitan growth. The fiscal issue is how much of that increase returns to the public institutions responsible for growth. The OECD identifies betterment levies, developer obligations and charges for additional development rights as distinct land-value-capture tools [14]. These instruments can finance part of metropolitan expansion, provided valuations, rules and spending are credible. A metropolitan fund should combine predictable national or state transfers, a portion of own-source property revenue, charges linked to new capacity, and a transparent share of public-land proceeds. It needs binding service and affordability outputs. Funds should follow infrastructure catchments and commuting corridors rather than existing administrative boundaries alone. A county that hosts households for the metropolitan labor market should receive support for schools, local streets and drainage; a central jurisdiction should contribute when it benefits from workers who live elsewhere. Each financing instrument still needs a project-level feasibility and revenue test.

Charges and rental revenues should be appraised using feasible obligations and realistic collection and maintenance assumptions. Too high a charge can deter projects, reduce serviced supply and pass costs to buyers when competition is weak. Too low a charge grants private windfalls. Auctions for additional rights, published standard rates, and residual land-value models can help reveal the market. The baseline rights must be clear and predictable, or officials can create artificial scarcity and then sell permission to escape it. The same discipline applies to public land. Government should record its market value before committing it, publish the lease or tender, price any below-market transfer as a subsidy, and require long-run housing services in return. A thirty-year lease with enforceable affordable rents can create a different public return from an outright sale. Each must be appraised with a full fiscal and household balance sheet. Private finance requires identifiable revenues and an explicit allocation of risk. Municipal bonds depend on collectible revenue and credible reporting. Guarantees transfer risk to public budgets. Refinancing can extend loan terms without reducing land prices or increasing tenants' incomes. Rental projects still need realistic vacancy, maintenance, and collection assumptions. The political assessment must also recognise the informal housing through which households already accommodate growth.

Informality, segregation, and development opposition

Informal production contributes housing capacity that formal strategies must understand before changing rules or priorities. Small landlords add rooms and floors. Shared taxis connect peripheral neighborhoods. Vendors create local jobs near transport hubs. Informal settlements place workers within reach of the labor market, often under dangerous service and tenure conditions. A plan that erases these functions can raise measured housing quality and reduce household welfare. The political interests differ by place. Central homeowners and landlords can gain from scarce housing and oppose apartments; young renters and employers gain from more supply. Peripheral landholders favor roads and rezoning that raise their land

values; incumbents who fear acquisition may resist the street grid. Developers favor predictable permissions and infrastructure; some benefit from negotiating exceptional projects. Intermediaries may control information about claims and allocations. Investigate any alleged coercion or unlawful interference through documented complaints and legal findings. Municipalities seek revenue and fear service costs. Construction counts are visible sooner than many gains from infrastructure maintenance. Compare budget allocations and performance measures to assess whether that timing affects priorities. Social exclusion and local vetoes can prevent that capacity from becoming safer and more accessible.

Metropolitan reform should address documented local impacts without giving each established neighbourhood an indefinite veto over wider access. In India, access to rental accommodation can be affected by social screening even when household income is adequate. South Africa's spatial legacy remains racialized. In African cities, customary rights, informal occupation and patronage can give different groups unequal access to legal plots. Any digital or legal reform that recognizes only registered owners can exclude tenants and others whose claims are missing from the registry. A credible metropolitan coalition should include renters, small landlords, employers, transit agencies, construction workers, utilities, resident associations and representatives of settlements facing upgrading or eviction. The coalition must be built around concrete gains: legal additional rooms, reliable water, protected tenants, faster permits, local public revenue, cheaper commutes and jobs. Citywide housing targets can give future residents a proxy voice. Local consultation is needed for impacts; it cannot give each established neighborhood a permanent veto over the metropolitan housing supply. The responsibility map below identifies where agreements and protections are required.

Interests and responsibilities in metropolitan reform

Actor	Potential gain or rent	Veto / failure mode	Credible exchange
Central homeowners / landlords	Higher scarcity rents; protected neighborhood	Hearings, politics, litigation; tenant exclusion	Fund services; by-right rules; tenant protection
Peripheral landowners / customary claimants	Windfall from rezoning, roads and pipes	Holdout, claims, speculative withholding	Fair readjustment; verified rights; public uplift share
Developers / builders	Faster volume, margins, exceptional rights	Delays, lobbying, premium-only supply	Predictable broad rights; competition; phased obligations
Small landlords / tenants	Rental income / shelter near work	Informal construction / displacement	Safe infill, small finance, lease and grievance rights

Actor	Potential gain or rent	Veto / failure mode	Credible exchange
Municipalities / utilities	Rates, fees, reliable networks	Fear unfunded service burdens	Formula transfers; O&M; funding; shared capital plan
Transit operators / employers	Riders, workers, less congestion	Route resistance / relocation of jobs	Feeder contracts, station catchments, better access
Brokers / local power	Queue, land and exception rents	Coercion, fraud, capture of titles and lists	Auditable records, appeals, independent enforcement

Analytical actor map. The rents and vetoes are mechanisms to investigate, rather than an assertion that every actor behaves alike in each city.

Use the stakeholder map to test who can deliver or block each option and what obligations an agreement must resolve. Benefits of abundant housing are spread across future households, firms and the urban economy. Costs fall on landholders, neighborhoods, institutions and contractors that are already organized. A reform that ignores those losses can stall at the first permit or court case. A workable sequence gives local actors predictable rules, compensates genuine infrastructure and displacement costs, and discloses which private gains would be preserved by the proposed compensation rules. Compare accessible infill, expansion and in-situ improvement with the same household and service-cost assumptions. Shared evidence can support those negotiations if it is linked to lawful authority and safeguards.

Digital information and institutional authority

Common metropolitan data can clarify capacity and costs while creating new risks for occupants with unrecorded rights. Agencies hold different parcel IDs. Jobs data age quickly. Developers know available sites that planners cannot see. Utility capacity is recorded in engineering files, while housing plans use generalized zones. Lenders and governments often lack records of informal rents. Satellite images, digital addressing, parcel services, transport feeds and interoperable permit systems can reduce the cost of observing and coordinating these decisions. A tool should help resolve a specific decision, with a named agency responsible for acting on its results. A machine-learning score that clears a backlog can create new exclusion if an applicant cannot correct the record. A parcel-level tax system can finance services and can expose informal occupants to enforcement before they gain legal protection. Initial digital applications should therefore support specific, reviewable public decisions.

Use	Constraint and magnitude to test	Who could capture the gain?	New power / risk
Satellite change detection	Months to find new structures; % field-verified	Tax authority, developer, community	Surveillance, wrong classification, eviction targeting
Parcel + permit ledger	Median decision days; exception share; rework	Applicants and faster intermediaries	Central gatekeeper, discriminatory automated refusal
Travel-time/job maps	Jobs reachable in 45/60 min by income group	Renters or owners via land-price uplift	Location labels stigmatize settlements; data bias
Infrastructure capacity map	Connections secured before occupancy; outage rate	Utilities, serviced landowners, builders	Sensitive asset exposure; planned projects unfairly prioritized
Rental/ occupancy register	Vacancy, rent change, unit take-up, displacement	Tenants, landlords, housing agency	Tenant profiling, privacy loss, eviction and social screening

Every digital deployment should report a baseline, a measured improvement, distribution of gains, decision rights and an appeal mechanism.

Early applications can focus on public decisions: a shared growth map, a dated infrastructure capacity commitment, a decision clock for permits, and an audit of public-land allocation. Predictive tools can then help direct inspections or model urban expansion. Governments should validate models in the field, report false-positive and false-negative rates, protect personal information and retain accountable human decision-makers. The implementation sequence should use that evidence to establish corridor-level commitments.

Implementing metropolitan coordination

Corridor planning should combine informal livelihoods, care networks, feasible housing capacity, and funded infrastructure. Assign capital and operating costs across jurisdictions before site selection is final. Transport, housing and service authorities need a joint spatial investment schedule and an authority able to resolve cross-boundary disputes. Track household travel expenditure and time, reliable job and school access, retention, earnings where measurable and the location of added affordable homes. As an indicative timetable, aim in the first year to negotiate a five-year growth agreement with binding near-term actions. The agreement should start with a common functional urban area defined from settlement patterns and travel, followed by an agreed housing capacity map. Each corridor should show land

available for safe infill, expansion and renewal; the legal floor space likely to be delivered; trunk and local service costs; rental and ownership demand by income group; and the public rights required to secure infrastructure. Each resulting investment then needs a responsible agency, budget, and delivery date.

A metropolitan programme should update its commitments when observed demand or the binding constraint changes. Give local governments transparent transfers for servicing growth. Tie a portion of funding to verified added legal floor space, households occupied, service reliability and job access. Reserve public land near high-capacity transport for mixed-income housing before announcing the investment. Introduce predictable by-right rules in suitable areas; handle hazardous sites and heritage through specific safeguards. Tenant rights must enter at the same time as land-value change. Enumerate informal occupation before infrastructure raises the stakes. Protect return rights where upgrading displaces residents temporarily. Budget rental assistance and relocation costs when hazards require permanent moves. Monitor rents and evictions inside affected catchments. Provide appeal routes for land claims, permits, beneficiary allocation and automated decisions. A five-year agreement also needs an exit condition. If jobs do not arrive in a proposed satellite center, slow residential expansion there and redirect public capital toward locations with proven access. If infill approvals rise while construction stalls, investigate site assembly, developer competition, building costs and credit. If housing completions rise but vacancy grows, examine the price and location match. Reform should follow the constraint that has moved. Readiness review must still stop options with unresolved safety, displacement, or occupation risks.

Metropolitan housing assessment

The metropolitan diagnostic should expose fatal gaps rather than allow them to disappear within an average score. Give each gate an evidence owner. Treat a red gate that threatens safety, displacement or occupancy as a stop condition; a high average score does not erase a fatal flaw. Approval depends on housing, services, and jobs being available on compatible schedules.

Gate	Decision question	Evidence and threshold to set locally
1. Functional market	Where do residents work and travel across borders?	Commuting flows, built-up footprint, origin-destination survey
2. Demand	Who needs which tenure at what price?	Household payment capacity and rental/ownership demand
3. Accessible supply	How many legal serviced homes can be delivered?	Parcel rights, FAR, hazards, infrastructure and completion probability
4. Job access	Do added homes reach actual employment?	Jobs reachable in 45/60 min; fare/time by income group

Gate	Decision question	Evidence and threshold to set locally
5. Infill	What blocks safe additional floor space?	Rules, utilities, title, small-landlord capacity, homeowner vetoes
6. Expansion	Are rights-of-way secured before settlement?	Grid, drainage, schools, plots and claims in growth areas
7. Fiscal contract	Who pays capex and receives recurring revenue?	Public-land value, transfer formula, tariff and property-tax forecast
8. Building market	Can builders deliver diverse safe products?	Input costs, competition, permits, inspections and staged finance
9. Tenure / subsidies	Will the lowest-income households actually enter?	Rental and incremental channels; subsidy per occupied household
10. O&M;	Who owns and maintains the networks and buildings?	Asset owner, annual costs, collection and replacement reserve
11. Existing residents	Who faces higher rent or displacement?	Tenure census, return rights, relocation and rent monitoring
12. Inclusion	Which groups fail at each gate?	Gender, caste, race, class, disability, migrant and tenant outcomes
13. Political agreement	Who captures uplift and can block reform?	Actor-rent-veto map; coalition commitments and disclosure
14. Digital rights	Who can use data to tax, deny or remove?	Purpose limits, field validation, correction and independent appeal
15. Results	Does the compact create cheaper access?	Additional occupied housing; generalized cost; job access; service uptime

Approve the metropolitan plan only when added legal housing capacity, funded service capacity and accessible jobs are available in the same locations and on compatible schedules. Require a full household-and-public location balance sheet, a transparent agreement over the land-value uplift, and enforceable safeguards for existing residents. Review those locations and schedules annually against observed growth. The plan should then be judged against how households and builders actually accommodate growth.

Implications for urban growth

A metropolitan plan must guide existing housing processes through explicit location, service, financing, and protection commitments. Public policy can influence these choices by clarifying rights, securing infrastructure corridors lawfully, and permitting safe homes in locations with access to work. A metropolitan agreement should bring location costs, public-service obligations, and the distribution of land-value gains into one appraisal. It should state which jurisdiction will accommodate growth, where housing can be built, which services are funded, who pays, and how existing residents are protected. Applying that test requires respecting the different limits of projections, observational studies, and causal evidence.

Limits of the evidence

The evidence supports a locally populated options comparison rather than a universal metropolitan model. The main estimates are drawn from official projections, an Africa-wide price study, spatial job-access and labor-market research, and quasi-experimental work on Auckland. Their methods differ. Population and infrastructure needs are forecasts. Nairobi and South Africa provide observational spatial evidence. Auckland's rent estimate uses a synthetic counterfactual, with uncertainty around the choice of comparison cities. Ahmedabad and São Paulo are institutional and spatial case studies. The hypothetical location-cost calculation demonstrates a decision method and has no claim to observed project costs. The metropolitan agreement, actor map and fifteen-gate test are policy judgments built from those mechanisms. Their value depends on local parcel data, developer costings, household surveys, infrastructure capacity, public finance and field verification. Fieldwork should supply the household journeys, project costs, network capacity, and decisions needed for that comparison.

Prepare the local decision

Compare metropolitan options with consistent household, service, and accessibility evidence verified in the field. Map homes, rents, jobs by occupation, travel times and fares, network capacity, hazards, and planned investments across municipal boundaries. Measure peak and off-peak journeys, including informal transport. Compare infill, expansion, and satellite locations using the same household and service-cost assumptions. The Nairobi and Auckland discussions above raise different questions: coordination across jurisdictions and the response to broader development rights. Bring municipal finance and planning teams, transport agencies, utilities, employers, and representatives of commuters and tenants into the appraisal. Record which jurisdiction receives land revenues and which must pay for services. Agree how costs and receipts will be shared before endorsing the location. Test route-model estimates against actual journeys and verify whether mapped jobs match residents' skills and working hours. Publish the assumptions behind each location score rather than a single opaque ranking. Stage investment against observed demand and committed service capacity. Review occupied homes, travel reliability, employment access, municipal costs, and displacement; revise an expansion plan if employment and transport fail to materialise. The CDR workflow in Chapter 13 can help identify commuting corridors for independent verification. Chapter 11 adds climate exposure and the distribution of protection costs to that location decision.

Metropolitan policy should expand the opportunities households can actually reach and afford to use. Transport, school places, services, and the costs of moving and remaining determine whether relocation improves access. Cheap peripheral sites can lose their apparent advantage once travel and lost social support are counted. A foreign mobility programme cannot resolve fragmented local service finance without an agreement among the responsible municipalities. At metropolitan scale, test congestion, school places, network capacity and the distribution of fiscal costs across municipalities.

Ethiopia: service access and the limits of advance reservation

Ethiopia's Urban Expansion Initiative provides evidence of improved service access near new roads and practical limits to advance land reservation. A March 2026 International Growth Centre working paper analysed 4,000 households in eight selected cities where road construction permitted household-level evaluation. In matched cross-sectional comparisons, households near new arterial roads were 7.4–8.4 percentage points more likely to have piped water and 11.5–13.7 points more likely to have improved toilets than comparable households farther away. Clinic journeys were about three minutes shorter. These estimates account for observed household differences. They cannot eliminate unobserved differences or selection into better-connected neighbourhoods. [224, pp. 5–7, 16–22]

The study did not establish consistent gains in poverty reduction or formal employment. Its separate income analysis used retrospective reports of conditions roughly a decade earlier. Those difference-in-differences estimates depend on accurate recall and comparable trends between groups. It did not estimate changes in rents or property prices. A coauthor helped develop the initiative, so the study should not be described as an independent replication. [224]

Implementation evidence shows why reservation needs a credible budget. In the five cities covered by qualitative interviews in the original evaluation, none acquired arterial rights-of-way substantially beyond immediate projects. Limited compensation budgets and uncertainty about keeping unused corridors clear constrained advance acquisition. Municipal teams also reported that urban expansion displaced farming households whose transition to sustainable urban livelihoods remained difficult. Respondents considered compensation and support insufficient despite improvements. [225, pp. 12–14]

I would phase corridor reservation and acquisition with funded protection of existing rights. Authorities need to identify affected households, fund compensation and livelihood support, and provide access to affordable land and services. Monitoring should track who benefits and who loses access, alongside roads opened and plots supplied. These are implementation requirements suggested by the evidence; the studies did not test this complete package.

Sources and evidence notes

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11. Disaster risk, housing reconstruction, and affordability

Select housing risk measures by the safety they deliver and whether households and institutions can sustain them. Compare protection, adaptation, and relocation against actual alternatives, including recurring costs and livelihood access. Risk information should trigger an affordable action with a responsible agency, rather than merely label a site. Experience after disasters shows why this institutional preparation must precede construction.

Disaster recovery is a central part of housing policy. In 2010, I wrote Safer Homes, Stronger Communities: A Handbook for Reconstructing after Natural Disasters with Jennifer Duyne Barenstein, Priscilla M. Phelps, Daniel Pittet, and Stephen Sena. We argued for reconstruction that restores livelihoods, gives affected communities a leading role, and combines household control with finance and technical support. We also warned against avoidable relocation and the exclusion of tenants and people without formal tenure. Those principles remain the starting point for the housing decisions examined here. [231]

We tested the handbook with practitioners because much of the available reconstruction experience had not been documented well. While writing it in 2009–10, we drew on experience from Pakistan, Indonesia, Gujarat, and Colombia, among other places. We ran field tests with practitioners to check whether the messages were useful and grounded in the realities of implementation.

1 Reconstruct homes so households can recover

Reconstruction should restore safe, affordable occupancy and the means to earn a living. A programme needs to ask how long a family remains displaced, what it pays while waiting, and whether its rebuilt home supports its daily life. A completed structure is one milestone. The public objective should include working services, manageable household debt, and access to employment. These distinctions matter when the pressure to announce construction targets arrives before officials have agreed who qualifies for assistance or how rebuilding will be supervised.

Decide what to repair, rebuild, or relocate

Repairable homes should receive an explicit assessment before demolition or replacement is approved. Classify structural damage separately from the safety of the site. Compare repair and strengthening, rebuilding on the same plot, and relocation against the same life-safety requirements. Record the engineering judgment, the estimated cost, and the household's preference. Give disputed classifications a second technical review. A repair grant should pay for the identified strengthening work. Replacing a repairable building should require an explanation of why repair cannot deliver an acceptable result.

Relocation decisions should account for the household's income as carefully as the replacement building. Compare the costs of transport, temporary accommodation, interrupted work, and reconnecting services. A home-based producer needs usable workspace. A trader needs access to customers. Where a site cannot be made acceptably safe, prepare the move with affected residents, document their claims, and establish what support will cover the transition. Publish the alternatives considered and the remaining risks. Cheaper land at the edge of a city is insufficient grounds for choosing it.

Put households in charge and fund the support they need

Household-led reconstruction requires a public delivery system. Residents can choose layouts, engage builders, and supervise spending while engineers set safety requirements and inspect critical work. The Independent Evaluation Group's review of reconstruction experience found advantages in homeowner control, alongside a need for specific assistance to vulnerable households. [236] Governments should offer an assisted route for people unable to organise construction themselves. Contractor delivery may be appropriate for complex buildings or households choosing that option. Compare each route on quality, total cost, completion, and resident satisfaction.

Gujarat's recovery after the 2001 earthquake shows both the reach and the limits of owner-driven reconstruction. The World Bank's 2009 completion report recorded 125,781 houses reconstructed and 41,751 repaired under the project phase it assessed. The programme supported training and linked payments to progress. Its evaluation also found that construction deadlines excluded some households unable to keep pace. Rising costs then made completion harder. These are administrative outputs and implementation findings, rather than a controlled comparison with contractor delivery. The report also records material banks closing and project technical staff being absorbed into regular administration before all homes were finished. Any decision to withdraw dedicated support should include a funded completion plan for the remaining eligible households. [232]

Nepal made inspection a condition for releasing reconstruction finance. Following the 2015 earthquakes, eligible households received grants in three instalments, with later payments tied to approved construction at the plinth and roof-band stages. The World Bank's 2024 completion report recorded 319,621 reconstructed core houses under its project by June 2023. It also described technical assistance and inspections, including at completion. This establishes substantial delivery under a monitored construction system. It does not establish how every house will perform in a future earthquake, or whether every household recovered its income. The figure is the project total, not the total for Nepal's national reconstruction programme. [233]

A technically compliant house can still leave its owner with an unaffordable financing gap. A 2021 study published by the United Nations Development Programme found that only 8.4 percent of surveyed homeowners in eight municipalities of Gorkha district had built within the housing grant. Vulnerable households progressed more slowly, and the study described delays between construction stages and payments. Its findings concern those municipalities during reconstruction, rather than the national programme at closure. They support a practical

safeguard: price the minimum adequate dwelling locally and establish how a poor household will finance each stage before requiring it to begin. [234]

The grant schedule should follow a realistic construction and cash-flow plan. Provide an initial payment sufficient to start, publish inspection deadlines, and track the interval between a passed inspection and receipt of funds. Establish an appeal route when certification is refused. Reserve targeted assistance for households whose labour, disability, care responsibilities, or lack of savings prevents completion. Review grant adequacy when verified material and transport costs change. Technical advice should arrive before work is concealed or expensive to correct. This is a proposed operating standard, to be costed against local staffing and fiscal capacity.

REKOMPAK's community model depended on sustained technical and organisational support. In Indonesia, household groups managed rebuilding with block grants and help from facilitators responsible for construction, finance, and community organisation. The programme's 2012 account describes training, supervision, and checks on materials and workmanship. It also records shortages and turnover of facilitators in Aceh, with resulting delays and weak quality control in some communities. The evidence supports investing in the support system alongside household responsibility. It does not justify transferring construction risks to residents while removing the staff who help them manage those risks. [235]

Reconstruction capacity should be budgeted in workers and inspections as well as money. Estimate how many active sites each engineer can visit, how far staff must travel, and which tests local laboratories can undertake. Train masons in the techniques the programme will actually fund. Publish material specifications and check suppliers rather than assuming that a purchased input meets them. Compare procurement costs after counting transport, storage losses, and household labour. Community contributions should be recorded and valued, so an apparent public saving does not conceal an unaffordable demand on residents' time.

Include residents and restore the neighbourhood

Eligibility rules should distinguish housing need from proof of property ownership. The 2010 handbook explicitly included owners, tenants, and people without legal status in its reconstruction framework. [231] A recovery census should therefore record occupants as well as structures. Establish a documented procedure for corroborating claims when records are missing, with independent review of contested cases. Protect privacy when publishing eligibility decisions. Recognition for emergency assistance should not silently become a final judgment on a disputed title, nor should a title dispute automatically suspend all help with safe accommodation.

Tenants need a funded recovery route of their own. Programmes should budget for temporary rental assistance and the repair of affordable rental rooms, alongside support for owner-occupiers. Assistance to a landlord can carry clear conditions on former tenants' return, rents, and minimum habitability, subject to local law and enforceable agreements. Track whether those tenants actually return. Record people who shared rooms, sublet informally, or lacked a written lease, and give them access to review. Otherwise, an apparently successful rebuilding programme may restore a landlord's asset while leaving its former occupants displaced.

Housing completion and service restoration need a common schedule. The REKOMPAK account records houses in Aceh that remained unoccupied while water or electricity connections were missing. [235] Each neighbourhood plan should therefore name the agency, funding, and completion date for water, sanitation, access roads, drainage, and electricity. Check the services people can use, including their charges and reliability. Align reopening of schools, markets, and transport with return. Assign maintenance responsibilities before the construction team leaves and record who will fund recurring costs.

An empowered reconstruction authority needs clear obligations to residents and other agencies. In my 2019 account of lessons from disaster recovery, I stressed capable institutions, protection for vulnerable people, and strategies fitted to local conditions. [214] Applied to housing, that means published assistance rules, delegated spending authority, a route for resolving cross-agency delays, and accessible complaints. Central coordination should settle obstacles that municipalities cannot resolve alone. Local teams still need staff and money to implement the decisions. Any temporary authority should have a funded handover of records, unfinished claims, and inspection responsibilities.

The recovery scorecard should follow households after the building is handed over. Report completed and occupied homes separately. Track time displaced, unresolved defects, service outages, rent, debt, earnings, and travel costs against a documented baseline. Separate owners from tenants and identify who remains outside the programme. Independent sample checks should test construction records and household accounts. Future events can provide evidence of performance, while engineering compliance remains an intermediate measure. These proposed indicators test whether a household can occupy a safer home, meet its recurring costs, and remain connected to work.

2 Identify household exposure and the costs of safer housing

Climate-adjusted housing policy must decide whose risk evidence governs and how protection costs and rights are allocated. A settlement close to jobs may sit in a floodplain; a safer site can be two bus rides away. A metal roof may cost less than a cool, durable one; higher indoor temperatures may increase illness and lost work. Insurance can protect a lender while excluding the household whose home is hardest to insure. Climate measures can change land values, bargaining power and the boundary between places where poor households may live and places from which they are excluded. Affordable housing therefore needs a climate-adjusted location agreement. The state must estimate future hazard and service costs, finance safety where improvement is feasible, offer credible nearby alternatives where it is not, and protect renters and informal workers whose claims are easy to erase. Better maps can make these choices clearer. Implementation requires deciding whose map governs, who bears the bill and who can challenge the decision. The Msimbazi basin illustrates how that decision combines flood management with livelihood access.

Flood risk and the cost of relocation

Flood-reduction projects should distinguish expected protection benefits from verified outcomes after delivery. In Dar es Salaam, the Msimbazi River basin connects an expanding city to its center. The same corridor holds busy roads, public transport, market activity and neighborhoods where people live close to their livelihoods. The valley also carries floodwater, creating a conflict between occupied land and flood-conveyance capacity. The World Bank describes a basin containing about 1.6 million people, with significant floods in seven of the ten years preceding its 2023 account.[3] The project combines engineering, land, and resettlement measures. Tanzania and its partners planned river-channel works, flood retention, a safer Jangwani transport crossing, a city park, institutional reforms and preventive resettlement. A US$200 million IDA credit anchors the 2022-2028 program. The published project objective is to lower flood exposure for more than 300,000 people; this is an expected benefit, subject to delivery and verification.[3] The households affected by land and river works also need a livelihood and tenure assessment.

Flood-management design should resolve compensation, occupancy, and livelihood questions alongside hydraulic requirements. The publicly disclosed March 2022 draft resettlement and livelihood plan reports that over 90 percent of the trades described in its livelihood assessment were informal and unregistered. Petty trading, vending, employment, and gardening were the main activities reported. The relocation appraisal therefore needs to include livelihoods. A family that gains a safer roof but loses a market stall, customers or a reliable commute may become poorer. Appraisal should examine both hazard-related land-price discounts and the livelihood value of proximity, rather than assume either determines the price. A river project can also create a new development rent. Regraded land, improved transport and restored public space raise the value of adjoining parcels. Which parcels can be built upon after the works? Does the public retain land for affordable rental housing? Who receives compensation, and who is recorded as a tenant? How will the city stop settlement from returning to land reserved for floodwater? Resolve these land, compensation, and occupancy questions alongside the hydraulic design. The wider risk assessment must distinguish the physical hazard from the development decisions that create exposure.

Urban development and disaster exposure

Climate hazards become housing risk through settlement patterns, building conditions, poverty, and the capacity to respond. Urban policy determines how many people and assets sit in their path. A drainage channel blocked by fill raises upstream flood depth. A wetland converted to housing removes storage. A bus depot built in a river corridor creates a public asset that becomes a physical barrier when floodwater rises. A zoning decision that excludes low-cost rental rooms from safe neighborhoods sends households toward precarious land. A peer-reviewed 2022 global analysis using flood maps and poverty data across 188 countries estimates that 1.81 billion people, or 23 percent of the world population, live in places exposed to more than 15 centimeters of water during a 1-in-100-year flood. India accounts for around 390 million of them. Eighty-nine percent of exposed people live in low- and middle-income countries; of the 170 million people facing both extreme poverty, defined in the study as

living on less than US$1.90 per day, and significant flood exposure, 44 percent are in sub-Saharan Africa.[1] Modelled exposure estimates are useful only when their uncertainty and decision purpose remain explicit.

Exposure models identify potential risk rather than settle which intervention or institution should address it. They tell us where to look; they do not predict that 1.81 billion people will be flooded in any single year. They also omit much of the damage a flood causes through failed sewers, closed bridges, interrupted wages and disease. Poor households can lose several days of income from water that never enters the dwelling. A useful causal chain has five links. Climate and upstream land use change the hazard. Zoning and land prices influence exposure. Building quality and services influence physical vulnerability. Savings, insurance, and social networks influence recovery. Government decisions redistribute the loss among homeowners, tenants, utilities, taxpayers and future residents. At each link, a different actor may hold the power to reduce risk. Households' reasons for accepting exposure must also enter a stay-or-move comparison.

HAZARD	EXPOSURE	VULNERABILITY	RECOVERY
Rainfall, river flow, heat, drought	Land price, zoning, access to jobs	Materials, drainage, indoor heat, water	Income, tenure, credit, insurance, transfers
Hydrology and forecasting	Planning and land policy	Standards and local services	Social protection and finance

Households may accept physical risk to retain affordable access to work, so safety and livelihoods need joint appraisal. Such sites can lie outside formal land markets and offer small, flexible rental units. Declaring them unsafe without providing accessible alternatives can exclude low-income residents from employment locations. Risk policy should apply consistent standards and account for the different protection available to higher-income developments. An accurate risk map is therefore a necessary input and an incomplete housing policy. The other map shows jobs, transit, fares, tenure, rent, schools, employers, household debt and informal commerce. Every stay-or-move decision requires both. Chennai shows how disputed land and environmental records can complicate that appraisal.

Land records and wetlands in Chennai

Chennai's wetland disputes illustrate the need to distinguish reported legal proceedings from settled rights and final judgments. The 2015 floods exposed how much of Chennai depends on wetlands, tanks, channels and the unobstructed flow of water across a low-lying plain. A remote-sensing study estimated that 18 percent of the metropolitan area was inundated, directly affecting around 21 percent of its population. Those are model-based estimates for a particular event and spatial method.[4] The Pallikaranai marsh has become a test of state capacity. Hydrologists see flood storage and feeder channels. Developers and landowners see registered parcels. The planning authority relies on zoning and revenue records. The

environmental authority must delineate an ecologically connected wetland. These records have not always agreed. According to the news reports cited here, following a National Green Tribunal order on 24 September 2025, Chennai planning authorities paused construction approvals in and around the Ramsar wetland pending a scientifically grounded assessment of its influence zone. Developers challenged the restrictions in the Madras High Court in March 2026, arguing that the blanket buffer lacked adequate legal and scientific basis and restricted land already classified for urban development.[5] These are reported procedural developments and submissions by the challengers; they do not establish the legal merits or a final judgment. The planning response must reconcile the records and make the allocation of costs reviewable.

Inconsistent land and environmental records require legal review and engineering evidence before restrictions or compensation are applied. A hydrologically important channel can cross land that was recorded years ago as private, developable property. Existing rights create legitimate expectations and political power. Sudden prohibitions can destroy value and strand projects; permissive approvals can shift flood costs downstream onto households who were never part of the transaction. The technical task is to map flow paths and retention needs at parcel level. The legal task is to give those maps authority through clear procedures, hearings and review. The fiscal task is to price any restriction, acquisition, restoration or development-right transfer. Advance expansion planning should identify safe buildable land alongside land that requires protection. Angel's public planning framework calls for protecting high-risk areas, drainage channels and ecological assets while preparing land for urban growth. Each proposed restriction should have a stated legal basis and an assessment of the people and livelihoods affected. The World Bank's guidance distinguishes broader land-use planning from project-induced displacement; the applicable assessment and safeguards depend on what is financed and implemented. I would include compensation or assistance where required, accessible remedies, and affordable alternatives for affected households in the plan's costs. Chapter 10 explains how this assessment connects with road reservations and local subdivision. [228, p. 131; 230, pp. 6–8] Heat and water stress require a different set of housing and service responses.

Heat, drought, and housing conditions

Heat exposure reaches households through the dwelling, home-based work, and the ability to recover at night. Heat and water scarcity cross nearly every neighborhood, although their effects vary sharply by income. Insulation, ventilation, and affordable cooling can reduce indoor heat. Small rooms under metal roofs may offer less protection, particularly when nights remain hot. Exposure can continue through home-based work and sleep. Ahmedabad's heat response helps distinguish warning systems from improvements to housing conditions.

Heat warnings and housing quality in Ahmedabad

Heat-action plans need practical resources for protection near vulnerable households as well as warnings. Ahmedabad adopted South Asia's first city heat-action plan after the deadly 2010 heatwave. The plan joined forecasts to public warnings, health-system preparation and outreach. An ecological evaluation comparing 2014-2015 with a 2007-2010 baseline estimated an average of 1,190 deaths avoided annually in the post-plan period, with a wide 95 percent

confidence interval of 162 to 2,218. This is a before-and-after association, with possible confounding from population, health systems, and other changes. When the authors excluded the unusually hot year 2010 from the baseline, the estimated decline remained but was no longer statistically significant.[6] A warning helps a household only when someone can act. A construction worker may lose wages by staying home. A child or older person inside a poorly ventilated room may have no cooler place to go. A landlord receives no automatic return from coating the roof of a short-term rental. A municipality may have a heat plan yet lack a budget for public shade, reflective surfaces, drinking water and cooling shelters near informal settlements. Cool-roof evidence should be tested through indoor exposure and health over repeated seasons.

Measured temperature reductions from exploratory cool-roof research require local testing before a wider performance claim. An exploratory 2024 Ahmedabad study found cool-roof treatments associated with indoor temperature reductions of roughly 1-1.5 degrees Celsius on peak summer days. The study also found that ventilation, ceiling height, adjacent trees and open space mattered. The observed effect should be treated as site-specific; it is distinct from measured causal effects on mortality or earnings.[7] The policy logic is to combine low-cost passive improvements with labor protection and neighborhood design. A government can subsidize reflective roofs on rented rooms, set performance standards for new low-cost housing, protect shade and public space, and trigger worker protections through clear heat thresholds. The final test is indoor exposure and health, measured over seasons, with special attention to older residents, women working at home and children. Cape Town's water crisis provides another example in which exposure and capacity to respond differ by income.

Water conservation and inequality in Cape Town

Cape Town's water-demand evidence shows why income and household use patterns matter for conservation policy. Cape Town's 2018 "Day Zero" crisis threatened emergency rationing as reservoir levels fell. The city combined warnings, restrictions, tariff changes and conservation; per-person limits reached 50 liters per day during the emergency period. The political conflict arose from differences in consumption and access. Some households had lawns, pools and private tanks. Others had shared taps and had used little water long before the drought made national news.[8] A 2021 study analyzed 14.9 million monthly billing observations covering roughly half a million households from 2014-2018. It found that the top 10 percent of household water users in the billing sample consumed 31 percent of the water delivered to that sample before the drought, and that citywide consumption fell about 50 percent during the crisis. The link between income and water use rose strongly in outdoor-watering seasons.[9] Housing approvals and utility planning need to reflect those differences in their demand assumptions.

Water-saving measures should be designed around household use, affordability, and a shared utility operating forecast. A flat per-person restriction appears even-handed while imposing different costs on households with private storage and those carrying water from a communal point. New housing projects add long-lived demand to constrained water systems. Developers may sell homes that connect to an unreliable network, while the utility and tenants inherit the shortage. Resilience requires water balance sheets at the growth-corridor scale: firm supply, nonrevenue water, demand by income group, drought triggers, reuse, wastewater services

and the marginal cost of connecting new homes. Water conservation reduces billed volumes in the short run and can weaken utility revenue unless tariffs and transfers adjust. Resilience investments fail when a water utility has incentives to sell more water to finance its existing debt. Housing approvals and utility capital plans must therefore share a demand forecast and an operating model. The flood, heat, and water cases therefore support different mechanisms rather than one general adaptation prescription.

EVIDENCE AND LIMITS The global flood counts are modeled exposure estimates [1]. Ahmedabad's mortality findings are observational [6], and its cool-roof study is exploratory [7]. Cape Town's billing analysis draws on actual household-level records [9]. These three types of evidence answer different questions and should never be presented as though they carried the same causal certainty.

These outcomes answer different questions. Cape Town's conservation result does not establish equal access or equal sacrifice. Ahmedabad's heat-plan evaluation associates implementation with fewer deaths, but its before-and-after design cannot isolate the plan's effect. The Chennai evidence concerns flood exposure and disputed development controls; it is not a matched test of either response. Adapt the specific supported mechanism, whether water-demand reduction, heat protection, or risk-sensitive land use, and measure its distributional effects. [9,6,4–5] A project appraisal must select among protection, adaptation, and relocation using the mechanism relevant to the site.

3 Compare adaptation, protection, and relocation

Comparing protection, adaptation, and relocation

Compare physical risk-reduction options using explicit safety thresholds and independent review of borderline cases. Each answer changes land rights, income and public finance. The practical choice sits between four strategies: reduce exposure through drainage or restoration; reduce building vulnerability through retrofits; create safe new housing in connected locations; and move households when residual life-safety risk remains intolerable. The first step is to sort risk by whether it can be credibly managed through the life of the dwelling. A home above an unstable slope or inside a flood-conveyance channel may be impossible to make safe at acceptable cost. A home exposed to shallow, infrequent surface water may be protected through drainage maintenance, local elevation, safe electrical installation and better emergency routes. Building permits and subsidy eligibility should reflect this distinction, with an independent review route for borderline cases. Beira demonstrates how engineered works and ecological assets can form part of that comparison.

Drainage infrastructure in Beira

Combined drainage and ecological protection require both capital works and continuing control of land use and maintenance. A World Bank account published in March 2025 describes drainage rehabilitation, flood-control stations, a retention basin, mangrove restoration and a 17-hectare green park. The program reports reduced flood risk for over 234,000 people, close to 40 percent of the city's population.[11] The numbers describe program reach and

reported risk reduction; they do not establish avoided losses in a randomized comparison. A retention basin needs land protected from new construction. Drains need regular clearance. A park can defend that land through public use, while open space raises surrounding property values. The city must decide who maintains it, who can vend there, which edge parcels may be developed and how flood safety and new land value will be shared. Nature-based solutions can be cheaper and more flexible in some basins, yet they carry real land and maintenance costs. A wetland or retention park without a legal boundary and a budget can be filled when the next housing shortage arrives. Protecting flood storage requires enforceable land-use rules and continuing maintenance funding. Resident knowledge can help establish how the system actually operates within occupied neighbourhoods.

Community risk information in Mukuru and Freetown

Community risk information can improve statutory planning and negotiations when residents can influence its interpretation. Residents, the county and more than forty partner organizations worked on a plan covering roughly 100,000 households. The structure of local consultations moved from cells of ten households to neighborhood clusters and larger forums. Residents could identify flooded passages, absent sewers, dangerous industrial edges and the households most exposed to displacement.[12] A participatory plan changes what government can know and agree upon. It does not, by itself, provide land titles, funded drains or completed upgrades. Mukuru's fragmented ownership and service arrangements still demand fiscal and legal decisions. Public claims about plan coverage should be distinguished from measured completed-service outcomes. The IPCC records a related experience in Freetown, where the urban-poor federation and local research partners mapped flood and coastal hazards. In Cockle Bay, community organizations used this knowledge to agree on limits to further land reclamation, mark a settlement boundary and build drainage channels. Negotiation with the environment ministry reduced eviction threats.[13] The same information needs safeguards against uses that expose those residents to exclusion.

In both cases, residents help explain why water moves where it does, who bears the risk and what land is actually used for. Community mapping also creates a disclosure risk: a detailed occupancy database can help an authority identify whom to evict. Data governance must set limits on disclosure and use, and give residents control over corrections and challenges. A larger flood programme also requires durable land, finance, and enforcement institutions.

Flood management and public finance in the Netherlands

The Netherlands' flood programme illustrates institutional requirements that must be distinguished from its unusually strong fiscal capacity. Between 2006 and 2019, national and regional authorities undertook interventions at 34 locations for approximately EUR 2.3 billion, improving flood safety for about four million people. Measures included moving dikes, lowering floodplains and creating flood bypasses, with residents involved in local design and compensation processes.[14] Adaptation in African and Indian cities requires legal authority and funding. Giving a river space requires government to acquire rights, compensate some owners, protect new boundaries and keep the waterway functioning over decades. A poor city can adopt

the principle of flood storage and planned growth on safer land. It also needs financing, a land-rights process and institutions capable of protecting the result against future encroachment. Housing finance must account for the residual risk that remains after public protection.

Climate risk and housing finance

Climate exposure affects property value and financing over time, so appraisal should use scenarios rather than a single precise forecast. Developers look at the cost of capital, expected sale price and delivery time. Renters calculate monthly payments and proximity to work. Banks examine collateral, insurance and repayment probability. Municipalities see infrastructure expense and property-tax revenues. Climate risk reaches each through a different channel. A developer who can buy exposed land at a discount may still earn a high margin if risk disclosure is weak or public flood defenses are expected. A builder using cheap metal roofs may sell quickly while the household bears higher future heat costs. A mortgage lender may insist on insurance, pushing high-risk borrowers outside formal credit. A municipality that extends sewers into a floodplain creates both a service liability and an expectation that the land is safe to build on. The correct price of a home includes its lifetime hazard exposure and the cost of recovery, in addition to construction, land and finance. It also includes the social cost of moving away from employment. These amounts need scenario ranges. Twenty-year projections based on a single flood-frequency map or a single climate pathway can create false precision. Insurance prices and availability provide another indication of how that risk is allocated.

Insurance cost and availability

Risk-based insurance can signal danger while making protection unaffordable or unavailable to exposed households. They can also become a barrier to ownership when premiums rise above a low-income household's budget. An Australian actuarial study found that, by March 2024, 15 percent of households, or 1.61 million, faced home-insurance premiums exceeding four weeks of gross income. An estimated 5 percent of mortgaged households were under insurance affordability stress, covering about A$57 billion in loans.[15] Australia has deep insurance and mortgage markets, making the effect visible. Many African and Indian low-income households have little or no formal disaster cover. They self-insure through savings, rotating credit, family networks, emergency sales and forgone consumption. Without insurance, households still bear disaster losses, often when their income and savings are already under pressure. Subsidising that protection creates a trade-off between immediate assistance and future development incentives.

Insurance support should protect vulnerable households without concealing incentives to continue building in dangerous locations. A purely actuarial price may force the poorest households to move before safe alternatives exist. Research by Carolyn Kousky and Howard Kunreuther argues for means-tested vouchers paired with mitigation finance, rather than broad premium discounts that hide the risk signal.[16] The fiscal design should separate frequent losses, rare catastrophic losses and immediate liquidity. Households can cover small repairs with savings and targeted grants; cities can fund maintenance and contingency reserves; governments can arrange emergency liquidity or risk transfer for large shocks. Insurance payouts should be

linked to repair capacity and social protection so that liquidity reaches households, renters and small landlords after a disaster. Parametric insurance adds basis risk: an event can cause household loss without triggering the index, or trigger a payout where little damage occurred. Mortgage and capital-market contracts also need to recognise how exposure changes over their term.

Credit and capital-market exposure

Long-term credit should account for changing climate exposure and the division of residual losses between owners and public budgets. A 25-year mortgage on a low-lying home may face rising losses or insurance costs. Appraisals should test those changes and consider how abrupt withdrawal of lending could affect existing owners. Green or resilience finance changes incentives only when verified physical performance underpins the label. A cheaper bond for a "climate-smart" project may produce little benefit if the developer captures the interest saving while household prices stay fixed. A publicly subsidized retrofit loan should document lowered indoor heat, flood entry, expected loss or insurance premium, alongside the distribution of savings among owners and tenants. The state has a balance-sheet problem too. Public guarantees shift loss to taxpayers. Flood defenses create ongoing commitments to maintain and upgrade them. When land values rise behind a defense, government has an opportunity to recover part of that uplift through betterment charges, land leases or property taxation. Appraisal should identify how public protection changes development incentives and how residual disaster costs are shared between owners and public budgets. Adaptation also changes which locations are desirable and who can afford safer land.

Access to safer land

Safer land and public protection can create gains and exclusion pressures that need a distributional assessment. A new risk map can reduce the value of a parcel by placing it in a no-build zone. A new dike or reliable water connection can raise adjoining property values. A relocation program can create a construction pipeline and new public-land opportunities. Political conflict follows these changes in asset value. Four sets of interests should be assessed. Exposed owners and developers ask for protection, compensation or permission to build. Some incumbent owners in safer, high-opportunity neighborhoods may resist density that would let displaced or migrating households move in. Poor renters and informal workers need safety without losing location and networks. Utilities, transport agencies and municipalities bear costs that national housing budgets often omit. Identify which groups receive the benefits and which pay the costs. A workable reform must therefore provide safe capacity near jobs as well as restrict dangerous development.

Actor	Asset or institutional interest	Veto or influence	Possible agreement
Exposed landowner / developer	Public defenses and rezoning can restore land value	Land litigation, permits, political access	Publish risk and value; allow safe intensification elsewhere; transparent compensation
Safe-area homeowner	Scarcity supports asset price and neighborhood control	Local hearings, elections, zoning objections	Fund services and offer citywide, by-right infill with design protections
Informal renter / worker	Cheap access to jobs; flexible space; weak documentation	Collective action, local knowledge, low formal power	Right to return, portable rental aid, nearby serviced alternatives, earnings protection
Structure-owning landlord	Rental income from low-cost rooms; benefits from upgrading	Can raise rent or withhold occupancy records	Co-fund safer retrofits with rent and habitability conditions
Utility / municipality	Revenue, network performance, tax base, liabilities	Connections, rates, maintenance, permits	Long-run O&M; transfers and revenue from added safe density
Contractor / insurer / lender	Construction margin, premium, loan security	Procurement, underwriting, rate setting	Open bids, shared risk data, additionality checks, independent verification
Politician / land broker	Control over relief lists, safe parcels and exceptions	Allocation, enforcement, legal change	Public transaction log, appeals, independent audit and multi-year funding with legal authority

The first test of a reform coalition is whether it can produce safe housing capacity in locations with jobs. The second is whether the original poor households can use that capacity. A new planning rule that allows apartments but leaves financing, water and landlord discrimination untouched may preserve exclusion. Relocating traders from a river corridor to unaffordable premises can reduce flood exposure while increasing livelihood and rent insecurity.

Race, caste, class and migration status matter because social exclusion changes access to the safer housing stock. South African spatial separation makes long commutes costly. Indian renters can encounter discrimination by caste, religion and family status. In African cities, migrants and tenants may be absent from compensation records because their claims are neither titled nor readily documented. Every climate-resettlement program needs a record of these people before announcing the intervention. A technically sound prohibition can also be politically fragile. Without fair notice, compensation, an appeal mechanism and nearby affordable alternatives, affected owners organize to reverse it. Without protection against tenant displacement, environmental restoration can displace the residents it is meant to protect. The durable agreement finances safer supply, respects due process and retains the public benefit from restored land. Digital risk information can support that agreement only when its classifications are checked and contestable.

4 Fund and verify a feasible risk reduction programme

Digital risk information and public decisions

Risk models need field engineering and implementation capacity before they can guide housing decisions. Flood models describe depths at one scale; parcel records describe legal rights at another; utility plans show capacity in separate systems; household surveys observe livelihood ties that satellite images cannot see. A city may hold all four databases and still approve projects with agencies using inconsistent records. Modern imagery, remote sensing and AI can help join these records. The World Bank's Global Program for Resilient Housing describes a five-step method: map structures; infer features such as materials and openings from imagery; screen vulnerability; connect households to subsidy or finance; and verify safer works. Its own description emphasizes field engineering checks and local implementation partners as part of the process.[17] Parcel classification makes the consequences of model error particularly direct for households.

Decision	Potential information gain	Minimum empirical test	Power and safeguard
Flood and site selection	Parcel-level inundation, past events, return-period uncertainty	Blind field audit; independent hydraulic review; scenario comparison	Hazard labels can devalue land; publish uncertainty and allow challenge
Heat retrofits	Roof materials, indoor exposure and local shade	Temperature loggers and treatment-comparison design	Tenant may receive heat benefit; landlord may capture capital gain

Decision	Potential information gain	Minimum empirical test	Power and safeguard
Retrofit and construction grants	Damage class, quantities, progress, repair quality	On-site structural samples and defect checks	Automated fraud flags need human review and right of correction
Rental and relocation	Occupancy, rent, commute, earnings, school access	Household panel before and after moving	Tenants must be counted; limit reuse of personal data for eviction
Utility and water planning	Network leakage, service outages, flood-drain blockages	Meter and field verification; time-to-repair tracking	Shared data must carry one accountable maintenance owner
Insurance and finance	Hazard-adjusted collateral and expected loss	Calibrate by realized claims; test bias in approvals	Avoid opaque exclusion by neighborhood, income or tenure

Parcel risk classifications need documented uncertainty, field checks, lawful data use, and correction procedures. A model may identify structures inside a likely floodplain in hours. Its value depends on ground-truth accuracy, whether hydraulic design supports the classification, whether occupants are legally recorded and whether government can offer a lawful, funded remedy. A wrong flood label can make a bank reject a mortgage; a missed channel can place another family in danger. High-consequence classifications deserve independent checks. The right government product is a shared, versioned risk-and-rights record: parcel and occupancy IDs, flood depth and uncertainty, heat exposure, utility capacity, livelihoods, planning status, tenure claims, public investments and decision history. Public layers should be open where safe. Identifiable household records need purpose limits, access logs, consent or clear legal authority, and a route to correct errors. The choice of intervention should then compare full household and public costs under those risk assumptions.

Comparing the costs of safety options

Adaptation costs should be compared with relocation and residual losses using common assumptions and separate safety gates. A housing minister asks how many units can be provided. A city manager asks whether the utility can support them. A low-income household asks whether it can still get to work and pay the rent. All four need the same appraisal. For a defined group of households, compare three classes of options: (A) in-situ upgrading with

drainage, safer electrical systems, cool roofs and maintenance; (B) relocation to an available peripheral site; (C) relocation or infill on safer, well-connected land. For each option, count public capital and operations and maintenance, private building costs, expected disaster losses, recurring transport and income effects, insurance or recovery cost, and the value of any asset surrendered. Test credible future climate scenarios and severe tail events. Exclude options that fail life-safety standards even if their present-value cost looks low. Here is an illustrative per-household calculation for twenty years using a 5 percent real discount rate. All figures are constructed examples in constant US dollars, not measured costs for a named city. Assume the options deliver equivalent basic housing services and that upfront values are net of any residual asset value. The annuity factor is approximately 12.46. The worked example below shows how travel and livelihood costs can change the apparent ranking.

Incremental cost per household	A: protect on site	B: move far	C: move near jobs
Net capital / relocation now	$5,000	$13,000	$15,000
Annual residual risk + O&M;	$400	$180	$200
Annual livelihood / travel penalty	$0	$820	$0
20-year present value	$9,985	$25,462	$17,492

A location that looks safer in a flood model may impose household costs that require a different feasible solution. The added transport and income burden makes it the most expensive option in this constructed example. Protecting the present site costs least if residual life risk can be brought to an acceptable level. A near-jobs move may be necessary when a flood channel or landslide slope cannot be made safe; its larger capital bill purchases both safety and access. Present value alone cannot settle the decision. The model should include fatalities and serious illness, disruption during transition, uncertainty in future hazards and the distribution of costs. A single expected-loss number can conceal an extreme event that the household cannot survive. The option screen should use a separate life-safety gate, sensitivity to climate change and a household-consent or due-process gate. The same calculation can help test how much extra cost an accessible safe site could justify.

The calculation also helps price safe land. At a 5 percent real rate over twenty years, paying an extra US$5,000 for an adequately serviced safer site costs about US$401 a year when annualized. If the cheaper site imposes US$450 in expected disaster losses and US$500 in added transport burden, the safer site may be a good public investment even before health and avoided disruption are considered. This is an arithmetic illustration, with no claim that those figures describe a particular market. Financing should then assign the cost to the beneficiaries and institutions able to sustain the protection.

Financing risk reduction

Risk-reduction finance should match the benefit and distinguish institutional progress from completed housing outcomes. Developers can pay for site drainage and code compliance through transparent standards; households can finance affordable repairs when tenure and cash flow are secure; municipalities can fund catchment drains and operations and maintenance from transfers, tariffs and property revenue; national government can finance major flood defenses and compensate households asked to surrender rights for public safety. Donor and climate finance can support public goods and the early costs of better risk information, with explicit mechanisms to keep the gain from bypassing low-income residents. Kenya offers a fiscal example. The Financing Locally-Led Climate Action program runs across all 47 counties with roughly US$294 million in total program funding according to the National Treasury, combining national and partner finance with county climate institutions and ward-level planning.[18] A 2024 World Bank account reports counties meeting key institutional conditions rising from nine to 45 within two years. These are institutional results, not proof that the program has already produced affordable climate-resilient homes.[19] Shared platforms can fund the network and maintenance components that individual housing projects cannot provide alone.

Such a platform can finance the shared parts of the housing adaptation problem: drainage, local hazard mapping, ward investments and maintenance capability. Housing agencies should then bring tenure, building safety and relocation or retrofit finance. A shared outcome for climate and housing budgets is additional low-income household-years of safe occupancy within reach of work. The assessment must verify that these shared commitments meet the social and safety conditions for proceeding.

Climate and housing assessment

Investment review should make risk evidence, household consultation, and routes to challenge decisions explicit. A team seeking a housing investment or flood-mitigation loan should present a household-level record, an engineering model with uncertainty, a livelihood model, the public-land implications, a credible operating institution and a political coalition. It should show how rejected alternatives compare. Affected households need a documented consultation process and an accessible way to contest errors or proposed treatment. Hard safeguards must remain binding even when another part of the appraisal scores well.

Gate	Question for the approving authority	Proof required
1. Hazard	What event and future scenarios define risk?	Model limits, return periods, climate ranges, upstream effects
2. People	Who is exposed, including renters and migrants?	Verified occupancy and tenancy census; correction and appeal

Gate	Question for the approving authority	Proof required
3. Safety	Which parcels can be made acceptably safe?	Independent engineering and life-safety thresholds
4. Location	What happens to job and school access?	Household travel time, fares, earnings, social networks
5. Land	What claims, rights and rents change?	Parcel, public-land, compensation and value-uplift ledger
6. Planning	Is replacement safe floor space legally available?	Permitted density, serviced inventory and affordable-rent pipeline
7. Housing	Can incremental and rental homes be improved?	Retrofit designs, landlord terms, tenure rules, workmanship checks
8. Utilities	Will water, drainage, power and sanitation last?	Network capacity, hazard service standard and maintenance owners
9. Construction	Can firms deliver at transparent cost and quality?	Competitive bids, material compliance, supervision and repair markets
10. Finance	Can owners, tenants, and small landlords pay?	Grants, repayment, insurance, guarantees and disaster liquidity
11. Public finance	What is the full 20-year public obligation?	Capex, O&M;, contingent liability, tax and land opportunity cost
12. Distribution	Who gains from safer land and who can be displaced?	Incidence by income, owner/tenant, gender and social group
13. Politics	Who can veto and what coalition will sustain reform?	Actor-rent-veto map, compensation, hearing and escalation rules
14. Digital	Which model or map influences an adverse decision?	Field accuracy, bias test, explanation and human appeal
15. Outcome	What result will be independently verified?	Safe occupied household-years; commute, service and loss tracking
16. Adaptation	What happens when the hazard or the city changes?	Review triggers, future upgrade funding and reversal or retreat plan

Treat gates 2, 3, 4 and 12 as hard social safeguards. A project can score well on flood depth and construction speed while harming displaced renters, removing market access or pricing poor

residents out of improved neighborhoods. A red life-safety risk may justify relocation. It cannot justify leaving livelihoods out of the compensation and placement plan. The implementation sequence should resolve those safeguards before commitments become irreversible.

A staged implementation sequence

Begin implementation with local hazard, livelihood, and tenure evidence under published data-use rules. Fund drainage maintenance, health response and utility revenue stability as well as protective construction. Planning, utilities, health and emergency services must rehearse responsibilities and disclose failures. Track exposure and actual losses separately, service continuity, heat outcomes, recovery time, displacement and total household costs. Use the following stages to prepare a bounded operation. The responsible agency should first cost staff time, specialist work, consultation, and verification, then choose the number of sites it can handle. Set dates after that review. Disputed rights, procurement, or statutory approvals may extend the schedule; do not advance until the applicable legal and funding conditions are met. Stage 1: prepare linked hazard and household-access maps. Name a city-level accountable lead. Combine hazard and service layers with jobs, transit, tenure, rents and occupancy. Select a manageable shortlist of neighbourhoods where risk and poverty overlap, and record what information remains uncertain. Make the data-sharing and privacy rules public. The next stage should trace households and agree treatment, funding, and remedies.

Household tracing should establish compensation, placement, service access, and emergency support responsibilities. Stage 2: trace households and decisions. Conduct sampled engineering checks, tenant-inclusive household fieldwork and a process trace for selected locations: a site suitable for upgrading, a site requiring significant drainage, and a site where relocation is a serious option. Record every approval and agency veto. Stage 3: price alternatives and agreement. Prepare a twenty-year balance sheet for each option. Publish public-land values and relocation costs. Test whether safe, well-connected land can accommodate additional rental and incremental homes. Negotiate utility operating commitments, landlord conditions, financing and independent verification. Stage 4: prepare measured demonstrations. Launch a limited adaptation or safe-housing pilot with a feasible comparison design only when required consultation, safeguards, financing, and procurement are complete. Record baseline indoor heat, flood occurrence, commute, rent, earnings, water reliability and grievance outcomes. Reserve an external review and adjustment point before scaling. Design an emergency liquidity and post-disaster livelihood support channel. The result should be a funded agreement whose obligations can be checked by residents and government.

The sequence is intended to produce a funded and reviewable implementation agreement. Households see choices and enforceable rights. City agencies get a funded plan. Developers receive clear buildable land and rules. Owners know how compensation and protection work. Treasury can assess the lifetime public commitment alongside the initial capital cost. The same transaction record allows citizens to check whether the agreed obligations are met. That agreement provides a basis for adapting the investment as risk and housing conditions change.

Implications for housing investment

Climate-sensitive housing policy must combine safer locations, viable protection, and the institutions that sustain affordability. Poor households will still need to live near jobs. The practical task is to expand safe, connected housing supply while protecting the social and economic value that existing neighborhoods already create. Appraisal should assess flood exposure, heat, water reliability, disaster losses, access to work, and the households likely to remain after improvements. These measures allow planners to compare physical protection with its full household and fiscal consequences. It assesses the rules, finance, and services needed to improve safety and affordability in this stock. Fieldwork should test those conditions against observed housing and hazard experience.

Prepare the local decision

Compare protection and relocation using local engineering, household, and service evidence, with a trigger to revisit failed assumptions. Combine hazard scenarios with building condition, drainage capacity, past damage, heat exposure, household losses, insurance terms, and access to refuge. Record uncertainty and the date of each map. Walk drainage routes with residents and check whether maintenance failures explain observed flooding. The Ahmedabad, Cape Town, and Chennai cases above support different mechanisms and should not be treated as interchangeable evidence of resilience. Engineers should assess physical options with emergency services, utilities, public-health staff, residents, insurers, and municipal finance. Compare protection, building adaptation, and relocation using household costs and livelihood effects as well as engineering costs. Identify who can maintain a drain, enforce a safe setback, fund temporary accommodation, and resolve disputed eligibility. Validate mapped high-risk and low-risk locations in the field, including missed settlements. Use forecasts to trigger a specified inspection or warning response, with an accessible non-digital channel. Do not turn an uncertain risk classification into automatic clearance or credit refusal. Record avoided disruption where defensibly estimated, actual service downtime, household cost, and the outcomes of relocated residents. Revisit the design when observed events exceed its assumptions. Chapter 12 applies the delivery and safety tests to incremental homes and small rental buildings already accommodating urban growth.

Risk information should lead to an affordable protective action that an identified institution can deliver. Warnings require trusted communication, feasible household responses, and functioning health and utility services. A copied hazard threshold, uniform water cap, or distant relocation plan can shift burdens onto people least able to respond. Permits, maintenance, and emergency arrangements must reinforce the protective action. Test correlated shocks across the whole city, not just the protected project. Upstream development and simultaneous emergencies can invalidate pilot performance.

Sources and evidence notes

The sources below report modeled exposure, administrative program data, project targets, and observational research. Figures labeled “estimated,” “reported,” “projected” or

“illustrative” retain those qualifications. Climate exposures are modeled; project targets are prospective; program figures are generally administrative; observational studies are described with their identification limits.

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12. Incremental and small-scale rental housing

Support small-scale rental and incremental housing through improvements that are safe, financially feasible, and protective of existing tenants. These producers need proportionate approvals, technical help, suitable finance, and services that can accommodate additional rooms. Enforcement must distinguish immediate life-safety requirements from improvements that can be staged. Understanding the households and producers comes before choosing the support package.

Small-scale construction and rental markets supply housing that official programs may not record. A household adds a room behind its home. A plot owner rents six bedsitters above a shop. A worker pays for a single room close to the factory. A mother extends a roof before a child is born. These acts add space, stage payments and keep jobs within reach. They can also expose renters to collapsing buildings, fire, polluted water, unlawful eviction and monopoly prices for basic services. Small builders supply inexpensive rooms in locations where formal developers may not serve low-income demand. Prohibiting this supply can reduce housing access; leaving it unregulated can expose tenants to unsafe construction. Policy must make safe building and fair rental practices viable for small producers. Delivering that support requires coordinated changes to services, finance, permissions, and tenant protection.

Small-rental policy should make safe production and fair occupancy feasible through a coordinated package. A plot needs a claim strong enough to justify investment, permission to add floor area, a drain that can take the extra load, a mason who follows a safe design, a lender able to fund a modest building stage, a landlord who maintains the property and a tenant whose rights survive a rent rise. Delivery requires the responsible agencies, builders, and lenders to share relevant information and make timely, reviewable decisions. Incremental housing describes a construction process: households and small builders add quality or space in stages. Informal rental describes a tenure and market arrangement: occupancy is bought from someone whose land claim, building permission or lease may be weak. The two overlap, yet neither guarantees poverty, low quality or illegality. Diagnose the specific construction, tenure, service, or finance constraint before selecting a reform. Support small rental buildings through safe construction, feasible finance, enforceable tenant rights, and funded connections. Mukuru shows why that package must distinguish the many claims attached to a single structure.

1 Understand the households and producers in small rental markets

Small rental units and their occupants

Mukuru's rental economy depends on access to nearby work and on overlapping land, structure, and tenancy interests. Its location keeps workers close to factories, markets and transport. The building in which a family rents may sit on a parcel held by one legal owner, carry a metal structure erected by another person, receive water from an intermediary and hold tenants who have little proof of residence. When the city improves a road, the distribution of benefits

depends on land ownership, structure ownership, service arrangements, and tenancy. Enumeration is needed to identify those interests before public investment changes their value.

Resident enumeration makes visible the occupants and claims that ownership records alone can miss. Resident enumeration undertaken with Kenyan universities and the Muungano Alliance mapped about 23,000 structures and identified roughly 100,000 households. Around 94 percent were tenants. Many paid structure owners who had built shacks on land owned by other, sometimes absentee, proprietors. The same research uncovered a system of service brokerage and threats of eviction as land values rose. The International Development Research Centre described court action that halted evictions and the subsequent effort to build a shared record of land, structures and occupants.[1][2] For compensation, identify affected households separately from title holders. A building record cannot tell who receives rent. A beneficiary list cannot show who will lose customers when a lane closes. If government compensates registered owners alone, public investment can reward the strongest claim and exclude households whose occupancy or livelihood claims are unrecorded. A statutory planning route can then connect that information to decisions about services and rights.

Mukuru's planning process illustrates the need for a treatment rule for each type of claim. Nairobi County's 2017 declaration of the Mukuru Special Planning Area created a legal route to coordinate infrastructure, land, health, environment and community groups. The planners organized approximately 100,000 families into 13 planning segments, with smaller clusters for consultation. Such organization converts dispersed knowledge into usable planning evidence. Implementation must then address a further problem: residents, structure owners and landowners have claims to different pieces of the same place. A plan needs a rule for each claim.[3] The resulting agreement must accommodate how households and landlords build in stages.

Incremental housing supply

Incremental housing responds to changing income and household needs through staged investment. A worker renting near a market can save on transport despite cramped shelter. A young couple may rent a room while testing job prospects. A family with some security may build a first floor, add a toilet later, then rent a room to pay school fees or finance a second floor. A pensioner may depend on rent from a backyard unit. Housing is simultaneously consumption, collateral, a workplace and a family savings instrument. The conventional housing pipeline offers a finished unit financed with a long mortgage and delivered to one registered household. The incremental pipeline runs differently: savings, wages, a small loan and a building-material purchase arrive in uncertain order. Construction pauses during a job loss. A tenant moves before a structure is complete. A builder uses rent from the first two rooms to pay for the next two. A standard mortgage rarely fits this risk pattern. That flexibility can expand supply while making the size and use of the resulting stock difficult to measure.

Staged construction allows later investment to be deferred, but detected structures should not be treated as verified occupied homes. Each stage preserves the right to postpone the next investment until household income, demand and legal security justify it. That flexibility can be valuable where earnings fluctuate. It can also be expensive. Small purchases lose

bulk discounts; repeated mobilization raises labor costs; poorly planned foundations make future floors dangerous. A policy that accelerates building without improving design can turn incremental finance into a subsidy for structural risk. Satellite-based analysis by the Gauteng City-Region Observatory found that backyard structures increased from 266,929 in 2001 to 813,224 in 2016. Over those fifteen years, backyard structures accounted for about 546,000 of the 1.3 million net additional residential structures reported in the same dataset: roughly 43 percent. These are detected structures, some with uncertain residential use; they are not an independently counted stock of occupied rental dwellings.[4] Service planning must account for the actual demand added within existing plots.

Figure 12.1. Backyard housing structures in Gauteng



Source: Gauteng City-Region Observatory / GeoTerraImage; structures, not confirmed households [4].

Additional rooms can increase service demand well beyond that assumed for the original dwelling. Backyard rooms add density in townships where formal public housing and services already exist. They help young households enter the city and give owners a revenue stream. Yet the infrastructure attached to one household often serves several more. Service planning based on the original house can therefore understate the plot's later demand. The producer's cash flow must therefore include realistic connection and capacity costs.

Small-landlord finances

Small landlords differ in assets, control, financing, and risk, so support should be based on their actual business model. Policies that call them all "informal" conceal differences in wealth, power and incentives. Their financing needs, maintenance practices, rent levels, and compliance costs should be investigated separately. Absentee ownership alone does not establish unsafe practices or unlawful conduct. The small landlord's real choices depend on expected net rent, the probability of losing the land, the price of materials, the right to add rooms, the chance of fire or prosecution, and the ability to collect rent. A reduction in approval delay has value only if permission survives future political changes. A loan has value only if the extra rooms attract tenants at a rent that services debt. A public sewer connection has value only if the charge and network capacity are credible. A worked investment account can show which charges and repayment terms are feasible.

A small-rental investment should be tested through its full construction cost, net rent, and debt-service capacity. A six-room rental project costs R1.2 million to construct, excluding the value of the already-owned land. The landlord finances 60 percent with a ten-year loan at 12 percent annual interest. At R3,000 rent per room, 90 percent collection and occupancy combined, and operating and reserve costs equal to 20 percent of collected rent, the property generates

roughly R12,960 in monthly net operating income. Monthly debt service is about R10,330. The resulting debt-service coverage ratio is 1.25. Reduce rent to R2,600 and the ratio falls to 1.09. Raise the borrowing rate to 14 percent at R3,000 rent and it falls to 1.16. These are illustrative assumptions, not observations from a South African project. The difference between a bankable and a marginal property is small. Construction delays, unpaid rent, failed drains or a higher loan rate can erase the surplus. The illustration shows how cash-flow constraints can reduce funds for maintenance or increase reliance on rent advances. It does not establish how common either response is. An expensive small loan can make an otherwise low-cost design unaffordable. The example below makes those assumptions explicit before financing is considered.

Monthly rent per room	12% loan: DSCR	14% loan: DSCR	Interpretation
R2,600	1.09	1.00	Little margin for repairs or shock
R2,800	1.17	1.08	Thin resilience
R3,000	1.25	1.16	Potentially financeable; verify market rent

Illustration: capital cost R1.2m, debt R720,000, amortization 120 months, combined occupied/paid share 90%, and operating/replacement costs 20% of collected rent. DSCR = net operating income / scheduled debt service. Land opportunity cost, income taxes, origination fees and major shocks excluded.

A lender should finance the verified building and rent stream, with staged drawdowns and inspection, while checking the rights and income evidence available to the borrower. A wholesale capital-markets vehicle may fund thousands of such loans if loss data, collateral rights, servicer performance, concentration risk and tenant outcomes can be audited. Securitization transfers risk; it cannot make unviable rents viable or unsafe buildings safe. Cape Town's additional rental rights provide a policy setting in which to test that production and financing model.

2 Compare support, regulation, and tenant protection

Backyard rental development in South Africa

South African rental-rights and financing initiatives create a testable opportunity rather than proof of completed affordable supply. Cape Town's amended Municipal Planning By-law, gazetted in August 2025, created an additional use right for small affordable rental flats in designated areas. The rule permits up to eight additional rental units alongside a house, or twelve units on a site without a dwelling house, subject to conditions. It also added selected incentive overlays and powers to stop unlawful construction. This is a specific exchange: more legal floor space for small builders, paired with greater ability to require safe connections

and enforce standards.[5] In July 2026 Rand Merchant Bank and INDLU announced a R1 billion blended-finance program for micro-developers. Its announced structure includes a R30 million concessional construction-warehouse commitment from the FirstRand Foundation, long-term funding from FNB, and participation by FSD Africa Investments in catalytic finance. INDLU reported that its earlier model had helped deliver more than 2,200 rental units and had facilitated approximately R311 million in finance. These are company-reported delivery figures and a new funding commitment; independent measures of tenant rents, occupancy, defaults, subsidy cost and building performance are still needed.[6] The evaluation should follow each stage from usable permission to sustained tenant occupation.

Evaluate the rental-development mechanism through permissions, finance, construction, occupancy, and tenant costs separately. Zoning clarity raises the value of compliant plots. Engineering standards and formal connections lower catastrophic risk. Staged finance turns an owner's future rental income into current construction funding. Digital rent collection creates a repayment history. A larger lender can then fund a portfolio. Each change also affects the distribution of costs and returns. Permits may raise land prices. Loan terms may raise the minimum feasible rent. Formal utility charges may land on tenants. A portfolio sold to outside capital may give the financier new influence over property management and eviction decisions. A proper evaluation would compare eligible and otherwise similar plots near the new regulatory boundaries. Did rental rooms increase relative to untreated plots? Did rent per square meter, structural compliance, connection quality and household transport cost improve? What was the net gain for tenant households after utility bills? How many small landlords borrowed safely, and how many became overleveraged? The announced R1 billion program offers an opportunity to test these outcomes as lending and construction proceed. India's vacant stock illustrates why even a completed dwelling may fail to become usable rental supply.

Vacant housing and rental demand in India

Vacant homes and unmet demand can coexist when legal risk, location, or operating conditions prevent letting. Census 2011 recorded roughly 11.09 million vacant urban census houses. The figure is old and its category includes dwellings that may be poorly located, unusable, unfinished or otherwise unavailable for rental; it should never be read as 11 million immediately habitable units. OECD analysis drew attention to the concentration of vacancies in peripheral developments with weak transport and services. Both geography and law can keep a building vacant even while workers crowd into central rooms.[7] A careful empirical study by Sahil Gandhi, Richard Green and Shaonlee Patranabis used district panel data and changes in rent-control provisions in four Indian states. It found that restrictions on rent revision were associated with an additional 2.8-3.1 percentage points of vacancy. The authors modeled a possible 7.5 percent reduction in the housing shortage in reforming states, with faster contract enforcement also linked to lower vacancies. The result concerns legal and judicial incentives; it does not imply that removing tenant protection alone will make housing cheap. The vacancy measure pools several forms of vacancy, and model estimates require care in new places.[8] Refurbishment and rental-conversion programmes need to resolve those specific obstacles.

Rental conversion requires a viable agreement over ownership, refurbishment, operation, and demand at the site. The Model Tenancy Act, approved in 2021 for states to adapt, seeks clearer rental contracts, rules for deposits and faster dispute resolution. Because tenancy law depends on state legislation, the national model has no uniform automatic effect. The Affordable Rental Housing Complexes program seeks to put publicly funded vacant housing into rental use and to stimulate purpose-built rental stock on other land.[9][10] The program's official public-housing conversion page currently lists 83,534 government-funded vacant houses as available for its first model and 5,783 as converted, about 6.9 percent of that administrative pipeline. Those figures are a web snapshot accessed in October 2026; they do not establish how many converted units are occupied, in which locations, or by what income group. The page itself shows large nominated stocks in Delhi and Maharashtra with no conversions recorded in the summary. The basic institutional obstacles include who holds the asset, the cost of refurbishment, the ability to bundle it into a viable operating contract, and demand at the actual site.[9] The test is whether a household can use the resulting home without losing access to work.

An empty apartment is an asset with a location, a claim, a repair bill and a future operating duty. A rental project works when the owner, contractor, municipality, operator and tenant can sign a credible contract for those facts. A national dashboard that counts conversions cannot reveal the household value of a unit if its commute makes a job impossible. That use also depends on legal protection that supports tenants without unnecessarily withdrawing supply.

Tenant protection and rental supply

Tenant protections should be assessed together with landlords' maintenance, entry, and withdrawal decisions. Landlords need predictable possession rules, rent collection and a credible way to resolve disputes. One-sided systems can unravel. When a landlord expects a five-year court struggle over a room, keeping the unit vacant may be rational. When a tenant expects expulsion after a repair request, reporting a dangerous building may be irrational. The sustainable agreement separates three functions. First, set clear written rights: habitability, notice, deposits, repairs, rent changes and lawful eviction grounds. Second, make small disputes fast and affordable: mediation, an accessible tribunal, inspection and enforcement. Third, expand supply so market tightness cannot erase statutory rights. Rent stabilization for incumbents may prevent displacement; controls that destroy returns to new safe rental units can shift investment away from the sector. Design and context determine the net effect. The Indian evidence supports measuring legal risk and household welfare as distinct outcomes.

Rental reform needs separate measures of vacancy, supply, and tenant welfare. The Indian vacancy evidence identifies a real supply response to legal risk. It does not establish that weak protection is welfare-improving. A proper analysis must value the landlord's willingness to rent, the tenant's stability and bargaining power, and the fall in overall rent if new supply enters. The evaluation should report all three outcomes rather than infer tenant welfare from supply alone. Advance-rent requirements create another barrier that operates through household liquidity rather than the monthly rent alone.

Advance rent payments in Accra

Accra's advance-rent system finances landlords partly through tenants' upfront payments. A low-income worker can afford monthly rent and still fail the entry test. Landlords often demand one or two years of rent before handing over a key. A 2024 mixed-methods study of 362 Ghanaian renters found the norm of advances of two years or more created repeated borrowing pressures and disrupted savings. Section 25(5) of Ghana's Rent Act distinguishes monthly or shorter tenancies, for which more than one month in advance is prohibited, from tenancies exceeding six months, for which the limit is six months. The cited news report describes officials announcing renewed enforcement of the six-month ceiling for longer tenancies beginning in April 2026; evidence that the new enforcement has changed market behavior remains limited.[11][12] If monthly rent is GHS 1,000, a two-year advance requires GHS 24,000 before moving. Six months requires GHS 6,000. The difference, GHS 18,000, is a liquidity barrier, equal to eighteen months of rent. This calculation does not imply that the landlord loses GHS 18,000 in economic rent. The landlord loses access to an interest-free pool of working capital and transfers financial risk back onto itself. Removing that requirement needs an alternative way to finance the transition and enforce the new terms.

A prohibition without replacement credit can push the same demand into illegal fees, higher rent or tenant selection. A package to test would enforce the applicable cap, assess whether working-capital finance against verified leases reduces rent advances, support deposit escrow, and resolve arrears disputes while protecting tenants from summary eviction. The fiscal question then becomes whether government should subsidize the transition, and whether tenants receive the savings. Safety upgrading presents a similar need to make compliance practically feasible rather than rely only on prohibition.

Improving safety in stages

Staged safety improvement should distinguish immediate life-safety requirements from upgrades that can be sequenced. A more useful choice is an ordered set of safety duties. Small rental buildings that pass structural and site-safety screening may be improved in stages. Prioritise immediate risks to life while setting deadlines and funding for the remaining work. A weak foundation cannot safely carry a subsidized extra floor. A single exit cannot safely evacuate a crowded tenement. A water tap connected to a failing sewer can increase exposure to disease. Begin with a risk screen before inviting investment: floodway, industrial hazard, unstable slope, fire access, electrical load, structural capacity and contested rights. A red life-safety condition triggers urgent enforcement and a credible housing alternative. For other buildings, the city can offer standard designs for one, two or three floors; pre-approved reinforcement details; a licensed engineer's inspection at critical stages; tested materials; and an accessible permit for a modest improvement. The next stage brings separate metering, toilets, drainage, ventilation and durable weather protection. Builders need a published route through those stages, with evidence on what each improvement achieves.

A usable compliance schedule should specify duties, costs, inspections, and the outcomes supported by the evidence. Requirements designed for large towers may make

compliance infeasible for a small rental extension. Keep structural and life-safety requirements enforceable, and allow dated improvement plans for other defects where law permits. Independently inspect decisions where officials retain discretion to demand additional work or fees. Randomized shelter evaluations distinguish physical improvements from income effects. Randomized evaluations of basic prefabricated shelter improvements in El Salvador, Mexico and Uruguay found higher housing satisfaction and general well-being, with improvements in child health in two settings. The researchers found no consistent employment or durable-goods gains. These studies support housing and well-being benefits without establishing general wage or neighborhood-service gains.[13] Shelter improvements should not be treated as proof that a complete rental-upgrading package will raise livelihoods.

Stage	Public obligation	Small-builder obligation	Tenant safeguard
0 Map the claim	Record land, occupiers, tenants, risk and service capacity	Disclose rooms and intended use	No registry shared for eviction without due process
1 Prevent catastrophe	Fire access, hazard rule, independent inspection	Load-bearing safety, wiring, exits, drainage	Emergency remedy and temporary shelter if unsafe
2 Connect and improve	Predictable utility connection and standard plans	Ventilation, sanitation, repairs and materials	Written terms, notices, rent disclosure
3 Add safe rooms	Quick permit, staged finance, verification	Engineered extra floor and compliant meter	Rules for displacement during works
4 Sustain the asset	Tax, inspections, services, complaint path	Maintenance reserve and tenancy records	Renewal rights and remedies

These shelter studies do not validate a complete rental-upgrading package. Accra's advance-rent barrier remains even when construction becomes cheaper, and physical improvements need not raise earnings. Set separate objectives for safety, affordable entry and livelihoods, then judge each against the evidence. [13,11–12] Additional safe rooms also need an area-level service plan.

Density and municipal services

Small rental growth uses shared municipal capacity and requires a fair rule for network costs. Its owner may receive more rent and pay more property tax. Yet water and sewer networks may have been designed for a single house. Municipalities often treat

incremental rental construction as a private land-use matter and omit its future load from service planning. Service plans and budgets should record that additional demand. The fiscal solution has three parts. First, forecast people and water use from observed structures and tenancy counts, with a check on whether satellite imagery mistakes storage sheds for dwellings. Second, publish a neighborhood capacity schedule for sewer, electricity, water and access roads. Third, pair extra legal building rights with targeted network upgrades and a transparent tariff or development charge. Charging the first compliant builder the entire cost of a trunk main would punish compliance and reward those who built earlier without permission. An area agreement can spread those costs without penalising the first builder who seeks compliance.

The instrument is an area-level agreement. City and utility fund common capacity. Builders fund fair local connection costs. Tenants pay a regulated, metered service bill where possible. National government can pay performance grants for verified additional safe rental rooms and connections, especially when municipalities bear costs while the metropolitan labor market gains workers. Plan operations before construction begins. A new block of six rooms adds solid waste, wastewater, roof runoff, fire load and potential arrears. A permit should specify who maintains each common part and what reserve pays for replacement. Link room counts and asset records to funded maintenance responsibilities. Service access alone will not resolve discrimination in who can rent the improved rooms.

Discrimination in low-cost rental markets

Low rents do not establish accessible housing where landlords or allocation practices exclude particular households. Some landlords refuse Muslims, migrants, unmarried adults, single women, people in stigmatized occupations or those lacking formal pay slips. In Delhi and its suburbs, a controlled online rental audit found a 35 percent landlord response rate for upper-caste Hindu applicants and 22 percent for Muslim applicants. The study did not find equally clear callback penalties for Scheduled Caste or Other Backward Class names in that design. It measured first contact; it did not observe final lease awards.[14] Exclusion works through discretion at several gates. A broker withholds a listing. A housing society adds a membership rule. A lender treats cash wages as unreliable. A utility refuses a meter to an occupant without title. A public program counts a sanctioned unit before learning who is allowed to occupy it. These separate choices cumulate into segregation. Supply support and enforceable rights must address those barriers together.

The housing-policy remedy needs supply and rights together. Require nondiscriminatory rental listings and written reasons for refusals in public-assisted housing; test landlords and brokers through lawful audits; allow alternative income evidence; provide a confidential complaint path; and ensure migrant households can access services without surrendering personal information to their landlord. Tenure security for the structure owner should never become a license to evict the sitting tenant after public upgrading raises value. The resulting reforms will redistribute income and discretion among landlords, tenants, and agencies.

3 Sequence improvements around viable projects and resident rights

Interests in small-scale rental housing

Small-rental reform should identify whose income, costs, or control changes before designing the agreement. Legal floor space creates development value. Sewer connections reduce private service sales. Loan guarantees move default risk onto public balance sheets. A tenancy register shifts bargaining power from a landlord to an enforceable contract and may give the tax authority new information. Stronger building controls threaten builders who profit from shortcuts. Blanket demolition empowers owners of scarce legal homes while harming renters forced farther out. The map below helps locate the conflicts that require a practical remedy.

Actor	Potential interest or risk to investigate	Veto or leverage	Possible agreement
Tenant and migrant worker	Job access, stable rent, safety; weak papers	Mobility, collective action, electoral voice often weak	Protected occupancy, repairs, legal address, fast appeal
Resident small landlord	Rental cash flow and asset appreciation	Refusal to register or invest	More legal rooms and finance for safety, phased costs
Absentee structure owner	Unrecorded rent and control over access	Broker networks, eviction threats	Publicly verified claims; tenant protection; tax enforcement
Land title holder / public agency	Capital gain from tenure recognition or rezoning	Litigation, disposal authority	Value capture, clear land settlement, disclosed transfers
Municipality / utility	Rates, fees, network costs and future deficits	Permits, metering, enforcement	Capacity grants and recurrent revenue for verified safe density
Lender / investor	Interest, data, collateral and step-in rights	Credit rationing and loan covenants	Risk-sharing tied to rent, safety and consumer protection
Homeowners / local elite	Property value and social control of neighborhood	Zoning, court action, politics	Infrastructure assurance and area-wide rather than selective density
Broker / informal service seller	Payment for access, scarcity and exceptions	Control of information, intimidation	Public queues, competitive connections, enforceable complaints

Compliance incentives and tenant protections need to resolve documented conflicts without driving the market further from registration. The welfare gains from an extra thousand safe rooms are diffuse and arrive later. Changes to service charges or approval discretion can impose immediate losses on existing providers or intermediaries. The homeowner who fears densification turns up at hearings. The migrant renter who would occupy the future room may have neither an address nor a vote in that municipality. Completed estates and new titles are more visible than routine drainage, building safety, or dispute resolution. These differences can affect public spending priorities. Rental reform needs support from tenants, small landlords, builders, lenders, and service providers who benefit from additional safe, occupied housing. A feasible coalition links small landlords seeking credit, renters seeking safety, builders seeking steady work, utilities seeking legal customers, and municipalities seeking sustainable density. The agreement must also address lawful objections and assign decisions clearly. Grandfathering existing occupancy for a fixed transition, paying for trunk networks, retaining relocation near jobs where a building is unsafe, and creating fair dispute resolution can reduce the immediate cost of compliance. A purely punitive campaign can discourage registration or reduce rental supply. Success must still be tested for tenants with the weakest incomes and documentation.

Test whether low-income tenants can enter and remain after improvement, including households without formal employment or documentation. Digital records can support that test only if they represent tenancy as well as structures.

Digital records and tenant rights

Rental information should connect physical structures, enforceable rights, financing decisions, and tenant remedies. A satellite image can identify an extra roof. It cannot tell who pays rent, who holds the key, whether the roof is safe, or who would lose a job if the room vanished. The proper digital system links a physical building record to protected, tenancy and service records collected under a valid legal basis, with purpose and access limits. A single official parcel identity should never be mistaken for a single economic beneficiary. Four applications deserve priority. First, service planning: map rooms, occupancy and water demand to direct capital upgrades. Second, staged permits: let a small landlord submit a standard design and receive a timed, reviewable decision. Third, finance: use verified construction milestones and consented rent collections to underwrite a small loan. Fourth, protection: provide tenants with receipts, a repair channel, deposit escrow and an appeal record outside landlord control. The same data can facilitate enforcement, making purpose and access rules part of the intervention.

Property discovery should be commissioned around safer occupied housing rather than the number of structures mapped. Automated rent scoring helps a lender assess an irregular cash flow and gives a platform influence over the landlord's pricing. A tenant registry improves legal proof and can expose undocumented migrants or politically vulnerable households. A sound program limits purpose, access and retention; requires reasons for adverse decisions; gives correction and human appeal; and keeps safety inspections and emergency relief available offline. Evaluate the system's housing outcome alongside technical accuracy: how many

additional safe, occupied rooms did the system cause, at what total cost, without higher rent burden or exclusion? The state should commission technology around that outcome. Implementation should sequence support, recognition, and enforcement around that objective.

Sequencing support and enforcement

Begin with local layouts, tenant conditions, and service capacity while retaining essential safety standards. Use targeted grants for non-recoverable safety or network costs and loans only where realistic cash flows support repayment. Municipal building control and utilities must offer a usable compliance path; tenant dispute resolution must remain accessible. Track safe occupied rooms added or retained, rent changes, closures, original tenant retention, service reliability and borrower distress. A city can move quickly if it separates evidence gathering from punitive enforcement and publishes a predictable path to compliance. The first ninety days should produce a representative room-and-tenant sample, a map of severe hazards, a count of eligible plots within existing infrastructure, and a cash-flow model for small landlords. It should also identify who profits from current informal water, power and permit markets. A sample based solely on advertised online listings will miss the poorest renters. The pilot can then test a limited set of approved designs, inspections, and recognition procedures.

Pilot expansion should follow demonstrated inspection and delivery capacity rather than a fixed calendar. Over the next 120 days, the city can designate pilot areas, approve a limited set of safe standard room and stair designs, publish utility capacity and fees, recruit qualified local inspectors, and open a voluntary tenancy-recognition channel with strong privacy safeguards. Treasury can offer temporary, conditional support for fire egress, sanitation and structural safety. Lenders can pilot staged loans with independent certification, clear repossession rules and a limit on landlord indebtedness. Tenants need funded grievance support from the first day. In the following twelve months, verify changes against comparison areas or a credible phased rollout. The indicators should include rental additions, net rent and utilities, occupied room-months, tenant turnover, structural defects, fire incidents, the share of units with safe sanitation, water uptime, permit time, lender losses and public cost per additional safe household-year. Monitoring displacement is essential: a project can improve average quality by forcing out the poorest residents. Pre-agreed stop rules are needed if rents, evictions, defects, or collection practices worsen.

A pilot should stop expansion when rents rise sharply, tenant evictions increase, safety inspections fail or collection complaints indicate unlawful or coercive practices. It should expand when additional occupied rooms appear, habitability improves, and the combined cost of rent, utilities and travel remains within the target household's means. Publish the stop and expansion criteria before the pilot begins. The assessment framework should make those conditions visible before approval.

Small-rental housing assessment

Assess the market in which poor workers actually rent, with evidence and responsibility assigned to each decision. The following fourteen questions can be applied to

a neighborhood, a serviced-site scheme, a landlord-finance program or an informal-settlement upgrade. Each gate must have a named owner, evidence and a result date. No aggregate score should override an essential safety or rights failure.

Gate	Question that must be answered	Minimum evidence
1. Demand	Who rents and why here?	Tenant enumeration, job access, rents and turnover
2. Land claims	Who holds title, structures and occupancy rights?	Distinct maps of plots, structures, owners and tenants
3. Safety	Which hazards require immediate action?	Engineer/fire inspections; risk classifications
4. Rules	Can a safe extra room be permitted predictably?	Permitted templates, fees, clocks and appeals
5. Materials	Can local builders deliver verified quality?	Input costs, tests, skilled workers and procurement
6. Capacity	Can water, sewer, power and access support density?	Network model and funded expansion schedule
7. Landlord economics	Will net rent cover costs and reserves?	Project cost, DSCR, rent sensitivity, stress test
8. Household cost	Does rent remain affordable after access costs?	Rent + utility + transport time and cash
9. Finance	Does capital reach viable micro-developers?	Terms, defaults, staged inspections, consumer rights
10. Tenancy	Can renters seek repairs and resist illegal eviction?	Lease terms, tribunal, grievance and remedy
11. Inclusion	Who faces screening or displacement?	Audits by income, tenure and social group
12. Public finance	What will the city and state pay over time?	Land value, subsidies, guarantees, trunk O&M;
13. Politics	Who earns rents from the status quo and can veto?	Named actor-rent-veto-coalition map
14. Learning	Does the reform produce net additional safe room-years?	Credible counterfactual, tracking, stop/scale rule

A small-rental programme should proceed only when its essential safety, tenure, service, and financing conditions are met. A building with no structural safety certification cannot be scaled through cheap credit. A title record that excludes tenants cannot justify eviction or rehabilitation compensation. A financing program that improves units while pushing renters out of reach has missed its target. A municipality that grants extra floors without paying for sewer capacity has exported the cost to future residents. Measure additional safe occupancy alongside household cost, tenant retention, and the public cost of services. The broader test is whether the combined reforms improve the production system already serving households.

Implications for incremental housing

Supporting existing producers requires testing the combined effect of rules, finance, services, and protection on safe affordable occupancy. Millions of households already create housing in small decisions, close to work, with risk and finance distributed across family, tenant, supplier and landlord. Governments can make those choices more productive by supplying rules, networks, credit and enforceable rights. They can also reduce supply by trying to replace every existing room with a complete formal apartment and a mortgage. Small rental housing supplies needed rooms but can expose residents to unsafe buildings, unreliable services, and insecure occupancy. Evaluate improvements against both the housing supplied and the risks residents bear. The proposed sequence is as follows. Legal small-scale density raises potential supply. Predictable permits and verified engineering lower construction risk. Staged credit reduces liquidity barriers. Funded utilities make added rooms habitable. Enforceable tenant rights can help tenants retain benefits. Protected administrative data and aggregate reporting can support oversight. Test whether the combined reforms add safe, affordable occupancy, rather than assuming each proposed link works in the city concerned. That package must change the incentives sustaining poor conditions rather than assume they will disappear after registration.

Persistent unsafe housing reflects the options available to participants, which reform must change without removing affordable access. Households accept insecure rooms to stay near work; landlords supply what those rents can finance; officials may tolerate non-compliance because enforcement would remove shelter without replacing it. Changing one rule can leave the other incentives intact. A title alone will not make a costly building standard affordable, and cheap credit will not create a legal service connection. Use a function-by-function transition plan. Preserve small payments, incremental investment, rental flexibility, and access to customers where residents value them. Remove unsafe wiring, fire traps, coercive eviction, contaminated water, and arbitrary extraction through funded, enforceable measures. For each requirement, identify the improvement cost, who pays, the likely rent response, and the option for a tenant who cannot absorb it. The South African backyard-rental and Indian incremental examples in this chapter are tests of this transition, not reasons to romanticise informality. Fieldwork should therefore measure both production and the tenancies retained after improvement.

Prepare the local decision

Evaluate small-rental support through net safe occupancy, repairs, tenant retention, full costs, and arrears. Sample occupied rooms and potential extension sites. Measure room area, rent, advance payments, water and sanitation charges, electrical and structural safety, vacancy, and landlord cash flow. Record connection fees and the time to obtain permission. In the South African backyard examples above, test whether financing, safe building space, or municipal capacity prevents an additional room. Inspect sites with small landlords and tenants separately, plus a building professional, utility staff, and a lender familiar with small loans. Specify which defects require immediate protection and which can be corrected through a staged plan. Agree how tenant notice, temporary accommodation, and rent changes will be handled during works. Use standard digital plans and photographed inspection records to reduce repeat paperwork, with physical checks for concealed hazards. Link any loan or grant disbursement to verified work rather than a modelled property value. Test the package on a defined group of buildings. Count net additional safe occupied rooms, repairs to existing rooms, tenant retention, full monthly costs, and arrears separately; larger floor area can coexist with fewer affordable tenancies. Chapter 13 develops the information and evaluation methods needed to establish those results.

Small rental improvements should proceed through safe steps that owners can finance without displacing tenants. Practical technical support, sufficient tenure security, and service capacity allow staged works to improve conditions. A model micro-unit or formalisation deadline does not establish those conditions. Unaffordable reconstruction standards, advance-rent barriers, and simultaneous rent increases can reduce access even where the buildings improve. At tenfold volume, check inspector workloads, transformer and sewer capacity, loan arrears and displacement from simultaneous upgrading.

Sources and evidence notes

The numbers and causal claims in this chapter come from official statistics, academic studies and identifiable program records. A study estimates a mechanism within its sample; an administrative figure records activity; a planned financial commitment reports an intention. The text keeps those categories apart.

[1] International Development Research Centre (Canada), “Data and activism drive dramatic improvements in struggling Nairobi neighbourhoods,” 2025. Mukuru enumeration, structure and tenant counts. [source] [2] Muungano wa Wanavijiji, “About the Mukuru SPA.” Resident organizations’ account of tenure, service brokerage and 94% tenancy. [source] [3] David Dodman, IIED, “A special approach to slum upgrading: the Special Planning Area in Mukuru, Nairobi,” 18 October 2017. Legal designation, consortiums, planning segments. [source] [4] Gauteng City-Region Observatory, “Backyard and informal dwellings (2001-2016),” 2018. GeoTerraImage structure classification; counts are buildings, not occupied households. [source] [5] City of Cape Town, “Cape Town puts affordable housing into high gear as revised Municipal Planning By-law is Gazetted,” 12 August 2025. Legal additional-use rights for designated small rental developments. [source] [6] Rand Merchant Bank, “RMB and INDLU Launch R1bn

Blended Finance Programme," 15 July 2026. Funding plan and company-reported prior delivery; independently verified tenant outcomes not supplied. [source] [7] OECD Economic Surveys: India 2019, chapter on Housing for All. Census-based 2011 vacancy and geography of empty units. [source] [8] Sahil Gandhi, Richard K. Green and Shaonlee Patranabis (2022), "Insecure property rights and the housing market: Explaining India's housing vacancy paradox," Journal of Urban Economics 131. District panel estimates and modeled shortage effect. [source] [9] Ministry of Housing and Urban Affairs, India, ARHC portal, "Details of Govt. Funded Vacant Houses." Administrative snapshot accessed 11 October 2026; occupancy not verified. [source] [10] Government of India, Cabinet approval of the Model Tenancy Act, 2021; MoHUA Affordable Rental Housing Complexes program FAQ. [source] [11] Richmond Juvenile Ehwi, Lewis Abedi Asante and Emmanuel Kofi Gavu (2024), "Understanding the norms and difficulties with advance rent payments in Ghana," International Journal of Urban Sustainable Development 16(1):163-181. Survey of 362 renters. [source] [12] Ghanaian Times, 20 March 2026, "Collect 6 Months' Rent or Face Prosecution," via AllAfrica. Reports April 2026 enforcement announcement; outcome evidence remains to be established. [source] [13] Sebastian Galiani et al. (2017), "Shelter from the storm: Upgrading housing infrastructure in Latin American slums," Journal of Urban Economics 98:187-213; randomized interventions in El Salvador, Mexico and Uruguay. [source] [14] Saugato Datta and Vikram Pathania (2016), "For whom does the phone (not) ring? Discrimination in the rental housing market in Delhi, India," UNU-WIDER Working Paper

55. Online audit experiment. [source]

Illustrative analysis: the six-room pro forma and Ghana advance-rent example use explicitly stated assumptions; they are calculations, not measured market effects. Recommendations on the compliance ladder, fiscal transfers, grievance protection and experiment design are authorial judgments.

13. Housing information and policy evaluation

Build housing information systems around decisions that can be changed and outcomes that can be verified. Begin with the operational question, establish a baseline, and specify who will act on the resulting information. Administrative records, surveys, imagery, and AI each have limits; better coverage also requires correction routes and controls on access. The first task is to distinguish the people and rights hidden behind a property record.

A housing information system can be assessed by its effect on six decisions: where growth is legal, which infrastructure is funded, which households receive aid, which buildings receive credit, which assets are maintained, and which policy is stopped after it fails. A housing information system should connect observations to a responsible decision-maker, an operational change, and a measured household outcome. Test each handoff. Missing occupants, unshared records, inappropriate targets, and unverified completion reports can each prevent useful action. Each proposed application should specify the decision it changes and the remedy when its information is wrong.

Evidence	What it establishes	What remains uncertain
In a Ghana field experiment, digital maps raised tax-bill deliveries 27% and collections 103%.	Information changed worker behaviour and fiscal yield.	The authors' 2022 account reported household-survey indications of more bribe activity. These are earlier-version survey measures, not audited bribe transactions. [GHA2022]
In Freetown, mapped properties doubled and tax collections tripled in the first year of reform.	Local data can support a more progressive tax base.	Central-local power and ownership of the system remained contested. [4]
India reported 99.8% computerised records of rights by January 2026.	Digitisation has reached substantial scale.	Surveys, corrections, pending cases, and customary rights still require institutions. [5]
Mukuru's planning documentation targeted about 100,000 households; a separate enumeration account estimated that 94% of residents were renters.	Enumeration changes who appears in the policy problem.	Formal title or building maps alone cannot determine who should gain. [1-2]

Evidence hierarchy: the Ghana estimate is experimental; the Freetown and India numbers are reported administrative measures; Mukuru figures come from enumeration and project documentation.

For each proposed use, specify the bottleneck, the measurable improvement, the beneficiary, the new holder of power, and the remedy for error. Resolve these requirements before procuring a system. A short specification should identify them, with detailed safeguards in the contract. Distinguishing occupants from owners is a first test of whether the system represents the decision correctly.

1 Collect information that changes a housing decision

Distinguishing owners from occupants

A roof map cannot establish the distribution of occupancy, ownership, and livelihood claims within a settlement. It can help identify structures for a service survey and support preliminary drainage planning, alongside terrain and engineering data. It cannot establish who occupies each room, who collects the rent, whose land claim is disputed, or how relocation would affect a resident's earnings. Those questions require household, tenure, and livelihood evidence. This example illustrates the different information needed for physical planning and resident protection. Mukuru covers roughly 689 acres and was designated a Special Planning Area in 2017. A city–community partnership brought together more than forty organizations and the residents themselves. Project documentation puts the target at roughly 100,000 households, with over 5,000 residents joining planning forums in 2018–19. In a separate account of the enumeration, residents and their partners found that around 94 percent of people were tenants and that underlying land rights sat with a much smaller set of owners. [1, 2] Planning needs these different records together before determining benefits and obligations.

Tenure and population records should be linked without treating any single source as a complete account of rights. A land-record modernization project that recognizes only titled owners can protect the claims of landlords and landowners while displacing the tenants who finance the settlement through rent. An upgrading subsidy paid to the structure owner may become a rent increase. A relocated household may receive a sound apartment and lose the clients or employers who made its old room worth living in. The unit of observation affects which households and claims appear in a housing assessment. Plot records describe land interests. Roof maps describe structures. Registered mortgage records cover borrowers who obtained formal finance. None is a household census. Surveys of households, rents, livelihoods, walking times, rights claims, and service failures can identify housing needs that those administrative sources omit. Mukuru's enumeration illustrates why these records need to be used together. Community enumeration also needs safeguards and a clear route from information to action.

Enumeration should include review and correction as well as an institutional commitment to use the findings. Local leaders may favor long-term residents over migrants, owners over renters, or men over women. A settlement register can become the basis for access to a future subsidy; people then have incentives to alter the list. Enumeration must record who supplied the information, when it was checked, which claims remain contested, and how someone can appeal. These checks are necessary when the register determines access to a public benefit or service. Information must be connected to service delivery. A survey identifies renters and structures. A rights process defines who is entitled to service or compensation. Utility

planning specifies connections. A funded works program installs them. A new survey checks rent, water cost, flood exposure, and the proportion of the original households still living there. A missing step can prevent the mapped households from receiving the intended service. The same coverage problem affects citywide housing-market data.

Gaps in housing-market information

Administrative housing records often omit informal production, failed applications, and households outside programmes. A city's administrative sources may record the government projects it has approved. It may know recorded sale values for titled plots, building plans filed at city hall, bank mortgages, and a sample of formal market advertisements. These sources may omit housing used by poor households: unregistered rooms, family subdivisions, houses built in stages, backyard rentals, informal contracts, shared water points, and households that move repeatedly in search of work. Changes in the measured population can distort reported results. A city can report that new home construction is rising while rents rise faster in the cheapest locations. It can report falling average rents because low-income renters have been pushed outside the boundary being measured. It can count more titled homes while low-income tenants lose protection. Report changes in population coverage alongside changes in the indicators. Household surveys can test both that coverage and the costs people actually bear.

Affordability measurement should combine recurring costs with realistic income and independently checked data coverage. For an owner it also includes loan installments, service charges, maintenance, property taxes, and the price of reaching work. Conventional price-to-income ratios remain useful but incomplete. A low-priced flat at the urban edge may impose a larger weekly cash burden than an expensive central room. The city therefore needs repeated observations, including households that left. A panel of sampled renters can track monthly rent, deposits, crowding, utility bills, commuting costs, tenure security, eviction and household composition. Transaction records and online listings provide much larger samples at lower marginal cost but overrepresent better-connected households and formal listings. Use independently sampled surveys to check the coverage of administrative and market data. It must also measure the set of homes the household can actually choose.

Count feasible housing options rather than theoretical capacity alone. Distinguish existing supply, including completed formal units, rooms added by small owners, and usable vacant homes, from the future pipeline of unbuilt permissions and land where services and legal rights allow construction. Counting only the first creates a policy bias toward large developers and public projects. Tracking all five can help identify where planning, services, or credit constrain additional housing. Property-location experiments show how a narrowly defined information gap can be evaluated.

2 Test what administrative and digital data can establish

Property-tax information in Ghana

A property-location tool can change revenue administration, but the evaluation must also consider residents' treatment. The Ghana study uses random assignment to estimate the effect of a property-location tool. In Ghana, four researchers worked with a municipality to test whether a digital map would help tax collectors find properties. Collectors were randomly assigned the use of tablets with geospatial records. Each was given roughly 135 property-tax bills to handle over six weeks. Treatment and comparison collectors worked in otherwise similar areas. [3] The tablets increased bill deliveries by 27% and tax collections by 103% in the experiment. Collector surveys and the authors' model support their explanation: faster navigation allowed more time for follow-up and learning which households were likely to pay. The collection effects are experimental estimates; the precise contribution of each behavioural channel depends on additional evidence and modelling assumptions. [3,115] In their 2022 account, the authors reported household-survey indications of modestly more bribe activity in treatment areas. These are survey-based indications from the earlier version, not audited bribe transactions or evidence that every collector acted improperly. Keep this evidence separate from the February 2025 paper's revised collection estimates and behavioural model. The account supports monitoring resident treatment alongside revenue; it does not establish the same outcome in a housing application. [GHA2022] The instrument's institutional use determines whether better location information produces a useful public decision.

A precise register supports several administrative tasks without eliminating discretion or abuse. The same information can also expose informal residents to enforcement or demands for unofficial payments where safeguards and accountability are weak. Outcomes depend on the rules governing how the information is used. The operational response begins with how money moves: direct digital payment to the treasury; no cash collected at the door; a published schedule of charges; a receipt linked to the legal demand; random integrity checks; a protected complaint channel; and meaningful penalties. Mapping does not remove opportunities for abuse if collectors retain broad discretion without an auditable record. The Ghana evidence should therefore be interpreted as a test of a specific tool and setting.

Tax mapping outcomes and safeguards

Tax-mapping results do not establish improved housing services without evidence on spending and resident outcomes. It increased bill delivery and collections, but did not establish subsequent housing improvements. The authors' 2022 account reported household-survey indications of more bribe activity. [GHA2022] A housing application should therefore measure resident treatment alongside administrative performance. Independent checks of receipts and payment complaints are proposed safeguards, not interventions evaluated in this experiment. [3,115] The Ghana experiment found higher measured collections. In their 2022 account, the authors also reported household-survey indications of more bribe activity; this earlier-version finding is distinct from the revised collection analysis. [GHA2022] Freetown's broader reform adds institutional considerations, including valuation, payment channels and political agreement.

Before extending a digital system, verify residents' treatment as well as revenue; the experiment did not measure whether additional receipts improved housing services. [115,4] Freetown illustrates a broader reform combining information with valuation, payments, and institutional arrangements.

Property-tax reform in Freetown

Freetown's reported results reflect a combined administrative reform rather than an isolated causal effect of mapping. By 2018 the city had not formally revalued its property-tax base for more than twenty years. The administration led by Mayor Yvonne Aki-Sawyerr introduced a reform intended to increase locally controlled revenue for municipal services. The ODI account describes fiscal constraints and uncertainty about national support. [4] The reform linked geospatial property identification, a simple points-based valuation, billing, bank payment, and a software system that recorded the process. In its first year, registered properties roughly doubled, the estimated revenue potential quintupled, and actual collections tripled. The distributional design helped win consent: the average tax liability of the least valuable 60 percent of properties stayed stable or fell, while that of the most valuable fifth increased by 242 percent. A follow-up survey found support from roughly 70 percent of property owners. These are reported reform and survey results, rather than randomized causal estimates. [4] The agreement over data, money, and authority is part of the explanation to investigate.

Information reform requires an explicit allocation of administrative control and responsibility. Mapping expanded information on taxable properties. A formula made valuation easier to explain and contest. The progressive design limited increases for lower-value properties while raising liabilities on higher-value properties. Tax payments were meant to finance the local services that make added housing density possible: drainage, roads, waste collection, public space, and routine repair. Control of the software and property data also affected authority over revenue collection and city administration. The system began under municipal leadership. In 2023–24 it spread to other cities. During 2024–25 the national government adopted an upgraded version as the basis for a national municipal property-tax system. The central state could finance maintenance and scale; it also gained access to local fiscal information. The ODI account describes unresolved questions about central and local control of the system. [4] For a housing data system, this experience suggests the need for an accountable implementing institution. It needs enough technical help to start and an operating budget after the donor leaves. It needs shared definitions so national and municipal agencies can cooperate. It must settle control over data, money and authority before expanding the system. Additional revenue then needs to be traced into the neighbourhood services it is intended to finance.

If a data initiative raises land and property tax receipts, the next test is whether that revenue supports safer and more affordable neighborhoods. A housing–fiscal pact could publish how much of incremental revenue pays for connections, street drainage, maintenance, or infrastructure in fast-growing low-income areas. Without such a pact, housing policy may improve tax collection while leaving the tenant's daily life unchanged. This final claim is a proposed reform design, not an observed result from Freetown. India's digitisation experience shows another limit: improved records may leave disputed rights unresolved.

Digital land records and disputed rights in India

National digitisation can expand access to records while leaving physical documents and adjudication necessary for disputes. In its report presented on 11 August 2026, a parliamentary standing committee stated that by January 2026 99.8 percent of records of rights had been computerized, 97.4 percent of cadastral maps digitized, 95.7 percent of sub-registrar offices computerized, and 88.6 percent of registration offices linked with revenue offices. [5] These figures measure administrative coverage. They do not establish that the recorded rights are complete or uncontested. The same committee pointed to incomplete modern record rooms, surveys, links between textual rights and spatial maps, and unresolved registration-to-mutation steps. Auto-triggered mutation, which begins an update in the revenue record when a sale or inheritance is registered, had been implemented in seventeen states. Physical records remained important for dispute resolution. [5] Transaction speed must be assessed alongside the validity and distribution of the rights being transacted.

Faster transactions cannot resolve disputed geometry, ownership, or exclusion from legally recognised claims. A parcel identifier is useful when there is a procedure for correcting the attributes linked to it. Linking agencies to an incorrect parcel record can spread the error unless corrections are also shared. Different findings on Bhoomi Karnataka's Bhoomi program has been assessed differently by researchers. A Princeton study of the 1998–2003 reform describes how roughly twenty million rural land records were digitized, village accountants lost some ability to delay copies and solicit small bribes, and people obtained records through a new kiosk network. The study also noted access barriers for farmers who had to travel to a kiosk. [6] An ethnographic study published by the Indian Institute for Human Settlements identifies distributional and access concerns. Its authors argue that digitization shifted control upward, increased the cost and delay of some land transactions, and made it easier for powerful actors to aggregate land during Bengaluru's expansion. [7] Different studies of digitisation should therefore be compared through their specific mechanisms and affected groups.

Digitisation evidence should preserve differences in setting and distinguish a recorded owner from adjudicated rights. They do not provide experimental estimates. The findings point to different outcomes: a simpler record-copying service can benefit a farmer even while a centralized correction process favors a well-connected developer. A good evaluation must separate retrieval, mutation, dispute resolution, conversion, and distribution of land gains. Government should measure each rather than treating a system's successful launch as proof that tenure security has improved. A geographic-discontinuity study of Rwanda's land-regularization pilot found improved recording of women's land rights and greater investment in soil conservation, particularly among female-headed households. The program changed legal recognition and implementation practices as well as information. It studied rural land and cannot be read as evidence that scanning urban titles will lower rents or increase mortgage lending. [12] Urban housing creates further complications. The person with recorded title, the holder of development rights, the tenant, the structure owner, and the occupant eligible for compensation may be five different parties. A data platform designed around one "owner" field conceals those relationships. A single-owner field may conceal unresolved claims and should not be treated as

proof that they have been adjudicated. Programme dashboards create a related distinction between administrative progress and household outcomes.

Measuring occupied housing and programme outcomes

Administrative totals need cohort definitions and cannot be read as vacancy, additionality, or equal-maturity completion rates. On the public portal checked in October 2026, the scheme reported 12.768 million homes sanctioned, 12.103 million grounded, and 9.907 million completed. These are important production milestones, and the portal offers more detail by state and city. [8] The reported totals do not establish an equivalent reduction in housing need. Sanctions span different years and project types. “Grounded” refers to a stage in the pipeline; “completed” refers to a physical milestone. The ratio of completed to sanctioned homes across the cumulative stock is around 77.6 percent. It is not a completion rate for projects that have had equal time to mature, a vacancy rate, or a measure of additional housing available to households who otherwise lacked a home. Follow-up records are needed to establish whether completed homes remain occupied and usable.

A learning system should connect programme milestones to subsequent household costs, occupancy, and complaints. By which household? Was this an additional unit or the rebuilding of an existing dwelling? What happened to monthly housing-plus-transport costs? Did the family leave its job, school, or support network? Was the drainage working after three monsoons? How much central subsidy and city infrastructure expense was required for each durable occupied home? An Indian investigation reported in July 2025 that, in part of Chhattisgarh, a single house photograph was uploaded under multiple beneficiary records. The report involved a specific local inquiry and cannot establish the prevalence of fraud nationwide. The reported allegation illustrates why a geotagged image needs to be checked against a unique unit record and independent evidence of occupancy. [9] The proposed response is independent verification of unit identity, construction, and occupancy. Each home needs a persistent project and physical-unit identity; each construction-stage event needs a date, source, and verifier; and occupancy needs independent sampling. Randomly selected on-site visits, repeat imagery, utility connections, tenant or owner contact, and time-series checks can identify discrepancies. A complaint should trigger review, not automatic deletion from a beneficiary list. Its reporting must answer different users’ questions without allowing one measure to define success for all of them.

The records serve different users. Treasury cares about the cost of each verified result. Lenders care whether income and collateral support repayments. Developers care about permits and sales absorption. Utilities care about connected users and maintenance. Residents care about rent, travel time, heat, flood, safety, and the risk of losing the home. The same administrative system must produce distinct answers for each user without allowing any one actor to define success for everyone. Imagery can add repeated physical observations, but its classifications require a separate accuracy test.

The capabilities and limits of satellite imagery and AI

Satellite and AI classifications need field validation, dated observations, and attention to errors on rare features. Flood exposure requires additional terrain and hazard data.

Compare the full cost of imagery, validation, and updating with the fieldwork it would replace. Research published in 2024 developed models for deprived areas in Lagos, Accra, and Nairobi using freely available Sentinel-2 imagery. In city-specific tests, some models reported F1 scores above 80 percent. The authors also found differences across settlement types and warned that transferring a model to a new city remains difficult. [10] A July 2026 International Growth Centre brief describes promising transfer tests using training data from locations in India and Burkina Faso. The researchers still stress local mapping for training and validation. [11] These findings support using satellite predictions to help prioritise field investigation. Surveyors, residents, and the responsible authorities must verify consequential claims; legal rights require the applicable adjudication process. Accuracy reports deserve close reading. A model that classifies pixels can score well while failing to identify tenants, income, building safety or tenure. An algorithm trained on tightly packed metal roofs may miss poor households in formal-looking flats. Even measured classification accuracy can mislead when the feature sought is rare. A worked example shows why a high accuracy percentage can still produce many incorrect flags.

Figure 13.1. Classification accuracy and the prevalence of deprivation



Classification performance must be judged through the consequences of false results as well as aggregate accuracy. In this hypothetical example, 10,000 of 100,000 map cells contain the feature being sought. The model identifies 9,000 correctly and misses 1,000. It incorrectly flags 4,500 of the other 90,000 cells. Thus 4,500 of 13,500 alerts are false. These are illustrative calculations, not field estimates. Training labels require scrutiny. Record who classified a neighbourhood as informal, vacant, or unsafe, using what evidence. Check whether clearance sites dominate the sample and whether roof or vegetation patterns stand in for social identity. Restrict the use of predictions to the agreed purpose and provide a route to correct consequential errors. AI can also be tested on defined administrative tasks: reconciling parcel identifiers across agencies, spotting repeated construction images, identifying long permit queues, estimating demand for utility connections, finding likely unregistered buildings for a neutral field audit, and comparing maintenance reports with actual repairs. None requires giving an opaque score legal authority over a resident. My own use of satellite information has reinforced the importance of the decision that follows the image.

What I want from better information

My experience obtaining imagery after the Manila floods showed how much the value of information depends on timely use. In September 2009, after flooding in Manila, a colleague asked me to help obtain satellite imagery. Working from Washington, DC, I contacted the United Nations Office for Outer Space Affairs, which asked for the coordinates needed. As I

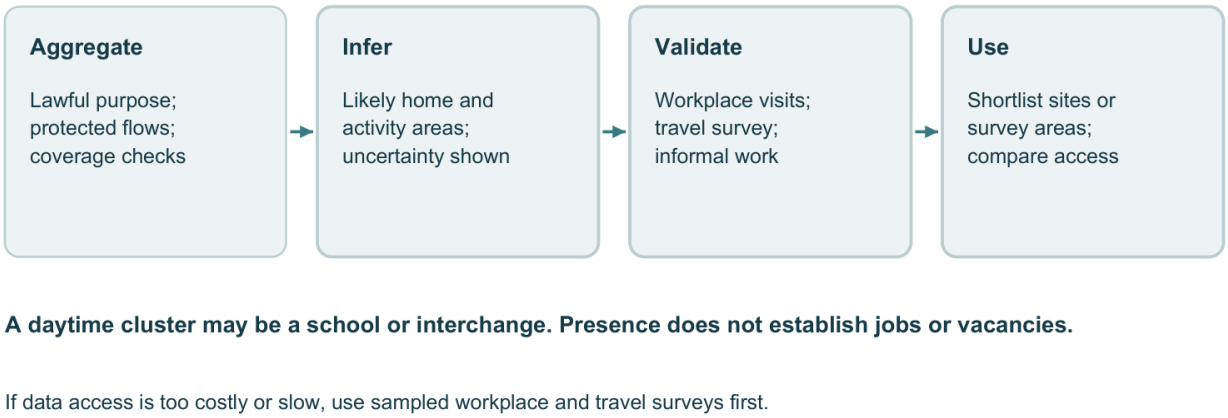
recorded at the time, the imagery available through that process would be more useful for relief and recovery than for the immediate rescue request. A subsequent call concerned imagery as the storm moved towards Vietnam. [214] I now specify the decision, location, and deadline before choosing the data. Satellite capabilities have changed since 2009; the need to match information to a usable decision remains. I am interested in digital tools when they change a decision that is currently made badly. A map of likely work locations may help compare housing sites. A linked asset register may reveal that no agency has budgeted for a pump. An application log may show that small builders face much longer waits than large ones. The value lies in the decision made with that information and the household consequence. That lesson connects directly to the distinction between recorded delivery and household benefit in IAY.

Better information should test household consequences and support accountable public judgment. Digitising an approval process can make it faster without making it capable of recognising a safe alternative design. Before automating a decision, examine the categories and evidence it accepts, who can challenge a rejection, and who has authority to correct the rule. A model can assist that work; it cannot supply the public judgment or accountability. Mobile phone data illustrates both the potential and the limits of inferring activity from a digital record.

Using mobile phone records to locate activity and commuting

CDRs can help identify activity and commuting patterns, but they are not a census of jobs or workers. Repeated patterns can help identify likely residential areas, daytime activity centres, and travel between them. Kreindler and Miyauchi use mobile phone records from Bangladesh and Sri Lanka to study commuting and urban economic activity; their estimates of workplace wages rely on a model of commuting choices. The records do not directly observe employment, vacancies, occupations, or wages. [207] Validation can change the interpretation. An Office for National Statistics research exercise compared mobile-derived commuting with census estimates around three London local authorities. Larger inter-area flows correlated strongly, but their levels were generally underestimated; within-area flows were overstated. The report identifies non-standard work schedules and students misclassified as commuters among the problems. It explicitly labels the outputs experimental and unsuitable for policy decisions. This is a useful warning against treating a strong correlation as accurate job counts. [208] A local workflow needs representative periods and independent validation before those patterns inform housing sites.

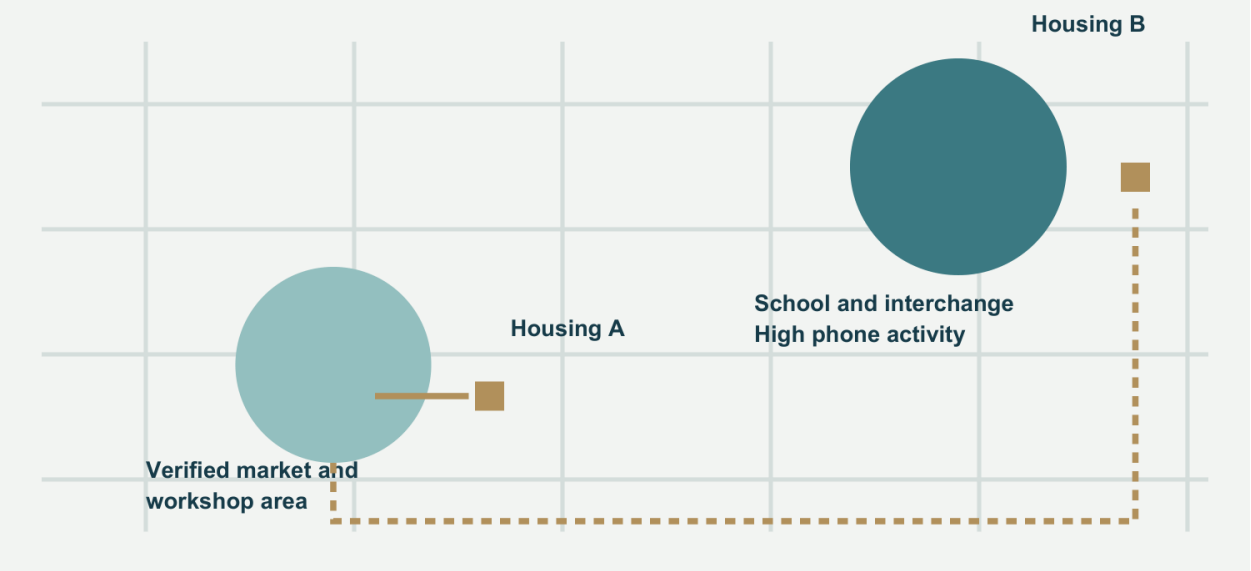
Figure 13.2. Using mobile records to guide field investigation



Proposed workflow informed by the studies in this section. Validation precedes a housing or transport decision. [207-208]

Validate aggregate mobile flows against independent employment and travel evidence before treating them as accessible work. Agree the legal purpose, permitted access, retention period, and disclosure controls with the operator and responsible statistical authority. Keep individual movement histories outside the housing team's working dataset; pseudonymous identifiers alone do not eliminate re-identification risk. Document operator coverage, active-phone selection, tower density, shared phones, and multiple subscriptions. Weighting can address measured differences, but cannot recover every unobserved group or trip. Overlay the aggregate activity signals with business registers, market and industrial-area surveys, transport routes, and actual fares. Visit a sample of high-activity and apparently low-activity areas: a hospital, school, or interchange may attract many phones without offering suitable jobs to prospective residents. Check shift workers, home-based enterprises, street traders, and people with limited phone use through independent fieldwork. Distinguish existing employment from vacancies and from jobs residents can realistically obtain. The validated result should support a defined choice, with a simpler fallback when data access is delayed.

Figure 13.3. Activity signals and verified access to work



A is close to verified work. B is close to a strong activity signal that is not itself evidence of suitable jobs.

Illustrative schematic only. Shapes, locations and routes are invented; there is no map scale.

Illustrative map, not a real-city analysis. A large daytime phone cluster can mislead site selection if its activity is not suitable employment. Chapter 19 illustrates the associated household-cost comparison.

Use the validated result to shortlist housing sites, target a business survey, or test a transport connection. Record how the ranking changes when uncertain activity estimates are reduced or excluded. Compare travel costs for the intended household groups before committing land or infrastructure. Evaluate against independently observed travel and employment patterns. CDR analysis can reduce the cost of deciding where to investigate; housing eligibility, eviction, and credit decisions should never depend on an inferred individual movement pattern. If operator access is unavailable or disproportionately costly, begin with employer and market listings, sampled workplace visits, and origin-destination travel interviews. Check branch locations rather than company headquarters, and include informal work. Do not delay a necessary field assessment while seeking a data-sharing agreement. A housing observatory should apply the same decision-led discipline to its wider data programme.

3 Build evaluation and safeguards into implementation

Designing a housing observatory

Begin by identifying decisions for which missing or unreliable information is a documented constraint. In one city that may be which wards receive sewer capacity. In another it may be the floor-space rules that stop small landlords from adding safe units. In a third it may be why subsidized homes stand empty. A useful observatory begins with those questions and works backward to the minimum data required. The design requires several linked but separate records. One describes parcels and rights, including disputed claims. One describes buildings and rooms. Another records households and tenancies, with strong privacy rules.

Others record planning rights and permit events, network capacity and failures, project and subsidy stages, market rents and sale prices, and public assets with their operating costs. Each record has its own lawful steward. Before linking records, establish a lawful purpose, necessary fields, authorised access, and correction procedures. The system need not pool every personal detail in one database. Its records need dates, definitions, uncertainty, and a clear account of who may act on them.

Record	Minimum fields needed for a decision	Main owner and check
Parcel and rights	Geometry; public/private status; rights holders; encumbrances; unresolved claims; update date.	Land agency; correction process; community challenge.
Building and units	Footprint; floors; unit type; structural-risk flag; unique physical-unit ID.	City building control; field audit of model estimates.
Household and rental	Anonymous panel ID; rent, utilities, crowding; access to work; move or eviction; tenancy type.	Statistics office or independent survey partner; privacy steward.
Planning and permits	Legal floor space; application events; refusal reasons; officer; clock; appeal.	Planning department; public audit of exceptions.
Networks and assets	Capacity, outage, flood risk, capital cost, service standard, maintenance owner.	Utilities and city finance; independent service checks.
Public projects and aid	Site; cost; subsidy; procurement; stage; verified occupancy; aftercare.	Housing agency and treasury; sample verification.
Market and credit	Repeated rents and sale prices; vacancy samples; loan terms and rejections.	Statistics unit, landlord/tenant panel, lenders; sample bias checks.

Table 1. The minimum viable information architecture. The critical fields are the ones attached to decisions and appeals. A universal platform can wait.

Useful housing records should make their coverage and uncertainty explicit and support a feasible response. It records a valid time (when a fact was true) and a recording time (when an agency learned it). It distinguishes verified facts from claims and model estimates. It preserves old versions. It shows the person or institution responsible for corrections. It allows lawful access without granting every agency the right to copy personal files. Test whether these design choices prevent incorrect decisions and make corrections effective across agencies. A city should also publish the market facts that private actors need: permitted floor space, transparent fees, service-capacity zones, average approval time, anonymized local rents, transport access, and infrastructure plans. More predictable information can reduce a developer's holding cost and help small builders seek finance. Government must then check whether developers pass savings into rents or prices. Supply constraints and market power still govern how the benefits are shared. Market incentives still determine how the benefit of better information is distributed.

Information and market incentives

Better development information can reduce uncertainty while raising land values and changing who captures public gains. The developer needs answers from five offices: whether title is secure, what can be built, what the utility can connect, how long approvals take, and what comparable homes sell or rent for. Uncertainty can increase required returns or prevent a project from securing finance. Some developers pay brokers to compress that uncertainty; others leave the market. The costs may fall more heavily on firms without established information channels or sufficient working capital. Publishing legally reliable parcel-level rights, infrastructure capacity, and permitting clocks can reduce this advantage. Developers can bid for land with a clearer sense of cost. Banks can underwrite time risk. Competition may strengthen if small and new builders can obtain the same information as incumbents. Yet the price of developable land can rise as uncertainty falls. If the landowner captures the gain, households see little benefit. The city must measure permitted and delivered affordable floor space after reform, and consider land-value capture or public-land conditions where the state creates large windfalls. The implications differ for a small landlord seeking finance or permission for a few rooms.

Information errors should be evaluated through their effects on small producers and affected households. The household needs a simple safe building standard, permission to connect toilets and electricity, a loan matched to rent, and a way to resolve tenant disputes. Data about average rents and structural risks help lenders price a loan. But a registration system that automatically triggers back taxes, demolition notices, and steep infrastructure fees may drive landlords back into invisibility. Offer a time-limited compliance route and finance safety improvements before applying punitive enforcement to noncritical violations. A citywide rental register also changes the bargaining balance with tenants. Information about comparable rents can aid dispute resolution and improve subsidy design. Named lists of tenants, irregular migrants or households in informal settlements can expose people to landlords or enforcement agents. Share aggregate prices with the market; protect identities and grievances. Assess errors by their consequences for affected households, as well as by their frequency. Investors further along the financing chain need reliable records of the asset and its operating performance.

Capital markets introduce another information chain. Pension funds or bond buyers need credible evidence of the cash flows and risks behind housing assets, including construction milestones, occupancy, rent collection, maintenance reserves, and insurance exposure. A shared record can lower due diligence costs. It cannot cure a project whose residents lack income to pay the contracted rent. Better information can deepen finance after there is a viable asset and an enforceable operating model. Evaluation is needed to establish whether those information improvements caused the intended policy result.

Testing and evaluating housing policy

Evaluation needs a credible account of what would have happened without the intervention. Resident moves, spillovers, changing land prices, and long delivery periods complicate that comparison. Choose the method and collect the baseline before expanding the programme. Start with a specific causal claim. "Faster permits will lower the price of low-income rentals" contains several links: shorter approval time → lower holding cost → more projects reaching financial close → more finished rooms in accessible areas → slower rent growth → households retaining more cash after housing and travel. Any link can fail. A permit reform can lower developer costs while site scarcity allows the landowner to raise the land price. Randomize where rules permit. A municipality might phase a permit-assistance service across equally eligible wards or assign added inspection support at random to a queue. For legal changes that must apply citywide, use matched comparison cities, staggered rollout, event-study designs, or carefully chosen boundary comparisons. Publish the pre-policy trend, all outcome definitions, and the planned analysis before results appear. Watch spillovers: cheaper rents in one ward can attract households from another; a tax enforcement intervention can alter decisions in comparison wards. Implementation records help explain the mechanism behind an estimated outcome.

Use implementation data to distinguish improved access from costs shifted to households or other locations. Track the time spent at each approving office, the number of projects that reach construction, and the median housing-plus-transport cost for eligible households. A null effect on rent can still teach: perhaps construction finance, not permits, became the next barrier. A policy that raises occupancy but increases commute time shifts costs rather than an unambiguous gain. Evaluation also needs authority and funding to change the programme when results fall short.

THE THREE NUMBERS TO REPORT TO A MINISTER

1. Household: change in monthly housing + utilities + travel cost and access to work for the target population. 2. Supply: net additional safe occupied units or rental rooms in accessible locations. 3. Public value: full public cost per additional safe occupied household-year, including future maintenance. Publish the comparison, sample size, uncertainty, and which households exited the sample.

Assign a budget and a responsible official to evaluation and follow-up. Quarterly meetings should review the latest constraint, the independent outcome estimate, and the action required. Decisions should name a rule that changes, a person who owns it, a deadline, and a

future test. Reforms that fail should be stopped or revised, with the decision published. The reporting process should specify who can change the policy and budget when results fall short. Vietnam taught me why that evaluation design must begin before the investment changes the baseline.

Vietnam: establishing the evidence before the investment

Vietnam's upgrading experience made baseline design a central part of my approach to programme preparation. Vietnam's upgrading programme taught me to take baseline design as seriously as engineering design. I remember frustration that gains in land values, household investment, and neighbourhood life were harder to demonstrate than the physical improvements. As a manager, I take from that experience an obligation to ask, before approval, what evidence we will need to judge the result. The published 2016 IEG review of the Vietnam Urban Upgrading Project (P070 197) rated its outcome Moderately Satisfactory. It noted missing baselines and targets, reported property-price increases that could not be attributed to the project alone, and discussed additional evidence supplied by the regional team. It ultimately found substantial achievement of the objectives. The review also recorded development of a National Urban Upgrading Program in 2009. Expansion and observed improvements did not establish which results were attributable to the project. [215] A price rise can reflect better services, a city-wide boom, new construction, or expectations of redevelopment. Owners may gain while tenants face higher rents. A defensible evaluation must follow people as well as properties and distinguish these mechanisms. The guide below is my proposed commissioning standard for a housing or upgrading programme; it is not a claim about procedures used in Vietnam. The commissioning guide translates that lesson into decisions the programme director and evaluation lead should settle early.

A practical baseline and follow-up guide

Agree the decisions the evaluation must support and collect the baseline before programme expectations change behaviour. Before approving the survey, the programme director should agree with the evaluation lead on three decisions the evidence must support: whether to change the design, whether to expand it, and whether the original households benefited. Write a short results chain for each main investment. For drainage, for example, test whether works reduce flood days, then whether reduced disruption changes workdays lost, school absence, or repair costs. Select a few primary outcomes and publish their definitions before results are known. Standard impact-evaluation guidance explains why the comparison strategy, sample, and data collection must be designed together. [216] Collect the baseline before works begin and, where feasible, before a credible project announcement changes land prices or investment behaviour. Record announcement, survey, eligibility, relocation, and construction dates separately. If the programme has already started, retrieve dated administrative records and earlier surveys, identify what can be reconstructed, and label recall data. Do not describe a survey taken after expectations or displacement changed as an unaffected baseline. Separate household, property, and business records are needed to track who benefits and who moves.

Keep separate linked records for households, properties, and businesses. A property may improve while its former tenant moves away. Sample renters, owners, informal occupants, women-headed households, and people facing relocation; an allocation register alone will miss some of them. Use consent-based recontact information stored separately from research data. Give people a clear assurance that survey answers will not decide eligibility or trigger enforcement, and establish rules that make that assurance credible. A comparison design must then be chosen before allocation and phasing remove feasible options.

Question and measure	Collection method	Decision supported
Who is living here, and who leaves? Tenure, household composition, rent, reasons for moving.	Household survey linked to a confidential recontact register; follow movers as well as current residents.	Detect exclusion or displacement that an improved neighbourhood average could conceal.
What does occupation cost? Rent or debt payment, utilities, transport, repair costs, and income volatility.	Short expenditure and income modules using consistent recall periods; validate a sample against bills where available.	Adjust subsidy, location, or service charges before arrears and forced moves rise.
Do services work? Flood days, water reliability, lift outages, indoor heat, travel time.	Combine resident reports with operator logs and targeted physical measurements; repeat in comparable seasons.	Locate the failing asset or service and assign a repair or operating response.
Do daily outcomes change? Workdays lost, illness expenses, school absence, satisfaction.	A small household panel with locally tested questions; consented administrative linkage where feasible.	Test the programme’s stated welfare mechanism, without assuming changes are caused by housing.
What happens to property and investment? Sale and rental values, building area, condition, private spending.	Dated transactions or independent repeat valuations with plot attributes; distinguish assessed tax values, asking prices, and completed sales.	Separate market appreciation, construction, and service gains; assess owner benefits and renter pressures.

Question and measure	Collection method	Decision supported
What would have happened otherwise? Outcomes in a defensible comparison group.	Evaluation specialist selects a feasible design before rollout; record pre-existing differences, other investments, and spillovers.	Estimate additional effects rather than crediting the programme for the wider city trend.

Suggested baseline modules. The evaluation lead should adapt the sample and frequency to the intervention, budget, and decisions; this is not a universal indicator requirement.

Commission the comparison design before allocation and phasing decisions become fixed. A phased rollout may permit comparison with eligible areas awaiting investment, but earlier sites may be systematically different. Where random allocation is appropriate and fair, agree it transparently; elsewhere, an evaluator should test the assumptions of the chosen non-random method. Never delay urgent safety works solely to preserve a comparison. If attribution cannot be established credibly, report change and contribution with that limitation. For land values, record location, plot and floor area, tenure, building condition, access, hazard exposure, transaction date, and whether a price is observed or estimated. Compare like with like and adjust monetary values consistently for inflation. Survey new private investment separately: money spent on an extension is a cost as well as evidence of confidence. Do not add a capitalised property-value gain to the full stream of service benefits it already reflects. Present effects on owners, renters, and displaced households separately. The programme must fund repeated follow-up and tracking of households who leave.

Fund follow-up before financing the works. For a multi-year programme, surveys at roughly one, three, and five years after occupation may be useful; the timing should follow expected effects and seasonality, not a fixed template. An evaluator should calculate sample requirements, including expected loss of respondents. Publish how many original households were recontacted and how those lost differ from those retained. A replacement household at the same address cannot stand in for the household that left. Use a modest core survey and targeted modules rather than collecting everything. The programme director should name a data custodian, budget fieldwork and independent spot checks, and approve a dictionary of definitions, identifiers, and version dates. Mobile forms can flag impossible entries; geospatial records can locate service gaps; administrative links can reduce repeat questioning. None substitutes for checking that the register includes the people the programme is meant to serve. Release anonymised findings with missing-data rates, methods, and limitations; restrict identifiable locations and personal records. Each review should connect the findings to an action that an identified official can take.

At each review, connect findings to an action and an owner. Rising transport costs may require a route or site-design change. Poor thermal comfort may require a construction detail to change. High turnover among original tenants may require rental protections or different eligibility rules. My test is whether the measurement can alter a decision while there is still time

to improve the programme, as well as support an honest assessment after completion. Technology procurement should be governed by the same requirement for a testable decision improvement.

Specifying and evaluating AI procurement

Use case	Binding constraint and test	Capture / power / safeguards
Satellite settlement mapping	Time and cost to identify new deprived areas; field-verified missed-settlement rate.	City saves survey effort. Residents face new visibility; map errors and purpose limits require community review.
Parcel matching and title records	Days to identify conflicts; correction time; share of records with contested claims.	Banks and landowners may gain; informal holders risk exclusion. Keep legal adjudication and appeals separate.
Digital permitting	Median and 90th-percentile approval time; additional safe units completed.	Developers save finance cost; landowner may capture gain. Open queues, exceptions and prices.
Construction verification	False payment and false rejection rates; cost of independent inspections.	Treasury gains; beneficiary may lose aid after a model error. Human review and redress.
Rental data and credit scoring	Reduction in loan-processing cost; new safe rooms financed; terms by income group.	Small landlords and lenders may gain; tenants may face rent pass-through; audit discrimination.
Tax, billing and service targeting	Net revenue, leakage, service improvements, bribe incidence by income.	City gains revenue; staff gain targeting power. Cashless payments and complaint audits.
Predictive maintenance	Emergency failures avoided; life-cycle cost per occupied unit.	Residents gain reliability; contractor incentives change; audit forecast misses and work orders.

Table 2. Technology contract. Every row is a hypothesis requiring measurement. None is an established effect size outside the cited experiments.

Specify operational outcomes, data rights, and validation requirements before procuring AI capabilities. Separate model development from independent evaluation. Require portability of core records, reproducible logs, staff training, correction routes, and a tested exit plan if the vendor withdraws or the contract ends. Restrict supplier use, onward disclosure, and model training on household or confidential client records to explicitly authorised purposes. Software requires a funded operating plan. Every system requires data stewards, a help desk, cybersecurity, periodic re-surveys, human review, and the ability to amend wrong records. Outdated maps can mislead users if the interface does not display observation dates and uncertainty. The resulting system will redistribute information and authority, which the appraisal must address explicitly.

Access to information and decision-making power

Information reform changes who can influence decisions and should be evaluated for unequal coverage and errors. Review whether publishing fees, allocation rules, processing times, and claims would reduce anyone's discretion or income. Establish this through local evidence. Tenants may reasonably resist disclosure if addresses expose them to eviction; purpose limits and protected access are part of the implementation agreement. Homeowners and neighborhood associations hold another kind of informational power. They can attend hearings, commission traffic studies, and present precise objections to rental apartments. Prospective residents may have no equivalent opportunity to participate. A citywide information system can help document that imbalance by publishing forgone homes, expected service costs, and who has been excluded. The same statistics can be used to stigmatize a poor neighborhood. Test whether registry coverage and treatment of claims differ by caste, race, or class. Formal job histories privilege salaried workers. A title-focused register can disadvantage customary and collective rights. Credit data can transmit the history of neighborhood segregation into present lending decisions. A rent database that covers formal advertisements may exclude migrant workers and poorer neighborhoods. A city should publish coverage and error rates by settlement type and, where legal and ethically sound, test differential error and outcomes across social groups. Those changes can create resistance from actors who lose control or income.

Actor	Information rent and veto	Feasible coalition or bargain
Public land agency and landowners	Control of inventory, valuation and disposal; can delay transfers or release selected parcels.	Independent public-land register; auction or allocation rules; appeals; parliamentary audit.
Planners and established homeowners	Discretion over permitted density and local objections; future residents lack a vote.	Publish infrastructure capacity and opportunity cost; share some land-value gains with local services.

Actor	Information rent and veto	Feasible coalition or bargain
Large developers and construction firms	Privileged information on approvals, infrastructure timing and public procurement.	Common APIs, published queues and bid data; time-bound approvals and competitive entry.
Small landlords and renters	Landlords control occupancy records; tenants fear exposure, rent rises and eviction.	Safe compliance path, affordable upgrade finance, anonymous rent sampling, rights enforcement.
Municipal officials and utility managers	Control records, service permissions and asset budgets; performance can be hidden.	Performance grants, public service standards, correction procedures, protected audit roles.
Central government and local government	Authority over revenue, data access, procurement, and system operation.	Shared standards with clear lawful access, fiscal returns, system operating responsibilities.
Data firms, enumerators and civic groups	Control over methods, software, classifications and who joins registers.	Portable systems, audit access, independent verification and funded community appeals.

Table 3. Institutional interests and decision powers. These are hypotheses for local investigation, not findings about named institutions.

Anticipate conflicts over data and discretion while protecting claims absent from existing registers. A new public-land register can trigger litigation by beneficiaries of opaque disposal. A property tax map can provoke wealthy owners and national–local rivalry. A rental survey can prompt tenants to refuse interviews if no protections exist. A digital permit can leave old agency vetoes untouched. Implementation planning should establish who loses discretion, what lawful compensation or alternative incentives are offered, and who has enough authority to enforce the new rule. A feasible sequence begins with lawfully disclosable public information: building permissions, public infrastructure plans, official fees, service standards, and aggregate rental indicators. Add parcel and project identifiers with correction rights. Introduce tax and subsidy uses only after the appeals and privacy systems work. Include informal tenants and community enumerators in validation teams, with clear access limits, conflict-of-interest rules, and independent review of disputed entries. Then tie fiscal transfers or performance grants to improvements verified outside the implementing agency. Measure inclusion and exclusion errors by income, tenure and locally relevant identity groups; do not treat absence from a registry as absence of a claim. Accuracy alone cannot determine whether a proposed use of the information is legitimate.

Data accuracy and resident protection

Assess the proposed use of a map as well as its accuracy. Suppose an AI model correctly identifies rooms built on a disputed public parcel. The information might support safer drainage, an inclusive redevelopment plan, fair land-value recovery, or a mass clearance. The lawful purpose, procedural rights, and allocation of gains decide whether the use is legitimate. Privacy should therefore be designed around the decision. Publish aggregate rents, neighborhood vacancy estimates and infrastructure constraints. Restrict named tenancy and household files. Separate eligibility checks from immigration enforcement and unrelated policing unless law explicitly authorizes a use with proportional safeguards. Set retention dates, log every access to sensitive records, and notify residents when an adverse decision relies on data or an algorithm. Give people a route to correct errors before an irreversible action occurs. Use "disputed," "estimated," and "verified" as distinct data states. Government should be able to deliver emergency water and sanitation while land adjudication proceeds. A household must be able to contest a boundary without losing its place on a service list. Community representation cannot substitute for individual appeals, since local committees can reproduce exclusions by gender, religion, tribe, caste or migrant status. Correction and appeal must be resourced as operating functions rather than left as unfunded promises.

Safeguards require an explicit budget wherever records inform consequential decisions. They should appear in project budgets beside aerial imagery and servers. Budget for accessible correction and appeal procedures wherever a register informs public decisions. A hypothetical rental-room programme shows how to incorporate these duties into a defined information intervention.

Identifying 20,000 safe rental rooms

A rental-room information pilot should specify the housing decision, comparison design, and safeguards before data collection. Consider a hypothetical African metropolis that wants 20,000 additional affordable rental rooms in three years. The minister proposes a large subsidized apartment program. The city should first investigate the binding constraint. Perhaps thousands of households already build or rent rooms in fast-growing districts, while permit rules prohibit additions, utilities deny connections to untitled properties, and lenders cannot verify rental cash flow. The city commissions a stratified survey of rents, structures and utilities, cross-checks roof change against high-resolution imagery, and maps legal building capacity within a commute of major job centers. It distinguishes approved rooms, rooms built without approval, rooms vacant or unsafe, and households who moved after a rent increase. It consults builders, renters, utilities and local savings groups before choosing the intervention. The city then pilots a package in matched neighborhoods: a clear small-building code; a published permit clock; safe utility connections; structural inspections; loan products based on staged works and verified rental income; and a tenant grievance process. The control neighborhoods retain existing services during the pilot and are promised later access if the intervention works. This design is illustrative; it is not a description of an existing experiment. Its quarterly results should distinguish new safe rooms, household costs, and municipal obligations.

Separate outcome accounts prevent a mapping or approval improvement from being mistaken for an overall housing gain. Its quarterly report has three independent outcomes: number of net new safe occupied rooms, change in the rent-and-transport burden of target households, and added municipal operating cost per room. It also reports unsafe units removed from service, rent increases, evictions, loan defaults, and permit and tax harassment. If landlords build more rooms while tenant burdens worsen, the reform is incomplete. If the rooms appear mainly in remote wards, the location constraint remains. If public service bills rise faster than city revenue, fiscal rules need repair. An illustrative operating budget of US$1.5 million per year for a city of 500,000 households is US$3 per household annually. It might finance a continuous rental panel, independent field checks, systems integration, appeals, and a small evaluation team. This is a design assumption, not a documented market price. The right budget depends on field costs, security, data access and the cost of correcting errors. The economic case should compare that expense with avoided wasted subsidy, faster lawful delivery, improved local revenue and safer occupied homes, without double counting those benefits. The pilot should expand only when those results justify its operating costs and risks.

A staged information pilot

Adapt and validate the information process locally, measuring correct decisions and resolved cases rather than system use. Cost data updates, field checks, licences, model monitoring and appeals after the pilot grant ends. The service owner must act on findings, the data custodian must correct records, and independent oversight must test abuse. Track correct decisions and resolved cases, cost per verified improvement, false exclusions, appeals, distribution of benefits and harms. The decision test below makes expansion conditional on useful results and acceptable errors.

Stage	Decision and evidence	Accountable lead
Prepare	Choose one decision and a bounded pilot. Confirm staff time, funding, lawful purpose, data access, privacy, correction and appeal rules before collecting or linking personal records.	Programme director, legal adviser, data custodian and finance officer.
Establish the baseline	Use existing records first. Cost missing data and an appropriate comparison design. Test linkage and coverage on a small sample; retain unmatched records for review.	Evaluation lead and field team, with service agencies and resident representatives.
Test readiness	Check security, access controls, offline participation and correction routes. Confirm delivery capacity and the response to a harmful or incorrect decision. Do not proceed merely because a date has arrived.	Data custodian and responsible service manager; independent checks proportionate to risk.

Stage	Decision and evidence	Accountable lead
Run a limited pilot	Freeze primary indicators before analysis. Record processing time, costs, take-up, errors, complaints and immediate harms. Stop or adjust if safeguards or the operating budget fail.	Service manager and evaluation lead.
Review and follow up	A six-month review may judge implementation feasibility. It usually cannot establish effects on supply, income, health or long-term maintenance. Fund follow-up at intervals suited to the expected effects before deciding wider expansion.	Programme director and budget authority, advised by the evaluator.

A public review should test operational value and safeguards without exposing personal records. Score each gate red, amber or green. A red flag at a rights or verification gate blocks automated action until corrected. A limited pilot may proceed on other amber gates only after lawful authority, funded safeguards, an accountable decision-maker, and stopping rules are established. Publish aggregate risk information without exposing personal records. Fieldwork should establish the source quality, legal authority, and control arrangements behind that judgment.

Gate	Question to ask before spending
Decision	Which government action will the new information change within six months?
Household	Which rent, housing quality, safety or access outcome will improve, and for whom?
Coverage	Which renters, migrants, informal occupants and unregistered units are missing?
Ground truth	What independent sample will test errors in maps, permits and beneficiary records?
Legal status	Are fact, claim, dispute, and model estimate kept visibly separate?
Incentives	Who gains money, discretion or bargaining power when the data become available?
Finance	Who pays to run, update, secure and audit the system after the pilot?
Interoperability	Can the records be exported with meanings, histories and open interfaces intact?

Gate	Question to ask before spending
Rights	Who can see personal records, and how can people correct or appeal them?
Competition	Can small landlords and developers access the same public rules as incumbents?
Fiscal use	How will added revenue or avoided waste support services and maintenance?
Causal learning	What credible comparison reveals whether the intervention worked?
Adaptation	Who can stop a failing policy and reallocate money to the next constraint?

Table 5. The 13-gate Housing Learning Test. Use a named owner, evidence link and deadline for every gate; avoid an average score that conceals a serious risk.

Prepare the local decision

Audit data provenance, coverage, permissions, and responsibility before linking housing records. Prepare a field dictionary covering definition, unit, source, date, lawful purpose, update frequency, and responsible custodian. Distinguish a structure detected in imagery from a dwelling, an occupied dwelling, and a resident household. Draw an independent field sample that includes places the model classifies as absent or low risk. This is necessary to measure missed households as well as incorrect flags; the Ghana case above also requires monitoring how officials use the information. Housing officials, statisticians, municipal teams, community enumerators, data-protection staff, and independent evaluators should agree which decision each field supports. Residents need a route to inspect and correct their own record. Record who can access, link, change, export, or delete identifiable data, and who investigates misuse. A small independently checked pilot can then test value and supplier dependence before expansion.

Technology and action. Compare manual review, simple rules, and any proposed AI on the same held-out cases. Report false exclusions and false inclusions by relevant population and location, with uncertainty and missingness. Test the complete workflow, including appeals and staff time; model accuracy alone omits these costs. Keep payment and eligibility decisions under accountable review. Expand only after independent checking shows a useful operational gain without unacceptable errors, and retain a tested export and exit route from the supplier. Chapter 14 examines how the resulting evidence should enter the full public cost and long-term affordability account.

Better information is valuable when it improves a decision and limits the additional discretion it creates. A specific search problem, staff able to act, linked records, and measured follow-through provide a defensible starting point. Dashboards, satellite models, and warehouses are instruments. Unlogged discretion, inaccessible corrections, or collection outside the treasury can convert improved information into additional power without a corresponding household benefit. At national scale, test model drift, vendor dependence, appeal queues, cybersecurity and the ability to inspect a sample independently.

Sources and evidence notes

Evidence and scope. The chapter distinguishes randomised findings, administrative monitoring, case studies, and illustrative policy models. Programme statistics describe programme records and should not be read as causal estimates of household welfare. Source dates and study versions are stated in the references.

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14. Public finance and long-term affordability

Approve public housing support only after accounting for land, capital, recurring services, risk, and the duration of affordability. A construction grant or discounted site captures only part of the public commitment. Consolidate the obligations across agencies, test the revenue assumptions, and disclose how benefits and costs are distributed. A public-land transaction illustrates why promised delivery and realised household benefit require separate accounts.

Cape Town's Salt River transaction illustrates why a land commitment must be distinguished from completed housing. In June 2026, Cape Town handed over the Salt River Market site, close to the city centre, for a planned 970-unit rental development. The scheme promised 300 regulated social rental homes and 670 other apartments described as affordable-market rentals. The municipality said it had discounted the land by 90 per cent, a concession it valued at roughly R95 million. The social-housing developer, Communicare, expected about R127 million in capital support from South Africa's Social Housing Regulatory Authority; it also planned to put in R40 million of its own funds. These were reported commitments and plans. They did not establish that the flats were finished or occupied. [1–3] The value of the land concession belongs in the subsidy calculation from the start.

1 Account for every public contribution and obligation

Competing claims on public land

The public-land contribution creates a subsidy whose distribution extends beyond the intended tenants. The land concession alone amounted to about R98,000 for each of the 970 planned apartments, averaged across the whole project. The separate R127 million social-housing grant amounted to about R423,000 per planned social apartment. The two calculations answer different questions: the first spreads a site-wide land concession across every apartment; the second assigns a stated grant to its 300 intended units. Neither tells us who will ultimately capture the benefit. A developer may absorb a subsidy through lower financing costs. A tenant may receive it through lower rent. Allocation rules may favour households with better information or easier access to documentation. A landowner nearby may gain when redevelopment raises local prices. Existing users and traders also have claims that the transaction must address.

A public-site appraisal should compare alternatives while accounting for the rights and livelihoods of existing users. Cape Town reported that 70 informal households had been moved to upgraded accommodation in Maitland in May 2026, following a long effort to resolve occupation of the property. The redevelopment also had to deal with long-standing market traders; the planned scheme reserved space for four produce traders. The relocation creates a question that a construction target cannot answer: what happened to the households' journeys to work, their costs, their claims, and their choice of where to live? The municipality's description of the process and independent reporting establish the relocation and dispute. They do not provide a

verified account of household welfare after the move. [1, 4] The central site offers access to transport and creates competing public choices: sale, rental income, housing, facilities, or continued use. Compare these options while accounting for existing occupants and traders. Changes in the published project terms show why the complete agreement needs continuing review.

A housing transaction should be appraised through its final obligations rather than a single initial announcement. An earlier city announcement in 2022 described a different proposed unit mix (723 affordable units and 216 social units) before the revised 970-unit, 300-social-unit proposal was announced in 2026. Development decisions changed while the land remained public. Later reporting described the project's long history and repeated announcements. Land release remains a separate milestone from completion and occupancy. Development rights, social agreements, finance, design, approvals, building, and occupancy require separate entries in the project record. [1, 5, 6] An agency's capital budget can omit costs paid elsewhere. A national subsidy may leave municipal service obligations unfunded. A public land concession may leave households facing connection fees and long journeys. Appraisal should consolidate these contributions and recurring costs before comparing alternatives. A housing programme needs a funded agreement across public agencies and private actors. It must value public assets, assign service costs, specify how publicly created gains are shared, and protect affordability after occupation. Six linked accounts can make the public, private, and household consequences visible.

Six accounts for a housing investment

A housing appraisal should cover land, construction, sales, subsidies, and recurring public costs, alongside the household's ability to pay, reach work, and remain in the home. Consider six ledgers that must be reconciled before calling a housing investment affordable. The cash account records grants, taxes, loans, debt service, rents collected by public owners, utility charges, and operating subsidies. Extend it beyond the annual budget to show later repair and service commitments. The land ledger records the market value of the parcel under lawful alternative uses, the value of a leasehold or freehold transferred, any change in its development rights, and the value retained by the public. A free transfer belongs in this ledger even when cash spending is zero. Valuation should use the rights actually available or credibly proposed under the counterfactual, with sensitivity tests for zoning; assigning the highest imaginable speculative price creates a different distortion. The infrastructure ledger covers access roads, bulk water, sewers, electricity capacity, drainage, waste collection, schools, clinics, and transport. Some expenditures are upfront; others recur each year. The wrong comparison places only the project's internal pipes on the developer's side and leaves the trunk network outside the calculation. The consolidated account must also capture contingent risks and transfers outside the project budget.

Risks, tax concessions, public assets, and future receipts belong in the full public-cost calculation. A contingent liability can be small in current cash terms and costly in a downturn. The household ledger follows rents or mortgage payments, deposits, energy bills, water tariffs, repairs, transport fares, travel time, lost work, school access, and eviction risk. A flat with a low

price and a two-hour commute may be unaffordable once the household's travel costs are included. The distribution ledger asks whose welfare changes. Owners of appreciated nearby land, tenants, informal occupants, registered bidders, women living alone, people without formal identification, contractors, local politicians, and future taxpayers belong on separate lines. A land concession can be a good investment in inclusion while still producing large windfalls for actors the program never meant to subsidize. The fiscal calculation should therefore begin with a consolidated view, alongside agency-specific cash flows: Net present public cost = cash subsidies + value of public land committed + public infrastructure and operating costs + expected guarantee losses + tax concessions – public receipts – retained land or housing value – credible incremental public revenues. Each entry needs assumptions and responsibility that auditors and residents can inspect.

Every fiscal entry should disclose timing, valuation, uncertainty, and the institution responsible for the obligation. Do not count the land concession twice: if an appraisal already includes a below-market land price as a subsidy, its market value must be adjusted before adding a separate opportunity-cost line. Do not treat an anticipated rise in property-tax collections as a certainty; specify how much is truly incremental and which government is legally entitled to collect it. Nor should uncertain future house-price appreciation substitute for cash needed to maintain drains next winter. A full economic appraisal then adds welfare effects that fall outside the fiscal account: safer buildings, fewer flood losses, time saved reaching work, cleaner air, and the costs of displacement. Fiscal affordability and social value answer separate questions. Appraisal should establish both fiscal feasibility and expected net social benefit. Evidence and judgment. The six-ledger framework is an analytical proposal. The cited cases establish that capital grants, public-land concessions, municipal revenues, and long-lived affordability obligations are material and observable; they do not prove that this particular accounting template has been evaluated in a randomized trial. The template makes assumptions explicit for review by auditors and residents. A public-land example shows how the choice of contract changes what the contribution purchases.

Public land and developer bids

A land concession should be assessed against the additional housing benefit it secures and the alternatives forgone. Suppose a city owns a site worth 100 million in local currency if sold under its existing legal rights. A developer offers to build 500 apartments if the site is transferred for 10 million. The implied land concession is 90 million. That figure may be justified. It becomes a subsidy to low-income households only if contracts and market conditions cause households to receive the gain through better locations, lower housing costs, or rights that endure. The same logic explains why cash subsidies sometimes raise land prices. When scarce, legally buildable sites limit supply, a subsidy that increases what buyers can pay can also increase what developers bid for those sites. Part of the subsidy then moves backward through the chain, from buyer to developer to landowner. Where serviced sites and legal density can expand quickly, the market may instead produce additional homes. The size of each transfer is an empirical question. A city should compare land transactions and buyer prices before and after a subsidy, test supply responses, and distinguish the value of extra homes from increases in the price of

existing sites. That benefit must be stated as an enforceable outcome before a legal form is chosen.

Public-land support should purchase specified affordability, occupancy, service, and protection obligations. A vague promise of "affordable housing" lets the recipient set rents near the market price, choose relatively wealthy tenants, or sell units after a short control period. The public then loses the asset and has few tools to keep the agreement. A clear contract instead states the number of homes reserved by income group, the rent formula, the period of restriction, the selection rules, the tenant's protections, service standards, the sanctions for breach, and the treatment of the asset at expiry. Different contract forms allocate the authority to enforce those obligations differently.

Choose the contract form after defining the outcome and the risks each party can control. A freehold sale with covenants brings cash immediately and leaves the city with limited future control unless covenants survive each resale and can be enforced against successors. A long ground lease keeps title public and can provide periodic income and reversion rights, but investors must believe that renewal and compensation rules will survive political changes. A development agreement with cross-subsidy uses profitable market units to help fund restricted homes; the private partner will negotiate the market-rate share and may seek higher density later. A public or limited-profit housing owner can retain affordability for decades if it has patient capital, sound operating revenue, and independent oversight. [7–9] Compare the enforceable rights under each instrument. A ninety-nine-year lease with unrestricted subletting may yield weaker public control than a properly recorded sale agreement. Public ownership with no maintenance budget may preserve title while allowing the asset to deteriorate. The state needs to specify the enforceable housing outcome before choosing the legal form. Procurement can then make bidders compete on a consistent public-benefit specification.

A fair procurement asks qualified teams to compete over measurable public outcomes at a disclosed subsidy level: number of deeply affordable homes, rent ceilings, access for excluded groups, quality, carbon and flood performance, operating reserves, and date of occupancy. A bid that offers the highest land premium may win a public auction while delivering the fewest affordable homes. A bid that promises the most units may be selected without adequate scrutiny of its operating costs. The residual-land-value method is useful here: estimate revenue from allowable rents and sales, subtract verified building, finance, operating, and risk costs plus a reasonable developer return, and identify the land value that remains. Publish the assumptions and challenge them with independent comparables. Changes after award need the same scrutiny because they can alter the value and risk of the original agreement.

Public-land amendments should reconcile institutional responsibilities and update the financial model before approval. Ministries can hold title while cities must service the site. Officials may prefer a one-off sale because receipts help the current budget. Developers and intermediaries may seek a private allocation before new planning rights are granted. Nearby homeowners may support the public housing label and then oppose the density required to make it feasible. A transparent auction alone cannot fix these problems: it must be preceded by disclosure of rights, claims, valuations, and the public obligations attached to each bid. Changes

after a land transfer can give a developer benefits beyond those priced into the original agreement. A city might discount land in exchange for low rents, then grant additional floor space while allowing those rents to rise. Contracts should specify how gains from rezoning, infrastructure, or changes in the affordable-housing mix will be shared. An independent valuer should update the financial model before amendments are approved. India's housing commitments illustrate why that reconciliation must extend across budgets and levels of government.

2 Test revenue instruments and continuing affordability

Housing commitments and fiscal capacity in India

National targets and grants should be reconciled with land, infrastructure, and recurring obligations before estimating subsidy per home. India's PMAY-Urban 2.0 sets national targets and financing commitments. Approved in August 2024, it envisaged assistance to ten million urban families over five years, total investment of about INR 10 trillion, and public assistance of about INR 2.30 trillion. Its routes include beneficiary-led construction, housing in partnership, rental housing, and interest subsidies. The official framework offers up to INR 250,000 in assistance for an eligible household under specified ownership and construction routes. These figures describe design and authorization; they are no measure of homes occupied or of poor households' welfare. [10, 11] The next question is how those commitments are delivered locally.

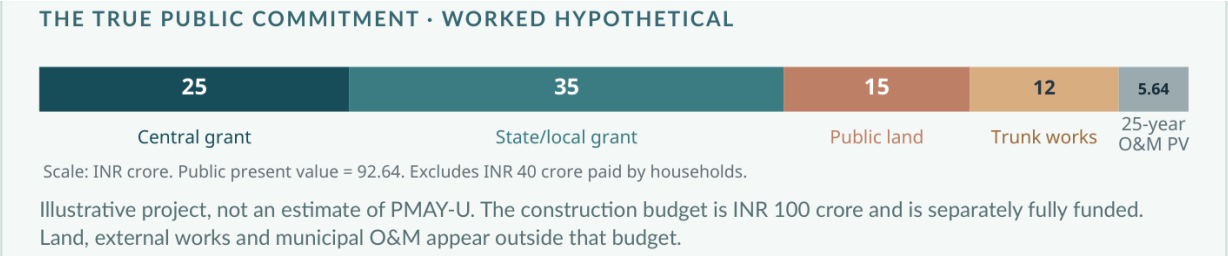
National housing grants depend on state and municipal services to produce usable homes. The scheme assigns different fiscal responsibilities to national and subnational governments. The national government can transfer funds for a dwelling. States and urban local bodies must enable trunk infrastructure and social facilities. The programme explicitly requires state or local resources to bring services to project sites. A flat can satisfy a central construction milestone before a municipality has the means to deliver reliable water, sewage, drainage, a safe road, or bus access. [10] The ability to fund these services depends on municipal revenue.

Municipal revenue limits the services that can accompany new housing. Municipal finances constrain the ability to meet these obligations. In its November 2024 Report on Municipal Finances, the Reserve Bank of India reported municipal-corporation revenue receipts of only 0.6 per cent of GDP in 2023–24; own taxes were around 30 per cent of total revenue. Municipal bonds outstanding in March 2024 were about INR 42.04 billion, only 0.09 per cent of outstanding corporate bonds. These totals cover the reporting municipal corporations, rather than every form of urban local government, and the receipts figure refers to budget estimates. These figures indicate the limited aggregate revenue available to the reporting corporations for services, including those required by new housing. [12] A consolidated project account makes the resulting obligations visible.

A consolidated appraisal includes land, infrastructure, and recurring municipal costs alongside the construction grant. Follow a hypothetical project of 1,000 homes. Suppose each home requires INR 1 million of construction expenditure. The INR 1 billion construction bill is

paid through INR 250 million of central assistance, INR 350 million of state or local support, and INR 400 million from households. Suppose public land worth INR 150 million is contributed outside the development budget. Suppose bringing mains, roads, and drainage to the site costs the city another INR 120 million. Recurring municipal service and asset-maintenance costs amount to INR 4,000 per unit each year over twenty-five years. Discounted at 5 per cent, the recurring bill has a present value of approximately INR 56.4 million. The consolidated public commitment is therefore about INR 926.4 million, or INR 926,000 per planned home, before any offsetting tax receipts, charges, residual public asset values, or wider welfare gains. The worked figures illustrate budget boundaries rather than a forecast for an actual project.

A hypothetical consolidated account exposes obligations that can be hidden when agencies report separately. They do not describe PMAY's actual average cost. The construction bill is fully funded by the INR 250 million + INR 350 million + INR 400 million sources; land, external infrastructure, and maintenance sit outside it. The public total adds INR 250 million + INR 350 million + INR 150 million + INR 120 million + INR 56.4 million once each. At an assumed co-payment of INR 400,000 per home, a household with annual income below INR 300,000 still needs savings or credit that may be difficult to obtain. The national grant can therefore coexist with exclusion at the point of purchase or construction completion. In this hypothetical, payments and responsibilities are poorly sequenced. The ministry pays its share, the state contribution may be delayed, and municipal infrastructure depends on uncertain revenues. Financing costs rise while households wait for services. No budget holder has reserved money for repairs. The remedy is to reconcile the funding schedule and assign the remaining obligations before construction. The remedy is a synchronized funding and delivery schedule.



Synchronise housing and service budgets while allocating support across tenure forms according to need. National capital commitments should be paired with signed state and municipal service plans, land valuation, named financing for last-mile connections, and a twenty-year maintenance agreement. Funding tranches should depend on independently verified readiness and occupancy. Equalisation grants should reflect fiscal need: a poorer city with the deepest need and weakest tax base should receive a larger predictable infrastructure grant, while every city should face incentives to disclose land, approve suitable density, and maintain assets. PMAY's beneficiary-led route also makes an important distributive choice. Assistance to build on already available land is useful to households that possess a buildable claim. Renters, recent migrants, informal occupants, women excluded from family title, and households whose land cannot pass the permit test need a different instrument. Housing rental and upgrading programs should receive attention in proportion to need, rather than appear as residual categories after ownership targets are set. [10, 11] A dedicated housing levy creates a different financing route but still requires the same distributional account.

Housing levy finance in Kenya

Kenya's levy finances housing through a tax whose contributors may differ from its beneficiaries. Under the Affordable Housing Act, 2024, the Kenya Revenue Authority collects a levy of 1.5 per cent of an employee's gross salary from the employee and another 1.5 per cent from the employer. The statutory design also reaches specified nonsalaried income. On a gross monthly salary of KES 50,000, the worker's deduction is KES 750 and the employer's assessed contribution another KES 750. The levy creates a recurring revenue source, with receipts dependent on earnings and compliance. It is also a tax claim on earnings, whose beneficiaries may be different from the people financing it. [13] The economic burden also needs to be distinguished from the party legally required to remit it.

The burden borne in the economy need not match the party named in the law. Some employers may absorb the employer contribution; others may eventually reduce wage growth, hiring, or other compensation. The extent depends on labor-market conditions and cannot be derived from the statutory split. A low-paid salaried worker may pay the levy for years while failing to qualify for a mortgage, raise a required deposit, or secure a unit near work. A richer buyer may be better placed to make the initial payment for a nominally affordable home. A public program that relies on formal payroll withholding also raises practical questions about reaching self-employed households and protecting precarious workers from arbitrary enforcement. Legal validity and distributional value are separate questions requiring their own evidence.

Litigation and policy claims should be reported accurately without treating a dedicated fund as inherently progressive. According to the cited reporting, on 25 September 2026 Kenya's Court of Appeal upheld the Affordable Housing Act and the levy, rejecting challenges to its legal basis. Reporting on the judgment also recorded the court's view that the Commission on Revenue Allocation should have been consulted before parliamentary passage and that public-land allocations remain subject to the National Land Commission's lawful role. The reported issues concern authority to collect revenue, allocate public land, and consult relevant institutions across national and county responsibilities. [14, 15] A potential benefit of a dedicated fund is that governments can commit revenue to a long-lived task. Earmarking does not resolve the choice between construction and other housing-related costs. A levy can finance construction while leaving schools, transport, water, and maintenance to county budgets. It can reassure investors that cash will arrive, yet weaken the political pressure to demonstrate who ultimately lives in the homes. Earmarking also carries an opportunity cost: revenue reserved for one goal cannot fund another urgent service. Those choices require public scrutiny rather than the claim that a housing fund is self-evidently progressive. Annual accounts should connect taxpayers, project spending, and sustained household benefits.

A credible levy-backed housing agreement would publish annual accounts that connect each class of taxpayer to each class of benefit. It would show contributions by income and employment status; the number of homes for renters versus owners; the income distribution of applicants, successful tenants or buyers, and rejected applicants; the full public subsidy per occupied home; loan arrears; resale restrictions; and where the units sit relative to jobs and existing infrastructure. Administrators should not expose named household income

records to the public. Independent auditors can publish distributions and conduct confidential checks while retaining rights of appeal for individual cases. The distributional case requires evidence: a national tax can help people who never obtain a unit themselves if it reduces the costs of urban poverty, improves labor mobility, and expands access to safe, accessible homes. That claim needs measurement. The governing test is whether the levy buys a housing service that low-income families could not otherwise secure, at a lower fiscal and social cost than realistic alternatives. Land-value capture provides another financing source whose distribution depends on the transaction design.

Land-value capture from infrastructure

An illustrative readjustment account shows how owners and the city can share value created by servicing. A plan changes its boundaries and permissions and connects it to roads and drainage. Appraisal should estimate the value created by those actions and specify how the owner's return and public infrastructure costs are shared. Gujarat's town-planning schemes offer a practical answer. In Ahmedabad's land-readjustment model, owners may surrender up to 40 per cent of their original land for roads, public facilities, and land that can be sold to help finance infrastructure; they retain reconstituted serviced plots, often around 60 per cent of the original area. Owners accept a smaller area in exchange for more valuable, buildable land. The city acquires corridors and sites that would be difficult to assemble through plot-by-plot purchase. [16] The same exchange can exclude people whose claims are not represented in the ownership bargain.

Readjustment needs separate treatment of tenants, workers, and other affected users to avoid concentrating benefits among owners. Researchers Bishwapriya Sanyal and Chandan Deuskar found that registered landowners were the participants recognized by the process. Renters, landless households, and informal settlers had weaker channels of voice and claims. State procedures could delay schemes, while local authorities lacked control over key approvals and land auctions. Land readjustment worked as a mechanism for coordination among legally recognized owners; it could also harden the exclusion of people whose lives and livelihoods occupied the land without recognized title. [16] These risks call for changes to the agreement and a comparison with feasible alternatives. Before drawing the final plots, authorities should enumerate occupants as well as owners; map informal livelihoods and tenancies; assess lost income and access; establish an independently funded grievance process; and reserve well-located plots or homes for households without title. The reconstituted plot register should be matched with a social-claims register. Compensation can combine cash, serviced land, rental assistance, replacement market space, and legally protected return rights according to the harm suffered. No one method fits every household. Infrastructure charges also need to align the agency collecting value with the one paying for services.

Land-value gains do not automatically reach the municipality that bears the infrastructure cost. Bogotá has long used betterment contributions to help finance infrastructure, and the OECD reports that improved public transport can be capitalized into nearby land values; estimates cited in its 2025 assessment suggest uplifts of about 15–20 per cent for transit access in Bogotá. These are estimates in a specific urban setting, not a reliable

percentage a city can levy on every parcel. The same assessment notes disputes over assessed gains and the difficulty of implementing charges perceived as unfair. [17] Capture instruments face three practical tests. First, additionality: did a public decision create or help create the gain, and can it be separated from background market appreciation? Second, liquidity: can the owner pay the levy without selling a modest home or driving out a tenant? Third, assignment: does the revenue reach the public agency financing the improvement? A city that creates value through a new road but cannot retain the proceeds may have too little revenue or incentive to finance such roads. A national agency that collects development charges while a municipality pays for drainage creates the same mismatch. A published sharing formula and payment schedule can address part of that mismatch.

Set feasible value-sharing rules before investment changes prices, while accommodating cash-poor owners and recurring service costs. A one-off charge is poorly suited to recurring operations. Recurrent land or property taxes, with protections for vulnerable owners, can support services over time. Betterment levies and plot auctions are better suited to identifiable capital works. Housing subsidy, infrastructure finance, and property taxation should complement one another, each matched to a different fiscal burden. Property taxation addresses a different part of that recurring financing requirement.

Property taxation and neighbourhood finance

Municipal revenue reforms need to be assessed through functioning services as well as reported collections. The question of municipal revenue therefore belongs inside the housing strategy. Freetown's property-tax reform is instructive. The city combined property mapping, computer-assisted valuation, better billing, and new payment methods. According to a May 2026 ODI review, registered properties doubled, potential revenue increased fivefold, and actual collections tripled within a year of reform. The least valuable 60 per cent of properties saw average liabilities remain stable or fall; average liabilities for the most valuable 20 per cent rose by 242 per cent. A follow-up survey found about 70 per cent support among property owners. These are reported reform and survey outcomes, rather than a clean causal estimate of long-run housing improvements. [18] Tax design should distinguish property value from household cash income.

Property-based revenue can be designed with progressive protections and measured service outcomes. High-value commercial and residential property can shoulder larger bills while low-value housing receives thresholds, circuit breakers tied to income, or deferrals for asset-rich but cash-poor owners. The aim is to fund drains, roads, waste services, and maintenance that make denser neighborhoods viable. Yet enforcement can be abused. Chapter 13 describes Ghana’s experimental increase in collections. The authors’ 2022 account also reported household-survey indications of more bribe activity, which should be distinguished from verified payment records. [GHA2022] A more complete tax roll needs independent appeals, clear receipts, predictable valuation cycles, and public oversight. Freetown’s experience also raises a fiscal-assignment question. A larger assessed tax base produces usable municipal resources only if the city can collect and retain the relevant revenue. Test the effect of exemptions, appeals, collection arrangements, and transfers between levels of government. The viable coalition depends on a

credible account of who pays and the services the money buys. Report service improvements, flood losses, and appeal outcomes alongside additional revenue and system use. Intergovernmental grants can reinforce this effort without imposing equal matching requirements on unequal municipalities.

A housing-oriented fiscal agreement could link national infrastructure grants to transparent local tax effort, while avoiding a crude requirement that poor municipalities match grants at the same rate as rich ones. A practical grant formula would include expected serviced households, poverty, exposure to climate hazards, local fiscal capacity, and verified asset-maintenance plans. Performance payments should reward functioning services and safe occupancy after two and five years. Rewards based purely on the number of construction approvals can induce local governments to approve unserviceable sites. The housing development itself also needs financing that preserves affordability after the capital grant is spent.

Maintaining affordability over time

Initial affordability should be converted into obligations that can be checked through rents, turnover, arrears, and asset condition. What happens fifteen or thirty years later depends on ownership rules, rent regulation, debt, repairs, and land values. South Africa's Social Housing Regulatory Authority offers an instructive blend of subsidy and operating conditions. Its published standard consolidated capital grant was R426,308 per unit, with the amount adjusted according to the share reserved for lower-income households. Its stated primary beneficiary income band is R1,850–R6,700 per month, and its subsidy design requires a minimum primary-market share. This encourages mixed-income schemes to support deep affordability while preserving revenue from somewhat better-off tenants. It also creates a risk: if financing costs rise, providers may press for a tenant mix that shifts away from the poorest eligible groups. The grant is a contractual starting point; actual outcomes require rent rolls, turnover records, arrears, and asset-condition data. [8] Those obligations require several continuing revenue and reserve streams.

Post-completion finance must cover household payments, replacement reserves, municipal services, and operating shocks. A grant that closes the development financing gap cannot automatically close an operating deficit forever. Reserves should be sized against component lifetimes, including roofs, lifts, pumps, fire systems, and water networks, then held in ring-fenced accounts with independently reviewed withdrawals. Tenants need accessible information on charges and a working dispute mechanism. Vienna and Austria's limited-profit housing sector demonstrate how institutions can stretch public support through time. Vienna's publicly posted rental-housing financing terms provide provincial loans of approximately €910–€1,250 per square metre, generally at 1 per cent interest, with repayment structured to finish within 40 years. Resident contributions and other private finance supplement those loans. The OECD's 2026 Austria review finds that for Vienna homes newly built since 2011, private for-profit rents were 91 per cent higher than rents in the limited-profit sector. This comparison is descriptive and reflects differences in selection, regulation, and finance; it is no randomized estimate of the loan program's effect. [9, 19] The durability of restrictions and reinvestment rules determines whether that arrangement lasts.

Affordability depends on continuing financing rules and must be reassessed before restrictions expire. Limited-profit providers face controls on profit distribution and use surpluses to maintain and add homes. The model requires enforceable rules, competent management, reliable rent collection, cheap long-term funding, and a housing market large enough to maintain specialized providers. A city with weak land records, unstable tariffs, and no credible rental regulator would need to establish these functions before adopting the model. The United States illustrates the risk of time-limited affordability obligations. Low-Income Housing Tax Credit developments usually carry affordability requirements for at least thirty years, subject in some cases to routes for earlier exit. Using a database of active units, the Federal Reserve Bank of Chicago estimated in 2025 that roughly one-third of then-active tax-credit units were scheduled to reach the end of their restrictions by 2035; allowing for certain early exits raised the share at risk to about 40 per cent. These are projections of restrictions ending, not counts of tenants actually displaced or units already converted to market rent. [20] New contracts can specify what happens at expiry rather than defer that decision.

A city preparing a new subsidized project can reduce the risk of losing affordability when restrictions expire. It can keep land in a public or mission-locked trust, require recorded affordability covenants that bind successor owners, provide first-refusal or buyback rights, restrict conversion gains, create refinancing plans before covenants expire, and report household outcomes over decades. Each safeguard has a cost and may change bids. The appraisal must show the price of preserving affordability, as well as the price of losing it. The contrast concerns the duration and enforcement of obligations. Vienna's observed rent differences do not isolate the effect of one rule; the US estimates identify future expiry risks rather than observed displacement. Test whether affordability, financing and renewal duties survive the original subsidy period. [118,119] Private investment then needs a credible revenue source consistent with those continuing obligations.

Revenue sources for private investment

Each participant needs finance suited to its expenditure timing and a credible source of repayment. These are separate markets with separate credit risks. Calling all of them "housing finance" hides who owes whom. A housing bond is a claim on future cash flows. If a municipality issues a bond to fund trunk infrastructure, bondholders need credible tax, tariff, lease, or transfer revenue for repayment. Selling municipal bonds against projected housing demand does nothing to create that revenue. India illustrates the scale of the constraint: the RBI's INR 42.04 billion municipal-bond figure was tiny relative to the corporate bond market in March 2024. Sound municipal finance requires audited accounts, predictable transfers, service costs, realistic revenue forecasts, and a way to handle default long before a bond prospectus is written. [12] Delays and guarantees change that financing cost and determine who ultimately bears it.

Public risk support should address risks government can manage without shielding private parties from their contractual responsibilities. A social landlord can pledge rental income, but a lender may prefer tenants with secure wages and high rents, undercutting the target group. A housing fund can offer guarantees, yet a guarantee transfers risk to taxpayers if the underlying loan fails. A public-private partnership can shift construction risk toward the

contractor when the contract contains an enforceable output specification; governments still retain land, regulatory, political, and much demand risk. International finance adds a further hazard. Borrowing in dollars or euros for projects whose rents arrive in shillings, rand, or rupees creates a currency mismatch. A devaluation can exceed the apparent saving from cheaper foreign loans. Guarantees and blended finance are useful when they address an identified risk that private lenders cannot efficiently bear, with the subsidy priced and disclosed. They become costly when they shield developers from risks they are paid to manage. A funding waterfall makes the order of payments and losses explicit.

The finance ministry should ask for a funding waterfall for every proposal: what money reaches the project first, which losses absorb private equity, when grants are released, whether government guarantees are called before or after lenders take losses, and where cash goes after sale or refinancing. The answer shows who holds bargaining power when costs rise. A developer that can renegotiate after the state has cleared land and relocated households may have greater bargaining power than the government that nominally owns the parcel. The allocation of those losses is part of the political agreement behind the project.

3 Agree the budget, distribution, and delivery commitments

Interests and constraints in fiscal reform

Housing reform distributes costs and gains across different groups and periods, creating interests that should be documented. Existing owners and developers can participate early in planning decisions. Future tenants and occupants with undocumented claims often have less influence. Municipalities and utilities may inherit long-term costs after the original project budget closes. The actors below illustrate the rents, veto points, and bargains a project team should identify. “Rent” here means a gain created by scarcity or control over a right, including legal gains; it does not imply criminal conduct. Public-land objectives need to be ranked explicitly when they compete.

Actor and source of power	What the actor may gain	Likely veto or distortion	Proposed agreement or control
Public land custodian; title and disposal authority	Sale receipts, allocation discretion, control of future use	Withholds land or transfers it without disclosed rights	Independent appraisal, published alternatives, lawful approval, retained public interest

Actor and source of power	What the actor may gain	Likely veto or distortion	Proposed agreement or control
City, county, and utility; permits and services	Tax base, density, service charges, delivery obligations	Delays connections or shifts costs to another treasury	Funded service covenant, shared revenues, measur- able occupancy payments
Developer and lender; capit- al and delivery expertise	Land discount, density uplift, fees, stable returns	Underbids then renegoti- ates; targets solvent house- holds	Competitive residual bid, open- book change rules, equity at risk, clawback
Landlords, nearby owners, and incumbent voters	Scarcity rents and land ap- preciation; protected amen- ities	Oppose density, affordable tenancies, or local taxation	Infrastructure benefits, law- ful consultation, citywide zoning, progressive taxes
Renters, informal occupants, women, migrants, and ex- cluded groups	Secure location, lower total housing cost, safer services	Often lack a formal veto; face removal or screening	Enumeration, return rights, rental supply, legal assist- ance, monitored allocation
Politicians, brokers, and in- formal power holders	Credit, contracts, alloca- tions, influence over land claims	Discretion over waiting lists or concessions; ability to delay procedural changes	Public lotteries with audits, conflict disclosures, appeals, local watchdogs

Public-land choices require an explicit comparison of fiscal receipts, housing access, and distributional effects. One objective is to maximise sale revenue for immediate budget needs. A second is to provide an ownership asset, potentially leaving less support for renters and migrants. A third is to secure long-term affordable access to neighbourhoods with jobs and services. Each objective involves distributional choices. An auction may exclude households unable to bid. Ownership can transfer public support into private resale gains. Permanent restrictions may protect affordability while limiting a family's ability to build wealth. Choices need explicit distributional accounting. The cases show different agreements for reconciling those objectives before construction.

Project agreements should resolve the specific coalitions and claims in each setting before irreversible commitments. Cape Town's negotiated land concession brought a

nonprofit provider, a municipal landowner, and social-housing finance together; the prolonged site history and relocation show the difficulty of final implementation. Ahmedabad enabled recognized owners to gain from serviced plots and the municipality to assemble land, while landless people remained outside the core agreement. Kenya enacted a dedicated revenue law, while legal challenges concerned the contested powers of taxpayers, Parliament, counties, and land institutions. Vienna built long-lived providers and financing rules that recycle public value; its approach depends on institutions and politics developed over many decades. None of the four cases is a controlled comparison. They identify different responsibilities and distributional questions for local appraisal. [1, 14, 16, 19] A reform can raise total welfare while imposing a concentrated loss on an organised group. Landowners may lose a speculative entitlement. Developers may lose the option to renegotiate an ambiguous contract. A ministry may lose discretion over allocations. A city may inherit operating costs without new revenue. Excluded renters may gain later, but they have little political standing today. Implementation should therefore follow a specified sequence: establish who has rights, compensate verified losses, share gains visibly, assign service costs, then alter the planning and fiscal rules. Trying to settle all disputes after construction begins invites litigation, delay, and repeated renegotiation. The full cost of inclusion extends to location, service charges, and the duration of affordability.

Include the cost of securing access to valuable locations and preventing exclusion through service charges or expiring restrictions. Digital tools can improve that account only when their own costs and decision effects are included.

Digital information and fiscal accountability

Decision	Constraint technology could change	What to measure, and who captures the gain	New power and safeguard
Assemble public land for housing	Fragmented inventories and disputed boundaries	Share of parcels with audited title, claims, valu- ation, and lawful disposal decision; gains can accrue to the landowner or bidder	Data custodian can erase occupants; pair geospatial maps with community claims and public appeal
Value land or property tax fairly	Missing buildings, stale val- ues, costly field inspection	Verified additions to taxable rolls, appeal outcomes, col- lection net of administra- tion; city gains revenue	Assessor gains enforcement reach; protect low-income owners and prevent abusive collection

Decision	Constraint technology could change	What to measure, and who captures the gain	New power and safeguard
Set affordable rents and subsidies	No reliable rents by location, unit quality, or occupancy	Audit a sample of actual leases and total household costs; tenants gain only if rents fall or quality rises	Platforms can discriminate or expose incomes; privacy and independent tenant rights
Procure and manage de- velopers	Cost estimates opaque; scope changes untraceable	Change-order rate, delivery delay, total subsidy per safe occupied unit	Large contractors gain in- formation advantage; pub- lish tender inputs and permit audits
Finance building and main- tenance	Lenders cannot observe milestones, arrears, or reserve balances	Completion time, loan losses, reserve adequacy and repair times	Credit scorers can exclude informal incomes; contest- able decisions and manual review
Check long-term public value	Resale restrictions, occu- pancy and service gaps hid- den	Affordable occupied dwell- ing-years, travel cost, and maintained asset condition at years 2, 5, 10 and 20	Central registry can facilitate eviction or surveillance; pur- pose limits, access logs, tenant representation

AI-supported fiscal administration should be evaluated through auditable decisions and the same full-cost accounts used for other investments. The size of any improvement must be established against the current process. A claimed 90 per cent reduction in mapping time does not establish faster lawful decisions if the registrar cannot resolve competing claims or the appeals office remains closed. A model may raise nominal tax potential while reducing trust, or identify overcrowded areas that become targets for eviction. Those risks follow from institutional power, rather than the model's advertised accuracy. Procurement should therefore specify the decision that will change, the baseline cost or error rate, an outcome target, the authorized data users, the financial beneficiary, an exclusion audit, and the remedy when the system is wrong. Consider an AI-supported valuation pilot: pre-register whether net collections improve after appeals, whether effective tax rates rise more on expensive land than on poor housing, and how much cost falls on newly identified renters. Keep a representative manual audit sample, publish error bands, and allow correction before collection. The technology's public return belongs in the same six ledgers as a road or land concession. The preparation sequence should turn those accounts into a legally enforceable financing agreement.

Preparing a municipal finance agreement

Prepare a fiscal agreement through readiness conditions rather than fixed deadlines that ignore unresolved rights or approvals. Name the budget, tariff or operator revenue paying each obligation; stress-test collections and fiscal transfers. Treasury, housing agency and operator must reconcile commitments and verify compliance throughout the affordability period. Track public cost per safe affordable home-year, affordability retained at covenant expiry, renewal funding and contingent liabilities. A city that wants to improve its next affordable housing project can begin with one corridor, ten candidate public sites, and a joint team drawn from planning, finance, lands, utilities, housing, social protection, and the operator or lender. The mandate should be signed by the mayor or chief executive and the finance authority. It should carry authority to obtain records, engage current occupants, and stop a bad proposal. Use the following stages to prepare a bounded operation. The responsible agency should first cost staff time, specialist work, consultation, and verification, then choose the number of sites it can handle. Set dates after that review. Disputed rights, procurement, or statutory approvals may extend the schedule; do not advance until the applicable legal and funding conditions are met. The first stage establishes the rights, alternatives, and public contribution.

Establish rights and compare alternatives before assigning the present value of land and service obligations. Stage 1: establish the rights and the alternatives. Shortlist sites within the team's appraisal budget, then map them and their legal owners, leases, occupants, traders, easements, flood risks, and utility access. Commission an independent valuation for the shortlisted transaction and a second review where material uncertainty or transaction rules warrant it. List the realistic alternative use of each site. Publish a parcel-by-parcel decision sheet except for personal or commercially protected details; create a claims desk staffed with local civil-society observers. A disputed site can stay under review without halting work on the others. Stage 2: reconcile the six ledgers. Finance and utility teams prepare the full twenty-five-year cost and revenue model for each candidate. Use at least three rent or sale-price scenarios, two interest-rate cases, and a cost-overrun case. Estimate low-income eligibility and ability to pay from actual household data, not a generic percentage of median income. Show what happens to the people occupying each site now. Publish the present value of public land committed and the agency responsible for each service. The procurement terms should then fix the affordable outcome and treatment of affected residents.

Set the agreement before bidding and test it with affected residents while respecting procurement confidentiality. Stage 3: set the agreement before the bid. Agree the affordable unit mix, allocation rules, lease or agreement length, permitted density, rights to return, maintenance reserve, tariffs, and beneficiary-protection rules. Decide who receives public revenues from land-value gains, and who carries any operating deficit. Resolve which approvals and service connections must be ready before a developer signs a binding contract. Agree escalation and dispute rules in advance. Stage 4: test bids against household outcomes. Invite competing development and operating proposals on a common subsidy and service baseline. Disclose a reference financial model and evaluate bidders on independently costed life-cycle performance. Reject offers that depend on an unspecified later zoning waiver or subsidy increase. Have tenant and disability representatives examine layouts, tenure documents, access and

grievance procedures; keep commercially sensitive bids confidential until the proper publication stage. Approval should be limited to obligations that can be financed, enforced, and monitored.

Authorize a project only when its fiscal, household, and service commitments are jointly credible. Stage 5: authorize only what can be financed and enforced. Sign a multi-agency service agreement; commit ring-fenced maintenance provisions; complete resettlement planning with affected households; document third-party review of costs and concessions. Establish a public record of the promises attached to each parcel. Release construction funding in stages tied to services and verified work. Baseline rents and household travel costs before anyone moves; repeat the survey after occupation. A practical cabinet or MDB decision sheet can then show the following three numbers side by side: the net present public cost per additional safe occupied household, the share of benefits received by the intended income groups, and the number of affordable occupied dwelling-years expected over twenty-five years. The first is a fiscal estimate, the second an incidence estimate, and the third a long-term outcome. Show uncertainty intervals. A project can look strong on one number and weak on the others. The appraisal questions below test for gaps that a favourable headline subsidy figure can conceal.

Twelve appraisal questions

Use a complete appraisal to test additionality, full public cost, household access, risk, and the conditions for changing course. 1. Is every public parcel valued against a defensible alternative and every occupant or claim recorded? 2. Which homes are actually additional to supply that would occur without public support? 3. Are the required rents, deposits, or loan payments affordable to the identified households, including informal earners? 4. What are the full public costs of roads, water, drainage, schools, transport, and climate protection? 5. Which agency funds operations and major repairs in years 5, 15, and 25? 6. Who takes losses when sales fail, financing costs rise, or the developer is late? 7. Can resale, conversion, or refinancing remove affordability; who can enforce restrictions? 8. Are renters, informal occupants, and traders protected, and are their outcomes independently monitored? 9. Does the municipality retain or receive adequate revenue from increased land value and activity? 10. Has the procurement fixed the affordable housing output before allocating the land concession? 11. Can every applicant contest an erroneous data record or discriminatory allocation decision? 12. What evidence would trigger cancellation, redesign, or a change in subsidy instrument? Unknown answers require a documented response from the responsible institution.

A negative or unknown answer requires a documented response; an average score must not override an unresolved legal, safety, or funding condition. Assign review duties explicitly: the finance ministry checks the full fiscal cost; the municipality checks operating responsibilities and service funding; the housing ministry verifies occupation; and residents must be able to challenge failures in access, affordability, or service. The resulting account should measure sustained housing benefits rather than the capital transaction alone.

Measuring sustained housing benefits

Consolidate the opportunity cost of public assets and continuing obligations before approval. Capital subsidies also create future service and maintenance obligations. Housing appraisal should therefore assess continued affordability, municipal operating revenue, and access for future households alongside the initial transaction. The remedy begins with documented property claims and complete accounts. Each parcel requires a lawful use and a disclosed value. Each subsidy requires a named beneficiary and an enforceable duration. Each service needs a revenue source and a maintainer. Each project needs a household test that counts those displaced as carefully as those newly housed. Each public gain from new infrastructure or development rights creates an opportunity to finance the next neighborhood, provided law and politics permit that gain to be shared. Cheaper capital is one part of housing finance. Appraisal must also identify the commitments made against future public assets and income. Land allocation limits alternative uses. A tax levy commits a share of household earnings. A loan commits rent or public revenue to repayment. A poorly specified contract can leave future governments liable for payments even when delivery falls short. These obligations should appear in the consolidated account before approval. Reporting must then show who receives the benefit and for how long.

Report the full public contribution, the households served, the duration of affordable occupancy, and the costs borne by existing residents. Update the appraisal when delivery, service costs, or eligibility changes. Next: Chapter 15, "Project readiness, procurement, and delivery," examines procurement, program management, administrative capability, and the contract failures that separate housing policy from homes occupied. A published distributional annex makes those benefits and costs traceable by affected group.

Publishing a distributional assessment

A distributional assessment should publish evidence and assumptions by group without exposing personal records or asserting unsupported motives. The housing team and finance analyst should prepare a group-level table before approval, publish assumptions and uncertainty, and update it when the site, subsidy, or contract changes. Use existing household samples, administrative records, and valuations first; commission additional work only where it could alter the choice. There are documented precedents for this form. The UK housing ministry's appraisal guide includes distributional analysis and an appraisal summary table. Transport Scotland's published National Speed Management Review 2024 contains a distributional assessment of modelled user benefits across income groups. These are appraisal and disclosure practices, not evidence that publication itself produces fair outcomes. Their transferable feature is a traceable comparison by affected group with stated assumptions. [212-213] A housing-specific format can make the comparison usable in an appraisal and later review.

Record the costs, benefits, uncertainty, and remedies for each affected group in a dated annex. In each row report the number affected, the counterfactual, expected change in income or costs, timing, evidence quality, and proposed remedy with its payer. Show unquantified effects such as loss of care networks separately. Do not add asset-value gains to capitalised rent benefits if that counts the same effect twice. Provide a short plain-language summary and an accessible

local-language version where needed, with a date and channel for corrections. Publish aggregate results through the actual approval process, such as the council paper or disclosed project appraisal, and place the full calculations in an accessible annex. Track unresolved objections and the authority's response. Keep personal compensation files confidential while making eligibility rules and aggregate obligations public. The municipality also needs a marginal budget for the additional households it will serve.

The municipal budget for additional households

I would assess the municipal cost of additional households alongside their own affordability calculation. I would put a marginal municipal budget alongside the household affordability calculation. Estimate the additional capital and annual service costs, the revenue legally available to the municipality, the timing of transfers, and the cost of future replacement. Separate a temporary connection cost from a permanent operating gap. Do not count a national grant as municipal income until the terms make it available for the required purpose. The fiscal instruments change different incentives. Property tax can give a municipality a continuing stake in a larger serviced tax base, but collection and valuation determine the actual revenue. A transfer linked to population or service need can help pay for additional residents, if it arrives predictably. An infrastructure grant can cover an initial network extension without paying for its operation. A service charge can fund use while excluding poorer households unless affordability support is explicit. Compare these instruments against the same marginal budget rather than treating them as interchangeable funding sources. A wider agreement may be needed where one municipality benefits from workers while another funds their services.

Metropolitan inclusion may require intergovernmental financing rather than a neighbourhood negotiation alone. Test a metropolitan arrangement that assigns feasible housing capacity across well-served areas and shares service costs. Require a published account of additional homes permitted, affordable homes occupied, operating revenue, and access to schools and transport. The arrangement must address genuine service costs without paying municipalities merely to abandon an exclusionary veto. Chapters 10 and 23 examine the coordination and infrastructure-financing constraints. The worked budget below shows how to distinguish the initial capital balance from the recurring gap.

Worked municipal budget for additional households

A marginal municipal budget should separate infrastructure capital from the continuing cost of serving additional homes. Consider 1,000 additional occupied homes within an existing municipality. The figures below are assumptions in millions of local currency units, at constant prices. They are not estimates for a particular city or recommended tariffs. The scope covers additional municipal access roads, drainage, public lighting, waste services, administration, and renewal. Water, electricity, schools, and health services sit in separate provider budgets in this example; those budgets must also balance before approval. Housing construction itself is financed separately. The two accounts below use explicit illustrative assumptions rather than a project forecast.

Table 14.3. Illustrative municipal capital account. Positive figures are resources available for the stated works; the investment requirement is shown separately.

Capital item	Million local currency units	Budget treatment
Additional municipal works required	60.0	Costed capital requirement, including the defined local network works.
National or state capital grant	40.0	Available only for eligible capital expenditure under the assumed grant terms.
Developer contribution	10.0	Collectible contribution; test its effect on project viability and household prices.
Municipal capital allocation	10.0	Existing funded allocation, with its opportunity cost recorded. No new borrowing is assumed.
Total capital financing	60.0	Balances construction finance. It does not finance future operations.

Table 14.4. Illustrative annual municipal account at full occupation. Renewal is a funded annual provision; do not also charge the same replacement cost to another line.

Annual item	Million local currency units	Assumption or result
Waste, local drainage and lighting operations	2.4	Additional operating cost within the stated municipal scope.
Administration and inspection	0.6	Additional staff and operating expenditure.
Asset renewal provision	1.0	Annual contribution derived from the assumed component replacement plan.
Total recurring requirement	4.0	2.4 + 0.6 + 1.0.
Property-tax receipts	1.2	Cash collected after exemptions, collection losses and applicable relief.

Annual item	Million local currency units	Assumption or result
Service-charge receipts	0.8	Cash collected after affordability support and collection losses.
Locally financed resources	2.0	1.2 + 0.8; the remaining annual gap is 2.0.
Required additional recurring transfer	2.0	A proposed funding commitment, not an existing entitlement. Equivalent to 2,000 currency units per home annually.

A balanced capital account does not establish that recurring services are funded. The capital account balances, but the municipality still needs 2.0 million each year. Higher employment might benefit households and national revenue without generating an equivalent municipal receipt. Count only revenues the municipality is legally entitled to receive, at the time it can use them. Do not count national income, consumption, or corporate tax as local resources unless the applicable revenue-sharing arrangements actually transfer them. Stress the cash account before signing. If tax and service-charge collections are 25 percent below the assumed 2.0 million, local receipts fall to 1.5 million. An unchanged 2.0 million transfer leaves a 0.5 million annual shortfall. A late transfer can create a cash shortage even if the full-year budget balances. The finance officer should identify an affordable contingency and payment schedule; an unapproved future grant or an unaffordable fee increase is not a funding plan. South Africa offers an institutional comparison for linking service obligations to performance-based support.

Institutional funding arrangements should be tested against the marginal cost and actual quality of municipal services. The explanatory memorandum to the 2025 Division of Revenue Bill describes a basic-services component of the local government equitable share that supports services to households below an affordability threshold. Its formula incorporates maintenance, and its household estimates need not update immediately with new arrivals. [218] This is evidence of a transfer design, not proof that every municipality receives enough or delivers the intended service. The transferable requirement is to connect service responsibilities to predictable revenue and report allocation lags. Calculate how to finance the services required by additional residents. Treasury and the municipality should compare a recurring transfer, broader revenue sharing, legitimate cost reductions, and an alternative serviceable location on the same household-access assumptions. A capital-only grant cannot close a permanent operating gap. Conversely, a locality should not receive compensation for inflated costs or unnecessary standards used to exclude households. Check the marginal cost against spare capacity and efficient service provision, then record the funded agreement and review it against actual receipts and service quality. Fieldwork should verify those obligations before the housing grant is approved.

Prepare the local decision

Reconcile land, grants, guarantees, service commitments, and household costs before approving public support. Reconcile land value, grants, tax relief, guarantees, infrastructure, operating transfers, and replacement obligations across agencies. Record timing and avoid counting the same transfer twice as it passes between governments. Distinguish committed, paid, and contingent amounts. Apply the chapter's six-account approach to an actual project, including the cost of the best feasible alternative use of public land. Treasury, the municipality, utility, public landowner, housing provider, and auditor should sign the reconciliation and identify disputed assumptions. Residents should verify charges and service promises. Ask each agency which budget line will pay after initial project finance ends, and what happens if land receipts or rent collection fall short. Link contract, parcel, project, and payment identifiers in a controlled reconciliation file. Automated checks can flag inconsistent totals or missing obligations; an accountant must resolve them. Compare instruments using discounted public cost and a separately reported measure of occupied affordable housing, with sensitivity to rates, vacancies, and repairs. Do not approve a capital grant while a necessary operating subsidy has no identified payer. Chapter 15 follows that financing agreement through readiness, procurement, handover, and safe occupation.

Public support should purchase a costed stream of affordable housing services over time. Durable covenants, viable operators, reinvestment rules, and provision for future obligations connect today's subsidy to continuing access. A foreign tax credit or loan rate cannot substitute for those arrangements. Expiring restrictions, omitted land values, unpriced guarantees, and deferred renewal create commitments that headline subsidy figures conceal. Aggregate expiring restrictions, refinancing needs and major repairs across cohorts; a national portfolio can face simultaneous liabilities.

Sources and evidence notes

Evidence and scope. Administrative announcements and programme guidelines describe commitments or rules. Observational comparisons establish associations; projections and worked models are identified as such. A land handover, unit target, or subsidy authorisation does not establish an occupied-home outcome. Source dates are stated in the references.

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15. Project readiness, procurement, and delivery

Procure housing against a verified delivery plan that connects construction to lawful, serviced occupation. Resolve land, approvals, financing, utilities, allocation, and management before committing to milestones that depend on them. Contracts should assign measurable responsibilities and remedies, while performance reporting distinguishes completion from use. Programme totals and project audits show why that distinction matters.

National construction totals and project audits show why sanctioned, completed, and usable housing must be distinguished. By 30 September 2026, India's official PMAY-Urban website showed 12.768 million homes sanctioned, 12.103 million grounded, and 9.907 million completed across the programme. These are reported programme milestones. The displayed headline figures supply no matching count of homes verified as occupied, safe, serviced, and accessible to livelihoods. Construction and sustained occupancy require separate measures. [2] South Africa offers a second view of the same problem. In March 2026, its Auditor-General reported findings on 23 of 24 housing projects examined, with a combined value of R5.67 billion. Twenty of the 24 projects had suffered delays; 15 had poor construction quality; six had shortcomings at commissioning. Some housing was signed off without electricity, water, or sanitation. These are results from an audit sample, selected for scrutiny; they do not establish that 96 per cent of all South African housing projects have failed. They identify service and quality shortcomings in the audited projects after substantial public commitments. [3] Correcting the gap requires a diagnosis of the particular decision that remains unresolved.

Capacity assessments should identify the decision, responsible institution, resource, and remedy preventing safe occupation. Review which decision could have prevented a delay, why responsibility moved between offices, whether incentives encouraged premature approval, and what remedies residents had when a project was defective. A ministry may release funds and record progress while land registration, utility capacity, inspection, or household finance remains unresolved. These interdependent decisions should be examined together. This chapter examines how to convert approved land, budgets, and contracts into safe occupied homes, with responsibility for delays and remedies specified before procurement. The delivery sequence contains several interdependent processes that must be tested together.

1 Establish readiness before procuring construction

Coordinating seven delivery processes

A project is ready when all essential legal, construction, and service conditions for safe occupation form a workable sequence. The land process concerns title, competing claims, resettlement, and the legal right to build. The planning process concerns density, building permission, environmental approval, and the route by which neighbours or officials may block a permit. Infrastructure delivery covers trunk water, sewerage, drainage, power, streets, and public transport. Construction covers design, tendering, materials, labour, inspection, and defects. Financing includes budget release, contractor payments, buyer deposits, mortgage approval, and

the cost of working capital. Allocation determines eligibility, selection, appeals, contract signing, and handover. Long-term neighbourhood management covers: waste collection, maintenance, water bills, repairs, safety, and the rules for the next household. These processes may follow different schedules. The housing department may control one or two. A state treasury, county executive, municipality, water utility, contractor, bank, residents' group, and land office control the rest. A project is ready only when all essential legal and service conditions for safe occupation have been met. An illustrative reliability calculation shows why several individually strong handoffs can still produce weak overall delivery.

Figure 15.1. Responsibilities in the housing delivery chain

Secure site →Approved plan →Funded services →Safe structure →Fair allotment →Occupied home

Each stage needs an accountable institution, an enforceable decision, independent evidence, and a funded remedy. Track the stages separately and verify safety, services, lawful possession, and occupancy before reporting a usable home.

Multiple required handoffs can reduce delivery reliability, subject to the stated assumptions about their dependence. Consider an illustrative calculation in which each of six essential handoffs succeeds with probability 90 per cent, independently of the others. Of 1,000 homes announced, the expected number clearing all six would be 1,000 × (0.9)^6 ≈ 531. This is a model assumption, not an estimate for a national programme. Actual failures may be correlated: insecure title can delay utility agreements and finance together. Without independence, the joint completion probability depends on that relationship and cannot be calculated by multiplying the six marginal probabilities. Resolving one shared constraint may address several risks. Starting physical work before essential legal and service conditions are settled can increase later costs. If construction begins before title, service capacity, and relocation terms are settled, sunk costs change the parties' bargaining positions. A developer can demand an extension because the water utility has delayed a connection; an official can argue that stopping now would waste public money; residents can face eviction because the project has already mobilised contractors. Assign these risks and remedies before works begin and become costly to stop. A common agreement should assign each handoff and its remedy before construction starts.

The correct starting point is a common delivery agreement. It should name the landholder, the municipality, the infrastructure operator, the implementing agency, the funder, and the resident representatives; state each party's contribution; and make each party's signature conditional on the relevant budget and legal authority. A coordination memorandum should specify enforceable obligations. A signed obligation to install a functioning water main by a specified date, backed by an appropriation and a dispute procedure, changes the developer's risk and the household's prospects. India's programme experience shows how recorded construction and actual use can diverge when responsibilities remain divided.

2 Learn from completion and occupation failures

Completions and occupation in India

A national housing mission can mobilise construction while leaving site-level obstacles to occupation unresolved. Its centrally supported structure can set goals, release funds, record beneficiary-led construction, and encourage states to build at scale. Different programme components, however, demand different kinds of state capability. A family improving a house on land it occupies needs staged cash, structural advice, and service connections. A new apartment development on public land needs a lawful site, a qualified developer, roads, utilities, a completed allotment list, and long-term management. A slum-redevelopment project must also negotiate the claims of tenants, owners, traders, street vendors, and people excluded from old enumeration lists. Central assistance for the AHP and slum-redevelopment components could support construction even where state governments lacked the money or authority to provide trunk services on time. The parliamentary committee found exactly this: states had yet to provide required infrastructure in many schemes, and allotment remained incomplete in others. The result was substantial recorded construction with a lag in actual use. [1] Those obstacles need distinct certification and accountability arrangements.

Certification should separate construction, legal occupation, and service readiness and remain independent of the contractor's invoice. This division of responsibilities can separate reported performance from the costs borne by other parties. A national ministry may be assessed on sanctions, investment, and completion. A state agency can award construction contracts while pushing the cost of off-site connections onto another budget. A municipal body may inherit water, roads, and solid-waste obligations without a matching revenue stream. Each office can defend its formal compliance. Households can bear the combined cost in rent paid elsewhere, longer travel, waiting, and lost income. The operating rule should be simple: approval of a housing block must carry an infrastructure financing plan signed by the entities that will build, own, and operate the services. Design review should test whether a water utility can supply the expected population at peak demand and where wastewater will go in the dry season and the wet. The project budget should show last-mile connections, utility deposits, access roads, pumping power, open-space upkeep, and the cost of service until full occupancy. A building-completion certificate and an occupancy permit should trigger separate entries in the management system; neither should be issued by the same contractor who invoices the works. A hypothetical project illustrates the cost of allowing these obligations to fall between budgets.

Delayed occupation can create substantial public and household costs even after construction expenditure is recorded as complete. Consider a hypothetical 1,000-home scheme costing INR 1 billion to build. Suppose the homes are physically complete while connection to trunk sewerage takes six more months. At an illustrative annual financing cost of 10 per cent, carrying the full INR 100 crore during that period costs INR 50 million in simple interest. This assumes that the full amount remains financed throughout the six months; actual carrying costs depend on the drawdown and capital structure. Families bear their own costs during the wait, and unoccupied buildings may require protection and maintenance before lawful occupation. Compare this delay cost with the cost of coordinating sewerage before completion

and with other construction savings. The amounts here are worked assumptions, offered to show the mechanism rather than to describe a particular Indian project. Recovery should therefore diagnose empty homes by cause rather than pressure households to accept unsuitable allocations.

An occupancy-recovery portfolio needs different remedies for location, services, rights, defects, and household constraints. An occupancy recovery programme would treat the December 2024 backlog as a portfolio of distinct problems. Buildings with functioning services and missing allotments need a different response from buildings where the water main is five kilometres away. Buildings that residents reject because jobs are too distant require a transport and economic-access test before another household is pressed to accept the unit. Some projects may merit adaptation to rental use, institutional housing, or mixed tenure. Others will need legal remedies and, in exceptional cases, an admission that an unsuitable site represents a sunk loss. A ministry should publish the cost per additional safe occupant-year recovered, rather than treating every empty flat as a failure that can be solved by additional pressure to accept an allotment. Administrative tracking can support that diagnosis when it is checked against inspections and residents' accounts.

The official PMAY portal's geo-tagging and beneficiary tracking can verify steps along the way. Coordinates can show where a building stands. A time-stamped photograph can document visible works. Neither proves that a named woman can use the water point at night, that a disabled resident can enter safely, that the landlord has not taken possession, or that the person allotted the flat actually lives there. A verifiable system combines administrative records with independent inspection and a right for residents to contest the state's account. [13] South African audit findings identify further failures across quality, utilities, and commissioning.

Housing project audit findings in South Africa

Project audit findings should be traced to the specific delivery stage and institution able to correct them. The Auditor-General's detailed 2024–25 account records 20 of 24 projects delayed, 13 with cost issues, 15 with poor build quality, and six with commissioning shortcomings. Different projects appear in more than one category. The prevalence of defects in the audited portfolio reflects failures in planning, contracting, monitoring, and intergovernmental coordination. A grant assessed chiefly by expenditure can miss these delivery failures. [3] A provincial human-settlements department may procure a settlement under a conditional grant. Municipal engineers must accept the roads, pipes, streetlights, and drains, even where staffing is thin and maintenance funding is scarce. An electricity distributor must confirm capacity and connect households. A property office must establish legal rights. An inspection team may cover dozens of scattered construction sites. The building contractor can finish the bill of quantities while the settlement remains unusable. Even when the builder bears responsibility for part of the works, a missing water connection can delay certification of everything else. Contracts must distinguish the milestones on which payment and acceptance depend.

Construction completion, commissioning, handover, and occupation need separate acceptance conditions. A road laid with asphalt has reached construction completion. It becomes a functioning service after drainage works, safety checks, access, and formal acceptance.

A water network reaches commissioning after pressure tests, water-quality checks, legal connection and an operator's sign-off. Homes reach handover when households receive lawful keys and know who pays for which service. Occupation can then be checked after three, six, and twelve months. Reporting the stages separately allows officials to identify delays and failures after construction. Corrective action requires funding and enforceable authority. A municipality that refuses to accept a defective network must have the capacity to inspect it and a legal path to require correction. A province must be able to withhold final payment without triggering an unmanageable dispute over work performed. The national government must face the possibility that annual disbursement targets will suffer when unsafe or unusable projects are stopped. Performance assessment should allow a delay needed to correct unsafe work, while tracking the resulting occupation and durability. The incentives facing certifiers and contractors should support correction of defects rather than premature closure.

Review payment and certification incentives alongside independent quantity, quality, and utility checks. Contractors are often paid by quantities installed; municipal residents care about services that work. A contractor can earn more from a variation order when the design was incomplete. A supervising engineer may face pressure to certify against a delivery deadline despite unresolved defects. Restricting variations by decree can lead builders to walk away or conceal problems. Better design, third-party checks, published change orders, and predictable dispute resolution reduce the room for opportunism on either side. Independent verification has experimental support outside housing. Across more than 600 Indonesian road projects, raising audit probability from 4% to 100% reduced measured missing expenditure by about eight percentage points. Greater grassroots participation had smaller overall effects and did not reduce missing material expenditure. This supports technical verification alongside resident voice, but does not establish the same effect size in housing. The South African findings show where to apply that discipline: quantities, quality, utilities and commissioning, as well as approvals. [116,3] Kenya's audit experience shows why the same discipline must begin before tender.

Project readiness in Kenya

Site-readiness audits should distinguish unresolved preparation conditions from evidence of a finished, usable project. In a special audit issued in March 2026, covering project status at 31 March 2025, the Auditor-General examined legal documents, procurement, site progress, and programme preparation. Subsequent reporting on that audit described unverified land records, pending approvals, delayed payments, contractor and consultant overextension, and gaps in supervision. The audit also questioned the absence of a consolidated programme appraisal, a formal affordability-threshold analysis, and documented cost-benefit assumptions before rollout. [4] [5] One widely reported audit figure deserves careful handling. 269 of 394 projects examined lacked the specified land-ownership documents at the audit stage. These 394 included markets, lighting, and other works alongside housing and social-housing projects. The number cannot be used as a claim that 68 per cent of housing projects alone lacked title. Land-document deficiencies may also have changed after the audit: officials subsequently reported progress in assembling records. The finding nevertheless identifies a serious source of preventable implementation and buyer risk. [6] Published project lists should reflect that distinction.

Keep non-ready sites in preparation until the responsible authority resolves their recorded legal and service conditions. Announcement can create pressure to proceed before those conditions are met. A county official may hope that a future budget will bring the access road. A developer may proceed in expectation of future approvals. A land claimant may demand compensation when works begin. Contractors with many simultaneous awards may stretch supervisors across sites. These are distinct conflicts, yet each can arise because someone with little power to stop the announcement had the decisive information about readiness. The solution is a pre-tender readiness certificate with evidence attached. It should establish the legally recognised land interest; disclose unresolved customary, community, tenant, and public-land claims; record design approval; identify affected households and traders; confirm environmental and safety permissions; attach signed service-capacity letters; and identify who will finance connections. The certificate should carry an expiry date and an accountable signatory. A public portal can publish a redacted version without exposing residents' personal data. Projects that fail the readiness check should remain in preparation until the responsible authority resolves the recorded conditions. Procurement review must also test whether competition and contract packaging support credible delivery.

Procurement concentration requires evidence on market participation, capacity, and quality rather than assumptions about motives. A single skilled contractor working on several sites can cut costs through shared suppliers and labour. Excessive concentration may overwhelm supervisors and increase the disruption and cost of replacing a failing supplier. An award committee should therefore assess aggregate committed capacity, recent delay rates, cash flow, site staff, and common beneficial owners across bidders. A tender that records competitive prices while ignoring the contractor's existing pipeline may underestimate delivery risk despite competitive bid prices. The most useful measure for Kenya's programme is a cohort record: every planned development should remain traceable from announcement to land clearance, procurement, serviced construction, independent certification, sale or rental contract, verified occupation, and defects repaired. A programme can then tell Parliament how many households moved into safe homes from each annual cohort, and what public value, including land and infrastructure, was spent per household. Such evidence also protects competent officials from blame when they refuse to approve a defective project. The contract then needs to allocate each delivery risk to a party able to manage it.

3 Make contracts, allocation, and handover work together

Contract responsibilities and remedies

A contract decides who has authority, who carries risk, and the remedies when an obligation is not met. These decisions determine whether housing becomes affordable before construction starts. Poorly specified risk either raises bid prices or reappears as a claim later. Contract wording cannot give a builder control over state decisions, such as resolving public title or deciding disputed zoning. Rational bidders will price it, refuse to bid, or seek to renegotiate after mobilisation. A design–bid–build contract can work where the government owns detailed designs, has a ready site, and can supervise quantities. It struggles when utilities, soil conditions,

and rights-of-way remain unknown. Design–build can place greater responsibility for technical integration with the contractor, provided performance requirements are clear and the government has the skill to test them. A developer-led partnership can mobilise finance, land, and marketing, yet creates opportunities to renegotiate unit sizes, affordability rules, tenure restrictions, and public-land value. A community-managed upgrading contract gives residents more control over design and sequencing; it also needs accounting systems, technical support, and safeguards against local elite capture. Contract choice should follow the capacity to prevent or absorb each risk.

Allocate risks according to control and capacity, with public guarantees justified by an explicit expected benefit. The land authority should bear responsibility for the lawful site it provides. The developer should bear the cost of defective design it controls and of workmanship that fails an objective test. The utility should commit to dates and capacity over which it has authority. The government should retain transparent responsibility for changes in law and for relocation that only it can authorise. Residents should hold enforceable rights concerning selection, accessibility, temporary accommodation, and remedy. Lenders and insurers should price credible residual risk, with public guarantees used only where the expected public gain can be shown. Procurement and audit evidence can test which oversight arrangements improve actual delivery.

Table 15.1. Allocating delivery risks and responsibilities

Failure point	Best placed to control it	Required evidence before money moves
Unclear title or community claim	Land authority and independent adjudicator	Search, claims register, legal instrument, grievance window
Insufficient trunk water or sewer	Utility, city, and project sponsor	Capacity letter, funded works, commissioning date
Unsafe foundations or fire design	Designer, contractor, independent checker	Ground tests, approved drawings, inspection hold points
Escalation in material prices	Shared under a transparent formula	Named indices, thresholds, capped contingencies
False beneficiary list or exclusion	Implementing agency with independent appeals	Public rules, protected audit trail, resolved challenges
Unfunded building maintenance	Future operator and public sponsor	Reserve policy, fees/grants, service contract, handover inventory

Proposed contract-design principles. Allocation is context specific; this table is a decision aid, not a universal standard form.

Procurement technology and independent audits address different risks and should be assessed through their measured effects. A study of electronic procurement in India and Indonesia found no clear reduction in government prices following adoption. It did find improvements in the quality of Indian roads and fewer delays in Indonesia, consistent with wider entry by higher-quality firms from outside local contracting networks. The evidence concerns public works; housing programmes should test the same mechanism rather than assume that a portal will reduce home prices. [7] A separate road-procurement study examined political connections. In an Indian rural-roads study using close elections, the share of awarded contractors with surnames matching the elected politician rose from about 4 to 7 per cent after the politician took office. The researchers linked political influence to greater costs and a higher chance of missing roads. Shared surname is an imperfect proxy for social or kinship ties, and the result relates to road procurement. It supports testing housing-procurement safeguards through monitoring of beneficial ownership, repeat bidding patterns, sudden contract amendments, and conflicts of interest. [8] A useful anti-corruption system has at least three parts: open competition, competent engineering checks, and credible sanctions. Indonesia's randomised experiment on more than 600 village roads found that raising the probability of a government audit from 4 to 100 per cent cut measured missing expenditures by roughly eight percentage points. Greater community monitoring alone had smaller average effects. For housing, this supports testing independent quantity and quality audits while giving residents accessible ways to report defects; community voice and technical verification address different problems. [9] Housing allocation requires additional safeguards because construction oversight cannot establish fair access.

Allocation rules and appeals

Allocation rules need scrutiny for exclusion and unequal access to information or appeal. Allotment determines who receives a subsidised asset, and the value of that asset may be substantial. Unchecked discretion can permit preferential allocation or unofficial charges. Landowners can classify tenants as temporary occupants and seek compensation for themselves. A developer can persuade a city to raise prices, waive affordable-unit obligations, or release units to a higher-income market. Incumbent homeowners can delay denser or mixed-income housing through planning objections. Test whether caste, class, ethnic origin, race, disability, gender, or migrant status affects access to allocation and appeal procedures. Eligibility, selection, and possession should be treated as separate decisions.

A fair allocation process should make eligibility, selection, and possession independently reviewable. First, which households qualify under a published objective, such as very low income, displacement by public works, rental need near jobs, disability-related housing needs, or disaster risk? Second, which of the qualified households receive scarce units, and in what order? Third, who verifies that the intended household can take and keep possession without coercion? A digital registry can help match applications and detect duplicates. It also gives the operator new power to freeze eligibility, infer identity, and make errors hard to challenge. A family without a formal address or accepted document may be excluded unless alternative evidence and assisted applications are available. For in-situ upgrading, a census must distinguish landowners, structure owners, tenants, subtenants, traders, home-based workers, and

people who use shared spaces. Eligibility dates create sharp incentives for brokers to buy claims or add names. The remedy combines advance publication of rules, independent grievance hearings, community-supported verification, and protection against retaliatory eviction. An appeals body should have access to underlying records and the authority to correct the list. Women should be able to register independent claims; documentation rules must recognise lawful but non-standard forms of tenure. The location offered must also be suitable for the household's work and social needs.

Location is part of fair allocation. A household offered a remote apartment with poor transit may refuse it rationally. Penalising refusal can push the poorest into accepting high commuting costs and lost work. Programme managers should record offers declined, with stated reasons, as useful evidence about product design. A home that requires a two-hour commute can be inexpensive to construct but unaffordable once travel costs and lost earnings are included. The governing outcome is the household's total cost of housing and access to work, with safety and social ties included, rather than the administrative act of issuing a key. Singapore and Thailand illustrate different ways to coordinate those delivery and allocation obligations.

Two approaches to delivery coordination

Singapore demonstrates coordinated authority while retaining distributional and political questions that integration alone does not resolve. Its public housing agency and the wider state have long connected land assembly, planning, infrastructure, finance, household allocation, and estate management. In 2023 about 80 per cent of resident households lived in Housing & Development Board flats, with homeownership among those households above 90 per cent. Supply-chain shocks still disrupted construction during the pandemic. By September 2024, the government reported that roughly 94 per cent of delayed flats had been completed after coordinated action on contractors and materials. [10] Singapore assigns responsibility for linked decisions from land development through maintenance. Its conditions are distinctive: unusually strong administrative and fiscal capacity, extensive state control over land and infrastructure, a high share of formal employment, and institutional arrangements developed over decades. An agency's name or organisational chart does not provide the land powers, finance, or operating capacity on which this arrangement depends. Nor does integration remove the political risks of compulsory acquisition, control over neighbourhood composition, or subsidising those already well placed to purchase. Thailand offers a contrasting division of authority between residents and public institutions.

Community-led delivery requires inclusion, viable finance, technical support, and enforceable rights. Under Baan Mankong, the Community Organizations Development Institute (CODI) channels subsidies and loans to organised residents, who negotiate land tenure, choose designs, and manage much of the work through cooperatives. CODI describes a traditional subsidy package of roughly 80,000 baht per household, split among infrastructure, construction, community management, and technical support; cooperative loans complement savings. A 2026 programme summary reported more than 130,000 households assisted in over 1,300 communities across about 200 cities. These are programme-reported cumulative reach figures, not evidence from random assignment that each intervention caused a given increase in

income or housing quality. [11] [12] Residents can negotiate as a group through a representative organisation. The community becomes a buyer of technical assistance, a manager of subsidies, and a counterparty in land talks. This gives poor households bargaining power, though cooperatives can exclude tenants, migrants, or dissenters unless their rules and leadership are contested. Progress can also slow when many negotiations require local consent, landowners hold out, or recurrent public finance is thin. The model changes control over decisions and information. Its effectiveness depends on inclusion, financial viability, technical support, and enforceable rights. A city can combine those functions without copying either institutional form wholesale.

A city can combine parts of these two approaches. It may centralise land and infrastructure finance while devolving neighbourhood design and upgrading choices to residents. It may create a single government programme office that signs enforceable service agreements, while municipal inspectors and independent engineers retain the authority to stop unsafe work. The choice depends on which institution holds useful information, who can be trusted with discretion, and what accountability can survive a change of government. Digital tools should support the selected responsibilities through reliable shared records and verification.

Digital tools for delivery problems

Start digital delivery support with consistent project identifiers and verifiable records before procuring more complex models. Its value varies by the bottleneck it changes. A parcel map helps when land boundaries are missing from project preparation; it cannot adjudicate competing customary or tenancy claims. Remote sensing can flag a building that has not progressed between two dates; it cannot certify electrical safety or a household's consent to relocation. An AI model may flag abnormal variation orders; it may also be trained on flawed records and direct scrutiny toward small firms while overlooking firms poorly represented in the training and validation data. A city should first establish a durable project identifier linking site, approvals, contracts, inspection records, service assets, complaints, and occupancy. It should keep an append-only record of changes and explain who has authority to amend each field. Linking records is necessary because the land department's map, the procurement system's contract, and the utility's connection usually describe the same place in different ways. Test whether an open location standard and a consistent project number resolve the coordination problem before procuring additional model capabilities. AI can then be tested on specific tasks with controlled access and review of consequential decisions.

AI applications should be tied to defined verification tasks and safeguards against incorrect or harmful decisions. Use a shadow period in which the model makes recommendations and humans record when it is right, wrong, or unhelpful. Publish false-positive and false-negative rates across large and small firms and across formally and informally occupied areas. Require an appeal path whenever an automated flag affects eligibility, payment, land rights, or enforcement. Procurement contracts must secure data portability, audit access, security obligations, and the city's right to change suppliers. The political economy of information is inseparable from the engineering. Publishing the locations of potential additional structures may improve municipal revenue and expand the tax base; it can also expose residents to new charges,

harassment, or eviction. The right default is to make public decisions transparent while protecting household identity and sensitive tenure claims. Increased data coverage should be accompanied by access controls, correction rights, and oversight of consequential decisions. The process still needs an agreement that prevents actors from shifting the cost of unresolved problems to others.

Table 15.2. Digital tools for specific delivery constraints

Decision to improve	Tool and testable gain	Capture risk and safeguard
Can land be tendered?	Linked parcel, claims, and planning records; measure days from identification to independently cleared site.	Wrongful exclusion of unregistered occupants; include ground verification and appeal.
Can a contractor deliver?	Open contract data linked to beneficial owners, site workload, performance; measure failed awards and delay-days.	Established firms gain an algorithmic advantage; offer transparent scoring and bidder challenge.
Does work match invoices?	Time-stamped site photos, satellite change detection, quantity checks; measure discrepancy rates confirmed by engineers.	Contractors optimise visible progress; inspect hidden materials, safety, and quality independently.
Can residents move in?	Shared service-commissioning and allotment ledger; measure days from physical completion to verified occupancy.	Pressure to mark households occupied; use consent-based follow-up and spot audits.
Are homes still safe?	Repair tickets linked to asset registers and paid maintenance; measure defect closure and repeat failures.	Tenants fear retaliation for reporting; protect identity and publish aggregate outcomes.

Illustrative performance tests. No percentage improvement is assumed; each intervention needs a pre-specified baseline and comparison design.

Interests and constraints in stalled projects

Readiness rules need recorded authority and remedies because the party causing a delay may not bear its cost. Announcements may precede funded services. Incomplete designs can lead to paid variations. Unresolved tenant claims can affect compensation. A municipality may postpone accepting assets with unfunded operating costs, while a national ministry continues reporting committed funds. Procurement and claims records should establish which

mechanisms apply locally. Future residents often have limited influence before allocation, although they bear costs from delayed occupation. Technical advice needs an enforceable decision process. If water capacity is unresolved, identify who can withhold authorisation to start, what evidence would release that hold, and who bears delay costs. A readiness rule that can be bypassed without a recorded decision provides little protection. A delivery coalition must connect each participant's benefit to evidence that the agreed conditions are met.

Support for delivery should be converted into funded obligations and independently checked readiness evidence. Their interests can align around ready sites, service connections, clear permits, transparent allocation, and durable maintenance. Incumbent homeowners and speculative landowners may resist the extra supply. A credible agreement offers stronger local services and lawful compensation where rights are taken, while refusing permanent vetoes over exclusionary land-use rules. Three reform sequences differ. Where corruption in contracts dominates, begin with independent engineering audits, beneficial-ownership disclosure, and credible sanctions. Where lack of utilities dominates, start by putting infrastructure financing and commissioning ahead of construction awards. Where residents reject completed homes, audit location, transport, tenure, and allocation before spending more on similar buildings. Uniform national checklists conceal the very information a reform needs. A completed checklist needs supporting evidence that each condition has actually been met. Representation must include groups whose interests are absent from owner-dominated meetings.

Examine how caste and class in India, the legacy of racial exclusion in South Africa, and customary or other local authority arrangements affect participation in each project. They shape the location of public land, the legitimacy of surveys, the cost of objections, and whether women or migrants can challenge a list. The institutional response must give those groups enforceable channels into design and dispute resolution. Participation meetings attended mainly by structure owners and local notables can deepen exclusion when most occupants are tenants. Acceptance criteria should reflect those households' access and service needs.

Table 15.3. Interests and constraints in delivery

Actor	Payoff or rent	Veto / credible bargain
National leaders and ministries	Quick project numbers and visible openings.	Control grant timing; reward independently verified occupation and maintenance.
States, counties, municipalities	Allocation authority; possible future property revenues; recurring service costs.	Control land, permits, connections; fund recurring obligations and share revenue.
Landowners, brokers, informal power holders	Compensation, rezoning gains, access fees, speculative uplift.	Control claims and site access; negotiate early with audited claims and safeguards.

Actor	Payoff or rent	Veto / credible bargain
Developers, consultants, contractors	Fees, margins, change orders, control of contractor lists.	Can withhold bids or delivery; offer bankable sites, fair risk pricing, open competition.
Existing homeowners and landlords	Scarcity rents, control of neighbourhood access.	Planning objections and political influence; link density with services and tenant safeguards.
Prospective residents and tenants	Secure homes, access to work, safety, and community ties.	Usually weak formal veto; give appeals, consent rights, compensation and occupancy checks.

Analytical actor map. Rents and incentives differ by country and project; investigation must establish them locally.

Allocation, utility readiness and neighbourhood access belong in acceptance criteria, alongside the building itself. The worked delivery plan applies these requirements to a hypothetical programme before tender.

Delivery plan for a 1,000-home programme

Adapt packaging to local capacity while retaining independent verification and readiness conditions. Fund supervision, defects and dispute resolution upfront; compare these costs with credible expected losses. The employer must clear sites and utilities, certify work honestly and enforce remedies; residents need an independent complaints route. Track occupied serviced homes, defect closure, delays by responsible agency, verified quantities and exclusion at allocation. Imagine a secondary city offered a national grant for 1,000 apartments. A developer proposes public land on the urban edge. The ministry has construction finance, the city promises access roads, the utility expects a treatment plant upgrade, and a state bank says mortgages can follow allotment. The proposal still lacks confirmed service and financing arrangements and should remain in preparation before tender. The first preparation step is a common set of site, service, household, and financing records.

Prepare the maps, claims, funding, and demand evidence needed to test a complete delivery plan. The maps show lawful land interests and disputes; employment access by affordable transport; existing and planned trunk-service capacity; and flood, heat, slope, and environmental risks. The lists name every institution with power to delay handover and each household group likely to gain or lose. Those six objects define the actual investment proposal. A layout of 1,000 apartments comes later, when the land and household market have been tested. Next, write a single project balance sheet covering public-land value, direct grant, infrastructure, temporary relocation, private finance, buyer equity or rent, operating subsidies, maintenance reserves, and contingent liabilities. It should contain three cases: timely delivery, a twelve-month

service delay, and a 15 per cent construction-cost increase. It should model which actor runs out of cash, what happens to affordability, and the government's maximum exposure. An overly optimistic sales forecast can leave an unfunded gap, unfinished homes, or pressure for additional public support. Residents should then be able to understand the commitments and the remedies for failure.

Make delivery obligations visible to residents and retain enforceable rights to correct or replace defective work. A sample of eligible renters and prospective buyers should see the actual travel time and total monthly cost of moving. Ask whether they would live there, what they would lose, and which alternatives they prefer. Verify the answers against observed prices and behaviour where possible. Enrolment numbers collected before people see the transport and payment terms are weak evidence of effective demand. The procurement should be released only when the land, environmental, and service conditions are contract-ready. Bid evaluation should test experience delivering occupied serviced homes, financial capacity across the bidder's entire contract portfolio, track record on safety and defects, and credible prices for standardised quality. The winning bid should be published with the reasons for award and the contract's future adjustment formula. During construction, an independent certifier should check foundations, structural stages, utilities, accessibility, and safety before payment. The authority must retain documented rights to suspend, correct, and replace works when tests fail. Handover also requires an operating and reserve-funding agreement.

Finally, sign a handover and operations agreement. It names the title or lease instrument for the household, the utility account, operator for shared spaces, defects procedure, monthly service charges, water and sewer obligations, expected tax or service subsidy, and a resident representation mechanism. Inspection should occur before occupation, again after twelve months, and thereafter on a risk-based schedule. The programme must budget for the first years of operation and reserve funds for long-lived assets. An accountable delivery lead should coordinate those obligations while remaining separate from verification.

Delivery readiness assessment

STOP: Site rights are disputed; safety or resettlement duties lack lawful remedy; trunk services have neither confirmed capacity nor funded delivery; procurement papers cannot assign material risks; or intended households have no fair route to claim a home.

PREPARE: The constraint is solvable with specified engineering, finance, consent, or legal work. Name the owner, due date, evidence, budget, and the official who can lift the hold.

TENDER: All critical preconditions have been independently documented, households have tested the product, and the full public value and lifecycle obligations are disclosed.

PAY: An independent inspector verifies the relevant milestone, safeguards are met, prior defects are corrected, and the contractor's security remains valid.

COUNT: A household has accepted lawful possession; the home is safe and connected; basic services work; and its actual cost and access to employment meet the programme's stated rules.

This is a proposed management tool. Its gates can be built into a city–state–national compact or an MDB project implementation manual.

A delivery officer needs authority to resolve problems and stop non-ready projects, with independent checks on the solution. The role should carry a published performance agreement, a small multidisciplinary team, and access to the finance ministry. It must remain accountable to an independent inspection and grievance process. Concentrating all decisions in one office without outside checks would concentrate discretion and weaken independent verification. The governing discipline is a narrow common mission, delegated authority to solve problems, and separate authority to verify the solution. Public performance reporting should follow each cohort through to sustained safe occupation.

Measuring delivery performance

Track cohorts from planning through commissioning and continued occupation rather than reporting completions alone. The public dashboard should show each annual cohort as a funnel: planned, site-cleared, tendered, started, physically complete, independently commissioned, legally allotted, occupied after 90 days, and safely occupied after twelve months. Display the number and share at each stage, median time between stages, reasons for delay, and the agency responsible. Show results separately for rental and ownership units and for households of different income groups, gender, disability, and recognised forms of tenancy where lawful data collection permits. Report relocation, grievance outcomes, and homes lost from existing low-cost stock alongside homes added. The main efficiency indicator can be public cost per verified safe, accessible household-year of housing. The numerator includes grants, discounted land, financed infrastructure, expected guarantees, administration, and maintenance subsidies; the denominator counts occupied, safe, serviced years attributable to intended low-income households over a stated period. A credible calculation uses a discount rate and tests changes in occupancy, asset life, and public upkeep. No single number can capture loss of community ties or exclusion, so publish access-to-jobs, travel time, household housing-cost burden, safety defects, and tenure security separately. A low initial cost per unit may coexist with a high cost per year of safe occupancy. Lenders and development partners should use the same distinction when assessing disbursement and progress.

Evaluate finance and delivery through verified occupation, quality, services, and costs that persist after the original team leaves. A programme with slow initial disbursement and a credible serviced pipeline may outperform one with high spending and unusable buildings. Performance reporting should distinguish these outcomes rather than reward disbursement alone. An evaluator should ask which changes came from the reform rather than from the economy, weather, or a new mayor. Some interventions permit phased or random rollouts: digitised inspections across comparable districts, independent engineering checks across lots, or alternative resident-notification methods across sites. Others, such as a single metropolitan

water-main agreement, need careful before-and-after process tracing and comparison with similar projects. Publish negative results and implementation costs. Learning requires knowing when an attractive reform has failed. Delivery requires enforceable commitments among agencies with different responsibilities. The proposed arrangements identify who can prevent an unready project from starting, resolve service connections, correct defects, and respond to residents' complaints. Their performance should be assessed through occupation, safety, service reliability, and costs over time, including after the original programme team has left. Physical completion and safe use should remain separate entries in that assessment.

Report physical completion separately from safe occupancy in a functioning neighbourhood. Both measures are needed to identify the remaining work and guide later investment. Fieldwork before procurement and handover should identify the remaining conditions between those two milestones.

Fieldwork before procurement and handover

Verify readiness, certification, household access, and operating responsibility before accepting handover. Review title, permits, utility commitments, designs, bills of quantities, financing, allocation readiness, and contractor performance. Reconcile certified quantities with paid invoices and independently inspected work. Include cancelled tenders and contract amendments. The South African audits and the Indonesian verification evidence above make checking delivered work a distinct task from checking whether procurement procedures were followed. The procuring agency, engineers, utilities, land authority, finance team, and future operator should sign a readiness register. Record each unresolved condition, its owner, and the consequence for tendering or mobilisation. Residents should review access, allocation, and complaint arrangements before the contractor begins work. Use version-controlled designs and dated inspection records to connect variations to cost and schedule. Flag unusual bids or ownership links for investigation, without treating a flag as proof of misconduct. Release payments against independently verified milestones. Before occupation, test services, fire access, accessibility, as-built records, defect remedies, and management funding. Track days from physical completion to safe occupancy and explain the longest delays. Chapter 16 examines the management agreement that must work once residents move in.

Resident knowledge should complement independent engineering and enforceable contract management. Independent measurements, credible audits, and consequences for non-compliance connect payment to delivery. Community participation cannot replace engineering checks, and paperwork cannot establish service readiness. Copying an audit rate or concentrating construction authority will not resolve disputed utility handoffs or absent contractual remedies. When project volume increases, test the supply of independent engineers, auditor conflicts and the authority to stop payment without freezing all delivery.

Sources and evidence notes

Published evidence and official data are distinguished from analytical proposals and clearly labelled illustrative calculations. Unless otherwise stated, figures reflect the dates in the cited sources.

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16. Neighbourhood management after occupation

Treat handover as the start of an enforceable service arrangement with funded maintenance and resident remedies. Assign asset ownership and operating duties, test the affordability of charges, and provide for major renewal. Resident organisations need authority and technical support proportionate to their responsibilities. The first decision is whether each asset has an institution able to accept and maintain it.

Handover is incomplete when ownership, operating duties, and repair funding remain unresolved. The immediate decision is who can accept each asset and sustain its service. Chapter 9 established the cost of losing usable housing; this chapter turns that problem into operating duties, accounts, and remedies. The first test is whether the service and renewal costs are affordable and funded.

1 Specify what must function after handover

Costs after handover

Set service charges from the household's full cost and the actual service obligation. The handover agreement must reconcile three accounts: what residents pay, what the estate operator must spend, and what the municipality or utility must fund. Record each transfer once and show any remaining gap. Test charges against irregular income and foreseeable major repairs, as well as the current monthly bill. An association that can collect routine dues may still lack authority or reserves to replace a lift. The management agreement needs an affordable contribution schedule and a funded response when essential costs exceed it. All-in private cost = rent or mortgage + energy + water and sanitation + compulsory dues + expected repairs + incremental travel. Report public operating grants, road and utility renewal, and administration separately in the fiscal account. An economic appraisal should count the underlying resource costs and relevant lost earnings, excluding transfers between households, providers, and government and avoiding double counting. A lower purchase price can coexist with a larger lifetime household burden. Payment timing can create hardship even when the lifetime cost appears reasonable.

For a low-income household, the terms of payment can be as important as the present value. Electricity may be prepaid, a pump may fail unexpectedly, and income may arrive daily. A monthly fee that is modest against annual earnings can cause default if collection falls on the wrong date. A credible operating system therefore needs predictable charges, flexibility for irregular earnings, an emergency process and a source of subsidy for the residents whom a government explicitly sought to house. The choice of assets determines maintenance requirements. A four-storey walk-up with simple gravity-fed water can impose a different maintenance bill from a high-rise that depends on lifts, pressure pumps, fire equipment and constant electricity. Concrete quality, waterproofing, corrosion resistance, roof slope, tree roots, access for sludge removal, and the availability of replacement parts set future costs at the design stage. A procurement process that rewards the lowest initial bid without pricing renewal can lock poor households into a building that needs more money than they possess. Ahmedabad's

experience shows why assigning those charges to a resident association does not establish capacity to collect or manage them.

2 Match neighbourhood management to local conditions

Resident associations after relocation in Ahmedabad

Resident associations can inherit maintenance duties before services and municipal obligations are resolved. Ahmedabad provides a qualitative study of the handover problem. Under the Basic Services to the Urban Poor programme, families displaced by infrastructure and urban development were allocated flats at resettlement sites. The rules called for resident welfare associations to take part through the life of the project and assume operating costs. In practice, the city sought to organize many of these associations after construction and allocation. The Mahila Housing SEWA Trust was asked to form associations in eight sites, ranging in size from 192 flats at Kesar-e-Hind Mill to 2,464 at Sadbhavna Nagar, Vatwa. Researchers Darshini Mahadevia, Neha Bhatia and Bijal Bhatt documented the association arrangements and residents' experiences. The residents had not chosen their neighbors. Some had been moved far from familiar networks; an allocation process had left empty flats scattered among occupied ones. Residents had grievances about water, theft and repairs. The municipality expected the new associations to collect fees, run water pumps, maintain lights, sweep common areas and pay for minor repairs. The study describes residents' reluctance to pay fees while municipal obligations remained unresolved. A legally demanding organisational form may then exceed residents' preparation and actual influence over decisions.

Formal association requirements should be matched by practical capacity and participation before design and allocation are fixed. The water network influenced association boundaries. An association was formed around the underground water tank it served. Sadbhavna Nagar had six tanks, and therefore six associations. The water network gave the organization a sensible service boundary. Yet the residents had been assembled by an administrative allotment rather than by ties of mutual obligation. The service boundary did not resolve the difficulties of organising households after administrative relocation. The city had already paid operating costs and wanted to withdraw. Residents were being asked for fees while repairs remained undone; many held an allotment letter without secure evidence of final ownership. The researchers found that creating associations after relocation shifted obligations to households least able to discharge them. Tenants were excluded from committee membership under the applicable cooperative form. The city could comply formally with a participation requirement while depriving residents of a credible voice at the stage when choices about design, location and allocation were still open. 1 Qualitative evidence identifies these mechanisms without establishing their prevalence in every estate.

Association performance should be judged through working services, reconciled accounts, repairs, and resident retention. This is a case study based on qualitative observation and interviews across eight sites. It does not estimate an experimental effect. It documents the sequence of decisions and the limits on each actor's authority. The municipal agency could stop supporting operations; the NGO lacked authority to repair defects or change

allocations; the residents could withhold money but could not compel the city to finish its work; the banks and cooperatives demanded formal compliance. The associations received operating duties while their authority, funding, and residents' ownership documentation remained unresolved. An improved sequence would have begun before displacement. Households could have helped choose the location of shared facilities, the size of the service area, the rules for tenants and the first committee. The municipality would have identified which pipes and drains it retained, kept a defects reserve under its control, and offered transitional operating support declining only as the association proved capable of collection and oversight. Registration would be one milestone; working water, reconciled cash, repaired defects and retained vulnerable households would determine success. A statutory transfer timetable is therefore insufficient without preparation and funding.

Policies for transferring maintenance should be tested against the recipient organisation's actual readiness. In April 2024, the Delhi Development Authority adopted a policy for certain new sale-scheme estates that envisaged transferring common-area maintenance and the remaining reserve to resident associations within a year of possession or earlier association formation. Previous arrangements had the agency maintain some common facilities for longer against an up-front contribution. This report concerns DDA sale-scheme flats, rather than the Ahmedabad resettlement programme, and offers no outcome evaluation. It shows that the timing and funding of the handover are still policy choices that deserve independent scrutiny. Baan Mankong offers a useful institutional contrast to these eight Ahmedabad sites. Organised groups help negotiate land and spending before occupation, with public finance and technical support. Ahmedabad's associations were often formed late and inherited unclear service boundaries. This is not a matched impact evaluation and does not prove that cooperatives always govern fairly. It supports preparing and funding an organisation before assigning it maintenance duties. [112,111] Predominantly rented settlements require a different agreement because owners, occupants, and service payers may be different people.

Infrastructure, land claims, and rents in Nairobi

Mukuru shows why infrastructure decisions must account for separate ownership, tenancy, and service-control interests. Across its three principal settlements, roughly 94 per cent of residents are tenants. Many rent one small room from a structure owner who lacks title to the land. The landlord controls the room, the tenant pays for the water, an informal supplier may control the pipe, and a public company controls part of the main network. The African Cities Research Consortium describes residents paying 400–800 per cent more per cubic metre for unsafe water than official county rates. These figures are a reported local price comparison, not a citywide household expenditure survey. Water supply involves several formal and informal arrangements. Thin hoses cross drains; a supplier manages the connection; a tenant buys small quantities at a kiosk; a structure owner can charge for water, shared sanitation or access. Some of these arrangements deliver a service where the public network fails. Others sustain market power, contamination, and coercion. Changing the pipe means changing the income of the people who control it. Changes in income and control can generate resistance; assess these interests alongside engineering feasibility. The price and welfare effects of a new connection need direct measurement.

Completed infrastructure should be assessed through remaining service gaps and the prices residents actually pay. The account reports completed infrastructure and remaining service gaps. In a June 2025 account, researchers reported about three kilometres of lateral sewers connecting 1,248 plots and benefiting about 12,500 families; they estimated that around 80 per cent of residents still lacked plot-level sewer-connected toilets. The estimates have different denominators and should not be treated as a single calculated coverage series. Planned extensions required additional pipes, toilet interfaces, finance, and coordination among Nairobi City County, the water company, community organizations and other agencies. A pipe laid through a landlord-controlled plot creates a bargaining problem. Suppose public investment cuts the monthly cost of safe water by KES 500 for a tenant. If the landlord raises rent by KES 400, most of the benefit has moved to the structure owner. If a connection fee or mandatory improvement contribution excludes the poorest renters, the service may improve for those who stay while excluding those displaced. Both outcomes are hypotheses to test on the ground; the price effect cannot be assumed from the existence of a new connection. A household and plot ledger can establish who controls each obligation and benefit.

Record land, structure, occupancy, payment, and connection rights separately before assigning service duties. A contract should specify who pays installation, who maintains the lateral pipe, who empties or replaces a sanitation facility, how the water tariff reaches tenants, and what happens when a landlord blocks access. Municipalities can license and regulate kiosks where direct connections remain impractical. They can set standards and fair resale margins while offering a lawful path for informal suppliers to become service agents rather than treating every intermediary as a criminal. Mukuru also illustrates coordination among residents and service agencies. Community mapping gave tenants an organized voice and an evidence base. The Special Planning Area created a formal vehicle to bring agencies together, while community groups and NGOs supplied local knowledge and pressure to deliver. This coalition can still fail: landowners may contest rights, commercial landlords may capture improvements, utility budgets may falter, and residents may have no enforceable claim when prices rise. A sewer connection needs a service agreement and tenant protection if its welfare gains are to persist. [10,11] Backyard housing presents a related challenge when the original plot's service account covers additional residents.

Backyard rental housing and municipal costs

Backyard rental policy needs to recognise additional service demand without destroying affordable rooms and owner income. Subsidized houses have given many low-income families a valuable asset, even where jobs are distant and building quality varies. A homeowner may let an informal backyard room to help pay for food, school fees or municipal bills. Researchers Andreas Scheba and Ivan Turok describe a rental economy that has become increasingly commercial, alongside older forms of household support. In one cited Cape Town study, backyard tenants paid rents of R200–R250 a month and about R50 toward service charges at the time of fieldwork. Those are historical observations, not 2026 market rents. The income can improve a poor owner's ability to retain the main home. The added population also increases demand for water, toilets, refuse removal and electricity. A city that sizes its networks around the original subsidized dwelling may face more households than its records show. Charging the main

titleholder for every service cost can be politically difficult and may lead to disconnections that harm tenants first. Exempting the entire property from charges can leave the utility without revenue for expansion. Trying to ban backyard accommodation may destroy affordable rooms and remove the owner's small source of income. A feasible agreement can combine recognition, safety, connection charges, and tenant monitoring.

There is a feasible agreement: recognize safe additional rooms, permit them in suitable serviced areas, require basic fire and sanitation standards, and state the cost of additional demand. Municipalities can offer upgrades and simpler approvals conditional on a service agreement and minimum tenant rights. A utility can place a lawful meter or submeter and provide an affordable lifeline tariff with clear resale rules. Lower-income landlords may need staged credit and technical advice; tenants need recourse when deposits, water charges or threatened eviction become sources of unaffordable charges or coercion. Developers, plumbers and local builders gain work from incremental upgrading; incumbent homeowners may resist if added density brings congestion or is associated with unwanted residents. An established homeowner may have clearer participation rights than a tenant in a backyard room. Zoning can protect property values for one group while rationing access for another. Include small landlords, tenant groups, builders, utilities, and employers when agreeing implementation responsibilities. Monitor whether new connections improve service and security, raise rents, or expose households to unofficial charges. Service quality should then be measured through residents' daily experience rather than installations alone.

Monitoring service quality

Convenience, reliability, privacy, and durability should be measured alongside engineering outputs. In Tangiers, Morocco, researchers randomly offered households an easier way to acquire a private water connection on interest-free credit. Take-up was 69 per cent. Families already had access to public taps, often nearby. The new connections raised water use and wellbeing, yet the experiment found no improvement in waterborne illness, consistent with the lack of a change in drinking-water quality. The measured value came substantially through time, convenience and household satisfaction. The Tangiers experiment is about the adoption of private connections, rather than housing-estate maintenance. It found benefits from a service improvement without a corresponding change in the measured health indicator. A lift that works lets an older resident leave home; working streetlights can support safer journeys before dawn; an emptied drain can prevent missed days of paid work. Benefits should be measured against what households actually experience and value, with the cost and durability of the improvement tested alongside them. Work orders and installed assets need to be checked against the service they provide.

Count functioning services and resolved access disputes rather than equipment installed alone. Count the hours they function where pedestrians walk, the time before a reported fault is fixed, and the share of women and residents with disabilities who feel able to use the route. Counting toilets built says little about whether they remain clean, open, safe and affordable. A governing body may prefer a visible new facility; residents may prefer that an existing one work reliably every day. Some goods require choices that divide residents. A small

play area, a community clinic, a place for street vending, a prayer room, parking and waste containers compete for scarce space. Established owners may want walls, gates and restrictions; poorer renters may need access for customers, domestic workers and deliveries. The rules for a shared courtyard decide whether the neighborhood remains a place of work and social exchange or becomes an enclave of private titleholders. Neighbourhood management must address access and disputes alongside physical maintenance. Neighbourhood management also affects the work, care, and social networks that residents depend on.

Social networks and economic opportunity

Housing management should preserve access to livelihoods and networks as well as building condition. Relocation may save a family from flooding or give it a sound roof, while imposing longer journeys, lost customers, weaker childcare arrangements and unfamiliar schools. The Ahmedabad study records the loss of neighborhood ties after relocation and difficulties in forming functioning associations. Neighborhood inclusion starts well before handover, with the decision about where and with whom people live. For neighbourhood management, the implication is to check whether location and service continuity support residents over time. Track schools, journeys, care networks, and tenant retention alongside repairs; building condition alone does not measure neighbourhood opportunity. [5] Rules on representation and service access need to address exclusion within the neighbourhood.

Caste, race, religion, migration status, gender and class are part of this mechanism. Resident committees can ration access to common spaces, veto tenants, exclude domestic workers from entrances and treat migrants as temporary guests. In India, cooperative rules and local social hierarchies can make participation unequal. In South Africa, apartheid's spatial legacy keeps many low-income households distant from economic centers. A formal apartment title alone cannot undo these restrictions. Inclusion must be reflected in by-laws, enforcement, an accessible grievance system, and the data collected about service outcomes, with continuing checks that the rules work in practice. Maintaining these commitments requires explicit provision for future asset replacement.

3 Fund renewal and enforce service responsibilities

Funding asset renewal

Replacement reserves should be based on the building's costs and financing rather than a percentage copied from another system. Pumps fail, paint fades, roofs leak, wiring deteriorates and lifts reach the end of their economic life. A renewal reserve spreads those future obligations across the years during which residents enjoy the assets. The alternative is a sudden special assessment, a plea to the municipality or a decision to leave the equipment broken. Such crises tend to harm the lowest-income residents first, because wealthier households can pay, move, or arrange private substitutes. Singapore is a useful institutional comparison. Its town councils manage common property in public housing estates and keep separate operating, ordinary sinking and lift replacement funds, subject to legal rules and government oversight. Under the Town Councils (Minimum Contributions to Sinking Funds) Financial Rules 2017,

minimum quarterly allocations include 26 per cent of relevant service and conservancy charges and grants into the ordinary sinking fund, and 14 per cent into the lift replacement fund, subject to the rule's specified calculations. These figures describe a financing discipline backed by law and grants. They provide no evidence that 40 per cent is the right percentage for an Indian or African estate. Audited accounts and external oversight can help distinguish current services from future renewal obligations.

Reserve rules need credible collection, accounting, and verification before their design can support a claim about results. Singapore also has resources, a high degree of public control and municipal institutions unlike those of most low-income urban districts. Copying a reserve percentage without copying the revenue, accountability and enforcement arrangements would achieve little. Thailand's Baan Mankong programme offers a different path. Community cooperatives pool savings, negotiate secure land arrangements and manage infrastructure and housing investments with technical support. Its published design includes an infrastructure grant per household, a community-management allowance of five per cent of the infrastructure subsidy up to a ceiling, and concessional loans to cooperatives. Borrowing groups are required to maintain savings equal to at least ten per cent of the loan. These are programme design rules and reported financing terms, rather than causal impact estimates. Collective ownership adds the need for representative authority, savings, and technical advice.

Collective ownership can strengthen bargaining power when a community has clear rights, a credible savings culture and access to engineering advice. It can also burden poorer members, exclude renters from decisions, or empower a few cooperative leaders to allocate scarce plots. Each cooperative needs open accounts, transparent entry rules, conflict resolution, and independent oversight. Baan Mankong's design budgets for social organisation and land negotiation before construction; this is a design feature, not an isolated estimate of its contribution to programme outcomes. A worked estate budget shows how these obligations translate into household charges and public support.

Maintenance finance for a 1,000-home estate

A maintenance budget should define the estate and municipal cost boundary before calculating household charges. Consider a hypothetical estate of 1,000 low-income apartments with pumps, common lighting, drains, security, waste collection and shared roofs. The figures below are illustrative assumptions in Kenyan shillings, chosen to expose the financing mechanics. They describe no actual Kenyan project. The municipal authority is responsible for streets and trunk networks; the residents' association is responsible for the building's shared property. The cost boundary is fixed to prevent either side from billing the other for the same service. The example below separates routine operations and renewal requirements under stated assumptions.

Annual item	KES million	Who is accountable?
Staff, lighting, water pumps, cleaning, routine repairs and administration	23.4	Estate operator, audited by resident board
Accrual for future roofs, pumps, electrical renewal and major repairs	7.8	Protected renewal reserve; qualified trustee
Total recurring requirement	31.2	Residents + transparent, funded public support
Revenue at full payment: 1,000 × KES 2,600 × 12	31.2	Uniform-fee scenario; affordability untested
Revenue at 85% fee collection, same billed rate	26.52	Annual shortfall: KES 4.68 million
Targeted fees: 600 × KES 1,200 + 400 × KES 3,000 per month, at 95% collection	21.888	Annual shortfall: KES 9.312 million

Occupancy and collection assumptions determine whether the calculated charge covers the estate's actual costs. If every household paid, the annual cost would require KES 2,600 a month per home. At 85 per cent collection, that fee produces a KES 4.68 million annual shortfall; a uniform charge of about KES 3,059 would be required to cover the same cost if the payment rate remained 85 per cent. Higher charges may cause more arrears, so that arithmetic does not prove the revised tariff would work. A means-tested arrangement might instead charge 600 poorer households KES 1,200 monthly and the other 400 households KES 3,000. At an assumed 95 per cent collection rate, receipts equal KES 21.888 million, leaving KES 9.312 million a year to fund. If the gap is expected to remain constant in real terms for 25 years, its present value at a five per cent real discount rate is approximately KES 131.2 million. This is an illustrative fiscal commitment with no escalation in real cost, no terminal value, and no allowance for unexpected failures. Inflation, construction defects or lower occupancy could increase it materially. Any remaining gap needs an identified source rather than an expectation of eventual payment.

Close the operating gap with an explicit financing arrangement while retaining essential access and safety features. Each has distributional effects. A municipal grant relies on future political consent; an endowment is exposed to investment and governance risk; commercial rent can be volatile; a property charge may fall partly on tenants; a central transfer may give municipalities weak incentives to manage costs. The financing instrument must be chosen after assessing these risks. One should also test whether the original design caused the gap. Could a walk-up replace a high-rise without sacrificing access to jobs or safe density? Would better waterproofing reduce future roof repairs? Does the public operator have a cheaper, reliable

bulk electricity contract? Can a local sanitation service be run safely at smaller scale? Lifetime engineering choices, financing terms and the location of the estate must be evaluated together. Value engineering that strips out access ramps, safe stairs, drainage or adequate structural design simply postpones costs and shifts them to vulnerable residents. The resulting recurring subsidy belongs in the public appraisal alongside capital support.

Under the stated assumptions, this hypothetical estate requires KES 9.312 million a year in additional support for common services. The budget decision should compare that recurring obligation with other housing spending and identify a payer over the full operating period. Agreeing that subsidy requires a clear account of who controls services after occupation.

Interests and responsibilities after occupation

Post-occupation management should be based on verified authority, accounts, and operating capacity. After occupation, the center of power moves toward associations, utility agents, landlords, maintenance contractors and the municipal treasury. Reliable services and shortages create different financial interests. The water vendor can charge a scarcity premium; the private tanker gains business from an unreliable pump; a contractor may hold exclusive access to spare parts; an estate chairperson may decide which arrears are forgiven and whose complaints are heard. Resident owners, landlords, and tenants may have different preferences over fees and repairs. Tenants can bear service failures while lacking a vote. A municipality may transfer upkeep without resolving approvals or funding. Verify these arrangements in contracts, accounts, and interviews before deciding whether to strengthen the association, appoint an operator, or retain municipal responsibility. Tariff and management changes must then be negotiated with those able to implement or obstruct them.

Actor and gain	Veto or leverage	A possible agreement
Owners and small landlords: asset values, rent, future resale	Block dues, association elections, planning appeals	Visible service standard, graduated fees, audited reserve
Tenants and informal workers: water, safety, access to work	Limited formal voice; rent and eviction exposure	Tenant representation, fee disclosure, protection against pass-through abuse
Utility and municipality: collections, public health, fiscal space	Connection permits, budgets, tariff decisions	Asset boundary, performance payments, shared reporting
Informal vendors and local brokers: scarcity rents, customer access	Control of taps, pipes, local enforcement, information	Regulated resale, transition contracts, competitive entry

Actor and gain	Veto or leverage	A possible agreement
Developers and maintenance firms: contracts, warranty and service income	Defect disputes, captive parts, delayed commissioning	Open parts, warranty escrow, independent condition audit
Women, migrants, excluded caste/race groups, disabled residents: safe access	Often excluded from committees and grievance hearings	Accessible services, representation, appeal with enforceable remedies

A service agreement needs enforceable remedies for payment, defects, and appeals rather than promises of future coordination. A municipality may announce universal household metering without identifying a landlord who resells water to twenty rooms. An engineer may recommend a fee sufficient to replace every pump without explaining how households with irregular income will pay. A digital bill can make arrears more legible while allowing disconnection at greater scale. Each proposal needs affordability, rights, and implementation checks alongside its technical assessment. Reform coalitions have to be built around tangible gains. Small landlords may support legal permits if safer rooms attract paying tenants. A water company may accept a lower tariff for a larger volume of metered sales. Tenants will support upgrades if they can see the charge, keep access, and challenge eviction or abuse. A resident committee will collect dues more readily after the city has completed overdue repairs. Each agreement must specify what happens if the other party fails to perform. Promises of future coordination have little value unless an actor with authority can withhold a payment, correct a defect or hear an appeal. Those remedies must remain usable by tenants and residents with disabilities.

Management must protect tenants and residents with disabilities from being priced out or trapped by service failure. Digital management systems should make these duties more accountable without exposing residents to new harms.

Digital management systems and accountability

Metering and administrative coverage do not establish reliable service or eliminate losses. The Water Services Regulatory Board's Impact 18 report, released in June 2026 for FY2024/25, records non-revenue water of 48 per cent and a metering ratio of 97 per cent. These are national regulated-sector indicators, rather than figures for Mukuru or Nairobi alone. Their coexistence shows that metered customer connections do not guarantee a well-managed distribution network. Bulk water balances, pressure zones, leakage repair, billing integrity and accountable operators remain essential. For housing estates, the high-value digital applications are often modest. A geocoded asset register specifies who owns and maintains each valve, pump, stair, light and drain. A public work-order log records when a defect was reported, whether it was fixed, the cost, and who signed off. Transparent billing and a ring-fenced renewal account reduce opportunities to invent arrears or spend tomorrow's reserve today. Leak sensors can prioritize crews. Satellites and street-level imagery can identify new buildings and flood exposures, subject

to validation. Resident feedback should remain available through low-tech routes for people without smartphones or formal identification. Occupancy and payment information also need controls against harmful secondary use.

Management data requires lawful purpose limits, access controls, testing, and a route to challenge errors. A private supplier could gain monopoly control of the metering platform. A city could raise property assessments as soon as it maps informal improvements, before clarifying tenants' rights. A risk score could direct service disconnections toward poor neighborhoods whose arrears are easy to observe while ignoring losses in areas poorly covered by reliable billing and loss data. Training data may contain biased repair responses, causing an algorithm to recommend more attention where complaints have historically been heard. The proposed safeguards include law, access controls, independent testing, community checks and a route to challenge a wrong record. Phased evaluation can test whether the system improves actual repairs and treatment of residents.

Mechanism to change	Measure the gain	Capture, power and risk
Asset inventory and spatial register	Share of assets with verified owner, condition and repair schedule	City controls visibility; tenants and informal residents need safeguards
Leak detection and pump alerts	Hours of service restored per KES spent; water volume billed	Utility or landlord may retain savings; protect household prices
Billing and arrears prediction	Net collection gain at equal service and fair appeals	Operator gains disconnection power; audit bias and hardship cases
Open work orders, procurement and complaints	Median repair days and repeat faults; costs against benchmarks	Contractors lose information rents; avoid manipulated close-out data
Rental and occupancy mapping	Tenure stability, rent changes, household access	Can expose tenants to eviction, taxation or targeted exclusion

Comparable estates could phase in public work-order systems, independent inspections or alternative payment schedules. Track service uptime, household spending, arrears, participation, eviction and complaints for at least two rainy seasons. Where random rollout is inappropriate, compare credible matched locations and trace the exact decisions taken after an alert. Report implementation costs. Validate recorded repair closure through inspections and resident follow-up. The management agreement should incorporate the duties that those tests verify.

Neighbourhood management agreement

Choose the operating arrangement for the tenure and service setting, with separate tenant representation where needed. Cost routine management, emergency repairs and replacement separately, with targeted help for households unable to contribute. The operator must maintain accounts and repairs; the municipality must honour public-service boundaries; residents must have appeal rights. Track service uptime, repair completion, reserve adequacy, household fees, tenant representation and retention after upgrading. A government can require an agreement at the point of first occupation and revisit it at year one, year three and every major renewal cycle. Its purpose is to identify future obligations early enough to revise the project and its financing. It is a proposed governance tool, not a finding from an evaluation. Five conditions should be binding. The first binding condition is clear ownership and responsibility for every essential asset.

1. Name the owner of every shared asset

Record each asset's owner, operator, acceptance condition, and duty before handover. Map the boundary between the building association, utility, developer and municipality. Record design life, warranty, spare parts, condition, and a responsible named official or legal entity. A gap in ownership is a reason to withhold the handover of that asset. The second condition is a funded operating and renewal forecast.

2. Fund life-cycle costs and test affordability

Stress-test long-term maintenance finance and show expected arrears and the support required for essential services. Set protected reserves and published audited accounts. Show the full private monthly cost separately for the bottom two income quintiles, informal workers, persons with disabilities, female-headed households and renters. If the scheme requires fees beyond the target household's means, redesign the product or provide a funded, transparent subsidy. Show expected arrears explicitly and identify the funding needed to maintain essential services. The third condition is representation and access to remedies for every occupied unit.

3. Give tenants and weaker residents a voice with remedies

Every occupied unit should have a path to representation and complaint, including renters and residents whose tenure is contested. Protect residents from retaliatory eviction after reporting a defect. Make meetings accessible, decisions public, and management accounts reviewable. Establish a separate appeal route for charges, discriminatory use rules, service cutoff and forced relocation. Ensure that women and residents with disabilities can inspect facilities in safe conditions before they are declared fit for use. The fourth condition is a measurable service contract with independent checks.

4. Contract for service, performance and renewal

Tender maintenance against observable service standards and publish remedies for missed performance. Identify responsibilities for emergency repair. Pay against inspected results; keep warranty security and competitive access to parts. Require the municipality and

utility to certify off-site capacity and take legal custody of transferred works. Publish remedial action when a standard is missed. The fifth condition is follow-up of the households and assets after handover.

5. Measure whether beneficiaries remain

Follow resident cohorts and stop transfers whose essential safety or operating obligations lack lawful funding and remedies. Distinguish voluntary moves from displacement. A rising property value is valuable information, yet its meaning depends on who holds the title, who pays more rent and whether original low-income beneficiaries still enjoy the neighborhood. An upgrading scheme that pushes out tenants can report better pipes while delivering weaker inclusion. GO when shared assets have legal owners, an affordable and funded life-cycle plan, inclusive management rules, reliable service tests and a credible right of appeal. REPAIR when the deficit or unclear boundary can be corrected through a time-bound, financed agreement. STOP a proposed transfer when critical safety, utility ownership, relocation obligations or structural renewal lacks a lawful and funded remedy. The handover review should resolve the boundary and emergency authority for each service.

Handover should settle asset responsibility, acceptance, emergency powers, and funding before expanding collection sanctions. Begin with existing drawings, contracts, accounts, and a joint site inspection; commission missing specialist work according to risk. The housing agency should assign a responsible officer and budget to unresolved items. Test water, drainage, lifts, fire access, and complaint routes before accepting the relevant works. Keep construction retention and warranty remedies enforceable. For occupied estates, apply Chapter 24's repair and financial triage before attempting a comprehensive renewal plan. This tool needs one lead institution with a defined authority to call meetings and enforce commitments. It also needs an independent inspector and an avenue for residents to challenge that lead institution. In a metropolitan system, a national housing ministry can condition new grants on municipal service acceptance and renewal funding; a municipality can offer matching support for low-income associations that meet disclosure and inclusion standards; a regulator can supervise service tariffs; and an MDB can finance capital works alongside time-bound improvements in operating discipline. The reform sequence should begin with verified service and trust, proceed to affordable dues and reserve accounts, and only then expand sanctions for persistent nonpayment. Long-term review should then test whether the arrangement preserves affordable, connected occupation.

Assessing long-term outcomes

Deferred maintenance should be evaluated through lost years of safe affordable use, not only the eventual repair bill. When service charges, reserves, and municipal responsibilities are unclear, repairs accumulate and residents face larger emergency bills. Later public grants may then pay for failures that could have been addressed through routine management. Effective management requires understandable bills, a process for correcting errors, accountable resident representation, clear ownership of shared assets, and funded replacement schedules. These arrangements determine whether homes remain safe and affordable. Assess neighbourhood management by the years of safe, connected, affordable occupancy it supports for

intended residents, relative to its full public cost. Fieldwork can identify the operating and affordability gaps before arrears make the arrangement unworkable.

Prepare the local decision

Inspect assets, accounts, representation, service performance, and household charges together before confirming management readiness. Assemble the asset register, warranties, ownership boundaries, resident roster, charges, arrears, supplier contracts, and a replacement schedule. Check actual service quality with residents, including tenants without association votes. Use the Ahmedabad association evidence above to test whether a committee has both the authority and resources needed for duties assigned to it. Hold a handover review with the municipality, utility, developer, operator, resident owners, and tenant representatives. Assign emergency authority and spending limits for failures affecting water, drainage, lifts, or fire safety. Identify which obligations remain with the developer and which pass to an association; obtain evidence that the receiving body can fund them. Start with a work-order and cash-reconciliation system that residents can use through assisted channels. Check completed repairs with a sample of affected households. Compare service contracts on actual downtime, cost, and repeat failures, rather than software features. Reforecast the reserve annually from condition and replacement prices. If affordable charges cannot meet essential costs, obtain an explicit subsidy or redesign the service arrangement before arrears make it unusable. Chapter 17 returns to planning implementation and the controls needed for safe additional housing.

Delegated neighbourhood duties require matching authority, affordable revenue, and technical support. Organisation before handover, clear service boundaries, credible dues and assistance, and external expertise make resident management feasible. Mandatory bylaws or savings percentages cannot repair fragmented communities, tenant exclusion, or unclear municipal obligations. The management arrangement must remain workable when responsibilities exceed what volunteers can provide. For 50,000 households, provide professional support and independent oversight without making every repair depend on a central office.

Sources and evidence notes

Evidence and scope. Research findings and official indicators are cited below. Illustrative figures, hypothetical households, and proposed tools are identified as such. Programme descriptions and financing rules do not, by themselves, establish effects on residents. Source dates and reporting periods are stated in the references.

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3. Scheba, A. and I. Turok (2020). “Informal rental housing in the South: dynamic but neglected.” Environment and Urbanization 32(1). Reviews backyard rental markets and associated municipal services, with historical field observations. [source]

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6. Singapore Statutes Online, Town Councils (Minimum Contributions to Sinking Funds) Financial Rules 2017 (SL 185/2017), current public legislation as at September 2026. Legal framework for ordinary sinking and lift replacement funds. [source]

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8. Community Organizations Development Institute (CODI), Thailand. “Baan Mankong: City-Wide Upgrading.” Programme description, grant and savings design; reported policy rules are not impact estimates. [source]

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11. International Development Research Centre (2025). “Data and activism drive dramatic improvements in struggling Nairobi neighbourhoods.” Reporting on infrastructure improvement and resident experiences in Mukuru, 31 January. [source]

12. Delhi Development Authority maintenance handover change, reported in Hindustan Times (13 April 2024), describing transfer of common-area management and residual funds to resident welfare associations. Contextual reporting; policy terms should be checked against scheme documentation. [source]

[111] Community Organizations Development Institute. Baan Mankong: City-Wide Upgrading. Programme design, community finance and implementation responsibilities; not a randomised impact evaluation. [source]

[112] Mahadevia, D., Bhatia, N. and Bhatt, B. (2016). Decentralized governance or passing the buck: the case of resident welfare associations at resettlement sites,

Ahmedabad, India. Environment and Urbanization 28(1). Qualitative study of eight sites. [source]

17. Implementing planning reform

Planning reform succeeds when lawful development rights become safe homes that the intended households can use. Test project economics and services alongside approval times. Retain independent safety verification and accessible remedies, then follow applications through construction and occupation to find the next constraint. A construction failure shows why formal permission cannot serve as the final assurance.

The South C collapse illustrates why a permit is not a substitute for verified construction safety. On 2 January 2026 a multi-storey building under construction in Nairobi's South C district collapsed before dawn. The National Construction Authority reported that it had registered the project in November 2023, after receiving county approvals, an environmental licence, and a geotechnical report. Four quality-assurance inspections had been recorded as accepted; the last had observed work at the third floor. After the collapse, investigators found that construction had added four floors beyond the approved structural design. Two bodies had been recovered by 7 January. The agency had yet to determine the precise technical cause. The county separately reported that it had intervened at the site in May, July, and December 2025 over breaches of building rules.[1][2] The South C case illustrates the need for continuing oversight after a permit is issued. Approval records, the structure actually built, inspections, and enforcement must remain connected. This single case does not establish the causes of other building failures. Planning systems can also fail by preventing safe feasible homes from being built.

Planning reform should address unsafe construction and unnecessary supply restrictions through distinct controls. On some plots a sound, simple rental room may be blocked by a density rule, a prescribed setback, an uncertain title, or an approval process designed for a much larger developer. Cities ration legal homes while struggling to regulate the homes that appear outside the rules. Planning reform should assess both restricted legal supply and unsafe construction. Before construction, public agencies determine lawful uses, development locations, safety evidence, objection procedures, and enforcement authority. These decisions affect land prices, small-developer entry costs, tenant safety, and infrastructure finance. Planning reform must coordinate them to expand safe housing near employment. Development permission creates value that makes both its allocation and its conversion into housing consequential.

1 Link planning permission to feasible development

Development permission and land value

A development right changes the options and value attached to a parcel, but does not guarantee construction. Its market price reflects what a buyer believes can be done there. If a city permits two homes where it previously allowed one, the owner receives a new option. If the city allows an extra floor, removes parking requirements, or permits rentals on a residential plot, it changes the plot's earning power. The option can raise the selling price of land immediately, even if no builder pours concrete for years. Changes to planning rules can therefore create gains

for landowners before construction. The revenue that new rooms can earn, discounted for vacancy and operating costs, gives them a possible capital value. The builder subtracts construction, finance, fees, risk, and a normal return. What remains is the maximum that can be paid for land and development rights. A rule that makes more floor space possible can enlarge that residual value. If enough plots become buildable, competing landlords may eventually lower rents. If only a few sites obtain discretionary exceptions, existing owners may receive much of the increase in land value. Evidence on density restrictions should therefore be connected to the full conversion process.

Modelled benefits from relaxing restrictions require local testing of the path from permission to occupation. The 2005 analysis by Alain Bertaud and Jan Brueckner examines the travel and welfare costs of building-height restrictions. Restrictions on building height shift households outward from employment centres, lengthening travel. Their stylised simulations for Bangalore implied welfare losses of roughly 1.5–4.5% of household income. These are model results, sensitive to assumptions about the city and transport; they illustrate a modelled relationship between height restrictions, travel, and household welfare.[4] Additional permitted floor area still requires title, services, construction, and occupation. A plot may be upzoned while its title remains disputed; a developer may receive permission while the sewer is full; a building may be completed while the power connection or occupancy certificate is withheld. Measure occupied homes that meet safety and accessibility requirements. Regulatory capacity should therefore be judged as a conversion process. The denominators at each stage are needed to establish who proceeds and who disappears from the records.

1 Claim Who holds defensible rights?

2 Use Which dwelling types are lawful?

3 Hazard Can people live there safely?

4 Service Will networks accept the load?

5 Capital Will anyone finance construction?

6 Build Is the approved design followed?

7 Live Is the home occupied and affordable?

EXHIBIT 17.1 Seven gates between a legal possibility and a liveable room. Each gate has a different owner; failure at one can erase the value produced at all earlier gates.

Count all eligible parcels and application attempts as well as approved projects. An administration may process most files promptly because poorer owners never apply, because applications are labelled incomplete for months, or because only simple, low-risk cases reach its system. A credible statistic must track the entire stock of parcels and the full sequence from first inquiry to occupancy. Report which applicants and households are absent from the administrative records. Zoning comparisons show why additional permissions and additional dwellings need separate measures.

Comparing zoning reforms

Zoning reform should be evaluated through actual supply responses rather than land-value increases alone. Two reforms show how much the answer depends on scale, expected demand, and the surrounding institutions. Chicago raised permitted density and cut parking requirements near transit through changes in 2013 and 2015. Yonah Freemark used differences in the timing and location of those changes to estimate their effect. Transaction prices rose on parcels that gained rights, including higher values for existing condominiums. Across five years he could detect no additional permitted dwellings attributable to the changes. The study found higher property values without a detectable increase in permitted dwellings during that period. [5] Auckland provides a comparison involving much broader geographic coverage.

Broad zoning reform requires separate measures of permits, completed homes, rents, and household access. Its 2016 Unitary Plan increased capacity across roughly three-quarters of residential land. Townhouses and attached housing followed in volume. Ryan Greenaway-McGrevy and Yun So, writing in Economic Inquiry in July 2026, use a synthetic comparison with other New Zealand urban areas and estimate that Auckland rents were 23% below their no-reform counterfactual in 2024, eight years after the change. Their analysis also finds a marked rise in housing permits. The estimate is a modelled policy effect, with uncertainty about the chosen comparison cities and other policy shifts; 23% should never be read as an observed fall in nominal rents.[6] These studies differ in markets, size of treatment, methods, and time. They still offer a useful test of the mechanism. Broad legal capacity makes more plots competitors in the housing market and can support a supply response. A narrow entitlement may first appear as a higher bid for scarce land. The result depends on finance, service availability, construction costs, and demand. Policy language should distinguish “more capacity”, “more permits”, “more homes” and “lower rent”. Each claim requires a separate measure. The comparison with narrower reforms suggests conditions to investigate rather than a universal coefficient.

EVIDENCE AND INFERENCE

Observed study findings: Chicago’s targeted reform raised property values without a detectable increase in permits over

the study window; Auckland’s broad reform is associated with a large supply response and a 23% lower modelled rent

level by 2024. Judgment: a city that wants lower rents should liberalise enough well-serviced land to create competition,

then protect the building and occupancy steps. A credible causal estimate for Chicago cannot simply be subtracted from

Auckland’s estimate.

Reform breadth affects the distribution of development opportunities, while fair access remains a separate test. The Chicago and Auckland studies do not isolate the political effects of that difference. Before implementation, identify which residents and builders gain usable

capacity, which neighbours face added service demands, and how the proposed rules address those costs. Treat the effect on support for reform as a local question to investigate. The Auckland –Chicago comparison in Chapter 2 should guide the implementation test: additional rights must become starts, completions and occupied homes. Access needs separate verification. In Delhi's rental audit, otherwise comparable inquiries received responses of about 35% with upper-caste Hindu names and 22% with Muslim names. That is evidence about responses, not completed leases. More physical supply cannot by itself establish fair access, and these studies do not measure the local size of both constraints. [102,103,120] Implementation depends on the actors who control the permissions and complementary services.

Authority over development decisions

Planning decisions require agreements among parties with different powers, costs, and timetables. The owner wants the increase in land value. A builder wants a predictable, saleable project. A bank wants enforceable title and a usable completion certificate. A utility wants payment for connections and capacity. Existing neighbours seek quiet, access, and protection against flood or fire risk. Tenants want more choices near work; they may also fear eviction when their block is rebuilt. Councillors must respond to both housing demand and concerns about congestion and service failures. These interests require explicit comparison. These actors receive benefits at different times. Permission lifts a landowner's option value immediately. A contractor is paid during construction. The utility pays for capacity before future customers arrive. The city receives property taxes only after assessment and collection. A tenant benefits from competition only when enough comparable dwellings are ready to rent, and may lose a current room while waiting. Different timing and responsibilities can leave each party waiting for another to act. Unequal political voice can prevent those agreements from representing prospective residents.

Published safety and service evidence should address local concerns without creating an indefinite veto over conforming homes. A low-income tenant seeking a room near a job has no clear plot on which to object or advocate. A homeowner whose street faces a proposed building can attend a hearing next Tuesday. In a study of planning meetings in the Boston metropolitan area, Einstein, Palmer and Glick found that attendees were more likely to be homeowners, older, male, politically active and opponents of new housing than the people living in the wider area. This is US evidence, with different institutions from Cape Town or Bengaluru. Participation in a hearing may not represent the wider population.[16] Sometimes objections are sound. A narrow road cannot accommodate fire engines; an extra basement can worsen flooding; an unreinforced structure can kill its occupants. Elsewhere, traffic, heritage and "neighbourhood character" are elastic claims that allow decision-makers to exclude poorer tenants or defend the scarcity value of existing homes. The remedy is to make safety and service standards explicit, publish their evidence, and avoid granting an open-ended veto over every conforming home. The agreement must also assign the cost of accommodating growth.

The political agreement must assign both the right to build and the cost of growth. In many African and Indian cities a municipal planning department can permit density, while a separate water utility, electricity distributor, state road agency, district land office, or

environmental regulator controls essential inputs. An announcement of new homes may precede agreement on responsibility for drainage and other services. The agencies that must provide extra capacity may receive no share of the new property-value gains. An intergovernmental fiscal mismatch can prevent additional housing even when the zoning change remains in force. Cape Town's small-rental reform illustrates how an entitlement can be designed around a specific producer and service setting.

Institutional constraints

Actor	Rent or stake	Veto	Possible agreement
Incumbent landowners	Scarcity premium and future uplift	Withhold plots; litigate boundaries	Predictable rights; staged charges on realised value
Existing homeowners / resident groups	Amenity and property prices	Hearing, court, electoral pressure	Clear safety tests; network investments; broad rules
Large developers	Land banks, privileged approvals, scale	Lobby for exceptions; sequence projects	Uniform entitlements; publish approvals and completions
Small builders / landlords	Rental income from extra rooms	Exit into informality; avoid permits	Simple legal designs, verified tenure and staged finance
City, utilities and professionals	Fees, taxes; professional work; discretion	Refuse connection; request new studies	Service compact; timely payment; independent audit
Tenants and excluded groups	Affordable access to jobs; security	Limited formal veto; organise collectively	Tenant standing, anti-displacement rules, allocation evidence
Customary authorities / brokers	Land sales, informal levies, control of claims	Competing title, coercion, obstruction	Joint claim validation and public escrow; legal redress

Exhibit 17.2. Institutional interests and planning authority. These are hypotheses to test through local records and consultation. Adapt implementation to local ownership and service arrangements.

2 Compare reforms through project economics and rights

Small-rental planning reform in Cape Town

Cape Town's additional rental right creates a legal route that must still be converted into inspected, serviced homes. Its lower-income neighbourhoods have long supplied housing through backyard structures, converted houses and small rental buildings. Much of this densification has taken place where households are far from the city's highest-value jobs. Research using neighbourhood and satellite data finds that much of Cape Town's overall densification between 1998 and 2019 came from poorer households crowding into already dense peripheral areas, while formal, investment-led densification favoured more affluent neighbourhoods. Apartheid's spatial legacy survives in these uneven property and labour markets.[9][18] On 26 June 2025 the city approved amendments to its planning by-law. A new additional use right permits up to eight small affordable rental units alongside an existing dwelling, or up to twelve if no dwelling remains, on plots covered by the designated map. Related incentive overlays support more development in selected corridors. The rental provisions began operating on 1 September 2025; much of the wider package followed on 1 October. The city subsequently published location maps, development guides and prototype layouts for micro-developers. The change creates a legal route for small rental development that can be inspected and serviced.[7][8] The benefit depends on whether each stage in that conversion becomes feasible.

Small-rental reform should connect predictable permission to construction, finance, services, and tenant protection. A homeowner may already own the land, borrow against another asset, use a local contractor and add a few rooms. That owner may lack the legal and design capacity to build a conventional apartment complex. A map that states the allowable use, an intelligible standard plan, and an affordable inspection route could change the decision from "build outside the code" to "build under it". A lender can then verify the right to let the rooms. Utilities can identify connections before an overload occurs. A tenant may gain a sound structure, sanitation, and a lease that can be enforced. A new entitlement is evidence of a rule change, not a completed housing outcome.

Evaluate the reform by following projects and tenants through the agencies that must act after permission. The city's 2025 change is evidence of a new entitlement. It does not by itself establish completed rooms, lower rents, improved fire safety or reduced displacement. Legal "affordability" also needs to be tested against the earnings of households at the bottom of the rental market. A planning category can generate units called affordable while the final asking rent remains beyond their means. If formalisation brings larger bills for water, power, tax, insurance and debt, the owner will try to pass them on to renters. The same rule that makes eight rooms legal can raise the value of the parcel, rewarding the owner before any tenant benefits. A credible Cape Town coalition therefore reaches beyond the property owner. The municipality offers a pre-agreed safe envelope and quick review. The water and electricity departments publish the connection and reinforcement costs. Engineers provide an affordable route to certify stairs, loads and fire access. Credit providers offer loans whose repayment fits rent collections. Tenants receive written occupancy rights and a clear repair process. If these parties share the same

timetable, the legal change can result in usable supply. If each waits for the others, builders may delay work or proceed outside the formal process. A rental-project calculation can test whether the permitted design is financially usable.

Rental project viability

A small-rental appraisal should separate construction expenditure from land already held and its opportunity value. The numbers below are a teaching model in South African rand; they are not bids from Cape Town and make no claim about market rents or construction rates. Suppose the new rooms total 110 square metres of gross floor space, at R11,000 per square metre to build, with R180,000 for water, power and site work, and R110,000 for drawings, compliance and finance. Total new capital is R1.50 million. The land is already held; any increase in its opportunity value appears as a separate cost if the plot is sold to a new builder. Occupancy, operating costs, and financing assumptions then determine the feasible return.

Project viability and tenant affordability require separate calculations under explicit assumptions. Assume 90% paid occupancy, operating costs of 20% of collected rent, and an 8.5% capitalisation rate. At R3,000 a month per room, the annual net income is R103,680, capitalised at about R1.22 million. The project is short of its R1.50 million cost. It needs an illustrative rent of R3,689 per room to support that cost at the assumed yield. At R4,200, its imputed value rises to roughly R1.71 million. That larger rent can make the project viable while still failing an affordability test for many households. Assess prospective tenants' affordability separately from the return to the owner and any effects on existing occupants. Changing one input helps identify which constraint actually determines whether the project proceeds.

ILLUSTRATIVE 4-ROOM INFILL	Lower rent	Break-even	Higher rent
Monthly rent / room	R3,000	R3,689	R4,200
Annual net income	R103,680	R127,491	R145,152
Capital value at 8.5%	R1.22m	R1.50m	R1.71m
Value minus R1.50m cost	−R0.28m	≈R0	+R0.21m

EXHIBIT 17.3 Explicit assumptions: 4 rooms; collection 90%; operations 20% of collected rent; capitalisation 8.5%; capital cost R1.50m. Rounded to the nearest R1,000 where appropriate. These are hypothetical calculations, not reported prices or a forecast.

Sensitivity analysis shows whether additional rights, lower costs, or public support make the project feasible. A nine-month delay while R1 million of site and design capital is committed at 12% simple annual finance cost adds R90,000. Spread at the same assumed yield across the four rooms, that would require about R221 more in rent per room each month. A city may save the applicant a modest official fee and still impose a much larger bill through slow decisions. Conversely, a faster permit has little value if the site cannot connect to the water main

or the builder cannot prove lawful possession. A transparent, conditional capital grant of R300,000 would reduce the required supportable rent from roughly R3,689 to R2,951 under these assumptions, provided the owner did not pay more for the land or face new charges. If the entitlement raises the asking land price by a similar sum, much of the public grant accrues to the seller. A fair programme should therefore test the rent, the extra occupied rooms, the quality and the land transaction together. The subsidy chapter and the public-land chapter established how to account for the transfer; permission shows the point at which the transfer first becomes possible. Approval deadlines address a different constraint and must be measured from the applicant's first attempt.

Statutory approval deadlines in Hyderabad

Statutory deadlines change the legal consequences of delay but do not establish actual processing performance. Telangana's building approval and self-certification law of 2020 requires ordinary applications accompanied by valid, complete documents to be decided within 21 days. The statute provides for deemed permission if an order does not arrive in time, with powers to revoke a deemed approval obtained by false statements or contrary to rules. The online TG-bPASS process combines submission, scrutiny, downloadable approvals and complaint handling. The statute changes the legal consequences of missing the deadline; actual processing times require separate measurement.[10] Incomplete applications and pre-filing steps can remain outside the reported clock.

Measure the full interval from first filing attempt to lawful occupation, including amendments, withdrawals, and inspections. If an official can keep a file outside the clock by seeking another plan, land certificate or clarification, the statutory period may say little about actual delay. Self-certification also shifts risk. A small owner with a lawful plot may gain. An applicant on a floodplain, or a builder whose structural design is wrong, may gain an unwarranted paper approval unless independent checks operate at the right construction stages. The Act and portal prove what the system is designed to do. They supply no causal estimate of how many extra affordable homes were occupied as a result. A city should publish the interval from first filing attempt to lawful occupancy, including withdrawn cases, amendments and inspections. Household access can still be blocked after the building receives permission.

A completed lawful unit is not an accessible opportunity where landlord selection excludes eligible households. Under PM-UDAY, the Delhi Development Authority offers rights of ownership, transfer or mortgage to eligible residents in 1,731 unauthorised colonies. Its own account links property recognition to potential bank loans and approved building plans. That may enable investment on some plots. Yet recognition depends on competing documents and on the person who holds the claim. It says little, on its own, about how a tenant gains protection when a newly valuable site is rebuilt, whether streets can accommodate fire engines, or which agency will pay for upgraded sewers. Appraisal should include the public-service obligations alongside the private rights created.[11] A field experiment by Saugato Datta and Vikram Pathania sent otherwise comparable inquiries to online rental listings in Delhi and surrounding areas. Landlords responded to 35% of upper-caste Hindu applicants and 22% of Muslim applicants. The 13-percentage-point gap was measured at the contact stage; the study did not observe final

leases or the whole housing market. It identifies differential treatment at the inquiry stage in the sampled market. A new apartment in the right neighbourhood expands the physical supply. Landlords can prevent that unit from becoming a housing opportunity for particular households.[12] Planning implementation needs records of land claims, usable permission, and occupant rights.

A successful permission reform in India therefore needs three records that often sit in different offices: the landholder's claim, the building's usable permission and the occupant's rights. A fourth record, the service connection, determines whether a family can remain there. Digitising the first two can improve lending and inspection while deepening exclusion if tenancy, caste, religion, disability and migrant status disappear from the questions government asks. Any inclusion rule must be enforced through lawful, privacy-preserving channels; a public tenant registry could expose vulnerable households to retaliation. Accra illustrates why even the authority to grant development permission may be contested.

Land rights and development permission in Accra

Contested land authority can delay development beyond the formal permit process. Customary owners, family heads, competing chiefs, the Lands Commission, district planners and informal land guards can each assert claims over a plot. Legal forms alone will not reveal who can stop the work. A study by Divine Mawuli Asafo, published in Housing Studies, followed the experiences of 22 housebuilders and interviewed customary authorities and land sellers in several peri-urban communities. Buyers described repeated payments when new people claimed ownership, violence or threats from land guards, interruption of construction, and repeated attempts to secure the same site. One household had begun seeking land in 2010 and had yet to begin building in 2018. The qualitative sample cannot estimate citywide prevalence, though the accounts identify possible mechanisms linking contested claims to delay.[13] Resolving that risk requires documenting rival claims and the consent and revenue-sharing agreement.

Land-conflict remedies should establish claims, consent, benefit sharing, and appeals before credit is secured against the site. When savings are tied up in disputed land, the household may abandon a qualified contractor, build in fragments, reduce the quality of materials, or start without complete plans. A partially built house then becomes evidence of possession. Construction may be used to assert possession even when starting work increases financial risk. Informal authority becomes a parallel approval market with its own fees and coercive power. Research on land governance in Kumasi describes the overlap of ownership and administration by traditional leaders, whose decisions can govern use, resolve disputes, and affect infrastructure location alongside statutory planners. Where the state treats customary institutions as obstacles to be bypassed, the promised permit may lack local acceptance. Where it delegates every decision without oversight, land sales and arbitration can become sources of private rent. The workable agreement starts by documenting rival claims, setting rules for consent and revenue sharing, protecting customary members and women with weaker bargaining power, and offering impartial appeal before credit is secured against the parcel.[14] Digital records must preserve disputes rather than erase them through a single authoritative entry.

This is where digital maps can do harm as efficiently as good. A satellite footprint proves that a roof existed on a date. It cannot decide which family sold the land, whether a chief had authority to sign, or who was threatened into leaving. Recording the first claimant as a digital "owner" can record a disputed claim as settled ownership. Georeferenced evidence should identify discrepancies and support a public adjudication process; it should not silently adjudicate them. The right to challenge the record is part of the information system. Mukuru provides an alternative route for bringing residents and owners into statutory planning.

Tenant participation in Mukuru

Area planning can coordinate infrastructure and claims that cannot be resolved one plot at a time. Around 94% of residents in the study area were tenants, many renting structures on private land with contested or irregular histories. Organised residents, Muungano wa Wanavijiji, lawyers, planners and county officials used the 2017 Special Planning Area designation to create space for a settlement-wide plan. The process drew together separate work on land, infrastructure, health, housing and public participation. It recognised that a drainage line, a road, or a new rental block could not be planned one plot at a time while informal owners and tenants remained outside the decision.[15] Meaningful participation still needs remedies independent of registered ownership.

Participation should give affected tenants and occupants enforceable routes to influence and challenge decisions. Conventional consultations often begin with registered owners. Mukuru's resident-based organisation gave tenants a stronger route into decisions that could otherwise raise their rent or displace them. Its organisers described the settlement's economy and residents' payments for costly informal services as evidence that residents could support better public networks. This helped form a coalition around service delivery rather than a simple contest between "legal landowners" and "encroachers". The process also encountered resistance from officials, structure owners and informal providers with income at risk.[15] Participation did not ensure delivery of services or secure leases. A special planning area can suspend ordinary development while the integrated plan and finance remain unresolved. Uncertain eligibility lists can reward early participation while excluding later arrivals. Landowners may demand compensation for rights that cannot be reconciled with the settlement plan. Where affected residents have weak property claims, the process needs representation and independent remedies that do not depend on registered ownership. Those rights complement safety controls rather than justify removing them.

3 Implement and evaluate a proportionate approval system

Safety requirements and supply restrictions

Essential safety rules protect occupants who cannot independently inspect structural and fire risks. A failed retaining wall, inaccessible fire stair, narrow evacuation route, blocked drain or building over a river has costs that the original owner may never pay. Kenya's Architectural Association, reporting on 2024, recorded thousands of unsafe or dangerous structures identified in inspections; its findings are from buildings selected for inspection and

cannot be read as a national prevalence rate. The South C chronology shows why the inspection system must continue after the first permit. These rules protect renters who cannot inspect reinforcement steel or underwrite structural failure themselves.[17][1] Each restriction should nevertheless be tested for evidence, proportionality, and distribution of costs.

Test safety requirements against the hazard, feasible compliance options, and the households affected. First, can an engineer state the harm and the measurable standard? Second, is there a lower-cost design that delivers the same performance? Third, can an inspector verify it with reasonable effort and authority to stop work? A mandated parking bay may consume space with little safety benefit in a walkable rental district; a second exit from a crowded building may save lives. Some provisions should be non-negotiable. Others can be performance-based, graduated for small extensions, or replaced with safe standard plans. Protecting safety through a rule that nobody can afford to obey often produces the very uninspected buildings the city fears. The same distinction applies to environmental and cultural protection. A floodplain or natural wetland can be a reason to refuse density. A heritage district can preserve genuine common value. But a city should make the evidence public and assess the distribution of the burden. If the valuable centre is closed to new homes under loose definitions of "character", the poor may be forced into more hazardous peripheries. Report the locations and household groups that bear the costs and receive the benefits of protection. Digital permits should record that reasoning and the agencies accountable for it.

Digital permits and accountability

Digital approval systems can improve traceability only within the limits of incomplete and uneven source records. Their value depends on the constraint being addressed. Faster review leaves restrictive density rules unchanged. A public development map may omit customary claims. AI may detect an additional floor without detecting defective concrete. Each application needs a defined decision, an accuracy test, and a remedy for affected residents. The Kenya Housing Survey found that 20.5% of counties reported digitised building-plan applications and 23.3% reported georeferenced land records. These are institution-reported adoption indicators in the 2023/24 survey, not evidence that digitisation reduced rents or corruption. The reported coverage identifies a data limitation: building and land information are incomplete and uneven across the agencies asked to govern construction.[3] Procurement should specify the identifiers, timestamps, and human review needed to resolve those gaps.

Use	Binding constraint	Verify	Gain	Power / risk
Public buildability map	Conflicting rights, hazards and zoning	Resubmission rate; share of plots with a clear safe envelope	Owners and builders first; buyers through competition	Officials who control the authoritative map; false precision on disputed title

Use	Binding constraint	Verify	Gain	Power / risk
End-to-end permit log	Unseen delay and informal discretion	P50/P90 days from first attempt to usable approval	Applicants save carry cost; owners may raise prices	Portal operator and reviewer; hidden “incomplete” queues
Construction -sta ge change detection	Building drifts from approved design	Verified deviations corrected before occupancy	Tenants and neighbours through lower risk	Inspectors gain surveillance power; image error and selective enforcement
Occupancy and rent follow-up	No view of who receives homes	Completed, safe, occupied rooms and rent by income group	Tenants if services and protections hold	Landlords or officials may identify and displace vulnerable tenants

EXHIBIT 17.4 Technology appraisal under the book’s five-question test. Measures are proposed evaluation variables; the table reports no causal effect from these technologies.

Require interoperable records, logged decisions, and accessible correction and appeal before automating approvals. An effective procurement specification would demand interoperable parcel and building identifiers, logged changes to the official plan, timestamps for each agency’s action, and human review of disputed rights. It would prohibit automatic approval of hazards and automatic enforcement based only on image classification. The city should retain its data in a documented schema and make redress available offline. It should procure independent tests of accuracy, racial or spatial bias, and cyber security. An approval label is insufficient unless the authority can inspect the applicable rules, evidence, and decision record. The same records can support accountability. Publish anonymised counts of all first attempts, the reasons for rejection, median and 90th-percentile decision times, subsequent construction starts, illegal additions found, completed rooms, occupancy and rent bands. Track neighbourhoods and plot sizes, while suppressing personally identifying information. An application with an incomplete title should remain in the record and denominator as an unresolved claim. Assign responsibility for delays and give applicants a usable route to correct errors and challenge decisions. The reform sequence should begin by testing the actual constraint in a limited area or application type.

Implementing planning reform

Diagnose authority, density, safety, network, finance, and enforcement constraints before selecting the reform. Pay for inspection and network expansion through transparent budgets and feasible charges, not informal negotiation. Planning must decide consistently, utilities must connect, inspectors must verify safety and an accessible body must address exclusion. Track conversion of rights into starts and completions, decision time, safety compliance, net rents and access by excluded groups. Suppose a mayor, a planning director, a water utility and a coalition of small rental builders agree to test reform in two urban corridors. The temptation will be to announce a new density limit and launch a portal. A better first step is to assemble a sample of plots, inspect the land records, walk the water and drainage network, identify hazards, and interview tenants as well as owners. The sample should include rejected applications and plots where nobody has applied. Use the results to assess whether the principal barrier is permitted density, title uncertainty, network capacity, cost, financing or fear of enforcement. Pilot locations should then allow the relevant changes and protections to be evaluated.

Select pilot areas and professional responsibilities that make the proposed supply and safety mechanism testable. In a serviced location with jobs and an existing rental market, grant an automatic entitlement to a limited class of safe small additions. On adjacent blocks where networks are near capacity, publish a funded reinforcement schedule and a transparent connection charge. Require an engineer to certify structural loads, fire access and drainage before the permit can become occupancy permission. Standard plans may reduce design cost for simple plots; complex or hazardous sites retain individual review. Every denial must identify a specific rule and an appeal route. Specify the benefits and obligations for each participant. Small builders receive predictable rights, standard design support and a path to loans. Utilities receive a capital plan and payment for incremental service costs. Existing homeowners receive documented traffic, drainage and safety safeguards. Tenants receive lease terms, repair channels and a chance to participate before redevelopment displaces them. Councillors receive reports on occupied homes against agreed targets. Landowners face predictable charges on realised increases in value with exceptions subject to published criteria and review. Professionals receive fee-paying work tied to verified safety, with sanctions for false certification. An evaluation baseline is needed before the new process changes applications and behaviour.

A 12-month evaluation framework

Establish rules, rents, occupancy, service limits, and processing times before tracking the reform's effects. The planning lead should cost the rule changes, staff training, inspection workload, and appeals before announcing a start date. Begin with low-risk applications that can be handled under existing powers; support actual projects only after lawful permissions and service commitments are in place. Track implementation during the first year and compare similar untreated locations where credible, recording spillovers and demand differences. Starts, completions, rents, and displacement may require longer follow-up than faster permit processing. At each step ask a narrow question. Did more low-income owners and small builders submit complete applications? Did the worst delays fall? Did finance become cheaper? Did starts turn

into completed, connected rooms? Did rents decline relative to comparable units, or did rising land prices absorb the gain? Did unsafe additions and evictions increase? Did women, migrants, lower castes and minority tenants gain access to the new stock? An aggregate rise in approvals can conceal worsening exclusion, and the early figures may show developers exercising existing options rather than truly additional homes. Results need denominators that distinguish eligible opportunities from the projects that successfully entered the system.

THE PERMISSION-TO-HOMES DECISION RULE

Go forward when legal capacity rises and the conversion rate into safe occupied units improves without serious

displacement, service failure or price capture. Revise when applications accelerate but construction stalls: test credit, land

disputes, infrastructure and inspection. Stop or redesign where increased density exposes residents to unmitigated fire,

flood or structural hazards, or where identifiable groups bear eviction and discrimination. These are proposed governance

thresholds; each city must set measurable local values before the pilot starts.

The 12-month evaluation should publish five denominators: all eligible parcels; all first application attempts; all approved plans; all completed and inspected units; and all units occupied by the intended income groups. It should separately publish the share of tenants who retained their neighbourhood after redevelopment. The fiscal ledger records the cost of added water, drains, roads, schools and inspections alongside revenue from property taxes and connections. Compare neighbourhood construction and service costs with the public revenues and land-value gains received elsewhere; an unaddressed imbalance may undermine implementation. Those measures allow the authority to judge whether additional rights have become safe accessible housing.

Implications for development control

Development control should be judged through delivery, household access, and the distribution of public costs and land-value gains. Appraisal should distinguish returns to private investment from gains created by a public decision. Discretionary exceptions require transparent reasons and review. Broad rights still need service capacity and safety checks. Digital parcel records should preserve unresolved claims until the responsible authority determines them. Planning reform should make lawful construction predictable. Give households and builders clear requirements, use verifiable safety and service tests, and provide a route to contest official mistakes. Test whether modest homes can be built lawfully at prices intended residents can afford. Apply consistent rules while retaining the authority to stop dangerous work. Evaluate support for the reform among existing residents and prospective occupants. The cases support different parts of this test and should not be pooled into one effect estimate.

Different studies illuminate safety, supply, rights, and access without establishing a common effect size. Cape Town's rental-planning reform is intended to give small builders a route to lawful development. Accra shows the limits of a state permit when competing authorities control the land. The Delhi study identifies unequal treatment in rental inquiries. Auckland and Chicago show that legal capacity becomes lower rent only through a supply response. Together, the cases identify decisions to test: development rights, services, safety, tenure, supply, and equal access. Their different methods and settings do not establish one common effect size. Fieldwork should identify which mechanism is operating in the proposed reform area.

Prepare the local decision

Test real applications and affected households against rules, network capacity, safety, and access constraints. Extract a cohort of applications with timestamps at every review stage, fees, revisions, reasons for refusal, appeals, starts, and completions. Include informal builders who did not apply and record their expected costs. Compare simple extensions with larger developments. The Hyderabad and Cape Town examples above require separating faster decisions from an increase in safe completed housing. Planners, inspectors, utility reviewers, small builders, tenants, and neighbouring residents should examine anonymised delayed files together. Identify which review protects a measurable safety or service requirement and which repeats another office's work. Agree the authority to resolve conflicting requirements, while retaining an independent route for appeal. For each well-served area receiving additional planned capacity, examine refusals, withdrawals, exceptions and connection delays alongside approvals. The planning lead should explain persistent differences between planned capacity and completed homes using documented constraints. Correct unsupported administrative barriers through the lawful rule or appeal process. Do not override genuine safety limits or resident rights merely because a district misses a numerical target. If a digital pilot accelerates approvals without construction, investigate the remaining constraints before buying more features.

Technology and action. Test automated checks only on rules that are explicit enough to encode. Log overrides and return a specific correction request rather than an unexplained rejection. Track the median and longest delays, staff time, rework, compliance, starts, and net new occupied units. If approval time falls but construction does not rise, inspect financing and connection constraints before adding another digital feature. Chapter 18 examines the protections needed when additional development affects existing residents.

Planning reform should make the sequence from lawful capacity to inclusive occupancy observable and accountable. Feasible rights, reliable approvals, funded services, and effective access rules connect permission to housing outcomes. A deadline or map alone can leave discretionary vetoes, financing gaps, unsafe construction, and discrimination intact. Monitoring should identify where the next constraint appears as applications rise. Test approvals and inspections at tenfold application volume; faster approvals can increase queues for utility connections or completion checks.

Sources and evidence notes

Evidence and scope. References identify public reports, laws, and research papers. Dates refer to publication or policy commencement as indicated. Academic estimates, administrative counts, statutory requirements, and illustrative calculations provide different kinds of evidence. The chapter's capital model is a hypothetical calculation, not a field-validated estimate.

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18. City of Cape Town (2022). Integrated Human Settlements Sector Plan, 2022–27; spatial inequality and infrastructure mandates. [additional source]

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Method note. The South C account relies on the NCA's report of 7 January 2026 and the county's reported enforcement chronology; neither settles legal liability or the structural cause. The rent and capital example is a transparent hypothetical with no empirical rent benchmark. The Auckland result is a synthetic-control estimate for 2024 relative to a no-reform counterfactual; the Chicago result comes from a different quasi-experimental design. The Accra and Mukuru studies explain mechanisms and coalitions through qualitative evidence; they cannot support national prevalence claims.

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18. Redevelopment and resident protection

Agree enforceable treatment of existing residents before committing land to redevelopment. Record overlapping claims, compare feasible alternatives, and fund temporary arrangements, livelihood protection, and return rights where applicable. Evaluation must follow affected people who leave as well as those who receive the new homes. The Joe Slovo case illustrates how delivery promises depend on enforceable decisions over time.

The Joe Slovo case demonstrates why redevelopment decisions must distinguish a housing promise from enforceable treatment of existing residents. In June 2009, South Africa's Constitutional Court authorised a supervised removal of residents from Joe Slovo, an informal settlement beside Cape Town's N2 highway. The eviction order came with a timetable, minimum standards for temporary accommodation and duties to consult. Roughly 20,000 people were affected. The government said the land had to be cleared so permanent housing could be built. In March 2011, the same court discharged the eviction and supervisory provisions while retaining the costs order. The state had changed its plan. It now intended to improve the site while residents remained there. The removal order no longer matched the government's revised plan [7–8]. The judgments illustrate why redevelopment must assess outcomes for existing residents. A new building can improve living conditions. It can also make an existing tenant homeless. Removing a settlement from a hazardous strip of land can save lives. Relocation to a distant estate can reduce residents' access to the work that paid their rent. A government may announce a right to housing while the rules used to allocate new homes exclude the claims of the people most dependent on the site. An appraisal therefore needs a separate record of their rights, uses, and potential losses.

Redevelopment requires a separate assessment of existing residents' claims, including those of tenants and occupants without registered ownership. The relevant question is how those claims will be protected when public action raises land values. Assessment should cover tenure, rent, work, family networks, discrimination, finance, construction, and the authority's ability to meet its obligations. Alongside land and new floor space, record residents' access to jobs, schools, customers, and support networks. These benefits and losses may not have market prices, but omitting them can distort the comparison between upgrading and relocation. Occupancy and overlapping claims provide the starting point for that record.

1 Establish rights and household exposure before redevelopment

Occupancy and overlapping land claims

Settlement records must include the livelihood and care relationships attached to occupied space. In Nairobi's Mukuru, residents and researchers documented about 300,000 people living on some 650 acres. Muungano's account reports that around 230 landowners held virtually all the land and that 94 per cent of residents were tenants [1]. A cadastral system showing 230 owners would still fail to describe the housing market. Between the title holder and a household paying monthly room rent may stand a structure owner, a caretaker, a rent collector

and a service broker. Their rights and responsibilities require separate records; a deed will not identify them all. The room is also a workplace in many households. A shop may open on the pavement; tools and stock may be stored beside the bed; a worker may depend on an employer reachable by foot. A child's school and an aunt who provides childcare make the location productive. Clearance can disrupt these livelihood and care arrangements even when a household holds no land title. Mumbai illustrates similar layers of claims that a simple ownership entry cannot represent.

Eligibility surveys can reveal overlapping claims while increasing official control over who receives assistance. The official Dharavi redevelopment website says its survey assigns each structure a 15-digit identifier, uses lidar mapping of lanes and collects documentary proof from households. The data help the authority decide who qualifies for rehabilitation. The survey therefore informs eligibility decisions [4]. A newly recorded building can strengthen the state's ability to plan services. The same survey can give officials and developers greater control over which households qualify for assistance. They should therefore supplement, rather than be replaced by, the formal property register.

Exhibit 18.1. Claims and interests on an occupied parcel

Person or body	Asset or dependency	What conventional property data can miss
Registered owner	Title, land rent, option to redevelop	May live elsewhere; a title says little about occupants.
Structure owner or landlord	Rent from shacks or rooms; control of access	May occupy land informally and still collect substantial rent.
Tenant or subtenant	Affordable location and continuity of occupancy	Often no recorded lease or entitlement to new ownership.
Home-based worker	Foot traffic, inputs, customer ties, storage	Loss of livelihood can exceed the value of the walls.
Neighbourhood / public body	Road, drainage, school, networks and access	Public investment may raise private rents before costs are recovered.

The categories are analytical. They may overlap in one household. A defensible inventory records people, uses and claims separately; ownership

is one dimension of exposure.

A property register is not a complete record of people at risk from redevelopment. Its omissions have distributional consequences. Compensation tends to follow legible assets: land, permanent walls, a registered shop. Households with weaker documents may receive a moving allowance while losing a daily wage, an informal workshop or the right to return. The problem deepens when a scheme ties compensation to a date: rights become a contest over old utility bills,

voting records, ration cards and testimony from intermediaries who may charge for helping produce the evidence. Public spending or increased permitted density can raise expected returns to land. Owners seek a higher sale price; structure owners seek a higher rent; developers bid for redevelopment rights. Tenants hold a shorter contract, face higher costs of organising and can be replaced one household at a time. Land-value gains may occur before residents receive the promised housing and services. Displacement can occur through several channels even when no household is directly cleared.

Four channels of displacement

Resident protection must address physical removal, price changes, eligibility exclusions, and other mechanisms that shift access and bargaining power. A second channel is the market. A paved road, safe drainage and legal electricity make a neighbourhood more desirable. Without tenant protections or greater housing supply, landlords can increase rents and replace poorer tenants. The third channel is administrative: a household keeps living there until an eligibility cutoff, a missing document or an impossible contribution excludes it from the new scheme. The fourth is the slow exit. School fees, commuting costs, service charges and building maintenance can make a new unit unusable even when its transfer was formally free. These channels need different instruments. A court can stop an unlawful demolition and still leave rising rents untouched. A tenure certificate can protect an owner against eviction and leave the owner's tenants exposed. A relocation grant can pay for a truck while a household loses its customers for years. A price cap may help an existing tenant and deter a landlord from maintaining or adding rental rooms. Each instrument changes bargaining power; the policy must track where that power moves. Household relocation evidence helps value losses that the project budget can overlook.

Household costs of relocation

Relocation appraisal should include households' valuation of access and neighbourhood conditions alongside dwelling quality. Takeuchi, Cropper and Bento used data from around 5,000 Mumbai households surveyed in 2003–04 to estimate how families value housing quality, travel to work and the composition of nearby households. In their simulated policies, providing good roofs and piped water in place was worth about INR 500 per household per month, roughly 10 per cent of income for the affected group. For one group considered for relocation, the model put the gross housing improvement at INR 813 a month, offset by INR 290 from greater distance to work and INR 490 from changes in neighbourhood composition [3]. These are estimates from a stated model of location choice, in historical rupees; they are neither measured changes in earnings after relocation nor universal welfare prices. The model includes costs omitted from a floor-area-only comparison. A building project may report a gain of one room with a toilet. The household's account contains that benefit alongside the cost of lost time, lost work, lost customers, new fees and disrupted social ties. An assessment that counts floor area while omitting these costs biases the comparison. It will often favour distant land because distant land looks cheap in the government's ledger. The interpretation must distinguish measured preferences and heterogeneous outcomes from a general rule about relocation.

Relocation effects vary across households and destinations, so average findings require careful interpretation. In India, religion, language and caste affect social networks and can also reflect discrimination. A statistical preference for living near people of the same group cannot justify exclusion or segregation. It warns policymakers that household networks and barriers to equal access belong in the welfare model. The same reasoning applies to racialised urban space in South Africa, where the location of a formal home can reproduce the old separation between work and residence. Brazil supplies stronger causal evidence on the value of location. Belchior, Gonzaga and Ulyssea studied a large public-housing lottery linked to administrative employment records. Winning improved housing quality and lowered housing spending, while average formal employment probability fell. Effects differed across neighbourhoods; measured access to labour markets explained 82–93 per cent of the between-neighbourhood differences in their design [13]. The study links outcomes to differences in access to labour markets. It does not prove every relocation lowers employment: effects differed by initial poverty and by neighbourhood. The relevant decision is whether the proposed destination preserves or replaces the opportunities each affected group needs.

These findings do not support a universal rule for or against relocation. The Ahmedabad lottery improved housing without income gains; the US experiment found benefits for children exposed to better neighbourhoods young, with different results at older ages. Neither justifies forced displacement or guarantees gains for an individual household. Before clearance, test whether the destination preserves livelihoods and networks or provides usable alternatives, and whether residents can sustain the move. [107,106] Mumbai's redevelopment rules show how eligibility can determine which households obtain that opportunity.

2 Compare protection arrangements and distributional effects

Redevelopment eligibility in Mumbai

Redevelopment financed through development rights creates incentives that need to be examined without presuming misconduct. Mumbai's Slum Rehabilitation Scheme has offered a development-rights financing arrangement since the mid-1990s. A private developer finances rehabilitation units and receives development rights for saleable space. The World Bank's description of the early scheme records free on-site units for eligible occupants, consent requirements and restrictions on resale [5]. In economic terms, the developer uses the market value of a scarce building right to pay for some of the existing residents' new homes. The city can add formal units without writing a construction cheque for each one. The arrangement creates a distributional trade-off. Revenue comes from the floor space that can be sold at high prices. The costs of providing rehabilitation rise with the number of people who receive rights. The scheme therefore has strong financial reasons to draw an eligibility boundary and to retain valuable land for the sale component. It also gives a developer an interest in accelerating clearance and a resident association bargaining power through consent. Discretion over beneficiary lists creates a risk of preferential allocation or unofficial charges. These are incentives implied by the structure; they do not establish misconduct in every scheme. Published eligibility rules must specify the treatment of each occupancy and business category.

Eligibility cut-offs can create sharply different outcomes for households sharing the same redevelopment area. Dharavi's official programme site states that ground-floor residential tenements existing on or before 1 January 2000 qualify for 350 square feet within Dharavi. It sets different terms for ground-floor tenements dating from 2000–11 and for later or upper-floor structures. The entitlement page offers 300 square feet under specified payment, metropolitan-area or hire-purchase arrangements to other categories, while the general FAQ describes eligible tenements in Dharavi and others outside it [4]. The official pages should be reconciled with binding resolutions for individual cases; an online FAQ cannot substitute for a household's written determination. As a programme design, the cutoff dates and distinctions between floor levels do three things. They help establish a finite fiscal obligation; they favour households able to document older occupation; and they create sharp differences in value between people who may live next door, or above and below each other. A home-based garment workshop can be financially and socially different from a legal residential room. The public authority must therefore publish the treatment of tenants, subtenants, upper floors, mixed residential-commercial premises and those who lack standard paperwork before construction makes old claims hard to recover. Credible protection requires rights and funding before removal rather than a later promise of assistance.

Resident organising and litigation illustrate the consequences when relocation terms fail to provide acceptable alternatives. Rishi Jha's 2025 fieldwork on Mumbai resettlement documents how households sent to hazardous peripheral locations used community organising and legal action to press for alternatives [6]. The study uses qualitative research, with more than 100 interviews, to examine relocation and residents' responses; it does not estimate an average treatment effect. Once an agency has paid to clear a site and a developer has sold the future right to build, the bargaining power of displaced residents falls. Credible rights must be agreed and financed before removal. The project's financial incentives help explain why treatment of claims must be settled before clearance.

Land values and eligibility incentives

The residual value available for redevelopment depends partly on which claims and obligations are recognised. An additional rehabilitation household increases the project's cost and reduces that residual. Tighter development rights, weak off-site infrastructure or high interest rates raise the temptation to narrow eligibility. More saleable floor area can cover more rehousing, although the new floor area also demands water, sewage, schools and open space. Public land or an infrastructure contribution that the state calls "free" still has an opportunity cost, as Chapter 14 showed. Developers bear construction and financing risks; titleholders and courts can affect acquisition; eligibility certificates can change residents' interests in redevelopment. Renters and newer arrivals may lack comparable representation. Test these decision powers and claims alongside engineering feasibility before approving the scheme. Payment milestones should therefore include verified rehousing and return where promised.

A fair agreement must specify the treatment of every claimant before land is cleared. It can offer in-place reconstruction in phases, nearby temporary accommodation with a binding return right, rental tenure that does not demand a mortgage, protected ground-floor trading

space, and a funded mechanism to correct exclusion errors. The developer should face verified occupancy and return targets in addition to construction milestones. Payment conditions tied only to physical completion omit whether eligible residents have been rehoused. In predominantly rented settlements, owner compensation alone cannot establish that upgrading benefits occupants.

Distribution of upgrading benefits in Nairobi

Upgrading in a rental settlement needs explicit evidence on tenants' costs, safety, and ability to remain. In Mukuru, the tenants who would benefit from safe water and sanitation may be renting structures on private land. Public infrastructure can make these rooms more valuable. As long as there are too few legal, serviced alternatives, the landlord may have the power to convert a large part of the gain into rent. The tenant receives the new tap and loses some of the income saved by no longer buying expensive water. The title holder may benefit through land value even when the household using the infrastructure pays for the connection. Mwau and Sverdlik's research in Mathare shows why Nairobi's rental market cannot be treated as an accidental feature of informal settlement. The authors describe growing multi-storey tenements, alongside single-storey shacks, and the role of caretakers, investors and informal intermediaries. Their 2020 paper estimated that nearly 70 per cent of Nairobi residents were renting single rooms, using the sources available to the authors; the figure should not be conflated with the 94 per cent community-specific estimate for Mukuru [2]. The market responds rapidly to demand for centrally located rooms, yet weak regulation can leave tenants in dangerous buildings. The operating agreement must connect infrastructure gains with tenure and fair access.

Upgrading requires coordinated land, service, fiscal, and resident-protection arrangements rather than connections alone. Wainaina and co-authors found that weak routines for learning from implementation reduced the programme's ability to adapt; another study of upgraded facilities traces why residents accepted some new infrastructure and struggled to sustain other parts [19–20]. These qualitative studies do not estimate how much a new sewer raises rent. They show that an effective upgrading coalition has to survive the handover from consultant to county, from county to utility, and from the neighbourhood committee to the landlord and tenant who use the facility. The practical solution starts with a roster of occupants, updated independently of land ownership, and a rent baseline before the investment is announced. Community organisations need both tenants and structure owners represented; otherwise a committee can turn into an owner-dominated body. Residents need rights to challenge the roster, report retaliatory evictions and receive utility bills that show who pays which charge. Landlords need a credible path to lawful investment and cost recovery. The city needs a means to recover a share of uplift through its fiscal system without allowing informal brokers to become monopoly gatekeepers. A hypothetical neighbourhood account shows how rent changes can redistribute the benefit.

Model of a 1,000-household neighbourhood

An upgrading appraisal should test how higher rents affect the many households that do not own the improved assets. Consider 1,000 occupied rooms in a hypothetical Nairobi upgrading area, with 940 renter households, mirroring Mukuru's reported tenant share. Assume every renter pays KES 3,000 a month before the works. After drainage, lighting and a water connection, suppose rents rise by KES 800 a month. The extra rent received from 940 tenants would be KES 752,000 a month, or KES 9.024 million a year. At a 5 per cent real discount rate over ten years, that stream has a present value of about KES 69.7 million. These are illustrative assumptions, not measured rents or a forecast for Mukuru. Inflation, vacancies, quality improvements and changes in the supply of rooms would alter the result. The resulting rent transfer must be distinguished from payment for genuine additional quality and maintenance.

A connection count cannot establish affordability when rent changes and landlord investment are unmeasured. Estimate the rent increase attributable to each component. Suppose a tenant's water spending and monetised lost time previously totalled KES 1,000 a month, and the new service cuts that measured burden by KES 700. A KES 800 rent increase would raise the combined monthly burden by KES 100 before counting unmeasured health, safety, and reliability gains. Report cash savings and time savings separately. Landlords could be gaining, tenants could be gaining in non-cash ways, and the city could be receiving little of the value it created. A design that looks successful by counting connections can fail an affordability test. Policy should test alternative protections rather than presume a universal rent cap will resolve the trade-off.

Exhibit 18.2. Distributional outcomes of a public improvement

Illustrative outcome	Tenants' monthly monetised net benefit	Implication for the public agreement
Rent rises KES 800; service saves KES 700	−KES 100	Measured burden rises by KES 100; unmeasured health, safety, or reliability gains may remain.
Rent rise limited to KES 400; service saves KES 700	+KES 300	Tenant gains, but maintenance and landlord incentives need a funding plan.
New competing rental supply limits rent rise to KES 200	+KES 500	Supply response allows more savings to remain with tenants.

All values are hypothetical sensitivity tests, not observed outcomes. Service savings combine water spending and monetised time; report the two components separately. The same

service benefit may have distinct distributional effects because of rent regulation, competition and landlord bargaining power.

Compare locally feasible tenant protections with their enforcement costs and supply effects. A time-bound rent-transition agreement may help existing tenants while the city adds serviced rental supply. A targeted cash payment can offset short-term shocks if rents are monitored and households can actually find rooms. A formal lease and a grievance system can lower arbitrary eviction risk. Each option has costs and enforcement problems. Permanent rent restrictions can protect incumbent tenants while affecting maintenance and entry into rental supply; relying only on market adjustment can expose current tenants to rent increases after public investment. Government must specify the sequence, costs, and enforcement arrangements for its chosen approach. San Francisco's evidence illustrates why incumbent protection and future rental stock require separate measures.

Rent regulation can protect current tenants while changing the housing available to future renters. After a 1994 change in rent-control coverage, protected tenants were almost 20 per cent more likely to stay in their homes, while affected landlords reduced their rental stock by about 15 per cent through conversion or redevelopment. A structural model in the same study estimated a 5.1 per cent increase in citywide market rents from the reduced stock [23]. These are estimates for one unusually expensive US city; they establish a mechanism, not a predicted effect for Nairobi or Mumbai. Assess incumbent retention alongside changes in the stock available to new renters. Mumbai's regulatory history provides a different setting in which to test those competing effects.

Rental reform should measure effects on vacant rooms and new entrants alongside sitting tenants. Tandel and co-authors describe how the city's long-standing rent-control regime favoured some sitting tenants while weakening incentives for maintenance and formal rental investment [22]. Attempts to rewrite these terms meet organised resistance: incumbents fear losing their homes, landlords seek higher returns, and small tenants fear informal eviction even when a legal safeguard exists. A feasible reform sequence protects low-income incumbents during a defined transition, opens lawful space for modest new rental housing, and makes landlords responsible for basic safety and upkeep. A city should measure the effect on vacant-room supply and new migrants alongside the effect on current leases. Formalisation also changes security and investment incentives, with results dependent on the rights actually created.

Tenure formalisation and housing security

Tenure interventions can affect investment and work, but the cited effects belong to specific programmes and populations. Erica Field's study of Peru's urban titling programme found that housing renovation increased by more than two-thirds relative to its baseline level after titles strengthened tenure security. Much of the additional investment came without bank credit. In separate work, she found that titling was associated with about 17 per cent more household work hours, alongside reductions in home-based work and child labour [10–11]. These estimates draw on programme rollout patterns and comparison groups; implementation and local labour markets matter. A natural experiment in the suburbs of Buenos Aires points in the same

direction. Galiani and Schargrodsky compared families whose land became titled with otherwise similar families whose former owners contested expropriation compensation. Titled households invested more in structures and their children's education. Credit access did not explain the main effect. The authors' analysis reported around 12 per cent more built space and a 37 per cent rise in a housing-quality index in the treatment group [12]. They do not establish that every titling instrument is preferable to other forms of security.

Choose tenure instruments according to local claims and enforcement rather than presume that one title resolves every conflict. Households invest when their expected right to enjoy the return becomes credible. That assurance can come from different legal forms: individual ownership, collective lease, occupancy agreement or an enforceable right to return. A title can also make a plot mortgageable, saleable and attractive to outside investors. An owner may then sell voluntarily to meet a crisis or be pressured to sell. If the tenant has no parallel right, formalising the owner's claim may raise the likelihood of the tenant's displacement. The choice of instrument should follow the problem. Where eviction risk prevents repairs, durable occupancy is valuable. Where the land has a dangerous flood exposure, indefinite tenure in place may entrench loss of life. Where a city owns well-located land, a long lease with affordability conditions can keep part of the uplift public. Where land belongs to a customary authority or several disputing families, a registry cannot adjudicate every conflict by assigning the first name it finds. Legal remedies can also change the redevelopment option itself.

Legal remedies and alternative redevelopment plans

Legal review can preserve alternatives and collective negotiating capacity before removal becomes irreversible. The 2009 court permitted removal only with detailed safeguards, including safe temporary accommodation and a process of engagement. By 2011 the state had adopted an in-place approach and the court discharged the removal provisions, retaining the costs order [7–8]. The case involved resident organisation, judicial proceedings, and a revised public development plan; the judgments do not isolate the contribution of each to the change. The case does not prove that every community can be rebuilt while fully occupied. It demonstrates that the supposed engineering necessity of removal can depend on a contestable project design. The Lagos evictions illustrate failures of resident protection. Amnesty International documented the forced eviction of more than 30,000 residents from Ilubirin and Otodo-Gbame during 2016–17, including destruction of homes, loss of livelihoods and actions taken in defiance of court orders [9]. A Lagos High Court ruled against further evictions without alternative arrangements and called for consultation [21]. Where courts cannot bind implementing agencies and land is highly valuable, compensation rules on paper offer thin protection. Eviction can destroy the collective organisation needed to negotiate a future project. Baan Mankong offers another way to recognise residents within land and finance negotiations.

Community-led redevelopment requires recognised negotiating authority and duties to include less powerful households. CODI reports that its community-led programme had covered 142,763 households by 2026 [15]. Communities organise savings groups, negotiate collective tenure, choose among upgrading and nearby relocation, and borrow through cooperatives. A separate railway-side programme reported 3,229 supported households across 76

communities as of August 2025 [16]. These are administrative programme counts, not estimates of net household welfare. They show what changes when the community becomes a contracting and planning party, rather than a list of future allottees. A coalition can still exclude people. A cooperative may prefer long-established members to late arrivals; male household heads may control membership; an owner-occupied solution may exclude renters who cannot service debt. Baan Mankong itself recognises the risk of borrowers falling behind after illness, lost wages or disaster, and has experimented with community housing insurance [17]. The transferable feature is the way residents gain a recognised seat in land and finance negotiations, coupled with duties to include the least powerful households. The political appraisal should identify losses that are otherwise absent from the project account.

3 Enforce resident protection throughout delivery

Interests and constraints in redevelopment

Redevelopment gains and losses are recorded unevenly. Owners and contractors can estimate land gains and payments; tenants may lose customers, childcare, and support that compensation schedules omit. Survey those losses and provide participation routes for people without titles, time, or legal representation. Claims made in favour of clearance should be tested against safer, less disruptive alternatives and enforceable resident treatment.

Actor / rent at stake	Veto or leverage	Possible agreement and verification
Landowner / higher ground rent, sale value	Title dispute, litigation, political influence	Transparent land valuation, phased lease or purchase, clawback of public uplift.
Structure owner / existing room rents	Blocks access; resists tenant registration	Legal room upgrades, fair return on proven capital, tenant-protection conditions.
Tenant, migrant, home worker / location	Collective action; weak individual legal claim	Verified occupancy register, return option, rent bridge, livelihood restoration.
Developer, lender / sale rights and cash flow	Financing pause, bid withdrawal, renegotiation	Bankable phasing, escrowed rehabilitation costs, pay for verified occupancy.
City, utility / service costs and land tax	Withholds connections; can delay approval	Incremental network plan, fiscal sharing, enforceable service covenant.

Actor / rent at stake	Veto or leverage	Possible agreement and verification
Elected representative / delivery commitments and allocation authority	Controls lists, notices or mobilisation	Public dashboards, independent grievance process, cross-party oversight.
Women, caste/race minorities / equal access	Possible exclusion from eligibility or representative bodies; verify locally	Joint records where appropriate, separate hearings, bias checks and appeal rights.

Rents and veto points are hypothesised institutional incentives. Actual control, ownership and legal authority must be established for each site.

A clearance proposal needs evidence of necessity and a complete sequence from claims recognition to verified rehousing. Residents describe a right to the city and decades of labour invested in the place. Homeowners nearby invoke crime, congestion or property values. Developers stress that extra floor area is the only way to finance free rehabilitation. City treasuries point to tax potential; utilities point to overloaded sewers. Assess each claim against evidence and record the costs imposed on affected groups. The proposed process makes distributional obligations explicit at each stage. First, recognise the affected people and the genuine safety risks. Second, set a publicly funded minimum for those who have the weakest claims. Third, price the developer's rights and the city's infrastructure together. Fourth, pay for verified household outcomes through escrow or retained payments. Fifth, protect an independent dispute channel through construction and after occupancy. Skip the third step and the project may be financially impossible. Skip the first or fourth and many former residents may never enter the finished buildings. Longer-term monitoring is needed because announcements and compensation precede many livelihood effects.

Project announcements and compensation payments are visible early, while effects on health, livelihoods, and commuting costs emerge over years. Appraisals should therefore assess compensation alongside recurring costs borne by relocated households. Intergovernmental finance can separate benefits from responsibilities: a national ministry may fund construction while the city pays for sewer operations and landowners receive capital gains. Delivery agreements need to specify who pays each bill and who can enforce each commitment. Track original households wherever they move; rising average neighbourhood income can reflect replacement rather than gains for residents. Rights of return must also account for discrimination and independent claims within households.

Discrimination and rights of return

Resident protection should test unequal access by identity, tenure, and household position rather than price alone. South Africa's long history of racial dispossession placed many workers far from jobs; a relocation that is affordable to build can reproduce that pattern. In

a Delhi online rental audit, landlords replied to 35 per cent of upper-caste Hindu inquiries and 22 per cent of Muslim inquiries. The same study found no similarly clear callback penalty for Scheduled Castes or Other Backward Classes [26]. A migrant without older documents may also fail a redevelopment cutoff [4]. In a redevelopment ballot, a formally equal vote can conceal unequal consequences: a property owner may gain an appreciating flat while a renter loses an affordable room. Welfare analysis must separate these positions. It should also test whether women can hold a tenancy, title or cooperative membership independently. A legitimate safety need still requires comparison with feasible less disruptive options.

This does not make an existing settlement beyond reform. Overcrowding, unsafe structures, contaminated soil and blocked drains impose severe costs, especially on children. In-situ renewal can sometimes be impossible at present densities or hazards. A government must then demonstrate the physical constraint, the alternatives considered, and how affected households will retain access to livelihoods and services. A claimed safety emergency should face independent technical review. The review should distinguish a documented safety need from redevelopment choices that can be addressed through less disruptive alternatives. Digital records should support that comparison without accelerating exclusion before due process.

Digital records and displacement risk

Digital redevelopment systems can improve knowledge while making errors and exclusion harder to reverse. An image can help locate an unrecorded building. It cannot identify every family renting a room, decide who has a moral claim, or measure what a home-based business earns. A roof footprint is evidence of a structure, not a determination of occupancy or a right to remain. Dharavi's official programme describes lidar-supported structure numbering and document collection [4]. The immediate constraint improved is locating and tracking structures at scale. Evaluate whether the system reduces omitted households, speeds appeals, prevents double allocation, and improves residents' outcomes. A private survey contractor and the authority gain new power over eligibility data. Errors, biased documents, opaque scoring and exposure of residents' private information are new risks. Digital speed without due process can make exclusion harder to reverse. Notice, hearings, and auditable procedures remain essential where consequential decisions rely on those records.

Exhibit 18.4. Assessing digital interventions

Application / constraint changed	What to measure; who may gain	Power gained and risk created
Community enumeration: unrecorded tenants	Share of baseline renters found, independently verified; residents gain standing	Enumerators and local leaders can manipulate lists; protect challenges.
AI image matching: hidden structures	False positives/negatives by settlement type; owners and tax authorities gain coverage	Agency gains enforcement power; image cannot prove occupancy or title.

Application / constraint changed	What to measure; who may gain	Power gained and risk created
Job-access map: weak relocation appraisal	Travel time to actual jobs, total fare, lost customers; workers gain better offers	Analyst controls assumed job set; informal/night work may vanish.
Digital rent register: unrecorded rent changes	Change in rents, renewals, evictions and room supply; tenants gain warning	Landlords and agencies gain household data; surveillance and rent collusion risk.
Eligibility portal: slow, opaque claims	Decision time, appeal overturn rate, exclusion by sex and identity	Authority sets rules; automated denial may lock in old bias.

Each technology should be piloted with a counterfactual and published errors. Distributional effects require household follow-up rather than platform activity counts.

Demolition procedures and digital trails must be applied within their legal scope and with accessible routes to challenge errors. India's Supreme Court addressed one part of this asymmetry in its 2024 directions on demolitions: for covered cases it required prior notice, an opportunity to be heard and a digital trail, while identifying exceptions for certain public-place encroachments and court-ordered removals [18]. These procedural protections have limits. A digital portal can preserve evidence of a notice; it cannot guarantee that a person with no smartphone, legal help or formal address can defend the case. Provide accessible offline filing and a timely route to seek a stay from the competent authority, consistent with applicable law. The resident-protection agreement should consolidate those duties with treatment of overlapping claims.

Resident protection agreement

Recognise overlapping claims through locally valid procedures and specify remedies before redevelopment commitments. Fund interim rent, compensation, moving costs and ongoing services as part of the project, with provision for delay. The redevelopment authority must track claims and delivery; an independent mechanism must enforce remedies before irreversible clearance. Track original-household retention or voluntary relocation, time to return, rent and travel burdens, livelihood recovery and unresolved claims. A city should begin by replacing the vague phrase "beneficiary household" with a list of people and enforceable claims. The agreement is a pre-investment contract among government, the project company, utility providers, recognised resident bodies and an independent grievance authority. It governs existing occupants during construction, the allocation of finished units and the cost of staying afterwards. It is a proposed operational design, subject to local law and fiscal capacity. International project standards offer relevant protections whose applicability must be stated accurately.

The World Bank's Environmental and Social Standard 5 distinguishes physical from economic displacement and addresses project-related livelihood losses, including for affected people without formal land rights. It applies to covered land acquisition and land-use restrictions in Investment Project Financing under the Environmental and Social Framework, with the scope and exclusions specified in the standard. The accompanying guidance explains implementation, including the risk that relocation improves tenure while worsening access to livelihoods. Governments considering similar protections elsewhere must establish the applicable legal requirements and funding. [14] A staged operation can translate the applicable duties into site-specific decisions and funding.

A staged implementation sequence

Begin with a bounded operation, resident records, and specialist assessment of identified hazards. The responsible agency should first cost staff time, specialist work, consultation, and verification, then choose the number of sites it can handle. Set dates after that review. Disputed rights, procurement, or statutory approvals may extend the schedule; do not advance until the applicable legal and funding conditions are met. Stage 1: establish the denominator and the authority. Have the competent authority establish lawful protection against retaliation and contested removal while a full list of households, tenants, subtenants, businesses and vulnerable people is prepared. Create the parcel–structure–occupant link, with community enumerators, third-party checks and offline appeals. Collect sex, migration status, and disability information only where lawful and necessary, with informed consent where applicable, protected access, and a documented retention period. Screen for fire, flood, structural and contamination risks, then commission the specialist assessments indicated; publish hazards without revealing personal identities. The next stage compares actual alternatives and tests their financing and occupancy conditions.

Compare feasible options before tying financing and payments to the promised resident outcomes. Stage 2: compare real options. Model on-site phasing, nearby rehousing, serviced rental additions and distant resettlement against each other. For a representative sample of households, record door-to-door job travel, fares, earnings, childcare, school and health access, commercial turnover, current rent and service costs. For every site, calculate housing welfare under at least two plausible employment scenarios. Draw up a rent and vacancy baseline in nearby markets. Stage 3: finance the full agreement. Have an independent adviser calculate the developer's residual land value and the public value of land, density rights and new infrastructure. Price temporary accommodation, rental bridges, livelihood assistance, affordable commercial space, replacement assets, maintenance and operating funds. Set aside protected financing for these obligations before financial close. Test whether lenders will accept release of some payments only after verified return and occupancy. Rights, services, and remedies then need enforceable agreements before works proceed.

Make the treatment of residents enforceable within one auditable delivery plan. Stage 4: make rights enforceable. Publish eligibility categories and their evidence requirements; grant a dated appeal decision and a stay mechanism for contested removal. Sign a return or relocation offer with each affected household, identifying the floor space, tenure, maximum initial housing

cost, services, time frame, remedy for delay and whether a business can reopen. Provide separate representation for tenants, women, migrants and disabled residents, since their interests may diverge from those of title holders. Stage 5: contract construction and stewardship. Tender or amend works in phases that minimise forced movement. Require verified service capacity and safety inspections before occupancy. Set a condominium, cooperative or rental management structure and a funded reserve. Put the city, utility, estate manager, residents and developer under one auditable delivery plan, with specific remedies if a promised service fails. Public monitoring should follow the original households through the resulting moves and occupation.

A redevelopment result remains unknown without records of affected households' subsequent housing and livelihoods. Stage 6: create the public test. Launch the first phase only after an external verification confirms baseline coverage, compensation funding, agreed design, grievance independence and viable alternatives for those whose original homes are unsafe. Budget regular aggregate follow-up of the baseline cohort, including renters who left; choose intervals and duration from the expected effects, and report attrition; protect identifiable household records. A completed building without this record is an output whose social result is unknown. The evaluation should include people who leave, lose eligibility, or do not return.

Monitoring resident outcomes

Exhibit 18.5. Indicators for evaluating redevelopment

Outcome measure	Operational definition	Failure signal
1. Baseline coverage	Share of independently estimated residents, renters and enterprises rostered	Structures counted; tenants absent.
2. Tenure/security	Written enforceable claim and appeal available by claimant type	Approval status depends on a broker.
3. Neighbourhood retention	Share of original households within agreed travel-time catchment at 1, 3, 5 years	Success reported only for eligible owners.
4. Affordability	Housing + service + commute cost as share of disposable income, by decile	Free unit with unaffordable monthly dues.
5. Livelihoods	Real earnings, hours, travel time, lost trading days and business survival	Stable physical shelter, falling incomes.
6. Renter market	Rent distribution, vacancy, lease renewal and new room supply	Sharp rent rise after infrastructure works.
7. Distribution of uplift	Land sale value, rents, tax, developer return and net tenant benefit	Land gains opaque; service costs public.

Outcome measure	Operational definition	Failure signal
8. Claims and fairness	Appeal time, reversal rate and outcomes by sex, caste/race where lawful	High rejection and no effective remedy.
9. Stewardship	Water/sanitation uptime, verified structural safety, funded reserves	Occupancy before networks or operations and maintenance.

Suggested management indicators; local baselines, legal safeguards and thresholds must be set before approval. All rates use the full baseline population as a tracked cohort.

The strongest test follows the people the project originally affected, including households that chose a different address, lost eligibility or moved out after rent increases. It compares outcomes with similar, independently sampled communities where possible. A randomised phasing experiment may be feasible for some infrastructure or rent-support instruments; it would be unethical to withhold emergency safety work from a known high-risk site. A credible evaluation can use staggered delivery, matched comparison locations, independent rent surveys and transparent attrition reports. The aim is a plausible counterfactual, not a neat causal claim from weak data. The cases can then inform a conditional approach to improvement rather than a general preference for clearance or preservation.

Improving neighbourhoods while protecting residents

Redevelopment should combine evidence of physical need with funded rights, alternatives, and follow-up of affected residents. Their practical use is to test whether affected households have enforceable claims, usable alternatives, and access to decisions before redevelopment proceeds. Record the remedy and responsible institution for each unresolved claim. Check the omissions that can distort redevelopment appraisal: tenants absent from title records, lost customers excluded from compensation, commuting costs beyond the relocation grant, and rent increases after service improvements. Record these effects alongside land, construction, and infrastructure costs. Residents need a route to challenge both their own record and the assumptions used to assess the project. A city must create space for newcomers as well as protect people already there. Incumbent rights can be drawn so tightly that no new worker can enter, and redevelopment can be carried out without adequate protection, displacing low-income residents. A durable settlement gives existing residents a fair claim, adds competing housing supply, assigns the costs of services, and lets households choose between credible ways to improve their lives. It also accepts that some dangerous sites must change, with evidence, compensation and workable alternatives. Refugees, displaced households, and host communities require additional attention to durable access outside designated camps.

Refugees, displaced households, and host communities

Urban displacement responses should follow housing outcomes and unresolved claims across renters, hosts, and informal settlements. Start by checking the rights that apply locally to residence, work, renting, assistance, and documentation. Distinguish refugees, internally displaced people, other migrants, and host households; their legal positions and assistance channels differ. Assess the rental market, available services, and host-community pressures before choosing rental assistance or construction. UNHCR's integration guidance treats housing access as a combination of suitable accommodation, support, information, and links with housing providers. Its resettlement context is not a universal model for every displacement emergency. [211] The proposed local task is to establish lawful tenancy evidence, a safe complaint route, access to schools and work, and a plan for assistance expiry. Check rents and evictions for both recipients and comparable host households. Keep protection records separate from public housing lists and restrict onward sharing. Where documents were lost, obtain specialist advice on acceptable alternative evidence. After a disaster, distinguish temporary shelter from repair, rebuilding, and voluntary relocation; verify land claims and fund the transition before temporary support ends. Report durable housing outcomes and unresolved claims, not just emergency places provided. Fieldwork must track those households at their destinations as well as within the original project boundary.

Prepare the local decision

Verify claims, options, compensation, livelihoods, school continuity, and grievances before and after relocation. Enumerate households, tenants, subtenants, businesses, shared facilities, and overlapping claims before eligibility decisions. Record income sources, customers, schools, care arrangements, travel costs, and housing payments. Retain a confidential baseline and a way to follow households who leave. The relocation evidence in this chapter requires comparing the original residents' outcomes with those of later entrants. Meet owners, tenants, traders, women, and residents with disabilities separately before negotiating a common plan. The implementing agency should publish eligibility evidence, cutoff rules, remedies, and responsibility for temporary housing and return. An independent adviser should help residents understand choices and contest omissions; attendance at a meeting is not proof of informed agreement. Use geotagged records to support a claim, with alternative evidence for people absent during enumeration. Preserve disputed entries and their history. A model must not determine entitlement from roof type or a missing digital identity. Authorise relocation only after the agreed replacement and assistance are usable. Follow costs, livelihoods, school continuity, tenure, and unresolved grievances at the destination, including households that do not return. Chapter 19 examines municipal rental policies that can widen access without requiring households to become owners.

Redevelopment should preserve or improve residents' feasible access to housing, livelihoods, and services as land values change. Recorded claims, genuine choices, accessible destinations, and enforceable interim and return arrangements support that result. A tower ratio or eligibility cutoff can instead exclude users without ownership documents. Uncosted livelihood losses and weak return enforcement will remain hidden if monitoring follows only

those who receive the completed units. At 100 times pilot size, test the stock of interim homes, claims-resolution capacity, school continuity and the time households wait to return.

Sources and evidence notes

Evidence and scope. Administrative figures describe the reporting programme. Causal estimates retain the populations and settings of the cited studies. Simulations and worked financial examples are labelled illustrative. Court judgments establish the decisions reported, not subsequent household outcomes. The references identify public websites, judgments, working papers, and journal articles.

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Evidence hierarchy. Programme pages and audits report their own activities; they rarely measure net welfare. The Peru, Argentina, Brazil and San Francisco studies use quasi-experimental or experimental methods, with distinct estimands. The Mumbai welfare figures are predictions from a fitted discrete-choice model, and the Mumbai/Mathare/Joe Slovo/Mukuru cases also draw on qualitative and legal evidence. Transfers in Exhibit 18.2 and the Nairobi example are hypothetical sensitivity tests, with no claimed field calibration.

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19. Rental markets and municipal policy

Municipal rental policy should improve entry, habitability, and security while keeping additional supply and maintenance financially viable. Diagnose the local barriers facing tenants and producers before choosing regulation, credit, or subsidies. The municipality then needs responsibilities and revenue consistent with the density and services it permits. Evidence on rental occupancy establishes the scale of that task.

Nairobi's rental evidence makes tenancy central to housing policy while leaving the effect of job proximity to be tested. The Kenya National Bureau of Statistics' 2015/16 household survey put the share of rented dwellings in Nairobi at 86.4 per cent. India's 2018 National Sample Survey found that 63.8 per cent of urban households occupied a dwelling they owned, leaving a large minority with some other tenure arrangement. South Africa's 2022 census recorded 23.2 per cent of households renting nationwide and 36.6 per cent in Gauteng, its main urban jobs region [1-3]. Different dates, definitions and geographies limit comparison. The figures show that renting is a major housing tenure in these locations. They do not, by themselves, establish how proximity to employment affects rental demand. Different household needs require rental products and protections suited to their circumstances.

1 Understand rental demand, costs, and producers

Rental demand and ownership-focused policy

Rental policy should accommodate varied household incomes, mobility, and living arrangements. A new worker may need a room before considering a mortgage. A street vendor may need to move after the market changes. A couple with uncertain wages may choose a short lease rather than commit its savings to a distant flat. An older owner may depend on rental income from two rooms at the back of her plot. A housing programme that offers each of these households the same subsidised ownership product will fail some of them even if its contractor delivers on time. Rental supply depends on agreements among tenants, landlords, utilities, lenders, planners, and the state. This chapter examines how to support additional safe rooms while keeping entry costs and recurring payments affordable. The small landlord's investment decision determines whether those options can expand.

Additional rental supply depends on viable permissions, services, finance, and allocation of gains between owners and tenants. Consider a landlord with a legally occupied plot near a bus route. She could add six rooms in the yard. Prospective tenants would pay monthly. A bank, however, wants titled collateral and a clear repayment stream. The city wants a plan, a drainage connection and a development charge. The landlord wants the option to recover a room when rent goes unpaid. Tenants want protection against a sudden rent increase or eviction after they have paid for improvements. The electricity distributor wants a safe meter. No single actor can settle the whole agreement. If the transaction cannot be completed, households still rent: in overcrowded houses, unsafe backyard structures, or distant districts where the commute consumes their wages. This chapter treats rental housing as a system of contracts

backed by land and services. It links the household's weekly cash flow to the developer's financing terms and the city's legal and fiscal choices. Renters, owners and the public can receive different benefits and bear different costs when supply expands or a neighbourhood improves. The household comparison should begin with what rent actually purchases.

Rent, services, and household costs

Rent purchases shelter, services, location, and security whose combined value differs across households. The first is shelter: floor space, roof, ventilation, water and safety. The second is location: access to work, school, care, customers and transport. The third is flexibility: the right to leave without selling a large asset. The fourth is protection against risk, including eviction, service failure, violence and unpredictable charges. A low rent can conceal a large cost in travel time, illness, danger or insecurity. Equally, a modest room near work may yield better household welfare than a spacious subsidised flat far away. Residual income after these costs provides a more useful affordability test than rent alone.

A useful starting measure is the residual income after housing and access, not an isolated rent-to-income ratio. Suppose a worker earns KES 24,000 a month. A room at KES 6,000 within walking distance of work, with no commuting fare leaves KES 18,000 before utilities and other essentials. A room at KES 4,000 that adds KES 3,000 in fares and two hours of unpaid travel a day can be economically dearer. These figures are an illustration; household surveys must measure the actual trip pattern, including the value of time, uncertain earnings and the cost of childcare. A fixed 30 per cent affordability rule is a screening device. It is a weak substitute for estimating whether a household can pay for food, transport, education and emergencies after its rent. Landlords need a corresponding calculation of income, costs, and risk for the product being supplied.

Figure 19.1. Lower rent can be offset by transport fares



Author's hypothetical calculation using the rent and fare assumptions in the text. These are not observed Nairobi market data.

Different landlord types require different financial and regulatory responses. Contract rent becomes cash only if a unit is occupied and payments arrive. Gross receipts must cover

cleaning, security, repairs, water shared in common areas, waste, insurance, property charges, arrears and periodic roof or pipe replacement. Debt is paid from the remainder. Lenders and tenants both need to test whether payments remain manageable in costly or low-income months. Ignoring these cash-flow risks can leave a project unfinanceable or require rents beyond the intended households' means. There are at least four rental submarkets in an African or Indian city. A room in an informal settlement may rest on disputed land and be let by a structure owner who has no title. A backyard room may sit behind a formal homeowner's house. A small walk-up block may have permits and utility connections but depend on an individual investor's savings. A social rental estate may use public land, capital grants, regulated rents and professional management. Each has different costs, tenants, collateral, political power and enforcement risks. One law and one credit product cannot serve all four equally well. The city needs tenancy information as well as ownership records to assess those responses.

Exhibit 19.1. Rental market segments

Segment	Real constraint	Who may take the gain	Main tenant risk
Informal settlement rooms	Disputed land; unmetered utilities; weak proof of tenancy	Structure owners; intermediaries; informal water sellers	Eviction; high utility mark-ups; fire and flood
Backyard rooms	Serviced plot; plans; staged building credit	Existing homeowner; land seller; contractors	Crowding; unequal bargaining; overburdened services
Small formal walk-ups	Land, approvals, long-term debt and lease-up	Micro-developer; bank; landowner	Higher asking rents; screening and deposit costs
Regulated social rentals	Public land/subsidy; professional management; collections	Tenant where subsidy is enforced; otherwise suppliers	Allocation bias; poor upkeep; insolvency

Rental policy requires information on actual tenancies as well as ownership. A property map can reveal a roof and sometimes its floor area. It does not reveal the number of paying households, the chain of subletting, the rent, the arrears history, the condition of a toilet, or whether a tenant can leave safely. These are facts that households and landlords may have reasons to conceal. A programme that equates registered ownership with housing need risks excluding tenants or directing benefits to owners whose needs differ. South Africa’s small-producer experience shows how these relationships shape financing and construction.

Small landlords as housing producers

Small-rental production depends on staged finance, local relationships, and enforcement arrangements that must be examined rather than idealised. A homeowner trades some yard space, privacy and construction savings for monthly income. Jan Brueckner, Claus Rabe and Harris Selod used a model and satellite-derived counts of backyard structures to test this choice; they found results consistent with more backyarding where jobs were easier to reach [4]. Their finding is evidence about the association predicted by a model, rather than proof that any given planning relaxation will create a specified number of safe rooms. It does identify the scarce good many tenants pay for: proximity to work. Field research by Scheba and Turok finds a sector that can improve physical buildings as informal landlords reinvest rental proceeds. It also documents persistent safety, infrastructure and legal risks [5]. Research in eight Cape Town neighbourhoods finds that family ties, safety and relations between homeowners and tenants affect the market alongside rental yields [6]. A landlord may accept a relative who pays late; another may select tenants perceived as less likely to challenge authority. The financial model must therefore include trust and informal enforcement without assuming that these arrangements are fair. Public support should then be judged by additional safe occupied rooms rather than participation alone.

Lower entry barriers should be evaluated through net additional affordable and compliant occupation. Cape Town approved a R20 million Development Charges Fund in March 2024 for qualifying micro-developers in 194 targeted lower-income areas. Its stated design covered 50-90 per cent of applicable development charges for developments of three to 30 units, subject to readiness conditions [7]. In August 2025 the city gazetted planning amendments, with the new affordable-rental-flat provisions taking effect in September. In designated areas, the 2025 framework enabled up to eight small rental units alongside a house, or twelve where there was no main house, subject to conditions [8]. These are policy provisions, rather than measured housing outcomes. The net addition of occupied, affordable, code-compliant units must be measured separately. Predictable rights can change land values and returns as well as the number of rooms built.

Permission and finance affect the distribution of benefits through supply response and market power. Relief from a development charge cuts the cash needed before rent begins. Technical support can lower the cost of meeting fire, structural and service rules. Together, they may shift investment from one informal room at a time to a safer, planned building. Yet several claims compete for the saving. The homeowner may keep a higher return; the seller of a well-located plot may raise its asking price; contractors may charge more; or tenants may see lower rents if other landlords can also build. The distribution depends on supply response and market power, not on the programme's intention. Producer coalitions therefore need tenant participation when rules are designed and reviewed.

Cape Town’s micro-developer coalition illustrates an institutional route for smaller producers to influence municipal decisions. A 2025 study of Cape Town's emerging micro-developer alliance describes a coalition of small property entrepreneurs, civic groups, officials and other specialists that opened channels into municipal decision-making [9]. Such representation

can affect which proposals officials consider. Larger developers may have more resources to prepare projects and seek changes to rules, while small builders may have limited contact with officials outside inspections. Organised micro-developers can make a credible case for clear rules, affordable finance and safe infrastructure. Tenant organisations should participate in setting and reviewing the rules so that the reform addresses tenant interests alongside those of property owners. Access to credit remains a separate constraint requiring a viable project and shared lender risk.

Guarantees can help a sound small-rental project cross a funding barrier but cannot make its rents viable. Interviews with South African small-scale developers reported by Viruly and co-authors, and further work published in 2026, identify lending products poorly matched to how these builders acquire land, construct in phases and collect rents [10]. A conventional mortgage finances a finished asset against a salary or title. A small landlord needs staged construction money before the completed rooms generate cash. The lender must understand permits, cost overruns, lease-up, payment histories and a possible title dispute. Short unsecured loans force high monthly debt payments; expensive equity pushes up the rent needed for a project to proceed. A guarantee cannot make an unviable project viable or provide missing utility capacity. It can help a sound project cross a collateral or tenor barrier if the lender shares losses and has reason to monitor it. A worked building account shows where that distinction matters.

Financial model of a twenty-room building

A rental appraisal should test debt coverage at affordable rents before assuming that credit will unlock construction. Take a hypothetical twenty-room building on a small urban plot. The owner already holds the land. The proposal costs KES 16 million in construction, service connections, design, compliance, contingencies and initial letting expenses. The land could be used for another purpose and has an assumed economic value of KES 4 million. Thus, the full economic investment is KES 20 million even though the new cash outlay is KES 16 million. None of these amounts is a market estimate for Nairobi. They are deliberately transparent assumptions for testing a deal. The owner secures a hypothetical KES 6.4 million construction-to-term loan, equal to 40 per cent of the new cash cost. Suppose the lender agrees to a fixed 12 per cent nominal annual rate and fifteen years of level monthly payments. Monthly debt service is about KES 76,811. Assume 92 per cent physical occupancy and an operating-and-renewal provision of 25 per cent of rent actually collected. At KES 6,000 per occupied room, annual rent collections are KES 1.3248 million; monthly net operating income after costs is KES 82,800. Debt-service coverage is only 1.08. A bank asking for 1.25 coverage would reject the project at this rent even before testing the landowner's return. The owner’s equity and land contribution must also be included in the return calculation.

Exhibit 19.2. Illustrative cash flow for a twenty-room building. Assumptions, not a Nairobi market survey.

Monthly rent	Occupancy / costs	Monthly NOI	Loan payment	DSCR
KES 6,000	92% / 25%	KES 82,800	KES 76,811	1.08
KES 7,500	92% / 25%	KES 103,500	KES 76,811	1.35
KES 8,500	92% / 25%	KES 117,300	KES 76,811	1.53
KES 7,500	80% / 30%	KES 84,000	KES 76,811	1.09
KES 6,000 + KES 3m capital grant	92% / 25%	KES 82,800	KES 40,806	2.03

NOI = collected rent after operating and renewal provision; DSCR = NOI / debt service. Base loan: KES 6.4m at 12% nominal with 180 equal monthly payments. Grant case replaces KES 3m of debt. All values rounded.

Project returns should account for land, equity, delays, and operating costs alongside loan payments. The owner has KES 9.6 million of cash equity in the build plus KES 4 million of land at risk. At a rent of KES 7,500, net cash after debt service is about KES 320,000 per year, before income taxes and any major surprise. That is roughly 2.4 per cent of the KES 13.6 million owner contribution on a simple annual cash basis. Debt principal repayment adds equity over time, and land values can change; these benefits remain outside the cash-yield measure. The calculation explains why credit availability alone may yield little low-rent supply. With the assumptions above, rent must reach roughly KES 6,958 for debt-service coverage of 1.25. That is 29 per cent of the illustrative KES 24,000 income, before utilities. If occupancy drops to 80 per cent and operating costs rise to 30 per cent of collections, a rent of KES 7,500 produces coverage of about 1.09. A six-month construction delay on the original KES 6.4 million loan adds roughly KES 384,000 in simple interest at 12 per cent, before capitalised fees and inflation. Administrative delays increase financing costs and can create pressure to raise rents or reduce construction spending. A capital subsidy can then be evaluated by the obligation it removes and the benefit it purchases.

A grant replacing debt can change cash flow, but its accounting cost per occupancy-year is not proof of additional impact. Now suppose the public sector contributes KES 3 million as a one-time, verified capital grant and requires it to replace debt rather than private equity. Loan principal falls to KES 3.4 million and monthly debt service to about KES 40,806. At KES 6,000 rent, coverage rises to 2.03. Yet the grant may simply lift the owner's return if rents are left unregulated. The public investment works only if the project is genuinely additional, rents are capped for a stated period, maintenance is funded, selection is fair and repayment or clawback follows breach. Over ten years, KES 3 million spread mechanically across twenty rooms amounts to KES 1,250 per potential room-month before discounting, vacancy or displacement. That

accounting figure is a way to compare instruments, not proof of impact. Tenant-side assistance provides an alternative that should be compared under the same assumptions.

Compare producer and tenant support through additional access, public cost, risk, and the capacity to sustain supply. Compare a tenant-side payment of KES 1,500 a month for twenty eligible households over ten years. Its undiscounted cost is KES 3.6 million; discounted at an illustrative 5 per cent annually with payments at year-end, the present value is about KES 2.78 million. A capital grant and a rental voucher have different uses. A supply grant might create stock where a building otherwise fails to clear the financing threshold. A voucher can help families choose from existing safe units where vacancies and landlord participation permit. The public buyer should fund the constraint it can identify, and test whether the benefit reaches tenants rather than being absorbed in land prices or higher rents. Cheaper blocks or prefabricated walls change one component of KES 16 million. Savings help most when land, permits and infrastructure remain available and competing builders can enter. A digital design tool that cuts errors by a measurable percentage may improve the cash flow; it cannot repair an absent sewer, confer title or guarantee that the landlord offers an affordable lease. Chapter 20 will address these financing and risk-sharing choices across ownership and rental housing. India's migrant-rental programmes illustrate why available units and administrative allocations can still differ greatly.

2 Compare policy choices through access and supply

Rental housing for migrants in India

Migrant rental programmes require agreements over operating losses, eligibility, and complaints as well as identified buildings. The government responded in 2020 with Affordable Rental Housing Complexes (ARHCs). One model sought to convert existing government-funded vacant homes through public agencies or partnerships. A second supported construction and operation on public or private land. The official scheme envisaged one-bedroom units and dormitory beds, with rental use preserved for at least twenty-five years [11]. The stated beneficiaries included industrial workers, street vendors and other low-income urban migrants. The conversion model offered a logical agreement: government-owned units already stood vacant, and workers needed somewhere to live. But a vacant flat has little value to a worker who cannot afford the journey to a job, cannot assemble the deposit or would be separated from co-workers. The costs of retrofitting, operating shared areas, screening residents and managing turnover still have to be paid. An allocation plan must therefore identify who will cover operating losses, decide eligibility and resolve complaints. Allotment figures help reveal how much of the identified stock has entered use.

Vacant government stock should not be counted as effective rental supply before it is made usable and allotted. A written parliamentary reply reported in December 2024 indicated that 5,648 units from the stock of 83,534 identified vacant government-funded houses had been allotted to migrant workers - about 6.8 per cent of the identified stock. This is a dated administrative snapshot, not an occupancy estimate for all ARHC models or for 2026. The same reporting distinguished new-build sanctions and completions from converted and allotted stock [12]. The large gap between identification and allocation points to a series of gates: legal control,

conversion cost, tenders, usable services, local demand and the authority to allocate and manage tenancies. Counting identified stock as housing supply would obscure all of them. New rental programmes must assess the destination's services and suitability for intended residents.

Rental design should be tested against transport, safety, work schedules, and household use rather than bed counts alone. PMAY-Urban 2.0, approved in 2024, retained an Affordable Rental Housing vertical alongside beneficiary-led construction, partnership housing and interest subsidies. The current official design recognises urban migrants, women workers and the need for civic and social infrastructure [13]. Its results will depend on whether states and cities can identify employment-linked sites, package viable management contracts, keep rents affordable and measure actual occupied room-months. A lease for a dormitory bed entails different privacy, safety and gender needs from a family home. A design that optimises the number of beds can still fail for women working late shifts if transport and security are missing. A lawful tenancy must also be inexpensive enough to obtain and enforce.

Rental-law implementation should be assessed through the time, cost, and accessibility of obtaining a remedy. India's 2021 Model Tenancy Act proposes written agreements, security-deposit limits, specified obligations and dedicated rent authorities and courts [14]. It is a model circulated for state adoption, without automatic force across India. Government statements in December 2023 reported that four states had revised tenancy laws along its lines; adoption and operational performance must be checked state by state [15]. Even where registration is mandatory, a contract that requires a landlord and tenant to navigate an expensive process may push low-value tenancies further off the record. A useful implementation test is how much time and money a woman renting one room spends to obtain, enforce or terminate a lawful agreement. Transitions from older regulation need particular protection for vulnerable sitting tenants.

Rental reform should fund repairs and protect vulnerable incumbents while giving new investment clear rules. Tandel and co-authors describe how stringent historical rent controls protected incumbent tenants while reducing incentives to maintain buildings and invest in new formal rental supply. Informal key-money transactions and other workarounds emerged [16]. The study is a detailed institutional and historical analysis, not a causal estimate of the size of the supply loss. It nonetheless identifies a credible mechanism: when rents cannot cover renewal costs and contracts are hard to adjust, owners try to avoid the regulated activity. Removing protection overnight would shift the cost onto incumbents who have built their lives around an old contract. A feasible reform needs a transition that protects vulnerable existing renters, finances building repairs, and gives new rental investments clear rules. South Africa's informal and regulated sectors show why several production and operating models may coexist.

Two rental housing systems in South Africa

Different rental institutions require separate subsidy and operating accounts, including the value of public land. Each fills a gap left by conventional ownership programmes. Their finance differs sharply. The backyard owner may use savings and small loans to erect rooms on land already held by the family. A social housing institution assembles land, a

capital subsidy, equity and debt, then manages tenants and common areas for years. The second route can provide professional maintenance and access to central locations; it requires an organisation able to collect rent, address grievances, service debt and renew assets. The Social Housing Regulatory Authority describes a capital-grant instrument intended to make regulated rents viable in well-located areas. Its published investment information gives an indicative development cost of R426,000 per unit and a grant quantum of R328,867, with the balance financed through debt and equity; these are programme benchmarks, not verified costs for every project [17]. This large public share makes the conditions of allocation, rent setting and long-term upkeep decisive. Cheap public land adds a further subsidy that financial reports should value separately. Their performance can nevertheless be compared through common household and service outcomes.

Evaluate rental models by occupied safe homes, household costs, location, tenure, and public cost per year of use. South Africa's Department of Human Settlements reported 1,364 social rental units delivered in FY2024/25 against a target of 2,699, or 51 per cent. It cited approval and contracting delays, difficulty securing debt, bulk-service availability and construction-cost pressures. Institutions also tenanted 2,661 units against a distinct annual tenanting target; those tenancies can include units completed in earlier periods [18]. Targets, completions and letting therefore describe different processes. They cannot be combined into a single conversion rate. Regulated estates and backyard rentals require different forms of oversight. A well-managed regulated estate can give tenants predictable repairs and an independent institution to hear complaints. It can also fail if collection systems, security costs or the maintenance reserve weaken. An informal landlord may build quickly and know every tenant; that same personal control can turn into harassment, discriminatory selection or unsafe repairs. A city needs multiple tenure options and different oversight instruments. It should evaluate each by a common set of outcomes: occupied safe homes, rent-to-residual-income, location, tenure security, service uptime and the public cost per year of occupancy. Distressed operators also need a credible agreement on utilities, payment, and dispute enforcement.

Operating stability requires reciprocal duties from rental providers, residents, and public agencies. The 2025 national policy statements acknowledged rent boycotts, criminal interference with construction and the need to stabilise distressed rental institutions [19]. These problems require attention to revenue collection, service performance and lawful enforcement. A capital grant cannot protect a lender if a building cannot collect revenue. Enforcing payment obligations also requires a process for resolving documented service failures. A landlord who receives public support should therefore accept transparent billing, independent complaints, audited reserves and a tenant voice in service decisions. The public agency must in return provide predictable utility service and dispute enforcement. The ordinary lease should therefore be supplemented by clear municipal service responsibilities.

Tenant rights and municipal responsibilities

Rental governance should include service duties and accessible receipts and remedies for households without digital access. A city requires the landowner to pay for bulk water and sewer connections. A landlord passes part of the charge through rent or a separate utility bill.

Tenants may pay an informal reseller above the utility tariff and still receive poor service. When a toilet backs up, the resident blames the landlord; the landlord blames the utility; the utility says the drain is private. Each actor may have a technically defensible account of why it has failed to act. For small rental blocks, link the lease to an agreement on service responsibilities. The owner should demonstrate lawful control of the land, structural and fire safety, capacity for water and sanitation, metering arrangements, maintenance responsibility and a route for complaints. The municipality should disclose the network's spare capacity, connection time and charge before a builder commits money. Tenant notices should state the rent, deposits, separate utility charges, annual adjustment rule, service standards and a dispute route in a language the resident understands. Digital payment records may strengthen proof of tenancy and income. A paper receipt and an offline appeal must remain legally effective for tenants without a bank account or smartphone. Compliance and additional network costs need a proportionate financing rule.

Share compliance and service costs so that neither newcomers nor existing households face an unworkable burden. A building permit that costs the same administrative effort for six rooms as for six hundred will deter the small investor. Proportionate, risk-based standards can protect residents while allowing standardised plans, staged inspections and shared technical assistance. Structural calculations, fire egress and sanitation capacity still require professional judgment; a planning amnesty alone cannot make an unsafe building sound. Municipalities can pay part of the cost of network upgrades where the public benefits from density. They must then decide how to share those costs among old and new households so established owners do not subsidise newcomers indefinitely and newcomers are not priced out before moving in. Affordability conditions attached to public support should be distinguished from general rent regulation.

Publicly purchased affordability obligations and market-wide rent ceilings are different instruments with different supply effects. A city can allow more rooms near transit, improve drainage and reduce development charges. Existing landowners are well placed to capture the first rise in land value. A tenant's wage does not rise when a plot's value does. To preserve a public return, cities can attach time-bound rent conditions to targeted grants, require independent verification of new units, fund portable assistance for tenants in severe need and invest in neighbourhood services with clear operating budgets. Blanket rent ceilings imposed on an expanding private market are a different instrument, with consequences for the incentive to supply and maintain space. Regulation should be evaluated separately for sitting tenants and prospective entrants.

Rent regulation and rental supply

Rent regulation affects current and prospective tenants differently. Sitting tenants can identify an immediate rent increase and organise around it. Prospective occupants of housing that has not yet been built are harder to identify and represent. Planning and rent-setting decisions should therefore report both protection for existing tenants and the expected effect on new supply, then test those expectations against actual construction and withdrawals. Chapter 5 sets out the San Francisco rent-control estimates and their limits. For municipal implementation, use that evidence to test owners' options to convert, redevelop, or withdraw rooms, and separate

protection for sitting tenants from access for new tenants. The local supply response must be measured. [20] The menu of protections is broader than an unrestricted market or a permanent nominal freeze.

Choose tenant protections that households can use without costly litigation and test their maintenance and supply effects. They include predictable rent adjustments linked to verifiable inflation and maintenance, limits on arbitrary eviction, time to cure arrears, temporary shock assistance, deposit guarantees, protections against discrimination, and an explicit transition for tenants under historic controls. Each rule needs an affordable enforcement route for low-value tenancies and an assessment of its effect on future rental supply. Protections that depend on expensive litigation may be difficult for low-income tenants to use, especially when eviction is imminent. Landlord selection can still exclude applicants even when the advertised price is affordable.

Rental access requires evidence on discriminatory selection as well as price and legal supply. Datta and Pathania ran an online audit experiment of the Delhi rental market. Identical inquiries with upper-caste Hindu names received landlord responses about 35 per cent of the time; those with Muslim names received responses about 22 per cent of the time [21]. The 13-percentage-point gap was produced by differential signals of identity rather than differences in the advertised ability to pay. The result measures this audit setting, and the authors note limits in matching all replies to listings. Still, it demonstrates that a subsidy can fail before a price is negotiated if landlords deny access to the intended group. Caste, religion, race, gender, marital status and migration status can all shape the informal rules of access, although their relative effects require local evidence. Enforceable complaints need independent low-cost testing and privacy protections.

Making discrimination complaints enforceable needs a low-cost testing and remedy system, independent of landlords and property agents. A rental-matching platform that simply reproduces owners' preferences can deepen segregation. An eligibility algorithm trained on past lettings may reproduce earlier exclusion if its data or predictors encode landlord selection rather than repayment risk. Any public marketplace needs a documented policy on protected attributes, a way to audit rejection rates, a human review channel and real penalties for repeated exclusion. Protect privacy: a complete electronic record of where informal migrants live is a valuable administrative asset and a possible instrument of harassment. A durable reform agreement must include those remedies alongside the interests of providers and municipalities.

3 Implement a funded municipal rental programme

Interests in rental reform

Rental reform needs institutions able to withstand disputes, budget pressure, and defects without presuming misconduct from opposition. In Cape Town, small builders sought legal routes to add rooms, civic groups supplied technical and social knowledge, and officials saw a way to expand housing through private capital [7-10]. Homeowners gained a possible income stream. Contractors gained work. The city could gain rates and metered utility

revenue over time. Renters could gain supply and quality if competition and oversight worked. This is a plausible reform coalition, documented as a policy process; the final effects on rents, safety and displacement remain an evaluation question. Potential disagreements include tenant protections, permit thresholds that exclude small builders, deposit rules, and the cost of additional utility connections. Identify affected income streams and decision powers through local evidence. Record threats or interference through appropriate reporting channels; do not infer misconduct from opposition to a proposal alone. Documented control over permits, fees, inspections, and connections provides the basis for that agreement.

Before reform, document who controls permits, allocations, inspections and utility connections, and what fees each decision permits. Use audits, complaints and other reliable records to investigate alleged unofficial payments or preferential treatment. A digital portal changes only part of this process; discretion may remain at inspection or connection. The reform needs a responsible authority able to resolve disputes and enforce published decisions, with safeguards for affected households. A bounded sequence can then demonstrate benefits while containing implementation risks.

Exhibit 19.3. Interests and constraints in rental reform

Actor	Benefit / rent at stake	Power to obstruct	Credible reform exchange
Existing landowners	Plot uplift; extra rental income	Refuse sites; litigate	Predictable rights, conditional on safety and leases
Small builders & lenders	Rent yield; fees; collateral	Withhold projects and loans	Staged loans with audited approvals and cash flow
Incumbent tenants	Stable rent; proximity to work	Protest; legal action	Cure periods, repair rights, lawful adjustment rules
Migrant & excluded renters	Access; flexible tenure	Limited representation in local decisions	Portable aid, no-discrimination testing, grievance desk
Utilities & municipal finance	Connection income; rates	Deny capacity, delay connections	Pre-funded upgrades, transparent tariffs and revenue sharing
Intermediaries	Fees and allocation influence	May influence permits, allocations or inspections	Publish awards and complaints; independent oversight

Begin rental reform where gains can be verified and protection and service commitments can be enforced. Legalise a narrow class of safe small rental units on adequately serviced plots; publish the standards and fees; make one-stop technical assistance available; offer finance only to applicants with verified plans and utility capacity; require simple leases and tenant notice; and monitor both rents and displacement in nearby streets. Organised tenants and women workers should see the data before new powers are given to landlords or enforcement officers. An area with disputed ownership, frequent forced eviction or no drainage calls for tenure mediation and infrastructure first. Before expansion, confirm that the pilot can maintain the services its new tenants need. Measure access to jobs and schools for new tenants, including those excluded from owner-focused programmes. Subsidy choice should follow the specific remaining barrier to household entry or viable production.

Choosing the recipient of rental subsidies

Specify the constraint before choosing whether assistance should go to the tenant, provider, or service network. A family may need help with a deposit and the first three months' rent while changing jobs. Another faces a permanent gap between disability benefits and the cost of a safe room. A small developer may have enough tenants at an affordable rent yet lack a fifteen-year loan or water connection. An estate in a central location may need a land concession and operating support because commercial returns will never cover the target rent. These are different problems with different instruments. Deposit guarantees, operating support, capital grants, and public leases address different parts of that problem.

Match the rental instrument to the barrier and disclose its valuation, enforcement, and renewal obligations. Its value depends on the landlord accepting it and on the tenant's ability to meet ongoing rent. Short-term emergency assistance can interrupt a spell of arrears after illness or job loss. A longer-lived rent supplement can stabilise very poor households while preserving choice among safe landlords. A targeted capital grant can make additional units viable, provided the state buys measurable affordability and keeps that condition on the property. A public-land lease can provide even more value, but requires published valuation, transparent award and a credible plan for renewal at the end of the lease. Evidence on long-term assistance can inform the choice without establishing identical effects elsewhere.

Rental-assistance evidence should be adapted to local supply, rents, tenancy law, and fiscal capacity. The US Family Options Study randomly assigned 2,282 families in shelters to priority access to different interventions. Long-term rent subsidies improved housing stability and other dimensions of wellbeing relative to usual care over thirty-seven months, at approximately 9 per cent higher cost within the study's measured package of services [22]. Chilean quasi-experimental research finds that rental vouchers improved housing conditions and mobility before the pandemic; receipt during the shock helped families limit debt and maintain stability, while evidence of moves to substantially better neighbourhoods was weaker [23]. These findings establish plausible benefits for households facing large affordability shocks. Local rent levels, supply, informal tenancy law and fiscal capacity will determine whether a version works in an African or Indian city. The evaluation must then determine how much support reaches an occupied household.

The public sector also needs to know how much of each subsidy reaches the tenant. Where new units cannot be added, landlords may capture some voucher value through higher asking rents or tighter selection. Where safe vacant housing exists, assistance can raise occupancy with little new construction. Where construction is responsive, predictable subsidies may draw in new supply. An evaluation should therefore compare recipient and non-recipient rents, vacancy, construction permits, occupied units, tenant mobility and rent burden in matched places. Programme agencies often report the number of vouchers issued; an unused voucher buys no housing. Digital records and communication can help only when they resolve a specific leasing or verification barrier.

Information systems for rental policy

Use rental information systems to reduce a measurable cost or strengthen a household's evidence and bargaining position. A geospatial inventory can locate roofs, plot boundaries, flood exposure, transport and nearby utility networks. Building-permit records can identify where small additions are allowed. A tenant-friendly lease register can supply evidence of rent and deposits. Payment histories can help responsible landlords demonstrate cash flow to lenders, potentially lowering credit risk. An inspector's mobile record can link defects to accountable owners and completion certificates. These changes are useful when they lower a measurable delay, improve verification or shift bargaining power toward the household that previously lacked proof. Chile's voucher experiment tests one such information barrier.

Information assistance should be tested against landlord participation and available homes rather than assumed to resolve all take-up barriers. In a randomised evaluation of 11,149 unused rental-voucher recipients, basic information about the subsidy and lease-up steps, plus reminders, increased take-up by 6.4 percentage points in the short run, roughly 35 per cent relative to the comparison level. Early reporting found effects still present after twenty-four months; richer neighbourhood information had more complex effects because finding a willing landlord remained difficult [24]. A March 2026 Inter-American Development Bank analysis examines the heterogeneity and fiscal welfare implications of this experiment and reports improved use without higher observed rents in that setting [25]. This supports testing information against a specific leasing barrier, then checking whether landlord participation and available homes limit further take-up. Follow-up must trace the path from a message or award to an affordable contract and continued residence.

Publish replicable aggregate evidence while protecting households and distinguishing information from completed access. An identifiable tenancy registry deserves strict limits on access and retention. Informal households must be able to contest a record without exposing themselves to enforcement retaliation. The best early digital investment may be a simple case-tracking and payment-receipt system, accompanied by a staffed desk and a clear remedy. Estimates based on weak field data need explicit uncertainty ranges and independent validation. Chile's information intervention and Delhi's rental audit identify different points where an eligible household can fail to secure a lease. The populations and methods differ, but both require the agency to follow the whole leasing process. A successful message or subsidy award must be

followed by landlord acceptance, an affordable contract and continued residence. [24,25,120] A municipal pilot should establish the baseline needed to measure that conversion.

Municipal rental reform and evaluation

Adapt the pilot to actual leases and records, distinguishing new rooms, replacements, permissions, and occupied supply. Fund targeted support and inspection over time; do not use a short pilot grant to imply permanent affordability. The municipality must coordinate rental standards, landlord assistance and tenant remedies with housing-benefit administration. Track lease-up and renewal, new safe rooms, total rent and fees, discriminatory refusals, repairs and access to opportunity. A city could test these ideas through a defined eighteen-month experiment covering, for example, three well-serviced employment-linked neighbourhoods and three comparable areas. In the first ninety days, a team of planners, utilities, tenant groups, local builders and independent researchers would map existing rooms and rents, verify service capacity and identify land-tenure conflicts. The team would record the distribution of migrant status, household type and disabilities using consent-based interviews rather than inferring personal attributes from satellite images. The baseline must distinguish a new room from a replacement, and an occupied room from a permitted one. Publish the complete package of rights, charges, inspections, leases, and remedies offered to eligible plots.

A pilot package should make obligations predictable and report retention, new entry, and repairs. A financing partner would offer construction-to-term loans linked to verified milestones, with any public first-loss exposure capped and audited. Separate assistance would be offered to very poor tenants for deposits or rent, with landlords paid against verified tenancy and service standards. A local tenant body would monitor arbitrary eviction and the effect of improvements on rents. The municipality would report each quarter how many old tenants stayed, how many newcomers found rooms and which defects were repaired. Independent evaluation should compare those benefits with the full household and municipal costs.

Expand a rental intervention only when its benefits persist after costs and displacement are counted. First, did the package produce additional safe occupied room-months relative to the comparison areas, net of replacement and displacement? Second, did the monthly cost of rent, utilities and access fall for households in the bottom part of the income distribution? Third, what was the fiscal cost per safe and well-located occupied room-year, including land, grants, administration, utility capacity and expected maintenance? It should also record landlord returns, loan default, female tenants' safety, discrimination complaints and the time required to resolve disputes. The city could expand the intervention whose benefits persist after these costs are counted. The public commitment should state the specific benefit purchased and how it will be verified.

Exhibit 19.4. Rental investment assessment

Gate	Minimum evidence	Red flag that stops the deal
1 Land and services	Control and tenure checked; capacity letter from water, sanitation and power; flood and fire review	Unresolved eviction claims; no service or flood plan
2 Build and finance	Costed safe typology; stress-tested monthly cash flow; repayment and major repairs accounted for	Project survives only through unrealistic occupancy or unpaid maintenance
3 Tenants and rights	Baseline of actual tenants; published rent/deposit/notice terms; accessible complaint remedy	No protection for incumbent renters or excluded groups
4 Public return	Counterfactual occupied room-years; subsidy value and clawback; independent evaluation	Public spending buys only permits, loans or registries

Specify the benefit purchased with public money, the households intended to receive it, and how delivery will be checked. Permissions, loans, and registrations are intermediate steps; the evaluation must follow their conversion into affordable occupancy. The long-term institution must sustain that benefit through maintenance and continued supply.

Institutions for rental housing

Rental institutions should support mobility and affordable entry while preserving provider viability and tenant benefits. An enforceable contract should protect a tenant's belongings and deposit, assign fire-safety and drainage duties, and establish remedies for unpaid rent and disputes over possession. Evaluate rental policy through safe, occupied, well-located housing available to low-income households. Include public land and service costs, measure benefits retained by tenants, and test whether landlords and lenders can finance continued supply and maintenance. Fieldwork should test that agreement against actual rents, costs, rights, and operating performance.

Prepare the local decision

Compare actual tenancies and providers using household costs, safety, services, disputes, and a credible comparison where feasible. Combine a repeated sample of actual tenancies with asking-rent listings, vacancy checks, deposits, utility charges, repair records, evictions, and landlord operating accounts. Listings omit many low-cost rooms and may differ from agreed rents. The San Francisco evidence above makes supply responses part of the appraisal, while the small-building model requires local costs before it can inform a financing decision. Compare the positions of small landlords, larger operators, current tenants, prospective tenants, employers, utilities, and dispute-resolution bodies. Determine who can enforce deposit,

notice, and repair obligations at a cost proportionate to a low-value tenancy. Specify how a proposed subsidy reaches tenants and what prevents its conversion into higher rent. Check listing data against tenant interviews and repeat observations of the same dwelling. Test a rental-matching service on successful lease-up and retention, including applicants without conventional documents. Audit rejection patterns with legally appropriate methods and privacy controls. Evaluate regulation or support against net safe rental supply, full household cost, displacement, arrears, and repair quality. Retain a comparison group or clearly state why a causal estimate is unavailable. Chapter 20 examines the credit assessment and household risk behind both rental and ownership finance.

Rental policy should remove the actual entry barrier while preserving tenant rights and viable maintenance. An existing usable benefit, clear leasing steps, available stock, and willing landlords supported the search-assistance results. A message campaign will not overcome unaffordable upfront payments, discrimination, or uncertain enforcement by itself. Foreign rent rules and standard landlord loans also need testing against local supply and costs. At citywide scale, monitor rent responses, landlord withdrawal, counselling capacity and displaced unsubsidised tenants.

Sources and evidence notes

Reference numbers in the text point to documents below. Research, official administrative figures, institutional analysis and illustrative calculations are distinguished in the chapter. Titles are clickable to reach public source material; datasets may be revised.

01 [source] 02 [source] 03 [source] 04 [source] 05 [source] 06 [source] 07 [source] 08 [source] 09 [source] 10 [source] 11 [source] 12 [source] 13 [source] 14 [source] 15 [source] 16 [source] 17 [source] 18 [source] 19 [source] 20 [source] 21 [source] 22 [source] 23 [source] 24 [source] 25 [source]

Scope and evidence note. Statements about the effects of a specific law or project are qualified by the available evaluation design. National statistical shares are not directly comparable across countries. All Kenya-shilling financial examples are hypothetical, with no assumed conversion to 2026 purchasing power or actual lender terms. The proposed municipal diagnostic is a policy design, not a completed evaluation.

[120] Datta, S. and Pathania, V. (2016). For whom does the phone (not) ring? Discrimination in the rental housing market in Delhi, India. UNU-WIDER Working Paper 55/2016. Correspondence audit; response rates, not completed leases. [source]

20. Credit assessment and household risk

Assess housing credit through the household's capacity to sustain the whole housing commitment, including shocks and recurring costs. Verify irregular earnings, choose finance suited to the property and stage of construction, and specify remedies when income or subsidy payments fail. Digital underwriting should improve this assessment without making adverse decisions unchallengeable. The loan contract is the starting point for allocating these risks.

Housing credit commits a household to payments over many years. The lender must assess income shocks, property rights, and collateral risk; the household must consider fees, transport, services, and repairs. Long-term funding and functioning local services are necessary to sustain the agreement. Kenya's Central Bank counted 30,762 outstanding residential mortgage loans at the end of 2025. Their average size was about KES 10 million (about US$77,340 at the 2025 annual-average rate of KES 129.3 per US dollar, author's conversion). The average interest rate was 13.5 per cent, and 75.6 per cent of loans carried variable rates. The ratio of non-performing mortgage balances to gross mortgage loans was 16.3 per cent. The stock of loans had risen by just 746 during the year. These figures describe a relatively small, costly and stressed formal market, with substantial exposure to interest-rate changes. A typical low-income household seeking a room or a small starter home occupies a very different market. [1] India's larger market presents different institutions but still requires a precise diagnosis of the risk excluding a household.

1 Assess the household risks behind a housing loan

Mortgage lending and household risk

Housing-finance reform should change an identified cost or risk and establish who receives the saving and bears the loss. Its National Housing Bank reported INR 35.61 trillion (INR 35.61 lakh crore) in outstanding individual housing loans in March 2025. Within its detailed slab-wise portfolio excluding pooled purchases, loans of INR 1 million (INR 10 lakh) or less accounted for 29.28 per cent of accounts and only 5.73 per cent of outstanding value. The distribution reflects loan size, rather than a direct measure of borrower income. It shows that small-balance accounts represent a larger share of contracts than of outstanding value. [2] Credit design should match household repayment capacity, property rights, and public risk limits. A lower interest rate cannot resolve a disputed title, an unaffordable price, or a location that cuts earnings. This chapter examines the allocation of credit risk among households, lenders, developers, public agencies, and long-term investors. A useful reform changes the cost or allocation of a specific risk and establishes who receives the saving and who bears losses. That diagnosis must extend beyond the mortgage contract itself.

Five contracts in housing finance

Housing credit depends on several linked contracts governing earnings, rights, production, funding, and neighbourhood services. In practice it depends on five linked agreements. The household agreement converts daily or seasonal earnings into an obligation payable on a fixed date. Payslips make a salaried worker's recorded income easier to verify. A tailor, taxi driver, domestic worker, market trader or construction mason may have comparable average earnings but greater volatility, higher verification costs and longer spells without cash. A lender may respond by demanding a deposit, a guarantor, a higher rate or proof that excludes an otherwise creditworthy borrower. Women whose work and assets sit in another family member's name face an extra barrier. The land agreement gives the lender a claim it can enforce. That requires a parcel whose boundaries are agreed, a valid interest that can be transferred or mortgaged, a recorded chain of transactions and courts or registries that work. A title can settle part of this problem. Enforcement may still take years, especially where purchasers and occupants have competing claims. Households also need to decide whether they can accept the risk of losing the home pledged as collateral. The production and funding agreements determine whether the promised dwelling and loan terms can be delivered.

Credit requires a usable dwelling and a clear allocation of funding and financial risks. A developer needs construction credit before buyers take possession, then needs purchasers who can obtain mortgages at completion. A bank financing a purchaser worries about unfinished works, fake completion certificates, defects, unconnected utilities and land encumbrances. A bank financing the developer worries about sales, cost overruns and the allocation of delay risk. Advance sales can shift these hazards to families least able to absorb them. The long-money agreement links a bank's short-term deposits to a fifteen- or twenty-year home loan. Even a solvent borrower becomes a costly client if a bank cannot fund the mortgage through its life. Refinance institutions, covered bonds, long-dated deposits, insurance companies and pension funds can help bridge the mismatch. Each instrument raises a second question: whose balance sheet absorbs interest-rate, prepayment and default risk? Neighbourhood services then determine whether the household can afford to continue living there.

The neighbourhood agreement is part of credit viability because location and service failures can undermine repayment. Transport, water, safety, drainage, schools and upkeep influence both the household's ability to pay and the property's resale value. A lender's valuation may record the wall and plot while barely measuring the commute, local flood risk or whether a resident association has a funded maintenance plan. These omissions can contribute to arrears, forced sales or losses on the loan portfolio. Exhibit 20.1 shows why the same phrase, "lack of finance," can describe several different failures. Lower funding costs can increase approvals while leaving unresolved land, construction or service risks. A debt-service calculation should therefore be tested alongside lifetime household costs.

Risk or failure	Who first bears it?	What a lender can observe	A credible response
Earnings fall or arrive late	Household, then lender	Accounts, turnover, shocks	Cash-flow underwriting; buffers; workout rules
Tenure or boundary is disputed	Buyer, lender, occu- pants	Registry and ground checks	Adjudication; enforceable use rights; ap- peal
Developer fails before delivery	Buyer, lender, workers	Stage inspections, escrow	Milestone finance; independent certifica- tion
Interest rates rise	Floating-rate borrower	Contract, repricing schedule	Long-term local-currency funds; capped resets
Services or location fail	Resident, city, lender	Service tests, travel access	Readiness gates; funded maintenance ob- ligations
State guarantee is called	Taxpayer, future bor- rowers	Guarantee contract and port- folio	Priced cap; audited losses; fiscal reporting

Assessing affordability and debt service

A loan calculation should expose the payment implied by the price, deposit, rate, and term before support is designed. Consider an illustrative Kenyan buyer seeking a KES 3 million home. At a 10 per cent deposit, the mortgage is KES 2.7 million. At 13.5 per cent for 11.5 years, the monthly repayment is about KES 38,623. The rate and maturity correspond to the average rate and average maturity reported in the Central Bank's 2025 survey; the property price, loan-to-value ratio and household are hypothetical. At a screening threshold of 30 per cent of gross income, debt service alone requires earnings of roughly KES 128,745 each month. Service charges, insurance, transport and other debts would raise that requirement. [1] Hold the house price constant and finance the same loan at 10 per cent for 20 years. The payment falls to KES 26,056. This is a 32.5 per cent reduction. It can bring in some households previously excluded by the repayment test. Yet the household must still assemble KES 300,000 upfront, pay fees, meet the bank's documentation test and find a house at that price in a place where it can live and work. The longer loan also extends exposure to income loss and other risks. The appraisal must still test lifetime affordability. Comparing that payment with actual lending and household incomes shows the size of the remaining gap.

Average mortgage terms can imply payments far beyond the households a housing programme intends to reach. At the Kenyan system's reported average loan size of KES 10 million, a calculation using the survey's average rate and maturity yields about KES 143,050 per month. It would be a mistake to call this a national average monthly payment: banks lend at many rates, and the 10-million figure is the average stock of loan principal, not the amount

drawn at origination. The calculation instead shows why growth in aggregate mortgage lending can coexist with severe exclusion at the lower end. [1] The same logic applies in India. Imagine a household with income of INR 25,000 per month and a cautious repayment limit of INR 7,500. At a hypothetical fixed rate of 9 per cent over 20 years, the largest fully amortising loan is approximately INR 834,000 (INR 8.34 lakh). A bank may lend less after examining earnings shocks and other obligations. A home that costs INR 1.8 million (INR 18 lakh) remains far out of reach without substantial household equity, a cheaper plot, a grant or a smaller dwelling. An interest subsidy can bridge only a defined portion of that gap. The stress test below separates those calculations from lender offers and forecasts.

Loan illustrations should identify the binding deposit, price, and income constraint rather than imply that a lower rate resolves it. Exhibit 20.2 uses the two loan calculations as a stress test, not as lender offers or predictions. The binding constraint may be the deposit, the dwelling price or the month of lowest income, even after the advertised interest rate falls. The benefit must also be traced through funding costs, lender terms, and housing prices.

Illustrative buyer	Loan	Rate / term	Monthly payment	Gross monthly income at 30% ceiling
Kenya: conventional illustration	KES 2.7m	13.5% / 11.5 years	KES 38,623	KES 128,745
Kenya: longer, cheaper local-currency finance	KES 2.7m	10% / 20 years	KES 26,056	KES 86,852
India: maximum loan under INR 7,500 payment cap	INR 8.34 lakh	9% / 20 years	INR 7,500	INR 25,000

Credit support reaches households through changes in funding, loan terms, and the price of the home. First, a lower funding cost reduces the monthly price of borrowed money. Second, households submit more bids for scarce homes. Third, developers and landowners decide how much new supply to build and at what price. Where serviced land, permits and construction respond slowly, some of the credit benefit turns into a higher selling price. The borrower may receive only part of the subsidy's value if landowners or sellers capture the remainder through higher prices. Where documentation is the obstacle, underwriting rather than subsidy may be the appropriate response.

2 Match credit products to earnings and property conditions

Verifying irregular earnings

Income verification can address missing documentation, but operating practices alone do not establish additional safe lending. A payslip solves part of that information problem. Reliance on payslips can exclude earners without formal employment records. Two shopkeepers may earn equal profits; one uses a bank account and files tax returns, while the other receives cash from customers, mixes household and business spending and keeps no formal books. A documentation rule may favour the first shopkeeper even where repayment capacity is similar. The difference in paperwork can be larger than the difference in repayment capacity. Some Indian housing-finance companies have built a business around this gap. IFC described Aavas Financiers as serving low- and middle-income, mainly self-employed customers in semi-urban and rural India through individualised appraisal. The firm describes examining business turnover, bank records and field evidence where standard tax returns are incomplete. These are operating practices and claims about reach, rather than a randomized estimate of added credit access or default performance. [3] [4] The method's labour cost and the selection embedded in past loan records require separate attention.

Underwriting evidence should account for assessment costs and the borrowers absent from historical approvals. A loan officer visits the shop, checks the market's working days, estimates sales and purchases, looks at seasonal variation, examines other debt and asks what happened during the worst recent month. Household expenses are separated from business cash flow. Repayment is tested after transport and childcare costs. The verification team checks property papers independently of the sales officer. These steps can make the true risk legible while reducing the temptation to use an inflated income estimate to close a sale. Digital trails reduce some of those costs. Deposits, mobile-money receipts, merchant transactions, utility payments and rent records can show a pattern that tax filings miss. Their potential value is a lower cost of obtaining reliable earnings histories and detecting inconsistent claims. The gain is conditional on coverage, consent and a genuine out-of-sample test. A trader who takes cash, a woman who uses her husband's phone, and a migrant with a newly opened wallet may look risky to an algorithm for reasons unrelated to future repayment. A model trained on loans already granted cannot reveal the default rate among people the bank has always refused. Evaluation needs to observe previously excluded applicants as well as the existing loan book.

Historical lending data reflect earlier selection decisions. If nearly all observed borrowers hold registered property, the data provide little evidence about repayment among otherwise comparable people without it. A model may reproduce that exclusion if the lender fails to test performance for the missing group. A responsible pilot records rejected applicants, compares models against observed cash flows rather than proxies for class or caste, tests false rejection rates and gives each applicant an understandable avenue of appeal. It measures the additional safe loans at a stable default rate, alongside changes in interest, fees and foreclosure, rather than reporting a percentage gain in automated decisions. Collateral reform likewise requires evidence for each link between rights, investment, and credit.

Property rights and housing investment

Registered title may improve collateral and encourage investment, but each link requires evidence. The studies below distinguish stronger tenure security, investment, and actual access to credit. In suburban Buenos Aires, some land occupiers received titles because the original owners accepted compensation; others waited because owners contested it. Sebastián Galiani and Ernesto Schargrodsky used this uneven process to study the effect of property rights. Titled households increased housing investment, reduced household size and improved children's education. The study found no corresponding improvement in access to credit. In Peru, Erica Field found that the rate of housing renovation rose by more than two-thirds of its baseline level following an urban titling programme; much of that investment occurred without loans. Her separate study found increased labour supply after titling, consistent with less time spent defending possession. [5] [6] [7] Titles may improve security or investment without resolving lenders' remaining concerns.

A stronger recorded right does not establish affordable credit and can increase harm if errors become enforceable. A secure family may replace a roof because it expects to enjoy the investment. It may take a job away from home because it fears eviction less. Neither decision needs a mortgage. Credit also depends on repayment income, accurate valuation, foreclosure law, lender funding and the household's willingness to risk its principal asset. The value of a home does not by itself provide the cash needed for monthly repayments. African and Indian governments should draw two operational lessons. First, tenure programmes require an objective beyond mortgage counts: better security, safe construction, investment and reduced conflict are valuable outcomes in their own right. Second, registries must record real claims. A digital map can show a roof and an approximate boundary while leaving unanswered who inherits, sublets, occupies, holds customary rights or contests an old allocation. Recording one party as owner can transfer power away from spouses, tenants and communities. Faster digitisation may harden an error into an instrument of foreclosure or displacement. Adjudication, appeals, and fair enforcement must therefore accompany collateral reform.

Collateral reform should establish valid rights and borrower protection while testing repayment and treatment directly. It must specify what happens to tenants and other users when a lender enforces a charge. Evaluate tenure reform through security and the ability to improve a home, while testing access to mortgage credit separately. The evidence thus separates two objectives: securing investment in a home and obtaining a loan against it. Field-based appraisal by Indian lenders offers an operating alternative for borrowers without standard income records, but the institutional accounts cited here do not establish its causal effect or its performance at national scale. Test repayment capacity and borrower treatment directly rather than infer them from title issuance or loan growth. [3–7] Refinancing addresses a different constraint in the lender's funding structure.

Access to mortgage refinancing in Kenya

Refinancing can lengthen funding without ensuring wider geographic or income access. A mortgage-refinance company provides longer-term funding to lenders. It supplies

long-term funds against an eligible loan portfolio, allowing lenders to offer longer and sometimes cheaper mortgages without funding twenty years of repayments from volatile deposits. Kenya's Mortgage Refinance Company (KMRC) reported KES 13.9 billion in cumulative disbursements to primary mortgage lenders and 3,855 refinanced mortgages by December 2024. Its 2024 report says 77 per cent of loan disbursements went to Nairobi and the adjoining counties of Machakos, Kajiado and Kiambu. These are the institution's own administrative results. The reported 15,420 “households impacted” uses a wider definition and should not be presented as the number of directly refinanced mortgages. [8] The Central Bank's 2025 survey provides a separate measure of the wider mortgage market. The share of mortgages with fixed rates increased from 14.1 per cent in 2024 to 24.3 per cent in 2025, while the market still counted only 30,762 outstanding loans. Surveyed banks named low income, high property prices, costly land, property registration and access to long-term finance among the principal obstacles. These are responses from lenders, not estimates of each obstacle's causal effect. [1] KMRC addresses a defined funding constraint. Its refinancing can lengthen terms and reduce the funding-cost premium for borrowers who already meet mortgage criteria. Refinancing depends on eligible originated loans. Land disputes, unverifiable income, incomplete buildings and unaffordable purchase prices can still prevent origination. Common loan standards, lending through cooperatives, technical support and prudent guarantees may widen access, but their effects require separate measurement. Public support should therefore be assessed by who receives it and what alternatives the same resources could finance.

The distribution of refinancing benefits should be compared with other uses of scarce fiscal capacity. A national institution uses public and concessional resources, while participating banks choose which loans to originate. Existing bank relationships, property records and concentrations of lending activity may make metropolitan borrowers easier to serve. The contribution of each factor requires local evidence. A subsidy may therefore reach borrowers who would have obtained credit anyway. That creates a genuine public benefit if maturity improves and interest risk falls. Its additionality is smaller when the money merely refinances contracts without changing access, terms or the number and location of homes delivered. A useful appraisal must report the difference between the refinanced borrower and the next eligible borrower denied finance. It must also compare the public cost of a refinance against the cost of improving a serviced site, guaranteeing a small initial loss, or funding an incremental building loan. These are competing uses of scarce fiscal capacity. A guarantee-backed capital-market structure requires the same accounting of public exposure.

Capital-market funding should disclose guarantees and losses that are otherwise hidden from the apparent cost of credit. The Nigeria Mortgage Refinance Company states that its business rests on uniform underwriting standards and bonds sold to investors. Its official account reports NGN 33 billion in cumulative mortgage refinancing through December 2024. A rating note identifies an unconditional federal guarantee behind a fifteen-year bond. Investors receive stronger protection because the sovereign stands behind the instrument; the government takes a contingent liability that may become a budget cost if the portfolio or issuer fails. The total guarantee ceiling is therefore a public-finance decision, even where the budget records little spending at issuance. [9] [10] A pension fund needs a lawful, well-priced, local-currency asset. A family needs payments that survive shocks. A mortgage-refinance institution can connect them,

subject to a demanding set of conditions: transparent loan pools, reliable servicing, audited arrears, realistic prepayment assumptions, sufficient loss absorption and a fiscal authority willing to disclose the guarantee. Funding backed by an unpriced guarantee can understate the public cost of credit and leave taxpayers exposed to later losses. India's financing and subsidy arrangements show how several instruments can operate within one larger system.

Housing credit and public support in India

India's housing-finance system requires separate evaluation of funding, borrower eligibility, subsidy timing, and remedies. The NHB's March 2025 reporting placed individual housing loans at INR 35.61 trillion (INR 35.61 lakh crore), with a national outstanding-loans-to-GDP ratio of 10.81 per cent. The same report found wide differences across states and loan sizes. Maharashtra accounted for 21.88 per cent of the reported state-wise outstanding stock, while the regional spread remained uneven. These are administrative aggregates, not a census of households still unable to borrow. [2] The distribution of loans clarifies which households the system reaches.

Loan-size distributions show who receives housing credit more clearly than aggregate lending does. Examine the loan-size distribution separately. In the detailed loan-size table, balances up to INR 1 million (INR 10 lakh) accounted for 5.73 per cent of the covered value and 29.28 per cent of covered contracts. Loans above INR 2.5 million (INR 25 lakh) accounted for 69.87 per cent of value. Large loans take a great deal of capital even though the number of large-loan accounts is lower. The figures do not prove that a small loan belongs to a poor household or a large loan to a wealthy one; loan balances change with repayment and property prices vary across districts. They do show why a housing-finance reform that increases total lending can leave the bottom of the market largely unchanged. [2] Subsidy rules require a separate assessment of eligibility and timing.

An interest subsidy changes loan costs only for households that meet its conditions and receive it. The Union government's Pradhan Mantri Awas Yojana–Urban 2.0 interest-subsidy rules illustrate the distinction between a financing instrument and a house. For eligible applicants, the announced 4 per cent interest subsidy applies to the first INR 800,000 (INR 8 lakh) of a housing loan, for up to 12 years, subject to a INR 2.5 million (INR 25 lakh) loan ceiling, INR 3.5 million (INR 35 lakh) house-value ceiling and INR 180,000 (INR 1.80 lakh) maximum subsidy. The programme specifies payment in five annual instalments. These are policy rules, and their reach depends on approvals, lender processes and actual disbursement. [11] The payment schedule determines which immediate financing constraints remain.

Delayed subsidy payments do not remove the need to finance the initial deposit. A benefit released over five years has a different effect from a grant delivered at purchase. It may reduce the effective burden after credit has been approved. A family unable to pay the initial deposit still needs financing or assistance for that payment. A lender assessing the repayment obligation must decide how to treat benefits that arrive later and might be delayed. An agent can advertise the maximum advertised subsidy while a buyer confronts the full sale price and immediate loan payment. Implementation rules should specify when money reaches the loan account, how arrears or early prepayment affect the benefit, and what recourse a borrower has if

a subsidy claim stalls. Specialised incremental products address needs that the headline mortgage rate does not capture.

Assess finance through product fit and borrower outcomes rather than one average rate. A small self-employed borrower may need a INR 300,000 (INR 3 lakh) roof loan, then INR 500,000 (INR 5 lakh) for an extension, before ever purchasing a completed apartment. The lender can examine earnings through field appraisal. A city may have to certify that the structure is safe. A land authority may need to recognise a lease or occupancy right without forcing a speculative sale. A national interest subsidy designed around a completed home can miss this common sequence of building and repair. The most consequential test for India is whether the finance system can reach a larger set of safe housing paths: incremental building, shared ownership where lawful, small rental construction, modest formal home purchase and repairs that reduce disaster risk. A single mortgage rate is too crude a measure of that progress. Staged construction requires payment and inspection arrangements matched to smaller improvements.

Financing incremental construction

Incremental-credit evidence supports testing product use and housing benefits while retaining limits on causal interpretation. They buy the plot or secure an occupancy claim, build a room, improve the roof, connect water and sanitation, and extend later as savings permit. A conventional mortgage for a finished unit arrives at the wrong point in this sequence. A shorter housing-improvement loan can match the next step, provided the instalment fits uncertain earnings and the builder gets sound technical advice. A study of Habitat for Humanity's housing microfinance programme in Kenya and Uganda examined 2,341 existing clients, with 1,213 offered the programme. The programme combined small labelled loans with housing-support services, including advice on construction. The researchers reported that 95 per cent of clients used the loan for the intended housing improvement and found improvements in housing conditions and satisfaction. Rollout varied across branches, with earlier treatment in more profitable or larger branches; the researchers used a difference-in-differences approach to compare changes over time between groups. The study is useful evidence of product use and plausible benefits, with limits on causal interpretation because branch rollout was non-random. [12] Shared services and safe construction may remain more binding than the price of the loan.

Small improvement loans need complementary services and technical support to produce safe usable space. An inspection or adviser helps the borrower avoid a dangerous construction choice. Repayment begins before the whole home is finished, so the payment schedule must reflect the household's income cycle. Loan officers need to verify that the financed improvement sits on land where the family can remain. A sound product makes each stage habitable and safe. A poor product locks borrowers into high-cost revolving debt or funds an unsafe additional floor. The lending institution has an incentive to finance improvements with quick repayment and visible assets. A household may need drainage or a safer access path that serves several plots, with no single borrower able to capture the gain. That remains a municipal or collective-finance problem. Developers face a similar indivisibility: roads, sewers and land assembly have to be financed before a small project can sell. Public money may yield greater

gains by funding these shared inputs than by buying down each purchaser's mortgage rate. A range of non-debt and debt options is therefore needed across household circumstances.

Different needs require different financing arrangements: savings and safe tenancy for new workers; small improvement loans with inspections; serviced plots and incremental building credit; properly supervised small-landlord loans; and long-term mortgages for households whose income and tenure can support them. Households should be able to move between these arrangements without losing all prior assistance. Lenders should be able to recognise a record of successful smaller repayments, while rules stop them from using that record to push borrowers into loans whose worst-month payments exceed capacity. Any public credit subsidy must then be traced through market prices to establish who receives its value.

Credit subsidies and their beneficiaries

A discount credited to a borrower's account can still be partly captured through higher property prices. Its economic benefit can pass elsewhere before the family enters the home. Cheaper borrowing raises what some buyers can bid. A developer with a pipeline of approved units may increase the asking price. A landowner negotiating with the developer may demand a greater share of expected sales revenue. A contractor with few competitors may absorb more of the margin. In a market where serviced sites and approvals are scarce, prices can adjust faster than additional homes are delivered. Empirical subsidy evidence gives reasons to test that mechanism rather than assume complete pass-through.

Subsidy incidence depends on supply and institutional conditions and requires evidence beyond the nominal transfer. Berger, Turner and Zwick studied the United States' 2009–10 first-time homebuyer tax credit with quasi-experimental methods. Their estimates link the programme to about 490,000 additional home sales and an increase in median prices of $2,400 for each standard deviation of programme exposure. The context was an exceptional post-crisis market, and the estimate cannot be copied onto Kenya or India. It establishes that buyer subsidies can influence both transactions and prices. [13] South Africa's First Home Finance scheme illustrates how eligibility rules determine access to support. The Western Cape government's published guidance describes a once-off subsidy ranging from about R38,911 to R169,265, depending on income, for qualifying first-time buyers in the R3,501–R22,000 monthly household-income band. The benefit can reduce the deposit or principal barrier for households with bank or approved finance. It also creates a boundary: applicants with incomes below the qualifying band, without adequate loan access or with an ineligible family arrangement must use other routes. Rules and application channels can change by jurisdiction; the figures describe the published provincial guidance. [14] An effective subsidy asks four questions. Which household receives it, including renters and workers with unstable jobs? Which location and housing product does it make accessible? How much is passed into a lower all-in cost, including fees and transport? How much of the fiscal transfer remains with the household rather than the land seller, developer, bank or broker? A ceiling on unit price has limited value when projects reduce floor area, move to cheaper peripheral sites or add informal charges. A public-land subsidy can leak in the same way

if the state disposes of serviced plots cheaply without an enforceable affordability agreement. Representation in programme design influences which products and beneficiaries receive support.

The allocation of subsidies depends partly on who participates in setting the rules. Mortgage lenders can administer support through approved loan accounts, while developers may seek support tied to their projects. Owners may gain when public investment raises land values. Tenants and informal builders are poorly represented in mortgage records. Programme design should document these interests and compare benefits for new borrowers, existing clients, renters and self-builders, including the costs of maintenance, guarantees and title adjudication. A credible reform coalition must pay attention to these incentives. Finance ministries should publish the estimated cost and beneficiaries of tax privileges, grants and guarantees on a comparable basis. Cities should condition publicly created land value on serviced-site delivery and lasting affordability. Consumer regulators should disclose loan fees and preserve legal recourse. Lenders should be rewarded for sustainable new access rather than merely refinancing safer borrowers. Tenants and self-builders need a seat where programme rules are decided. Borrower protection must also address the shocks that arise after approval.

3 Protect borrowers and assign the cost of shocks

Insurance, arrears, and financial shocks

Test housing credit against adverse income, rate, essential-cost, and care scenarios as well as a central forecast. Social protection and prudential rules matter when that forecast fails. An illness can cut earnings for three months. A major flood can destroy a family business and damage the same house pledged to the lender. A borrower with a variable interest rate can face a higher payment even if wages remain flat. In each case the borrower, bank, insurer or state must absorb a loss. Unclear loss-sharing terms can lead to litigation or place costs on households with limited ability to bear them. A prudent lender tests a severe but plausible month. For the KES 2.7 million mortgage, raising the hypothetical interest rate from 13.5 to 16.5 per cent over the same 11.5 years raises the payment from about KES 38,623 to KES 43,774, an increase of about 13.3 per cent. This is a sensitivity calculation, not a rate forecast or a claim about Kenya's contractual repricing rules. If a family starts at the maximum permitted debt-to-income ratio, the rise alone can force cuts to essentials. The test should also include food and transport inflation, unpaid care time and maintenance charges. Insurance and arrears rules should specify predictable treatment before a shock becomes default.

Insurance and default procedures need explicit coverage, exclusions, remedies, and household treatment. A flood affecting an entire district creates correlated claims. Insurance then depends on adequate capital, reinsurance, coverage terms and affordable premiums. A public guarantee can encourage credit where loan officers overestimate risk, but it can also weaken screening if banks retain little first loss. The government should retain an explicit maximum exposure, charge a fee where appropriate, require the originator to keep meaningful risk, and publish claims by borrower group and location. Any concessional foreign-currency loan to a local-currency mortgage institution adds exchange-rate exposure unless fully hedged; depreciation can increase costs borne by the institution, its borrowers or taxpayers. Workout

rules are part of affordable housing policy. Before foreclosure, a lender should have a time-bound process for income verification after a shock, temporary rescheduling, partial payments, independent advice and orderly sale where needed. A public guarantee should never require instant eviction as its first enforcement step. Equally, open-ended debt cancellation can weaken repayment incentives and create unfunded public obligations. Fairness requires predictable treatment before default occurs. Climate exposure also requires affordable renewal and maintained physical protection.

A climate-resilient home also needs affordable renewal. A flood-resistant plinth or sound drainage costs money today and reduces expected losses later. Financiers can price that benefit only with credible site-level hazard data, building inspections and evidence on actual losses. Satellite imagery helps locate exposure; it cannot prove that a hidden foundation, buried sewer or insurance contract will perform. Good finance joins engineering, land records and household data with accountability on the ground. AI-based assessment should support these protections without concentrating unreviewable decision power.

AI in credit assessment

Tool and decision	Constraint it can change	What must be measured	Who may capture the benefit / new risk
Cash-flow underwriting	Cost of verifying irregular earnings	Approval among previously excluded applicants, net defaults	Borrower or lender; hidden bias and surveillance
Linked land and occupancy records	Time to verify collateral and claims	Disputes found, completion time, re- versals	Owners, banks; tenants or spouses erased
Digital construction certificates	False completion and di- version of advances	Defects, delays, escrow release errors	Buyer, insurer; false assurance from bad data
Refinance loan-level reporting	Funding opacity and pool risk	Rates, maturities, arrears, additional borrowers	Existing banks; public backstop for cherry-picked loans
Hazard and service mapping	Unseen physical risk	Insured losses, service uptime, risk pricing	Buyers, city; redlining of exposed neighbourhoods

AI credit tools should be evaluated for sustainable access and borrower treatment as well as processing speed. A lender that sees years of merchant receipts can quote a better loan; it can also monitor a borrower's trade or discriminate through opaque proxies. A parcel platform can lower the cost of checking a charge; it can also make the sale of contested public land easier.

An automated inspection system can reduce discretion; it can also issue thousands of certificates without verifying the structure behind the uploaded photograph. Five safeguards follow. Borrowers must know what data are used and be able to correct errors. Decisions need reasons that a loan officer and applicant can understand. Human inspections must validate critical land and building claims. Regulators need disaggregated denial, pricing and loss data. A public institution that claims savings should show actual processing cost, loan price and borrower outcomes before and after introduction, using a credible comparison group. Faster processing alone does not establish improved housing access or affordability. Risk-sharing institutions must decide how the resulting costs and gains are allocated.

Institutions for risk sharing

Finance reform should allocate costs and risks across households, lenders, producers, and public institutions explicitly. A mortgage bank wants predictable recovery and low-cost funding. A small lender wants affordable wholesale credit and leeway to assess irregular incomes. A developer wants pre-sales, serviced land and timely release of construction funds. A pension fund needs a safe local-currency asset. A finance ministry must contain explicit subsidies and contingent liabilities. A city needs property revenue and affordable infrastructure; its officials may also control permits that create valuable scarcity. A low-income borrower needs to retain enough income for essentials, employment changes and shocks. Each actor can block a different step. One possible coalition joins large developers, banks, established landowners and an agency that announces mortgage subsidies. Where its members already hold approved plots, financeable properties and documented clients, it may be able to launch quickly. Its weakness is distribution: it risks allocating public help to transactions that would have happened anyway, raising land values while renters and self-builders remain excluded. An alternative coalition brings small landlords, housing microfinance lenders, utilities, tenant groups and municipal planners into the same agreement. It requires coordination across additional agencies and can serve housing types common among low-income households. Evidence on rejected applicants should determine the first intervention.

Begin with rejection reasons and coordinate credit reform with supply and rights before support raises land values. First, map household cash flows and feasible housing products in two employment corridors, including renters, women, recent migrants and informal workers. Second, audit the legal and physical readiness of plots, service capacity and the cost of commuting. Third, design distinct credit products for home repair, staged construction, rental additions and completed purchases. Fourth, commit public money to the demonstrable market failure: an infrastructure connection, lender verification costs, a capped first-loss guarantee, or interest support for a narrowly defined group. Fifth, publish aggregate subsidy allocations, rejection reasons, default experience and changes in land and housing prices, while protecting identifiable applicant and borrower data. Public-land allocation requires especially strong checks. If an authority transfers a valuable site to a developer at a concessionary price, publish the independent appraisal, competing bids, all costed infrastructure commitments, the eligibility agreement, the deadline for occupied homes and clawback terms. Where the state retains the land, long leases can preserve some public control over the agreed use. Where an owner sells, resale and rental protections need enforceable duration and realistic monitoring. Otherwise a

capital subsidy can become an immediate land windfall. The agreement must also address documented opposition to complementary reforms.

Before choosing a credit subsidy, test whether disclosure rules, small-builder entry, permit discretion, or density changes face opposition. Record its basis and the decision power involved. A workable agreement needs predictable enforcement for lenders, protected resident claims, feasible compliance for builders, and a specified limit on public exposure. A local assessment can test whether the proposed package changes the right constraint.

Housing credit assessment

Use a common application protocol and local household evidence while retaining safe non-debt alternatives. Publish expected guarantee losses, subsidy timing and the cost of arrears support; avoid implicit uncapped rescue commitments. Lenders must verify and explain decisions, regulators must monitor consumer outcomes, and title authorities must resolve claims without coercive shortcuts. Track additional sustainable access, rejection reasons, payment stress, arrears, foreclosure, location and public losses by borrower group. Suppose a city and a multilateral development bank (MDB) have financing for 2,000 low-income households. They can offer a buyer subsidy, fund a refinance line, back housing-improvement loans or pay for serviced land. The first decision should concern the households and the binding constraint rather than the instrument already favoured by the sponsoring agency. The city can run a 24-month, two-corridor pilot. It begins with a household sample and a full market inventory, including off-registry tenants and informal rooms. Independent enumerators document income variability, rent, debt, travel, tenure and the condition of potential homes. Separate public and private teams verify land claims, utilities and hazards. Participating lenders record approvals and rejections against a common application protocol, while retaining legally required credit discretion. A bounded pilot can then compare products and technical support under actual market conditions.

Evaluate pilot products through repayment, housing improvement, prices, and borrower treatment rather than approvals alone. In the other, they offer an otherwise similar product with access to a capped and priced risk-sharing facility, phased in across branches or applicants where operationally feasible. The design must protect borrowers and avoid withholding established entitlements. A separately identified sample of eligible completed-home buyers may receive a narrowly tested deposit subsidy; all arms face the same disclosure and consumer-protection rules. Researchers should pre-register the design and acknowledge that different neighbourhoods and lenders may limit causal comparison. Success has to be measured in outcomes that a policymaker can defend. Count net additional safe, occupied home-years among the intended income groups. Report the all-in monthly burden after transport and services; the size and timing of public transfers; loan approval and rejection by sex, income type, caste or locally relevant exclusion markers where lawful and appropriate; arrears at 30, 90 and 180 days; restructuring and forced sales; verified construction quality; displacement and evictions; and the public cost per additional household served. Track changes in nearby land and rent prices, because a subsidy can leak into the market surrounding the funded project. The decision screen should distinguish funding shortages from physical supply and affordability gaps.

Exhibit 20.3 is a decision screen for the first appraisal meeting. A "no" in an early row should redirect the intervention before funds are committed. It is deliberately short; its purpose is to require the appraisal to distinguish funding shortages from land, service and affordability constraints. Different failed conditions require different financing or non-financing responses.

Gate	Evidence the team must produce	What changes if the answer is no
1. Real affordability	Worst-month residual income after payment, fares and essentials	Smaller product, rental option, income-linked support
2. Location and services	Jobs access, utility capacity, hazard and maintenance plan	Reject or redesign site; fund shared works first
3. Transferable right	Ground-verified claims, occupants, enforceable lender charge	Resolve claims; use permitted alternative secur- ity
4. True credit gap	Reasons for denial among qualified applicants	Improve underwriting or origination, not cheaper funds
5. Additionality	Counterfactual loan price, supply response and likely land gains	Re-target subsidy; use competitive allocation
6. Public exposure	Budgeted grant, guarantee cap, stress losses and exit rules	Price or cap risk; stop hidden liabilities
7. Accountable delivery	Occupancy, defect remedy, rights of appeal, open results	Hold milestone payments; publish failure and re- pair

Choose the response indicated by the binding constraint and verify the benefit reaches intended households. Where households have money to repay but lenders cannot verify earnings, invest in better underwriting and a supervised credit product. Where land claims are contested, resolve them before lending. Where price exceeds income even at a fair interest rate, reduce the cost of the home or provide a transparent grant; do not stretch the loan into an unsafe repayment burden. Where construction costs are high because materials markets lack competition, act upstream. Where safe rental rooms are missing, finance small landlords and their infrastructure under tenant-protection conditions. Where lenders lack duration, a refinance facility may be the right instrument, with verified pass-through to new borrowers. Fieldwork should distinguish improved terms for existing borrowers from sustainable access for new ones.

Prepare the local decision

Assess applications, rejections, worst-month payments, arrears, remedies, and household outcomes together. With appropriate permission, reconstruct household cash flow across high- and low-income months. Include existing debt, fees, deposits, essential spending,

transport, and plausible income or interest-rate shocks. Check property value and enforceable rights independently of the loan seller. The credit and title cases above require separating collateral documentation from the ability to repay. Loan officers, risk staff, consumer advisers, valuers, and representatives of informal earners should review anonymised borderline cases. Record why applications are rejected and whether a smaller loan, staged construction, rental option, or grant would address the actual gap. The regulator and guarantee provider should agree loss-sharing and workout responsibilities before expanding credit. Compare cash-flow tools with a documented manual assessment and test errors for groups with sparse records. Use only data necessary for the lending purpose, with a correction route. Keep model scores advisory where their performance for new borrower groups is untested. Follow loan cohorts through arrears, rescheduling, foreclosure, and housing retention. Report improved terms for existing borrowers separately from sustainable access for previously excluded households. Chapter 21 follows public support through the accounts needed to verify who benefits.

Credit support should match the financing gap and assign each risk to a party able to manage it. Credible rights, an affordable loan size, verified cash flow, and capable servicing matter more than reproducing a score threshold or concessional rate. Titles do not create repayment capacity. Delayed subsidies, omitted fees, and automated rejection without recourse can defeat the intended access benefit. Test income shocks affecting many borrowers at once, model drift, fraud and servicing costs as lending expands beyond the pilot population.

Sources and evidence notes

Source numbers link to the evidence used in the text. Official aggregates, institutional claims, quasi-experimental research and illustrative calculations carry different levels of confidence. Report years refer to the period measured, not necessarily the year of publication.

[1] OFFICIAL STATISTICAL REPORT Central Bank of Kenya. Bank Supervision Annual Report 2025, pp. 27–29 (mortgage survey and Table 11).

[source] December 2025 stock, rates, arrears, loan maturity, obstacles; survey figures.

[2] OFFICIAL STATISTICAL REPORT National Housing Bank, India. Annual Report 2024–25, Box 3.4, Graph 3.8 and Annex II (published 2026).

[source] INR 35,61,277 crore system-wide individual outstanding; separate slab table excludes pooled loan purchases.

[3] INSTITUTIONAL STATEMENT International Finance Corporation. IFC and Aavas Financiers to Catalyze Affordable Green Housing Finance (2020).

[source] Describes lender client segment and approach; corporate programme evidence.

[4] LENDER OPERATING STATEMENT Aavas Financiers. What Is Assessed Income for a Self-Employed Home Loan? (2026).

[source] Describes appraisal methods; no independent causal outcome estimate.

[5] CAUSAL RESEARCH: NATURAL EXPERIMENT Galiani, S. and Schargrodsky, E.. Property Rights for the Poor: Effects of Land Titling. Journal of Public Economics 94 (2010): 700–729.

https://www.cedlas.econo.unlp.edu.ar/wp/en/doc-cedlas103-pdf/ Buenos Aires land allocation; investment and schooling gains without a credit channel. [additional source]

[6] CAUSAL RESEARCH: DIFFERENCE-IN-DIFFERENCES Field, E.. Property Rights and Investment in Urban Slums. Journal of the European Economic Association 3 (2005): 279–290.

[source] Peru urban titles; investment response largely financed without loans.

[7] CAUSAL RESEARCH: POLICY TIMING Field, E.. Entitled to Work: Urban Property Rights and Labor Supply in Peru. Quarterly Journal of Economics 122 (2007): 1561–1602.

[source] Tenure security and labour supply; does not estimate mortgage supply impact.

[8] INSTITUTIONAL ADMINISTRATIVE REPORT Kenya Mortgage Refinance Company. Annual Integrated Report 2024, pp. 11 and 21.

https://kmrc.co.ke/assets/file/e4f9034c--kmrc-annual-report-2024v2.pdf Cumulative refinanced mortgages, wholesale funds, geographic concentration; institution-reported. [additional source]

[9] INSTITUTIONAL ADMINISTRATIVE REPORT Nigeria Mortgage Refinance Company. Mortgage Refinancing: How it Works and Bond Issuances (accessed October 2026).

https://nmrc.com.ng/our-business/mortgage-refinancing/ Business model and total refinancing as of December 2024; institution-reported. [additional source]

[10] INDEPENDENT RATING OPINION Agusto & Co.. Affirmation of NMRC Series II 15-year bond rating (2025).

[source] Federal guarantee and sovereign contingent-risk structure.

[11] OFFICIAL PROGRAMME DECISION Prime Minister of India / Union Cabinet. Cabinet Approves Pradhan Mantri Awas Yojana–Urban 2.0 Scheme (2024).

[source] Announced interest subsidy amounts, caps, tenure and five annual instalments.

[12] QUASI-EXPERIMENTAL PROGRAMME EVALUATION Housing microfinance study. Improving Housing Conditions: Labelled Loans in Kenya and Uganda. Journal of Development Effectiveness (2022).

[source] 2,341 clients, 1,213 offered loans; non-random phased branch rollout limits causal precision.

[13] CAUSAL RESEARCH: QUASI-EXPERIMENTAL Berger, D., Turner, N. and Zwick, E.. Stimulating Housing Markets. Journal of Finance 75 (2020): 277–321.

[source] United States buyer tax credit; sales and price effects in a post-crisis market.

[14] OFFICIAL PROVINCIAL PROGRAMME GUIDANCE Western Cape Government, South Africa. Housing Subsidy for Income Bracket R3,501–R22,000 (accessed October 2026).

[source] First Home Finance / FLISP qualifying income and subsidy scale; confirm local application rules.

Scope and calculations. The Kenya and India repayment examples are hypothetical fully amortising loans calculated using A = P × (r/12) / [1 − (1 + r/12)^(-12*T)], with nominal annual rates and monthly instalments. They exclude fees, insurance, taxes, downtime, inflation and early repayment. Kenya's market figures concern outstanding residential mortgages reported by regulated lenders, and cannot be interpreted as all housing credit. India's small-loan percentages are calculated from NHB Annex II's loan slabs excluding pooled purchases; they do not classify household incomes. The text distinguishes causal findings from policy announcements, provider accounts and analytical judgments. Source dates refer to the periods covered by the underlying reports. Administrative figures, provider statements and causal study estimates require different interpretations.

21. Subsidy accounting and accountability

Account for housing subsidies through the full public contribution, the burden on payers, and the benefit retained by households. Track land, taxes, grants, guarantees, and continuing obligations together. Compare programme results with the actual alternatives available to intended beneficiaries, then make allocation and delivery verifiable. A levy-funded programme illustrates why legal liability and eventual benefit must be examined separately.

An illustrative levy comparison shows why legal liability, economic burden, and receipt of housing benefits are separate questions. A formally employed cleaner sees a housing levy deducted from her payslip each month. A contractor has been awarded work on a large housing estate. A family renting beside the estate applies for a flat. The levy payer, contractor, and applicant have different claims on the programme's resources. Kenya's Affordable Housing Act established a levy of 1.5 per cent of employees' gross pay, with an equal employer contribution; the law also provides for a levy on other gross income. For a worker earning KES 50,000 a month, the employee deduction is KES 750 and the matching employer payment is KES 750. Across twelve months their combined statutory payments amount to KES 18,000. The employer's ultimate economic burden may fall partly on wages or prices: the statute alone cannot tell us. [1] Collection and audit figures need to be interpreted within those distinctions.

1 Record the full contribution and intended benefit

Public contributions and household benefits

Reported levy collections and audit findings establish resources and implementation concerns, not final welfare or misconduct conclusions. Treasury figures reported in August 2026 showed roughly KES 206.5 billion in collections over three fiscal years, including KES 79.1 billion in 2025/26. Collections and housing delivery must be reported separately. A special audit of the programme's status as at March 2025 found serious gaps in readiness, coordination and documentation. The audit's affordable-housing project list contained 142 projects; physical verification classified 60 as under way and 78 as yet to start, with four not verified. Those counts describe the particular project list and audit date. They should never be mistaken for a current count of all housing units delivered under the programme. [2–4] The Auditor-General also reported that, among 394 projects examined for land documentation across a broader programme portfolio including markets and related infrastructure, 269 lacked adequate verifiable records at the time of review. The housing ministry subsequently reported improvements in the number with supporting documents. The audit questioned the absence of a consolidated feasibility study and a formal programme-wide affordability analysis before the rollout. These findings identify implementation risks; they cannot, on their own, establish corruption, the final fiscal cost or the number of future households helped. [3–5] The legal status of the levy also remains distinct from whether the programme provides value.

A legal decision on a housing levy does not establish who ultimately benefits from the spending. Contemporary reporting states that a five-judge Court of Appeal upheld the

Affordable Housing Act and levy on 25 September 2026. This account relies on contemporary reporting; the full judgment text was not available. It also noted a flaw in the earlier legislative process: the Commission on Revenue Allocation should have had an opportunity to consider the Bill before Parliament acted. That reported decision does not establish delivery or affordability. Programme accounts must connect public contributions to occupied homes and identify their beneficiaries. [6] Building on Chapter 7's analysis of subsidy incidence, this chapter develops the accounting and contractual arrangements needed to track public contributions and verify household benefits. Programme accounts need to trace the complete public contribution through to occupied homes.

Tracking the full public contribution

A subsidy is a public resource supplied on terms more favorable than a market transaction. It can take the form of a cash grant to a buyer, cheap public land, an interest concession, free trunk infrastructure, a tax exemption, a rent payment, a loan guarantee or a favorable development right. These instruments have different accounting treatment. Each consumes fiscal resources, creates a contingent liability or gives up an asset or revenue stream that could have served another purpose. A government may report that its cash contribution to a housing project was KES 1 billion. The economic contribution may be much larger if the project also received valuable public land, waived charges, a favorable loan guarantee or a new road. The converse can also occur: an expensive public intervention may induce new private construction, generate taxes and create benefits for neighboring residents. A credible assessment must trace all resources and all outcomes, rather than treat the treasury transfer as the whole cost or the announced unit count as the whole gain. The prices at each stage determine how that contribution is divided among households and other parties.

The distribution of housing support depends on land, development, household, and location prices. The land price affects the cost of obtaining a site in a particular location. The construction price rewards materials suppliers, workers, builders and developers. The financing price pays for capital and risk over time. The cost of residence includes rent, loan payments, service fees, transport, lost earnings and upkeep. Together these costs determine whether the household can stay. Public money can lower one price while pushing up another. Imagine a household offered a subsidized flat at a price below its construction cost. The offer has little value if the family cannot raise the down payment, the daily commute absorbs the saving, the building has no reliable water, or a management company collects fees that rise faster than income. Conversely, a modest grant for a household's existing plot can fund useful improvements when the family already has secure location, earning networks and water access. Equal subsidy amounts can yield different household benefits. The framework therefore separates financing, delivery, and welfare claims that require different evidence.

Exhibit 21.1. Points at which public support changes beneficiaries

Decision in the chain	Party with discretion or bargaining power	Typical route by which value shifts	Observable test
Site selection and land transfer	Landowner, planning authority, land agency	Site price rises; public land transferred below its opportunity value	Independent land valuation; title; comparable transactions
Density, use and permits	Municipality, local politicians, large neighbors	Permission creates private land value; delay favors connected builders	Rights granted; approval times; permit-to-completion rate
Roads, water, drainage	Utilities, public works agencies, landholders	Public works increase private site values and landlord rents	Before/after land and rent panels; network-service tests
Materials and contracts	Contractors, suppliers, procurement bodies	Concentrated supply and change orders absorb grant	Competitive bid spread; verified quantities; defects
Finance and guarantees	Banks, developers, fiscal authorities	Cheaper capital boosts bids for scarce land or shifts default risk to taxpayers	All-in interest and fees; stress losses; guarantee claims
Selection and occupation	Allocating agency, brokers, landlords, lenders	Connected or already-bankable households obtain the scarce unit	Applicant-to-occupant audit; income, gender and exclusion data
Upkeep and resale	Building manager, owners' group, utility, local government	Maintenance shortfalls erase value; resale restrictions push sales underground	Fee arrears; defects; resale and twelve-month occupancy

Analytical framework. These are channels to test, not findings that capture occurs in every project.

A programme's funding, additional housing, and household benefit should be evaluated as separate claims. Legal allocation asks who is entitled to a grant. Economic incidence asks whose welfare or profits rise after prices and behavior adjust. Additionality asks

what would have happened without the programme. A family can receive a legal grant while a landlord captures some of its value through higher rent. A developer can build a subsidized flat that a household would have purchased anyway. A project can help its direct occupants while raising property values for nearby owners. Each is a different claim and requires different evidence. Price-support evidence illustrates why the amount paid cannot be equated with the benefit retained.

Price support and housing costs

A grant's effect depends on supply, market visibility, and the opportunities for price adjustment. Suppose a government promises each eligible buyer a KES 900,000 grant. If serviced land is abundant, permits are predictable and builders can add homes, competition gives sellers a reason to reduce net prices and increase output. If the best locations have a fixed supply of permitted plots, some of the subsidy can be bid into land values and asking prices. Owners who purchased land before the policy then receive a windfall. Developers who bought after the rise may have normal profits and still sell expensive homes. The appraisal should record when the subsidy was announced and when land changed hands. The same reasoning applies to rent assistance. The relevant parameter is the response of rents to a change in benefit: how many additional cents of rent follow an extra unit of public aid, for the same dwelling and household, compared with a plausible no-reform path? A rise in housing quality or a move to a better neighborhood complicates the calculation. So does a benefit that is temporary or hard for landlords to observe. Rental-assistance studies provide differing incidence estimates under different designs.

Incidence estimates should preserve the difference between a benefit change, a ceiling change, and the market studied. Studies of rent assistance report different incidence estimates. Gabrielle Fack's study of a French reform estimated that 78 cents of each extra euro of housing benefit appeared in higher rents paid by new recipients. Scott Susin's earlier study of US vouchers estimated rent increases of 16 per cent in the low-quality housing market of the ninety largest metropolitan areas under his model. These findings come from specific countries, periods and methods; neither number is a universal landlord share. [15–16] Other designs yield very different results. A 2026 Journal of Public Economics study of a Finnish reform estimated a rent increase of about 3.6 cents for an extra euro of allowance, with a standard error of 2.8 cents. This small estimate was not statistically distinguishable from zero at conventional levels. An earlier Finnish study also found little evidence of price changes at allowance thresholds. The authors interpret the small changes in recipients' housing choices as one reason rent responses were limited. In the United States, Robert Collinson and Peter Ganong found that a one-dollar increase in a voucher programme's rent ceiling increased same-address voucher rents by 13–20 cents; a different Dallas ceiling design improved the neighborhoods participants chose. A change in a ceiling is not identical to a one-dollar increase in a household's cash benefit. [17–19] A local test needs information on vacancies, competition, household choice, and supply.

Exhibit 21.2. Evidence on housing assistance and rents

Evidence and design	What changed?	Reported rent response	What a policymaker should learn
France, Fack (2006), reform- based study	Additional housing benefit for new claimants	Around €0.78 per €1 benefit	Rapid supply response cannot be assumed; landlords may capture much of the gain
United States, Susin (2002), metropolitan comparison	Voucher programme exposure	Modeled 16% rise in low-end rents	Marketwide effects on nonrecipients deserve measurement; identification differs from modern experiments
United States, Collinson–Ganong (2018 research version)	Voucher rent-ceiling rules	$0.13–$0.20 per $1 ceiling increase for same address	Payment rules and ability to move shape incidence and neighborhood choice
Finland, Eerola and coauthors (2026), quasi-experimental reform	Increased allowances for affected units	€0.036 per €1; SE €0.028	Large transfers can accrue mainly to residents when housing choices respond little

These are different causal estimands and markets; figures should be read side by side, never averaged.

A subsidy designed to help poor tenants should be tested in a market with known vacancies, competing landlords, real tenant choice and a plan to increase well-located supply. Tight price ceilings may protect a treasury budget while preventing access to good areas; loose ceilings may strengthen landlords' bargaining position. Digital rent data can improve assessment, although incomplete records of cash tenancies and informal subletting will bias any model that observes only formal agreements. India's programme channels illustrate how the accounting varies with the route through which assistance is delivered.

2 Compare subsidy results and fiscal exposure

Comparing subsidy instruments in India

National programme totals should be reconciled with the household's remaining contribution and the obligations outside the grant. The programme's public dashboard, last updated on 30 September 2026, reported 127.68 lakh sanctioned homes, 121.03 lakh grounded for construction and 99.07 lakh completed, approximately 12.77 million, 12.10 million and 9.91 million homes, respectively. It reported central assistance released of INR 1.80582 trillion (INR 180,582 crore). These are administrative stocks accumulated over the scheme's life, not an estimate of the net number of households newly housed, the total economic subsidy or the quality of occupied neighborhoods. [7] The programme routes determine who can use that support.

Different programme routes reach different households and exclude others. Under the original programme, beneficiary-led construction helped eligible families improve or build on their own land. Affordable Housing in Partnership routed public support through multi-unit projects; a credit-linked scheme helped households with mortgages; and in-situ slum redevelopment attempted to combine land, relocation and construction. These channels solve different problems. Beneficiary-led construction can make incremental investment possible and preserve jobs and social ties. Its binding condition is access to an acceptable parcel. The landless renter in a dense ward can be excluded from that route even if her need is greater than that of a titled owner. [8–9] Households that qualify must still finance their remaining contribution.

The household contribution can remain large enough to prevent completion. In 2023 a parliamentary standing committee reported that an average economically weaker section house under PMAY-U cost about INR 650,000 (INR 6.5 lakh) and that the average beneficiary contribution was around 60 per cent. Those figures describe the programme and period examined by the committee; they do not represent every state, project or the later PMAY-U 2.0 design. At that benchmark, 60 per cent means INR 390,000 (INR 3.9 lakh) from the household and other resources beyond the central grant. A nominal subsidy can therefore leave a family with a payment large enough to stop construction. The committee reported that missing state shares in some states increased the household burden. [8] That gap must be addressed across the institutions responsible for delivery.

Construction-stage payments cannot substitute for coordination among governments, lenders, and households. The programme divides responsibilities across several institutions. The Union government can announce a uniform benefit and count sanctions. States must provide their shares and clear land. Cities must issue permits and connect services. Banks decide whether the household can borrow the balance. Families supply savings and labor, often before all agencies have delivered. Each actor can meet its own narrow target while a house remains unfinished. Release of national funds on verified construction stages reduces one information problem, but it does not by itself cover a household's remaining financing gap. Programme design rules then need verification through actual occupancy and use.

Eligibility and financing provisions specify intended delivery without proving that families occupy suitable homes. PMAY-U 2.0, issued in September 2024, makes some of these choices explicit. For beneficiary-led and partnership units in most states, the central government offers INR 150,000 (INR 1.50 lakh) and the state must provide at least INR 100,000 (INR 1 lakh); funding splits differ in some states and union territories. The interest subsidy reaches up to INR 180,000 (INR 1.80 lakh) under its applicable terms. The new rental vertical allows conversion of suitable publicly funded vacant housing and the construction, operation and maintenance of rental buildings. Applicants for beneficiary-led assistance must establish rights to the land, with states able to provide heritable, nontransferable rights to some landless applicants. None of these design provisions by itself proves that the intended family occupies a good home. [9] Relocation evidence shows why household outcomes can diverge from construction and allocation records.

Programme appraisal should include relocation costs and households excluded before becoming eligible applicants. Winners received the opportunity to move from an urban slum to improved homes on the periphery. Fourteen years later, the researchers found better housing among winners, but no gains in income or human capital and weaker access to family and caste support. Thirty-four per cent of lottery winners never moved in; another 32 per cent subsequently left. These were lottery winners in one programme, not PMAY-U beneficiaries. The results support measuring access to livelihoods and mutual help alongside physical housing quality when appraising relocation. [10] Construction grants can increase the value of property already held by titled owners. Contractors receive paid work, while state and city agencies face matching-fund and infrastructure obligations. These benefits and costs should be recorded explicitly. Renters, seasonal workers, migrants, households lacking documents and tenants discriminated against on grounds of caste or religion may enter the process late or never. An application portal can make these exclusions harder to see if the denominator is defined as eligible applicants rather than all housing-insecure households. Two distinct audits are needed to examine delivery and inclusion.

Audit both recorded delivery and the households omitted or unable to use the benefit. One traces money through central government, state, city, developer and household accounts. The other traces people: eligible households, applicants, sanctioned beneficiaries, completions, actual occupants and those who sell, rent out or leave. Link the two only under legal consent and privacy controls. Without both, impressive construction figures and genuine beneficiary gains can coexist with serious exclusion. Kenya's levy programme requires the same distinction across land, contracts, occupation, and maintenance.

Housing levy accountability in Kenya

A levy-financed project needs the economic cost of its site and obligations, not only their book value. The compulsory levy pools contributions and finances a national building programme. Its financing design combines contributions from many people with benefits distributed over time: many people contribute, a smaller number obtain a home now, and the stock of affordable housing expands over time. Its justification depends on delivery, fairness to low-income contributors, and an allocation rule that citizens can scrutinise. Public accounts should show how recurring contributions finance completed, well-located and occupied homes. At the planning stage, the first risk is site readiness. A selected parcel must pass a chain of tests:

public or private ownership, boundaries, rights of occupants, permissible use, access roads, trunk services, resettlement obligations, geotechnical conditions and the legal power to transfer sectional units. A public agency can list the land as free; if the land is disputed, unserved or occupied, a contractor inherits costs and delay that no single budget line properly owns. A site with a low book value can be expensive to society. Announced project counts must then be reconciled with readiness and contract commitments.

Delivery accounts should link prepared sites and construction contracts to the agencies responsible for use and maintenance. Cash can accumulate faster than projects absorb it. The Auditor-General's special audit found weak contract documentation, gaps in approvals and contractor supervision, and the lack of an integrated programme appraisal. Subsequent government work may correct these gaps. That possibility makes dated project-level reporting essential: publish the status as at the audit, what changed afterward and the independent evidence of correction. [3–5] Procurement choices matter because a contractor's payment is not automatically a household's subsidy. Awarding multiple projects to a small set of contractors may bring economies of scale and speed; it may also exhaust working capital and supervisory capacity. A building programme may gain support from contractors and officials responsible for delivery. Local utilities and county officials face unfunded connections and maintenance obligations. Tenants in informal settlements may fear relocation. Construction agreements must therefore include the agencies and residents responsible for occupation and maintenance. The welfare assessment must follow the entire investment rather than one reported milestone.

Follow levy finance through verification, delivery, occupation, and household benefit. Does a low-income buyer qualify for the home and for finance? What deposit and monthly payment are required? Will the resident pay for transport, water storage, repairs and management? Are the units truly additional to market production? How does the allocation affect households who do not receive one? If a nearby landlord raises rents after roads and water arrive, part of the public investment goes to that landlord, although the physical works may still create social value. One straightforward reform would give each project a public subsidy record. It would join audited cash flows to land opportunity cost, contract changes, independent certifications, public eligibility rules, verified household occupation and the funded maintenance plan. The identification number would stay with the site when officials or contractors changed. Aggregate information would be public; sensitive records of individual households would remain protected. The audit findings make verification, coordination and enforcement necessary parts of programme administration. South Africa's housing evidence shows why location can change that benefit even under the same grant model.

Housing assets and job access in South Africa

Housing-asset programmes require parcel-level scrutiny of location as well as evidence of asset delivery. Grant-funded ownership has therefore been a prominent route to redress. Yet inherited urban geography puts inexpensive land far from many places where jobs and incomes are found. A government may hand over a debt-free asset and still leave a household worse placed to earn a living. The Gauteng City-Region Observatory mapped publicly supported

housing against major economic centers using provincial data available in July 2008. Average distances between housing areas and the nearest selected centers varied sharply: near Midrand the mean was 7.8 kilometres; for sites on the outskirts of Pretoria and Johannesburg the mean reached 25.7 kilometres and 20.3 kilometres, respectively. These are distances to the chosen centers using an older project dataset, not measures of each resident's actual daily commute. They show the importance of checking location at the parcel level. [13] Different employment findings make household circumstances and destination conditions central to the comparison.

The welfare effects of subsidised housing vary with household selection, location, and access to work. Simon Franklin used longitudinal household data and spatial variation in the location of housing projects to study households who received free government housing. His research found income gains driven substantially by increases in women's employment, suggesting that secure housing can ease constraints on work. Nathalie Picarelli used a fuzzy regression-discontinuity design around eligibility rules to study relocation. She found a decline in household labor supply two to four years after relocation, driven mainly by women's hours, with greater distance from opportunity an important possible mechanism. The designs, places, time horizons and treatments differ. Their findings should prevent a sweeping verdict that subsidized ownership always helps or always harms employment. [11–12] The national programme must also respond to differences in location and household need.

An applicant register cannot establish the location and service requirements of every household. In March 2026 South Africa's housing minister reported approximately 3.4 million households on the National Housing Needs Register, while cautioning against treating that number as the single official housing backlog. Applicants may value a formal allocation, but their needs differ by household and location. Peripheral sites can reduce land-assembly costs while increasing the cost of access and services. Standardised greenfield projects may simplify construction; the appraisal must also include the roads and utilities the local authority will operate. [14] Rental institutions offer a different way to assign continuing responsibility for those requirements.

Regulated social rental housing connects asset management to continuing income, but its finances still have to work. South Africa also offers a useful counterpoint through regulated social rental housing, where a professional housing institution retains responsibility for the asset. Better locations and sustained operating income can improve access and care. Such institutions face their own risks: debt service, collection failures, subsidies paid late and pressure to dilute income targeting. A tenure model that works in one rental market will fail in another if its cash flows cannot cover repairs and management. [22] These alternatives should be compared through the household outcomes identified by the research.

Compare the value of the home with changes in earnings, access, costs, and mobility. A practical rule follows from the two studies. Before comparing a subsidized home with a rental room, value the asset and the change in earnings, household travel time, school continuity, informal insurance, health, upkeep and freedom to move. The household should make the trade-off with credible information. A grant of the same amount can finance homes with very different travel costs and employment access. Appraisal should retain both positive and adverse results when comparing the proposed home with the real alternative.

The contrasting employment results make household selection, care responsibilities and job access part of subsidy appraisal. They cannot be reduced to a clean experiment changing only location. Compare the proposed home with the household's real alternative and retain both positive and adverse findings when setting eligibility, location and tenure rules. [11,12] Public land introduces another contribution whose value and distribution can be missed in the cash account.

Public land contributions

Public-land subsidy should be valued explicitly and separated from restrictions that preserve scarcity returns. Suppose a government transfers a serviced city parcel to a developer for a nominal fee in exchange for 500 affordable units. The land could instead have been leased competitively, reserved for future schools or used to create more homes through smaller plots and higher density. Its opportunity cost equals the value of the best feasible alternative after taking account of public purposes and legal constraints. An independent appraisal should measure that cost before the agreement is signed. Land enters politics earlier than the building contract. A landowner benefits when zoning, water and a public road raise what others will pay for a site. Unequal access to information or discretionary allocation can give a developer favourable rights before competitors can bid. Public-land disposal can transfer decades of appreciation through a single cabinet decision. Planning committees, land registries, courts, incumbent occupants, neighboring owners and utility providers all hold veto points. The appraisal should distinguish the protection of legitimate rights from restrictions that preserve scarcity returns. Contract design can reduce capture while retaining feasible construction incentives.

Transparent valuation and enforceable obligations can preserve public value without assuming every concession reaches households. First, value land at its highest legally feasible use, with a recorded comparison to other social uses. Second, award sites through a transparent competition that specifies quality, lifetime affordability, location and number of homes, rather than choosing the developer and negotiating benefits afterward. Third, make affordable obligations enforceable for a fixed period, with clawbacks when units are diverted to higher-income buyers or price caps are breached. Public ownership or long leases can preserve land value for future beneficiaries, although those models require durable management institutions. Occupied sites also need representation and remedies for existing claims.

Land regularisation and redevelopment require explicit protection for tenants and other groups absent from formal ownership records. Counting legal titleholders alone misses the people who pay rent, earn locally and depend on the site. Thailand's Baan Mankong programme offers a contrasting institutional model: housing support and infrastructure money flow through community organizations, complemented by savings and collective loans. CODI describes a package of roughly 80,000 baht per household with components for infrastructure, construction and community process, and cooperative land and housing loans at 4 per cent before cooperative on-lending margins. Programme descriptions document its design; they do not provide randomized estimates of housing welfare. Independent scholarship shows both the achievements of collective bargaining and the conflicts within community groups. [20–21] The

recipient of a subsidy gains authority over how it is used. A developer-led contract concentrates authority in the firm and the public agency. A community-held budget lets organized residents negotiate with landowners and contractors, while creating risks of elite control within the community and exclusion of new migrants or tenants. Neither institutional choice automatically represents everyone. The rules should provide representation and appeals for tenants, new migrants and other groups at risk of exclusion. A worked account shows how to consolidate these contributions before claiming a subsidy amount.

Subsidy accounting for a 1,000-home development

A consolidated subsidy account should include the economic value of land and services alongside cash expenditure. Consider a hypothetical 1,000-home development in an African city. The benchmark cost of a completed unit at the same site and specification is KES 2.71 million, using competitively procured construction, the independent value of land and off-site services. Assume normal competitive margins are already included in the construction amount. The government supplies cash and noncash support. The final price paid by the eligible household is KES 1.65 million. The illustration below makes those entries comparable with the household price.

Exhibit 21.3. Public subsidy and household benefit: an illustrative model

Item	KES per unit	KES million, 1,000 units
Competitive construction and normal margin	2,250,000	2,250
Public land, at independent opportunity value	300,000	300
Public off-site roads and connections	160,000	160
Economic benchmark: all-in unit cost	2,710,000	2,710
Cash subsidy budgeted by government	900,000	900
Noncash public land and service contribution	460,000	460
Gross public resource commitment	1,360,000	1,360
Actual household purchase price	1,650,000	1,650
Apparent price advantage versus benchmark	1,060,000	1,060

Item	KES per unit	KES million, 1,000 units
Unexplained gap between public input and price advantage	300,000	300

All inputs are invented for teaching. The KES 300,000 gap signals a question for audit. It is not proof of overcharging or corruption: specification changes, financing and measurement errors must first be ruled out.

Cash-only reporting can understate the public contribution and misstate the value reaching households. First, public land and infrastructure make up KES 460 million of the economic contribution, even if the housing agency's budget shows only KES 900 million. Second, the price discount that can be demonstrated from the stipulated benchmark is KES 1.06 million per household, or about 78 per cent of gross public support. Third, an unexplained KES 300 million difference must be reconciled against real costs, quality, timing, developer returns and disposal terms. An account limited to cash omits the land and service contribution and can misstate the value received by households. Location and continued occupation add further tests of what that contribution purchases.

Add household access costs and continued use before interpreting an accounting subsidy as a benefit. Assume the flat requires KES 3,500 more a month in transport and KES 1,200 a month in new fees and repairs than the family's previous dwelling. At a 5 per cent real discount rate, those additional outlays have a five-year present value of approximately KES 244,000 per household. If the family's other benefits and costs were unchanged (a strong and deliberately unrealistic assumption), the apparent KES 1.06 million price advantage would fall to about KES 816,000 after these five-year costs. That calculation omits improved safety, services, tenure, forgone rent, financing payments, resale value, health and employment effects. It demonstrates which data are missing; it estimates no project's actual benefit. The denominator also changes when completed units are not securely occupied by eligible households.

Cost per completed unit should be reported alongside cost per additional year of safe eligible occupancy. Finally, suppose all 1,000 homes are completed but only 700 are securely occupied by eligible households throughout the first year. The gross public commitment becomes KES 1.94 million per occupied unit, compared with KES 1.36 million per completed unit. Vacancy, diversion and exits are therefore fiscal outcomes, rather than mere management statistics. At sustained full occupancy across ten years, gross subsidy is KES 136,000 per occupied home-year before discounting or upkeep; at 70 per cent sustained occupancy it becomes approximately KES 194,000. A serious project model would discount future years and add long-term public obligations. No single ratio settles whether the programme is worthwhile. Some households obtain an asset with far greater value than its sticker discount. Other households may receive a unit they would have purchased anyway. People outside the estate may gain from roads or lose through higher rents. The economic analysis must compare the project with credible alternatives:

upgrading existing housing, serviced plots, rental grants, small-landlord loans and in-situ redevelopment. The appropriate unit of public output is the additional year of safe, well-located housing for a low-income household, accompanied by measures of welfare and fiscal risk. Those measures alter the incentives of the parties funding and allocating support.

3 Make allocation and continuing obligations accountable

Interests in funding and allocating subsidies

Subsidy reform redistributes benefits and obligations among households, producers, lenders, landlords, and public agencies. Identify those changes using application records, project accounts, and interviews. Municipalities need a funded service obligation; banks need payment and collateral rules; households need a product and location they can use. Test these conditions before fixing the subsidy instrument. Households with limited representation or documentation can face several barriers to housing support. A migrant renter cannot supply a property deed. A woman whose work is informal may lack the documentation a lender's model expects. A Dalit or Muslim applicant can encounter discrimination in the private rental market or exclusion within local housing committees. In South Africa, the geography of apartheid remains embedded in the market for well-located land. Across many African cities, structure owners and landlords mediate access to rental shelter in informal settlements and can collect the gains from infrastructure while tenants bear higher rents. The release of funding should reflect the delivery outcome each actor controls.

Tie money and authority to specified obligations within the delivery sequence. A programme sets a target for affordable units. The housing ministry obtains a subsidy allocation. Land must then be assembled; a county or municipal government must allow density; a utility must promise connections; a builder must find finance; households must pass eligibility and credit tests. Each stage gives an actor the power to delay the rest. When the overall programme is judged on sanctions or contracts signed, these actors can negotiate for changes in their obligations without being answerable for resident welfare. The reform task is to change whose signature releases money and which outcome each actor must deliver. Evaluation must also consider effects on nearby households and people outside the recipient list.

Exhibit 21.4. Interests and responses in subsidy reform

Actor and gain sought	Effective veto or source of rent	Coalition-building response	Risk that needs a safeguard
Public land authority and incumbent users	Title, land release, compensation, occupancy claims	Independent valuation, consent and negotiated rights; reserve public share of uplift	Displacement of tenants or customary rights

Actor and gain sought	Effective veto or source of rent	Coalition-building response	Risk that needs a safeguard
Homeowners and neighborhood groups	Hearings, lawsuits, council votes	Publish service-capacity plans, infrastructure funding and design standards	Exclusion without a demonstrated safety benefit
Developers, material suppliers, contractors	Bids, change orders, insolvency, land banking	Compete on delivered affordable homes; milestone payments and open quantities	Cartels, related-party contracts, diluted affordability
Banks and mortgage institutions	Credit screening and release of construction finance	First-loss support only for verified additional borrowers; repair title defects	Guarantee subsidy to already- bankable clients
Local governments and utilities	Permits, rights of way, trunk capacity, O&M	Pay municipalities for real service obligations; earmark maintenance revenue	Unfunded operating liabilities and selective provision
Tenants, migrants, women, marginalized groups	Often lack formal veto rights	Independent enumeration; appeals; rental and stay-in-place options	Elite-led resident lists and automated exclusion
Politicians and fiscal authorities	Budget priorities, allotment announcements	Public project registers, independent audits, published counterfactuals	Unpublished allocation criteria and inflated pipeline counts

An increase in housing production does not by itself establish access for low-income households. New market-rate supply can ease rents over time, especially where cities allow substantial density. But the first homes in desirable locations may still be priced beyond low-income households' reach. Conversely, a fully subsidized project in an isolated location can provide immediate shelter while leaving nearby exclusive districts untouched. Governments need both additional supply and a credible distribution rule, with honest accounting of the trade-off between them.

Subsidy accountability should include spillovers and exclusion, with staff and enforcement assigned to the proposed remedies. Rebecca Diamond and Tim McQuade studied Low-Income Housing Tax Credit developments and found that new projects in low-income neighborhoods raised nearby house prices by about 6.5 per cent and reduced crime, while projects in higher-income areas lowered nearby prices by about 2.5 per cent and attracted lower-income residents. These are local estimates from US settings. They suggest that nearby owners may face different asset-value effects across locations. The study's price estimates alone do not establish which households support or oppose development. [23] These cases suggest a conditional implementation sequence. Begin where legal rights, utility capacity and land can deliver benefits quickly. Offer existing residents enforceable protections before announcing a project. Explain clearly who will receive homes and who will finance connections. Couple density permissions with timely utility investment. Put independent review and appeals around allocation. For expensive central land, require a public account of its value and the time for which homes will remain affordable. Expand only when measured occupation and household welfare justify it. These steps require staff, budgets and enforcement; their effect on allocation and household welfare must be evaluated. Reporting should retain separate results for each affected group.

Report outcomes for recipients, rejected applicants, displaced residents and future entrants separately. Digital accountability can help maintain those distinctions across a large portfolio.

Digital accountability and exclusion risks

Digital review should follow the decision and beneficiary at each stage rather than treat one database as a complete account. A city does not know every household's title status. A construction inspector cannot stand on every site. Remote sensing, digital registries, mobile payments and machine-learning tools can reduce the cost of matching public resources to actual assets and residents. They can also increase the authority of the agency or platform that controls the data. The relevant technology test has five parts: what constraint changes; by how much; who receives the saving; who gains power; and what new harm becomes possible? The answer changes at each stage of the subsidy. Discrepancies need an official able to act and an accessible route for affected people to challenge them.

Exhibit 21.5. Technology and the distribution of subsidies

Use in the subsidy chain	Constraint and measurable gain	Who could capture the gain?	New power and risk	Minimum control
Satellite and parcel screening	Fewer unbuildable sites; lower survey time; earlier flood and access flags	Landowners tipped off before public acquisition; connected developers	Land data can raise speculative bids or expose informal occupants	Publish criteria; restrict pre-acquisition access; verify rights on the ground

For example, automated satellite screening might cut the time to identify plausible sites by half. That is an illustrative target to test, rather than an established impact estimate. A pilot should record hours saved, later land disputes, rejected sites and final serviced-unit cost. If the savings merely enable a developer to bid more for favored land, the reform has improved information but failed to pass value to households. A risk model that rejects a woman because she has no formal salary can make loan processing faster and expand discrimination at the same time.

The information infrastructure should therefore follow the subsidy record. Assign persistent identifiers to land, planning rights, infrastructure commitments, contracts, units and anonymized household outcomes. Keep a historical record of valuations and contract revisions. Permit independent auditors to reconcile announced units, certified homes, legally allocated homes and verified occupation. Release aggregate statistics and methods, with purpose-based access controls, retention limits and legal safeguards against surveillance or retaliation involving identifiable tenant records. Better records only change incentives when somebody can act on discrepancies and when those who are wrongly excluded can challenge the record. The accountability agreement should convert that information into conditions for approval and payment.

Subsidy accountability agreement

Select the subsidy route against local tenure, livelihoods, services, and the household's total occupation cost. Include land, guarantees, tax concessions, infrastructure and maintenance support in one lifetime public account. Treasury and housing agencies must reconcile funds and benefits; independent sampling must verify households and non-recipients. Track sustained occupancy, full household cost, work and school access, distribution of subsidy, exclusion and public cost per outcome. A government considering a housing programme should require a subsidy pass-through agreement before committing land, capital grants, guarantees or tax benefits. The agreement sets requirements for approval, payment and enforcement. It links public money to additional housing and binds the rights of the intended residents to the contract. Gate 1: Define the household. Map the local distribution of incomes, irregular earnings, tenure, rents, deposits, commuting costs and household size. Count migrants, women-headed households, informal tenants and families who lack titles. Use a citywide denominator, then explain the eligibility rule. Reject a programme whose target household cannot pay the total cost of occupying the proposed home. The next condition is a complete valuation of the public contribution.

Price land, infrastructure, fiscal support, and risk before choosing the transaction and its public obligations. Gate 2: Price the public contribution. Record cash, land, trunk services, waived fees, tax expenditure, credit support and future maintenance obligations. Obtain an independent land valuation and compare a realistic alternative public use. Where a guarantee is offered, publish an expected-loss estimate and a severe-stress estimate. Gate 3: Establish the counterfactual. What would the market build without the subsidy, on the same site and over the same period? Would tenants be displaced? Could in-situ upgrading, rental finance or serviced

land help more households? Compare at least three credible options and explain why the preferred option offers the best estimated welfare gain per unit of public spending, stating uncertainty and distributional trade-offs. Gate 4: Publish a net price. Tender for the maximum verified reduction in lifetime housing cost to eligible households, conditional on standards, safety, service connections and location. Disclose normal developer margin and the method for adjusting construction costs. Disclose the land contribution and affordability obligations of any joint venture. The agreement must then bind inclusion, location, and maintenance outcomes.

Specify the households, locations, affordability period, and recurring funding that public support must secure. Gate 5: Bind inclusion and location. Make rent, deposit and service-fee limits enforceable where required. Count current occupants before clearing land. Provide appeals, resettlement protections, tenure choices and priority rules that can withstand independent scrutiny. Value livelihood and social networks during relocation decisions. Gate 6: Pay on verified outcomes. Advance funds for works where necessary, with independent certification and safeguards for contractor cash flow. Retain a performance-linked share until basic services work and the specified units are occupied. Record defects and assign responsibility for repair. Housing completions should be checked in the field as well as in the database. Gate 7: Fund the life of the neighborhood. Show a twenty-five-year plan for roads, water, drainage, buildings and common spaces. Identify who levies fees, collects them, subsidizes hardship cases and replaces major assets. Identify the funding source for maintenance after the construction grant ends. Incidence and additionality measures should determine whether the programme continues unchanged.

Give an accountable authority the power to act on measured shortfalls, with independent challenge and remedies for exclusion. Gate 8: Measure incidence and additionality. Within twelve months of occupation and again at three and five years, publish the share of intended households still present; total shelter and transport costs; changes in work, school access and safety; vacancies and resales; neighborhood rents; and public cost per additional occupied home-year. Use randomized tie-breakers when morally and legally acceptable, phased rollouts when feasible, or carefully matched comparisons. Publish why some evidence remains uncertain. The agreement should have an owner with authority over budgets and contracts. An interagency committee that merely receives presentations cannot halt a disbursement, remove a stalled parcel or secure a utility connection. The binding condition must be held by a minister, city chief executive or programme board with transparent powers, independent challenge and clear liability for wrongful exclusion. A first application can begin with one scheme whose records can be reconciled.

Exhibit 21.6. Required fields in a public subsidy record

Public field	Why it must be visible
1. Project and parcel identifiers	Reconcile land, contracts, funds and homes across agencies

Public field	Why it must be visible
2. Land rights and current occupants	Prevent disputed transfer and invisible displacement
3. Independent land value and alternative use	Price the in-kind public subsidy
4. Cash and in-kind budget by year	Prevent partial fiscal reporting
5. Buyer or tenant eligibility and appeals	Test whether the poorest can enter
6. Contract awards, related parties and amendments	Detect preventable procurement rents
7. Infrastructure connections and service standards	Distinguish completed walls from usable homes
8. All-in prices, deposits, fees and mortgage terms	Measure household payments and benefits
9. Physical completion and safety certificates	Stop payment for unfinished or unsafe dwellings
10. Allocation, initial occupancy and continuing occupancy	Count homes used by intended residents
11. Maintenance reserve, service provider and defects	Protect value after the handover
12. Independent evaluation, grievances and corrections	Review, contest and enforce the agreement

Personal identities and sensitive tenancy records remain in secured systems. The public passport provides aggregated eligibility and occupancy outcomes, not household-level dossiers.

Test the accounting system on a traceable project while distinguishing early reconciliation from long-term welfare evidence. Treasury assigns a finance officer to reconcile land, grants, tax support, guarantees, and operating commitments; the municipality checks permits and services; a risk-based sample of costs and occupied homes receives independent verification. Record unresolved balances and missing households rather than manufacturing a complete total. Use that reconciliation to correct the next budget or allocation decision. Wider procurement or legal reform should follow the identified problem, with its own staffing, consultation, and timetable. A first-year accounting improvement cannot establish a long-term welfare gain. Public reporting should then show the resources, additional housing, retained access, and other beneficiaries together.

Measuring the benefits of public spending

Public reporting should identify the resources committed, the housing added relative to a credible alternative, the households retaining access, and the other parties receiving economic benefits. Fieldwork should make these claims traceable across locations and household groups.

Prepare the local decision

Reconcile subsidies and outcomes by site and household group so that aggregate delivery does not conceal exclusion from valuable locations. Reconcile receipts, commitments, payments, land concessions, tax costs, guarantees, and operating support at project level. Link them to completed homes, occupied homes, beneficiary eligibility, and full household charges. Keep reporting dates and portfolio boundaries explicit. The Kenya example above illustrates why a project-start count and a revenue total cannot establish the subsidy received by an occupied household. The revenue agency, treasury, housing administrator, municipality, auditor, and utility must identify the records they hold and the officer responsible for discrepancies. Include unsuccessful applicants and residents displaced by the project. Require a documented explanation of changes to eligibility, rents, or land terms after the initial approval. Add location to the subsidy reconciliation. Publish aggregate public support, affordable homes occupied, and retention by well-served opportunity area and by peripheral location. Include off-site payments and the delivery they financed. Ask the auditor to reconcile promises with occupations and charges, not only payments with invoices. This makes it possible to identify a programme that meets its total unit target while leaving prosperous areas effectively closed. Unmatched balances and records should remain visible until the responsible authority resolves them.

Use digital matching to support reconciliation and investigation rather than hide unresolved balances. Publish aggregate reconciled accounts with version history, and restrict access to personal eligibility data. Match records to identify unexplained gaps for investigation; retain cases that cannot be matched rather than silently dropping them. Independently check a sample of both payments and occupied homes. Report public cost per sustained affordable occupancy alongside beneficiary income, location, exclusion errors, and appeals. Keep unreconciled balances in the report until resolved. Chapter 22 applies this accountability to the development agreements governing public land.

Subsidy accountability should follow durable household access and the capture of each public contribution. Security can improve welfare when it relaxes a real constraint and households retain job access, affordable recurring costs, and reliable services. Copying a grant amount or ownership target neglects those conditions. Remote locations, missing contributions, and price capture can reduce the benefit while a completion dashboard still records success. At national scale, test fiscal exposure, land-price responses, spatial concentration and the administrative cost of checking actual occupancy.

Sources and evidence notes

Numbers are attributed to the dated sources below. Administrative figures refer to the specified dates. Causal study results apply to the population, design and time described by each paper. Financial examples in Exhibits 21.3 and subsequent calculations are illustrative.

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22. Public-land development agreements

Use public land through agreements that specify the housing benefit purchased and the obligations needed to sustain it. Establish rights and opportunity costs before choosing sale, lease, or retention. Compare the transaction with feasible alternative uses, fund services, and make affordability and operator duties enforceable. A contested well-located site exposes the competing claims that this appraisal must address.

Tafelberg illustrates the competing public claims on a valuable site near employment and services. The former Tafelberg School at 355 Main Road in Sea Point, Cape Town, was a provincially owned site near jobs, buses, schools and clinics. The provincial government planned to sell the property to a private school for R135 million. For domestic workers, cleaners and other low-paid people who served the district, the proposed use raised a question about access to well-located housing. Housing groups sought social rental housing on the site. Provincial officials defended the sale as lawful and argued that its proceeds would serve other public needs, with housing supplied elsewhere. The dispute concerned both the use of a public asset and low-income workers' access to an expensive urban location. A legal decision about those claims still requires a financed delivery agreement to produce housing.

1 Establish the rights, alternatives, and value of public land

Public land and access to jobs

Meaningful engagement and spatial obligations must be followed through to finance, completed homes, and tenants able to remain. The South African Human Rights Commission's account of the Constitutional Court judgment of 2 July 2026 states that housing location is relevant to the state's obligations to address apartheid-era spatial exclusion. It also records a finding that public engagement over the proposed disposal was defective. Contemporary reporting describes an order for the province and city to report on programmes, budgets and delivery in central Cape Town. The ruling followed a 2024 Supreme Court of Appeal decision favouring the authorities. The full Constitutional Court judgment was not available; these propositions are attributed to the Commission and contemporary reporting. [1–3] The legal decision was followed by further planning and delivery work. The authorities filed reports in early October 2026. Cape Town said it had released 14 municipal sites and was preparing another 39, while critics questioned the lack of completed social homes in the central locations at issue. Plans for Tafelberg remained in design and approval. Reported unit mixes changed as planning advanced; a land-use application discussed in October envisaged roughly 440 homes of mixed tenure. These planning figures do not establish delivery. Verification requires signed finance, enforceable allocations, a completed building and low-income tenants who can stay. [4–5] The appraisal should therefore compare the site's uses through their enforceable outcomes.

Public-land decisions should connect valuation, resident rights, services, development duties, and long-term enforcement. A government ministry sees an underused asset that could finance schools or hospitals. A city sees a potential residential block that needs water,

drainage and public transport. A developer sees scarce development rights. A nearby owner may fear congestion or changing property values. A hypothetical household commuting two hours each day may value the opportunity to reduce travel time and costs. These claims on the same plot should be assessed in a common appraisal. This chapter examines the terms under which public land can support affordable housing: valuation, development obligations, resident rights, infrastructure commitments, and long-term enforcement. The first step is to establish which rights and responsibilities the term "public land" actually describes.

Rights and responsibilities attached to public land

Public ownership does not establish that occupancy, development permission, infrastructure, and disposal authority are resolved. It says little about the rights attached to the parcel, its physical condition, its best use, or the people who already rely on it. An old railway yard may be owned by a national corporation, occupied by tenants, used informally for workshops, marked as industrial in a planning map and located beside a transport stop. A vacant military compound may be contaminated or lack a drainage outlet. A former market may contain traders whose claims never entered the registry. Committing a site before resolving these matters can lead to disputes, delay or an unfinanceable project. There are at least five separate decisions. The government must establish what it owns and who else has rights. It must decide which public service deserves the site. The planning authority must define buildable use and density. Utilities must commit to service capacity. Finally, a developer and a housing provider must accept an economic agreement that households can live with for decades. Each decision rests with different officials, budgets and legal powers. A register can clarify some of these questions without answering them all.

Maps can narrow an information gap. They cannot settle a disputed inheritance, confer legal title, negotiate with a street vendor or compel a water utility to connect a project. A public land register that records ownership alone does not establish that occupancy claims, services or development obligations have been resolved. The appraisal must also distinguish physical land scarcity from scarcity of usable rights and accessible locations.

Figure 22.1. Conditions for public-land housing development

1. Rights	2. Place	3. Permission
Recorded owner, other claims, residents, rights of way, disputes and legal power to transfer.	Access to jobs, schools, transport, markets and existing social networks.	Land use, density, height, heritage, flood exposure and building requirements.
4. Services	5. Deal	6. Stewardship
Water and sewer capacity, roads, drainage, power, schools and who pays to connect.	Land price, finance, developers, affordable rent or sale terms, allocation and sanctions.	Occupancy, repairs, tenant rights, renewal costs and a body able to enforce the agreement.

The chain is sequential, yet failures travel both ways. A developer cannot finance a project without clear rights; a title is of little use when service capacity is absent. These six conditions form the chapter's implementation framework.

Different forms of land scarcity imply different public choices and cannot be resolved by a map alone. Physical scarcity arises when a well-located area lacks space. Regulatory scarcity arises when plots exist but rules prohibit enough housing on them. Institutional scarcity arises when no agency can release land on credible terms, supply infrastructure, settle disputes and monitor the resulting homes. Density changes, land assembly and administrative coordination address different forms of scarcity. A transfer of ownership alone does not establish usable housing capacity. Ownership also generates options. A city can sell now; lease for a fixed term; develop and retain the buildings; exchange land for a fixed number of affordable units; combine land with transport investment; or protect existing occupants and help them build incrementally. The choice should follow a comparison of lifetime social value. A high auction bid reveals a strong private willingness to pay for a legal interest. It gives the treasury useful information. It does not, on its own, tell the minister how many low-wage workers should be able to live near jobs. The value of those choices should be recorded even when the land is transferred without cash.

The opportunity cost of public land

A public-land contribution has an opportunity cost that must be compared with its housing and inclusion benefits. Project accounts may call the land free. The public has still surrendered something valuable: the chance to lease the plot at market rates, build a hospital, preserve a park, or save it for a future need. The relevant cost is the value of the best feasible alternative after legal restrictions, development costs and social obligations are considered. In public finance this is an opportunity cost. It belongs in the appraisal even when no money leaves the treasury. Government actions also change land values. A new station, a drainage channel, extra building rights or a permitted change of use can turn a low-value site into an expensive one. If the state transfers the plot before settling these rights, a buyer may capture much of the gain. If the state waits and sells at full market price, it can collect more revenue but lose a rare chance to secure inclusion in a central location. Four linked cost accounts can make that comparison explicit.

Compare land rights, serviced development, household payments, and location costs over consistent periods. A project that earns a high land receipt may move families to a remote suburb and create new public expenses for transport and extensions to utilities. A central rental project may surrender land revenue yet reduce the costs borne by workers and employers. These savings should be measured rather than assumed. The worked lease example applies those accounts to an accessible public site.

Calculating the public contribution of land

Public economic commitment = the independently appraised market value of the rights granted minus the market value of the consideration returned to the state, plus

cash grants, services and risk guarantees, minus verified public revenues or assets retained.

Household gain = the measured difference in shelter quality, cash housing costs, travel, time, employment access and security over a credible alternative, after fees and future upkeep. Land value and household gain use different measures; the analyst should show both rather than add them without care.

Valuing a public-land lease in Nairobi

A public lease should be valued against the additional targeted homes and credible alternatives it delivers. Take an illustrative two-hectare, well-located public site. Suppose an independent valuation puts a sixty-year unrestricted development lease at KES 1.20 billion. Under a tender with enforceable affordability conditions, qualified bidders offer KES 650 million for the same sixty-year interest. The proposal provides 600 apartments, of which 400 will be allocated to a verified low-income group at controlled rents. The public utility companies must invest another KES 300 million to serve the site. For simplicity, assume the bids, lease term, infrastructure scope and state residual rights are directly comparable. The apparent land concession is KES 550 million; the new network spending is KES 300 million. Combined public commitment equals KES 850 million, or KES 2.125 million per contracted low-income apartment. This is a resource cost, not a cash cheque to each family. If an independent review later finds that only 300 of the 400 apartments are additional homes occupied by households in the target group, the same amount equals KES 2.83 million per additional targeted apartment. It may still be good policy, especially if the homes are beside employment. The appraisal must compare this higher cost per additional targeted apartment with credible alternatives. Continuing affordability obligations determine whether the initial concession retains its public value.

Exhibit 22.2. Valuing a public-land transaction

Assumption or result	Value	Interpretation
Market bid for unrestricted 60-year lease	KES 1,200m	Comparator, including lease reversion on equal terms
Winning bid with binding affordability terms	KES 650m	Less cash received for the same land rights
Implied public land contribution	KES 550m	Difference in like-for-like bids
Additional trunk infrastructure	KES 300m	Government or utility investment

Assumption or result	Value	Interpretation
400 contracted affordable units / 300 additional units	KES 2.125m / 2.833m	Public commitment per contracted / additional unit
Illustrative household travel saving: KES 150 × 250 days × 15 years	KES 562,500 gross; ≈KES 321,000 present value	8% discount rate; excludes time, wages and transport rebound

Illustration, not an appraisal of any actual Nairobi parcel. Time savings, rent differences and occupancy must be measured; the travel line is an assumption and the present value uses an end-of-year 15-year annuity. The public-land contribution is sensitive to valuation, rights granted and market competition.

Lease appraisal needs long-term enforcement and comparison of feasible designs rather than a single initial price. A developer can promise 400 affordable homes and later seek permission to raise rents, alter the eligibility rules, reduce apartment sizes or shift maintenance costs to tenants. The land discount is paid at the outset; the social benefit arrives over decades. A weak contract gives the state little power once the building is occupied. The agreement needs enforceable, long-lived claims on the affordable units, transfer restrictions and independent checks. There is a further comparison. A treasury official may reasonably ask whether KES 850 million could buy more housing elsewhere. A housing minister may reply that the other site is an hour from work. Both should use a common appraisal: unit affordability, employment access, public fiscal cost, tenant security and future costs of new roads and services. Site-level procurement should permit alternative designs and allow evidence to decide. Cape Town's legal and institutional process shows why those duties require several actors to agree.

2 Compare how public land produces housing benefits

Legal decisions and housing delivery in Cape Town

Different public institutions can hold valid objectives without any one of them having power to deliver the housing outcome alone. The provincial department responsible for disposing of land sought cash for other government needs. The housing department had identified possible social use. The city controlled aspects of zoning and services; national authorities shaped the housing grant rules. Campaigners, supported by legal advocates, challenged the sale and argued that central location mattered for rights and opportunity. Heritage requirements and neighborhood opposition added constraints. None of these actors had unilateral power to deliver social rental homes. Engagement must therefore lead to an enforceable agreement on finance, services, and occupation.

Legal engagement requirements need to be translated into financed development and continuing tenant rights. The Commission's account of the July 2026 judgment establishes meaningful engagement as a requirement in the disposal process. Contemporary reporting also

describes an obligation to account for housing programmes in central Cape Town. Yet an October city submission also claimed a broader housing pipeline and described funding constraints, while campaigners pointed to persistent gaps in completed central-city homes. The dispute shifted from the legality of selling one site to the resources and delivery needed for an entire neighborhood programme. [1–5] The case distinguishes the legal conditions for land disposal from the finance and management needed for housing delivery. Litigation and resident representation can change a disposal decision, while land servicing, social-housing grants, operations and maintenance still require funded agreements. A successful second coalition must include housing institutions, national and provincial budget officials, developers able to deliver mixed schemes, the municipality and the workers intended to live there. The agreement must also provide tenants with enforceable service and affordability rights after occupation. Kenya's programme illustrates how divided budgets and land custodians can complicate that translation.

Public-land delivery in Kenya

Public-land delivery requires commitments across housing entities, land custodians, counties, and service agencies. Yet announcing a site and proving that it can support housing are separate steps. The Auditor-General's special audit, issued in March 2026 and covering project status as at March 2025, listed 142 affordable-housing projects. On the audit visit, 60 had started, 78 had not, and four were outside verification. This is a dated project-list finding; it says nothing conclusive about all completions later in 2025 or 2026. [6] The audit also raised questions about documentation and readiness across a broader portfolio. Each incomplete record creates a practical problem: a developer may discover an encumbrance after contract signature; compensation may remain unsettled; a utility may face a service demand outside its budget; or a title dispute may stop bank disbursement. National agencies can move cash faster than counties can supply approvals. County governments carry some infrastructure obligations, while land custodians and housing entities work under different legal and budget lines. Digitising the registry cannot substitute for those agreements.

Investment readiness should require resolved claims and documented agency commitments rather than platform registration alone. The Commission on Administrative Justice reported in March 2025 that incomplete digitization and shortcomings in the ArdhiSasa system were causing delays and leaving room for corrupt practices. Peer-reviewed research describes cases where official searches became inaccessible because conversion and georeferencing were incomplete; community groups used their own mapping to make residents and informal claims visible. [7–8] For a city manager, the reform sequence begins with each plot, rather than a national target. Verify the root of title and occupancy; obtain the service-capacity letters; reserve funds for off-site works; decide who owns and maintains the apartments; then tender. A central government can speed the process by putting those tasks under a joint performance agreement with named county and utility officials. Investment-ready status should require documented commitments from each responsible agency. Ethiopia provides a test of whether broad public land control itself guarantees affordability.

Public ownership and affordability in Addis Ababa

Public ownership can facilitate housing production without eliminating fiscal costs or household entry barriers. State control over urban land gives government powers to allocate and lease sites at scale. Addis Ababa used these powers to build large condominium programmes. By the time of a World Bank–government study published in 2019, roughly 383,000 units had been built under the programme. The study estimated that government recovered around 35% of the production cost of a typical unit. Its indicative cumulative subsidy for the programme was ETB 150 billion, using the prices and stock recorded at the time. These historical estimates should not be presented as current fiscal figures. [9] The distribution of apartment access must be examined separately from the scale delivered.

Formal housing gains should be assessed alongside the households unable to receive or sustain the benefit. The study estimated that producing formal urban land commonly took four to seven years because compensation, preparation, infrastructure and administrative handoffs stretched across agencies. It found that poorer households often could not afford the required down payment and mortgage; large condominium developments on the urban edge also carried costs of distance and service extension. State ownership had lowered a barrier to assembling land, while construction finance, affordability and infrastructure remained binding. [9–10] The allocation rules also affect who receives the subsidy. A government can underprice the first sale, absorb land and construction costs, and deliver homes to better-off households who can pay the deposit. In that case, completed housing can serve a narrower income group than intended. Strong allocation audits and a choice of incremental housing, well-located rental and serviced land are essential complements to mass public construction. Public land control enlarges what government can do. It does not ensure that the poorest receive the benefit. Delhi illustrates another tension between land-disposal revenue and affordable-housing objectives.

Land auctions and public costs in Delhi

Land agencies' revenue and planning roles need a consolidated appraisal of public costs and housing access. Auctions can disclose transfer terms and protect the public purse against arbitrary low-price deals. In a November 2025 auction, the Authority received bids totaling about INR 14.947 billion (INR 1,494.7 crore) for 81 residential, industrial, commercial, institutional and group-housing plots, against aggregate reserve prices of INR 6.292 billion (INR 629.2 crore). Two group-housing plots in Dwarka reportedly attracted bids of about INR 6.425 billion (INR 642.5 crore) against INR 2.549 billion (INR 254.9 crore) in reserves. These figures describe bids for the listed properties, not a measure of money available for affordable housing or evidence that the auction produced low-income homes. [11–12] A competitive price tells the state what developers think they can earn from the permitted use. It cannot decide the mix of homes a city needs. That decision belongs in a prior plan and budget. A development authority that depends on land sales to pay salaries, roads or debt service has a built-in reason to favor auctions and higher-value uses. A housing department may ask for a discounted plot; the land agency experiences that request as forgone revenue. The appraisal must reconcile revenue needs with affordable-housing objectives. The strategy for what is sold deserves scrutiny before the transparency of the auction process is assessed.

Before any sale, record the responsibilities and objectives of each agency. The treasury should show the market value of the rights being surrendered. The housing department should show the expected number, location and lifetime cost of additional affordable homes. The planning agency should publish the wider housing-supply plan and infrastructure needs. Without the strategy, the bidding portal can be perfectly transparent while the choice of what is auctioned remains hidden from public debate. Where a disposal involves relocation, the household cost comparison becomes an additional condition.

Relocation costs in Mumbai

Formal consent or title does not establish that off-site housing preserves livelihoods and fair access. The authorities have sought public and former salt-pan land outside Dharavi to house residents who are outside the eligibility rules for rehabilitation within the original settlement. In 2024–25, the state approved use of about 256 acres in Mulund, Bhandup and Kanjurmarg for the wider redevelopment effort. Officials argued that the parcels were suitable and would serve households otherwise excluded. Critics raised flood, ecological and displacement concerns; residents of receiving neighborhoods objected to the arrival of relocated families. Courts have considered disputes over the land. [13–15] Three allocations intersect. The original high-value land inside Dharavi supports redevelopment economics. Rules governing who qualifies for on-site rehabilitation determine which existing households retain access to that location. State-controlled land elsewhere provides a fallback for excluded residents, at the cost of moving households away from jobs, tenants, suppliers and kin. Objections in receiving neighbourhoods should be assessed against service capacity, resident rights and evidence of exclusion. Formal title or planning consent alone does not establish that relocation preserves livelihoods or provides fair access. The destination needs an actual budget and journey assessment for each affected livelihood group.

The test for off-site land is an actual household budget and journey, calculated for each livelihood group. A tailor with customers in Dharavi, a waste picker using local sorting routes and a worker with a fixed industrial shift face different relocation costs. Official allocations should include rental and informal occupants separately, publish the eligibility and appeals rules, and guarantee transit, schooling, shop space and a credible right to return where that has been promised. Map hazards before selecting a low-cost parcel; exposure to monsoon flooding may cause repeated losses. India's PMAY–Urban in-situ redevelopment framework itself treats land as a project resource: for eligible government-land schemes it provides for land costs to be waived by central agencies, central assistance for slum homes, and additional floor-area rights or transferable development rights to support cross-subsidy. The implementation agreement depends on local economics. In a high-value district, sale units may finance rehabilitation. Where sale demand is weak, a developer cannot pay for hundreds of low-rent homes out of rights worth very little. Informal tenants, who often hold no qualifying documents, remain at risk of losing out even when the cross-subsidy works. [16] Pooling can avoid some outright acquisition while still requiring a separate test of inclusion.

Conditions for effective land pooling

Land pooling can secure public uses and owner participation without proving affordable access for non-owners. Owners give up part of their area and receive smaller parcels made more valuable by roads and legal permissions. With credible maps, compensation rules and phased trunk works, the design can reduce the need for compulsory acquisition. It is an important complement to state-owned land for city expansion. [17–18] Owners inside the scheme can gain valuable serviced plots. Renters, agricultural laborers, vendors and people living on land without recognized title may gain little, and speculative expectations may increase the cost of the affordable land reserved for them. Successful land readjustment therefore requires a separate plan for low-income housing, protection for existing households, and a transparent account of how the public share of the value increase will be used. Completed land readjustment therefore needs a separate assessment of low-income households' access. Vienna offers a different model that links land acquisition to continuing housing institutions.

Land acquisition and affordability requirements in Vienna

Long-term land and affordability institutions require contract monitoring beyond design and construction. The city's wohnfonds_wien supports land acquisition and uses developer competitions for subsidized housing. Bids are judged across economic viability, social sustainability, architecture and ecology. Successful offers enter sale or building-right agreements; the procedures explicitly allow penalties, exclusion from later competitions and reversal of a transaction if contractual promises are broken. These arrangements make the price of land part of a longer public contract. [19–20] The model requires a housing finance system, experienced nonprofit or limited-profit developers and a capable institution that can monitor design and construction. It also requires continuity of funding and contract enforcement. In 2026 Vienna reported plans for approximately 2,600 homes at its Eibengasse district, around two-thirds of them subsidized. These are planned homes; their expected mix is an official programme intention rather than proof of completion or long-run affordability. [21] Adaptation requires rules for evaluating bids and enforcing the resulting agreements. A city can ask bidders to compete over the lowest lifetime rent and highest quality for a fixed land contribution, rather than compete only over the biggest payment for the plot. Contract monitoring must continue after design and construction. Singapore shows the reach and costs of a more extensive public land and delivery system.

Land assembly and distributional costs in Singapore

Singapore's housing scale depends on coordinated land, finance, services, and renewal, with substantial public obligations. A government history records that nearly 70% of residents lived in slums or squatter settlements in 1960 and more than 80% lived in public housing by 1985. The 1966 Land Acquisition Act gave authorities strong powers to acquire private land, historically including at below-market compensation. Housing & Development Board (HDB) construction was paired with finance through the Central Provident Fund and public mortgages. [22] Those gains came with coercive resettlement and distributional costs. They rested on a distinctive city-state administration, land law and financing system. They cannot be copied

into African or Indian cities without confronting present-day constitutional rights, courts, customary claims, resettlement safeguards and local-government capacity. Public housing also requires substantial fiscal support: Singapore's HDB reported a S$6.34 billion net deficit before government grants in FY2024, including substantial costs for home ownership and upgrading. The public landlord must still pay to build, repair and renew its stock. [23] The comparison supports transferring functions rather than assuming public ownership alone is sufficient.

In Vienna and Singapore, land policy operates alongside continuing housing institutions and finance. The state needs skills to price sites, issue rights, plan services, structure finance, supervise construction, allocate homes and care for neighborhoods. Ownership without these functions can produce idle land. Delivery institutions also need lawful access to suitable sites in locations households can use. The Ahmedabad comparison identifies two limits: reserved amenity sites can remain unbuilt when finance and coordination fail, and tenants may remain outside the land agreement. Public control over a site is therefore only one input. The agreement must connect its opportunity cost to financed services and enforceable access for the intended residents. [101] The actual transaction must still address the objectives and authority of the institutions controlling it.

3 Contract for delivery and preserve public value

Interests in public-land decisions

Public-land agencies may face revenue, audit, and service incentives that differ from the housing programme's objective. Developers need feasible permissions and financing; municipalities need service funding; current users need their claims resolved. Record these constraints and legal responsibilities before deciding how the site should be used. These incentives create veto points. A national agency can refuse land transfer. A heritage authority can delay demolition. A municipality can hold planning approval. A water utility can withhold a capacity certificate. A finance ministry can decline the operating grant. A local political broker can mobilize tenants against surveying if they fear eviction. Courts can stop an unlawful sale. An administration that ignores these sources of power may have a technically sound feasibility report and still deliver nothing. A transparent comparison of alternative public uses can make those trade-offs explicit.

Exhibit 22.3. Interests in public-land decisions

Actor	What they gain or fear	Power or veto	Proposed agreement
Treasury / land custodian	Sale receipts, audit compliance; fear of undervaluation	Transfer consent, reserve price, title release	Independent value, explicit land subsidy budget, open alternatives

Actor	What they gain or fear	Power or veto	Proposed agreement
Municipal government	Rates, jobs, voters; fear of service overload	Zoning, permits, local works	Funded service plan, fiscal compensation and delivery targets
Utility and transport agencies	Connection charges; fear of unbudgeted loads	Capacity and connection timing	Signed works schedule and funded service obligations
Developers / lenders	Land upside, approval speed, debt recovery	Bid, build, withdraw or seek renegotiation	Competitive financeable contract, performance bonds and predictable approvals
Existing occupants, renters, workers	Location, livelihood and tenure; fear of eviction	Petitions, courts, protest, informal occupation	Independent census, tenant rights, fair allocation and appeals
Homeowners and business groups	Amenities, property values, workforce access	Public hearings, elections, litigation	Transport and school plan, open design, evidence on costs
Politicians and brokers	Credit for homes, access to scarce allocations	Appointments, land lists, eligibility and contracts	Published awards, citizen oversight, sanctions and independent audits

Compare sale, housing reservation, and retention through their funded public benefits and obligations. Each option has beneficiaries and costs. The governing body must compare them in public, rather than decide first and use consultation to legitimize the decision. Specify the benefits and obligations for each participant in the reform. Employers benefit from a more reliable nearby workforce. Transport agencies may avoid costly peak-hour trips. Low-income workers gain time and opportunity. Municipalities can receive funded network upgrades and future tax income. Housing providers receive predictable development rights. National treasuries can demand an independently appraised contribution and measurable household outcomes. A negotiated agreement should specify each obligation, its funding and the evidence used to value it. The agreement should then identify where preparation, delivery, or continued affordability could fail.

Public-land failure can arise before, during, or after construction and requires remedies at each stage. During site selection, incumbent land users and homeowners may defeat a plan before poor residents gain standing. During procurement, special exemptions and

renegotiation may transfer public land value to favored developers. During occupation, rent increases, weak repairs and illegal resales may slowly turn subsidized homes into market-rate assets. The first moment calls for early public participation; the second for open books and competitive tenders; the third for a housing institution that remains responsible long after completion. Compare mixed-income development on valuable sites with sale and replacement elsewhere, including transport and social access. Digital systems should be assessed against the specific decision or error they help resolve.

Assessing digital land systems

Mapping and monitoring are useful when they lead to an enforceable change in a public-land decision. A shared map can help agencies reconcile their records. The relevant test is whether that shared view leads to a different, enforceable decision. Registry reforms show why processing speed must remain separate from the validity of the underlying rights.

Exhibit 22.4. Assessing digital land tools

Tool and constraint	Measurable change	Who can capture the gain or new power	New risk and control
Satellite / AI parcel screen: public sites are hard to find	Verified candidate sites per 100 searched; days to ground validation; fewer false “vacant” labels	Land custodian, developers with early information	Existing tenants disappear from the map; publish aggregate occupancy evidence and provide appeals
Linked rights register: titles and claims are fragmented	Percentage of parcels with reconciled registry, court, allocation and field records	Formal owners and agencies controlling records	Old illegal allocations gain digital legitimacy; independent root-of-title review
Online tender with appraised alternatives: opaque land terms	Bidder count, price spread, related-party disclosures and contract amendment rate	Experienced bidders, powerful landholders	Transparent auction of an exclusionary land use; disclose land-use decision first
Service-capacity and permit model: agencies approve in sequence	Months from site selection to serviced permission; frequency of unplanned utility works	Utilities and large developers with better data	Forecast errors transfer cost to public; human engineers certify critical assets

Tool and constraint	Measurable change	Who can capture the gain or new power	New risk and control
Rent and occupancy audit: affordability weakens after handover	Verified eligible room-years, tenant complaints, rent and repair outcomes	Managers and governments with access to tenant data	Surveillance, discriminatory scoring and eviction; aggregate reporting, appeal and data limits

Proposed programme indicators, not measured technology impacts. Each intervention needs a baseline and a credible comparison group before its effect can be claimed.

Faster land transactions can magnify errors unless disputed claims and service commitments are resolved and reviewable. India launched guidelines for DILRMP 3.0 in September 2026 to link land records, cadastral maps, registration, court data and geospatial parcel identifiers. The stated objective is greater clarity and fewer disputes. A high share of scanned records cannot prove that ownership is lawful, that a plot lacks tenants or that an affordable rental project has become financeable. Implementation requires claims adjudication, engineering sign-off and protection for people absent from formal records. [7–8, 24] A useful AI pilot should therefore predict or detect a specific bottleneck, such as sites likely to fail a title review or networks unable to serve planned density. Its value should be measured by fewer failed transactions, lower elapsed time and additional occupied affordable homes. Training labels need independent checks; parcel data should carry uncertainty flags. Affected residents need a way to correct errors before the state transfers rights. Faster processing can increase harm if the underlying rights or service decision is wrong. The development agreement must consolidate those conditions before site approval.

Public-land development agreements

Select a locally lawful land arrangement after documenting claims and comparing the public uses of the site. Compare forgone land receipts with the value and cost of the promised public benefit, including recurring services. The public landowner must enforce the agreement; the housing operator and utilities must deliver; independent valuation and appeals must remain available. Track public subsidy including land, homes delivered and occupied by target groups, access, covenant compliance and displacement. Land-disposal rules should require the state to explain the value of each proposed use. The duty need not force every urban plot into subsidized housing. Parks, hospitals, markets, freight facilities and schools also make cities function. It should force a transparent choice, especially where scarce central land could permanently alter who can live close to work. The approval gates below assign responsibility for the remaining legal, financial, and service conditions.

Eight tests for site approval

Exhibit 22.5. Public-land development assessment

Gate	Question requiring documentary evidence	Release criterion
1	What legal rights are being offered?	Independent title, encumbrance and occupant review; dispute register and remedy
2	Why this site?	Jobs and service access benchmarked against at least two alternatives; climate and hazard screen
3	What is the forgone public value?	Two independent appraisals of identical land interests; credible options for public use
4	What can actually be built and serviced?	Planning permissions, funded network capacity, permits and a timetable with owners
5	Which families benefit, for how long?	Income bands, tenant recognition, rent/price rules, independent allocation and appeals
6	Will the scheme finance and operate?	Competitive bids, verified costs, debt terms, 20-year maintenance and stress testing
7	Who captures land and planning gains?	Land-subsidy calculation, beneficial owners, resale and rent covenants, sanctions
8	Who is accountable five years later?	Independent occupancy audit, public outcomes, repairs reserve, complaints and clawback

Each site gate needs a named official, evidence, and a recorded decision before a contract is signed. High-risk failure at Gates 1, 2 or 4 should stop transfer of the site. A project with no verified rights can impose involuntary displacement. A project with poor location may never be an effective housing intervention. A project without funded services will produce unusable buildings. Gate 6 can fail for good reasons; renegotiate the public agreement or choose another housing model before signing. The contract must then state what both the developer and the public receive.

Terms of the land agreement

A strong contract defines both what the developer receives and what the public receives. Specify the exact parcel, lease term and rights. Attach the independent valuation and

the approved density. State how many homes must serve each income band, their initial rent or sales price, the permitted growth of those charges, and the length of the affordability period. Provide for priority or protection of existing residents when applicable. Define the standard and funding of schools, drains, streets, water and public space. Make legal completion, safety certificates and tenant occupancy conditions for staged payment or release of guarantees. Control what happens after the project opens. A long lease, recorded agreement or other locally enforceable mechanism must bind successors. Require the landlord to maintain an audited reserve for lifts, roofs, sewers and fire systems. Require the manager to report aggregate eligibility, vacancies, rents and complaints. Set penalties for misuse, clear triggers for step-in rights and a route to transfer management to another competent provider. Occupancy rights and remedies should remain enforceable when officeholders change. Bidding should test that outcome against affordability, quality, and lifecycle requirements.

Procurement should compare feasible public outcomes rather than maximise revenue before testing suitability and rights. One method fixes the public land contribution and asks who can provide the most long-term affordable room-years under verified quality standards. Another fixes the minimum affordable provision and asks who will pay the largest land premium. Both require fair comparators, independent cost estimates and real competition. Bid design cannot repair a site that is in the wrong place or whose occupants were excluded from planning. The land agency needs a funded preparation process to make that comparison credible.

Preparing a public-land transaction

Budget the valuation, legal work, consultation, and independent verification required before award. Use these stages to prepare a manageable transaction; disputed claims or statutory processes can take longer than an indicative schedule. Stage 1: select a small portfolio of diverse public sites. Establish legal custody, physical occupancy, climate risk, access to employment, and the inventory of existing utilities. Bring tenants, neighbors, utility managers and employers into the mapping. Publish uncertainty about site readiness, while protecting identifiable resident claims. Stage 2: commission an independent land valuation on the proposed terms, with a second review where transaction rules or material uncertainty require it. Complete the service-capacity and location appraisals. Estimate the fiscal cost of selling, leasing, retaining and upgrading. Reserve any national and municipal infrastructure funds. Publish a short options paper before political approval. Stage 3: agree the legal route and eligibility rules, hold meaningful hearings, settle major claims, and issue a competitive tender with lifetime affordability requirements. Publish bidder identities, all public contributions and the reasons for the selected scheme. Appoint an independent engineer and an independent tenant/ household auditor. Financial close and service works should follow only when their conditions are met.

Proceed through financial close, works, and occupation with common time horizons and explicit additionality assumptions. Stage 4: secure financial close and begin service works. Record contract changes and test a small tranche of construction and allocations before full expansion. Compare implementation with similar sites not yet released where feasible, documenting selection and market differences; do not call them a causal control group without a defensible evaluation design. Measure time to approvals, additional housing starts, total public

cost and confirmed rights of displaced or existing residents. Extend or redesign only after field validation. The operating dashboard should track five-year eligible occupied room-years per unit of public economic commitment, alongside median commute, household housing-and-travel burden, actual service uptime, repair costs and eviction complaints. It should also report the number of additional homes relative to credible alternatives; raw completions can overstate the effect of a subsidy or a land release. Comparisons need common time horizons and explicit assumptions about who would have built anyway. The continuing review should test whether the public asset retains its intended housing value.

Preserving public value over time

Public-land performance depends on both the value committed and the institutions that preserve affordable access over time. Public land has two qualities governments seldom measure together. It is a scarce asset that can finance other public needs. It is also a way to decide who gets access to the city's most productive places. Once a well-located parcel is sold with unrestricted rights, buying back that option may be prohibitively expensive. Once a costly housing block is built in a remote place, its commute and service burdens can persist for generations. The appraisal should consider these long-term effects alongside annual cash flows. The cases report different outcomes under different institutions. Ethiopia mobilized land for hundreds of thousands of condominiums and struggled with affordability. Delhi used auctions to capture substantial bids, which say little about who can live in the new buildings. Cape Town's activists changed the legal terms of public-land disposal, while still facing a difficult task of financing and managing homes. Vienna links land transfers to developer obligations and sanctions. Singapore built a vast public housing system by coupling land acquisition to finance and a strong delivery state, with large fiscal and historical costs. Digital systems should be judged by their contribution to that service and housing outcome.

Digital land systems should reduce a documented delay or error while preserving disputed claims and appeals. Measure the resulting service and housing outcomes, and record who receives the savings. Fieldwork should verify the underlying claims, costs, rights, and obligations before approval.

Prepare the local decision

Compare public-land alternatives through verified value, existing uses, funded services, and enforceable housing benefits. Establish the public owner's legal powers, competing uses, occupants, claims, condition, planning rights, service costs, and independent valuation. Compare sale, lease, direct development, and continued public use on the same assumptions. The Cape Town and Vienna examples above require distinguishing proposed housing, enforceable obligations, and completed affordable homes. The landowner, municipal finance and planning teams, housing provider, utility, current users, and intended tenants should review the alternatives before procurement. Obtain independent advice on tenure restrictions and remedies. Record which party bears delay, low demand, operating deficits, and failure to preserve affordability, including after a transfer to a new owner. Include the location obligation in the public-land agreement. Specify the homes and income groups to be served on the site, the

affordability period, shared-service access, and remedies for non-delivery. If the bidder proposes to move affordable homes elsewhere, require a new appraisal of access, household costs and public value before approval. The largest land receipt cannot by itself establish that the transaction meets the housing objective. Any later increase in public support or reduction in affordability duties requires the whole agreement to be reappraised.

Technology and action. Maintain a public parcel and contract summary with valuation dates, milestones, affordability terms, and approved amendments; protect identifiable resident claims. Use imagery to check construction progress, then verify occupation and rents in the field. Release concessions against enforceable stages and retain remedies for breach. Reassess the agreement if public contributions rise or affordable obligations fall; evaluate both changes together. Chapter 23 examines how to finance the neighbourhood infrastructure on which that agreement depends.

A public-land agreement should connect the value contributed to delivery and continuing affordable access. Clear objectives, lawful disclosed rights, funded services, and enforceable operator duties make that connection testable. HDB's organisational form, a land bank, or a fixed discount cannot reproduce those functions. Treating land as free, excluding occupants, or leaving amenities unfunded can undermine even a low-price transaction. At portfolio scale, test management capacity, fiscal dependence on land sales and the enforcement of hundreds of long-lived covenants.

Sources and evidence notes

This chapter draws on the public sources listed below. The Tafelberg legal summary relies on the attributed official account and contemporary reporting because the full judgment text was unavailable. Reported programmes and planning proposals are identified as such. The Nairobi financial example and proposed decision indicators are illustrative. The references distinguish judgments, agency releases, programme estimates, research and news reporting; published case counts do not prove causal effects.

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23. Financing neighbourhood infrastructure

Finance neighbourhood infrastructure by assigning each capital and operating obligation to a credible payer and capable provider. Development charges, land readjustment, taxation, and borrowing serve different purposes and depend on different institutions. Test collection, delivery, maintenance, and tenant effects together before assuming that rising land values will fund the network. Upgrading in Dar es Salaam shows why service gains and their distribution need separate measurement.

Infrastructure improvement can expand access while changing rents and the distribution of benefits. Between 2015 and 2023, Dar es Salaam expanded roads, drainage and local infrastructure. Through the World Bank-supported Dar es Salaam Metropolitan Development Project, the authorities delivered about 207 kilometres of improved roads, with streetlights, footpaths and roadside drains; about 75 kilometres of major drainage; and local improvements in low-income areas. The Bank reports improved mobility for more than four million people. New drainage was designed to protect 406 hectares from the kind of flood expected, on average, once in ten years. These works were intended to improve all-weather access and reduce flood exposure. [1] Infrastructure improvements create benefits for residents, property owners, and businesses. Better roads improve access; drainage reduces damage; water networks support additional housing. Higher land values and rents can transfer some gains to owners. A tenant may benefit from better services yet be unable to remain after a rent increase. Appraisal must account for both effects. Output reports need household and service-cost measures to establish the result.

1 Estimate investment, operating costs, and household effects

Infrastructure investment and rents

Infrastructure counts should be linked to the reliability, access, and recurring costs experienced by households. Public project reports usually count kilometres built, drainage completed and people reached. They cannot, by themselves, tell us how many low-income households gained a lasting right to the improved location, how much rent changed, whether the new network was maintained, or how much of the increase in site value returned to the public purse. Dar es Salaam's published figures establish delivery at substantial scale. They do not establish the distribution of the resulting land-value gains. The public debate on affordable housing often begins with the price of a house. A developer offers a two-room unit for a given sum. A minister asks whether construction methods, land concessions or mortgage subsidies can lower it. Yet a house sold at cost remains a poor housing option if it has no reliable water, a septic tank that overflows, a road that floods, an electricity connection the family cannot afford, or a three-hour daily commute. The appraisal must include these service and access costs alongside the purchase price. Delivery requires an agreement across the actors controlling land, networks, budgets, and occupancy.

Infrastructure delivery requires agreement among network providers, landowners, municipalities, and residents. That agreement must cover capital spending, maintenance, access, and the distribution of gains. If that agreement fails, land speculation can thrive beside empty serviced plots and residents can endure poor services beside costly completed apartments. Four distinct infrastructure costs need to be included in that agreement.

Four components of infrastructure cost

A housing connection requires local works and resolution of the land and rights-of-way they use. Consider a developer building 500 apartments on the edge of a fast-growing African city. The building contractor can estimate concrete, rebar, labor, windows and lifts. The developer knows what a bank will lend against presales. The scheme still depends on decisions elsewhere: whether a water utility can supply peak demand, whether a sewer has an outlet and a treatment plant, whether the access road remains passable during heavy rain, whether an electricity feeder can carry the load, whether schools and markets can absorb 2,000 more residents. Each decision sits in a separate plan and often a separate balance sheet. The first bill pays for land and corridors. Streets, drains, pipes, easements, substations and retention areas need space. Public rights-of-way must be legally protected before plots are sold. Compensation, resettlement and utility relocation can exceed initial estimates. A line drawn across an aerial photograph may cross a religious site, a tenant's dwelling or an unrecorded market. Such conflicts can require redesign, compensation or legal resolution before works proceed. The wider network and its maintenance create additional bills beyond the site boundary.

Network capacity and preventive maintenance must be funded alongside local construction. A small apartment block needs on-site pipes, roads and wiring. It also needs larger shared assets: water treatment, trunk sewers, stormwater outfalls, power substations, bus services and access roads. Engineers separate internal works, links to the existing system, and bulk capacity. Each has a different beneficiary and a different funding claim. A developer can rationally pay for a building's internal reticulation. Asking one affordable-housing project to finance a treatment plant serving an entire district can overcharge its residents if existing users also benefit from the new capacity. The third bill pays for lifetime operation. A drainage channel needs desilting; a road needs resurfacing; a water network needs pressure management and leak repairs; a treatment plant needs energy, staff and chemical supplies. Sometimes a donor or national ministry finances a capital asset that the city lacks the money, tariff authority or technical staff to run. Capital approval should therefore include funding for preventive maintenance and responsibilities during heavy rainfall. Household access to opportunity adds a fourth cost that project accounts often omit.

The fourth bill pays for access to opportunity. A distant site can offer cheap land and a cheap housing unit while forcing its occupants to spend more money and time travelling to work. Good public transport can partly reverse that penalty. Schools, clinics, markets, childcare and public space matter in the same way. These networks connect the house to the income and services that make it affordable. A project's real cost to a family therefore includes housing payments, utilities, transport, foregone work time and expected losses from flooding or unreliable

service. These costs fall at different times and require a coordinated financing and service schedule.

Exhibit 23.1. Full household and public costs of serviced housing

Household affordability: rent or mortgage ✦ utility charges ✦ transport ✦ expected loss from outages and floods ✦ household share of repairs and fees.

Public investment: rights-of-way and land ✦ trunk and link networks ✦ neighbourhood amenities ✦ climate protection ✦ resettlement and grievance handling.

Exhibit 23.1 - the full cost of a serviced home

Whole-life fiscal cost: upfront investment ✦ financing and guarantees ✦ annual operations ✦ periodic renewal - realistic tax, tariff and land-value receipts.

Distribution test: count the benefits separately for landowners, homeowners, renters, firms, utilities and taxpayers; measure who can still live on the site five years later.

Every connection creates a time problem. Land can rise in price when a project is merely announced. Developers need an assurance of future service before they borrow. The public agency wants firm demand before it builds the trunk. The utility may refuse connections until the developer pays a charge. The residents cannot prove demand until they have moved in. These dependencies can delay the project. A joint financing and service schedule should resolve them and account for gains to existing landowners. India's investment and revenue evidence shows why capital needs alone do not establish delivery capacity.

2 Match financing instruments to delivery conditions

Infrastructure needs and delivery capacity in India

Infrastructure appraisal should test investment execution and operating capacity alongside estimated capital needs. A 2022 World Bank assessment estimated urban infrastructure investment needs of about US$840 billion over fifteen years, or roughly US$55 billion each year, to 2036. The estimate covered urban transport, water, sanitation, drainage, roads and other city services; it excluded housing construction and slum upgrading. At the time of the assessment, central and state governments financed more than 75 per cent of city infrastructure, urban local bodies around 15 per cent from their own surpluses, and private sources only about 5 per cent. The Bank estimated that ten of the largest urban local bodies spent only about two-thirds of their capital budgets over three fiscal years. [2] Additional borrowing can help creditworthy cities, but it must be accompanied by the ability to clear rights-of-way, prepare tenders, coordinate utilities and supervise contractors. The appraisal needs to assess both investment execution and the capacity to operate assets after construction. Local revenue and funded service commitments determine whether completed assets become usable housing support.

Low municipal revenue capacity makes explicit service-funding commitments essential to housing targets. The same assessment reported urban property tax collections of only 0.15 per cent of GDP during 2011-18, compared with 0.3-0.6 per cent for low- and middle-income countries in the comparison cited by the Bank. The precise benchmark depends on tax definitions and jurisdiction. The low collections indicate limited use of property taxation; they do not measure how much value from public investment could feasibly be collected. [2] A state can sanction thousands of subsidised housing units while the municipality that must provide drainage and solid-waste collection has little new revenue for either. A municipality may have to fund connections and operations while another level of government funds construction. Reconcile both budgets before approving the scheme. State how tariffs, transfers, and local revenues will cover the additional service load. The answer requires a joint investment schedule. A serviced-housing agreement should specify the pipeline, water capacity, power load, road links, drainage outlet and recurrent funding before a construction award. Each agency should identify the decision it controls, its cost estimate, the date of service readiness, and the legal consequence of delay. A construction target without these commitments can overlook the conditions needed for usable housing. Ahmedabad's readjustment approach offers one mechanism for obtaining land and financing part of the infrastructure.

Land readjustment in Ahmedabad

Readjustment can help assemble public corridors and housing land while sharing serviced value with owners. Under Gujarat's Town Planning Scheme (TPS), landowners in an urbanising area pool their parcels. The planning authority reserves land for roads, utilities, open spaces and other public purposes. The remaining land returns to owners as smaller, more regular plots with development potential. Instead of purchasing every road corridor through compulsory acquisition, the city distributes part of the cost across the plots expected to gain value. The model offers several possible gains to participants. A farmer can surrender some land and receive a better-connected plot worth more per square metre. The municipality obtains continuous streets and service corridors. Developers receive clearer, buildable layouts. State rules give the planning authority powers that a municipality negotiating one lane at a time would lack. A World Resources Institute study found that this system helped Ahmedabad provide land for roads, trunk services and low-income housing, including land that enabled about 33,000 low-income homes under an earlier national programme. [3] The agreement's inclusion depends on how it treats tenants and other users outside the ownership bargain.

Owner participation needs credible appreciation and separate protection for tenants and informal occupants. A review of Ahmedabad's Bombay Hotel area, a low-income Muslim neighbourhood, found that local objections led planners to reduce land deductions and alter road alignments to limit displacement. The compromise reduced planned infrastructure in places while allowing more residents to remain. A scheme designed around registered owners could otherwise have excluded households whose claims lay outside the land records. [4, 5] Ahmedabad also illustrates the distinction between reserving land and funding services. The WRI researchers describe public spaces and amenity sites that remained unbuilt because the authority could not fund the works. Delay arose from state-level approvals and poor coordination among agencies. Plot appreciation could benefit owners before residents received the promised public amenities.

Where future urban demand was weak, farmers had less reason to accept a deduction from their land. The agreement works best when land values rise enough to make participation credible. [3] Longer international experience reinforces the need to distinguish property ownership from livelihood and occupancy rights.

Land readjustment requires local legal capacity and an independent account of claims beyond the cadastre. OECD research reports that Japanese schemes often reserve roughly 30-40 per cent of readjusted land for public improvements and use some plots for sale to finance works. Japan generally requires the consent of two-thirds of participating owners and leaseholders for voluntary schemes. The relevant conditions for transfer are that land value gains can replace some cash compensation, provided there is a credible legal map, expected demand, and a process for dissenters. [6] In neighbourhoods with overlapping customary tenure, leaseholds, tenants and informal building owners, a registry-only scheme can transfer power to whoever controls the recorded parcel. It can expose occupants to eviction by a newly empowered owner. A land-pooling authority therefore needs an independent occupancy census, protections for renters and small businesses, access to appeal, and a schedule for financing the services promised. Record occupancy rights and livelihoods separately from cadastral ownership. Comparing it with in-situ infrastructure investment clarifies the different financing mechanisms.

Use maintained connections, affordable bills, and resident retention to compare infrastructure-financing approaches. Land readjustment can secure space for services; upgrading can deliver roads and drainage in existing settlements. Neither administrative coverage nor land reservation establishes that tenants retain the gains. Use completed and maintained connections, affordable bills and resident retention to judge whether the financing mechanism works. [101,1,3–5,7] Dar es Salaam provides an example of investment within existing settlements.

Services and formalisation in Dar es Salaam

Infrastructure in occupied settlements can improve mobility before a greenfield district would attract comparable use. Much of the demand for upgrading was already present in built neighbourhoods. The Metropolitan Development Project improved roads, drainage and public space across existing areas, with a special focus on underserved districts. Its reported outputs included approximately 207.8 kilometres of roads, 75 kilometres of trunk and secondary drainage, and improvements in 152 informal settlements, according to the Bank's December 2023 account. The reported number of people benefiting from improved mobility was about 4.1 million. [1, 7] Those gains still need to be assessed alongside clearance losses and distributional effects.

Upgrading existing areas can provide immediate access benefits while removing some homes and uses. A shop gets deliveries. Children reach school in the rain. Water no longer stands at a doorway after ordinary storms. Public works can change the economics of incremental construction: landlords are more likely to invest in durable rooms when streets and drainage reduce risk. Kenya's KISIP2 appraisal made a similar observation about earlier upgrading, describing new multistorey construction and local investment following works in beneficiary

settlements. It also reported that flexible road standards - including narrower one-way streets where suitable - reduced potential displacement by up to 85 per cent in some settlements, relative to applying ordinary planning widths. That figure is a project appraisal comparison in selected places, rather than an independently evaluated average. The appraisal should count housing losses from road clearance alongside the access benefits. [8] Higher attractiveness may also change rents and the types of housing landlords provide.

The distribution of upgrading gains depends on maintenance and whether lower-income residents retain access. A family that rented before the road was built may have no legal claim to the gains. Better drainage can also encourage construction on nearby land; if building expands into floodways, some protection may disappear. The impact evaluation should therefore track resident turnover, rent trajectories, land transactions, new floor area and maintenance alongside journey times and flood incidence. Large capital grants or concessional lending can provide roads and drains where annual municipal budgets could never cover the initial outlay. Those grants do not supply the maintenance contracts, spare parts, enforcement against blocked drains or compensation procedures for resettlement. Dar es Salaam's reported achievements show what a coordinated capital operation can deliver. Whether the gains remain progressive depends on who maintains the assets and whether lower-income households retain access to the improved neighbourhoods. A follow-up design can distinguish better access from property appreciation and displacement.

Track original residents, prices, and services to test who retains the benefit of upgrading. Before construction, enumerate owner-occupiers, tenants, landlords, businesses and public-space users; record baseline travel, rent, water prices and flood exposure; publish the location of every outfall and the responsible maintenance unit. After completion, repeat the survey at 12, 36 and 60 months. Compare trends with similar neighbourhoods still awaiting works where feasible. A programme can then distinguish improved access from rising property wealth and investigate whether rent increases displace intended beneficiaries. Development charges address the separate question of how additional construction funds shared capacity.

Development charges in Cape Town

Development charges should make the cost of additional network capacity explicit rather than leave it outside project accounts. Cape Town's policy, in effect since July 2014, levies a one-off capital contribution on a land development to pay its proportional share of additional municipal engineering infrastructure demand. Johannesburg's published 2021 policy similarly ties contributions to a development's added demands on external services and excludes maintenance and historic backlogs from the charge. [9, 10] The economic case is strong when the charge is predictable. Growth creates demand for pipes, substations and roads. A formula linked to additional capacity specifies part of that cost before approval. A developer who uses extra capacity can pay a fair share; existing ratepayers avoid carrying the entire cost of new private development. Developers also benefit from a reliable timetable and a clear method of calculating what they owe. When additional capacity is not priced or funded, a project can leave the utility with costs absent from the developer’s accounts. Their level and exemptions determine whether that financing supports or excludes low-income housing.

A charge or waiver is feasible only when the resulting capacity obligation still has a payer. If a small backyard landlord must pay the same large upfront contribution as a luxury developer with similar added peak demand, the first may abandon formal permission and continue to build informally. If a city loads historic service failures into a new project's charge, the price of entry rises for everyone. A waiver granted to an affordable-housing developer can lower the sales price or rent only when competition and binding affordability terms pass the saving through. If the city grants a waiver without replacing the lost infrastructure funding, service capacity may fail later. The resulting service failures can impose additional costs on residents. An exemption therefore needs eligibility, replacement funding, and an enforceable affordability condition.

A feasible exemption therefore has three linked parts: a clear eligibility rule tied to actual low-income occupancy, an explicit fiscal source that pays the waived contribution, and a recorded obligation to keep the units affordable. The municipality must publish the cost of the exemption, the agency providing bulk capacity and the terms under which eligibility ends. The charge can be reduced or deferred for qualifying rental rooms in already serviced areas if engineering evidence shows spare capacity. A formula should also reward sites near existing transport and water networks, where marginal costs are lower. Existing and future residents can face different costs and benefits from growth. Current residents can participate in local decisions, while prospective residents may be absent. Developers need predictable approvals and charges; utility managers need funding for additional capacity. A workable coalition couples greater density with funded network capacity, published service standards and a fair charge that new residents can afford. This makes the debate about shared capacity and long-term costs rather than an argument over a single building's height. Recurring property revenue can finance a different part of the neighbourhood service obligation.

Property-tax administration in Freetown

Property-tax reform can recover value for services, but reported administrative gains do not establish a lasting housing effect. Property taxes are valuable because a site cannot be moved out of the city to escape the tax. They can also be deeply unpopular. Owners may question a bill when they cannot see how revenue supports services. Effective reform must establish a convincing exchange between taxes and better services. In Freetown, Sierra Leone, the city mapped properties, built a simpler points-based valuation system, digitised billing and offered mechanisms to appeal. Research organisations helping the reform report that the stock of registered properties doubled, assessed revenue potential rose fivefold, and actual collections tripled during the early implementation period. The least valuable 60 per cent of properties saw average bills remain unchanged or fall; the most valuable 20 per cent saw average assessments rise by 242 per cent. About 70 per cent of surveyed owners expressed support, according to the follow-up research. These are reform evaluations and administrative findings, not a randomised estimate of a lasting fiscal effect. [11] Distributional changes and resident treatment should be measured with collections.

Higher assessed liabilities and improved collections need a separate test of equitable enforcement and service spending. Public meetings linked the higher tax to visible city

priorities, and a participatory process allocated part of the revenue to neighbourhood choices. The city pledged to direct 20 per cent of property-tax revenue to participatory budgeting, according to the Institute of Development Studies. [12] The council also had to contend with the national government. In 2020, central authorities intervened and paused the collection process amid disputes over legal powers and rules; negotiations allowed reform to resume. The case shows why tax administration also requires agreement on legal authority. [13] Ghana offers stronger causal evidence for a narrower proposition. In a randomised study of local property-tax administration, Dzansi, Jensen, Lagakos and Telli found that a tool helping collectors locate owners increased successful delivery of bills by 27 per cent and tax collections by 103 per cent. The researchers observed that collectors changed how they spent their time, concentrating effort where payment was more likely. The intervention improved collections in the study setting; it did not by itself show that poor neighbourhoods received better services or that enforcement was equitable. [14] A location tool's measured effect should not be relabelled as evidence of AI or long-term maintenance success.

Evaluate coverage, ability to pay, enforcement, and expenditure together before claiming fair neighbourhood finance. This finding concerns a tax-location tool, rather than an evaluation of AI. It illustrates a mechanism relevant to other administrative technologies. Technology changed the cost of finding a taxpayer. Collection agents gained discretion and new knowledge about households' propensity to pay. Fiscal revenues increased. The distribution of burdens depended on how agents used the information. A municipal data platform can improve service finance while concentrating enforcement on households already recorded or willing to pay and missing unrecorded or harder-to-enforce liabilities. Provide independent appeals, regular valuation rules, and published collection coverage alongside the software. For housing, a recurrent property-tax agreement should specify its distributional provisions: updated values for well-served high-value property; affordable payment schedules and deferrals for cash-poor owners; protection for renters against arbitrary eviction; a transparent share of proceeds financing local networks; and published eligibility rules and reviewable decisions on tax exemptions. A city that increases the assessed value of low-income homes after a public upgrading project should test whether its tax rules encourage landlords to raise rents or sell. Assess coverage, ability to pay and expenditure together to determine whether the reform finances maintenance fairly. Land-value capture requires its own baseline and legal claim on the increment.

The potential and limits of land-value capture

Value capture needs an early valuation baseline, clear authority, and disclosure of the decisions creating the gain. The proceeds can contribute to the investment's cost. Its feasibility depends on identifying the gain, establishing a baseline, collecting cash and assessing who ultimately pays. First comes identification. Property prices change with national interest rates, new firms, private building quality, inflation, school reputations and expected future growth. An observed price rise is rarely attributable in full to one public work. The state needs a credible comparison area, a valuation date and a way to distinguish rising land value from new construction. Aerial images show roofs, streets and flooding. They do not automatically reveal arm's-length prices or legal ownership. Second comes anticipation. Land often appreciates when a

transit line is announced, years before inauguration. The people who buy just before the announcement may capture a gain when the project is commissioned. Advance knowledge of routes, land releases or compensation creates a conflict-of-interest risk. A public process should fix the valuation baseline early, disclose conflicts of interest, and preserve dated parcel histories and planning decisions. Liquidity and household tenure determine whether the assessed contribution can be collected fairly.

Rising value does not imply immediate cash, and rising rents may exclude the intended beneficiaries. Higher land values cannot pay today's contractor unless the state can levy a charge, sell development rights, borrow against reliable future revenue, or dispose of public plots. A poor owner may hold appreciated land without the income to meet a large betterment bill. Deferral until sale, instalments and exemptions can avoid distress sales. Such relief needs rules that prevent wealthy owners from claiming it through nominee holdings. Fourth comes incidence. A levy collected from a developer may be passed backward into the price paid to an original landowner, forward into the selling price or rent, or absorbed in developer profit. Which occurs depends on market competition, land supply, timing and the ability to move investment elsewhere. A city cannot assume that the person who writes the cheque bears the economic cost. Similarly, rising land values may signal better access while rising rents exclude the tenants who most needed it. Bogotá’s evidence helps distinguish an observed access premium from a fully recoverable tax base.

The value of transit access can be capitalised in property prices without establishing a universal recovery percentage. A propensity-score matching study of residential buildings near the TransMilenio bus rapid transit system estimated a 5.8-17 per cent price premium for properties with system access. The estimate is context-specific and the matching design cannot remove all unobserved differences. The premium indicates that accessibility can be capitalised in property values; it is neither a universal rate for African cities nor a tax base that government can capture in full. [15] Further evidence on transport and land-use interaction informs when the rules should be announced.

Coordinate infrastructure and inclusive land-use rules before expectations are fully reflected in land prices. A 2026 study of Bogotá in the American Economic Review took the analysis further. Nick Tsivanidis found that conventional time-savings calculations captured only 52 per cent of the modelled welfare gains from transit investment, after accounting for shifts in where people and firms locate. Distributional impacts in the model were modest. The model accounts for changes in the location of jobs and housing as well as journey times. A city's appraisal needs to measure new opportunities, rent effects and movement of activity. The 52 per cent figure comes from that study's model and setting; it is no general multiplier for other projects. [16] Transport investment should therefore be planned with housing permissions. A bus line serving a corridor of low-rise, tightly zoned plots may enrich incumbent owners while preventing the extra rental supply that could absorb demand. Allowing modest density, small rental units and mixed use near stations can spread benefits. A targeted land-value levy can then recover some windfall from owners who benefit, while protecting low-income renters and financing access for surrounding neighbourhoods. Publish the line and the inclusive land-use

rules together, before expectations of development are reflected in land prices. A worked district budget can then test the financing assumptions and collection risk.

3 Agree who funds, delivers, and maintains the networks

Infrastructure financing model

A district financing model should disclose assumed costs, land values, and scale rather than present them as a forecast. Consider an illustrative, deliberately simplified expansion district of 100 hectares, planned for 5,000 dwellings. The project includes modest ownership flats and rented rooms near transport and employment. Assume 60 hectares remain in developable private plots after streets, public facilities, drainage corridors and open space. The following costs and land prices are assumptions in Kenyan shillings, chosen to show the logic; they are not measured costs or forecasts for any Nairobi project. The table below shows how those assumptions determine the apparent funding balance.

Exhibit 23.2. Illustrative infrastructure finance for 5,000 homes

Capital investment (KES million): roads and bus connections 1,500; water bulk and distribution 650; wastewater capacity 850; neighbourhood electrical links and local works 700; stormwater and social facilities 1,050. Total: KES 4,750 million = KES 950,000 per dwelling.

Land value: 600,000 m² of private land rises from assumed KES 10,000 to KES 25,000 per m² after servicing. Gross modelled increment: KES 9.0 billion. A 25% effective capture rate yields KES 2.25 billion, or 47.4% of the infrastructure bill.

Remaining capital gap: KES 2.50 billion. For illustration, a transparently funded development contribution raises KES 0.50 billion and national/city grants provide KES 2.00 billion. The 25% land levy and contribution must cover different liabilities: count the same uplift only once.

Recurrent bill: assumed operations and renewal reserve of 2% of capital cost = KES 95 million each year, before debt service. If the city instead borrows KES 1 billion at 10% for 15 years, annual debt service is about KES 131.5 million on top of network operations, subject to the debt's actual use and timing.

Distributional illustration: if 3,000 of the dwellings are rented and better services permit landlords to raise rent by KES 1,000 a month with no new supply, the additional annual rent transfer is KES 36 million. This is a sensitivity case, not a prediction or a benefit estimate.

Infrastructure finance is sensitive to appreciation and collection, so household charges cannot be assumed to close every shortfall. Land appreciation is gross and uncertain. The public authority captures only a fraction, and the levy may be collected over years. In a weak-demand district, suppose the appreciation is just KES 5,000 per private square metre rather than KES 15,000. At the same 25 per cent capture rate, revenue falls from KES 2.25 billion to KES 0.75

billion. The gap grows; promising to finance the network from land alone becomes untenable. In a booming district, gains may be large, but the ability to capture them depends on law, assessment and enforcement. Cost allocation also requires explicit assumptions. A development charge can fund added capital capacity; it should not be used twice to claim the same betterment increment. Grants and cross-subsidies remain necessary where local land markets are weak or resident incomes are low. A road serving an entire district cannot be attributed solely to 5,000 new households. Nor should a renter pay a permanent premium for a subsidised pipe whose recurrent cost was never budgeted. The business case should compare locations using the same full-cost boundary.

A defensible business case compares at least three locations: vacant peripheral land, serviced infill near jobs, and upgrading an existing low-income area. For each, calculate the discounted public cost per additional safe, occupied, affordable household-year while holding access to work, resilience and service quality constant. Infill may have a higher land price and a much smaller bulk-infrastructure bill. Upgrading may serve fewer new dwellings but yield immediate benefits to existing residents. Compare the full location and service costs before selecting the site. The preferred option also needs an agreement among the actors able to enable or delay delivery.

Interests and responsibilities in infrastructure delivery

Document objections, rights, and authority across utilities, landowners, municipalities, and residents. Before procurement, identify their legal responsibilities, expected gains, costs and decision powers. Landowners may gain from higher site values while facing land contributions or betterment payments. Developers need predictable capacity and approvals, and discretionary waivers can create unequal terms among competitors. Existing residents may value improved services while objecting to congestion or higher charges. Renters can lose access when rents rise and may lack formal representation in land readjustment. Utilities need cost recovery and capacity commitments. Municipalities need feasible budgets and delivery schedules. The agreement should cover recurring plant costs as explicitly as construction costs. The appraisal should also examine decisions made outside formal approval meetings. Brokers may hold information about plots and occupancy; contractors may operate under incentives that favour reactive repairs over preventive work; and officials may have discretion over connections, valuations and routes. Audit these arrangements rather than assume that discretion implies misconduct. Residents may challenge compensation or rights-of-way through litigation or protest. Record the basis of each objection and the lawful route for resolving it. A common capacity and cost assessment can provide a basis for resolving those disagreements.

Use verified service capacity and costs to negotiate responsibilities rather than rely on broad promises of coordination. State the additional service demand, works required, payer, household charges, and resident protections. Record objections and the evidence used to accept or reject them, rather than assuming that all owners or tenants share one position. The experience of Ahmedabad shows a coalition that worked for landowners and parts of the city but remained incomplete for tenants and unbuilt amenities. Dar es Salaam shows a financing coalition that delivered substantial public works while leaving the long-run incidence of land gains to be tested.

Cape Town and Johannesburg make added infrastructure costs visible through development charges, with difficult choices over waivers and maintenance still ahead. Freetown shows a municipal revenue coalition that improved tax administration and fairness, then met resistance from a higher level of government. None offers a universal template. Each identifies a veto point and a way to negotiate around it. The reform sequence should retain attention to the households who remain after improvements.

Sequence the agreement around documented constraints and monitor resident retention and affordable entry. Secure legal rights and protect vulnerable occupants before announcing alignments. Publish the network's spare capacity and cost before granting new density. Negotiate charges and development rights in the same transaction. Separate capital payment from annual maintenance finance. Tie any developer or landowner concession to homes that remain affordable and occupied. Finally, collect evidence on the households who actually gain, because infrastructure outputs and housing outcomes can move in different directions. Measure who remains after roads and drainage improve and whether service charges are affordable for new entrants. Digital tools should support the rights and service records needed for those decisions.

Assessing digital systems and AI

Map claims and capacity with dates, errors, and transparent allocation rules before using the results to assign services. Parcel imagery, building footprints and network maps can locate missing streets, unserved buildings, floodways and likely connection demand. The constraint is the time and error involved in locating assets and households. The relevant measure is a verified reduction in design revisions, missing parcels or time from survey to construction. Land agencies and utilities gain information; owners of newly recognised buildings may gain title or face tax and demolition. A satellite image should trigger field verification, an appeal window and a separate occupancy record. Even an accurately detected roof does not establish ownership or occupancy rights. Allocating network capacity. Hydraulic models, digital twins and forecasts can test the cost of a new block against spare water, sewer, electricity and road capacity. The measure is avoided oversizing, fewer outages, faster safe approvals and lower marginal cost per connected household. Utilities and better-equipped developers gain negotiating power. The danger is a model that treats informal residents as zero demand and allocates capacity to the first applicant with a complete digital file. A public capacity register should include existing use, planned low-income homes, errors and the rules for allocation. Valuation and anomaly tools need accountable officials who can investigate and correct their outputs.

Digital valuation and warnings require auditable explanations, human responsibility, and recorded remedies. Automated valuation can reduce the cost of updating assessments; Ghana's randomised tax experiment shows that a relatively simple tool changed both bill delivery and actual collections. The public budget may gain, yet collectors and data holders gain power to target. The test must include appeal success, distribution of assessments by property value, collections from high-value holdings and residents' trust, as well as gross revenue. Machine estimates should never be the sole legal basis for dispossession. Coordinating projects and detecting failure. A shared register of tenders, right-of-way clearances, utility permits,

disbursements, relocation cases and maintenance contracts can reveal whether the road contractor is months ahead of the sewer contractor. Simple rule-based alerts may be enough; AI can help identify unusual patterns in invoices, schedules or imagery. The gain belongs to households and taxpayers if early warnings lead to decisions. Programme managers gain surveillance capacity over contractors and residents. False flags can delay works or stigmatise a neighbourhood. Every warning needs a named human decision-maker, an auditable explanation and a recorded remedy. Occupancy matching must also protect households whose names or records do not align.

Use household follow-up and correction procedures so that matching errors do not remove water, tenancy, or eligibility. Linking consent-based household surveys to anonymised dwelling and service records can show rent change, resident turnover, travel times and service reliability. The measure is the share of original low-income households who enjoy the upgrade three and five years later, alongside new entrants. Such systems can also create a list of vulnerable tenants, informal vendors and undocumented occupants. Independent safeguards, purpose limitation, controlled access and community review must precede data collection. No household should lose water, tenancy or eligibility because a machine failed to match a name. The infrastructure agreement should make those protections and operating duties binding.

Infrastructure delivery agreement

Adapt infrastructure design to occupied settlements while retaining safety, access, and livelihood protection. Use one-off land receipts for suitable capital obligations and recurrent taxes, tariffs or transfers for ongoing service. Utilities and municipalities must reconcile capital schedules, connection conditions and maintenance duties before housing contracts proceed. Track actual connections, uptime, lifecycle costs, affordable bills, travel time, rent change and retention of original tenants. A city should enter a major housing or upgrading operation with one binding schedule shared by the municipality, housing agency, utilities, transit authority, land administration and finance ministry. The schedule should record the asset and the household result together. It should specify the financial and service consequences of delay or non-delivery. The following nine gates form a practical approval test. A bounded service-area appraisal can establish whether the required agreement is ready.

Exhibit 23.3. Infrastructure delivery assessment

1. People and rights. Has the city counted tenants, owner-occupiers, informal businesses, public-space users and their claims independently of cadastral ownership?

2. Location. Do typical low-income households gain access to work, schools and services at affordable money-and-time costs compared with the best alternative site?

3. Capacity. Are measured water, sewer, power, road and drainage headroom available, or are named network expansions contracted?

4. Whole-life cost. Do costings include land and corridors, relocation, engineering links, bulk works, operations, replacement and climate risk?

5. Incidence. Who receives the uplift, who pays every levy or charge, and how do rents and sale prices change after the intervention?

6. Funding. Are capital sources legally appropriated, with double counting excluded, payment dates matched to construction milestones and contingencies disclosed?

7. Maintenance. Is a utility or city unit contractually responsible, funded and measured for ten years after handover?

8. Inclusion. Do tenant protection, grievance mechanisms, fair access, anti-discrimination and affordability covenants survive the first sale or lease?

9. Evidence. Will the programme publish an auditable parcel-to-service dashboard, resident outcomes and a credible comparison or phased rollout for evaluation?

Start with an authorised engineering lead and a funded readiness assessment rather than an arbitrary completion promise. The finance counterpart prices the necessary works and identifies who can legally commit each funding source. Use existing network maps and targeted field checks to establish the critical capacity and rights-of-way gaps; cost a wider survey only if these gaps prevent the decision. Share the proposed household charges, service dates, and unresolved obligations with affected residents and developers. The first output is a decision on whether the site is ready for a funded investment agreement, not a promise that all disputes and network designs can be completed in 100 days. Subsequent review should measure execution, service, household costs, and continued access separately.

Judge infrastructure execution through functioning services and household outcomes as well as financial performance. Sequence road and drain works before home occupation where they are necessary. Publish every development charge and exemption. Track whether utility connections were actually delivered, whether the operator can maintain them and whether the public budget received the expected revenue. Where politically feasible, stage similar interventions so that outcomes can be compared. Commission independent spot checks of parcel data and rents. Build in a corrective trigger: if promised capacity is late, freeze new approvals in the affected sector or fund interim service; if existing tenants face abnormal displacement, deploy the agreed protections before releasing the next tranche. Public dashboards should display five results separately: additional homes safely served, existing residents protected, minutes and money saved reaching jobs, full public cost per household-year, and service uptime five years after handover. An increase in assessed property values is useful evidence about benefits and tax capacity. It cannot substitute for any of these measures. Fieldwork must verify the cost allocation and revenue assumptions behind those results.

Prepare the local decision

Reconcile network capacity, beneficiaries, lawful charges, replacement funding, and maintenance before approving infrastructure finance. Price trunk capacity, local networks, connections, operations, and renewal separately. Record existing demand, committed developments, actual collections, arrears, and sensitivity of revenues to occupancy. Map who receives the land-value increase and which agency pays for works. The Ahmedabad and Freetown

cases above show why reserved land, assessed revenue, cash collected, and functioning services need separate accounts. Utilities, the municipality, treasury, developers, landowners, and tenant representatives should agree cost allocation and payment timing. Test whether proposed charges are affordable to the intended households and collectible from the people legally liable. Obtain commitments for rights-of-way and asset adoption before using projected receipts to support borrowing. Allocate network funding alongside the adopted distribution of housing capacity. For a prosperous area with a verified capacity constraint, identify the upgrade needed, additional homes it enables, payer and operating duty. Compare it with alternative investments and existing underserved areas. Any developer charge should reflect the lawful cost-allocation method; its level should neither impose the whole network bill on affordable homes nor subsidise high-value development without disclosure. If collections fall short, the financing plan must change rather than assume a larger assessment creates cash.

Technology and action. Connect the asset register to billing and cash receipts, with reconciliation by service area. Verify automated valuations through sampled inspections and accessible appeals. Stage works against realistic demand and finance while protecting minimum service obligations. Review cash collection, connection time, service reliability, tenant costs, and displacement. If value capture underperforms, revise the financing plan rather than assume that a higher assessment will produce immediate cash. Chapter 24 examines the municipal operating institutions needed to keep these services functioning.

Each infrastructure service obligation needs reliable finance and an operator capable of fulfilling it. Credible demand, secured rights-of-way, funded works, and identified operating revenue matter more than the chosen deduction rate or development charge. Uncertain land-sale proceeds, assets without operators, and promised but unbuilt amenities create delivery gaps. Tenant protection also matters where better services increase rents. At 100 times scale, account for trunk-network thresholds, common drainage outlets and simultaneous maintenance needs.

Sources and evidence notes

The sources below include administrative reports, research studies and programme descriptions. Notes distinguish project-reported outputs, research estimates and the chapter's hypothetical calculations. Click the blue link at the end of each note to open the source.

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[3] Mahadevia, Darshini, Madhav Pai and Anjali Mahendra (2018), "Ahmedabad: Town Planning Schemes for Equitable Development - Glass Half Full or Half Empty?" World Resources Institute. [source]

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[6] OECD / Lincoln Institute of Land Policy (2022), Global Compendium of Land Value Capture Policies, Japan country chapter. [source]

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[9] City of Cape Town, Development Charges Policy and public implementation guidance; policy in effect July 2014, accessed 22 October 2026. [source]

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[11] Prichard, Wilson and Evan Trowbridge (15 May 2026), "Local leadership for digitalising public administration: lessons from property tax reform in Freetown," ODI Global. [source]

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[15] Perdomo Calvo, Jorge Andrés and others (2007/2010), "Study of the Effect of the TransMilenio Mass Transit Project on the Value of Properties in Bogotá, Colombia," Lincoln Institute working paper; propensity score matching. [source]

[16] Tsivanidis, Nick (February 2026), "Evaluating the Impact of Urban Transit Infrastructure: Evidence from Bogotá’s TransMilenio," American Economic Review 116(2):418-463, DOI 10.1257/aer.20190874. [source]

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24. Municipal services and maintenance

Maintain affordable housing through a municipal service agreement that connects asset responsibilities, reliable revenue, and remedies for failure. Estimate operating and replacement costs, identify the responsible providers, and test whether their authority and finances match their duties. Technology can support inspections and repair scheduling but cannot supply missing funding or accountability. Reported conditions in Mumbai show the consequences of leaving those obligations unresolved.

Mumbai's reported maintenance findings show why the condition of resettlement housing needs review long after construction. In August 2026, a Mumbai-based organisation published a study of housing built for families displaced by public works roughly two decades earlier. Youth for Unity and Voluntary Action (YUVA) examined 38 cooperative housing societies in 53 buildings across four resettlement layouts. Thirty-six societies reported falling external plaster. Thirty-three reported blocked underground sewers. Six of the seven societies that had lifts reported that the lifts were out of service. Fifteen of the 38 societies said they had never received the maintenance corpus intended to support their buildings. A total of 204 of the 3,324 dwellings examined remained unallotted. The buildings had been built between 2004 and 2006. [1] The figures are retained as reported in contemporary coverage; the original report was unavailable for direct verification. These figures remain attributed to the coverage. The study's coverage and attribution limit the conclusions that can be drawn from its counts.

1 Define the service obligation over the life of housing

Building and service management in Mumbai

A purposive maintenance study identifies institutional failures without estimating their prevalence across an entire city. These findings describe a purposive study of four layouts, not a statistically representative estimate for all of Mumbai. The study identifies gaps in responsibility for shared assets. Each household received the responsibility for its dwelling. A cooperative society was to maintain shared building parts. A federation was meant to look after services across the layout. The city provided networks up to the edge of what it treated as private property. No institution held both clear authority and sufficient money for the whole place. Repairs accumulated where responsibilities were unclear. The scale of the institutional gap becomes clearer in the accounts. The societies commonly charged members between INR 200 and INR 500 a month. YUVA examined the audited accounts of three societies and found that fees failed to cover routine running costs in two. Some vacant units still generated tax and maintenance liabilities, but contributed no dues. Residents were expected to manage pumps, sewage, stairs, lighting and contracts, collect money from neighbours, and negotiate with agencies that disagreed over where public responsibility began. [1] The underlying problem is the recurring decision and spending requirement created by shared assets.

Apartment buildings require repeated decisions and spending on shared assets. Someone must notice a leaking water tank, identify the leak's cause, authorise a contractor,

finance the work, obtain access, check the result and pay for replacement when a repair no longer makes sense. If the lift fails, a resident on the seventh floor with impaired mobility faces a barrier to leaving home. If the drain fails, a household pays through illness, lost work and damaged possessions. These are housing outcomes, even though their costs rarely appear in the original construction budget. Programme reporting should therefore measure the duration of safe affordable use.

Measuring safe, affordable occupancy over time

Administrative milestones need to be complemented by repeated observations of safe, serviced, affordable occupancy. Together they still miss the length and quality of the housing benefit. Twelve months and twenty years of safe occupancy represent different durations of benefit. The difference grows when a repaired building stays affordable while a deteriorating one pushes its residents into debt or another settlement. The useful unit of output is the safe, affordable, occupied home-year. For each dwelling and month, a programme would ask whether it is occupied by an eligible household, safe to use, connected to reliable essential services, affordable after service charges and transport, and secure against arbitrary displacement. A failed condition should be separately recorded with its cause and duration. One year of adequate housing comprises twelve qualifying months. Aggregating these observations measures the duration of housing services provided by a programme. That measure requires household-level checks rather than reliance on a citywide coverage percentage.

Measure the service a household actually receives, with definitions and follow-up adequate to support the claim. A building might remain structurally sound while its lifts fail. Water may reach the compound and never reach an upper floor. A tenant may live in the right location but face a service-charge increase that forces departure. The test therefore belongs at the level of the household and dwelling, backed by observations of common assets and network reliability. A citywide water-coverage percentage alone cannot establish that a particular child has potable water at home. Handover must assign the responsibilities needed to produce that service.

Exhibit 24.1. Accounting for safe, affordable housing over time

Stock: usable, safe homes at the start of the year ✦ completed and repaired homes − homes lost to unsafe condition, prolonged vacancy or conversion.

Service: for each home, count months occupied by the intended income group with safe structure, basic services, an affordable total housing bill and secure access.

Value for money: discounted lifetime public cost ÷ additional qualifying home-years relative to a defensible alternative, reported alongside distribution by income, gender, tenure and neighbourhood.

Decision framework proposed in this chapter. No universal maintenance or affordability threshold is assumed: service standards must be legally defined, locally affordable and independently tested.

Assigning responsibilities at handover

Completion and key transfer do not resolve boundaries between developer, association, utility, and municipal duties. Yet contracts at this point often assign ownership of a thing without transferring the means to care for it. The defect period may end before design faults become visible. Drawings may show where the pipes were intended to go rather than where they were installed. Service manuals may be missing. The sinking fund may not have received its agreed contributions. The utility may refuse to adopt an internal network that does not meet its standards. A housing project contains several nested jurisdictions. A household is responsible for fittings inside its unit. A cooperative or social landlord may own roofs, tanks, pumps, stairways, lifts and common wiring. A federation, estate manager or municipality may be expected to maintain shared access roads, storm drains and sewage pipes. Utility companies control trunk networks. A developer, contractor and equipment manufacturer may remain liable for specified defects. Police, emergency services and public health agencies respond to hazards. Each boundary creates a place where a fault can be passed to someone else. Mumbai's internal-layout disputes show how these boundaries can leave services without a capable operator.

Service responsibility should follow the function and legal duty rather than assumptions about a layout's appearance or ownership. The Mumbai study's authors found that the municipal corporation treated internal layouts as private, while resident federations lacked the funds and technical capacity to maintain streets, drainage, streetlights and other shared services. A federation may be registered and still lack the legal power to charge every beneficiary, borrow for replacement works or require a city utility to accept an asset. The problem starts with the contract and law; training residents alone cannot resolve it. [1] One solution is an asset-adoption schedule signed before construction begins. It assigns each component a permanent owner, operator, payer, regulator and emergency decision-maker. It records geographic boundaries and identifies who pays for internal works, connecting pipes and trunk capacity. It also sets measurable conditions for adopting assets: pressure tests, drainage performance, as-built drawings, rights-of-way, fire access and an independently verified defect register. A city should refuse handover of a network it cannot lawfully own, safely operate or finance. It should equally refuse to abandon a public service merely because a drain lies within a private-looking layout. The responsible institution also needs a budget and power to use it.

Ownership must be matched with a bank account and the power to use it. Social landlords can charge rents; cooperatives can collect dues; municipalities can levy taxes and fees; utilities can charge regulated tariffs. These revenue powers differ. A society of families with irregular incomes may have legal authority to collect fees and no realistic ability to finance a large lift replacement. A city can possess sufficient aggregate revenue and still allocate none to the settlement. The operating plan must match statutory responsibilities with funded maintenance obligations. Asset information is a separate handover requirement. Before acceptance, a public buyer should demand an indexed asset inventory; component warranties; inspection results; manufacturer and maintenance instructions; original design loads; georeferenced locations of buried services; spare-parts specifications; and the schedule of work required over twenty years. It should record the initial condition of each asset. Where the developer is paid before defects are tested, the public buyer has less financial security for the

remedy of defects. Retentions, performance bonds, staged commissioning and warranty enforcement should reflect the real cost of the risks being transferred. Utility and municipal evidence shows why neither a connection nor revenue automatically establishes that capacity.

Exhibit 24.2. Asset ownership, operating responsibility, and funding

Asset or service	Likely accountable operator	Funding rule to verify	Failure that reveals the gap
Inside the unit	Tenant, owner or social land- lord	Lease / repair duty / afford- able allowance	Leak persists because the landlord blames the occu- pant
Roof, lift, tank, common wir- ing	Cooperative or professional estate operator	Fees + audited reserve + protected subsidy	Replacement deferred after routine costs exhaust dues
Estate road, drainage, light- ing	Municipality or contractually funded estate manager	Public adoption or enforce- able district contribution	City says “private”; society says “public”
Bulk water, sewer, power	Regulated utility and city	Tariff + transfers + funded capital renewal	Network exists; household service fails
Fire, flood and health re- sponse	Statutory emergency/public- health authority	Citywide operating budget and tested protocols	Emergency vehicle cannot reach building

Institutional template. Every row should name an actual entity, named signatory, budget code, inspection frequency and escalation path.

2 Compare maintenance costs and institutional capacity

Utility, municipal finance, and maintenance constraints

Utility reform should distinguish connections, actual service, losses, and affordable lawful access. The Water Services Regulatory Board's Impact 18 report, covering 2024/25 and launched in June 2026, reported non-revenue water of 48%, up from 44% the year before. The reported metering ratio remained 97%. Average daily hours of water supply were approximately 18. A utility can know which customers have a meter while failing to know accurately where half the water it produces is lost, consumed without billing or mismeasured. The regulator's figures cover providers across Kenya and cannot describe conditions in any particular housing estate. [4] Some water leaks through old mains and poorly maintained connections. Some is delivered through illegal or unrecorded branches. Some is produced but never billed accurately. In informal

settlements, a landlord or local broker may control a shared connection and profit from resale. A replacement pipe can reduce physical loss. A district meter may reveal where it occurs. Identifying an unrecorded connection can affect an intermediary's income and expose residents without formal tenure to disconnection or enforcement. A plan to stop water theft without arranging legal, affordable access can punish the intended beneficiaries. Revenue and maintenance expenditure need to be traced separately.

Municipal income does not establish that the next repair has funding, authority, and procurement arrangements. WASREB's 2024/25 assessment reported average operations-and-maintenance cost coverage of roughly 103%, while its median provider remained below full cost recovery. An average above one does not demonstrate that most utilities are financially sound or that they are funding future pipe replacement. A water firm that covers salaries, chemicals and electricity can still be unable to renew pumps or respond to a major rupture. An affordable housing project should require a connection-level service agreement and test the receiving utility's asset and financial capacity. [4] South Africa's Auditor-General's consolidated assessment of local government for 2023/24 found only 41 municipalities with clean audits. It reported that 113 municipalities, 44% of the total assessed, adopted budgets that were not funded, and found deficiencies at 87 of 113 infrastructure projects visited. These figures measure governance and implementation risks rather than a causal effect on any one housing project. They nevertheless show why a housing agency cannot safely assume that municipal maintenance will follow automatically once a subsidy-funded development is finished. [5] Housing and rights findings reinforce the need to verify service performance after capital spending.

Capital works can coexist with service failures when operating systems, parts, and maintenance duties remain weak. In November 2024, the South African Human Rights Commission reported on failures affecting residents of subsidised housing. Investigating incomplete and inadequate subsidised houses in North West province, it substantiated systemic shortcomings across municipalities. It identified contract disputes, poor oversight, deficient design, delays in titles and failures to meet quality standards. It required a funded corrective plan, including budgets and time frames, and regular progress reports. The finding identifies delivery and oversight duties that require funded corrective action. [6] Kampala's road and drainage backlogs illustrate the need to fund connected networks together. In April 2023, Uganda's minister responsible for Kampala told Parliament that KCCA needed about UGX 2.2 trillion to address a backlog of 353 kilometres of paved roads in fair or poor condition and about UGX 610 billion for its drainage master plan. These were official estimates at that date, rather than independently audited cost-to-complete figures. Legislators asked how new roads could survive heavy runoff if drainage received no budget. Roads can be rebuilt, but without drainage and routine works their useful life contracts. In housing estates, the same failure appears when a paved courtyard sits above a blocked sewer or a newly installed pump runs without spare parts. [7] Every new asset therefore adds an obligation that must be matched with operating capacity.

Additional capital investment can be essential, but it also increases the assets a city must maintain. These cases support linking new construction to verified operating capacity, clear duties and funded recurrent costs. Design choices determine the size and timing of those obligations.

Operating and replacement costs

Every housing design contains an operating-cost choice. A twelve-storey block with several lifts and mechanical ventilation is very different from a low-rise cluster with gravity-fed drainage and good natural airflow. Taller buildings can use scarce central land efficiently and reduce commuting costs; the economic gain may justify their mechanical systems. Yet the residents' ability to pay for lifts, pumps, security, electricity and eventual replacement must influence the design from the outset. An affordable construction price must be tested against the cost of occupying and renewing the building.

Compare lifetime cost per safe home-year while separately testing whether households can pay bills when due. A University of Cambridge research profile, reproduced in Selwyn College's Summer 2020 magazine, describes Ronita Bardhan's survey of 1,244 households in Mumbai slum-rehabilitation housing. It reports residents spending around 40% of income on electricity in some observed settings, alongside increases in air-conditioning ownership after relocation. This is a research team's reported sample experience, not an estimate for all rehabilitation residents. A possible physical mechanism is that tightly packed buildings can trap heat and cooking pollution and require lighting and cooling where better orientation, ventilation and shared space could reduce demand. [2][3] A low electricity bill is therefore a matter of housing design, public health and household finance. So are stairs that can be used when the lift stops, roofs that can be repaired without destroying occupied rooms, a water system that can run during a short outage and drains that can be cleaned without breaking a concrete slab. Compare the cost of incorporating these features during design with the cost and disruption of later retrofits. The relevant comparison is the discounted lifetime cost per safe home-year, with a separate check that households can pay the bills as they arrive. The maintenance account should distinguish routine work, preventive action, major repairs, and replacement.

Maintenance budgets need separate cost categories and stress tests for collection and simultaneous failures. Routine care pays for cleaning, inspections, clearing drains and servicing pumps. Reactive work fixes a broken door, leak or electrical fault. Cyclical renewal replaces roofs, equipment, waterproofing and lifts at intervals. Major rehabilitation corrects structural failure or past design errors. The last two bills can be large and fall due many years after construction. A society that uses every month's dues for cleaning may appear solvent until the lift reaches the end of its life. A reserve fund can manage predictable replacement. It requires an asset-specific cash-flow forecast, deposits that stay in a protected account and clear rules for expenditure. Include inflation, equipment availability, interest rates, and premature defects in the forecast. A flat rule such as "put aside 2% of construction cost" may be a crude starting assumption; the right reserve follows the cost and expected life of actual systems. A six-floor walk-up and a twenty-floor lift tower need different plans. Reserves should be stress-tested against a sharp cost increase, collections below plan and simultaneous equipment failures. The resulting reserve contribution must be affordable or supported by an explicit funding source.

A reserve needs an affordable contribution schedule and a funded response when household payments cannot meet essential costs. Test targeted support, design changes, and procurement savings. Report the remaining gap openly; deferring repairs does not remove

the obligation. The worked model below illustrates errors that can arise when those accounts are incomplete.

Exhibit 24.3. Operating and replacement costs for a 600-home estate

Illustrative parameter	Amount
Homes built; 90% occupied; 80% of billed charges collected	600; 540 occupied; 432 effective payers
Monthly costs per home: routine and utilities	KES 1,500
Reactive work and administration	KES 450
Safety and insurance	KES 150
Funded long-term equipment and fabric reserve	KES 900
Total monthly lifecycle requirement	KES 3,000 × 600
Total required annual revenue	KES 21.60 million
Break-even charge per occupied unit, at observed collection	KES 4,167/month
Revenue if billed fee is capped at KES 2,000	KES 10.37 million/year
Annual funding gap at that cap	KES 11.23 million

Illustrative model, not an estimate for a Kenyan project. All costs expressed in constant KES, ignoring inflation and investment returns. Effective payers = 600 × 0.90 × 0.80. Break-even fee = 21.6m ÷ (432 × 12). The reserve is a provision for future works, not money already spent.

Maintenance affordability depends on collection, timing, and institutional performance as well as building materials. First, a city may approve a fee of KES 2,000 because that seems affordable and then leave KES 11.23 million a year unfunded. Second, the manager may erase the KES 900 reserve to narrow the shortfall, reducing current expenditure while accumulating a KES 6.48 million annual renewal liability. Third, the government may treat vacancies and poor collections as minor administrative problems. In the model, they raise the charge needed for break-even from KES 3,000 under full occupancy and collection to KES 4,167, a 39% increase. The same building materials produce different long-term affordability because the institution's performance differs. A funding gap requires comparison of feasible subsidy, reserve, management, and design options.

Choose the funding mix against local markets, household means, and the ability to enforce the agreement. Each instrument has a different distributional effect. A grant paid per verified low-income home-year puts pressure on the operator to keep units safe and occupied. An unconditional grant tied to units built rewards an earlier decision, regardless of later performance. A one-time endowment exposes residents to investment risk and inflation. A cross-subsidy may require higher-value commercial space and create a conflict over whose amenities receive attention. The correct mix follows the area's market and the institution's ability to enforce agreements. Those choices also redistribute income and authority within the operating system.

Interests in maintenance finance

Maintenance appraisal should distinguish necessary costs and competitive returns from gains created by control or scarcity. Developers prefer designs and contracts that let them recover costs, receive payments and exit risk. Contractors gain from repair work and, under poorly specified contracts, from repeated remedial work. Residents want safety and low fees. Landlords want to preserve rental income and asset value; some can pass repairs and service charges to tenants. Utilities need revenue and prefer connections they can bill and access. Municipal managers must respond to emergencies under annual budgets and procurement rules. National housing ministries face pressure to report completions. Assign service responsibilities before the construction contract closes. These arrangements distribute returns through construction, operations and location. Control over land, contracts and development rights affects construction margins; weak defect obligations can reduce incentives to maintain quality. Procurement, waste handling, vending, service resale and maintenance contracts generate operating revenue, with weak competition or opaque billing creating scope for excess charges. Improved services can raise site values, which owners may capture through higher rents or resale prices. The appraisal should distinguish necessary costs and competitive returns from excess returns generated by scarcity or discretionary allocation. Implementation power extends beyond the public agency signing the contract.

Identify the legal and practical actors able to block repairs, collection, access, or renewal. A cooperative board may reject a fee increase. A utility can refuse to adopt uncertified pipes. A treasury can decline a recurring grant. An estate manager can delay access to financial records. A state agency can withhold a corpus. A resident association may fight nearby rental blocks or public access through the estate, arguing that outsiders threaten security or services. When those who vote and those who occupy differ, homeowners can wield more power than tenants; informal renters and live-in workers can disappear from beneficiary lists. In India, the absence of clear title, contested rights over public land and the resale of rehabilitation entitlements add further layers of conflict. Social divisions can influence who bears the resulting risks and whose complaints are heard.

Management and contracting should assess unequal effects on residents and workers rather than shift risks down the social hierarchy. An online audit experiment of Delhi rental listings found a 35% callback probability for an upper-caste Hindu applicant, compared with 22% for a Muslim applicant, holding application characteristics comparable. The study found less clear evidence in that experiment for discrimination against Scheduled Castes and Other

Backward Classes. This is causal evidence of discrimination in callbacks within the sampled online market; it cannot establish who ultimately rented a home. It shows why the mere presence of affordable apartments does not ensure access for every group. [17] Maintenance burdens can compound this exclusion. A building with unreliable lifts harms elderly residents, pregnant women and residents with disabilities disproportionately. A poorly maintained public toilet increases safety and health risks, especially for women. The burden of cleaning sewers may fall on precarious workers exposed to hazardous conditions. India's NAMASTE programme, launched in 2023, explicitly seeks mechanised cleaning, trained workers and zero fatalities in sanitation work. Contracting out maintenance without worker protections transfers risk down the social hierarchy. [18] A workable agreement must specify obligations, payment, and remedies for the affected groups.

Build the agreement around specific obligations and losses. Tenants need a remedy for service failure; residents unable to afford essential charges need a funded eligibility rule; operators need predictable payment for verified work. Other losses may be unavoidable: a supplier can lose an exclusive contract or an owner may face a lawful charge. Do not promise every opponent a gain. Publish the reason for compensation where it is justified, identify who can lawfully block implementation, and distinguish a protected right from a demand to preserve an advantage. Contract terms should permit withholding the relevant payment for unverified work, subject to notice and dispute procedures. Programme communications should disclose future charges as well as the construction subsidy. Residents need audited operating costs, affordable payment rules and a clear remedy when services fail. Information alone will not resolve conflicting revenue interests or unfunded duties. Reform must establish who benefits from reliable service, who pays for it and who has authority to enforce the agreement. Institutional comparisons show different ways to organise those duties without establishing one universal model.

Exhibit 24.4. Interests in maintenance finance

Actor	Gain sought / cost resisted	Veto or source of power	Possible agreement
Developers and contractors	Prompt payment; low defect exposure	Technical information; deliv- ery deadlines	Staged payment, transparent tests, credible dispute resolu- tion
Owners, cooperatives, land- lords	Low dues; asset value; rent income	Votes, property rights, fee approvals	Audited budgets, phased re- serves, targeted grants
Tenants and informal occu- pants	Stable total cost; safe ser- vices; secure stay	Collective voice often weak- er; litigation or protest	Enforceable service stand- ards and non- retaliatory complaints

Actor	Gain sought / cost resisted	Veto or source of power	Possible agreement
Utilities and municipalities	Revenue, manageable assets, budget control	Connection, adoption, tariff and permitting powers	Funded service agreements; clear asset boundary; meas- ured performance
Politicians and treasuries	Visible delivery; fiscal room; credit	Grant approvals, appoint- ments and regulations	Home-year dashboard with published fiscal liability

Analytical map, to be populated with names, payments, legal powers and credible opponents before any reform is proposed.

Four institutional approaches

Singapore's separate operating and renewal funds illustrate a function rather than a contribution percentage to copy. Singapore's separate operating and renewal funds, described in Chapter 16, provide an institutional comparison. The municipal task is to verify actual component costs, contributions, public support, and the authority to spend the reserve. The statutory percentages cannot establish the appropriate contribution for another estate. [10-11] Regulated operators and collective organisations provide other ways to assign continuing responsibility.

Social-rental and collective-management models require legal rights, accounts, technical support, and municipal commitments that can be verified. The Social Housing Regulatory Authority's rules require accredited providers to show financial sustainability, maintain a property portfolio plan, inspect stock at least every three years, operate a complaint system that records response times and costs, and prepare for reactive and periodic large-scale works. The regulator requires ongoing reports and can intervene in cases of poor administration. This design assigns continuing responsibilities to a regulated provider; its suitability elsewhere depends on provider capacity, funding and enforceable resident rights. The regulations establish obligations; compliance monitoring and the operator's finances determine whether they are met. [12] Delivery performance shows where these obligations meet capacity constraints.

Regulatory obligations do not remove financing and infrastructure constraints on delivery. South Africa's Department of Human Settlements reported that 1,364 social rental units were delivered in 2024/25 against a target of 2,699, citing delays in approvals, debt finance and bulk services among the constraints. An institution required to maintain its stock may have to move more slowly when a project's full financing is unavailable. Speed alone would be a poor performance measure if it left tenants in unsafe buildings. [14] Community-led provision offers another way to organise implementation, with different capacity requirements.

Baan Mankong gave organised communities an implementation role backed by infrastructure subsidies and housing loans. Baan Mankong began in 2003 by channelling infrastructure subsidies and housing loans through organised communities that planned improvements with local authorities. The programme combined savings groups, collective tenure choices, community networks and negotiation with landowners. By December 2004, an early programme report recorded 137 approved projects serving 14,642 households; more than three-quarters of households in that initial portfolio were covered through forms of on-site improvement or reconstruction, and 9.3% through nearby relocation. These are contemporaneous administrative figures, not a long-run causal evaluation. [15] Its early delivery record needs to be read alongside the conditions that sustained collective action.

Collective management requires support and safeguards against unequal influence within the community. The programme gave community organisations a formal implementation role. Neighbourhood organisations helped decide and manage works, giving residents a stake in common assets and a collective forum to challenge officials. The same source warns that politicians and intermediaries can divide settlements into patronage networks; citywide community forums can provide a broader negotiating body. Volunteer leaders may be influenced by particular groups, lack time for their duties, or be assigned technically complex works they cannot fund. Collective management works best when coupled with legal tenure, technical help, audited accounts and a clear municipal commitment to trunk infrastructure. A large public landlord can still face substantial repair gaps when those functions are underfunded or ineffective.

Choose the operating form by who can own, fund, inspect, enforce, and respond reliably to complaints. The New York City Housing Authority's 2023 physical-needs assessment put its twenty-year capital requirement at US$78.3 billion. The authority's later plans describe conversions of parts of the portfolio to funding and management models intended to raise repair capital while preserving long-term affordability and resident protections. The figure measures modelled repair needs, not an amount appropriated or an estimate of avoidable past costs. It shows that formal public ownership, a sophisticated municipal government and a large tax base cannot, by themselves, prevent the accumulation of unfunded maintenance. [9] These cases support assessing institutions against their responsibilities, finance and performance. A professional landlord can serve tenants well and can also neglect repairs. A cooperative can build ownership and can also be captured. A municipality can pool risk across neighbourhoods and can also ignore places with little political voice. The choice depends on what can be verified: who owns the assets, who pays, who can inspect, who can enforce, and how the system responds when a household complains. The Singapore–Mumbai comparison therefore concerns funded duties rather than the label or percentage of a reserve.

The Singapore–Mumbai contrast concerns funded duties rather than a reserve percentage. Mumbai’s study is purposive, not representative of the city, and Singapore’s funding rules operate within much stronger fiscal and administrative institutions. Before adopting those rules, cost the actual assets, identify public support and verify who can collect, spend and enforce. A separate fund helps only if it can meet the obligation. [10–11,1] Municipal revenue arrangements must sustain those duties beyond the estate boundary.

3 Match revenue and accountability to continuing services

Municipal revenue and service obligations

Capital transfers are incomplete when they create recurring municipal liabilities without fiscal space to meet them. Municipalities inherit the street, drainage, solid-waste route, building inspection burden and the complaints. In many African and Indian cities, they have limited control over the taxes and transfers that could pay for those functions. A capital grant that creates recurring liabilities without giving the city recurrent fiscal space is an incomplete transfer. Budget and revenue evidence should be examined together to identify that mismatch.

Capital allocations and reported revenue reforms do not by themselves demonstrate reliable neighbourhood services. The World Bank estimated in 2022 that India required about US$840 billion of urban infrastructure investment over fifteen years. It found that the ten largest urban local bodies spent only about two-thirds of their available capital budgets across the years studied, and that urban property tax had averaged about 0.15% of GDP during 2011–18. These are estimates for capital finance and fiscal capacity, not a measure of housing maintenance funding. They show how difficult it can be for a city to turn additional investment into sustained service. [8] Tax reform offers one route to a stronger local operating budget. Freetown's property-tax reform, launched after the 2018 mayoral election, mapped properties, changed valuation, digitised billing and improved payment channels. ODI's May 2026 review reports that, within a year, registered properties doubled and collections tripled; the least valuable 60% of properties generally faced stable or lower bills while the most valuable 20% experienced average increases of 242%. A follow-up survey found support from roughly 70% of owners. These are reported reform outcomes and survey findings, not a randomised proof that technology alone caused the change. Performance-linked support needs a rule that preserves essential service when intergovernmental conditions fail.

Revenue reform should link information, administration, and spending without making poor households bear failures between governments. Freetown’s reported reform combined mapping, valuation, administration, and intergovernmental negotiation. Later movement towards a national platform raised questions about control of records and systems. Evaluate sustained collections and service expenditure separately; higher revenue alone does not establish better housing. A useful intergovernmental agreement would make four commitments. The national or state housing agency finances the portion of lifecycle costs that the poorest households cannot bear. The municipality adopts specified public assets and publishes a service budget. The utility accepts connections only after independently tested handover, with agreed financing for capacity and renewal. The estate operator keeps separate, audited accounts and publishes service and repair performance. Payments should reward actual occupied and safe homes while avoiding rules that encourage harassment, eviction or artificial vacancy reductions. For a multilateral development bank (MDB), this implies a shift in disbursement logic. Capital construction remains necessary. Yet a housing operation can reserve part of its support for independently verified post-occupancy performance, asset adoption, audited reserve contributions and resident access to grievance redress. Such incentives require reliable baseline

data, fraud controls and a mechanism to protect residents when their operator misses a target. The poor should never lose essential service because a government-to-government performance condition fails. Maintenance technology should support this operating agreement through a usable asset register.

Assessing maintenance technology

An asset register should assign service duties without becoming a tool for unaffordable charges or removal of unrecorded occupants. A satellite image can identify a roof or lane, but cannot reliably establish legal ownership, pipe condition, resident rights or the age of a buried connection. Pair georeferenced parcel and building records with construction documents, utility maps, tagged equipment and resident-verified occupancy. The mechanism is fewer disputed assets and faster fault assignment. Measure how many items have an identified owner, condition grade and maintenance plan before and after the intervention. Municipal departments gain information; tenants may gain a clear service contact. There is a risk that a more complete register will be used to levy unaffordable fees or identify unregistered occupants for removal. Sensors and predictive systems add value only when maintenance teams can act on reliable signals.

Remote monitoring needs validation and funded response capacity so that past neglect is not reproduced in future allocations. Kenya's combination of high household metering and high non-revenue water shows why the level of measurement matters: zone-level flow and pressure data can distinguish suspected pipe loss from household consumption more usefully than another meter on an already metered customer. Test whether the technology reduces repair time, water loss and household service interruptions after accounting for installation, calibration and maintenance. Utility managers gain operational control. Residents need safeguards against sudden disconnection and charges based on faulty data. [4] Predictive analytics can help rank roofs, lifts and drainage for preventive work. Begin with a simple baseline: age, inspection grade, outage history, exposure to flood or heat and repair cost. Compare that rule with more complex AI using a held-out dataset and prospective field trials. Predictive accuracy alone proves little. The required measure is a reduction in failures or cost per reliable home-year. Poor neighbourhoods often have fewer recorded complaints; a model trained on historical tickets may systematically overlook them. Bias can turn past neglect into future resource allocation. Payment and ticket systems require the same attention to rights and follow-through.

Digital payments and work orders should improve verified service and accountability alongside revenue collection. They can also deepen exclusion where residents lack smartphones, documentation, reliable connectivity or the language of the interface. A working design permits cash or assisted channels, anonymised complaints, appeals and a clear duty to respond. If renters fear retaliation, reporting must separate the complaint from the landlord's ability to identify the complainant. Digitised grievance closure should require evidence that the fault was corrected, with a sample independently checked by residents. Property-tax systems present a final caution. Freetown's experience suggests technology can help expose taxable value and improve the city's fiscal base. It also gives the authority greater power to classify, tax and surveil landholders. Clear appeal rights, transparent valuation rules and legal protection for informal residents determine whether this power supports inclusion. Evaluate the effect on

residents' rights alongside revenue and administrative gains. [16] Implementation should start with a failure the city has authority and resources to correct.

Exhibit 24.5. Testing maintenance technology

Intervention	Binding constraint changed	Observed test / possible capture	Power and safeguard
Parcel + asset inventory	Unknown location, owner- ship and condition	% assets with verified cus- todian and funded plan; avoided disputes	City gains tax and enforce- ment capacity; validate rights with residents
Zone meters / pump tele- metry	Losses and failures detected late	Repair time; measured un- billed water; household supply hours	Utility controls more; inde- pendent meter checks and service rights
Risk-based inspection / AI	Scant staff, weak repair prior- ity	Outages and injuries averted vs simple rule; audit false negatives	Inspectors gain discretion; mandatory appeal and human review
Payments and complaint re- cords	Revenue leakage and invis- ible failures	Net collection, actual repair time, resolution verified by residents	Manager can monitor ten- ants; privacy, assisted access, anti-retaliation

Targets in this exhibit are proposed measures for an evaluation. None should be represented as a quantified benefit before field measurement.

Implementing maintenance reform

Begin with the failure that the city can act on: for example, repeated pump outages, disputed drain ownership, or an unfunded lift replacement. The following sequence is a proposed approach. Its scope depends on available inspectors, legal powers, records, and repair funds; a calendar deadline does not establish readiness. In the first month, the city manager should appoint an engineering lead and finance counterpart, estimate staff days and inspection costs, and select a manageable sample of estates by age, building type, complaints, and management arrangement. Use records and short site visits to triage urgent hazards, interrupted services, disputed responsibilities, and missing funds. This is not a full structural survey or a representative estimate of citywide defects. Refer suspected life-safety risks immediately to the competent authority; commission specialist inspections where required. Record gaps for follow-up and let residents correct occupancy or service records without making that process an eviction

exercise. The next stage should price urgent works and resolve the ownership and payment disputes blocking them.

Resolve the principal service and financing disputes in selected estates and fund follow-up of residents. The engineering lead prepares the repair scope; the finance counterpart identifies a budget line and verifies cash availability; the authorised asset owner signs the duty and acceptance terms. Start a replacement forecast with the critical components whose age and condition are known, showing uncertainty where records are weak. Do not describe an unfunded plan as an agreed maintenance programme. Cost the remaining inspections and expand the register as capacity permits. Over the first year, test one feasible operating change within existing powers, such as pooled pump maintenance with a response-time requirement or an accessible fault-reporting and repair process. Record baseline downtime, charges, arrears, repair completion, and complaints, and compare similar untreated assets where feasible. A before-and-after improvement alone cannot establish causality. Changing ownership, creating a regulated landlord, or transferring municipal assets may require a longer legal and fiscal process. Decide on those changes after testing their costs and authority, rather than promising to trial several institutional models within a year. Fund longer follow-up for resident retention and household well-being. The contracts must give capable providers clear obligations and timely payment.

Maintenance contracts need enforceable service standards, remedies, and a reliable payment route. A development and handover agreement fixes defects, records drawings and warrants performance. A service and asset-adoption agreement assigns public and estate-level duties and funds them. An affordability and resident-rights agreement protects eligible households, publishes charges, restricts arbitrary eviction and mandates remedy for unsafe conditions. A reserve and audit agreement sets aside money for future renewals and requires independent reporting. These should be enforceable documents attached to land and grant conditions, with a remedy when any party defaults. Procurement should favour reliable service over the lowest initial price. A contract for lift maintenance needs preventive visits, independently measurable uptime, safety response times, spare-parts availability, escalation rules and credible penalties. An unrealistically low five-year bid can leave routine maintenance underfunded and require later emergency procurement. Where the market for qualified providers is thin, pool estates to attract bidders, publish unit costs, and separate the engineer who certifies repairs from the contractor who bills for them. Ensure firms have a route to timely payment; delayed public bills can destroy an otherwise capable maintenance market. Evaluation must also examine fees, allocation, and evictions that affect residents' treatment.

Evaluation should include the politics. Survey whether owners, tenants, estate boards, municipal engineers and politicians consider the new rules fair. Identify who loses informal fees or contract discretion. Publish contractor ownership and conflicts of interest. Review service charges alongside household income and time spent reaching work. Monitor displacement and informal subletting, particularly after repairs make the area more desirable. Use the monitoring results to enforce tenant protections and investigate disputed fees, allocation decisions, or evictions. Lifetime accountability should connect those findings to ownership, operation, payment, and verification duties.

Exhibit 24.6. Housing operations assessment

1. Asset: Is there a verified inventory, location, condition grade and expected replacement date?

2. Duty: Does each component have one legally accountable operator and a documented escalation path?

3. Handover: Are commissioning tests, as-built plans, defects and warranties independently checked?

4. Money: Are routine, reactive, cyclical and major-repair costs funded over the full asset life?

5. Household: Can eligible residents afford rent or mortgage, utilities, fees and transport together?

6. Risk: Can the system survive vacancy, arrears, inflation, flood and a major early equipment failure?

7. Voice: Can tenants and owners report a fault, appeal a charge and obtain a time-bound remedy?

8. Inclusion: Do women, renters, informal workers and people with disabilities receive the service?

9. Proof: Is performance measured in safe occupied home-months and verified repairs?

10. Implementation: Who loses revenue or decision power, and which authority will enforce continuing duties?

Decision rule: withhold the final construction acceptance or recurrent grant agreement until Gates 1–4 have a signed responsible party and credible financing; apply emergency safety remedies immediately, regardless of handover status.

Accountability over the life of housing

Use institutional comparisons to assign and verify each service obligation over the life of the housing. The choice must remain affordable to residents and funded over the asset life. Chapter 25 brings these obligations together with access to jobs, schools, and neighbourhood opportunity. Fieldwork should identify unfunded renewal needs before they become failures.

Prepare the local decision

Inspect asset condition, budgets, work orders, household costs, and complaints before confirming municipal maintenance readiness. Inspect critical assets and reconcile their ownership, condition, service area, operating budget, work orders, invoices, and replacement dates. Separate reported completion from independently checked restoration of service. The Mumbai evidence above makes responsibilities and missing funds part of the diagnosis, alongside physical defects; the South African operator rules require checking compliance rather than assuming it. The municipal service manager, utility, building operator, treasury, regulator, and resident representatives should agree an asset-adoption schedule. Specify who responds to an

emergency, who pays, and who can replace a failing operator. Check that tenants and residents with limited mobility can report faults and obtain a remedy without relying on the owner. Use sensors only where staff can maintain them and act on their readings. Compare predicted failures with a simple inspection schedule, counting missed failures and unnecessary visits. Link contractor payment to verified service restoration and retain a record of repeat defects. Report downtime, safety failures, repair cost, reserve adequacy, and affordability of charges. Escalate unfunded replacement needs before equipment fails, with a named budget decision and deadline. Chapter 25 brings these obligations together with access to jobs, schools, care, and shared neighbourhood life.

Future repairs need a credible claim on revenue and residents need a remedy when services fail. Clear asset custody, protected renewal funds, technical support, and enforceable service rights connect fees to maintenance. Singapore's fund percentages or cooperative registration requirements cannot establish these conditions elsewhere. Missing corpus transfers, unsupported fees below cost, and disputed boundaries leave the system without a responsible payer or operator. Stress-test ageing across a 50,000-home portfolio, contractor shortages and correlated failures during floods or power outages.

Sources and evidence notes

The chapter uses four evidence categories. Administrative statistics describe reported outputs and fiscal indicators; they do not by themselves establish causality. Field studies and surveys describe their study populations and must not be generalised automatically. Audit experiments identify causal effects within the tested setting and outcome. Illustrative models make assumptions transparent and are not observed estimates. Normative proposals and implementation sequences are the author's policy judgments.

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[11] Singapore Town Councils Act and Financial Rules (2017). Legislation on sinking-fund contributions, with public explanation of the 14% lift and 26% ordinary allocation. [source] ; [source] . Legal minimums; verify current amendments for transaction design.

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[13] Aljunied-Hougang Town Council (2026). Town Council Funds. [source] . Operator description of routine, sinking and lift funds.

[14] South Africa Department of Human Settlements (2025). Annual Report 2024/25. [source] . Administrative report: 1,364 social rental units versus target 2,699.

[15] Boonyabancha, S. (2005). Baan Mankong: Going to Scale with Slum and Squatter Upgrading in Thailand. Environment and Urbanization, 17(1). [source] . Programme-internal early assessment and administrative tables; long-term impacts not inferred.

[16] Prichard, W., and Trowbridge, E. (2026). Local leadership for digitalising public administration: lessons from property tax reform in Freetown. ODI Global, 15 May. [source] . Case study and reported reform statistics; no independent randomised attribution.

[17] Datta, S., and Pathania, V. (2016). For whom does the phone (not) ring? Discrimination in the rental housing market in Delhi, India. UNU-WIDER Working Paper 55/2016. [source] . Correspondence audit experiment; outcome is landlord callback.

[18] Government of India, Department of Social Justice and Empowerment. National Action for Mechanised Sanitation Ecosystem (NAMASTE), launched 2023–24. [source] . Programme objectives; zero fatality is a target, not an observed result.

25. Inclusive neighbourhoods: housing, opportunity and social integration

Affordable housing policy should enable people to enter and remain in neighbourhoods with opportunities they can use. Pursue access to well-served areas, improvement of underserved areas with resident protection, and relocation where safety or household choice requires it. Singapore and the comparative evidence help identify the functions that must be coordinated; local institutions determine how to deliver them. The first step is to define inclusion beyond the number of dwellings provided.

A city can add affordable units while leaving access to its best-served neighbourhoods unchanged. Housing prices, planning restrictions, allocation rules, school boundaries, landlord selection and recurring service charges can each exclude a household. The problem is the joint availability of a suitable home, an affordable total cost and usable opportunity. Government should pursue three routes together: enable entry into existing high-opportunity areas, improve opportunity in underserved neighbourhoods while protecting residents, and provide relocation when safety or household choice requires it. Each route has a different failure mode. A mobility subsidy can fail at landlord acceptance. Upgrading can displace tenants. Relocation can sever jobs and support networks. The evidence should determine the package and its limits. The evidence on location helps define which opportunities an inclusive housing programme should seek to provide.

1 Define inclusion through usable neighbourhood opportunities

Access to neighbourhood opportunity

How location affects household outcomes

Location affects households through work, schools, safety, travel, stability, and social networks, with effects that differ by age and disruption. These channels work at different speeds and vary across adults and children. A higher average income in an improved area is insufficient evidence of benefit if poorer residents have left. The US Moving to Opportunity experiment provides evidence that neighbourhood exposure can affect children's later outcomes. For children younger than thirteen at random assignment whose families used an experimental voucher, estimated earnings in their mid-twenties were 31% higher than the relevant control mean. Results for older children were much weaker and could be negative. The age pattern supports attention to exposure and disruption; it does not establish that every low-poverty destination works or that forced relocation is justified. [106] Ahmedabad's relocation evidence shows why a better building can fail to deliver gains through those channels.

Improved physical housing does not establish improved livelihoods when relocation weakens access and support networks. Its relevance is not a universal argument against moving. A household must be able to sustain the move and use what the destination offers.

Voluntary mobility and in-situ improvement should be compared with the costs of losing existing connections. [107] Inclusionary housing seeks a different route by securing affordable homes within better-served areas.

Inclusionary housing and school access

Montgomery County's outcomes reflect a combined institutional and development arrangement rather than a transferable percentage requirement. Its housing authority also acquired dispersed homes for very-low-income residents. A moderately priced unit may still be unaffordable to the poorest household, so a separate public-housing route may be needed. The policy history and requirements have changed over time, so a percentage taken from one period should not be treated as a timeless design rule. [105] Heather Schwartz followed children in dispersed public housing during 2001–07, using the housing-assignment process to compare school environments. Children attending the lowest-poverty schools reduced their initial maths achievement gap with non-poor peers by about half by the end of elementary school. The study found weaker benefits as school poverty increased; not every estimate was equally precise. Housing stability, school access and exposure operated together. The result is not an estimate of the effect of adopting an inclusionary percentage in another city. [104] Comparisons with other inclusionary programmes help identify the feasibility conditions behind that arrangement.

Inclusionary obligations need predictable rules and sufficient development value to fund the intended access. HUD's comparison of Montgomery and Fairfax counties found that both programmes added affordable units, while emphasising predictable rules and economic feasibility. Programme age, market conditions, design, and exemptions differed. The comparison does not establish that adopting the same inclusionary percentage elsewhere would produce the same result. Evaluate actual completions, household access, and project viability under local conditions. [130] These findings suggest that inclusionary housing needs viable development, predictable requirements, durable affordability rules and institutions that connect eligible households to useful services. Public acquisition and school access helped reach very-low-income families in Montgomery County. The studies do not identify a separate causal effect for each condition. A city should test where that chain could fail: too few viable projects, exemptions that reduce delivery, units beyond the poorest households' means, or inadequate school and service access. These are local appraisal questions, not all effects established by the two-county study. Set affordable-housing obligations against the development value public action creates. Model several building types, land prices and market conditions before choosing a percentage. Publish the assumptions and allocation rules. Where the available value cannot fund the intended access, provide explicit subsidy or choose another feasible approach. Redevelopment evidence adds a further test of whether improved neighbourhoods benefit the households exposed to them.

Redevelopment and access to improved neighbourhoods

HOPE VI evidence should distinguish estimated outcomes from the proposed peer-interaction mechanism and the study's design limits. A January 2026 HOPE VI working paper uses matched comparisons and exposure analyses to study revitalised US public housing. Its main child results concern families moving into developments after revitalisation. The study

estimates better later outcomes for these children, but no detectable causal improvement in incumbent adults' earnings. The paper cites earlier programme research reporting that 27.6% of original households had returned by 2014; this is a programme return statistic, not the new study's causal estimate. The evidence therefore supports benefits for children gaining access to redeveloped places; it does not establish that redevelopment improved the welfare of all displaced residents. The authors present increased interaction with nearby peers as an important mechanism, rather than a separately randomised treatment. [131]

2 Housing displaced people near jobs

Connect housing decisions to local livelihoods

Housing for displaced people should provide secure shelter and practical access to local jobs. For internally displaced people (IDPs) settling in cities, assess the home together with the work, services, and support networks available from that location. Emergency accommodation may be essential for immediate safety. Longer-term decisions should compare rental support, neighbourhood upgrading, and new construction against households' ability to earn and meet recurring costs. Apply the same test to low-income residents already living in the receiving neighbourhood.

Secure tenure can improve housing stability while leaving access to jobs unresolved. Samuel Hall's 2026 evaluation of Norwegian Refugee Council housing, land, and property programmes in four Somali cities used 27 key-informant interviews and 16 focus groups with 96 participants. In Garowe's Hoodale settlement, the evaluation documented housing and tenure support alongside roads, water, lighting, a school, and regular transport connections. Participants reported better shelter and safety, but also continuing weaknesses in access to markets and income opportunities. In Mogadishu, participants identified paying rent after subsidy withdrawal as a continuing constraint. [241] These are qualitative findings, without an untreated comparison group or an estimated employment effect. They identify operating conditions to verify: the services and transport must function, households must be able to reach customers or employers, and rent must remain affordable after support ends. Completed dwellings and tenure documents should be recorded separately from sustained livelihood outcomes.

Refugees and internally displaced people require different legal checks within the same neighbourhood assessment. IDPs remain within their country's borders; internal displacement does not confer refugee status. Refugees cross an international border, and their recognition, residence, and access to employment operate under a different legal framework. OECD and UNHCR document barriers involving registration, identity papers, and access to services for both groups. [242] Establish the applicable rules before offering a tenancy, training place, or employment referral. Help households obtain missing documents without treating their absence as evidence that the household has no rights or needs.

Map actual earning opportunities before selecting land or a housing package. Ask households where they work, seek work, sell goods, obtain supplies, and receive help with care. Check employers' vacancies and recruitment requirements, alongside the places where casual

labour and small businesses operate. Compare those locations with available homes using actual fares, journey times, service hours, and safe walking routes. A short distance on a map can conceal an unaffordable or unreliable journey. Assess women's access separately, and include people with disabilities and households whose members cannot work. Income support remains part of the response where employment is unavailable or inappropriate.

Combine secure housing with access to work

Compare housing options using the household's full recurring budget and its ability to remain. Rental assistance may offer access to established neighbourhoods where suitable vacancies exist. Repairs or safe extensions may add rooms without requiring a move. New serviced plots or completed homes may be appropriate where tenure, infrastructure, and access are credible. Test each option against rent or repayments, water, energy, transport, care, and income interruptions. Require evidence of usable employment access before selecting a remote site because its land is inexpensive. Where relocation is necessary for safety, fund the connections and livelihood support needed at the destination before occupation.

Secure occupancy should be designed alongside any public investment in the dwelling. Record who owns, occupies, rents, or uses the property, including tenants and households with informal agreements. Choose locally enforceable arrangements that specify charges, notice periods, repairs, and dispute resolution. Legal assistance and a written tenancy may be more immediately useful than waiting for an ownership transfer. Check women's interests and competing claims before recording rights. For publicly supported landlord improvements, specify the agreed period of affordable occupancy and monitor compliance. Assistance at the destination should preserve lawful claims to property left behind and avoid compelling households to abandon a future voluntary return. For IDPs, the UN Guiding Principles provide a benchmark for property protection and voluntary return or resettlement. [245]

Transport and care arrangements belong in the housing programme's operating plan. Verify that buses or other affordable services run when shifts begin and end, and that routes remain usable during seasonal disruption. Budget for safe walking access, lighting, and necessary service improvements. Check the cost, hours, capacity, and location of childcare and schools with caregivers themselves. A training offer or vacancy is of limited practical use when the household cannot arrange the journey and care responsibilities. Identify the agency responsible for each gap, its funding, and the date by which the service will operate.

Employment support should respond to identified demand and existing skills. Assess prior work, literacy, recognised qualifications, and the skills needed by local firms before commissioning courses. Where justified, combine skills recognition or short training with recruitment, apprenticeships, transport, and care support. Include safe, lawful routes for home-based production, trading, and other small enterprises rather than assuming that every household will obtain a salaried job. Consider workspace, storage, electricity, customers, and access to supplies. Apply labour protections, and measure sustained work and earnings after assistance ends. Course completion and temporary construction employment should be reported separately from lasting livelihood outcomes.

Expand affordable supply and services for both populations

Colombia's experience shows why housing-market monitoring must distinguish income groups. Emilio Depetris-Chauvin and Rafael Santos studied IDP inflows and rents in 13 Colombian cities during 1999–2014. Their analysis estimated rising rents for low-income housing and falling rents for high-income housing. Their working paper also found a response in construction licences outside the social-interest category, but found no statistically detectable response for social-interest housing. [243] The study uses an instrumental-variables strategy; its interpretation depends on the validity of that design. It does not establish that additional building will reach displaced households. The planning implication is to track the price and availability of homes they and poorer established residents can actually occupy.

A randomised trial in Jordan shows why rental relief should not be assumed to produce employment gains. Tamim and colleagues evaluated housing assistance across 2,870 Syrian refugee households in Irbid and Mafraq. The programme combined roughly a year of rent support with housing improvements, generally in existing homes; it did not test moving families closer to jobs. Take-up was 33% among surveyed treatment households, with housing eligibility and landlord reluctance among the obstacles. The study found lower housing expenditure and some short-run physical improvements. Follow-up one to two years after assistance ended showed no detectable sustained economic improvement and a persistent adverse effect on children's socio-emotional well-being. Host-neighbour attitudes also worsened, although a causal link to the child outcome was not established. [244] Pandemic-period implementation, low take-up, and differential attrition limit interpretation. Before scaling rental support, verify landlord participation and eligibility, coordinate with other assistance, monitor children and neighbours, and test the employment component separately.

Public support should address the receiving neighbourhood's housing and service constraints. Pair household assistance with a local assessment of vacant homes, safe conversions, landlord repairs, serviced land, and infrastructure capacity. Set transparent eligibility rules that include vulnerable established residents, while recognising displacement-related losses and protection needs. Track rents, overcrowding, evictions, and service access for both populations. If assistance increases demand while affordable vacancies remain scarce, review supply measures and programme terms rather than assuming that a larger payment alone will solve the shortage. Treat these as locally testable responses, not interventions whose effectiveness the two studies have already demonstrated.

Assign continuing responsibilities before emergency funding ends. National authorities, municipalities, service providers, and aid agencies should agree who funds housing support, maintains infrastructure, resolves occupancy disputes, and updates population estimates. Publish the division of responsibilities and a funded transition plan. Involve displaced and established residents in reviewing performance, with an accessible complaints process. Use the book's programme tests to follow household costs, tenure stability, usable services, and sustained livelihoods after occupation. Report outcomes for people who leave as well as those who remain. Singapore offers a different institutional setting in which to examine sustained responsibility for neighbourhood provision.

3 Examine how Singapore coordinates neighbourhood provision

Singapore joined housing provision to neighbourhood access

Singapore offers an institutional example of housing delivered at scale with continuing responsibility for neighbourhoods. Housing & Development Board (HDB) flats accommodated about 3.19 million citizens and permanent residents on 31 March 2024. That figure excludes non-residents and should not be presented as coverage of everyone living in Singapore. [SG1] My reading of the history is that other cities should examine the functions connecting production, household access, and continued occupation, then establish who can perform each under local conditions. Its history began with a rental supply response before household savings became central to purchase finance.

Rental construction preceded CPF-financed purchase

Singapore's transition from basic rental construction to purchase finance depended on wider institutions and household savings conditions. HDB began in 1960 by building basic rental flats. The Home Ownership for the People Scheme followed in 1964. In 1968, households gained permission to use Central Provident Fund (CPF) savings for housing. [SG2] Initial construction therefore preceded this financing channel. The chronology supports retaining rental provision and other ways to meet urgent needs while institutions for ownership finance develop. It provides no basis for requiring every household to become mortgage-ready before receiving adequate housing. CPF linked housing payments to employment-related savings. Under the current employee system, employers remit the contributions due for eligible citizen and permanent-resident workers, including the employee share where applicable. Coverage includes qualifying part-time and casual employees. The self-employed have a different compulsory contribution arrangement centred on MediSave. [SG3] The mechanism to examine is reliable contribution collection. Part-time work alone does not imply exclusion from Singapore's employee system. For households with irregular earnings in India and African cities, I would compare rental assistance, serviced plots, and incremental construction finance against a finished-flat mortgage. Repayments should be tested against income interruptions and essential consumption. Diverting more savings into housing also requires an explicit assessment of retirement adequacy. Production capability also had to develop alongside the financing system.

Procurement helped develop construction capability

Procurement can help build contractor capability when the programme sustains learning, standards, and credible demand. A contemporaneous ministerial speech records that HDB introduced its core-contractors scheme in January 1982. Selected firms received improved terms and a guaranteed workload in exchange for commitments on annual production, professional staffing, and worker training. HDB also awarded prefabrication contracts to overseas firms while training Singaporean engineers and technicians. [SG4] These are documented institutional arrangements, without an identified estimate of how much each increased output or reduced costs. The reciprocal commitment is potentially adaptable: a funded pipeline gives firms a reason to invest in equipment and skills, while contracts specify what they must deliver. That is a

mechanism to test locally. Guaranteed work can also protect incumbents, so renewal must depend on independently checked performance and continued competition. Industrialised methods require sufficient demand to sustain the investment. Land assembly supplied another part of the system, with substantial distributional consequences.

Land assembly requires a distributional account

Singapore's acquisition arrangements reduced public costs partly by limiting compensation, a historical choice later changed. In his 11 April 2007 parliamentary speech, Deputy Prime Minister S. Jayakumar explained that compensation had been capped using the lower of values at the acquisition date and an earlier statutory date, with further restrictions. He described the caps as contentious and proposed compensation at prevailing market value. [SG5] The Singapore Land Authority's published process now provides market-value compensation, an inquiry at which owners can submit claims, and an appeal route. [SG6] Those fiscal savings must be recorded alongside the losses borne by people surrendering property.

The fiscal saving from a compensation restriction has a distributional counterpart: some of the cost falls on those surrendering property. That is an inference from the valuation rule, and it should be reported alongside the housing delivered. Acquisition also requires attention to tenants, businesses, and residents whose claims are absent from title records. For adaptation, the World Bank's Environmental and Social Standard 5 provides a published safeguard benchmark covering displacement avoidance, replacement-cost compensation, livelihood restoration, and specified assistance to eligible people without legally recognised land claims. Its entitlements differ by claim and loss category. [SG7] These standards govern covered World Bank Investment Project Financing; they are not a description of every country's law. Housing access also depends on the shared spaces and everyday institutions created within the neighbourhood.

Shared spaces complement affordable entry

My experience in Singapore and Tharman's Brookings discussion draw attention to shared neighbourhood life alongside affordable entry. I lived in Singapore for ten years, and I remember Tharman Shanmugaratnam's public discussion of housing and inclusion at Brookings. On 30 November 2015, the then deputy prime minister said, in the institution's uncorrected transcript, "It must be attractive to the middle class and the upper middle class to want to live in that neighborhood." His argument connected mixed-income housing with public spaces that different income groups would want to use. Earlier in the discussion, he described rental blocks near larger flats in his constituency, with shared amenities. [SG8] This is a policymaker's account of design and experience. It does not establish that proximity alone raised incomes or produced friendships across social groups. My inference is that mixed-income housing needs shared services residents can actually use. Fees, inaccessible routes, or restrictive admission rules can defeat that purpose. The neighbourhood studies earlier in this chapter address the evidence on usable access and exposure. Income mixing and ethnic allocation rules nevertheless need separate evidence and distributional review.

Allocation restrictions should be assessed for their distinct objectives, household costs, remedies, and legal basis. Singapore's Ethnic Integration Policy sets limits at block and neighbourhood levels. HDB acknowledged in its 2021/22 annual report that these limits could make it harder for some owners to sell at a reasonable price, and announced buyback assistance for eligible constrained owners. [SG9] That documented cost belongs in any assessment of the policy. Cities considering ways to reduce segregation should first appraise locally lawful options such as dispersed affordable rental homes and common services. Any allocation restriction needs a distributional assessment, a remedy for affected households, and legal review. Estate technology offers a different means of improving daily conditions that should be judged through operating results.

Testing technology through estate operations

Greenprint illustrates environmental improvement of existing estates with operating teams needed to act on the technology. Singapore's Climate Action Plan records a 2012–15 pilot covering 38 blocks and about 3,200 households in Yuhua, followed by extension to Teck Ghee. This establishes implementation and reach; it does not, by itself, measure the effect of every feature on household welfare. [218] The Smart HDB Town Framework announced in 2014 brought together planning, environmental monitoring, estate operations, and in-home infrastructure. It proposed modelling combinations of solar power, lighting, greenery, water, and waste systems, and using sensors to inform maintenance. The announcement distinguished proposed trials from established results. The transferable approach is to compare combinations by their cost and performance, then test them under operating conditions. Fund the team that will respond to sensor findings before purchasing the equipment. [219] Later programmes should be assessed separately rather than attributed retrospectively to the original pilot.

Different HDB initiatives address distinct service and environmental problems and require their own maintenance arrangements. HDB's February 2025 announcement reported results from cool-coating trials on about 130 blocks in Tampines, begun in 2021, and set out an extension to existing estates by 2030. It also announced electrical sub-meters for monitoring common services such as lifts, pumps, and lighting. These are an agency-reported pilot and a rollout commitment, not proof that all estates already received the improvements. [220] Nature also has an operating role. HDB's account of its Biophilic Town Framework considers people, plants and animals, outdoor comfort, water, and soil together. Its Punggol examples treat greenery as part of neighbourhood design rather than decoration alone. [221] For another city, the practical questions are which routes need shade, where water must drain, who maintains planted areas, and whether residents can safely use them. Do not adopt landscape features without funding their maintenance. An adaptation should start with one locally defined operating problem.

In adapting these examples, I would start with one service problem and its operating budget. For overheated common space, compare shade, ventilation, and surface treatment before buying sensors. For pump failures, first establish who receives an alert, holds spare parts, and can authorise a repair. Ask the operator and residents to verify the result under actual use. The Singapore examples show an institutional approach and successive trials; they do not justify

purchasing an entire smart-town package. That operating discipline must continue through the funding of repairs and renewal.

Maintenance is part of the housing commitment

Estate management is part of the housing commitment and requires lifetime support for necessary services and renewal. The Centre for Liveable Cities documents separate operating, sinking, and lift-replacement funds, supported by resident charges and government grants. [SG10] Chapters 16 and 24 examine this division of responsibilities. The transferable requirement is to assign and fund future obligations before occupation. A city should cost routine services, major repairs, and component replacement, establish who collects and controls the money, and make accounts available to residents. A reserve percentage cannot establish adequacy without those costs and revenues. Ownership also has a defined term. A typical new HDB flat carries a 99-year lease, with ownership rights lasting for that period. [SG11] A comparison with freehold housing must account for that difference. For any adaptation, households need understandable information about tenure, resale restrictions, maintenance charges, and the treatment of major renewal. Public support should be assessed across the life of the asset, including assistance for residents unable to meet necessary service costs. Other countries can adapt these functions through their own land and tenure institutions.

Adapt the functions to local rights and incomes

India can use existing land instruments to secure public functions, provided reserved land is actually financed and operated. Ahmedabad's town-planning schemes pool and reconstitute land, deduct space for roads and public purposes, and return smaller serviced plots to owners. A 2020 World Bank case study describes consultation and valuation procedures and also notes procedural delays. [SG12] This provides an institutional example of assembling serviceable neighbourhoods through land readjustment. It does not establish that all occupants benefit. As Chapter 22 explains, adaptation requires enforceable treatment of tenants and other non-owner claimants, credible infrastructure finance, transparent valuation, and a workable dispute process. Reserved land only provides an amenity when someone finances and operates it. African upgrading programmes may perform similar functions without requiring immediate purchase of a finished flat.

Upgrading can preserve access while improving services, with protected relocation where hazards cannot be mitigated. South Africa's published upgrading programme combines community participation, basic and permanent services, and tenure security, covering both in-situ upgrading and relocation. [SG13] This is evidence of a policy design, without establishing its outcomes. Where a site can be made safe and serviced, I would compare upgrading with relocation using household costs, livelihood access, tenure protection, and municipal operating requirements. Where hazards cannot be mitigated, relocation requires the safeguards discussed above and a destination residents can use. Both approaches need a realistic account of municipal and national powers.

Municipal powers and national institutions

A municipal housing agency cannot reproduce powers and fiscal resources held by a city-state through organisational design alone. Its planning system coordinates housing, transport, utilities, and national land requirements. [SG14] CPF rests on national legislation and contribution collection from covered wage employment. [SG3] Its land-acquisition history included compensation rules that an Indian or African municipality cannot simply adopt. [SG5] Public finance also remains essential: HDB's 2023/24 report states that a government grant covers its annual deficit, while estate-management grants supplement resident charges. [SG2, SG10] My conclusion is that municipal adaptation needs a division of powers. Where local law permits, land readjustment can assemble serviced neighbourhoods, community upgrading can improve existing settlements, and contracts with funded reserves can assign maintenance duties. These substitute for particular functions of the Singapore system. A CPF-style savings mandate requires the government responsible for employment and pension legislation. Public-land transfers need the consent of the competent land authority. Where affordable resident payments and municipal revenues fall short of housing and renewal costs, any required national or state subsidy must be secured before commitments are made. A new municipal housing agency cannot create those legal powers or fiscal resources through its organisational design. The resulting neighbourhood should therefore be tested through residents' daily access and experience.

Testing social integration in daily use

Income or tenure mix is a design input, not proof of integration. Examine whether residents can use the same entrances, gardens, play spaces, schools, and community facilities on affordable terms. Compare lease conditions and service charges across tenures. Separate entrances or restricted amenities can preserve hierarchy inside a nominally mixed development. Specify equal access in allocation and management agreements, provide a remedy for exclusion, and test actual use with residents after occupation. Race, caste, religion, class, gender, and disability can affect access even where formal eligibility is identical. Test the particular barrier locally through confidential interviews, application outcomes, and lawfully designed paired application audits where appropriate. Collect sensitive identity data only with a justified purpose and protection. Do not impose a universal demographic quota or assume that living nearby eliminates discrimination. Measure safety, equal service access, resident choice, and sustained occupancy alongside the income mix. Evidence on nearby households adds a distinction between neighbourhood change and gains for the original residents.

Neighbourhood effects should be measured for existing tenants and new entrants separately. Diamond and McQuade's study of Low-Income Housing Tax Credit developments estimated nearby house-price increases of 6.5 percent in low-income neighbourhoods and decreases of 2.5 percent in higher-income areas, with different population and amenity effects. These are setting-specific neighbourhood effects, not a universal forecast or a complete measure of tenant welfare. [222] The comparison helps explain why nearby owners may respond differently to the same instrument. It does not justify excluding affordable housing from prosperous areas. Read that evidence alongside Montgomery County, Seattle, and the relocation comparisons above. Investment in a poorer neighbourhood and entry into an already prosperous

one serve different households and can both be worthwhile. Protect existing tenants during improvement, open usable opportunities to newcomers, and report both outcomes. Their interests can conflict and require an explicit distributional assessment.

4 Adapt the approach and measure who can enter and remain

Who gains and who loses

Inclusion policy should assess residents' displacement risks and the barriers they may impose on prospective entrants. Distinguish owners seeking to preserve asset values from tenants seeking security, and distinguish legitimate safety concerns from rules whose practical effect is exclusion. Future residents need representation through citywide plans, transparent capacity requirements and allocation criteria, because they cannot all attend today's neighbourhood hearing. A mixed-income programme requires agreements among owners, developers, municipalities, residents, and service providers. Owners may accept more housing in exchange for reliable services and predictable rules. Developers need viable returns and timely approvals. Municipalities need resources for additional residents. Low-income households need enforceable affordability and access. Schools and transport providers need capacity before demand arrives. None of these obligations should depend on an unrecorded future negotiation. South Africa's contrasting employment studies, discussed in Chapter 21, show why location and household circumstances must be examined rather than presumed. India's relocation evidence adds the value of networks. These findings do not identify a single universal distance threshold. They require project teams to measure usable access, lost livelihoods and care responsibilities before treating a dwelling allocation as an opportunity gain. [133,107] Technology can help identify access gaps but cannot resolve these distributive choices on its own.

Technology and access

GIS and administrative systems should inform access decisions with payment controls, logs, and remedies for error. Administrative records can track whether original residents remain. Rental audits can test whether otherwise similar applicants receive different treatment. These methods can expose exclusion that a unit-count dashboard misses. AI can help prioritise inspections or identify inconsistent records, but it must not infer a resident's entitlement from roof shape, phone use or neighbourhood identity. A model trained on historically accepted borrowers or tenants can reproduce past exclusion. Evaluate decisions for people with missing records, and provide assisted applications, explanations and independent appeal. The Ghana tax-mapping experiment in Chapter 13 found that collectors using the tool delivered more bills and collected more revenue. In their 2022 account, the authors also reported household-survey indications of modestly more bribe activity in treatment areas. These are survey-based findings from the earlier version, not audited bribe transactions or evidence that every collector acted improperly. Evaluate payment controls, decision logs and remedies alongside the information system. [115] The programme design must combine feasible supply and durable obligations with assistance addressing actual entry barriers.

Designing and expanding an inclusive programme

An inclusive programme should compare voluntary mobility, improvement, and protected redevelopment through enforceable household access. Seattle's experimental search-support evidence supports tailored help in its setting; it does not establish that information alone or an underfunded national counselling service will reproduce the result. Review landlord participation, local supply and the fiscal cost of sustained access. [108] For improvements in existing neighbourhoods, agree tenant protection, service finance and eligibility before works raise expectations and land prices. Follow original households even after they move, distinguishing voluntary moves, household changes and displacement. Where local rights cannot protect a tenant's access, price that limitation into the decision and provide a workable remedy. For unavoidable relocation, demonstrate the safety case and compare feasible alternatives. Cost interim housing, travel, school continuity, livelihood recovery and delays. A signed compensation receipt does not establish restoration of opportunity. Give households credible information and enforce the agreed terms before irreversible clearance. Accessibility and care requirements must then be tested in daily use by different residents.

Accessibility, care, and safety in daily use

Test access with residents who use the space differently. Walk routes with people with disabilities, older residents, caregivers, and women travelling at relevant times. Check entrances, gradients, toilets, lift reliability, lighting, public transport, school and care opening hours, and the cost of assistance. A dwelling can meet a floor-area standard while remaining unusable for its intended occupant. Carry the findings into drawings, allocations, service contracts, and budgets. Record who will correct a barrier and when, then ask affected residents to verify the result. Provide a confidential route for harassment and domestic-violence concerns with appropriate support providers. Allocation and inheritance arrangements also need a check of women's enforceable rights. These are proposed design and service checks; the actual technical and legal standards depend on jurisdiction. The same test should determine the feasible contribution of prosperous, well-served neighbourhoods.

Assessing the contribution of prosperous neighbourhoods

Opportunity areas should contribute usable affordable capacity under transparent safety, service, and rights constraints. High property prices alone are an inadequate definition. The metropolitan or other competent planning authority should publish proposed housing capacity by area and show the evidence behind exclusions. Set the allocation through the applicable planning process, with resident protections and a route to challenge factual errors; this book does not prescribe a universal percentage. Report three stages separately for each area: feasible additional capacity in the adopted plan; net homes completed after demolitions; and homes occupied affordably by the intended lower-income households. Add retention and access to shared services. An affluent district can permit expensive apartments without widening low-income access, while a peripheral district can supply many subsidised homes with poor access to work. Compare obligations against developable land, infrastructure capacity, demand, and public support. A cash contribution elsewhere should not automatically discharge an area's

responsibility to provide access within its own boundaries. A staged programme can establish which route delivers that access and can be repeated.

A programme for the first year and the following decade

I would use the first year to improve a few consequential decisions and test a feasible route before expanding it. The horizons below organise commitments; they are not deadlines for solving land disputes, passing legislation, or completing major networks. Emergency repairs, essential services, and lawful protection against displacement must proceed while longer-term reform is prepared. Consider workers renting rooms near an employment area. If safe extensions are lawful and services have spare capacity, the first intervention may be small-landlord approval and connection support. If the rooms are unsafe but upgrading is feasible, repairs and tenant protection take priority. If rights are disputed, resolve the relevant claims before offering collateral-based credit. Where no viable supply can serve the poorest at an affordable price, an explicit grant or operating subsidy remains necessary. The diagnosis determines which branch of the programme starts first. A short decision register should keep the responsible authority, evidence, funding, and next review visible.

Table 25.1. A conditional reform programme. Proposed sequence, to be adapted to legal powers, budgets and the diagnosed constraint.

Horizon and decision	Responsible authority	Evidence needed to advance
First year: choose a bounded intervention; establish rights, household costs and service constraints; cost the baseline and test delivery.	A designated programme lead, with finance, planning, utility and resident-protection counterparts. Use existing staff where feasible and budget missing expertise.	One workable route from application to safe occupation or restored service; recorded costs, errors, exclusions and unresolved duties. Process results do not establish long-term welfare gains.
Years two and three: expand a proven route in stages; fund connections and support for producers; adjust planning, rental or subsidy rules where the diagnosis warrants it.	The agency with the legal power to change the rule, treasury for recurring commitments, and operators responsible for services.	Repeatable delivery beyond the first team or site; viable producer finance; tenant retention, affordable charges and an operating budget. Evaluate a credible comparison where feasible.

Horizon and decision	Responsible authority	Evidence needed to advance
Years four to ten: sustain the stock and extend networks; establish durable metropolitan finance, rental management, renewal and suitable long-term funding.	Municipal and national budget authorities, utilities, housing providers and the relevant regulators.	Actual collections and service performance through successive budgets; funded replacement; net affordable occupation and household outcomes. Capital-market finance follows assets, cash flows and oversight that can support it.

Maintain a decision register and test whether expansion serves the original target households rather than narrowing eligibility to preserve results. Do not launch a citywide census or data platform simply to populate the register. Use a small diagnostic sample first; commission representative data only for questions whose answer would change the programme. Chapter 13 explains how to protect the baseline before works and announcements alter behaviour. Expansion in years two and three should test whether delivery survives ordinary staffing and budgets. Increase the pipeline by a tranche that utilities, inspectors and providers can actually serve. Compare the second tranche with the pilot on final cost, delays, errors, rents and incumbent retention. If the benefit depends on exceptional subsidy, a single facilitator, unpaid volunteer work, or temporary spare capacity, price the replacement arrangement before proceeding. Tightening eligibility to preserve a success rate is not evidence that the original programme scales. Long-term reform must then secure the institutions that outlast the project team.

Predictable revenue, networks, property administration, rental institutions, and metropolitan cost sharing sustain the results of individual projects. Secure the necessary intergovernmental or statutory agreement instead of leaving these duties to a project coordination unit. Review the programme when service costs, land prices, household incomes or the governing legal framework materially change. Publish why an intervention is continued, altered or stopped, including the effects on households already relying on it. The cases explain why this sequence is conditional. Chennai supports judging serviced neighbourhoods over years, while also checking who could wait and who left. Vietnam shows why expansion cannot repair a missing baseline retrospectively. Singapore's rental provision preceded the use of CPF savings for housing; an ownership-finance institution need not be a prerequisite for all assistance. Cape Town's small-producer support offers an administrative entry point that does not require acquiring every site. These cases do not establish a common implementation timetable. Together they support starting with a feasible route, testing its distributional effects, and securing the institutions needed for continued use. My own experience leads me to assess a programme through these continuing responsibilities and household outcomes.

My experience with IAY, sites and services in Chennai, upgrading in Vietnam, and Singapore's HDB informs how I would assess a housing programme. I would identify the

households it should serve, agree and fund the responsibilities for delivery and maintenance, and collect evidence on household outcomes before and after the investment. The final field assessment turns that approach into a decision about the households and neighbourhoods the programme will serve.

Prepare the local decision

Assess inclusion through actual entry, continued residence, usable opportunities, service funding, and the treatment of affected households. Follow eligible applicants, successful entrants, original residents, and households who leave as separate groups. Record rent, deposits, services, travel, school admission, work access, and reasons for refusal or departure. The Montgomery County and Moving to Opportunity evidence above requires measuring usable services and exposure over time; the HOPE VI comparison requires keeping entrant and incumbent outcomes distinct. Housing agencies, landlords, schools, transport operators, employers, and resident organisations should identify each barrier between eligibility and sustained residence. Ask prospective residents which locations they can actually use. Assign responsibility for search support, landlord participation, school places, recurring charges, and remedies for discrimination. Combine accessibility maps with verified vacancies, admission rules, and household preferences. Do not infer entitlement or likely success from neighbourhood identity. Test assistance through lease-up, retention, total household cost, and service access, recording people whom the system fails to place. Before expansion, check whether willing landlords, school places, transport capacity, and operating funds can support the next cohort. Publish adverse outcomes alongside gains. Use Chapter 13's call-detail-record (CDR) method only to inform area-level access checks; verify opportunities for the intended residents. Use these findings to select the first funded intervention, assign its responsibilities, and set the evidence that will determine whether it expands.

Inclusive neighbourhood policy should coordinate affordable entry, usable opportunities, resident rights, and continuing management. Successful access requires suitable homes, schools, services, and social connections that households can afford to use over time. Physical mixing alone does not provide those benefits. Cheap peripheral land, landlord refusal, displacement, and unfunded operation can defeat inclusion despite an attractive plan. An inclusionary percentage, zoning map, agency, or cooperative form is therefore an incomplete model for adaptation. preserve inclusion and service quality at tenfold and hundredfold volume, with explicit fiscal costs, institutional capacity and representation for people who are not yet residents.

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Housing policy in practice

Affordable housing policy must account for the cost of the home, transport, services, and maintenance. Households also need access to work, schools, and care.

Housing the Next Billion examines how land, planning, finance, infrastructure, construction, and long-term management affect housing costs and access. Abhas Jha treats housing as a coordination and information problem embedded in political economy, examining who makes decisions, receives benefits, and pays the costs.

The book draws on the author's experience and cases from Africa, India, and other regions. It compares results and limitations, explains the conditions behind them, and identifies what can be adapted. Worked examples, fieldwork guides, and transfer tests help readers assess costs, responsibilities, implementation capacity, and household outcomes.

AI and digital technology are assessed throughout the book by their contribution to specific decisions and the risks they create. The book is intended for policymakers, planners, investors, and practitioners designing and implementing housing programmes.

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